



सत्यमेव जयते

# CEA ANNUAL REPORT

## 2023-24



**CENTRAL ELECTRICITY AUTHORITY**  
**MINISTRY OF POWER**  
**GOVERNMENT OF INDIA**

**THE AUTHORITY (As on 31.03.2024)**

**Sh. Ghanshyam Prasad**  
Chairperson



**Sh. Ajay Talegaonkar**  
Member (E&C)



**Sh. Hemant Jain**  
Member (GO&D)



**Sh. Ashok Kumar Rajput**  
Member (Power System)



**Sh. Praveen Gupta**  
Member(Thermal)

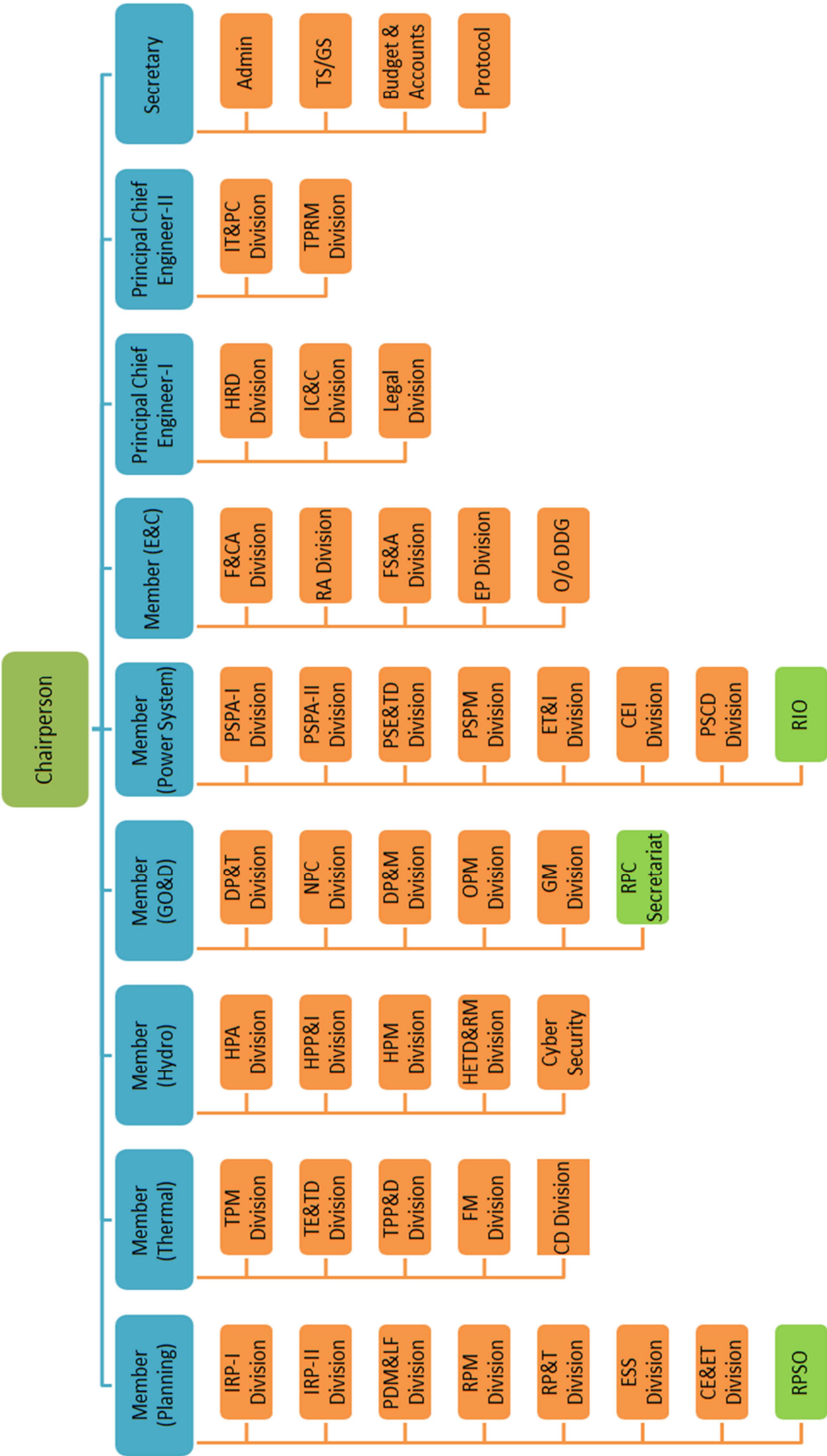


**Sh. A. Balan**  
Member (Planning)



**Sh. M. G. Gokhale**  
Member (Hydro)

ORGANISATION CHART OF CEA



**CENTRAL ELECTRICITY AUTHORITY**  
**Sewa Bhawan, R.K. Puram, New Delhi – 110066 CEA**  
**Website: [www.cea.nic.in](http://www.cea.nic.in)**

## **Sub-ordinate Offices**

### **Regional Power Committees**

- 1. Member Secretary, Eastern Regional Power Committee**, ERPC Building, 14 Golf Club Road, Tollygunge, Kolkata – 700033.
- 2. Member Secretary, Northern Regional Power Committee**, NRPC Building, 18-A, Shaheed Jeet Singh Marg, New Delhi – 110016.
- 3. Member Secretary, Southern Regional Power Committee**, 29 Race Course Cross Road, Near Anand Rao Circle, Bangaluru – 560009.
- 4. Member Secretary, Western Regional Power Committee**, Plot No. F-3, Opposite SEEPZ Complex, MIDC Area Marol, Andheri (East), Mumbai – 400093.
- 5. Member Secretary, North-Eastern Regional Power Committee**, Meghalaya NERPC Complex, 3<sup>rd</sup> Floor, Dong Parmaw, Lapulang, Shillong-793006.

### **Regional Power Survey Offices (RPSOs)**

- 1. Deputy Director, Regional Power Survey Office (East)**, ERPC Building, 14 Golf Club Road, Tollygunge, Kolkata - 700033.
- 2. Deputy Director, Regional Power Survey Office (North)**, West Block-II, Wing V, R.K. Puram, Sector-1, New Delhi– 110066.
- 3. Deputy Director, Regional Power Survey Office (South)**, Post Box No. – 38, 6th Floor, ‘F’ – Wing, Kendriya Sadan, Koramamgala, Bangaluru – 560034.
- 4. Deputy Director, Regional Power Survey Office (West)**, 5th Floor, Plot No. F-3, Opposite SEEPZ Complex, MIDC Area Marol, Andheri (East), Mumbai – 400093.

### **Regional Inspectorial Organisations**

- 1. Superintending Engineer, Regional Inspectorial Organisation (East)**, ERPC Building, 14 Golf Club Road, Tollygunge, Kolkata – 700033.
- 2. Superintending Engineer, Regional Inspectorial Organisation (North)**, NRPC Building, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi – 110016.
- 3. Superintending Engineer, Regional Inspectorial Organisation (South)**, Block-IV, Floor-III, Shastri Bhawan, Chennai – 600006.
- 4. Superintending Engineer, Regional Inspectorial Organisation (West)**, Ground Floor, WRPC Building, F-3, MIDC Area Marol, Andheri (East), Mumbai – 400093.
- 5. Superintending Engineer, Regional Inspectorial Organisation (North-East)**, NERPC Complex, 3<sup>rd</sup> Floor, Dong Parmaw, Lapalang, Shillong - 793006.

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### From the Chairperson

Electricity is one of the most vital components of infrastructure for the inclusive economic. Growth and development of the nations. The sustained growth with continuous transformation according to new challenges has been characteristic of Indian power sector. Central Electricity Authority (CEA) has played key role in this process since many decades. The journey of power sector on growth trajectory has continued in the year 2023-24 with focus on reliable, economic and quality power to all. CEA, as one of the apex organizations in the country, has been carrying out statutory functions including planning, specifying technical regulations, facilitating timely completion of schemes and advising Central Government, State Governments, Electricity Regulatory Commissions as well as other stakeholders on technical matters to ensure sustainable power sector development.

It is our pleasure to bring out this Annual Report of CEA for the year 2023-24. The Report gives an insight into the organization structure. Functions and activities of CEA highlighting the contributions made in the development of power sector in the country during the year 2023-24.

The key development during the year 2023-24 are highlighted below:-

1. The country witnessed the Peak demand of 243.271 GW during 2023-24 and the Peak met was 239.931 GW and the gap between Peak Demand and Peak Met was 1.4 % (3340 MW) only.
2. Various issues in respect of data submission procedure for First Time Charging/ ISTS connectivity and verification of Compliance to CEA (Technical Standards for connectivity to Grid) regulations related to K-factor, Short Circuit Ratio, Response time of LVRT/HVRT, Multiple Ride Through, Power quality Norms, Reactive and Active Power management etc. were resolved to maintain the safe operation, integrity and reliability of the Grid.
3. Electricity Distribution Network Planning Criteria was released by Hon'ble Minister of Power in January, 2024 for helping the DISCOMs to plan their distribution system optimally and efficiently.
4. Draft Distribution Perspective Plan-2030 to project the requirement of distribution infrastructure for the period 2022-2030 was published on CEA website for public comments in February 2024. The same has been revised upto 2035.
5. UPbhokta ADHIkar (UPADHI) portal was launched by HMoP on 11th July 2023 to serve as a dynamic, data-driven unified tool that gives a holistic assessment of performance of DISCOMs and SERCs in providing quality services to consumers.
6. The report of the Task Force for implementation of the AUFLS and df/dt scheme at National level was approved in 14th NPC held on 03.02.2024.
7. The Standard Operating Procedure (SOP) for (a) Third Party Protection System Audit (b) Grid Disturbances/Grid Incidents/Tripping's, (c) Communication Audit for Substations and (d) Communication System Outage Planning for bringing uniformity at National Level were approved in the 14th NPC Meeting held on 03.02.2024.
8. During 2023-24 "Operation and Maintenance (O&M) Guidelines and Standard Format for Memorandum of Understanding between New TSP and Existing TSP for projects to be awarded under TBCB route" were issued. These guidelines were further extended to the generating station switchyards and other than TBCB based transmission projects in ISTS substation with some modifications.
9. CEA had issued a Manual on Transmission Planning Criteria, 2023 which is effective from 1st April,

2023.

10. Guidelines for concurrence of DPR for PSPs was revised wherein timeline for concurrence of DPR of PSPs has been reduced from 125 days to 50 days;
11. Detailed Project Reports (DPRs) of 01 nos. of Pumped Storage Project (PSP) with Installed Capacity of 1350 MW, was concurred by CEA in record time u/s 8(1) of Electricity Act, 2003.
12. MoEF&CC on request of MoP/ CEA to ease the challenges faced by project developers in obtaining environment and forest clearances for Pumped Storage Schemes amended EIA Notification 2006 vide notification dated 18.05.2023.
13. PSP Guidelines were prepared and issued by MoP on 10.04.2023.
14. The Report on “The preference of selection between Variable Machine and Fixed Machine for Pumped Storage Projects” was published on CEA website in July’2023 as a guiding criteria for the Developers and End Users of PSPs on selection of one of the aforementioned available options as a type of machine for use in the PSP.
15. “CEA Guidelines for Field Efficiency Test in Hydropower Plants (including Pumped Storage Projects)” for field efficiency acceptance tests of Generating/ Pumped Storage Unit(s) comprising Generator/ Motor and Hydraulic Turbine/ Pump in Hydro Power Plants/ Pumped Storage Projects was published in June’2023.
16. National Framework for Promoting Energy Storage was issued in August 2023.
17. National Mission on Advance and High-impact Research was launched in June 2023.
18. A Side Event on “Cost effective Energy Storage System for Energy Transition and Integration of Renewable Energy” was organized on December 05, 2023 in COP 28” at Dubai, UAE.
19. The operational guidelines for the Viability Gap Funding Scheme for developing 4000 MWh of storage during the period 2023-26 were issued in March 2024.
20. Annual compilation and publication of database of CO2 emissions for all grid- connected power plants in India for FY 2022-23.
21. Guidelines for Medium and Long Term Power Demand Forecast was issued in July, 2023 to facilitate the demand forecasting exercise carried out by the power utilities & other stakeholders for achieving the objective of reliable and uninterrupted power.
22. National Electricity Plan (NEP) 2022-32 was published in May 2023. This NEP covers the review for the period 2017-22, detailed plan for the period 2022-27 and perspective plan for the period 2027-32.
23. Following studies were/are being carried out using the state of the art, sophisticated energy system planning Software “ORDENA” modeling tool:
  - a. Revised studies for the Optimal Generation Mix for the year 2029-30 was carried out and report was published in April 2023.
  - b. Resource Adequacy studies for 23 States, were carried out and the results were finalized after due consultation with the States and UTs. The detailed report on the results of the study was shared with the respective states.
  - c. A study was carried out to address the need for balancing the variation in RE power and to provide Round the Clock power to DISCOMS and report titled ‘Techno-economic Analysis of Renewable Energy Round the Clock (RE-RTC) supply for achieving India’s 500 GW Non-fossil based capacity target by 2030’ was published in Feb 2024.
24. During the year 2023-24, order for 8000 MW thermal capacity have been placed.
25. Publication of Guidelines on Design, Construction, O&M and Annual Certification of Coal Ash Ponds in June 2023.
26. Report on Ash Generation at Coal / Lignite based Thermal Power Station and its utilization in the Country

for the year 2022-23 in November 2023.

27. Statutory Guidelines for Training and Certification of Load Despatchers & Recognition of Training Institutes.

28. Guidelines for Recognition of Training Institutes in the Field of Generation, Transmission and Distribution of Electricity, Including Details of Training Curriculum for Engineers, Supervisors, and Technicians.

29. Phasing Plan for 40% Technical Minimum Load Operation for thermal generating upto the year 2030 prepared and notified in Gazzette in Dec 2023.

30. Final Report of the Committee Constituted for studying various aspects of R&M and LE of Coal Based Power plants.

Ghanshyam Prasad  
Chairperson, CEA

## CHAPTER – 1

### CEA AS AN ORGANIZATION

#### 1.1 Organization of CEA

**1.1.1** The Central Electricity Authority (CEA) is a statutory organization originally constituted under Section 3(1) of the repealed Electricity (Supply) Act, 1948 since substituted by Section 70 of the Electricity Act, 2003. CEA was founded in October, 1974 after carved out of erstwhile Central Water and Power Commission.

**1.1.2** As per Section 70(3) of the Electricity Act, 2003, the Authority shall consist of not more than fourteen members (including its Chairperson) of whom not more than eight shall be full-time Members to be appointed by the Central Government.

**1.1.3** CEA is headed by a Chairperson who as the Chief Executive of the Authority largely oversees the development of Power Sector in the country. A Secretary, appointed by the Authority with the approval of the Central Government under Section 72 of the Electricity Act 2003, assists the Chairperson in discharging of CEA's statutory functions. The Secretary also assists the Chairperson in all matters pertaining to administration and technical matters including concurrence of hydro power projects etc. There are six (6) Wings in CEA namely Planning, Hydro, Thermal, Grid Operation & Distribution, Economic & Commercial and Power System each headed by a Member of the Authority. Under each Member, there are technical Divisions, headed by an officer of the rank of Chief Engineer. At present, there are forty Divisions in CEA headquarter at New Delhi.

#### 1.1.4 Sub-ordinate offices of CEA

There are 14 subordinate offices of CEA viz. five (5) Regional Inspectorial Organizations, four (4) Regional Power Survey Offices and five (5) Regional Power Committees located in various parts of the country.

#### A) Regional Inspectorial Organization (RIO)

Under Chief Engineer (CEI) in Power System Wing, five (5) Regional Inspectorial Organization (RIO) offices, each headed by an officer of the rank of Superintending Engineer, function at New Delhi, Mumbai, Chennai, Kolkata and Shillong to inspect the HV/MV installations of the Central Government.

#### B) Regional Power Survey Offices (RPSOs)

Four (4) Regional Power Survey Offices (RPSOs), each headed by an officer of the rank of Deputy Director, function at New Delhi, Mumbai, Bengaluru and Kolkata under Chief Engineer (PS&LF) in the Planning Wing to carry out surveys to forecast the demand of power in their respective regions and are entrusted with the work of gathering information for Captive Power Plants.

#### C) Regional Power Committees (RPCs)

Five (5) Regional Power Committees (RPCs), each headed by a Member Secretary, an officer of the rank of the Chief Engineer, are functioning at New Delhi, Mumbai, Bangalore, Kolkata and Shillong to facilitate the integrated operation of the Regional Electricity Grids.

#### 1.2 Functions of CEA

The functions and duties of the Authority are delineated under Section 73 of the Electricity Act, 2003. Besides, CEA has to discharge various other functions as well under Sections 3, 8, 34, 53, 55 and 177 of the Act.

#### Section 73 - Functions and Duties of the Authority

(a) advise the Central Government on the matters

relating to the national electricity policy, formulate short- term and perspective plans for development of the electricity system and coordinate the activities of the planning agencies for the optimal utilization of resources to sub serve the interests of the national economy and to provide reliable and affordable electricity to all consumers;

(b) specify the technical standards for construction of electrical plants, electric lines and connectivity to the grid;

(c) specify the safety requirements for construction, operation and maintenance of electrical plants and electric lines;

(d) specify the Grid Standards for operation and maintenance of transmission lines;

(e) specify the conditions for installation of meters for transmission and supply of electricity;

(f) promote and assist in the timely completion of schemes and projects for improving and augmenting the electricity system;

(g) promote measures for advancing the skills of persons engaged in electricity industry;

(h) advise the Central Government on any matter on which its advice is sought or make recommendation to that Government on any matter if, in the opinion of the Authority, the recommendation would help in improving the generation, transmission, trading, distribution and utilization of electricity;

(i) collect and record the data concerning the generation, transmission, trading, distribution and utilization of electricity and carry out studies relating to cost, efficiency, competitiveness and such like matters;

(j) make public from time to time the information secured under this Act, and provide for the publication of reports and investigations;

(k) promote research in matters affecting the generation, transmission, distribution and trading of electricity;

(l) carry out, or cause to be carried out, any investigation for the purpose of generating or transmitting or distributing electricity;

(m) advise any State Government, licensees or the generating companies on such matters which shall enable them to operate and maintain the electricity system under their ownership or control in an improved manner and where necessary, in coordination with any other Government, licensee or the generating company owning or having the control of another electricity system;

(n) advise the Appropriate Government and the Appropriate Commission on all technical matters relating to generation, transmission and distribution of electricity; and

(o) Discharge such other functions as may be provided under this Act.

In addition to above functions and duties, CEA has to perform the following functions in terms of the under mentioned Sections of the Electricity Act, 2003: -

### **Section 3 - National Electricity Policy and Plan**

(1) The Central Government shall, from time to time, prepare the National Electricity Policy and Tariff Policy, in consultation with the State Governments and the Authority for development of the power system based on optimal utilization of resources such as coal, natural gas, nuclear substances or materials, hydro and renewable sources of energy.

(2) The Central Government shall publish the National Electricity Policy and Tariff Policy from time to time.

(3) The Central Government may, from time to time, in consultation with the State Governments and the Authority, review or revise the National Electricity Policy and Tariff Policy referred to in sub-section (1).

(4) The Authority shall prepare a National Electricity Plan in accordance with the National Electricity Policy and notify such plan once in five years.

Provided that the Authority while preparing the National Electricity Plan shall publish the draft National Electricity Plan and invite suggestions and objections thereon from licensees, generating companies and the public within such time as may be prescribed;

Provided further that the Authority shall –

(a) notify the plan after obtaining the approval of the Central Government;

(b) Revise the plan incorporating therein directions, if any, given by the Govt. while granting approval under clause (a).

(5) The Authority may review or revise the National Electricity Plan in accordance with the National Electricity Policy.

### **Section 8 - Hydro-Electric Generation**

(1) Notwithstanding anything contained in Section 7, any generating company intending to set up a hydro-generating station shall prepare and submit to the Authority for its concurrence, a scheme estimated to involve a capital expenditure exceeding such sum, as may be fixed by the Central Government, from time- to time, by notification.

(2) The Authority shall, before concurring in any scheme submitted to it under sub-section (1) have particular regard to, whether or not in its opinion:-

(a) the proposed river-works will prejudice the prospects for the best ultimate development of the river or its tributaries for power generation, consistent with the requirements of drinking water, irrigation, navigation, flood- control, or other public purposes, and for this purpose the Authority shall satisfy itself, after consultation with the State Government, the Central Government, or such other agencies as it may deem appropriate, that an adequate study has been made of the optimum location of dams and other river- works;

(b) The proposed scheme meets, the norms regarding dam design and safety.

(3) Where a multi-purpose scheme for the development of any river in any region is in operation, the State Government and the generating company shall co-ordinate their activities with the activities of the person responsible for such scheme insofar as they are inter-related.

### **Section 34 – Grid Standards**

Every transmission licensee shall comply with such technical standards, of operation and maintenance of transmission lines, in accordance with the Grid Standards, as may be specified by the Authority.

### **Section 53 Provision relating to Safety and Electricity Supply**

The Authority may, in consultation with the State Governments, specify suitable measures for-

(a) protecting the public (including the person engaged in the generation, transmission or distribution or trading) from dangers arising from the generation, transmission or distribution or trading of electricity, or use of electricity supplied or installation, maintenance or use of any electric line or electrical plant;

(b) eliminating or reducing the risks of personal injury to any person, or damage to property of any person or interference with use of such property;

(c) prohibiting the supply or transmission of electricity except by means of a system which conforms to the specification as may be specified;

(d) giving notice in the specified form to the Appropriate Commission and the Electrical Inspector, of accidents and failures of supplies or transmission of electricity;

(e) keeping by a generating company or licensee the maps, plans and sections relating to supply or transmission of electricity;

(f) inspection of maps, plans and sections by any person authorized by it or by Electrical Inspector or by any person on payment of specified fee;

(g) Specifying action to be taken in relation to any electric line or electrical plant, or any electrical appliance under the control of a consumer for the purpose of eliminating or reducing the risk of personal injury or damage to property or interference with its use.

### **Section 55 Use, etc. of Meters**

For proper accounting and audit in the generation, transmission and distribution or trading of electricity, the Authority may direct the installation of meters, by a generating company or licensee at such stages of generation, transmission or distribution or trading of electricity and at such locations of generation, transmission or distribution or trading, as it may deem necessary.

### **Section 177 Powers of Authority to make Regulations**

(1) The Authority may, by notification, make regulations consistent with this Act and the rules generally to carry out the provisions of this Act.

(2) In particular and without prejudice to the generality of the power conferred in sub-section (1), such regulations may provide for all or any of the following matters, mainly: -

(a) the Grid Standards under section 34;

(b) suitable measures relating to safety and electricity supply under section 53;

(c) the installation and operation of meters under section 55;

(d) the rules of procedure for transaction of business under sub-section (9) of section 70;

(e) the technical standards for construction of electrical plants and electric lines and connectivity to the grid under clause (b) of section 73;

(f) the form and manner in which and the time at which the State Government and licensees shall furnish statistics, returns or other information under section 74;

(g) any other matter which is to be, or may be, specified;

(3) All regulations made by the Authority under this Act shall be subject to the conditions of previous publication.

## **1.3 Broad Functional Areas of work of Chairperson and the Members of the Authority**

### **Chairperson**

Chairperson is the Chief Executive of the Authority.

### **Member (Planning)**

Formulation of National Electricity Plan; integrated resource planning; coordinating the activities of planning agencies for optimization of resource utilization; formulation of short, medium and long term power plans; long and short term demand forecast and sensitivity studies; material and manpower planning; surveys for power demand growth; identification and testing of co-lateral parameters for economic model for demand forecasting; collection, compilation and publication of statistics of Power Sector; securitization of resources/fuel availability and fuel efficiency with the support of emerging technologies; modernization of project management; concepts of skill development; pro-active technology forecasting approaches; research and development in Power Sector, co-ordination with multiple agencies involved in research and development activities, coordination of fuel oil/liquid fuel supplies; coal quantity and quality control; development of renewable energy resources for electricity generation etc.

### **Member (Thermal)**

Overall thermal power development in the country; updating, development and evaluation of thermal technologies; design and engineering of thermal projects; quality assurance standards and plans; preparation of model documents and standards; thermal projects investigation and ash utilization; coal, oil and gas linkages to power projects; energy conservation; energy auditing; environmental

aspects of thermal projects; monitoring of construction and stabilization of thermal projects and suggesting remedial measures to problems involved; renovation, modernization and life extension programs of thermal generating stations; making operating norms for thermal generating stations, development of Ultra Mega Power Projects (UMPPs) etc.

### **Member (Hydro)**

Overall hydro power development in the country; technical appraisal of hydro-electric projects; integrated planning for utilization of water resources; assessment of hydro potential; assistance to States on investigation and project report preparation; construction & investigation, monitoring of hydro projects and suggesting remedial measures to problems involved; updating, development and evaluation of hydro technologies; environmental aspects of hydro projects; quality assurance plans and standardization, design and engineering of hydro projects; renovation, modernization and up rating of hydro stations; co-operation with neighboring countries of Nepal, Bhutan and Myanmar for development of water resources for mutual benefits; etc.

### **Member (Power System)**

Planning and development of transmission system consistent with national power plans; studies for the purpose of appraisal of transmission projects; transmission technology development; design and engineering; standardization and preparation of model document; renovation and modernization of transmission schemes; construction monitoring of transmission projects; coordination of telecommunication system and power lines; matters related to communication, data acquisition and software support in power sector; inspection of existing electrical installations in Union Territories and Central Government Departments; investigation of accidents on electrical installations and suggesting remedial measures for their minimization and prevention etc.

### **Member (Grid Operation & Distribution)**

Formulation of policies for safe, secure and

economic operation of regional grids; integrated operation, co-ordination of five regional grids through Regional Power Committees(RPCs); monitoring of delivery of shares from Central Sector projects; intra and inter-regional exchange of power; regional energy accounting; load generation balance; investigation of grid disturbances; matters related to distribution planning, policy and regulations; monitoring of rural electrification programme and distribution schemes of the Central Government; all matters relating to power development in union territories; operation monitoring and performance review of thermal power stations; updating of maintenance procedures; generation data collection; performance analysis; maintenance monitoring etc.

### **Member (Economic & Commercial)**

Economic evaluation of power policies and projects; appraisal of tariff for Nuclear Power Stations; analysis of financial packages; financial parameters; interest during construction and completed cost; performance of power sector utilities, Examination of Power Purchase Agreement, advice and legal matters, amendments in Electricity Act, 2003 National Electricity Policy, Tariff Policy and Electricity Rules, etc.

### **Secretary**

The Secretary (CEA) appointed by the Authority with the approval of the Government of India, assists the Authority in discharge of CEA's statutory functions. The Secretary also assists the Chairperson (CEA) in all matters pertaining to administration and technical matters including techno-economic appraisal and concurrence of hydro power projects, planning of budget and expenditure control etc.

## **1.4 Personnel and Administration**

### **1.4.1 Staff strength of CEA**

The staff strength of CEA as on 31.03.2024 was 900 as against the sanctioned strength of 1209 leaving 309 posts vacant. The summarized position of staff strength is shown in the table below:

| Category                | No. of Govt. Employees | No. of Women employees In position | % age         |
|-------------------------|------------------------|------------------------------------|---------------|
| Chairperson/<br>Members | 07                     | 00                                 | 00            |
| Group-A                 | 498                    | 64                                 | 12.67%        |
| Group-B                 | 194                    | 40                                 | 20.62%        |
| Group-C                 | 201                    | 43                                 | 21.39%        |
| <b>Total</b>            | <b>900</b>             | <b>147</b>                         | <b>16.33%</b> |

#### 1.4.2 Representation of Scheduled Castes, Scheduled Tribes, OBC & Physically Handicapped Employees

| Category     | No. of Govt. Employees | No. of SC Govt. employees in position |              | No. of ST Govt. employees in position |             | No. of OBC Govt. employees in position |              | No. of Phy. H. Govt. employees in position |
|--------------|------------------------|---------------------------------------|--------------|---------------------------------------|-------------|----------------------------------------|--------------|--------------------------------------------|
|              |                        | SCs                                   | SCs%         | STs                                   | STs%        | OBCs                                   | OBCs%        | PWDs                                       |
| Group-A      | 505                    | 69                                    | 13.66        | 29                                    | 5.74        | 68                                     | 13.47        | 08                                         |
| Group-B      | 194                    | 22                                    | 11.34        | 05                                    | 2.58        | 32                                     | 16.49        | 05                                         |
| Group-C      | 201                    | 39                                    | 19.40        | 09                                    | 4.48        | 58                                     | 28.86        | 05                                         |
| <b>Total</b> | <b>900</b>             | <b>130</b>                            | <b>14.44</b> | <b>43</b>                             | <b>4.78</b> | <b>158</b>                             | <b>17.56</b> | <b>18</b>                                  |

#### 1.4.3 Representation of Physically Handicapped employees

| Group        | Total employees as on 31.03.2024 | Physically Challenged Employees |          |           |           | Percentage of Physically Challenged |
|--------------|----------------------------------|---------------------------------|----------|-----------|-----------|-------------------------------------|
|              |                                  | VD                              | HD       | OD        | Total     |                                     |
| Group A      | 505                              | 0                               | 3        | 5         | 8         | 1.58                                |
| Group B      | 194                              | 1                               | 2        | 2         | 5         | 2.58                                |
| Group C      | 201                              | 3                               | 0        | 2         | 5         | 2.49                                |
| <b>Total</b> | <b>900</b>                       | <b>4</b>                        | <b>5</b> | <b>09</b> | <b>18</b> | <b>2.00</b>                         |

## 1.5 Annual Budget

### 1.5.1 Budget of CEA during the year 2023-24.

| Rs. In Crores                                 |               |                                              |               |
|-----------------------------------------------|---------------|----------------------------------------------|---------------|
| BUDGET OF CENTRAL ELECTRICITY AUTHORITY (CEA) |               |                                              |               |
| Budget Estimate of CEA for the Year 2023-24   |               | Revised Estimate of CEA for the year 2023-24 |               |
| Salary                                        | 113.62        | Salary                                       | 115.18        |
| Non-Salary                                    | 21.41         | Non-Salary                                   | 25.74         |
| <b>Total</b>                                  | <b>135.04</b> | <b>Total</b>                                 | <b>140.92</b> |
| Allocation of Budget Estimate 2023-24         |               | Allocation of Revised Estimate 2023-24       |               |
| Admin. Of Electricity Act                     | 112.41        | Admin. of Electricity Act                    | 120.91        |
| Hydel Generation                              | 2.70          | Hydel Generation                             | 2.79          |
| Transmission & Distribution                   | 19.93         | Transmission & Distribution                  | 17.21         |
| <b>Total</b>                                  | <b>135.04</b> | <b>Total</b>                                 | <b>140.92</b> |
| Allocation after surrender 2023-24            |               | Expenditure Status as on 31.03.2024          |               |
| Salary                                        | 110.61        | Salary                                       | 110.40        |
| Non-Salary                                    | 25.15         | Non-Salary                                   | 24.93         |
| <b>Total</b>                                  | <b>135.76</b> | <b>Total</b>                                 | <b>135.33</b> |
| <b>Budget utilization %</b>                   |               |                                              | <b>99.68%</b> |

### 1.5.2 Consultancy services by CEA

CEA renders Consultancy Services for design and Engineering of thermal and hydro projects to various SEBs and power utilities. Bill raised by various SEBs and power utilities. Bill raised by various Departments / Organizations during the year 2023-24 is Rs. 2.23 Crores and payment received is Rs. 11.88 Lakhs.

### 1.6 Progressive use of Hindi in Official work of CEA

Central Electricity Authority was notified in the Official Gazette in pursuance of sub-rule 4 of Rule 10 of the Official Language Rules 1976 and under sub-rule 4 of rule 8, officers possessing proficiency in Hindi were specified to do all their official work in Hindi.

#### 1.6.1 Quarterly Meetings of Official Language Implementation Committee:

During the year following four meetings of Official Language Implementation Committee were held:

- 1st meeting - 21st Jun, 2023
- 2nd meeting – 26th Sept, 2023
- 3rd meeting – 29th Dec, 2023
- 4th meeting – 20th Mar, 2024

During these meetings, action are taken for implementation of official language policy.

- i. All works such as noting, drafting, office orders, letters, etc. in all the divisions/sections throughout the year were done as per Section 3(3) of the Official Language Act.
- ii. Full efforts were also made to meet the goal of Hindi correspondence in all divisions/sections.
- iii. All the letters received in Hindi were answered in Hindi itself. Thus, Rule 5 of the Official Language Rules, 1976 was complied with.

### 1.6.2 Letters sent in Hindi during the financial year 2023-24:

Quarterly percentage of Letters sent in Hindi during the year 2023-24 is as follows:-

| Quarter | Letters sent in Hindi Region ('A' + 'B') | % of Hindi letters |
|---------|------------------------------------------|--------------------|
| 1st     | 8219                                     | 96.97%             |
| 2nd     | 8482                                     | 97.78%             |
| 3rd     | 9461                                     | 97.93%             |
| 4th     | 10914                                    | 98.62%             |

- In the 'C' Region, the target of the letter sent in Hindi was 100 percent.

### 1.6.3 During the year, following Reports/ Documents were issued in bilingual form:

1. राष्ट्रीय विद्युत योजना के संक्षिप्त शब्द/विषय-वस्तु/मुख्य बिंदु/प्रस्तावना का हिंदी अनुवाद एवं उसमें किए गए सुधारों का हिंदी अनुवाद का योगदान। Contribution of Hindi translation of the acronyms/themes/main points/introduction of National Electricity Plan and Hindi translation of the amendments made in it.

## 2. केविप्रा की वार्षिक रिपोर्ट-2022-23

Annual Report 2022-23 of CEA

## 3. वितरण दिशा-निर्देशों का हिंदी अनुवाद

Hindi translation of distribution guidelines

## 4. 8वें सदस्य, केविप्रा के पद सृजन हेतु ड्राफ्ट कैबिनेट नोट

Draft Cabinet Note for creation of post of 8th Member, CEA

## 5. ऊर्जा संक्रमण रिपोर्ट

Energy Transition Report

## 6. केविप्रा अधिसूचनाओं का हिन्दी अनुवाद

Hindi translation of CEA Notifications

## 7. क्षेत्र दक्षता परीक्षण पर दिशा-निर्देशों का हिंदी अनुवाद

Hindi translation of guidelines on field proficiency test

## 8. उत्पादन के लिए अनुमोदित दिशा-निर्देशों का हिंदी अनुवाद

Hindi translation of approved production guidelines

## 9. अनुलग्नक प्रारूप अधिसूचना का हिंदी अनुवाद

Hindi translation of the Annexure Draft Notification

**1.6.4 Hindi Fortnight Celebrations, 2023:**

Hindi pakhwada was organized at Central Electricity Authority Headquarters from 14th September, 2023 to 29th September, 2023 as per the relevant instructions by the Department of Official Language, Ministry of Home Affairs, and Government of India. As per the instructions of the Department of Official Language, all the offices of the Central Government were to inaugurate the Hindi pakhwada from Hindi Diwas, 2023 and Third All India Official Language Conference organized in Pune (Maharashtra) on 14th and 15th September, 2023 under the chairmanship of the Honorable Union Minister of State for Home Affairs. Accordingly, 3 officers looking after the work of implementation of official language of CEA participated in the said program. Apart from them, as per the instructions of Chairman, CEA, first prize winners of last year's Hindi competition were nominated to participate in the function as an incentive. The senior-most award winning officer represented Chairman, CEA in the function.

**1.6.5 Publication of "Vidyut Vahini":-**

During the last year, the fourth, Fifth and sixth issues of CEA's official Hindi quarterly magazine "Vidyut Vahini" were published and distributed and for the third issue, the digital publication was done.

**1.7 Welfare Activities in CEA****1.7.1 Recreation and Sports:**

The employees of CEA are actively participating in the Sports & Cultural Tournaments/Competitions at All India Civil Services (A National Status), Inter-Ministry and Inter-CPSU levels every year regularly. For the year 2023-24, the following Sports Team/Individuals of CEA are participated in the AICS/Inter-Ministry/Inter-CPSU Tournaments and brought laurels to CEA by winning the medals. The achievements of these Sports/Cultural Teams/Individuals are as under:

**1. CARROM:**

CEA Men Team stood 2<sup>nd</sup> Runners-up in the Inter-CPSU Carrom Tournament 2023-24 organised by THDC at Rishikesh from 24.06.23 to 28.06.23 under the aegis of MoP. The team members are as under:

| S.No.           | Name & Designation<br>S/Shri/Smt./Km.                            | Section/Division        |
|-----------------|------------------------------------------------------------------|-------------------------|
| <b>MEN TEAM</b> |                                                                  |                         |
| 1.              | Shubham Garg,<br>Asstt. Dir.-----<br><b>(Player-cum-Manager)</b> | DM Division             |
| 2.              | Sumit Kumar,<br>Deputy Director-<br>-----<br><b>(Captain)</b>    | RPSO(W),<br>Mumbai      |
| 3.              | Saurabh Parth<br>Sarathi, Assistant<br>Director-I                | TPRM Division           |
| 4.              | Sanjeev Dhingra,<br>PPS                                          | O/o<br>Member(GO<br>&D) |
| 5.              | Sanjeev<br>Kumar Beri,<br>PPS                                    | HPP&I                   |
| 6.              | Rajesh                                                           | Protocol Office         |

|    |                                          |              |
|----|------------------------------------------|--------------|
|    | Kumar,<br>Professional<br>Assistant      |              |
| 7. | Nikul<br>Rohen,<br>Assistant<br>Director | NPC Division |

- Ms. Nisha Aatil, MTS has won the **Bronze Medal** in Women Singles Event in the Inter-CPSU Carrom Tournament 2023-24.

## 2. CHESS:

CEA Men and Women Chess Team stood Winners and Runners-up respectively in the Inter-CPSU Chess Tournament 2023-24 organized by REC Ltd. at Lucknow, Uttar Pradesh from 19.12.2023 to 22.12.23 under the aegis of MoP. The Men and Women Chess Team Members are as under:

| S.No.             | Name &<br>Designation<br>S/Shri/Smt./Km.                                        | Section/Division     |
|-------------------|---------------------------------------------------------------------------------|----------------------|
| <b>MEN TEAM</b>   |                                                                                 |                      |
| 1.                | Gurujit Medhi,<br>Deputy Director-----<br><b>(Captain)</b>                      | ESDD Division        |
| 2.                | Asif Iqbal, Deputy<br>Director                                                  | TPP&D Division       |
| 3.                | Lokesh Kumar,<br>Assistant Director-II                                          | Services<br>Division |
| 4.                | Pradeep Kumar,<br>Executive Engineer                                            | NRPC                 |
| 5.                | D. Tanoj<br>Kumar,<br>Assistant<br>Director-II-<br><b>(Playing<br/>Manager)</b> | RPSO(N)              |
| 6.                | Niteesh Kumar,<br>Steno.-----<br>---(Men Coach)                                 | O/o DS(Vig.)         |
| <b>WOMEN TEAM</b> |                                                                                 |                      |
| 1.                | Aarti Singh,<br>Deputy Director-----<br><b>(Captain)</b>                        | EPD Division         |
| 2.                | Deepa Sharma,<br>Stenographer                                                   | Admn.II<br>Section   |

|    |                                                                   |                            |
|----|-------------------------------------------------------------------|----------------------------|
| 3. | Sujata, PS                                                        | O/o<br>Member(Planni<br>g) |
| 4. | Rajesh, Deputy<br>Director-----<br>----- <b>(Women<br/>Coach)</b> | PSPM Division              |
| 5. | K. Neetu, PS -----<br>-----<br><b>(Playing Manager)</b>           | <u>CD Division</u>         |

## 3. BADMINTON:

Smt. Archana, PPS, PSPA-II Division, CEA has won the **Bronze Medal** in Women Veteran Doubles Event in the Inter-Ministry Badminton Tournament 2023-24 organized by CCSCSB, DoPT from 04.12.23 to 15.12.2023 at AIIMS Gymkhana, New Delhi.

## 4. Other Games:

Others games i.e. Inter-CPSU Athletics. Badminton, Cricket, Volleyball, Football and Kabaddi to be held shortly.

## 5. Other Recreational activities:

- To celebrate “Azadi ka Amrit Mahotsav” and “Fit India” initiative by Government of India, a Gymnasium has been set up in CEA premises at ground floor (near SBI ATM) for the officers/officials of CEA, where CEA Men and Women officers/officials have regularly use/exercise on the gym machines such as Tread Mill, Cycling, Bench press, various kg weights/dumbbells, work station, Pull string and gym balls etc. Further, proposal for procurement of some other gym machines such as Cross Trainer Machine, Bench press with steel rod, Dumbbells (5/10/15 Kgs) and weight machine is being in process for the year 2023-24.
- CEA/BEE Recreation Club has also been running at Ground Floor, West Block-II, R.K. Puram, New Delhi. At present Carrom, Chess and Table Tennis facilities are available for the recreation of CEA officers/officials during the lunch hours. For the year 2023-24, proposal of the Carrom Boards and other sports materials is being in process.
- It is also decided by the Sports & Cultural Committee of CEA to organise the SPORTS &

LAUREL FUNCTION 2023-24 to felicitate the Officers/officials of CEA, who brings the laurels to CEA by winning the medals/merit.

- d. Some observance days has also been organised in befitting manner in CEA such as A PLOG RUN (A UNIQUE WAY OF PICKING UP GARBAGE WHILE JOGGING)- a Special Campaign under Swachhta 3.0, JANJATIYA GAURAV DIWAS (a quiz has also been held during the JJGD), MISSION LIFE, CONSTITUTION DAY, Observance of “World No Tobacco Day (WNTD)” on 31<sup>st</sup> May, 2023 and Swachh Diwali Shubh Diwali signature campaign has been organised in the CEA premises.
- e. Apart from above, various sports team like Athletics, Badminton, Bridge, Carrom, Chess, Cricket, Football, Kabbadi, Table Tennis, Volleyball and Cultural has been selected every year to their participation in the Power Sports Control Board and Inter-Ministry Tournaments.

## 1.8 Vigilance Activities/Disciplinary Cases in CEA

1. The Vigilance Division, CEA is headed by Chief Vigilance Officer (CVO) and is the nodal point in Vigilance set up of the Authority and its Subordinate Offices. The Division deals with various facets of vigilance mechanism and functions for carrying out investigations into complaints, suggesting corrective measures for improving the control system, compliance of the laid down procedures and also for carrying out preventive vigilance exercise.

2. As part of preventive vigilance, the Vigilance Division conducts Periodic inspections of Subordinate offices under CEA for time to time. Scrutiny of Immovable Property Returns (IPRs) filed by Officers of CEA are being carried out by this Division at regular intervals. Workshop/training were conducted for CEA employees. Vigilance Awareness Week – 2023 was observed in Central Electricity Authority and its subordinate offices from 30.10.2023 to 05.11.2023. The Vigilance Awareness Week was observed to highlight the theme “Say no to corruption; commit

to the Nation”. “भ्रष्टाचार का विरोध करे; और राष्ट्र के प्रति समर्पित रहे”

3. Complaints received were handled and disposed of as per Government of India guidelines. As on 31.03.2024 no disciplinary case is pending in the Vigilance Division. Prescribed periodical reports are sent Central Vigilance Commission.

## 1.9 Electric Power Information Society

The Electric Power Information Society (EPIS) was established in June, 1996 under the aegis of Central Electricity Authority on no-loss-no profit basis for bringing out various CEA publications. These are also available for sale for general public.

### 1.10 Grievance Cell

Chief Engineer (ET&I) has been entrusted with the task of Grievance Officer of Ministry of Power. In this regard, the Progress Report of the grievances handled for the period 01/04/2023 to 31/03/2024 is as below:

Total grievances received during the period: **121**  
Total grievances disposed during the period: **120**

### 1.11 Right to Information Act, 2005

Under the Right to Information Act, 2005, the Chief Engineer (Coordination) acts as the Nodal Officer for RTI for CEA. 997 applications were received during the year 2023-24 (i.e. up to 31.03.2024), under the Act and were disposed of by various CPIOs in CEA. Out of 975 applications, 22 applications were received in hard copy format while the remaining were received through RTI MIS portal. Further, 49 applicants filed appeal to the First Appellate Authority (FAA), 4 of which were received in hard copy format while remaining were received through RTI MIS portal. All 49 appeals have been disposed off.

Section 4 of the RTI Act provides a comprehensive framework for promoting openness in the functioning of the public authorities. RTI suo moto disclosure in the format specified by the Central Information Commission (CIC) and approved by the Chairperson, CEA was uploaded on the website of CEA. The detailed information as per the specified format was also forwarded to CIC for

transparency audit.

Third Party Audit on “Proactive Disclosures” of CEA, as mandated by Central Information Commission for facilitation of suo moto disclosure of information under section 4 of RTI Act 2005, was conducted in the month of Sept, 2023 in CEA Head Quarter by National Power Training Institute (NPTI).

## 1.12 Parliament Questions/Assurances, VIP references

(A) Works relating to various assignments given below were carried out:

1. Parliament Questions
2. Parliamentary Assurances
3. Oral evidence
4. PMO/VIP/MOP references
5. Consultative Committees
6. Standing Committee on Energy
7. Material for Calling Attention Motion
8. Material for Economic Survey 2023-24
9. Major Achievements in Power Sector
10. Annual Report of the MOP for 2023-24
11. Material for interview of Power Minister and Secretary (power) to various press media
12. Monitor able targets for the year 2023-24 and Achievements
13. Power Ministers' Conference
14. Material for various speeches.
15. International Cooperation with various countries
16. Inputs for regional meeting relating to power matters of the regions
17. Action taken reports were prepared based on the inputs received from various divisions.
18. Niti Aayog Dashboard
19. Examination of DPRs
20. Material for President's Address to both the Houses of Parliament and Finance Minister's Budget Speech.
21. Compilation and processing of material for matters such as:
  - i) Power sector reform,
  - ii) Private Sector participation including action

taken reports, and

iii) Ministers meeting on power scenario etc.

**1.13 During the year 2023-24 (till 06.04.2024) there were three Parliament Sessions and the Admitted version of Questions were dealt with as follow**

| Sr. No. | Session              | Starred | Un-starred |
|---------|----------------------|---------|------------|
| 1.      | Monsoon Session 2023 | 11      | 94         |
| 2.      | Winter Session 2023  | 05      | 129        |
| 3.      | Budget Session 2024  | 00      | 30         |

## 1.14 Reports Prepared in Coordination

The CEA receives data regularly on various aspects of Indian Power Sector, such as generation, transmission and distribution of power. The information received is incorporated in the following regular reports:

- Summary report for Council of Ministers on important developments in Power Sector during the month.
- Monthly Executive Summary
- Annual Report of CEA

## CHAPTER – 2

### PLANNING FOR POWER DEVELOPMENT

#### 2.1 Power Planning

##### 2.1.1 National Electricity Plan

Section 3(4) of the Electricity Act, 2003 stipulates that the Authority shall prepare the National Electricity Plan, in accordance with the National Electricity Policy and notify such plan once in five years, after obtaining the approval of the Central Government.

National Electricity Plan (NEP) 2022-32 was published in May 2023. This NEP cover the review for the period 2017-22, detailed plan for the period 2022-27 and perspective plan for the period 2027-32.

##### 2.1.2 Generation Planning Studies

Following studies were/are being carried out using the state of the art, sophisticated energy system planning Software “ORDENA” modeling tool:

- i) Revised studies for the Optimal Generation Mix for the year 2029-30 was carried out and report was published in April 2023.
- ii) Resource Adequacy studies for 23 States, were carried out and the results were finalized after due consultation with the States and UTs. The detailed report on the results of the study was shared with the respective states.
- iii) A study was carried out to address the need for balancing the variation in RE power and to provide Round the Clock power to DISCOMS and report titled ‘Techno-economic Analysis of Renewable Energy Round the Clock (RE-RTC) supply for achieving India’s 500 GW Non-fossil based capacity target by 2030’ was published in Feb 2024.
- iv) A study was carried out to prepare a roadmap for greening of Andaman and Nicobar (A&N)

v) Islands and report titled ‘Study report of CEA on future roadmap for greening the A&N Islands’ was prepared.

vi) Under the Indo-Denmark collaboration, long-term generation planning studies were carried out for the formulation of the Indian Power Outlook- 2050 (IPO).

##### 2.1.3 Generation Capacity addition during the Year 2022-23 and 2023-24

- i) For the Year 2022-23, against a schedule of capacity addition of 5,410 MW, Capacity addition of 1,580 MW was achieved comprising of 120 MW Hydro, 1460 MW Thermal, and 0 MW Nuclear.
- ii) For the Year 2023-24, against a schedule of capacity addition of 18980 MW, Capacity addition of 6,864 MW was achieved comprising of 60 MW Hydro, 5404 MW Thermal, and 1400 MW Nuclear

##### 2.1.4 Participation of CEA as Committee Member /Interaction Meets etc.

- i) Collaboration with Danish Energy Agency (DEA) under ‘INDO Danish energy partnership’: CEA is jointly working with DEA in the field of energy planning, modelling and forecasting scenarios.
- ii) Chief Engineer (IRP) is Member Secretary for the committee for purchase of Resource Adequacy modelling tools.
- iii) Chief Engineer (IRP) is Member Secretary for the committee for development of Indigenous Resource Adequacy modelling tools.
- iv) Director (IRP) is member of the committee to give report on power demand/supply position and power procurement planning of Uttarakhand.

### 2.1.5 Reports Published

- i) Notified National Electricity Plan 2022-32 (Vol-I: Generation) was published in the gazette of India in May 2023. This NEP covers the review for the period 2017-22, detailed plan for the period 2022-27 and perspective plan for the period 2027-32.
- ii) Report titled 'Techno-economic Analysis of Renewable Energy Round the Clock (RE-RTC) supply for achieving India's 500 GW Non-fossil based capacity target by 2030' was published in Feb 2024.
- iii) A report on Optimal Generation capacity mix for 2029-30 (Version-2.0) published in April 2023

## 2.2 National Level Data Registry System

Section 74 of Electricity Act, 2003 and Regulation 4 & 5 of CEA (Furnishing of statistics, returns and information) Regulations, 2007, mandates every licensee, generating company, or person(s) generating electricity for its or his own use to furnish the statistics, returns or other information relating to generation, transmission, distribution, trading to CEA.

In accordance with the above provisions, a framework of National Level Data Registry System (NLDRS) has been devised to collate the statistics of the power generation projects. The framework provides for mandatory registration of each power generating unit of the country having installed capacity of 0.5 MW or above with CEA.

In order to facilitate the registration process, a web portal (<https://egen.cea.gov.in>) has been developed that is now operational in public domain. The registration with the portal is now one of the mandatory conditions for availing grid connectivity w.e.f. 20.11.2020 as per CEA Regulations "Central Electricity Authority (Technical Standards for Connectivity to the Grid) (Amendment) Regulations, 2019" issued vide Notification No. 12/X/STD (CONN)/GM/CEA/2018 dated

06.02.2019 and CEA Order No. CEA-PL-15-13(11)/1/2020-PSLF dated 05.11.2020.

## 2.3 Electricity Demand Forecasts

The electricity demand of the country is reassessed periodically, once in five years, for the medium term and long term period. The demand projection exercise is carried out by obtaining inputs from Regional Power Survey Offices located in various regions, along with data obtained from various organizations/ utilities. The electricity demand forecast is the basic input for the formulation of Developmental Plans and Programmes & Schemes concerning generation, transmission, trading, distribution and utilization of electricity.

The Volume –I report of 20th EPS covering the year-wise electricity demand projection for each Discom/State/ UT/ Region and the all-India for the years 2021-22 to 2031-32 as well as the perspective electricity demand projection for the year 2036-37 and 2041-42 was published in November, 2022.

The electricity demand projection of the National Capital Region (Volume-II of 20<sup>th</sup> EPS) was published in August 2023.

The Volume-III of 20<sup>th</sup> EPS report covering electricity demand forecasts of Mega Cities is under progress.

The report titled "Report on Twentieth Electric Power Survey of India (Econometric Method)", i.e. Volume-IV of EPS Report was published in January 2023

## 2.4 Crisis and Disaster Management Plan

CEA prepares "Disaster Management Plan" for the whole Power Sector on behalf of Ministry of Power to fulfil its obligations under the provisions of section 37 of the Disaster Management Act, 2005 and revises it on regular basis to keep it abreast with the new challenges and issues coming up with changing time. The document has recently been revised in accordance with the National Disaster Management Plan 2019 prepared by National

Disaster Management Authority (NDMA). The plan is consistent with the three landmark global agreements reached in 2015 – (i) the Sendai Framework for Disaster Risk Reduction, (ii) Sustainable Development Goals of United Nations and (iii) Climate Change Agreement (COP21) that together represent a nearly complete agenda for Disaster Risk Reduction. The plan also aims at achieving the contemporary national priorities set within Prime Minister's Ten Point Agenda for Disaster Risk Reduction.

Also, as per the Crisis Management Plan (CMP) of the Government of India prepared by the Cabinet Secretariat, each Central Nodal Ministry is required to prepare a detailed Crisis Management Plan for dealing with crisis situations falling in the areas of their responsibility. The plan indicates Ministry of Power as the nodal ministry for crisis situations arising out of disruption in generation, transmission, distribution and supply of electricity. Accordingly, CEA has also prepared "Crisis Management Plan for the Power Sector" on behalf of Ministry of Power. Apart from that, sector-specific generic documents on crisis and disaster management for thermal, hydro, renewable, transmission and distribution sector are also prepared and updated periodically by CEA. Crisis Management Plans for Cyber Security for each such sector have also been prepared separately.

These crisis and disaster management plans provide broad guidelines to the power utilities to prepare their own documents for crisis and disaster management encompassing the emergency situations to which their establishments are vulnerable. Crisis Management Plan and Disaster Management Plans for Power Sector have been reviewed in October, 2023.

## **2.5 Publications on All India Electricity Statistics – General Review & Growth Electricity Sector in India**

In fulfillment of its duties and functions under section 73 (i) & (j) and exercising powers vested

under Section 74 of the Electricity Act, 2003, CEA publishes following documents containing annual electricity statistics.

### **2.5.1 All India Electricity Statistics – General Review**

In General Review-2024, Nationwide electricity statistics relating to Generation, Transmission, Distribution, Consumption and Trading are included along with important information relating to growth of the Indian Electricity Sector, organizational structure of Electricity Supply Industry in India and reforms carried out by Utilities are incorporated.

The General Review incorporates important statistics/ data on installed capacity, electric energy generation and utilization of electric energy along with the transmission and distribution losses, per capita consumption.

This publication also contain energy utilization by various categories of electricity consumers like domestic, commercial, irrigation, industries (LV /MV, HV /EHV), public lighting, public water works, etc. The various Chapters/Tables of the publication indicate the above Information State wise/ Sector wise/ Category wise/ Mode wise etc.

In addition to the above, the GR-2024 containing information regarding Installed Capacity and generation of captive power plants (about 8408 Nos. of Industries having Captive Power Plants). General Review-2024 containing data for the year 2022-23 was published in May 2024. General Review for publishing data for the year 2023-24 is under the process of data collection and preparation.

### **2.5.2 Growth of Electricity Sector in India**

Publication titled "Growth of Electricity Sector in India from 1947-2024" was published in July, 2024 containing data up to 2022-23 and provisional /estimated data for the year 2023-24 in respect of Indian Electricity Sector. The data for these publications has been sourced from various Utilities and Non-utilities and various National & International sources. This publication illustrates the growth of vital

development indicators like installed generating capacity, electrical energy production, transmission and distribution network, captive power plants in industries and pattern of consumption of electricity etc. The important statistics have been compared with the International data with respect to some of the developed and developing nations. The publication also contains charts indicating state of basin wise and region wise Hydro Electric Potential development in the country.

The booklet contains maps and charts presenting a panoramic view of the growth of Indian Electricity Sector.

## 2.6 Implementation of initiative of Working Group III on NMEEE for retirement of old and inefficient Thermal Units

Ministry of Power, under National Action Plan on Climate Change (NAPCC) has initiated National Mission on Enhanced Energy Efficiency (NMEEE). Working Group -III under NMEEE had inter-alia recommended retirement of old and inefficient Thermal Units.

## 2.7 Standing Committee on Derating, Upgrading and Retirement of installed capacity of Generating Stations

A Standing Committee is constituted under the chairmanship of Member (Planning) for considering the proposals of de-rating, upgrading & retirement of electricity generating units. The Committee considers the performance of the units for de-rating & upgrading, analyses the performance data and the overall generation throughout the life of the plant/unit and carries out detailed scrutiny of technical parameters of proposed units.

A total of 18582.24 MW have been retired from 10<sup>th</sup> Plan onwards. Out of which 701.50 MW during 10<sup>th</sup> Plan, 2398 MW during 11<sup>th</sup> Plan, 5082.44 MW during 12<sup>th</sup> Plan and 10400.3 MW after 12<sup>th</sup> Plan (out of which 2550.38 MW during the year 2017- 18, 2409 MW during the year 2018-19, 2462.92 during 2019-20 797.50 MW during 2020-21, 1604.5 MW during 2021-22 356 MW during 2022-23 and 220 MW during 2023-24) was retired.

The list of the generating units retired during the year 2023-24 is given below: -

| Sl. No. | Name of Station/Plant | State         | Unit No. | Retired (MW)  | Retired on |
|---------|-----------------------|---------------|----------|---------------|------------|
| 1.      | PARICHHA TPS          | Uttar Pradesh | 2        | 220.00        | 29.04.2024 |
|         | <b>Total</b>          |               | <b>2</b> | <b>220.00</b> |            |

Plan wise and Fuel wise summary of retired capacity:-

| Plan            | Coal        |                 | Lignite     |            | Gas         |            | Diesel      |               | Plan wise Total |                 |
|-----------------|-------------|-----------------|-------------|------------|-------------|------------|-------------|---------------|-----------------|-----------------|
|                 | No.of Units | MW              | No.of Units | MW         | No.of Units | MW         | No.of Units | MW            | No.of Units     | MW              |
| 10th Plan       | 9           | 629.5           | 0           | 0          | 2           | 72         | 0           | 0             | 11              | 701.5           |
| 11th Plan       | 38          | 2135            | 0           | 0          | 11          | 261        | 5           | 2             | 54              | 2398            |
| 12th Plan       | 49          | 4721.5          | 0           | 0          | 7           | 205        | 9           | 155.94        | 65              | 5082.44         |
| After 12th Plan | 87          | 8979.38         | 11          | 740        | 24          | 353        | 10          | 327.92        | 132             | 10400.3         |
| <b>Total</b>    | <b>183</b>  | <b>16465.38</b> | <b>11</b>   | <b>740</b> | <b>44</b>   | <b>891</b> | <b>24</b>   | <b>485.86</b> | <b>262</b>      | <b>18582.24</b> |

## 2.8 Progress of Grid connected Renewable Energy Projects:

As on 31.03.2024, India has successfully achieved installed capacity of 190.57 GW in respect of Renewable Energy (including Large Hydro) generation projects comprising of 45.88 GW of Wind Power, 81.81 GW of Solar Power, 10.94 GW of Bio Power, 5.00 GW of Small Hydro and

46.92 GW of Large Hydro.

## 2.9 Generation from Renewable Energy Sources:

Generation from Renewable Energy (RE) Sources (including Large Hydro) and Conventional sources, for the years 2019-20, 2020-21, 2021-22, 2022-23 & 2023-24 and the share of RE to total generation for the above period, are as under:

| Year    | Conventional Generation | RE Generation (excluding Large Hydro) | Large Hydro Generation | Total Renewable Energy Generation (including Large Hydro) | Total Generation | % of RE Generation (including Large Hydro) w.r.t. Total Generation |
|---------|-------------------------|---------------------------------------|------------------------|-----------------------------------------------------------|------------------|--------------------------------------------------------------------|
|         | (MU)                    | (MU)                                  | (MU)                   | (MU)                                                      | (MU)             |                                                                    |
| 2019-20 | 1089220.31              | 138377.02                             | 161563.60              | 299940.62                                                 | 1389160.93       | 21.59                                                              |
| 2020-21 | 1075542.62              | 147247.51                             | 159065.02              | 306312.53                                                 | 1381855.15       | 22.17                                                              |
| 2021-22 | 1161826.54              | 170912.30                             | 159120.53              | 330032.83                                                 | 1491859.37       | 22.12                                                              |
| 2022-23 | 1252071.76              | 203552.17                             | 168841.17              | 372393.34                                                 | 1624465.10       | 22.92                                                              |
| 2023-24 | 1374486.34              | 225834.82                             | 138770.02              | 364604.84                                                 | 1739091.18       | 20.96                                                              |

Renewable energy generation share during the year 2023-24 was about 20.96 % of total energy generation in the country.

Year wise generation from renewable energy sources (RES) indicating the growth rates is given below:

| Year    | Generation from RE (including Large Hydro)<br>(BU) | Year-wise growth (%) |
|---------|----------------------------------------------------|----------------------|
| 2018-19 | 261.65                                             |                      |
| 2019-20 | 299.94                                             | 14.63                |
| 2020-21 | 306.31                                             | 2.12                 |
| 2021-22 | 330.03                                             | 7.74                 |
| 2022-23 | 372.39                                             | 12.84                |
| 2023-24 | 364.60                                             | -2.09                |

The graphical representation of the Source-wise All India Energy Generation and the source-wise breakup of RE generation (excluding Large Hydro), for the year 2023-24 is attached as **Annexure I**. The details of State/UT-wise and Source-wise RE Generation (excluding Large Hydro) for the year 2023-24 are given at **Annexure II**.

### **2.11 Development of Renewable Energy (RE) Sources:**

For sustainable development and economic growth, focus of the Government of India is towards de-carbonization of Indian Power Sector and shifting from the fossil fuels based energy to the renewable sources based energy which is cleaner, safer, environment friendly and more sustainable. The Renewable Planning and Technology (RP&T) Division in the Planning Wing of the CEA has been

entrusted with responsibility to assist in promotion of renewable sources of energy, development of new RE technologies, preparation of Construction Standards for RE power plants, to make Regulations and Guidelines for the smooth and rapid integration of RE in Indian Power Grid and to assist MoP, MNRE, States and other stakeholders on policy and regulatory matters. CEA is actively participating in faster development of RE sources and reduction of dependency of power sector on fossil fuels to ensure smooth energy transition and contributing to achieve India's international commitments. To achieve the aforesaid objectives, the tasks accomplished by RP&T Division of CEA during the year 2023-24 are as under

### **2.11.1 Formulation of Renewable Energy Power Plants Construction Standards:**

Central Electricity Authority (CEA), in exercise of powers conferred by subsection (2) of Section 177 of the Electricity Act, 2003, is formulating “Technical Standards for construction of RE Power Plants [(Solar Power Plants, Wind Power Plants and construction practices, facilitate smooth development of RE capacity in the country and help meet the Government of India’s target to achieve 50% of total installed capacity through non fossil fuel by 2030.

A Committee in this regard has been constituted in CEA under the Chairmanship of Member (Planning), CEA for the preparation of draft for the aforesaid Technical Standards.

### **2.11.2 RE Policy/Guidelines/references related matters:**

RP&T Division has handled various references pertaining to Policy/Regulation/Guidelines in respect of RE Sector and provided its valuable inputs on the same. Some of these are as below:

- i. Guidelines for Procurement and Utilization of Energy Storage Systems for various end-use applications including standalone storage capacity.
- ii. Waiver of Inter State Transmission System (ISTS) charges.
- iii. Renewable purchase obligations (RPO): MoP on 20.10.2023 issued Renewable purchase obligation (RPO) notification wherein RPO trajectory for wind, hydro, distributed and other renewable energy as a percentage of total consumption of electricity has been laid down for the years up to 2029- 30.

- iv. MoP on 27th Feb, 2023 has notified “Renewable Generation Obligation as per Revised Tariff Policy, 2016. As per this Order it has been decided that any generating company establishing a coal/lignite-based thermal generating station and having the Commercial Operation Date (COD) of the project on or after 1st April 2023 shall be required to establish renewable energy Generating capacity (in MW) i.e. Renewable Generation Obligation (RGO) of a minimum of forty percent (40%) of the capacity (in MW) of a coal/lignite-based thermal generating station or procure and supply renewable energy equivalent to such capacity.
- v. For promoting the use of Renewable Energy and replacing the costlier thermal/hydro power with RE, a revised scheme for “Flexibility in Generation and Scheduling of Thermal/Hydro Power Stations through bundling with Renewable Energy and Storage Power” was issued by MoP on 12th April 2022.
- vi. CERC issued a regulation dated 12.03.2022 which provide for a regulatory mechanism for treatment and settlement of deviation from schedule of drawal or injection of electricity in the interest of reliability, security and stability of the grid.
- vii. Minimum benchmark of land requirement for wind and solar-a meeting was held under Chairmanship of Member(Planning) on 23.08.2023 in which it was decided that minimum land requirement per MW for Solar and Wind standalone are 3 acre and 0.025 acre respectively.

### **2.11.3 Establishment of Renewable Energy Management Centres (REMCs) in RE Rich States and other parts of the Country:**

Ministry of Power (MoP) has entrusted the task of third party evaluation/ inspection of all commissioned/upcoming Renewable Energy Management Centres (REMCs)/EMCs across the Country to CEA.

Accordingly, RP&T Division carried out the third party evaluation/inspection in regard to REMC at Telangana and EMC at South Andaman and report for the same has been submitted to MoP in the month of April, 2023. Further, this Division is also collecting the monthly status report from all REMCs regularly.

### **2.11.4 Core Group for more precise weather data collection and forecasting for improving Renewable Energy (RE) forecast.**

A Core Group had been constituted in February, 2023 by Central Electricity Authority under the Chairmanship of Sh. H. S. Sawhney, Scientist-E from India Meteorological Department (IMD) comprising other Members from Government Organizations/ CPSUs, Indian Wind Power Association (IWPA) and Solar Power Developers Association (SPDA). This Group has been given a task to devise a methodology for more precise weather data collection and forecasting for improving Renewable Energy (RE) forecast. Subsequently, Members from IITM, Pune, NISE, NIWE and one member from CEA has been incorporated in the group.

Four meetings and three special meetings including two chaired by Chairperson, CEA of this group have already taken place during this year. NCMRWF has been entrusted the task of developing a model for accurate forecasting of all solar and wind generators. NCMRWF is being

facilitated by CEA by providing requisite real time and historical weather parameters through NLDC/RLDC/generators over FTP/API for more accurate forecast. One lecture on Numerical Weather Prediction by NCMRWF was also held in CEA. The specifications for new AWS and additional features required in existing AWS were also discussed in details. Stress was given on restoration of existing non-working AWS installed by NIWE.

### **2.12 Works related to Clean Energy, Energy Transition and International Matters**

CE&ET Division was formed in CEA keeping in view of the increasing importance of promotion of **Clean Energy and Energy transition from fossil fuel to non-fossil fuel** based Power.

In this pursuit, the reports published by CE&ET division are as follows: -

- i. Annual CO<sub>2</sub> baseline data-base for FY 2022-23.
- ii. Monthly power consumption details of Electric Vehicle- Public Charging Stations (PCS).

Brief details of these reports are given below:

#### **2.12.1 CO<sub>2</sub> baseline data base for FY 2022-23:**

CEA has been compiling and publishing database containing data on CO<sub>2</sub> emissions for all grid- connected power stations in India since 2006.

#### **Purpose of CO<sub>2</sub> baseline database:**

- i. Projects involving CO<sub>2</sub> displacement or CO<sub>2</sub> saving relating to grid electricity must calculate their emission reduction based on a grid emission factor, which needs to be determined in accordance with the rules set by CDM Executive Board.
- ii. Emission reductions from CDM projects in the power sector are calculated based

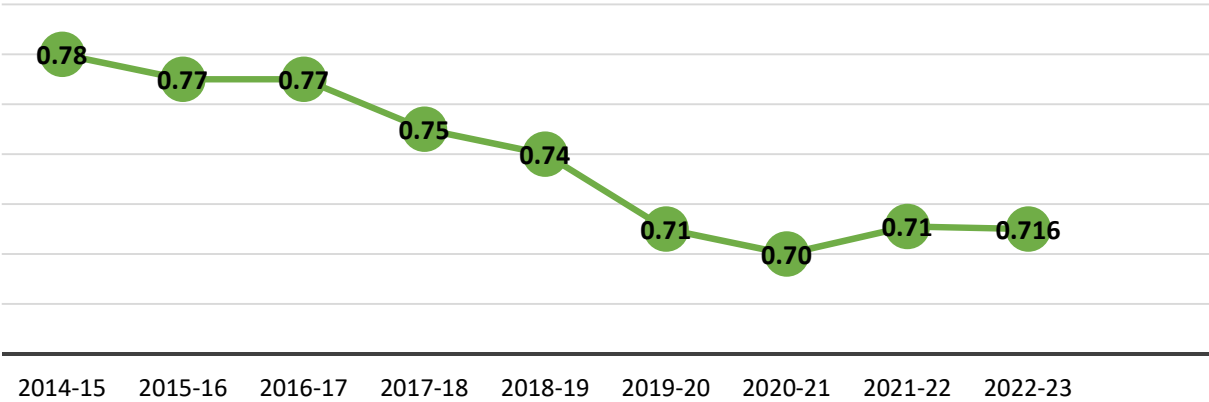
on the net electricity generated by the project and the difference between the emissions factors (in tCO<sub>2</sub>/MWh) of the baseline and the project activity.

- iii. The purpose of the database is to establish authentic and consistent quantification of the CO<sub>2</sub> emission baseline, which can be readily used by CDM project developers in the Indian power sector. This would enhance the acceptability of Indian projects and would expedite the clearance/approval process.

**Key points from CO<sub>2</sub> baseline database for FY 2022-23:**

**Emission Intensity Reduction:** - There is a gradual increase in total electricity generation (including RE generation) as well as total CO<sub>2</sub> emission from 2014-15 to 2022-23 i.e. around 47.4% increase in generation and 35.5% increase in total CO<sub>2</sub> emission. However, due to increasing share of RE in the Grid, the carbon intensity of grid is reducing. It can be seen that there is a significant decrease of about 9% (from 0.78 in 2014-15 to 0.71 in 2022-23) in average carbon emission of grid electricity in India due to RE.

Carbon Emission factor of Grid Electricity (including RE) (tCO<sub>2</sub> /MWh)



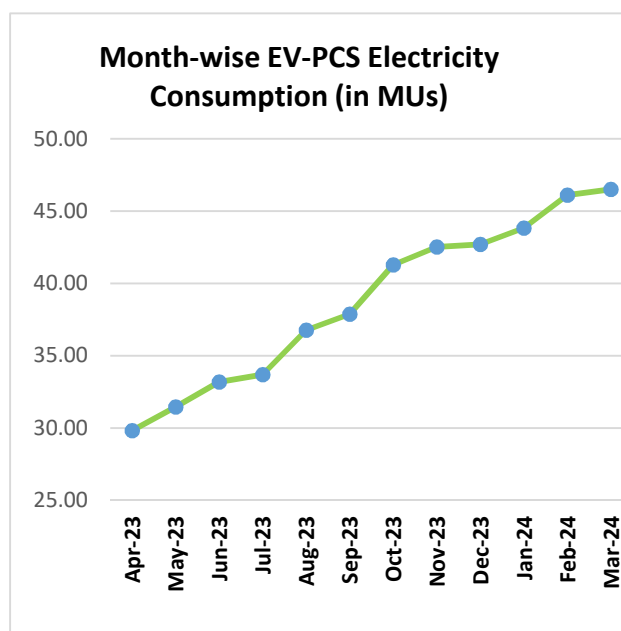
### 2.12.2 Collection of monthly power consumption details of Electric Vehicles-Public Charging Stations (PCS) and publication of Monthly Report.

The data of energy consumption in the EV charging stations is important for the load forecasting, distribution system planning and integrated resource planning in the country. Hence, CEA is collecting data from all DISCOMs regarding monthly power consumption details of Public Charging Stations (PCS) under their area of supply. Currently, EV Power Consumption report covers data received from 33 major DISCOMs and the follow up with the remaining is in progress.

As per information received so far, total power consumed by PCS of various DISCOMs during the period from 01.04.2023 to 31.03.2024 is 465.85 Million-units (MUs).

Month-wise power consumption details are tabulated below: -

| Month         | Total Electricity Consumption (in MU) |
|---------------|---------------------------------------|
| Apr-23        | 29.84                                 |
| May-23        | 31.46                                 |
| Jun-23        | 33.20                                 |
| Jul-23        | 33.70                                 |
| Aug-23        | 36.77                                 |
| Sep-23        | 37.89                                 |
| Oct-23        | 41.29                                 |
| Nov-23        | 42.53                                 |
| Dec-23        | 42.71                                 |
| Jan-24        | 43.83                                 |
| Feb-24        | 46.11                                 |
| Mar-24        | 46.51                                 |
| <b>Total*</b> | <b>465.85</b>                         |



*\*The actual consumption is more than the given figure as consumption details of few utilities are not received yet.*

### 2.12.3 Details of other Activities/ Reports:

#### i. Monthly Environmental Emission data of all Coal/Lignite/Gas/Diesel/Naptha Based Thermal Power Plant

CEA has notified "Central Electricity Authority (Furnishing of Statistics, Returns and Information) Regulations, 2007" on 19th April, 2007 under Section 74 and Sub-Section (3) of Section 177 of Electricity Act, 2003 to collect and record the data concerning generation, transmission, trading, distribution and utilization of electricity and to carry out studies relating to environmental issues and

competitiveness and such like matters.

CEA, in fulfillment of its statutory obligations, collects and compile the monthly environment emission data from thermal power plants as per FORMAT-60. The purpose of this report is to analyze the quantum of emission of Pollutant such as PM, SO<sub>x</sub>, NO<sub>x</sub>, Hg and Specific water consumption from various units of Indian Thermal Power Stations, while ensuring the compliance of the norms specified by MoEF&CC vide notification dated 07.12.2015 under Environment (Protection) Act, 1986 is the domain of CPCB/SPCB.

Such analysis will help in identification of major pollution emitter units so that concerned power producers can be sensitized to take immediate corrective and necessary action to reduce the emission within the norms. Thus it helps in achieving the broader objective of ensuring clean & healthy environment and reducing global warming.

## **ii. Verification of NO<sub>x</sub> emission data in 9 identified thermal power plant units w.r.t. Compliance of Hon'ble Supreme Court Orders in W.P.(c) No. 13029 of 1985**

MoEF&CC has notified NO<sub>x</sub> emission norms of 100 mg/Nm<sup>3</sup> for thermal units commissioned after December 2016 in the Environmental Norms on December 2015. The matter is pursuant to the Hon'ble Supreme Court's order dated 29.07.2020 in the matter of W.P. (c) No. 13029 of 1985 titled M.C. Mehta V/s Union of India and Ors. In this regard a Joint meeting with representatives of CAQM, MoEF&CC, MoP, CPCB and NTPC Ltd.

was held under the Chairmanship of Member Technical, CAQM on 09.09.2022 wherein it was decided to constitute a joint committee of 2 members each from MoP (including CEA), CPCB and concerned SPCB. This committee would visit 11 units in 9 Nos. of thermal power plants (TPPs) to verify & validate NO<sub>x</sub> emission level. The joint committee has submitted its report to CAQM of NO<sub>x</sub> emission analysis of 11 Units in 09 TPPs on 01-05-2023.

## **iii. Report on SCR operation status of Unit-10, 660 MW, Harduaganj Thermal Power Plant**

On the issue of finalizing the NO<sub>x</sub> emission norms for Thermal Power Units commissioned after 1st January 2017, several Joint Committee meetings of CAQM, MoEF&CC, Ministry of Power, CPCB and NTPC has been conducted under CAQM. In 6th Joint Committee meeting, held on 27th June, 2023 wherein Member Secretary, CAQM stated that CEA should get NO<sub>x</sub> emission data with SCR Operation for Harduaganj power plant Unit-10, where India's 1st SCR is installed and come out with interpretations based on the data collected it and other 11 units where NO<sub>x</sub> emission sampling exercise was conducted by a team of CEA and CPCB/SPCB.

CEA along with CPCB & NTPC officials visited Unit-10 Harduaganj TPP, for examining the status of SCR operation and verification of NO<sub>x</sub> data through digital analyzer/ manual sampling, based on the visit observations and the NO<sub>x</sub> emission data provided by Harduaganj TPP, CEA has submitted a detailed report to CAQM on 23.08.2023.

Sample photos regarding SCR operational Status taken during the site visit of Unit-10, Harduaganj TPP is depicted below: -



Figure 1: Front View of SCR installed at Unit-10 HTPS



Figure 2: Side View of SCR installed at Unit-10 HTPS



Figure 3: Ammonia Storage Tank at Ammonia Handling Plant for injection into SCR System at Unit-10 HTPS



Figure 4: NOx sampling with portable Analyser



Figure 5: Choking/Freezing issue in Ammonia Handling Plant

#### iv. Energy Transition from Fossil fuel to Non-Fossil:

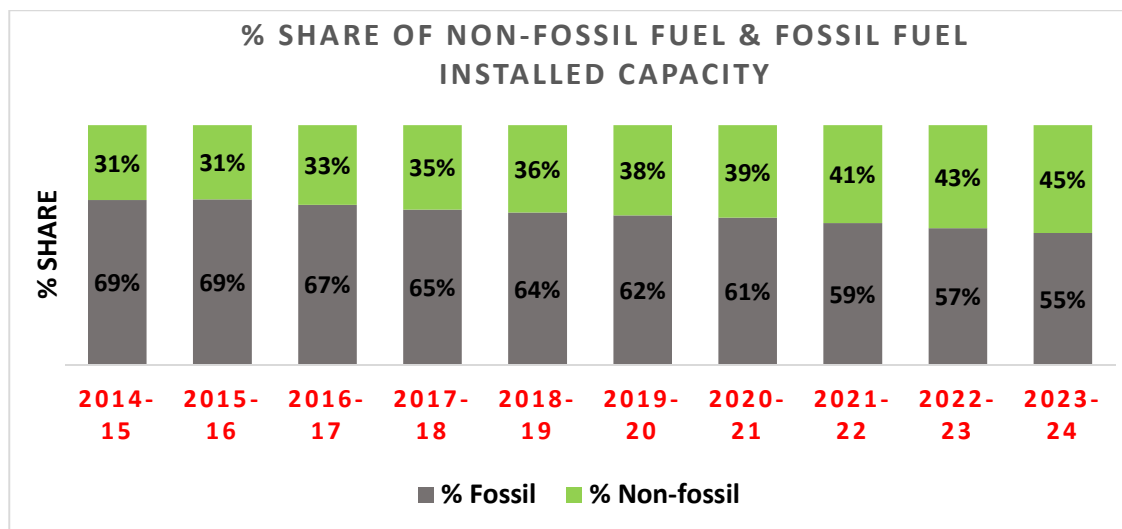
1) At COP-21, as part of its intended Nationally Determined Contributions (NDCs), India had committed to achieve 40% of its installed electricity capacity from non-fossil energy sources by 2030. However, India has overachieved this commitment almost nine years ahead of its commitment in Nov-2021. The share of solar and wind in India's energy mix have grown phenomenally. As on 31.03.2024, the non-fossil fuel installed capacity is about 45% (198.75 GW out of total 441.97 GW).

2) India communicated its **updated NDC** to UNFCCC in August 2022, which inter-alia includes-

- 'LIFE' – Lifestyle for Environment
- Reduction in Emissions Intensity of its GDP by 45 percent by 2030, from 2005 level.
- To achieve about 50 percent cumulative electric power installed capacity from non-fossil fuel-based energy resources by 2030, with the help of transfer of technology and low-cost international finance including from Green Climate Fund (GCF).
- To create an additional carbon sink of 2.5 to 3 billion tonnes of CO<sub>2</sub> equivalent through additional forest and tree cover by 2030.

3) A depiction chart of Year-wise % share of Non-Fossil based Power is tabulated below: -

| Year         | 2014-15 | 2015-16 | 2016-17 | 2017-18 | 2018-19 | 2019-20 | 2020-21 | 2021-22 | 2022-23 | 2023-24 |
|--------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| % fossil     | 69%     | 69%     | 67%     | 65%     | 64%     | 62%     | 61%     | 59%     | 57%     | 55%     |
| % Non fossil | 31%     | 31%     | 33%     | 35%     | 36%     | 38%     | 39%     | 41%     | 43%     | 45%     |



#### v. Monitoring and monthly reporting of Biomass co-firing status as per policy formulated by CAQM to abate air pollution in Delhi NCR

Every year, farm-stubble burning is one of the important factors of air pollution across the country especially in Delhi-NCR. Ministry of Power vide order dated 08.10.2021 has issued

revised policy on Bio-mass Utilization for Power Generation through Co-firing in Coal based Power Plants to use 5-10% blend of biomass pellets made, primarily of agro-residue along with coal after assessing the technical feasibility viz. safety aspect etc. Further, revised policy on 16.06.2023, for the use of agro residue-based biomass and mandated usage of minimum 5% biomass co-fired along with coal for thermal power plants w.e.f. F.Y. 2024-25.

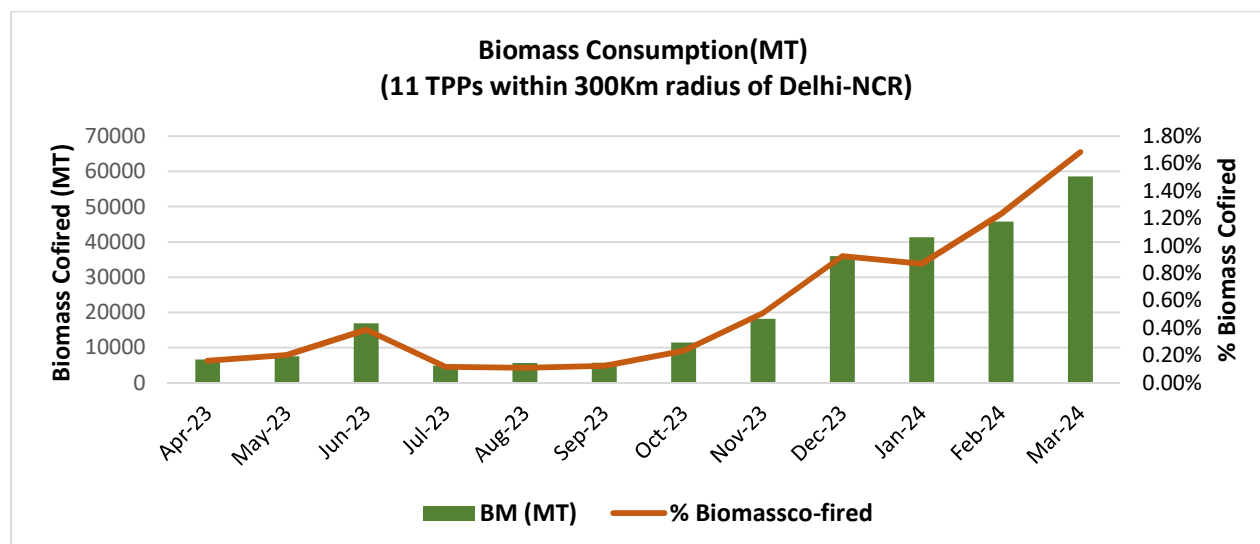
In order to address the issue of air pollution and to reduce carbon footprints of thermal power

generation, CEA under MoP has monitored and furnished monthly progress report of 11 TPPs within 300 km of Delhi-NCR on compliances of Biomass cofiring direction for onward submission to CAQM.

According to data collected for the period from 01.04.2023 to 31.03.2024, cumulative 258772 MT biomass has been co-fired in 11 TPPs (within 300 km of Delhi-NCR), out of which Indira Gandhi Super Thermal Power Plant and Mahatma Gandhi Thermal Power Plant have contributed to 1.7 % and 1.5 % respectively.

**Month-wise progress of Biomass consumption and Co-firing percentage is tabulated below:-**

| Month      | Apr-23 | May-23 | Jun-23 | Jul-23 | Aug-23 | Sep-23 | Oct-23 | Nov-23 | Dec-23 | Jan-24 | Feb-24 | Mar-24 | Total  |
|------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| BM (MT)    | 6691   | 7611   | 16878  | 4954   | 5702   | 5738   | 11474  | 18146  | 35958  | 41350  | 45737  | 58533  | 258772 |
| Coal (kMT) | 4138   | 3718   | 4334   | 4198   | 5124   | 4611   | 4885   | 3537   | 3849   | 4717   | 3664   | 3419   | 50193  |



vi. The Report on visits to Thermal Power Plants to verify Conversion from Once

through Cooling System (OTCS) to Closed Cycle Cooling System (CCTS)

A committee was formed in June 2021 under the Chairmanship of Chief Engineer, CE&ET Division (erstwhile TPE&CC) in which Director level officers from CEA (F&CA, TE&TD, TPRM, CD, and CE&ET) were the members and the purpose of the committee was to take objective based assessment on case-to-case basis for granting exemption from OTCS to CCCS conversion considering water conservation, useful

life of the plant, energy impact including technical feasibility and economic viability.

This committee held six committee meetings and carried out multiple site visits and verification to the thermal power plants. The committee completed its work and submitted its final Report on the matter to MoP on 16.01.2024 which was further forwarded to MoEF&CC.

**A short glimpses of visit to Thermal Power Plants for OTCS to CCCS site verifications:**



**vii. International and Bilateral Co-operation for development of affordable, reliable & sustainable Power Sector by way of promoting Clean Energy and Energy Transition in the country.**

1. Inputs were provided to MoEF&CC regarding COP-28 presidential action agenda, Climate change data Booklet for COP 28, draft Pledge on renewable and energy efficiency and various Declarations/Statements, which lay out India's vision and approaches towards reaching net-zero by 2070. The final report was submitted by MOEF&CC to the United Nations Framework Convention on Climate Change (UNFCCC) under Paris Agreement.
2. Inputs/suggestions were furnished to MoP & MoEF&CC regarding matters on bilateral/multilateral co-operation viz. India-

Czech, Shanghai Cooperation Organization (SCO) Member States Energy Cooperation Development, Indo-Pacific Economic Framework for Prosperity (IPEF)-Trade Pillar, Pillar-III (Clean Economy) and Clean Electricity Initiative (ICEI), Major Economies Forum Leaders Meeting on Energy, agenda of annual Asian Development Bank (ADB) meeting, Climate Investment Fund Accelerated Coal Transition Intermediate Mission jointly by World Bank (WB), Asian Development Bank (ADB) and International Finance Corporation (IFC).

3. On the direction of Hon'ble Minister of Power, a pledge on reducing per capita GHG emission is prepared and furnished to MoP. Moreover, inputs and detailed state wise power sector data were furnished to Nodal Ministry for preparation of national developmental indices viz. Network Readiness Index by NITI Aayog.

#### **viii. Participation in International Co-operation on Energy Transition**

1. Chairperson, CEA along with officers of CEA participated in meeting of Energy Transitions Working Group (ETWG) of G20 and contributed in Priority Area -1 (PA 1) –

Energy Transition through Addressing Technology Gaps and PA-3- Energy Security and Diversified Supply Chains and associated with World Bank in preparing presentation on Grid Interconnection for energy security.

2. CEA presented a concept paper on “Transnational Grid Connection for ensuring Energy Security during ETWG meeting held in Goa. The decisions taken in G20 ETWG were incorporated in the G20 New Delhi Leaders Declaration.

3. Chief Engineer, CE&ET Division, CEA participated in Dubai Climate Change Conference (COP 28) in November 2023, 8th Annual BRICS meeting in Johannesburg, South Africa in August 2023 and Bonn Climate Change Conference in June 2023.

4. With the efforts made by the Indian negotiation team in COP28 held in Dubai, the long-term interest of the country in terms of energy security and affordability of electricity were protected and the decision was made more equitable by incorporating all fossil fuels for emission reduction in place of coal only. CEA representative participated in the discussions on Global Stock Take, Technology Development, Response Measures and Just Transition.

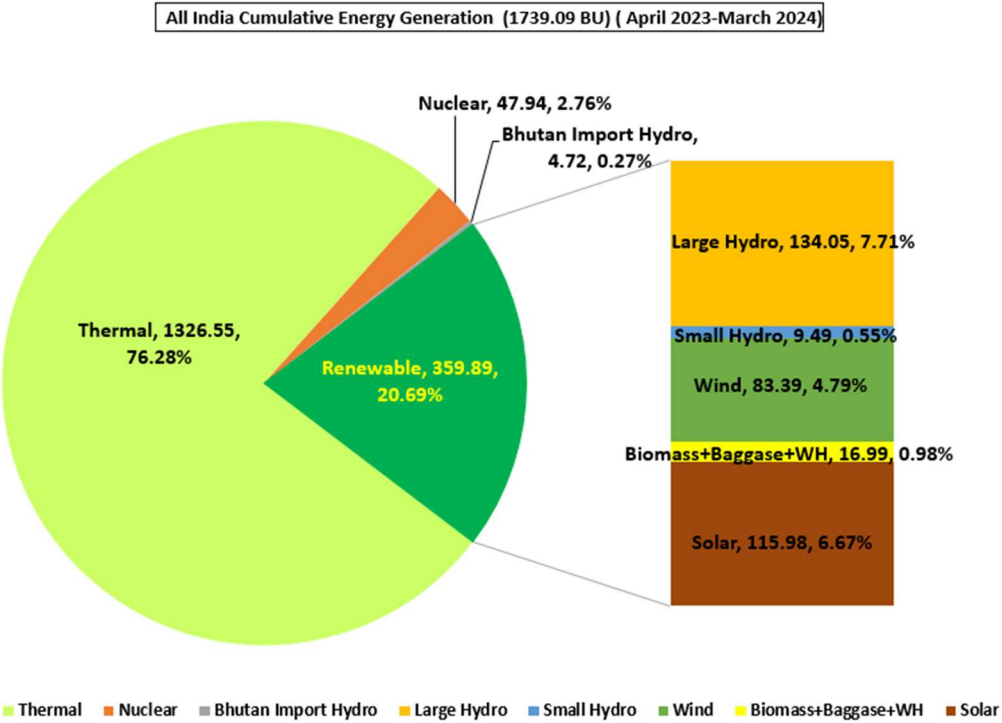


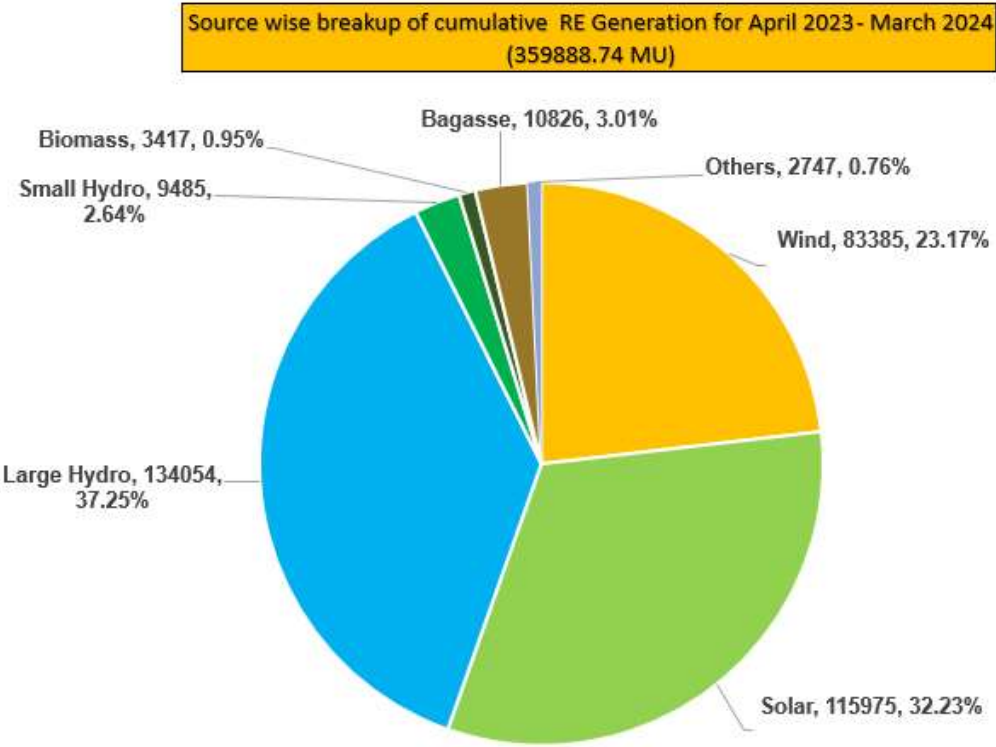
Shri Ghanshyam Prasad, Chairperson CEA & Shri Vijay Menghani (CE, CEA) at G20-ETWG Meeting, 2023



Indian Delegation at COP

Annexure-I





## Annexure-II

| Name of State/UT                       | Wind            | Solar           | Biomass        | Bagasse        | Small Hydro    | Others        | Total Renewable Energy |
|----------------------------------------|-----------------|-----------------|----------------|----------------|----------------|---------------|------------------------|
| Chandigarh                             | 0.00            | 11.70           | 0.00           | 0.00           | 0.00           | 0.00          | 11.70                  |
| Delhi                                  | 0.00            | 206.53          | 0.00           | 0.00           | 0.00           | 522.28        | 728.81                 |
| Haryana                                | 0.00            | 992.91          | 294.12         | 83.60          | 222.05         | 58.82         | 1651.50                |
| Himachal Pradesh                       | 0.00            | 59.54           | 0.00           | 0.00           | 2526.98        | 0.00          | 2586.52                |
| Jammu & Kashmir                        | 0.00            | 0.00            | 0.00           | 0.00           | 408.69         | 0.00          | 408.69                 |
| Ladakh                                 | 0.00            | 0.00            | 0.00           | 0.00           | 0.00           | 0.00          | 0.00                   |
| Punjab                                 | 0.00            | 2673.99         | 613.44         | 197.99         | 636.97         | 0.00          | 4122.40                |
| Rajasthan                              | 8390.67         | 38363.28        | 387.55         | 0.00           | 7.45           | 0.00          | 47148.96               |
| Uttar Pradesh                          | 0.00            | 3971.31         | 46.65          | 2923.55        | 175.24         | 84.85         | 7201.59                |
| Uttarakhand                            | 0.00            | 331.80          | 0.00           | 248.52         | 350.62         | 0.00          | 930.94                 |
| <b>Northern Region</b>                 | <b>8390.67</b>  | <b>46611.07</b> | <b>1341.76</b> | <b>3453.66</b> | <b>4328.01</b> | <b>665.95</b> | <b>64791.12</b>        |
| Chhattisgarh                           | 0.00            | 943.75          | 1368.61        | 19.54          | 145.54         | 0.00          | 2477.44                |
| Gujarat                                | 24794.50        | 13468.91        | 0.00           | 2.13           | 217.68         | 0.00          | 38483.22               |
| Madhya Pradesh                         | 4949.78         | 4025.19         | 84.62          | 97.01          | 469.60         | 28.83         | 9655.02                |
| Maharashtra                            | 8228.97         | 5814.13         | 304.61         | 3495.82        | 888.48         | 33.40         | 18765.41               |
| Dadra and Nagar Haveli & Daman and Diu | 0.00            | 15.74           | 13.12          | 0.00           | 0.00           | 0.00          | 28.86                  |
| Goa                                    | 0.00            | 59.99           | 0.00           | 0.00           | 0.00           | 7.96          | 67.95                  |
| <b>Western Region</b>                  | <b>37973.25</b> | <b>24327.70</b> | <b>1770.96</b> | <b>3614.50</b> | <b>1721.30</b> | <b>70.19</b>  | <b>69477.90</b>        |
| Andhra Pradesh                         | 8644.00         | 8300.03         | 18.75          | 66.63          | 127.10         | 307.97        | 17464.48               |
| Telangana                              | 304.63          | 6884.68         | 10.57          | 95.08          | 58.87          | 155.29        | 7509.10                |
| Karnataka                              | 10950.20        | 15404.09        | 47.45          | 2754.06        | 1370.76        | 0.00          | 30526.55               |
| Kerala                                 | 214.53          | 1195.28         | 0.00           | 78.08          | 716.31         | 0.04          | 2204.24                |
| Tamil Nadu                             | 16908.08        | 11737.48        | 129.14         | 622.61         | 206.00         | 0.00          | 29603.31               |
| Lakshadweep                            | 0.00            | 0.09            | 0.00           | 0.00           | 0.00           | 0.00          | 0.09                   |
| Puducherry                             | 0.00            | 12.24           | 0.00           | 0.00           | 0.00           | 0.00          | 12.24                  |
| <b>Southern Region</b>                 | <b>37021.43</b> | <b>43533.89</b> | <b>205.91</b>  | <b>3616.46</b> | <b>2479.03</b> | <b>463.31</b> | <b>87320.02</b>        |
| Andaman & Nicobar                      | 0.00            | 27.50           | 0.00           | 0.00           | 12.00          | 0.00          | 39.50                  |
| Bihar                                  | 0.00            | 195.19          | 0.00           | 140.98         | 5.92           | 0.00          | 342.08                 |

|                             |                 |                  |                |                 |                |                |                  |
|-----------------------------|-----------------|------------------|----------------|-----------------|----------------|----------------|------------------|
| <b>Jharkhand</b>            | 0.00            | 17.64            | 0.00           | 0.00            | 5.52           | 0.00           | 23.16            |
| <b>Odisha</b>               | 0.00            | 757.69           | 96.07          | 0.00            | 407.97         | 0.00           | 1261.72          |
| <b>Sikkim</b>               | 0.00            | 0.00             | 0.00           | 0.00            | 12.35          | 0.00           | 12.35            |
| <b>West Bengal</b>          | 0.00            | 168.32           | 2.49           | 0.00            | 204.46         | 1545.13        | 1920.39          |
| <b>Eastern Region</b>       | <b>0.00</b>     | <b>1166.33</b>   | <b>98.55</b>   | <b>140.98</b>   | <b>648.22</b>  | <b>1545.13</b> | <b>3599.20</b>   |
| <b>Arunachal Pradesh</b>    | 0.00            | 1.89             | 0.00           | 0.00            | 0.66           | 0.00           | 2.55             |
| <b>Assam</b>                | 0.00            | 316.31           | 0.00           | 0.00            | 64.20          | 0.75           | 381.26           |
| <b>Manipur</b>              | 0.00            | 7.73             | 0.00           | 0.00            | 0.00           | 1.23           | 8.96             |
| <b>Meghalaya</b>            | 0.00            | 0.00             | 0.00           | 0.00            | 66.55          | 0.00           | 66.55            |
| <b>Mizoram</b>              | 0.00            | 3.19             | 0.00           | 0.00            | 95.93          | 0.00           | 99.11            |
| <b>Nagaland</b>             | 0.00            | 0.00             | 0.00           | 0.00            | 81.14          | 0.00           | 81.14            |
| <b>Tripura</b>              | 0.00            | 7.01             | 0.00           | 0.00            | 0.00           | 0.00           | 7.01             |
| <b>North Eastern Region</b> | <b>0.00</b>     | <b>336.12</b>    | <b>0.00</b>    | <b>0.00</b>     | <b>308.48</b>  | <b>1.98</b>    | <b>646.59</b>    |
| <b>All India Total</b>      | <b>83385.35</b> | <b>115975.11</b> | <b>3417.19</b> | <b>10825.59</b> | <b>9485.04</b> | <b>2746.55</b> | <b>225834.83</b> |

## CHAPTER – 3

### POWER SYSTEM PLANNING AND DEVELOPMENT

#### 3.1 Transmission Planning

All issues relating to planning and development of Transmission System in the country are dealt in the Power System Wing of CEA. This includes evolving long term and short term transmission plans in coordination with central, state transmission utilities and generating companies. The network expansion plans are optimized based on power system studies. This also involves formulation of specific schemes, evolving a phased implementation plan in coordination with the Central and State transmission utilities and their implementation, issues pertaining to development and strengthening of national power grid in the country. Transmission planning studies are being conducted to identify power evacuation system from generation projects and to strengthen the transmission system in various regions.

In addition to this, Member (Power System) has been appointed as Designated Authority for facilitating the process of approval and laying down the procedure for import/ export of electricity under the guidelines for Import/Export (Cross Border) of Electricity, 2018. Designated Authority (DA) has been entrusted the role to coordinate with the respective authorities of the neighbouring countries for all purposes as stated in the Guidelines which, inter alia, include, planning, monitoring and commissioning of transmission lines for import/export of electricity as well as the grid security, safety and operation; accord approval to the participating entity(ies) proposing to import/export electricity from/ to India [including through Indian power exchange(s)]; and accord approval to an Indian generating station, supplying electricity exclusively to neighbouring country and proposing to build a dedicated transmission line for connecting to the transmission system of the neighbouring country.

The Electricity (Transmission System Planning, Development and Recovery of Inter State Transmission Charges) Rules, 2021, notified by Ministry of Power on 01<sup>st</sup> October, 2021, inter alia provides that;

*“3(1) The Central Electricity Authority shall draw up short term plan every year on rolling*

*basis for upto next five years and perspective plan every alternate year on rolling basis for next ten years for development of the electricity system and co-ordinate the activities of the planning agencies for the optimal utilisation of resources to sub serve the interests of the national economy and to provide reliable and affordable electricity in accordance with section 73 of the Act.*

*3(2) The Central Electricity Authority shall also draw up the perspective plan for development of transmission system after consultation with all the relevant stakeholders such as Central Transmission Utility, State Transmission Utilities, System Operators, generating and distribution companies, industry associations and the State Governments, etc., and after assessing the rate of growth in demand as well as the growth of generation in different areas of country.”*

Accordingly, for preparation of short term and perspective plan and for coordinating the activities of planning agencies, CEA vide letter dated 25.10.2022, constituted five (05) number of regional standing committees namely “Standing Committee on Short Term & Perspective Power System Planning (SCSTPPSP)” one for each region.

#### 3.2 Inter-regional transmission system in India – National Grid.

The national grid in the country has been developed in a phased manner. All the regional grids have been inter-connected synchronously to form One Nation-One grid -One frequency. Inter-regional transmission capacity by the end of the 9<sup>th</sup> plan was 5,750 MW which increased to 13,450 MW by the end of the 10<sup>th</sup> plan and to 27,150 MW and 75,050 MW by the end of the 11<sup>th</sup> and 12<sup>th</sup> plan respectively. Interregional transmission capacity added during the period 2017-22 (up to 31<sup>st</sup> March 2022) was 37,200 MW. Interregional transmission capacity added during FY 2023-24 is 6490 MW. As on 31.03.2024, inter-regional transmission capacity in the country was 1, 18,740 MW. Details of inter-regional transmission lines are given at **Annexure-3A**.

The increase in inter-regional transmission capacity would further facilitate the smooth flow of power from surplus to deficit regions.

### 3.3 Transmission planning related works carried out in 2022-23:

#### 3.3.1 Manual on Transmission Planning Criteria, 2023

The manual on Transmission Planning Criteria was first brought out by CEA in 1985, which was revised in 1994 and 2013. Considering large-scale renewable energy integration, growth of load, increasing fault level, right of way issues, technical advancements, notification of Transmission Rules, etc. CEA had issued a Manual on Transmission Planning Criteria, 2023 which is effective from 1<sup>st</sup> April, 2023.

The Manual is quite useful in planning the Inter-state transmission system as well as the Intra-state transmission system. The manual covers planning philosophy, information required from various entities, permissible limits of parameters, reliability criteria, broad scope of system studies, modelling and analysis etc. and provides guidelines for transmission planning.

#### 3.3.2 BIMSTEC Energy Centre

Memorandum of Association (MoA) among the BIMSTEC Member States (The BIMSTEC is a regional organization comprising seven Member States: five deriving from South Asia, including Bangladesh, Bhutan, India, Nepal, Sri Lanka and two from Southeast Asia, including Myanmar and Thailand ) for the Establishment of BIMSTEC Energy Centre (BEC) was signed on 22<sup>nd</sup> January 2011. According to the MoA, the **BIMSTEC Energy Centre (BEC)** has to be set up in India.

#### 1<sup>st</sup> meeting of Governing Board of the BIMSTEC Energy Centre (BEC)

The first meeting of the Governing Board of the BIMSTEC Energy Centre (BEC) was held in Bengaluru on 27.02.2023 and was chaired by Mr Ajay Tiwari, Additional Secretary Ministry of Power. The outcome of the meeting are as below:

- (i) Sh Ghanshyam Prasad, Chairperson CEA, Ministry of Power, Govt. of India has

been appointed as the first Executive Director of the BIMSTEC Energy Centre.

- (ii) The draft Host Country Agreement between the Government of the Republic of India and the BIMSTEC Secretariat was finalized and recommended to be put before the Seventh Meeting of the BIMSTEC Permanent Working Committee for consideration.
- (iii) The Office of BEC is being set up and is expected to be operationalized within four months. BEC will temporarily be housed in the premises of the Central Power Research Institute (CPRI), Bengaluru. Permanent infrastructure has also been identified at the Ground Floor of the new Southern Region Load Dispatch Centre (SRLDC) building (CPRI Campus) at Bengaluru which is also expected to come at an early date.
- (iv) Rules of Procedure for BIMSTEC Energy Centre, will be finalized in a Special Meeting of Governing Board of BEC.

#### 3.3.3 BIMSTEC Grid Interconnection Coordination Committee (BGICC)

A Memorandum of Understanding (MoU) for establishment of the BIMSTEC grid interconnection was signed between Member States on 31<sup>st</sup> August, 2018.

Article 3.2 of MoU provides the need for BIMSTEC Grid Interconnection Coordination Committee (BGICC), to actively coordinate, for the successful implementation of grid interconnections and trade in electricity. The First meeting of BGICC was held on 30<sup>th</sup> June 2021 (in virtual mode).

#### 2<sup>nd</sup> Meeting of BIMSTEC Grid Interconnection Coordination Committee (BGICC)

The second meeting of the BIMSTEC Grid Interconnection Coordination Committee (BGICC) was held in Bengaluru on 27.02.2023 and was chaired by Sh Ghanshyam Prasad, Chairperson CEA. The outcome of the meeting

are as below:

- (i) The Meeting considered India's presentation as a discussion paper for the preparation of the draft "BIMSTEC Policy for Transmission of Electricity".
- (ii) The Meeting considered India's presentation as a discussion paper for the preparation of the draft "BIMSTEC Policy for Trade, Exchange of Electricity and Tariff Mechanism".

### 3.3.4 Intra State Transmission System Strengthening requirement of North Eastern States and Sikkim by the year 2030

MoP vide letter dated 16.02.2022 had requested CEA to ascertain further requirements of North Eastern States to augment their transmission systems.

Accordingly, CEA had submitted a Report on "Intra state transmission system strengthening requirement of North Eastern States and Sikkim by the year 2030" to MoP vide letter dated 25.02.2023.

Subsequently, in the meeting held on 17.01.2024 under the chairmanship of Secretary (Power), it was directed to revise the NER 2030 report considering resource adequacy plan, evacuation plans of small hydro plants, plotting of existing as well as planned network on PM Gati Shakti (PMGS) portal National Master Plan, separate plan for required distribution network (downstream) and other requirements.

Accordingly, CEA is in the process of updating the Report.

### 3.3.5 Examination of proposals for intra-state transmission planning

41 proposals pertaining to the planning of intra-state transmission systems submitted by States/Union Territories (UTs) were examined.

## 3.4 Private sector participation in transmission sector

### 3.4.1 Brief Introduction

IV. In compliance with provisions laid down in Tariff Policy dated 6<sup>th</sup> January, 2006, MoP issued an OM on 9<sup>th</sup> December, 2010, which provides that since 6<sup>th</sup> January, 2011, all the ISTS transmission projects are to be implemented through Tariff Based Competitive Bidding (TBCB) route, except some specific category of projects of strategic importance, technical upgradation works, required to be done to cater to an urgent situation on a case to case basis etc.

V. The Revised Tariff Policy issued by Ministry of Power on 28<sup>th</sup> January, 2016 states the following: -

Clause 5.3: "The tariff of all new generation and transmission projects of company owned or controlled by the Central Government shall continue to be determined on the basis of competitive bidding as per the Tariff Policy notified on 6<sup>th</sup> January, 2006 unless otherwise specified by the Central Government on case to case basis.

Further, intra-state transmission projects shall be developed by State Government through competitive bidding process for projects costing above a threshold limit which shall be decided by the SERCs."

Clause 7.1(7): "While all future inter-state transmission projects shall, ordinarily, be developed through competitive bidding process, the Central Government may give exemption from competitive bidding for (a) specific category of projects of strategic importance, technical upgradation etc. or (b) works required to be done to cater to an urgent situation on a case to case basis".

VI. The Empowered Committee on Transmission (ECT) was subsequently dissolved vide office order dated 4<sup>th</sup> November, 2019 and in place of ECT, MoP has constituted National Committee on Transmission (NCT). The terms of reference of NCT inter alia is the recommendation of ISTS scheme to MoP for approval. To further streamline the

process of planning and approval of ISTS schemes, MoP vide its office order dated 28.10.2021 has revised the Terms of Reference of the NCT, delegating powers for approval of ISTS system costing between Rs 100 to Rs 500 crores to NCT and for ISTS schemes costing up to Rs 100 crores to Central Transmission Utility. The ISTS schemes costing above Rs 500 crores would require approval of MoP.

### 3.4.2 Status of the Transmission schemes notified through TBCB:

As far as Inter-State transmission system is concerned, till 31<sup>st</sup> March, 2024, one hundred and six (106) schemes have been awarded through Tariff Based Competitive Bidding. The status of the transmission schemes, as on 31.03.2024, is as below:

|                                                                     |            |
|---------------------------------------------------------------------|------------|
| <b>Projects under TBCB</b>                                          | <b>106</b> |
| Projects commissioned                                               | 53         |
| Projects under implementation                                       | 49         |
| Stalled projects                                                    | 04         |
| <b>Projects notified and are under bidding/bidding yet to start</b> | <b>38</b>  |

|                                                |           |
|------------------------------------------------|-----------|
| <b>Stalled projects</b>                        | <b>04</b> |
| Project cancelled by CERC                      | 01        |
| Projects not taken up & CERC cancelled license | 01        |
| Projects under litigation                      | 02        |

The name of schemes notified through TBCB are at **Annexure-3B**.

### 3.4.3 Meetings of the National Committee on Transmission (NCT) held during 2023-24:

In the year 2023-24, six Nos. of meeting of National Committee on Transmission were held which is given below:

| Sl. No. | National Committee on Transmission (NCT) Meeting No. | Date on which meeting was held |
|---------|------------------------------------------------------|--------------------------------|
| 1.      | 13 <sup>th</sup>                                     | 12.05.2023                     |
| 2.      | 14 <sup>th</sup>                                     | 09.06.2023                     |
| 3.      | 15 <sup>th</sup>                                     | 25.08.2023                     |
| 4.      | 16 <sup>th</sup>                                     | 30.11.2023                     |
| 5.      | 17 <sup>th</sup>                                     | 31.01.2024                     |
| 6.      | 18 <sup>th</sup>                                     | 05.03.2024                     |

The transmission schemes and relevant issues taken up in these meeting are given at **Annexure -3C**.

### 3.5 Examination of Detailed Project Reports (DPRs) / Feasibility Reports (FRs) of Hydro Electric Projects / Pump Storage Projects for processing of concurrence by CEA:

- No DPRs/FRs of hydropower projects/pumped storage projects is under examination for processing of concurrence by CEA.

### 3.6 Examination of DPR of Transmission Works for processing of clearance by CEA

9 Nos. of proposals/DPR pertaining to transmission works were examined for processing of clearance by CEA.

### 3.7 Grant of prior approval of Government to transmission proposals under Section 68(1) of the Electricity Act, 2003 during 2023-24

As per Section 68(1) of the Electricity Act 2003, an overhead line shall be installed or kept installed above ground with prior approval of the Appropriate Government. During the FY 2023-24, total of 94 Nos. of transmission schemes were recommended for grant of approval under Section 68 (1) of the Electricity Act, 2003.

### 3.8 Grant of authorization to Transmission proposals under Section 164 of the Electricity Act, 2003 during 2023-24

In 2023-24 (up to 9<sup>th</sup> June 2023), total 5 applications have been granted authorization by CEA under section 164 of Electricity Act, 2003. After 9<sup>th</sup> June 2023, in place of CEA, Ministry of Power (MoP) started granting authorization under section 164. 46 applications have been notified by MoP from 9<sup>th</sup> June 2023 till 31<sup>st</sup> March 2024 based on the recommendations of CEA.

### 3.9 Cross-Border power exchange

#### 3.9.1 Indo-Bangladesh Cross-Border Interconnection and Power Trade

Bangladesh is connected with both Eastern and North Eastern Region of India with power transfer capacity of 1160 MW from India to Bangladesh through following links:

- 1,000 MW through Baharampur (India) to Bheramara (Bangladesh) 400 kV 2x D/c line with 2x500 MW HVDC back-to-back station at Bheramara, and
- 160 MW through Surajmaninagar (India) to North Comilla (Bangladesh) – South Comilla (Bangladesh) 400 kV D/c link (presently operated at 132 kV).
- Implementation of the 765 kV D/c Katihar (India) – Parbotipur (Bangladesh) – Bornagar (India) cross border link has been agreed and modalities of its implementation are being finalized.

### **Import/Export**

In the financial year 2023-24 about 8.394 BU energy was exported to Bangladesh.

### **3.9.2 Indo-Bhutan Cross Border Interconnections and Power Trade**

India and Bhutan have MoU on cooperation for exchange of power between the two countries. Bulk power generated at Hydro Electric Projects at Tala HEP (1020 MW), Chukha HEP (336 MW), Kurichu HEP (60 MW) and Mangdechu HEP (720 MW) in Bhutan is being exported to India through 400 kV, 220 kV and 132 kV lines.

Presently, about 2070 MW power from the existing hydro projects in Bhutan is being imported by India from Bhutan. The associated cross-border transmission system for evacuation and transfer of power from these HEPs is being operated in synchronism with the Indian Grid. Further, associated transmission line works have been completed for upcoming Punatsangchu-I (1200MW) and Punatsangchu-II (1020MW) HEPs.

### **Import/Export**

In the financial year 2023-24 about 5.73 BU energy was imported from Bhutan and about 1.868 BU energy was exported to Bhutan.

### **3.9.3 Indo-Nepal Cross Border interconnection and Power Trade**

- Presently, power is being exported to Nepal through 11 kV, 33 kV, 132 kV voltage level

transmission lines and Dhalkebar (Nepal) – Muzaffarpur (India) 400 kV D/C line.

- For evacuating power from Arun-3 (900MW) HEP in Nepal and other hydro projects in vicinity in future, Arun-3 HEP (Nepal) – Dhalkebar (Nepal) – Sitamarhi (India) 400 kV D/c (Quad) line is under implementation.
- 400 kV Gorakhpur – New Butwal D/c (Quad) line to facilitate increased transfer of power between the two countries is also under implementation.
- Further, Inaruwa (Nepal) - Purnea New (India) 400 kV D/c (Quad Moose) line and Dododhara (Nepal) - Bareilly (India) 400 kV D/c (Quad Moose) line are also agreed.

### **Import/Export**

In the financial year 2023-24 about 1.725 BU energy was imported from Nepal and about 1.850 BU energy was exported to Nepal.

### **Interconnection with State Grid**

Nepal is also connected with the Indian State Grid of Bihar, Uttar Pradesh (UP) and Uttarakhand. Bihar, UP and Uttarakhand exports power to Nepal upto 350 MW, 10 MW and less than 1 MW respectively. The electricity export from Indian states of UP/Bihar/Uttarakhand to Nepal is taking place through links from State Grids within the Power Exchange Committee (PEC) Mechanism. Methodology for electricity transaction with Nepal through State links were agreed between CEA and Nepal Electricity Authority (NEA) in a meeting held on 19.09.2023. The methodology charts out the activities to be carried out for bringing the transactions through the 132 kV and below voltage level state transmission lines with Nepal under the ambit of Import/Export (Cross Border) Guidelines-2018 of India and Regulations/Procedures framed there under.

14<sup>th</sup> Power Exchange committee (PEC) meeting for the revision of power exchange tariffs for export of electricity from states of Bihar, Uttar Pradesh and Uttarakhand for the FY 2024-25 was held on 11.03.2024.

### **3.9.4 Indo-Myanmar Cross Border Interconnections and Power Trade**

- India is providing about 2-3 MW of power (Since 5<sup>th</sup> April 2016) from Manipur (India) to Myanmar through 11 kV transmission line

from Moreh in Manipur (India) to Tamu town in Myanmar.

- Further, brief scope of works for India – Myanmar high capacity electricity transmission interconnection through Imphal - Tamu 400 kV D/C cross-border interconnection between India and Myanmar with 500 MW HVDC back-to-back terminal at Tamu for drawl of power by Myanmar has been agreed.

### **3.9.5 Guidelines for Import/Export (Cross Border) of Electricity**

- Guidelines for Import/Export (Cross Border) of Electricity were issued by the Ministry of Power on 18.12.2018 to facilitate import/export of electricity between India and neighboring countries.
- Procedure for approval and facilitating Import/Export (Cross Border) of Electricity by the Designated Authority was issued on 26.02.2021.
- Verification Mechanism for export of power from eligible fuel by generating stations under Import/Export (Cross Border) of Electricity-Guidelines 2018 were issued by the Ministry of Power on 5<sup>th</sup> March, 2021.
- Amendment-1 to the Procedure for approval and facilitating Import/Export (Cross Border) of Electricity by the Designated Authority was issued on 31<sup>st</sup> July, 2023. The trading in Indian power exchange by and Indian power trader on behalf of entity of neighboring country has been allowed in RTM (Real time Market) by this amendment.

### **3.9.6 Approvals granted by Designated Authority for Import/Export (Cross Border) of Electricity during 2023-24.**

- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for export of up to 14.55 MW of power to India from Devighat Hydropower Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line for a period from 19<sup>th</sup> May, 2023 – 30<sup>th</sup> April, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for export of up to 23.28 MW of power to India from Trishuli Hydropower Project, Nepal through NVVN

in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line for a period from 19<sup>th</sup> May, 2023 – 30<sup>th</sup> April, 2024

- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for export of up to 68 MW of power to India from Middle Marsyangdi Hydropower Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line for a period from 1<sup>st</sup> June, 2023 – 31<sup>st</sup> May, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for export of up to 140 MW of power to India from Kaligandaki 'A' Hydropower Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line for a period from 1<sup>st</sup> June, 2023 – 31<sup>st</sup> May, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for export of up to 51 MW of power to India from Likhu-IV Hydro Electric Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line for a period from 1<sup>st</sup> June, 2023 – 31<sup>st</sup> May, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for export of up to 67 MW of power to India from Marsyangdi Hydropower Project, Nepal to India through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line for a period from 1<sup>st</sup> June, 2023 – 31<sup>st</sup> May, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for import of up to 21.44 MW of power from Chilime Hydro Power Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line from 19<sup>th</sup> May, 2023 to 30<sup>th</sup> April, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for import of up to 22.8 MW of power from Solu Hydro Power Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line from 1<sup>st</sup> June, 2023 to 30<sup>th</sup> April, 2024

- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for import of up to 24.25 MW of power from Kabeli B-1 Hydro Power Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line from 1<sup>st</sup> October, 2023 to 30<sup>th</sup> September, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for import of up to 19.4 MW of power from Lower Modi Hydro Power Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line from 1<sup>st</sup> October, 2023 to 30<sup>th</sup> September, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for import of up to 24.25 MW of power from Upper Dordi 'A' Hydro Power Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line from 10<sup>th</sup> November, 2023 to 31<sup>st</sup> October, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for import of up to 34.92 MW of power from Upper Balephi 'A' Hydro Power Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line from 16<sup>th</sup> February, 2024 to 31<sup>st</sup> October, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for import of up to 83.42 MW of power from Solu Khola (Dudhkoshi) Hydro Power Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line from 8<sup>th</sup> December, 2023 to 31<sup>st</sup> May, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for import of up to 26.19 MW of power from Dordi Khola Hydro Power Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line from 8<sup>th</sup> December, 2023 to 31<sup>st</sup> May, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for import of up to 83.42 MW of power from Solu Khola and 26.19 MW of power from Dordi Khola Hydro Power Project, Nepal to Haryana through NVVN through Muzaffarpur-Dhalkebar 400 kV D/c line for a period from 05.09.2023 to 31.10.2023, 01.06.2024 to 31.10.2024, 01.06.2025 to 31.10.2025, 01.06.2026 to 31.10.2026, 01.06.2027 to 31.10.2027.
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for export of up to 14.55 MW of power to India from Devighat Hydropower Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line for a period from 1<sup>st</sup> May, 2024 – 30<sup>th</sup> April, 2025
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for export of up to 23.28 MW of power to India from Trishuli Hydropower Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Muzaffarpur-Dhalkebar 400 kV D/c line for a period from 1<sup>st</sup> May, 2024 – 30<sup>th</sup> April, 2025
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for export of up to 32.7 MW of power to India from Upper Kalangagad Hydropower Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Tanakpur-Mahendranagar 132 kV S/c line for a period from 5<sup>th</sup> September, 2023 to 31<sup>st</sup> May, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for export of up to 37.3 MW of power to India from Upper Chameliya Hydropower Project, Nepal through NVVN in Indian Power Exchange(s) (PX) - Day Ahead Market and Real Time Market through Tanakpur-Mahendranagar 132 kV S/c line for a period from 5<sup>th</sup> September, 2023 to 30<sup>th</sup> June, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for export of upto 400 MW of power to Nepal Electricity Authority (NEA), Nepal, from Indian Power Exchange(s) - Day Ahead Market for a period from 1<sup>st</sup> April, 2023 to 7<sup>th</sup> November 2023 and from 7<sup>th</sup> November 2023 to 31<sup>st</sup> March 2024 (Both Day Ahead Market and Real Time Market) and modified approval of 650 MW for a period from 1<sup>st</sup> March, 2024 to 31<sup>st</sup> March 2024 through Muzaffarpur-Dhalkebar 400 kV D/c transmission line.

- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for export of upto 54 MW of power to Nepal Electricity Authority (NEA), Nepal, from Indian Power Exchange(s) - Day Ahead Market for a period from 19<sup>th</sup> May, 2023 to 14<sup>th</sup> November 2023 and from 15<sup>th</sup> November 2023 to 31<sup>st</sup> March 2024 (Both Day Ahead Market and Real Time Market) through Tanakpur - Mahendranagar 132 kV S/c transmission line.
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for export of upto 500 MW of power to Nepal Electricity Authority (NEA), Nepal, from Indian Power Exchange(s) - Day Ahead Market and Real Time Market through Muzaffarpur - Dhalkebar 400 kV D/C line for a period from 1<sup>st</sup> April, 2024 to 30<sup>th</sup> June, 2024
- Approval to NTPC Vidyut Vyapar Nigam Limited (NVVN) for export of upto 54 MW of power to Nepal Electricity Authority (NEA), Nepal, from Indian Power Exchange(s) - Day Ahead Market and Real Time Market through Tanakpur - Mahendranagar 132 kV S/c transmission line for a period from 1<sup>st</sup> April, 2024 to 30<sup>th</sup> June, 2024
- Approval to PTC India Limited (PTCIL) for import of upto 850 MW of power on behalf of Druk Green Power Corporation Limited, Bhutan (export from India to Bhutan) through Indian Power Exchange(s) - Day Ahead Market and Real Time market for a period from 1<sup>st</sup> December, 2023 to 30<sup>th</sup> April, 2024.
- Approval to PTC India Limited (PTCIL) for import of upto 450 MW of power on behalf of Druk Green Power Corporation Limited, Bhutan (export from India to Bhutan) through Indian Power Exchange(s) - Day Ahead Market and Real Time market for a period from 1<sup>st</sup> May, 2024 to 31<sup>st</sup> May, 2024.
- Approval to PTC India Limited (PTCIL) for import of upto 450 MW of power on behalf of Druk Green Power Corporation Limited, Bhutan (export from India to Bhutan) through Indian Power Exchange(s) - Day Ahead Market and Real Time market through Bihar state 132 kV lines for a period from 12<sup>th</sup> March, 2024 to 15<sup>th</sup> June, 2024.
- Approval to PTC India Limited (PTCIL) for import of upto 150 MW of power on behalf of Druk Green Power Corporation Limited, Bhutan (export from India to Bhutan) through Indian Power Exchange(s) - Day

Ahead Market and Real Time market through UP state 132 kV lines for a period from 12<sup>th</sup> March, 2024 to 15<sup>th</sup> June, 2024.

### 3.10 Miscellaneous works

#### 3.10.1 Green Energy Corridor

##### a) Transmission Works under Green Energy Corridors-I

The report on Green Energy Corridor had been prepared by PGCIL as a comprehensive scheme for evacuation and integration of the renewable energy (RE) capacity addition of 32,713 MW during 12<sup>th</sup> Plan Period. Total fund requirement of Rs 34,141 Crore was initially assessed for the development of the transmission system and control infrastructure for the addition of RE capacity in the renewable rich States of Andhra Pradesh, Gujarat, Himachal Pradesh, Jammu and Kashmir, Karnataka, Maharashtra, Rajasthan, Madhya Pradesh and Tamil Nadu.

Intra State transmission schemes are to be funded as 20% equity of the State Govt., 40% grant from National Clean Energy Fund (NCEF) and 40% soft loan, whereas, the Inter State transmission schemes are to be funded as 30% equity by PGCIL and 70% soft loan.

For the funding of green energy corridors in both intra and inter State transmission projects, under the framework of cooperation between Govt. of India and Govt. of Germany. KfW Germany is providing soft loan to the tune of Euro 1 Billion. For Inter-state transmission projects pertaining to Part A, B and C of Green Energy Corridor, Loan agreement for financial assistance of Euro 500 million from KfW, Germany has been signed by PGCIL. For implementation of transmission schemes under Green Energy Corridor-Part D, POWERGRID has taken loan from Asian Development Bank (ADB).

In Green Energy Corridor-I (GEC-I) scheme, under ISTS, around 17,000 MVA substation capacity and 3,200 ckm of transmission lines were planned and the same has been commissioned.

The InSTS GEC scheme with total target of 9,700 ckm intra-state transmission lines and 22,600 MVA sub-stations was approved by the

Cabinet Committee on Economic Affairs (CCEA) in 2015. The InSTS GEC scheme is currently under implementation by the State Transmission Utilities (STUs) of 8 RE rich States, i.e. Andhra Pradesh, Gujarat, Himachal Pradesh, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan and Tamil Nadu. The projects are being set up for evacuation of approx. 24 GW of RE power in the above 8 States. Out of the eight States, four have completed all the projects, viz. Rajasthan, Madhya Pradesh, Karnataka and Tamil Nadu. The remaining four states had requested for further extension up to 31<sup>st</sup> December, 2024.

#### **b) Transmission Works under Green Energy Corridors-II**

Intra- state transmission schemes for power evacuation and grid integration of RE generation projects in seven States (Kerala, Uttar Pradesh, Tamil Nadu, Gujarat, Himachal Pradesh, Karnataka and Rajasthan) have been approved by the Government with an estimated cost of Rs 12,031 Crore with Central Financial Assistance (CFA) @33% of project cost to the States. The scheme includes setting up of 10,753 ckm of transmission lines and 27,546 MVA transformation capacity of substations. About 20 GW RE capacity is planned to be integrated in to the intra-state system under Green Energy Corridor-II (GEC-II) schemes. The states are in process of award of packages for implementation of intra-state transmission schemes under GEC-II Scheme.

#### **3.10.2 Study, analysis and formulation of policies on specific issues relating to transmission**

##### **a) National Electricity Plan (Volume II: Transmission)**

As per Section 3(4) of the Electricity Act, 2003, Central Electricity Authority has been mandated to prepare a National Electricity Plan (NEP) in accordance with the National Electricity Policy and notify such plan once in five years.

The Draft National Electricity Plan (Vol II: Transmission) was published on CEA's website on 24.01.2024 for a period of 60 days (extended for 20 more days) for inviting comments from various stakeholders. The plan covers the review of transmission system addition during the period 2017-22, detailed planning for the period 2022-27 and perspective plan for the

period 2027-32. The National Electricity Plan (Vol II: Transmission) would be notified after approval of the Ministry of Power.

#### **3.11 Formulation/review of Regulations, Guidelines and audit.**

- a. The Operation and Maintenance (O&M) Guidelines and Standard Format for Memorandum of Understanding between New TSP and Existing TSP for projects to be awarded under TBCB route was issued.
- b. The Operation and Maintenance (O&M) Guidelines between New TSP and Existing TSP were extended to the generating station switchyards and other than TBCB based transmission projects in ISTS substation with some modifications.
- c. Draft Generic Technical Specifications of the Nitrogen Injection Fire Prevention and Extinguishing System (NIFPES) to be used as an Interim measure was drafted and circulated for public comments.

#### **3.12 Representation/ Nomination in the Committees**

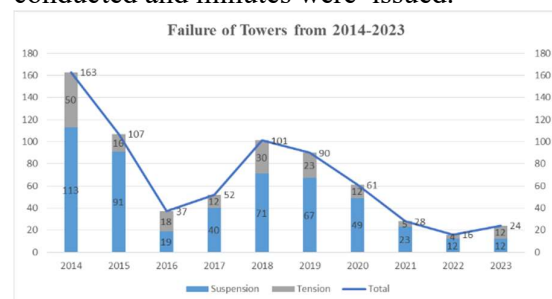
- a. Member (Power Systems) is Chairman of the Electro Technical Division Council (ETDC) of Bureau of Indian Standards (BIS).
- b. Chairman and Member Secretary of the Standing committee of experts to investigate the failure of:
  - Towers of transmission lines of 220 kV and higher voltages of power utilities
  - Equipment of 220 kV and above substation
- c. Chairman & Member Convener of the Committee for calculation of reduction of Right-Of-Way (ROW) width through technological options available like Insulated Cross Arms, Monopole Towers, and High Tension Low Sag (HTLS) conductor.
- d. Chairman & Member Convener of the Committee to conduct pilots for capacity expansion of existing lines to save Right of Way.
- d. Committee under the chairmanship of Member(Power System), CEA and Director (PSETD) as Member Convener to examine various technical issues raised by stakeholders on Standard specification and

- technical parameters for transformers and reactors (66 kV and above voltage class).
- Member of the Committee in CEA to suo moto scrutinize tenders floated for Public Procurement having value Rs. 500 crores and above.
  - Member of Sub group for Techno- Economic appraisal of DPRs for system improvement under PSDF funding.
  - Member of Technical Committee on Transmission Research for Review, recommendation and monitoring of R&D proposals under IHRD, RSOP, NPP schemes of MoP, Govt. of India.
  - Member of various Technical Committees of BIS pertaining to EHV transmission lines (Conductor, Earthwire, insulator & hardware and transmission line towers) and substations (surge arrestor, switchgear, transformer, HVDC, power electronics, high voltage engineering, battery etc.).
  - Member of Bid Evaluation Committee (BEC) for projects being awarded through Tariff Based Competitive Bidding (TBCB).
  - Chairman and Member Convener of the cost committee meeting for TBCB projects.

### 3.13 Analysis of causes of failure of transmission line towers and substation equipment.

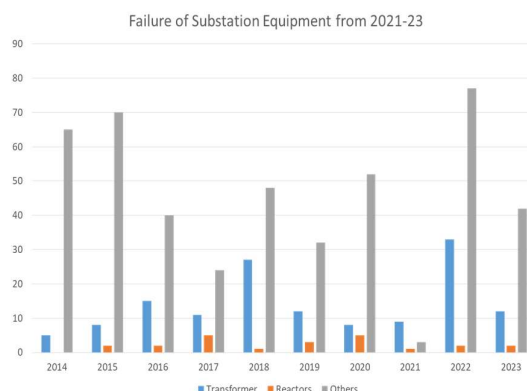
#### a) Transmission Line Towers:

As a part of the activity of the Standing Committee to assess the causes of failure of various Transmission Line Towers of 220 kV and above voltage levels, CEA officers visit failure sites, investigate the failure and based on these a report is published. The trend of failures as per the report published from 2014 to 2023 is given below. Meeting of the Standing Committee of Experts on Failure of the EHV Transmission Tower (January 2022 to December 2023) was conducted and minutes were issued.



#### b) Substation Equipment failures:

As a part of the activity of the Standing Committee to assess the causes of failure of various Substation Equipment of 220 kV and above voltage levels, CEA officer visited failure sites and investigation was carried out. A detailed report suggesting remedial measures is published. Meeting of Standing Committee of Experts on Failure of 220 kV and above Voltage Class Substation Equipment (January 2022 to June 2023) was conducted and minutes were issued. The trend of failures from 2014 to June 2023 of major equipment as per published report is given below.



### 3.14 Inputs on various Transmission schemes based on Tariff based competitive bidding

Inputs relating to Specific Technical Requirements for transmission lines and substations for Request for Proposal (RfP) documents of the various projects to be awarded through TBCB is provided. Clarification to bidders on various technical queries is provided and participation in bid evaluation committee is carried out. Cost committee which has been constituted under the Chief Engineer level officer of CEA estimate the cost of the Scheme. The work pertaining to technical specification/cost estimation/ bidder's queries/participation in bid evaluation was carried out for the following 46 transmission schemes under TBCB in the FY 2023-24:

| S. No. | Name of the Transmission Schemes under TBCB in the FY 2023-24                                            |
|--------|----------------------------------------------------------------------------------------------------------|
| 1.     | Transmission System for Evacuation of Power from Shongtong Karcham HEP (450 MW) and Tidong HEP (150 MW)  |
| 2.     | Transmission System for Evacuation of Additional 7GW RE Power from Khavda RE Park under Phase-III Part A |

|     |                                                                                                                                           |      |                                                                                                                             |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------|
| 3.  | Transmission system for evacuation of additional 7 GW of RE power from Khavda RE park under Phase III Part B                              |      | under Phase-III Part C1                                                                                                     |
| 4.  | Transmission System for Evacuation of power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part A  | 22.  | Transmission system for evacuation of power from REZ in Rajasthan (20GW) under Phase-III Part D                             |
| 5.  | Transmission System for Evacuation of power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part B  | 23.  | Transmission system for evacuation of power from REZ in Rajasthan (20GW) under Phase-III Part F                             |
| 6.  | Transmission System for Evacuation of power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part C  | 24.  | Transmission system for evacuation of power from REZ in Rajasthan under phase III –Part I                                   |
| 7.  | Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7GW): Part D   | 25.  | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-1), Part-A                                       |
| 8.  | Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase- IV(7 GW): Part E2 | 26.  | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-1), Part-B                                       |
| 9.  | Transmission System for Evacuation of power from potential renewable energy zone in Khavda area of Gujarat under Phase-V (8 GW): Part A   | 27.  | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-1), Part-C                                       |
| 10. | Provision of Dynamic Reactive Compensation at KPS1 and KPS3                                                                               | 28.  | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-1), Part-D                                       |
| 11. | Transmission System for Evacuation of Power from RE Projects in Solapur (1500 MW) SEZ in Maharashtra                                      | 29.. | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :5.5 GW) (Jaisalmer/ Barmer Complex): Part A   |
| 12. | Transmission system for Evacuation of Power from RE Projects in Rajgarh (1000 MW) SEZ in Madhya Pradesh-Phase II                          | 30.  | Transmission system for evacuation of power from Rajasthan REZ Ph -IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex) Part B    |
| 13. | Western Region Network Expansion scheme in Kallam                                                                                         | 31.  | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex): Part C".  |
| 14. | Western Region Expansion Scheme XXXIII (WRES – XXXIII): Part A                                                                            | 32.  | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :5.5 GW) (Jaisalmer/ Barmer Complex): Part D   |
| 15. | Western Region Expansion Scheme XXXIII (WRES – XXXIII): Part B                                                                            | 33.  | Transmission system for evacuation of power from Rajasthan REZ PH-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex): Part E    |
| 16. | Western Region Expansion Scheme XXXIII (WRES – XXXIII): Part C                                                                            | 34.  | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex) Part F     |
| 17. | Network Expansion Scheme in Navinal (Mundra) area of Gujarat for drawal of power in the area                                              | 35.  | Transmission System for Evacuation of Power from Rajasthan REZ Ph-IV (Part-2: 5.5 GW) (Jaisalmer/ Barmer Complex): Part H1  |
| 18. | Network Expansion Scheme in Gujarat for drawl of about 3.6GW load under Phase-I in Jamnagar area                                          | 36.  | Additional Transmission system for evacuation of power from Bhadla-III PS as part of Rajasthan REZ Phase-III scheme (20 GW) |
| 19. | Augmentation of transformation capacity at Bhuj-II PS                                                                                     | 37.  | Transmission system strengthening for interconnections of Bhadla-III & Bikaner-III complex                                  |
| 20. | Transmission system for evacuation of power from REZ in Rajasthan (20GW) under Phase-III Part B1                                          | 38.  | System strengthening at Koppal-II and Gadag-II for integration of RE generation                                             |
| 21. | Transmission system for evacuation of power from REZ in Rajasthan (20GW)                                                                  |      |                                                                                                                             |

|     |                                                                         |
|-----|-------------------------------------------------------------------------|
|     | projects                                                                |
| 39. | Transmission scheme for integration of Tumkur-II REZ in Karnataka       |
| 40. | Transmission Scheme for Solar Energy Zone in Bidar (2500 MW), Karnataka |
| 41. | North Eastern Region Generation Scheme-I (NERGS-I)                      |
| 42. | North Eastern Region Expansion Scheme-XVI (NERES-XVI)                   |
| 43. | Eastern Region Generation Scheme-I (ERGS-I)                             |
| 44. | Eastern Region Expansion Scheme-XXXIV (ERES-XXXIV)                      |
| 45. | Eastern Region Expansion Scheme-XXXIX (ERES – XXXIX)                    |
| 46. | Transmission system for evacuation of power from Luhri Stage-I HEP      |

### 3.15 Technical advice/inputs on Transmission lines and Substation

- Technical assistance/advice relating to transmission system in the Country provided from time to time to MoP/ Power Utilities/State Utilities/Other Ministries/ BIS/ CPRI etc.
- Technical advice to transmission utilities regarding re-routing of transmission lines.
- Examination of Cost Escalation aspect of PMRP-2004 Scheme.
- Technical inputs for preparation of RfP/RfQ document for TBCB projects and participated in pre-bid clarification meeting.
- The technical specifications of the state of the art LCC and VSC technology based HVDC projects which are being awarded under the TBCB based scheme has been prepared in association with CTUIL.
- DPRs submitted by utilities for funding under PSDF Scheme were examined and participated as member of techno-economic sub-group.

### 3.16 Construction Monitoring of Transmission Projects

The monitoring of construction of transmission lines and sub-station (220 kV & above) covered under various transmission projects under central/state/private sector is being carried out with a view to achieve timely completion of transmission projects to ensure evacuation of

power from new Generation Projects as well as strengthening of existing transmission network required for transmission of power to load centers.

The delay in execution of transmission projects are primarily due to ROW issues, compensation & forest issues, contractual issues, poor financial condition of the executing agencies, land acquisition for substation, delay in getting statutory approval from various agencies like Railways and State / National Highway Authority etc. and law & order problem.

In respect of transmission lines, 16,682 CKM was targeted for FY 2023-24 as per the details given below:

|                                | ±800 kV HVDC | ±500 kV HVDC | 765 kV | 400 kV | 230/220 kV | Total  |
|--------------------------------|--------------|--------------|--------|--------|------------|--------|
| Transmission Line length (ckm) | 0            | 0            | 2076   | 7326   | 7280       | 16,682 |

Against the above target of 16,682 ckm, total 14,203 ckm has been commissioned, during FY 2023-24 as per the details given below:

|                                | ±800 kV HVDC | ±500 kV HVDC | 765 kV | 400 kV | 230/220 kV | Total  |
|--------------------------------|--------------|--------------|--------|--------|------------|--------|
| Transmission Line length (ckm) | 0            | 0            | 2119   | 6088   | 5996       | 14,203 |

Thus, overall achievement in respect of transmission line is 85.14%. Details of transmission lines commissioned /completed during FY 2023-24 (as on 31<sup>st</sup> March 2024) are given in **Annexure-3D**.

Similarly, in respect of substations, 78,109 MVA of transformation capacity was targeted for FY 2023-24 as per the details given below:

|                                           | ±800 kV HVDC | ±500 kV HVDC | ±320 kV HVDC | 765 kV | 400 kV | 230/220 kV | Total  |
|-------------------------------------------|--------------|--------------|--------------|--------|--------|------------|--------|
| Sub-Station transformation capacity (MVA) | 0            | 0            | 0            | 21,000 | 32,160 | 24,949     | 78,109 |

Against the above target of 78,109 MVA, total

70,728 MVA transformation capacity has been commissioned during FY 2023-24 as per the details given below:

|                                           | ±800<br>kV<br>HVD<br>C | ±500<br>kV<br>HV<br>DC | ±320<br>kV<br>HV<br>DC | 765<br>kV | 400<br>kV | 230/220<br>kV | Total  |
|-------------------------------------------|------------------------|------------------------|------------------------|-----------|-----------|---------------|--------|
| Sub-Station transformation capacity (MVA) | 0                      | 0                      | 0                      | 18,000    | 32,185    | 20,543        | 70,728 |

Thus, overall achievement in respect of Transformation capacity in Sub-stations is 90.55%. Details of substations commissioned / completed during FY 2023-24 (as on 31st March 2024) are given in **Annexure-3E**.

Voltage-wise/Sector-wise program vs achievement for the financial year 2023-24 in respect of transmission lines and sub stations (220kV and above voltage level) are given in **Charts I to X**.

Total 14,203 ckm of transmission lines and 70,728 MVA of transformation capacity in substations (220kV and above voltage levels) have been added during the financial year 2023-24 resulting in all India transmission network of 485,544 CKM of transmission lines and 12,51,080 MVA of the transformation capacity (220kV and above voltage level) as on 31<sup>st</sup> March 2024.

Growth of Transmission lines (in CKM) and Transformation capacity in substations (in MVA) since 2014-15 to FY 2023-24 is as under:

| Year    | Transmission Lines (in ckm) | Transformation Capacity (in MVA) |
|---------|-----------------------------|----------------------------------|
| 2014-15 | 313437                      | 596100                           |
| 2015-16 | 341551                      | 658949                           |
| 2016-17 | 367851                      | 740765                           |
| 2017-18 | 390970                      | 826958                           |
| 2018-19 | 413407                      | 899663                           |
| 2019-20 | 425071                      | 967893                           |
| 2020-21 | 441821                      | 1025468                          |
| 2021-22 | 456716                      | 1104450                          |
| 2022-23 | 471341                      | 1180352                          |
| 2023-24 | 485544                      | 1251080                          |

For the FY 2024-25, Program for Transmission Lines and Transformation Capacity (Substations) is as under.

#### Annual Target for FY 2024-25:

|                         |         |
|-------------------------|---------|
| Transmission line (CKM) | 16,667  |
| Substation (MVA)        | 116,490 |

### 3.17 Inspection of Electrical Installation:

The Chief Electrical Inspector (CEI) and Electrical Inspectors appointed by the Central Government under section 162 of the Electricity Act, 2003 discharge the functions described in “The Qualifications, Powers and Functions of Chief Electrical Inspector and Electrical Inspectors Rules, 2006”. These rules stipulate the statutory inspection of electrical installations as per Central Electricity Authority (Measures Relating to Safety and Electric Supply) Regulations, 2023. The Chief Engineer of Chief Electrical Inspectorate Division is appointed as Chief Electrical Inspector to the Government of India and is assisted by the Electrical Inspectors and the officers from five Regional Inspectorial Organizations (RIOs) in discharging the various responsibilities, briefly described as under:

- Periodic inspection of electrical installations for compliance under Regulation 32 of Central Electricity Authority (Measures Relating to Safety and Electric Supply) Regulations, 2023.
- Inspection of new electrical installations under Regulations 45 of Central Electricity Authority (Measures Relating to Safety and Electric Supply) Regulations, 2023 for according approval for first time charging of electrical installation of voltage exceeding notified voltage.
- Amendment of Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023 as and when necessitated.
- Inquiry of fatal and non-fatal electrical accidents.
- Collection of Statistics, Return and information relating to electrical accidents in Format-19 and 20 under “Furnishing of Statistics, Returns and Information rules 2007”.
- Advising stakeholders in the matter related to the electrical safety.
- Conducting electrical safety awareness workshops/seminars.

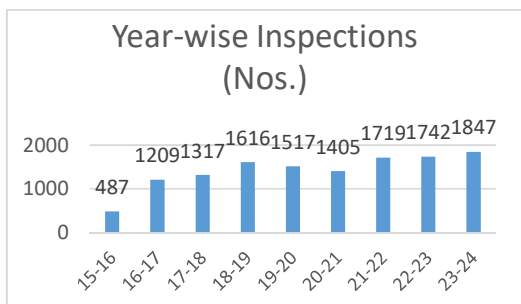
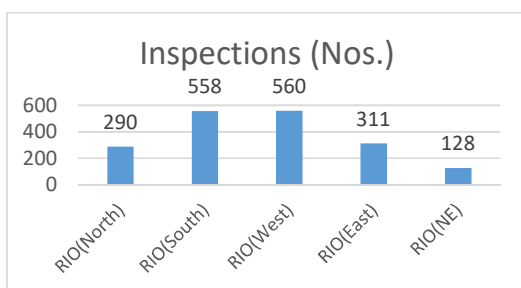
### 3.18 Review of Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2010:

Comprehensive review of the above mentioned regulations was performed and the Regulations were notified on 8th June 2023 as Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023.

### 3.19 Major achievement in terms of Inspections during the year 2023-24:

#### 3.19.1 New electrical installations/ apparatus inspected under Regulation 45 of Central Electricity Authority (Measures Relating to Safety and Electric Supply) Regulations, 2023:

A. The region wise summary of total inspections carried out during 2023-24 is given below: -



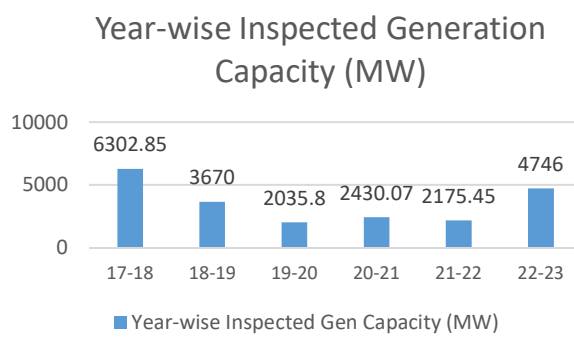
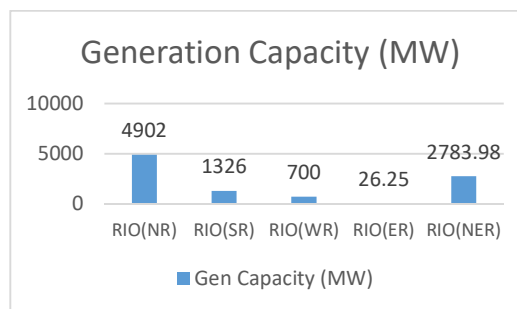
#### B. Major Power System Elements inspected:

##### 1) Substations:

- Substations of the capacity 765 kV, 400 kV, 220 kV, 132 kV and HVDC are being inspected by the Electrical Inspector. Power System elements such as Transformers, Reactors, Capacitors, Bus, and Bays STATCOM etc are inspected.

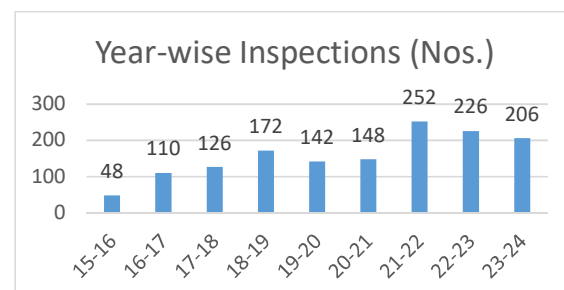
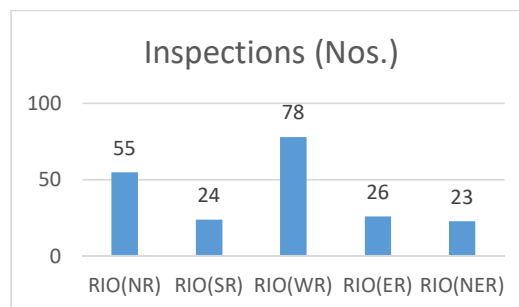
##### 2) Generating Units:

- The region wise summary of inspections carried out in 2023-24 is given below: -



### 3) Transmission Lines:

- Transmission Lines of the capacity of 765 kV, 400 kV, 220 kV, 132 kV and HVDC are being inspected by the Electrical Inspector.
- The region wise summary of inspections carried out in 2023-24 is given below: -



- Circuit kilometers of transmission lines inspected at different voltage levels (in ckm):

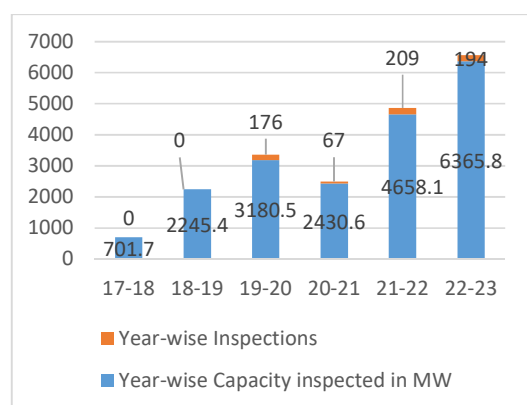
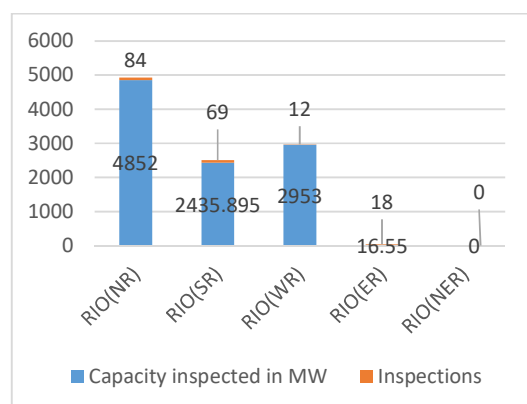
| RIOs   | NR  | SR   | WR  | ER  | NER |
|--------|-----|------|-----|-----|-----|
| 765 kV | NIL | 1658 | 613 | NIL | NIL |

|                    |      |        |     |      |         |
|--------------------|------|--------|-----|------|---------|
| <b>400 kV</b>      | 2185 | 299.5  | 509 | 1837 | 632.85  |
| <b>220 kV</b>      | 326  | 204.05 | 104 | 7    | 8.114   |
| <b>132 kV</b>      | NIL  | NIL    | NIL | 1.5  | 618.937 |
| <b>110 kV</b>      | NIL  | NIL    | NIL | NIL  | NIL     |
| <b>66 kV</b>       | NIL  | NIL    | NIL | NIL  | NIL     |
| <b>33 kV</b>       | 238  | 401.62 | NIL | 3.77 | NIL     |
| <b>11 kV</b>       | 4.5  | NIL    | NIL | 3.5  | NIL     |
| <b>800 kV HVDC</b> | NIL  | NIL    | NIL | NIL  | NIL     |

The data above are based on the cumulative inspections carried out by all RIOs

### C. Inspections done for Renewables:

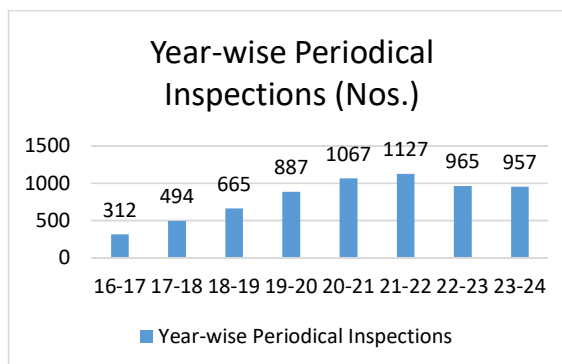
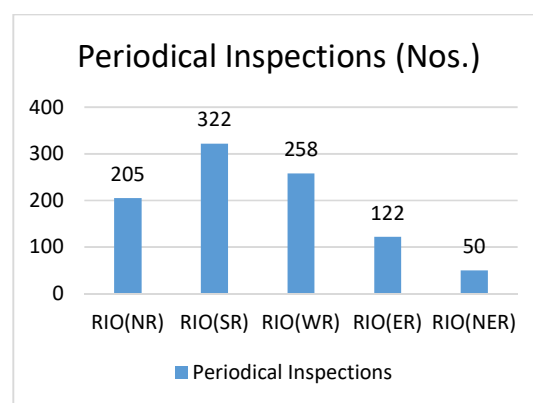
The region wise summary of inspections carried out in 2023-24 is given below:-



### 3.19.2 Periodic Inspections under Regulation 32 of Central Electricity Authority (Measures Relating to Safety and Electric Supply) Regulations, 2023

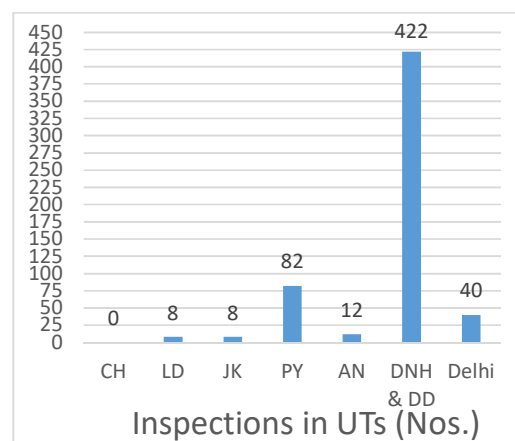
Summary of Periodic Inspections carried out

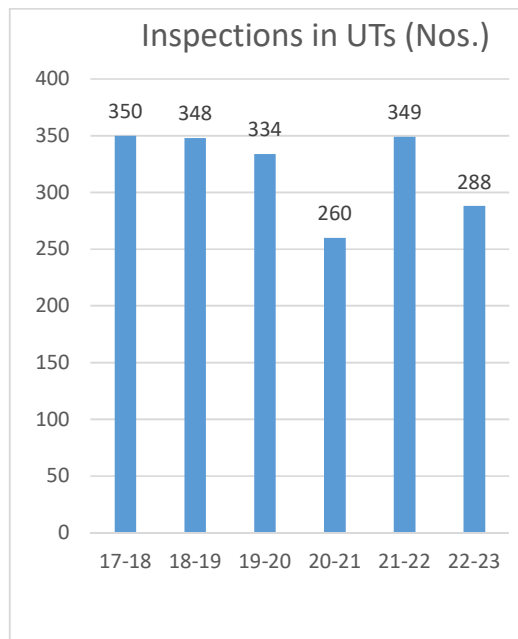
during the year 2023-24 is given below:



### 3.19.3 Inspections done in Union Territories

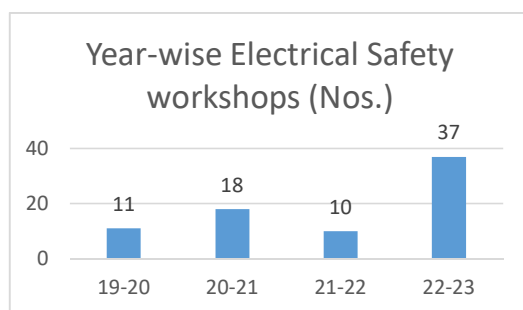
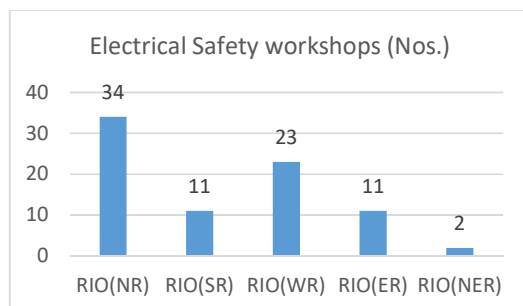
Summary of Inspections done in Union Territories during the year 2023-24 is given below:





### 3.19.4 Electrical safety awareness workshops/seminars

Summary of Electrical safety awareness workshops/seminars conducted during the year 2023-24 is given below:



(The above data are compiled based on the cumulative inputs by all RIOs)

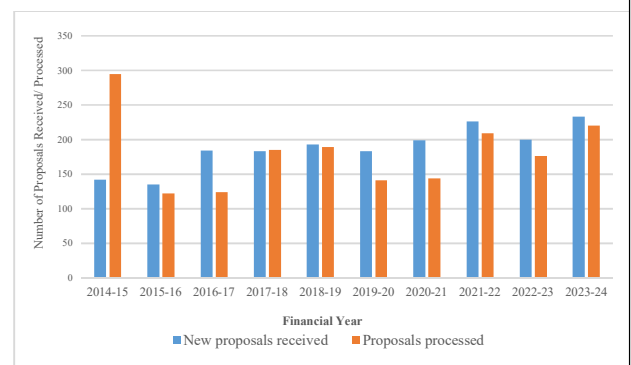
### 3.20 Power & Telecommunication Co-ordination Committee (PTCC)

Section 160 of the Electricity Act, 2003 provides all reasonable precautions to be taken by the operator in laying down and placing his electric lines or electrical plant so as not to injuriously

affect by induction, the working of telephonic and electric signaling lines.

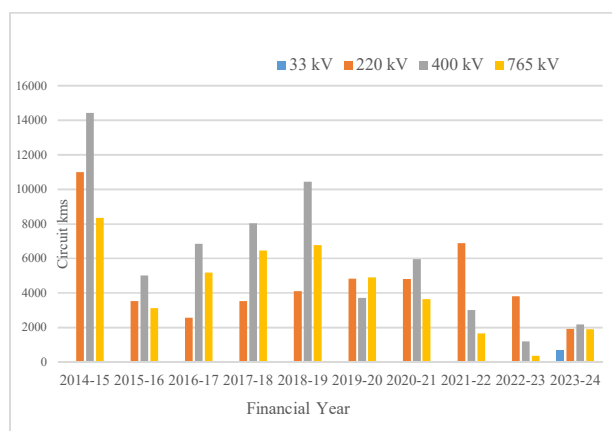
To ensure safe co-existence of Power & Telecommunication System, Central level Power and Telecommunication Coordination Committee (CLPTCC) was constituted.

During the year 2023-24, 233 nos. of new cases of EHV (220 kV and above) and ISTS power lines were received and Induced Voltage Calculation for 220 nos. of cases have been processed for PTCC route approval. A bar chart indicating the number of cases received/processed for PTCC route approval from 2014-15 to 2023-24 financial years is given below:



During the year 2023-24, Induced Voltage Calculation for about 1913.6 Circuit kilometers of 220 kV lines, 2181.4 Circuit kilometers of 400 kV lines, 1898.9 Circuit kilometers of 765 kV lines and 660.9 Circuit kilometers of 33 kV lines have been done. It is pertinent to mention that during 2023-24, due care has been taken to process PTCC cases of those transmission lines which were required to be charged on urgent basis; and with the result there has been no delay of charging of any line for want of PTCC approval.

A bar chart indicating the Circuit kilometers of 33 kV, 220 kV, 400 kV and 765 kV transmission lines, for which Induced Voltage Calculations were done from 2014-15 to 2023-24 is given below:



Power Systems Communication Development Division continues to strive for prompt computation of Induced voltages for PTCC clearance of EHV transmission lines of voltages 220 kV and above and ISTS lines. Regular follow-up has been done with Bharat Sanchar Nigam Ltd. (BSNL), Railways, Defense and SEBs/Power Utilities to expedite PTCC clearances. The Division also rendered assistance to the State Power Utilities in resolving PTCC cases of voltage level of 132 kV and below.

### 3.21 Providing technical advice for Reliable Communication System in Power Sector

PCD Division provides technical advices related to Communication System in Power Sector and its up gradation. PCD Division is involved in vetting of proposals from transmission licensees, finalization of RfPs and analyzing feasibility of various communication technologies suitable for power sector.

During the year 2023-24, finalization of following RfPs has been done:

1. Augmentation of transformation capacity at Bhuj-II PS
2. Eastern Region Generation Scheme-I (ERGS-I)
3. Eastern Region Expansion Scheme-XXXIV (ERES-XXXIV)
4. Eastern Region Expansion Scheme-XXXIX (ERES-XXXIX)
5. North Eastern Region Expansion Scheme-XVI (NERES-XVI)
6. Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part B"
7. Transmission system for evacuation of power

- from Rajasthan REZ Ph-IV (Part-2 :5.5 GW) (Jaisalmer/ Barmer Complex): Part E
8. Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part D
9. Transmission Scheme "Transmission system for evacuation of power from Rajasthan REZ Ph -IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex) Part D"
10. Transmission system for evacuation of power from Rajasthan REZ Ph -IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex) Part B
11. Transmission system for evacuation of power from Rajasthan REZ Ph -IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex): Part F (By clubbing Part F1 & F2)
12. Transmission System for Evacuation of Power from Rajasthan REZ Ph-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex): Part A
13. Network Expansion Scheme in Gujarat for drawl of about 3.6GW load under Phase-I in Jamnagar area
14. Transmission System for Evacuation of power from potential renewable energy zone in Khavda area of Gujarat under Phase-V (8 GW): Part A
15. Transmission System for Evacuation of power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part C
16. Transmission Scheme for integration of Tumkur-II Renewable Energy Zone in Karnataka
17. Transmission system strengthening for interconnections of Bhadla-III & Bikaner-III complex
18. North Eastern Region Generation Scheme – I (NERGS-I)
19. Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex): Part H1
20. Transmission System for Evacuation of power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part A
21. Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2: 5.5 GW) (Jaisalmer/Barmer Complex): Part C
22. Western Region Network Expansion scheme in Kallam area of Maharashtra
23. Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7GW) Part E2
24. Provision of Dynamic Reactive Compensation

at KPS1 and KPS3

25. Transmission System for Evacuation of Power from RE Projects in Rajgarh (1000 MW) SEZ in Madhya Pradesh - Phase-II
26. Transmission System under “Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-1), Part-A
27. Transmission system for evacuation of power from RE projects in Solapur (1500 MW) SEZ in Maharashtra
28. Transmission system for evacuation of additional 7 GW RE power from Khavda RE park under Phase-III Part B
29. Western Region Expansion Scheme XXXIII (WRES-XXXIII)
30. Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-1), Part-C

### 3.22 Frequency Allocation Coordination for Power Line Carrier Communication (PLCC)

PCD Division coordinates and follows up with Wireless Planning and Coordination (WPC) Wing of Department of Telecommunications (DoT) to achieve timely frequency allocation for PLCC links of new power transmission lines of power utilities in the country.

### 3.23 Research & Development in Power Sector

#### 3.23.1 R&D activities in Power Sector: -

The Central Electricity Authority (CEA) under Section 73(k) of the Electricity Act, 2003 is vested with the function to promote research in the matters affecting the generation, transmission, distribution and trading of electricity.

Thus, Research and Development (R&D) for the Indian Power Sector is promoted and coordinated by CEA. Research & Development Schemes of the MoP are implemented by Central Power Research Institute (CPRI) in coordination with CEA.

**Standing Committee on R&D (SCRD) in Power Sector:** The SCRD, under the Chairmanship of Chairperson, Central Electricity Authority, identifies and prioritize important strategic areas of research and development, which are to be implemented under various research schemes of Ministry of Power. The SCRD identifies leading Researchers and

Domain Experts in diverse areas of Power Sector and engage them in the Research Schemes.

The Technical Committees in specific fields of power, namely, Thermal Generation, Hydro Generation, Transmission and ‘Grid, Distribution & Energy Conservation’ assist the SCRD in evaluation of new research proposals as well as in monitoring of the on-going research projects till their completion. The committees have representation from Academia, Industry, Utilities and Policy making bodies as well as experts from CEA.

Further, Research projects from eminent Institutions across India like IIT Kharagpur, IIT Kanpur, IIT Madras, IIT Bombay, CPRI, etc. have been supported on various thrust areas pertaining to Generation, Transmission, Distribution, Clean Energy and Renewables.

CEA oversees and promotes the activities of research and development in the Power Sector through coordination with multiple agencies involved in research and development activities.

Currently, following are the “Research Schemes of Ministry of Power being implemented through CPRI” for power sector development that are being facilitated by CEA through the SCRD:

- 2 **R&D under National Perspective Plan (NPP)** aimed at improving design of an individual plant component, evolving cost-efficient overall process in the plant, improving control & monitoring for system performance parameters, etc.
- 3 **Research Scheme on Power (RSoP)** for need based research in power sector including solving of operational problems encountered in the power system.
- 4 **In-house Research and Development (IHRD)** scheme for Central Power Research Institute (CPRI).

#### 3.23.2 Action taken for implementation of R&D for power sector:

Eight (8) meetings of the Technical Committees were held during the FY 2023-24 for monitoring of the on-going research projects as well as for evaluation of new research proposals. Also, one (1) meeting of the SCRD was held during the FY 2023-24 wherein the research proposals cleared by the respective Technical Committees were further reviewed. A total of thirteen (13) research proposals with an outlay of Rs 11.16 crores were

approved (one under NPP Scheme, eleven under RSoP and one under IHRD) by the SCRD. These includes projects on some of the priority areas of research in Power Sector like Clean Energy, Robotics, Hydrogen fuel cell, Nano-materials, Recycling of Solar cells, Artificial Intelligence (AI)/ Machine Learning (ML) etc. The projects have been initiated by the Project implementing organizations.

### 3.23.3 Mission on Advanced and High-Impact Research (MAHIR)

- I. Ministry of Power and Ministry of New and Renewable Energy have launched 'MAHIR' in June 2023 with the objective of end-to-end development of advanced technologies indigenously to make the future power system in India robust and advanced so that India takes the lead in the disruptive technologies in the power sector, as to ensure self-reliance in meeting our energy needs with wide societal and commercial benefits.
- II. For implementation of the Mission one Apex Committee headed by the Hon'ble Minister of Power & RE finalizes the research proposals and one Technical Scoping Committee (TSC) headed by the Chairperson, CEA identifies, shortlists, and prioritizes the research areas of high impact.
- III. Four (4) TSC meetings and one (1) special meeting, under the chairmanship of the Chairperson, CEA, have been held so far with Academia, Industry, Utilities, etc., to discuss research areas and technologies that could be developed under the Mission.
- IV. The activities carried out by the TSC are as under:
  - Identification of Experts for Technical Scoping Committee
  - Finalization of Focus areas of Research
  - The following research areas of importance were prioritized:
    - Alternatives to Lithium-Ion storage batteries

- Geo-thermal energy
- Carbon Capture & Utilization
- Green hydrogen technologies (High Efficiency Fuel Cell)
- Automation and Artificial Intelligence: indigenous development of the eco-system
- Renewable Energy Technologies (Solar PV, Solar thermal, Wave Energy, Tidal Energy, Biogas, Biofuels, and Waste to Energy)
- Two-shift operation of old coal based power plants

- Identification of proposals for Technical Scoping Committee
- Evaluation of proposals by Technical Scoping Committee

- V. Five (5) proposals have been recommended by the TSC for presentation to the Apex Committee of MAHIR after evaluation.

### 3.23.4 Updation of thrust areas for R&D in power sector and identification of high priority areas of power sector:

The thrust areas for R&D in power sector are updated and high priority areas are identified/updated for dissemination to the power sector organizations under the Central /State/Private Sector.

In this context, the details of new areas related to power sector where R&D is needed are sought from concerned stakeholders of the power sector.

### 3.23.5 Other R&D initiatives in CEA (MoU with IIT, Delhi):

CEA, being an apex technical organization for the development of power sector, its human resources need to be developed inter-alia through enhancing their technical knowledge and exposure to R&D activities.

In this regard, Central Electricity Authority (CEA) has in place a Memorandum of Understanding (MoU) with Indian Institute of Technology, Delhi (IITD) for the development of CEA's Human Resources by virtue of enhancing their technical education/ knowledge and R&D

exposure.

Under the provisions of MoU, one Chair Professorship for CEA has been instituted at IIT Delhi. M. Tech/MBA/Ph.D programmes are being made available to the officers of Ministry of Power (MoP), CEA and NPTI & CPRI on the recommendation of CEA/MoP as per rules of the Institute. The fees and charges for the courses are borne by the respective organization.

Presently, seven officers of CEA are pursuing part time M. Tech/MBA/Ph.D programmes from IIT Delhi.

### **3.23.6 Other R&D initiatives where CEA played important role and provided comments/inputs:**

- I. Department of Science and Technology (DST)'s EFC Note on Central Sector Umbrella Scheme S & T Institutional & Human Capacity Building and Central Sector Umbrella Scheme Research and Development.
- II. Draft Cabinet Note of Department of Science and Technology (DST) seeking approval for the continuation of the unified scheme - 'Vigyan Dhara' during the 15<sup>th</sup> Finance Commission period (FY 2021-22 to FY 2025-26).

## **3.24 Standardization activities**

### **3.24.1 Standardization activities and efforts to enhance the implementation of standards in the field of Power Sector:**

- I. Under the initiative of Government of India through the Indian National Strategy for Standardization (INSS) for establishment of robust 'Quality Ecosystem' in the country, a Standardization Cell with respect to Power Sector has been established in CEA under the Chairmanship of Chief Engineer (ET&I), CEA, under the aegis of Ministry of Power.
- II. The Standardization Cell is envisaged to act as a channel of communication among the Government, Industry and BIS to facilitate the identification of new

subjects and relevant experts for standardization and enhance implementation of Indian standards.

- III. The Standardization Cell headed by Chief Engineer (ET&I), CEA, have members from state/private sector utilities along with members from respective divisions of CEA.
- IV. Two (2) meetings of the Cell were held during 2023-24 in order to identify the requirement of standards in new areas and for review/ revision of existing BIS standards.
- V. The Cell has proposed to BIS for formulation of Indian Standards in five (5) new areas and for updation in five (5) existing standards.
- VI. Regular follow ups with BIS regarding formulation/ revision of standards is also been carried out.
- VII. Participating and contributing in the meetings being organized by BIS under INSS (like Preparation of Annual Programme for Standardization (APS), etc.).
- VIII. Further, experts from CEA are contributing in formulation/revision of standards through various Sectional Committees of BIS constituted for developing national standards on products, processes and services related to Power Sector.

### **3.24.2 Implementation of BIS Standards and Quality Control Orders in power sector:**

- I. The issue regarding oil (silicone or mineral) to be used in HVR Transformers {i.e. Electrostatic Precipitator Transformer Rectifier (ESP-TR)} flagged by the BHEL was analysed and provided considered opinion/views to the MoP.
- II. In order to ensure that conductors/ cables being used in the power sector are of requisite quality, CEA in the month of March 2024 issued an advisory for implementation of the BIS Standards and Quality Control Orders related to

aluminium ingots, wire rods and wires used for the manufacturing of conductors / cables.

- III. The National Electrical Code of India 2023 (NEC 2023), a comprehensive electrical installation Code providing guidelines for regulating Electrical installation practices across the country was circulated amongst power sector stakeholders for compliance as well as for wider dissemination and general awareness.

### **3.25 Make in India (MII) and Aatma Nirbhar Bharat Abhiyan (ANBA) initiatives:**

- I. CEA is the nodal point for implementation of policy initiatives of Government of India (GoI) under Make in India (MII) & Atma Nirbhar Bharat Abhiyaan (ANBA) in power sector.
- II. The Chief Engineer (ET&I), CEA is the Nodal Officer in this regard. He is acting as an interface between the Ministry of Power and Department for Promotion of Industry and Internal Trade (DPIIT)/PSUs/Organizations/Autonomous Bodies under the administrative control of the Ministry of Power as well as the Industry Associations.
- III. Dealing with the implementation of MII Orders of M/o Power and Government of India. It includes handling of grievance against non-compliance of MII provision, recommendations on registration of bidders from land border sharing countries, relaxation/exemption in minimum local content (MLC)/Global Tender Inquiry (GTE) in public procurements, Production Linked Incentive (PLI) schemes, etc.

#### **3.25.1 Major activities undertaken by CEA in relation to the implementation of the provisions of Public Procurement (Preference to Make in India), Order 2017 (PPP-MII) Order, 2017:**

Some of the major activities undertaken in this regard are given as under:

- I. Dissemination of information: In line with the MII initiatives MoP/ CEA create awareness by circulating various Orders/Guidelines/instructions/SoPs issued by DoE/DPIIT/MoP so as to ensure compliance of MII provisions by all concerned procuring entities of power sector under MoP.
- II. To further promote indigenization and development of domestic manufacturing continuous efforts are being made to include more items/equipment/product in to the list of sufficiency and competition i.e. Annexure-I of the MoP Order dated 16.11.2021. These items listed in Annexure-I of MoP Order are to be procured from Class-I local supplier only.
- III. In July, 2023 the Smart Meters (with minimum local content 50%) has been added to the list (i.e. Annexure-I of MoP Order dated 16.11.2021).
- IV. On 20<sup>th</sup> Feb, 2024, a trajectory for enhancing minimum local content in Smart Meters from 50% to 60% w.e.f. 1<sup>st</sup> June 2024 and to 70% w.e.f. 1<sup>st</sup> June 2025 was issued.
- V. Further, MoP vide latest amendment dated 24.04.2024, 400 kV XLPE Cables (MLC 50%) was added to the list of self-sufficiency and competitiveness.
- VI. Recently, Department for Promotion of Industry and Internal Trade (DPIIT) vide its Order dated 19.07.2024 has revised the "Public Procurement (Preference to Make in India), Order 2017". In this connection, M/o Power is in the process of revising its PPP-MII Order dated 16.11.2021. In this regard, inputs/suggestions of concerned stakeholders are being sought so as to review and revise the MoP Order dated 16.11.2021.
- VII. Grievances: The grievances received from various companies against non-compliance of MII provisions in the tenders floated by Central and State sector power Utilities are being redressed (under

the Make in India initiatives, being done as a continuous activity).

Detailed examination/scrutiny in case of each grievance was carried out by the CEA in consultation with respective power sector procuring entities and other concerned stakeholders/State Governments, Industry Associations, etc. The comments/views in this regard were furnished to the Ministry. A total of six (6) number of grievances were received during 2023-24 which were appropriately disposed of in consonance with the PPP-MII provisions.

VIII. Registration of bidders/vendors/suppliers/contractors: Prior registration of bidder companies from countries sharing land border with India is mandatory with DPIIT, Government of India in terms of Department of Expenditure Order (Public Procurement No. 4) issued vide No. F.No.7/10/2021-PPD dated 23.02.2023 under Rule 144 (xi) GFR, 2017 to be eligible to participate in the bids for public procurements. The applications of such companies are analysed with regard to the demand supply scenario (sufficiency and competition) for the item for which registration is sought and recommendations to this effect are being provided to the MoP from time to time. Six (6) number of applications for registration were processed during the year 2023-24.

IX. Global Tender Enquiry (GTE) permission: DoE OM No F.12/17/2019-PPD dated 15.05.2020 regarding amendment in Rule 161(iv) of General Financial Rules (GFRs) 2017 stipulates that no Global Tender Enquiry (GTE) shall be invited for tenders upto Rs.200 crore, subject to the prior approval for relaxation from the competent authority.

The cases from procuring entities under Ministry of Power seeking such permission received during 2023-24 were examined and analyzed as per Standard Operating Procedure (SOP) prescribed by the DPIIT in this regard

and accordingly recommendations were forwarded to the Ministry of Power.

X. Also, references received seeking relaxation/exemption from Minimum Local Content (MLC) criteria (procuring entities not getting Class -I local supplier for the listed items or otherwise are seeking relaxation) were dealt in terms of Para 14 of the DPIIT Order No. P-45021/2/2017-PP (BE-II) dated 16.09.2020.

### **3.25.2 Action taken by CEA in relation to the scheme formulated by the MoP for creation of Manufacturing Hub/Zone for Power Sector:**

Creation of Manufacturing Zone, the scheme envisaged by the Ministry of Power for Power & Renewable Energy Equipment under the Phase Manufacturing Programme (PMP) in line with Make in India initiative.

Rs.400 crore budgetary outlay sanctioned by GoI in March 2022 for setting up of Manufacturing Zone on pilot basis.

The proposal of Madhya Pradesh Industrial Development Corporation (MPIDC) was selected by MoP and MNRE for setting up a pilot manufacturing zone at Mohasa Babai, Narmadapuram, near Bhopal, Madhya Pradesh for Power & Renewable Energy Equipment.

NISE, NIWE and CPRI would be setting up Common Testing Facilities (CTF) in the selected manufacturing zone.

CPRI Business plan covering establishment of test facilities for Cables and Batteries was approved.

MoU was signed between MPIDC and CPRI on 7<sup>th</sup> June 2024 for establishment of CTF.

### **3.25.3 Inputs/comments on various other references/ issues concerning Make in India and Aatma Nirbhar Bharat initiatives of Government of India were taken up from time to time are:**

I. Amendments proposed by the DPIIT in MLC requirement for local supplier in the Public procurement (Preference to Make in India) Order, 2017.

- II. FDI proposal no. 6606 of M/s TLG Energy Ventures Pte. Ltd. - Request for Government approvals for acquisition of company incorporated in India (M/s Lantau (India) Pvt. Ltd.) in compliance with the FDI Policy involved in the business of providing consultancy and advisory services related to transitioning energy markets and energy economics.
- III. Request of Apraava Energy Pvt. Ltd. for making approvals under Press Note 3 of the FDI Rules coincident with registration under General Financial Rules of the Ministry of Finance.
- IV. Reference received from New Northeast Electric India Private Limited (NHVS), a Manufacturer of Gas Insulated switchgear (GIS), requesting for grant of approval for Transfer of Technology (ToT).
- V. IEEMA representation (The Testing and Measurement industry) - Mandatory pre-bid demonstration in utilities.
- VI. NTPC Vidyut Vyapar Nigam Limited (NVVN)'s proposal for independent verification of self-declarations and statutory Auditor's certificates submitted by a bidder for being Class-I bidder.
- VII. Consolidation of the demand of Forgings and Casting and related products, under GFR 161 (iv), 2017-reg.
- VIII. Issue raised by the Indian Stainless Seamless Pipes Manufacturers Association (ISSMA) regarding restrictive conditions for local vendors/suppliers in the tender by BHEL for the supply of SS seamless (Grade SUP304H) boiler tubes for Talcher Thermal Power Project Stage-III of M/s NTPC.
- IX. Representation of M/s Jakson to Hon'ble Minister (Power & NRE) regarding use of Chinese generating sets for critical applications like Defense, Data Centres, Airports, Infrastructure.
- X. Concerns raised by Laghu Udyog Vidyut Nirmata Sangh against Chhattisgarh State Power Transmission Company Ltd. (CSPTCL) for non-compliance of provisions

of Make in India (PP-MII) Order and DoE Order in the tender (s) floated for Reliable Communication Project.

### **3.26 Works related to Electric Vehicle Charging Infrastructure (EVCI):**

Examined and provided comments on the following important references:

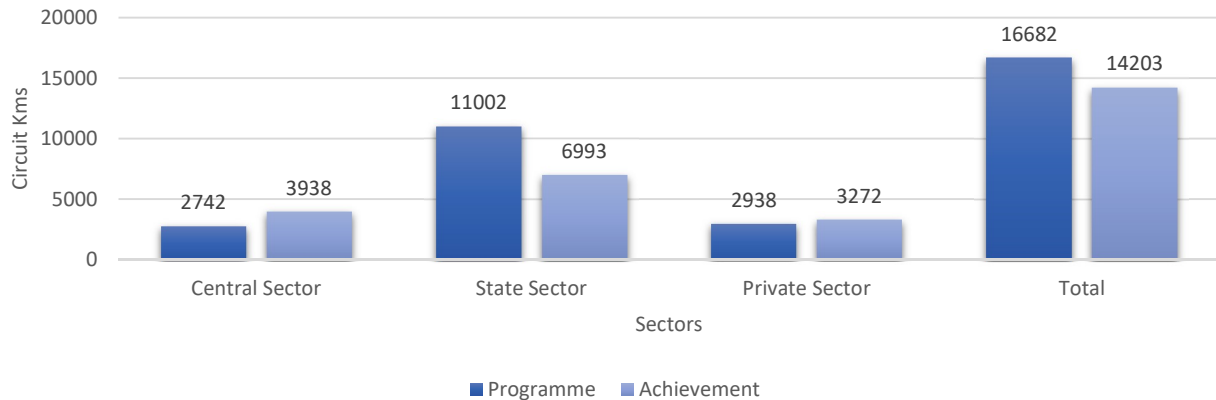
- I. Issues related to Behind-the-Meter (Upstream) power infrastructure in the PM e-Bus Sewa Scheme launched by Ministry of Housing & Urban Affairs (MoHUA).
- II. Consultation with BEE, in review and revision of the guidelines and standards for EV charging infrastructure (EVCI) issued by the Ministry of Power in 2022.
- III. Queries raised by the ESSAR Power on applicability of captive charging infrastructure and Battery Swapping Station (BSS) provisions of the Guidelines & Standards regarding Charging Infrastructure for Electric Vehicles (EV) issued by the Ministry of Power on 14.01.2022.
- IV. Clarification on whether the CCS2 type charger falls under the CCS (minimum 50 kW) charger specified in Annexure-II of the Revised Consolidated Guidelines & Standards dated 14.01.2022 issued by the Ministry of Power.
- V. Queries related to the issues associated with the charging infrastructure installed in Kerala.
- VI. Issue of "Lack of access to Government land for setting up EV Charging Infrastructure" has been raised by the Tata Power.

### **3.27 The following references were also dealt with:**

- I. Draft Cabinet Note of Ministry of Environment, Forest and Climate Change (MoEF&CC) for approval of the Central Sector Scheme "Control of Pollution" for the period from 2021-22 till 2025-26.

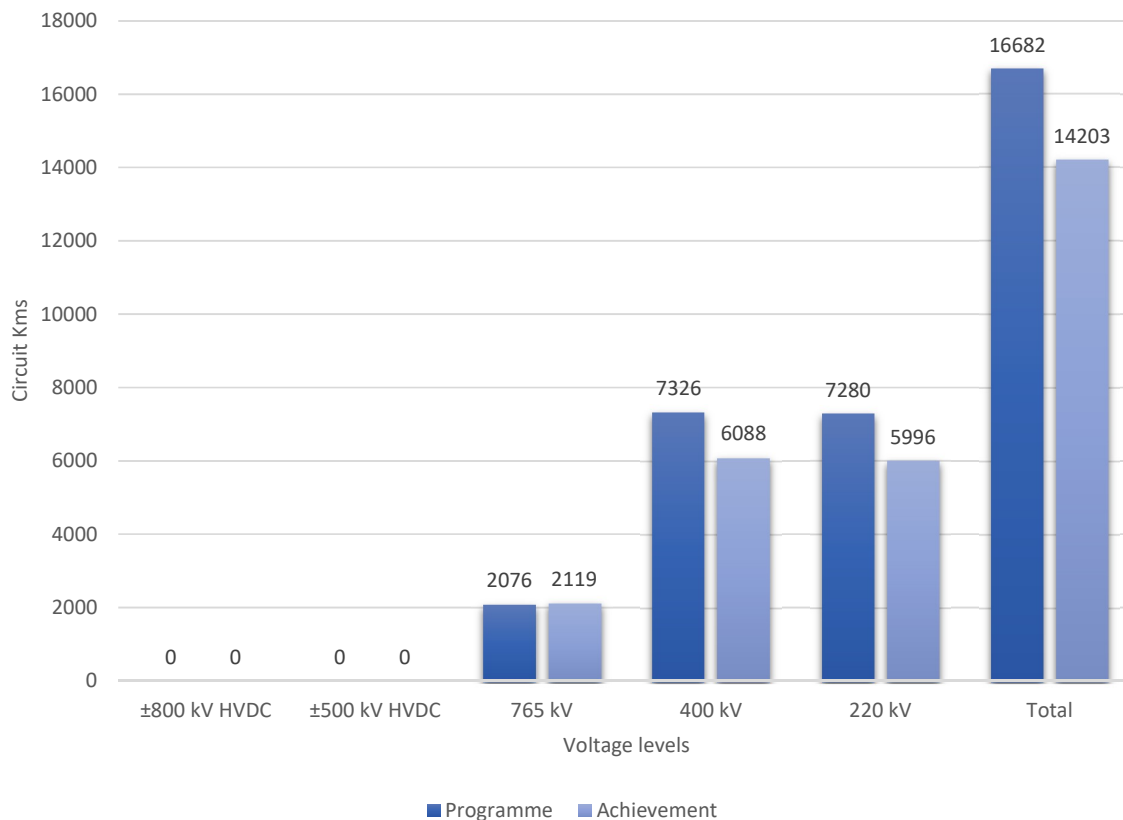
- II. EFC Memo on ADEETIE (Assistance in Deploying Energy Efficient Technologies in Industries & Establishments) having the objective to facilitate the financing support in deploying energy efficient technologies in key sectors of economy by providing interest subvention to eligible entities based on the defined energy savings. A total of Rs.1824.10 crore interest subvention (FY 2023-24 to 29-30) has been proposed under the ADEETIE scheme. Ministry of Power is the sponsoring agency whereas Bureau of Energy Efficiency (BEE) is the nodal agency for implementing the scheme.
  - III. The proposal of MHI for FAME III with an outlay of Rs. 12,600 crore for the period of two (02) years w.e.f. 01.04.2024 to 31.03.2026 along with a draft Concept Note, EFC Note, Scheme Guidelines and Scheme Notification was examined and provided comments thereof.
  - IV. Preliminary Project Report - (PPRID – 12225) of DPIIT, Min. of Commerce on “Alleviating Heat stress by enhancing production of Affordable Cooling Devices (AHEAD)” with total cost of the project is Rs.5810 Cr with 50% share from Central Government and remaining 50% from external assistance i.e. World Bank-IBRD.
  - V. Preliminary Project Report - (PPRID – 12106) of MNRE on Offshore Wind Development Programme to develop Offshore Wind eco system in the country. The total cost of the project was Rs.6864.4 crore including external technical assistance of Rs.3398.4 crore, Rs. 1320 crore by implementing agency, Rs.600 crore by Central Govt. and Rs.1546 crore by Others. The duration of the project is of five years and eleven months i.e. from 01.07.2023 to 30.06.2029.
  - VI. Proposal of MNRE PPRID - 12092 to seek Grant/TA from GIZ to support Energy Storage for Renewable Energy Integration in India (StoREin) with duration of the project seven (7) years and with 100% external assistance of Rs. 156.67 crores.
  - VII. Preliminary Project Report - (PPRID – 12135) by Industry, Energy and Labour department, Maharashtra on “Solar Photovoltaic Water Pumping Systems and Agricultural Feeder Solarization Project/Conventional Pump”.
  - VIII. Approval of Qualification Packs developed by Power Sector Skill Council (PSSC).
- 3.28 Nominations of officers of R&D Division, CEA in various Committees:**
- I. CE (ET&I) nominated as a member of a committee to draft the guidelines on reverse charging of grid from batteries of E-vehicle.
  - II. CE (ET&I) nominated as a member of a committee constituted for preparation of a project report/proposal for creating a state of the art common training infrastructure at NPTI.
  - III. Director (ET&I), CEA nominated as a member convener of the Committee constituted for preparation and finalization of the tender document for engagement of consultant for preparing a
  - V. Director (ET&I), CEA nominated as a member of the Committee constituted for considering ‘To develop of Energy Benchmarking parameters for EV & EVSE under E-mobility initiative of BEE’.
- IV. Feasibility report and tech-commercial analysis of different options for manufacturing and laying of Sub-sea cables in India.

### Sector-wise break-up of Program vs Achievement of Transmission Lines in FY 2023-24 (ckm)



**Chart-I**

### Voltage-wise break-up of Program vs Achievement of Transmission Lines in FY 2023-24 (ckm)



**Chart-II**

Programme/Achievement of 765 kV Transmission Lines in  
FY 2023-24 (ckm)

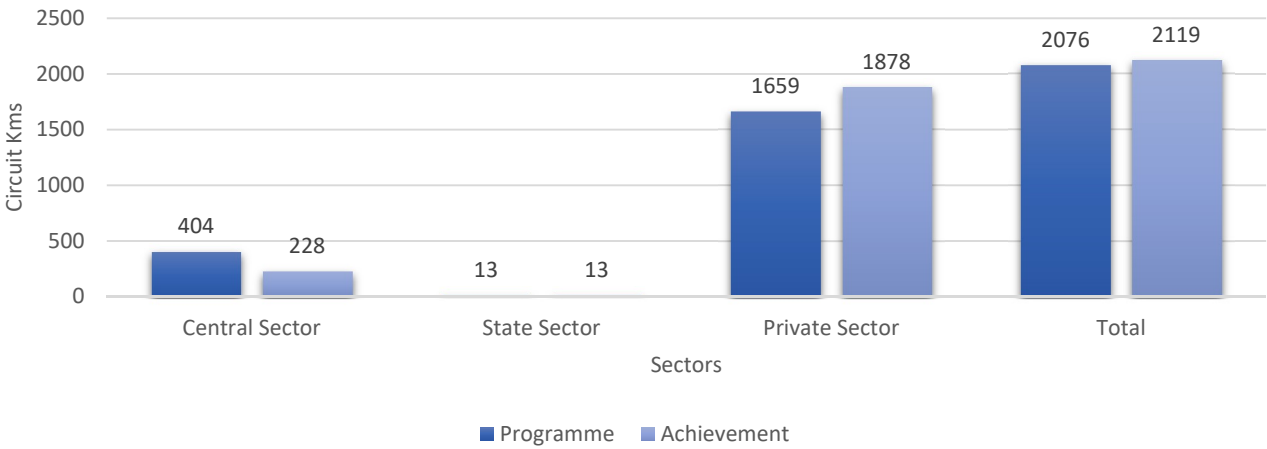


Chart-III

Programme/Achievement of 400 kV Transmission Lines in  
FY 2023-24 (ckm)

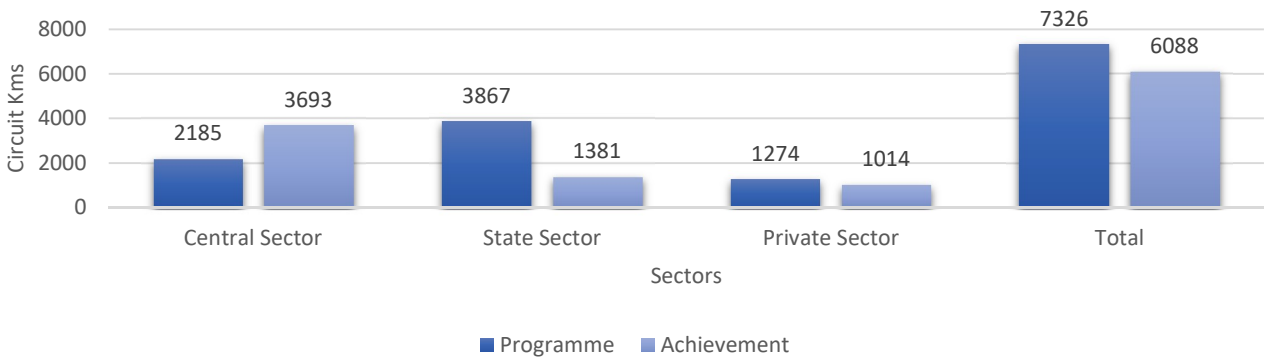


Chart-IV

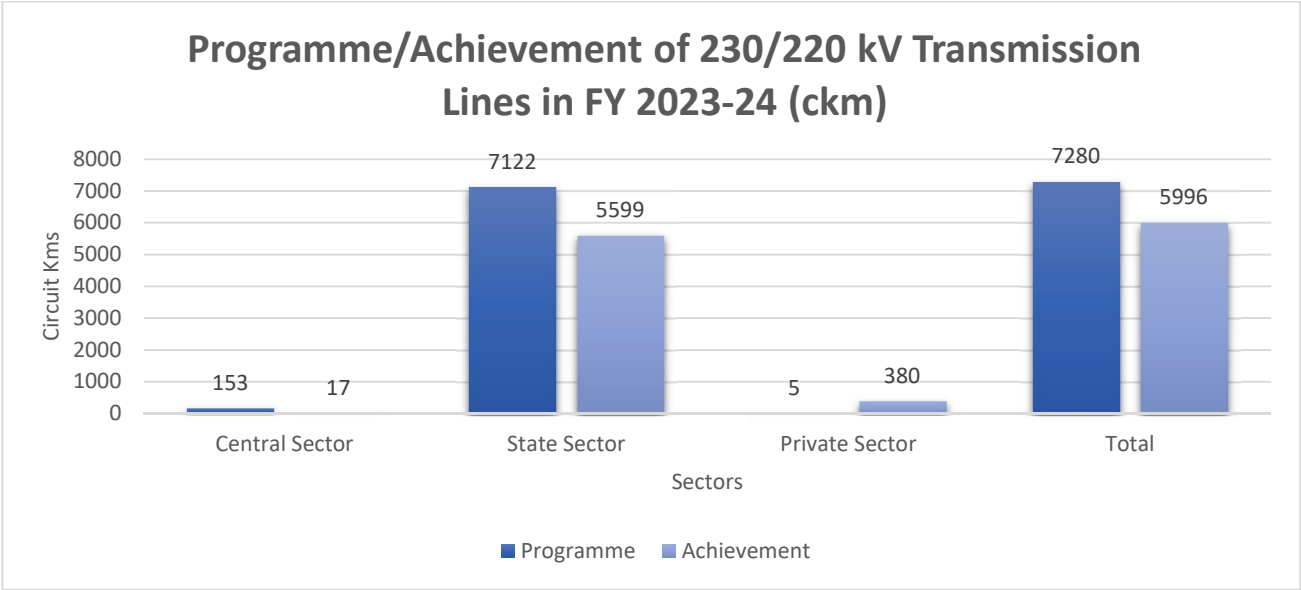


Chart-V

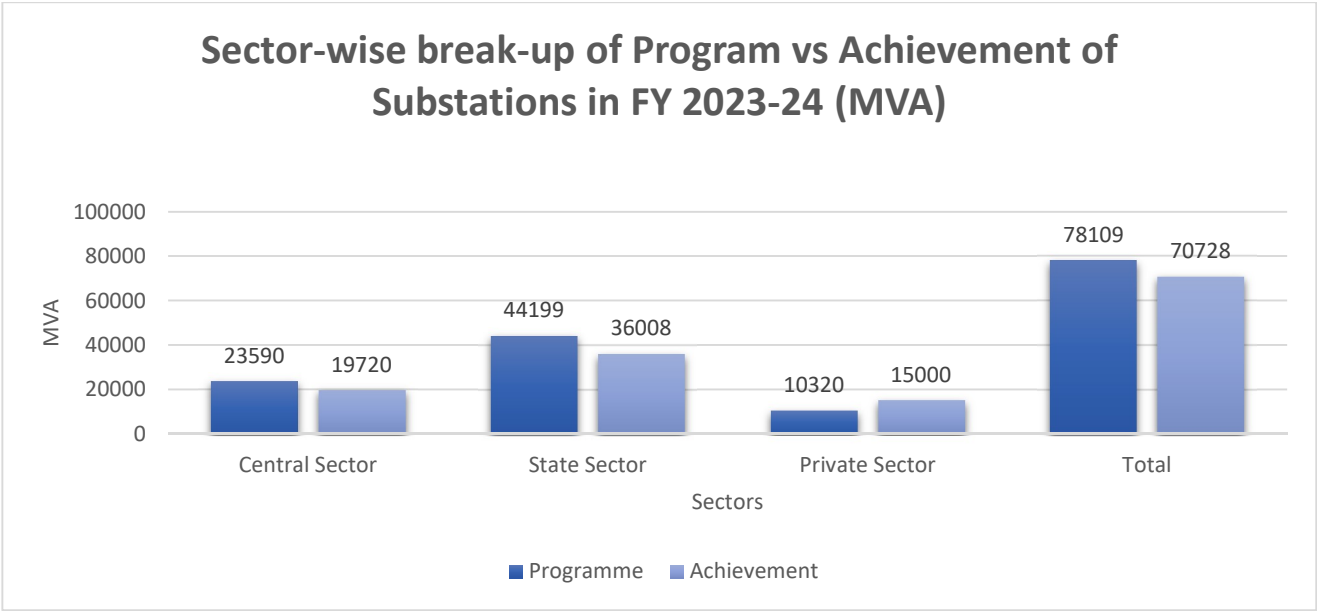


Chart-VI

Voltage-wise break-up of Program vs Achievement of Substations in FY 2023-24 (MVA)

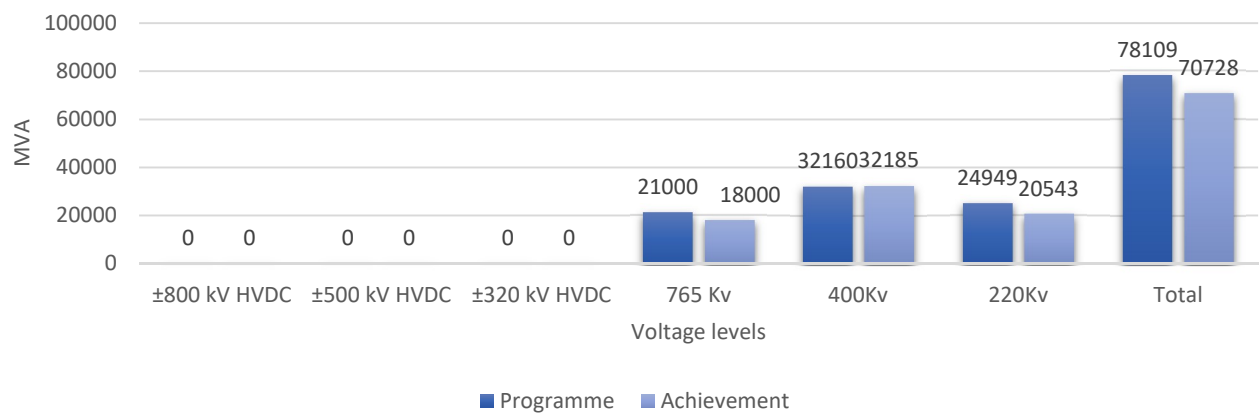


Chart-VII

Programme/Achievement of 765 kV Substations in FY 2023-24 (MVA)

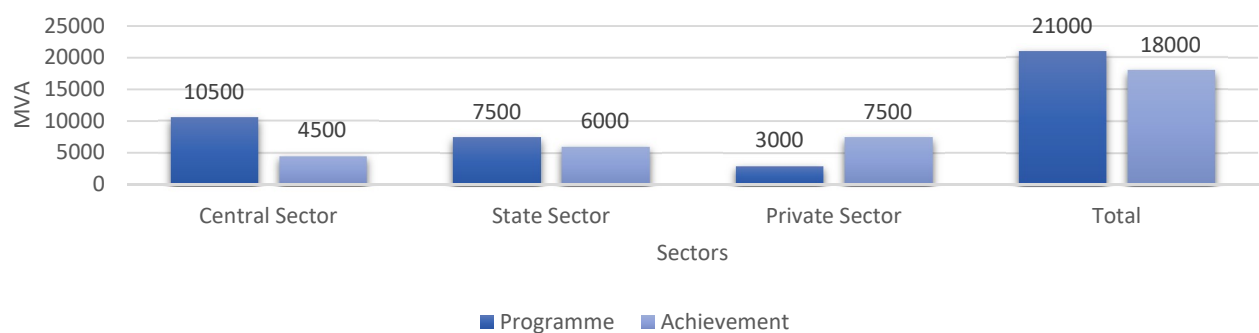


Chart-VIII

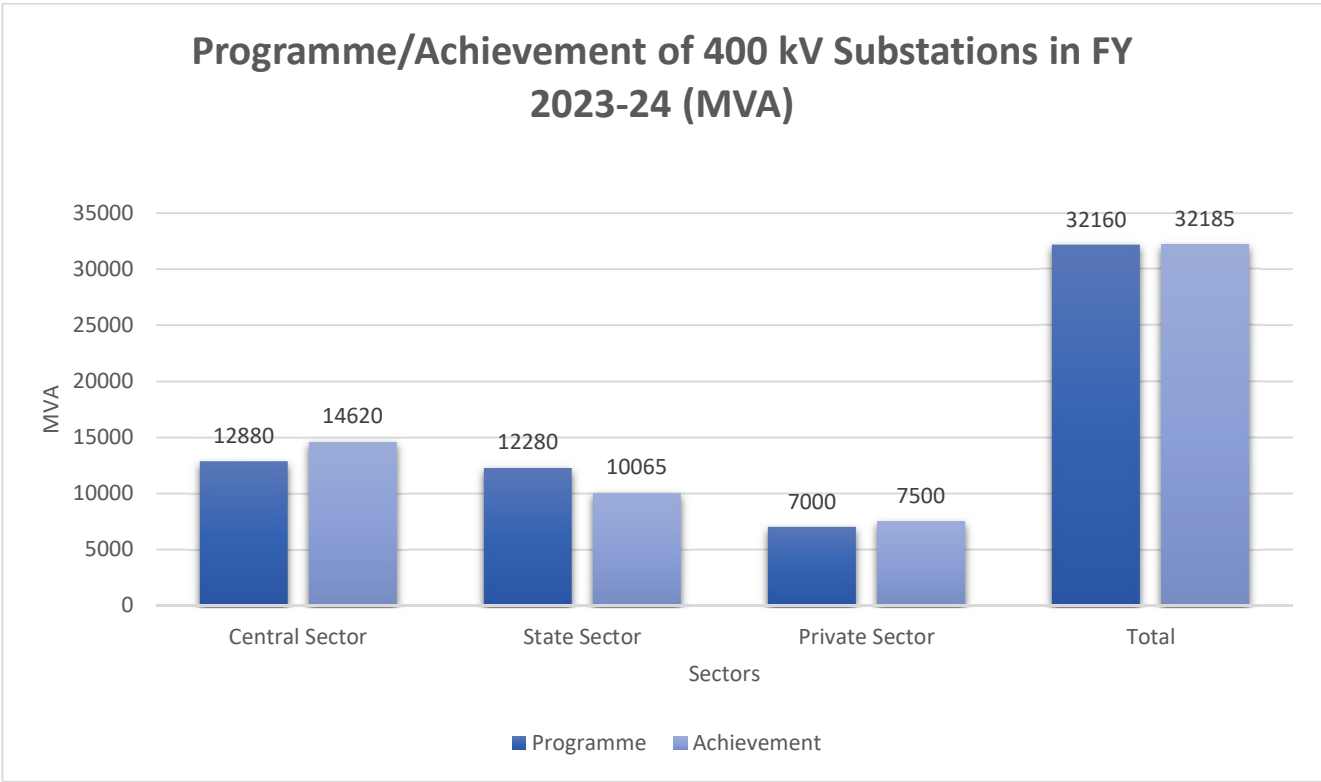


Chart-IX

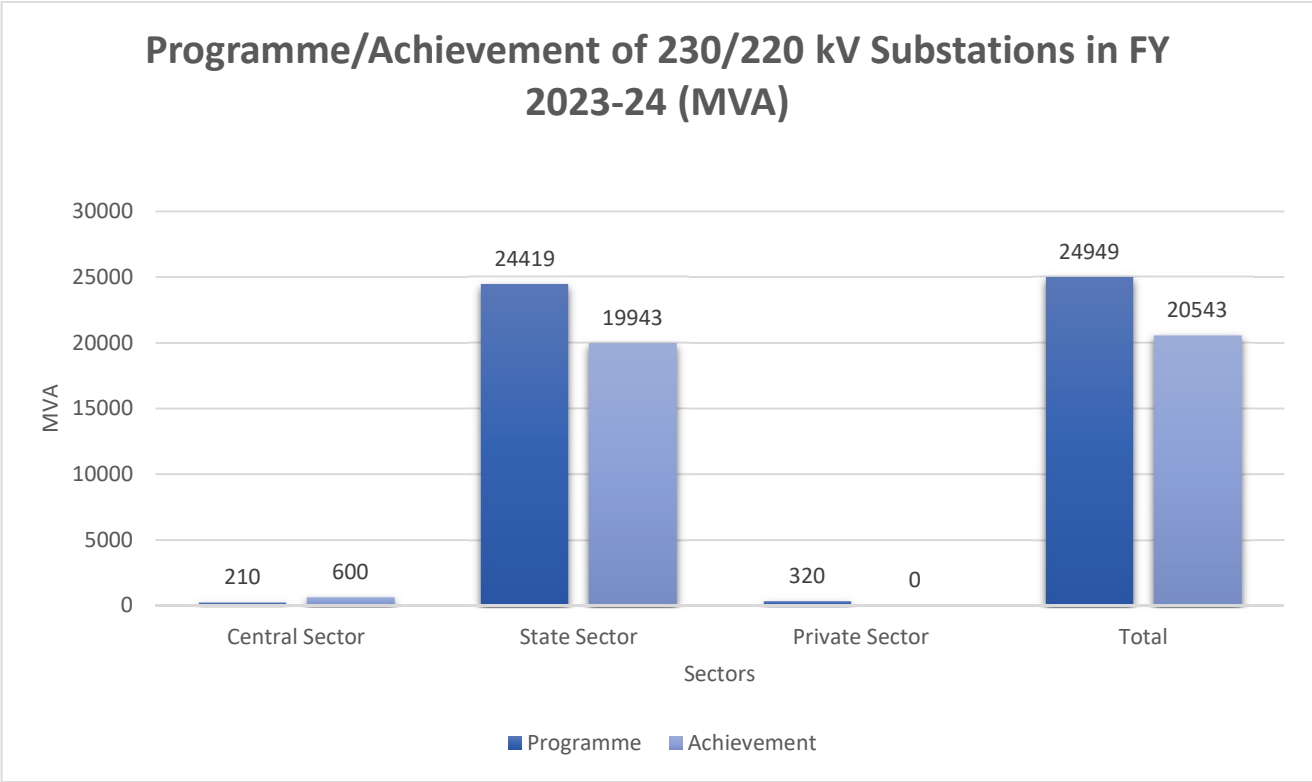


Chart-X

## CHAPTER – 4

# GRID OPERATION AND MANAGEMENT

### 4.1 Organizational Structure in Grid Operation and Management

The Central Government has established Regional Power Committee (RPC) in each region in accordance with the provisions of Electricity Act, 2003 to facilitate integrated operation of the power system in that region. The real time operation of the power system is looked after by the Regional Load Dispatch Centres (RLDCs) set up in the five Regions and at the national level by National Load Dispatch Centre (NLDC). The Regional Power Committee is a conglomerate of all the players partaking in grid operation, i.e. Regional Load Dispatch Centre, generating companies, transmission utilities, distribution utilities, power traders, etc. Its Secretariat is manned by the officers of Central Electricity Authority (CEA).

The Regional Power Committee (RPC) operates through a number of Sub-Committees, viz. Operation Sub-Committee, Commercial Sub-Committee, Protection Sub-Committee, System Studies Sub-Committee, and Technical Coordination Committee. The Operation Sub-Committee meets every month to review the grid operation in the previous month and plan grid operation for the next month. The Commercial Sub-Committee discusses commercial issues viz. energy accounting related matters, matters pertaining to Special Energy Meters (SEMs), settlement of dues, etc. The Protection Sub-Committee discusses and analyses the various trippings which took place since its last meeting and recommends/ monitors the corrective actions to avoid recurrence of such trippings. It also finalizes the various protection schemes including protection coordination. The System Studies Sub-Committee meets periodically for the purpose of system studies related to assessment of network elements for reactive compensation, operational load flow,

transient stability studies etc. The Technical Coordination (TCC) meets before the Regional Power Committee for deliberating on the various technical, operational and commercial issues and the decisions are placed forth for final resolution in the Regional Power Committee. The RPCs play an important role in planning grid operation, since they are responsible for protection coordination, outage planning of generating units and transmission system, planning reactive compensation etc. Member (Grid Operation & Distribution), CEA is also a Member of the Regional Power Committees and guides the Committees to arrive at amicable solutions with uniformity of approach through unbiased decisions. Apart from RPCs, the Ministry of Power (MoP) had vide Order dated 25th March, 2013, established the National Power Committee (NPC) to evolve a common approach to issues related to reliability and security of the grid.

CEA monitors the power supply position in the country, prepares the All-India monthly power supply position report, harmonizes all matters of grid operation and management between the five Regions, coordinates enquiry of grid disturbances, recommends to the Ministry of Power the quantum of allocation from Central Generating Stations and also facilitates the implementation of the allocation through the Regional Power Committees. The anticipated Power Supply Position for the next year known as Load Generation Balance Report (LGBR), is also prepared every year.

### 4.2 Power Supply Position

The Central Electricity Authority brings out the All India Power Supply Position on a monthly basis, both in terms of Energy and Peak, giving the Energy Requirement, Energy Supplied in Million Units (MUs) and Energy not Supplied in Million Units

(MUs) as well as in percentage and the Peak Demand, Peak Met in Mega Watt (MW) and Demand not Met in Mega Watt (MW) as well as in percentage. The total Energy Requirement in the country during the year 2023-24 was 16,26,132 MUs as against 15,13,497 MUs during the previous year 2022-23, registering an increase of 7.4%. The total Energy Supplied in the country during the year 2023-24 was 1,622,020 MUs as against 1,505,914 MUs during the previous year 2022-23, registering an increase of 7.7%. The Energy not Supplied during the year 2023-24 was 4,112 MUs (0.3%) against 7,583 MUs (0.5%) during the previous the year 2022-23. The Peak Demand during the year 2023-24 was 2,43,271 MW as against 2,15,888 MW during the previous year 2022-23, registering an increase of 12.7%. The Peak

Met during the year 2023-24 was 2,39,931 MW as against 2,07,231 MW during the previous year 2022-23, registering an increase of 15.8%. The Demand not Met during the year 2023-24 was 3,340 MW (1.4%) as against 8,657 MW (4%) during the previous year 2022-23.

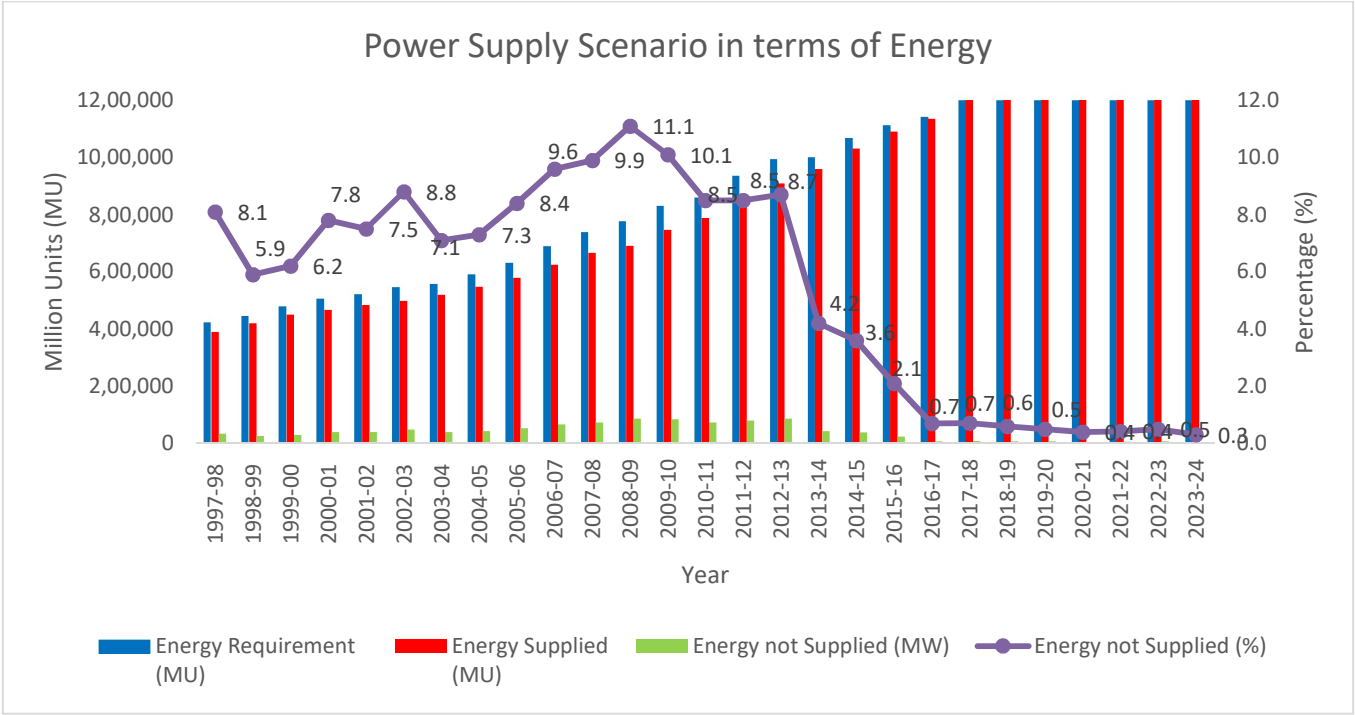
In the context of power supply, it may be mentioned that there is adequate availability of electricity in the country. The marginal gap between demand and supply of electricity is generally on account of factors other than inadequacy of power availability in the country e.g. constraints in distribution network, financial constraints, commercial reasons, forced outage of generating units etc.

The Power supply position since beginning of 9<sup>th</sup> Plan is as under:

### **ENERGY:**

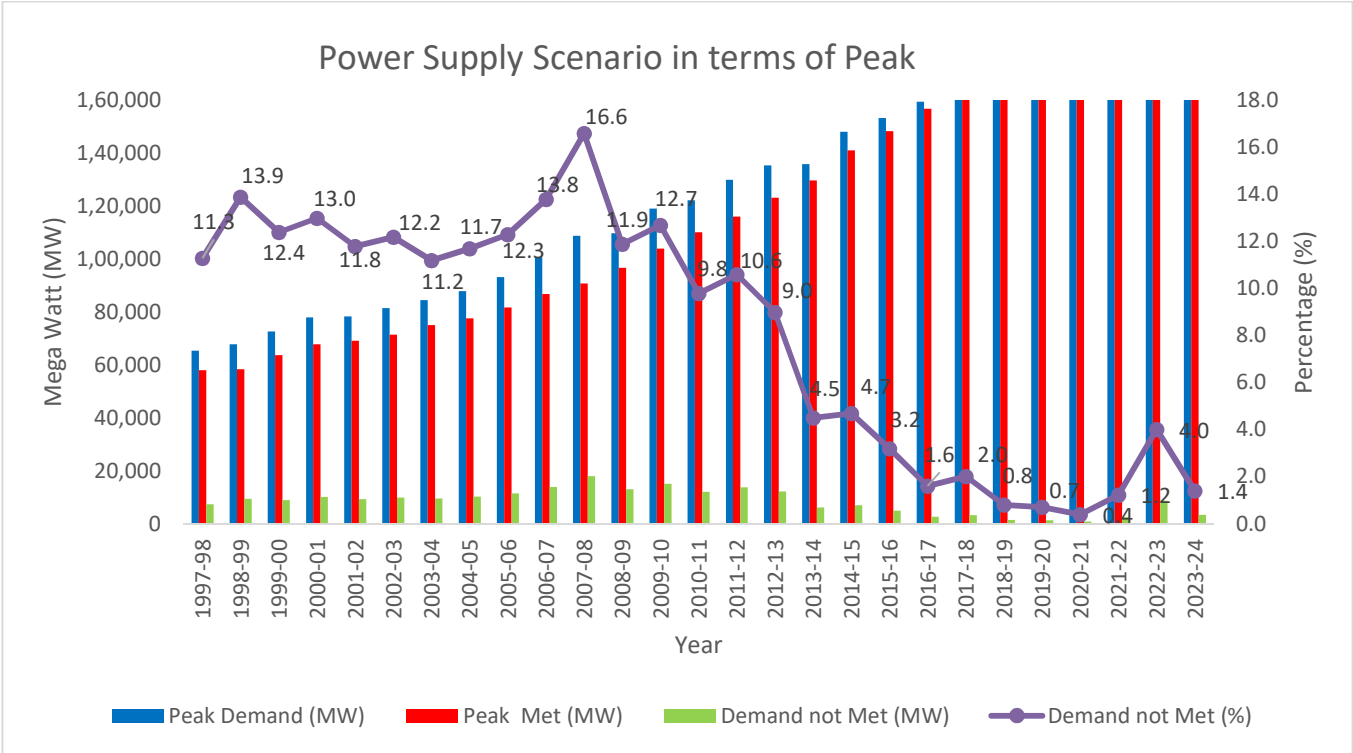
| Year    | Energy Requirement (MU) | Energy Supplied (MU) | Energy not Supplied (MW) | Energy not Supplied (%) |
|---------|-------------------------|----------------------|--------------------------|-------------------------|
| 1997-98 | 4,24,505                | 3,90,330             | 34,175                   | 8.1                     |
| 1998-99 | 4,46,584                | 4,20,235             | 26,349                   | 5.9                     |
| 1999-00 | 4,80,430                | 4,50,594             | 29,836                   | 6.2                     |
| 2000-01 | 5,07,216                | 4,67,400             | 39,816                   | 7.8                     |
| 2001-02 | 5,22,537                | 4,83,350             | 39,187                   | 7.5                     |
| 2002-03 | 5,45,983                | 4,97,890             | 48,093                   | 8.8                     |
| 2003-04 | 5,59,264                | 5,19,398             | 39,866                   | 7.1                     |
| 2004-05 | 5,91,373                | 5,48,115             | 43,258                   | 7.3                     |
| 2005-06 | 6,31,554                | 5,78,819             | 52,735                   | 8.4                     |
| 2006-07 | 6,90,587                | 6,24,495             | 66,092                   | 9.6                     |
| 2007-08 | 7,39,343                | 6,66,007             | 73,336                   | 9.9                     |
| 2008-09 | 7,77,039                | 6,91,038             | 86,001                   | 11.1                    |
| 2009-10 | 8,30,594                | 7,46,644             | 83,950                   | 10.1                    |
| 2010-11 | 8,61,591                | 7,88,355             | 73,236                   | 8.5                     |
| 2011-12 | 9,37,199                | 8,57,886             | 79,313                   | 8.5                     |
| 2012-13 | 9,95,557                | 9,08,652             | 86,905                   | 8.7                     |
| 2013-14 | 10,02,257               | 9,59,829             | 42,428                   | 4.2                     |
| 2014-15 | 10,68,923               | 10,30,785            | 38,138                   | 3.6                     |
| 2015-16 | 11,14,408               | 10,90,850            | 23,558                   | 2.1                     |
| 2016-17 | 11,42,928               | 11,35,332            | 7,596                    | 0.7                     |
| 2017-18 | 12,13,326               | 12,04,697            | 8,629                    | 0.7                     |
| 2018-19 | 12,74,595               | 12,67,526            | 7,070                    | 0.6                     |

|         |           |           |       |     |
|---------|-----------|-----------|-------|-----|
| 2019-20 | 12,91,010 | 12,84,444 | 6,566 | 0.5 |
| 2020-21 | 12,75,534 | 12,70,663 | 4,871 | 0.4 |
| 2021-22 | 13,79,812 | 13,74,024 | 5,787 | 0.4 |
| 2022-23 | 15,13,497 | 15,05,914 | 7,583 | 0.5 |
| 2023-24 | 16,26,132 | 16,22,020 | 4,112 | 0.3 |



**PEAK:**

| Year    | Peak Demand<br>(MW) | Peak Met (MW) | Demand not<br>Met (MW) | Demand not Met<br>(%) |
|---------|---------------------|---------------|------------------------|-----------------------|
|         |                     |               | (MW)                   | (%)                   |
| 1997-98 | 65,435              | 58,042        | 7,393                  | 11.3                  |
| 1998-99 | 67,905              | 58,445        | 9,460                  | 13.9                  |
| 1999-00 | 72,669              | 63,691        | 8,978                  | 12.4                  |
| 2000-01 | 78,037              | 67,880        | 10,157                 | 13.0                  |
| 2001-02 | 78,441              | 69,189        | 9,252                  | 11.8                  |
| 2002-03 | 81,492              | 71,547        | 9,945                  | 12.2                  |
| 2003-04 | 84,574              | 75,066        | 9,508                  | 11.2                  |
| 2004-05 | 87,906              | 77,652        | 10,254                 | 11.7                  |
| 2005-06 | 93,255              | 81,792        | 11,463                 | 12.3                  |
| 2006-07 | 1,00,715            | 86,818        | 13,897                 | 13.8                  |
| 2007-08 | 1,08,866            | 90,793        | 18,073                 | 16.6                  |
| 2008-09 | 1,09,809            | 96,785        | 13,024                 | 11.9                  |
| 2009-10 | 1,19,166            | 1,04,009      | 15,157                 | 12.7                  |
| 2010-11 | 1,22,287            | 1,10,256      | 12,031                 | 9.8                   |
| 2011-12 | 1,30,006            | 1,16,191      | 13,815                 | 10.6                  |
| 2012-13 | 1,35,453            | 1,23,294      | 12,159                 | 9.0                   |
| 2013-14 | 1,35,918            | 1,29,815      | 6,103                  | 4.5                   |
| 2014-15 | 1,48,166            | 1,41,160      | 7,006                  | 4.7                   |
| 2015-16 | 1,53,366            | 1,48,463      | 4,903                  | 3.2                   |
| 2016-17 | 1,59,542            | 1,56,934      | 2,608                  | 1.6                   |
| 2017-18 | 1,64,066            | 1,60,752      | 3,314                  | 2.0                   |
| 2018-19 | 1,77,022            | 1,75,528      | 1,494                  | 0.8                   |
| 2019-20 | 1,83,804            | 1,82,533      | 1,271                  | 0.7                   |
| 2020-21 | 1,90,198            | 1,89,395      | 802                    | 0.4                   |
| 2021-22 | 2,03,014            | 2,00,539      | 2,475                  | 1.2                   |
| 2022-23 | 2,15,888            | 2,07,231      | 8,657                  | 4.0                   |
| 2023-24 | 2,43,271            | 2,39,931      | 3,340                  | 1.4                   |



The State/UT/Region-wise Power Supply Position in terms of Energy and Peak during the year 2023-24 is enclosed at **Annexure-4A.**

The details of the State/UT-wise allocation from Conventional Central Generating Stations in the country as on 31.03.2024, is enclosed at **Annexure-4B.**

### 4.3 System Operation in the Regions

**4.3.1 Northern Region:** The gap between Energy Requirement and Energy Supplied was 0.4% in the region during the year 2023-24 as compared to 1.0% during the year 2022-23 and a gap of 0.5% between Peak Demand and Peak Met during the year 2023-24

as compared to 1.0% during the year 2022-23.

Further, the comparison between Power Supply Position in terms of Energy and Peak during the year 2023-24 and the year 2022-23 is outlined below:

#### Energy:

| State / UT             | April,2023 -March,2024 |                 | April,2022 - March,2023 |                 | % Change           |                 |
|------------------------|------------------------|-----------------|-------------------------|-----------------|--------------------|-----------------|
|                        | Energy Requirement     | Energy Supplied | Energy Requirement      | Energy Supplied | Energy Requirement | Energy Supplied |
|                        | ( MU )                 | ( MU )          | ( MU )                  | ( MU )          | ( % )              | ( % )           |
| Chandigarh             | 1,789                  | 1,789           | 1,788                   | 1,788           | 0.0                | 0.0             |
| Delhi                  | 35,501                 | 35,496          | 35,143                  | 35,133          | 1.0                | 1.0             |
| Haryana                | 63,983                 | 63,636          | 61,451                  | 60,945          | 4.1                | 4.4             |
| Himachal Pradesh       | 12,805                 | 12,767          | 12,649                  | 12,542          | 1.2                | 1.8             |
| UT of J&K and Ladakh   | 20,040                 | 19,763          | 19,639                  | 19,322          | 2.0                | 2.3             |
| Punjab                 | 69,533                 | 69,528          | 69,522                  | 69,220          | 0.0                | 0.4             |
| Rajasthan              | 1,07,422               | 1,06,806        | 1,01,801                | 1,00,057        | 5.5                | 6.7             |
| Uttar Pradesh          | 1,48,791               | 1,48,287        | 1,44,251                | 1,43,050        | 3.1                | 3.7             |
| Uttarakhand            | 15,644                 | 15,532          | 15,647                  | 15,386          | 0.0                | 1.0             |
| <b>Northern Region</b> | <b>4,76,852</b>        | <b>4,74,946</b> | <b>4,63,088</b>         | <b>4,58,640</b> | <b>3.0</b>         | <b>3.6</b>      |

#### Peak:

| State / UT             | April,2023 -March,2024 |               | April,2022 - March,2023 |               | % Change    |            |
|------------------------|------------------------|---------------|-------------------------|---------------|-------------|------------|
|                        | Peak Demand            | Peak Met      | Peak Demand             | Peak Met      | Peak Demand | Peak Met   |
|                        | ( MW )                 | ( MW )        | ( MW )                  | ( MW )        | ( % )       | ( % )      |
| Chandigarh             | 411                    | 411           | 407                     | 407           | 1.0         | 1.0        |
| Delhi                  | 7,437                  | 7,437         | 7,695                   | 7,695         | -3.4        | -3.4       |
| Haryana                | 13,088                 | 12,844        | 12,768                  | 12,768        | 2.5         | 0.6        |
| Himachal Pradesh       | 2,181                  | 2,181         | 2,071                   | 2,071         | 5.3         | 5.3        |
| UT of J&K and Ladakh   | 3,181                  | 3,133         | 3,137                   | 2,967         | 1.4         | 5.6        |
| Punjab                 | 15,293                 | 15,293        | 14,311                  | 14,311        | 6.9         | 6.9        |
| Rajasthan              | 18,128                 | 18,128        | 17,399                  | 17,206        | 4.2         | 5.4        |
| Uttar Pradesh          | 28,704                 | 28,284        | 27,369                  | 26,589        | 4.9         | 6.4        |
| Uttarakhand            | 2,635                  | 2,405         | 2,594                   | 2,594         | 1.6         | -7.3       |
| <b>Northern Region</b> | <b>80,980</b>          | <b>80,548</b> | <b>77,337</b>           | <b>76,561</b> | <b>4.7</b>  | <b>5.2</b> |

**4.3.2 Western Region:** The gap between Energy Requirement and Energy Supplied was 0.1% in the region during the year 2023-24 as compared to 0.1% during the year 2022-23 and a gap of 4.6% between Peak Demand and Peak Met during the year 2023-24 while Western Region was able to

meet its Peak Demand completely during the year 2022-23.

Further, the comparison between Power Supply Position in terms of Energy and Peak during the year 2023-24 and the year 2022-23 is outlined below:

#### Energy:

| State / UT                           | April,2023 - March,2024 |                 | April,2022 - March,2023 |                 | % Change           |                 |
|--------------------------------------|-------------------------|-----------------|-------------------------|-----------------|--------------------|-----------------|
|                                      | Energy Requirement      | Energy Supplied | Energy Requirement      | Energy Supplied | Energy Requirement | Energy Supplied |
|                                      | ( MU )                  | ( MU )          | ( MU )                  | ( MU )          | ( % )              | ( % )           |
| Chhattisgarh                         | 39,930                  | 39,872          | 37,446                  | 37,374          | 6.6                | 6.7             |
| Gujarat                              | 1,45,768                | 1,45,740        | 1,39,043                | 1,38,999        | 4.8                | 4.9             |
| Madhya Pradesh                       | 99,301                  | 99,150          | 92,683                  | 92,325          | 7.1                | 7.4             |
| Maharashtra                          | 2,07,108                | 2,06,931        | 1,87,309                | 1,87,197        | 10.6               | 10.5            |
| Dadra & Nagar Haveli and Daman & Diu | 10,164                  | 10,164          | 10,018                  | 10,018          | 1.5                | 1.5             |
| Goa                                  | 5,111                   | 5,111           | 4,669                   | 4,669           | 9.5                | 9.5             |
| <b>Western Region</b>                | <b>5,17,714</b>         | <b>5,17,301</b> | <b>4,77,393</b>         | <b>4,76,808</b> | <b>8.4</b>         | <b>8.5</b>      |

#### Peak:

| State / UT                           | April,2023 - March,2024 |               | April,2022 - March,2023 |               | % Change    |            |
|--------------------------------------|-------------------------|---------------|-------------------------|---------------|-------------|------------|
|                                      | Peak Demand             | Peak Met      | Peak Demand             | Peak Met      | Peak Demand | Peak Met   |
|                                      | ( MW )                  | ( MW )        | ( MW )                  | ( MW )        | ( % )       | ( % )      |
| Chhattisgarh                         | 6,148                   | 6,148         | 5,339                   | 5,285         | 15.2        | 16.3       |
| Gujarat                              | 24,829                  | 24,544        | 21,464                  | 21,382        | 15.7        | 14.8       |
| Madhya Pradesh                       | 18,252                  | 17,817        | 17,347                  | 17,238        | 5.2         | 3.4        |
| Maharashtra                          | 31,178                  | 27,996        | 30,935                  | 28,846        | 0.8         | -2.9       |
| Dadra & Nagar Haveli and Daman & Diu | 1,327                   | 1,327         | 1,243                   | 1,243         | 6.8         | 6.8        |
| Goa                                  | 776                     | 776           | 718                     | 713           | 8.1         | 8.8        |
| <b>Western Region</b>                | <b>76,050</b>           | <b>72,556</b> | <b>71,677</b>           | <b>71,677</b> | <b>6.1</b>  | <b>1.2</b> |

**4.3.3 Southern Region:** The gap between Energy Requirement and Energy Supplied was 0.1% in the region during the year 2023-24 as compared to 0.2% during the year 2022-23 and the Southern Region was able to meet its Peak Demand completely

during the year 2023-24 and the year 2022-23.

Further, the comparison between Power Supply Position in terms of Energy and Peak during the year 2023-24 and the year 2022-23 is outlined below:

**Energy:**

| State / UT             | April,2023 -March,2024 |                 | April,2022 - March,2023 |                 | % Change           |                 |
|------------------------|------------------------|-----------------|-------------------------|-----------------|--------------------|-----------------|
|                        | Energy Requirement     | Energy Supplied | Energy Requirement      | Energy Supplied | Energy Requirement | Energy Supplied |
|                        | ( MU )                 | ( MU )          | ( MU )                  | ( MU )          | ( % )              | ( % )           |
| Andhra Pradesh         | 80,209                 | 80,151          | 72,302                  | 71,893          | 10.9               | 11.5            |
| Telangana              | 84,623                 | 84,613          | 77,832                  | 77,799          | 8.7                | 8.8             |
| Karnataka              | 94,088                 | 93,934          | 75,688                  | 75,663          | 24.3               | 24.1            |
| Kerala                 | 30,943                 | 30,938          | 27,747                  | 27,726          | 11.5               | 11.6            |
| Tamil Nadu             | 1,26,163               | 1,26,151        | 1,14,798                | 1,14,722        | 9.9                | 10.0            |
| Puducherry             | 3,456                  | 3,455           | 3,051                   | 3,050           | 13.3               | 13.3            |
| Lakshadweep            | 64                     | 64              | 64                      | 64              | -0.3               | -0.3            |
| <b>Southern Region</b> | <b>4,19,531</b>        | <b>4,19,293</b> | <b>3,71,467</b>         | <b>3,70,900</b> | <b>12.9</b>        | <b>13.0</b>     |

**Peak:**

| State / UT             | April,2023 -March,2024 |               | April,2022 - March,2023 |               | % Change    |            |
|------------------------|------------------------|---------------|-------------------------|---------------|-------------|------------|
|                        | Peak Demand            | Peak Met      | Peak Demand             | Peak Met      | Peak Demand | Peak Met   |
|                        | ( MW )                 | ( MW )        | ( MW )                  | ( MW )        | ( % )       | ( % )      |
| Andhra Pradesh         | 13,237                 | 13,237        | 13,167                  | 12,293        | 0.5         | 7.7        |
| Telangana              | 15,622                 | 15,622        | 15,497                  | 15,497        | 0.8         | 0.8        |
| Karnataka              | 17,212                 | 17,212        | 15,828                  | 15,828        | 8.7         | 8.7        |
| Kerala                 | 5,284                  | 5,284         | 4,699                   | 4,370         | 12.5        | 20.9       |
| Tamil Nadu             | 19,045                 | 19,045        | 17,729                  | 17,724        | 7.4         | 7.5        |
| Puducherry             | 524                    | 524           | 501                     | 501           | 4.6         | 4.6        |
| Lakshadweep            | 12                     | 12            | 12                      | 12            | 0.0         | -0.6       |
| <b>Southern Region</b> | <b>68,094</b>          | <b>68,094</b> | <b>64,337</b>           | <b>64,337</b> | <b>5.8</b>  | <b>5.8</b> |

**4.3.4 Eastern Region:** The gap between Energy Requirement and Energy Supplied was 0.7% in the region during the year 2023-24 as compared to 1.0% during the year 2022-23 and a gap of 3.2% between Peak Demand and Peak Met during the year 2023-24

as compared to 3.7% during the year 2022-23.

Further, the comparison between Power Supply Position in terms of Energy and Peak during the year 2023-24 and the year 2022-23 is outlined below:

**Energy:**

| State / UT            | April,2023 -March,2024 |                 | April,2022 - March,2023 |                 | % Change           |                 |
|-----------------------|------------------------|-----------------|-------------------------|-----------------|--------------------|-----------------|
|                       | Energy Requirement     | Energy Supplied | Energy Requirement      | Energy Supplied | Energy Requirement | Energy Supplied |
|                       | ( MU )                 | ( MU )          | ( MU )                  | ( MU )          | ( % )              | ( % )           |
| Bihar                 | 41,514                 | 40,918          | 39,545                  | 38,762          | 5.0                | 5.6             |
| DVC                   | 26,560                 | 26,552          | 26,339                  | 26,330          | 0.8                | 0.8             |
| Jharkhand             | 14,408                 | 13,858          | 13,278                  | 12,288          | 8.5                | 12.8            |
| Odisha                | 41,358                 | 41,333          | 42,631                  | 42,584          | -3.0               | -2.9            |
| West Bengal           | 67,576                 | 67,490          | 60,348                  | 60,274          | 12.0               | 12.0            |
| Sikkim                | 544                    | 543             | 587                     | 587             | -7.4               | -7.4            |
| Andaman- Nicobar      | 386                    | 374             | 348                     | 348             | 10.8               | 7.4             |
| <b>Eastern Region</b> | <b>1,92,013</b>        | <b>1,90,747</b> | <b>1,82,791</b>         | <b>1,80,888</b> | <b>5.0</b>         | <b>5.5</b>      |

**Peak:**

| State / UT            | April,2023 -March,2024 |               | April,2022 - March,2023 |               | % Change    |            |
|-----------------------|------------------------|---------------|-------------------------|---------------|-------------|------------|
|                       | Peak Demand            | Peak Met      | Peak Demand             | Peak Met      | Peak Demand | Peak Met   |
|                       | ( MW )                 | ( MW )        | ( MW )                  | ( MW )        | ( % )       | ( % )      |
| Bihar                 | 8,049                  | 7,420         | 7,852                   | 6,631         | 2.5         | 11.9       |
| DVC                   | 3,451                  | 3,451         | 3,402                   | 3,396         | 1.4         | 1.6        |
| Jharkhand             | 2,193                  | 2,020         | 2,253                   | 1,918         | -2.7        | 5.3        |
| Odisha                | 6,443                  | 6,443         | 6,566                   | 6,391         | -1.9        | 0.8        |
| West Bengal           | 11,626                 | 11,626        | 10,125                  | 9,900         | 14.8        | 17.4       |
| Sikkim                | 133                    | 133           | 124                     | 124           | 7.0         | 6.7        |
| Andaman- Nicobar      | 65                     | 60            | 62                      | 62            | 4.3         | -3.7       |
| <b>Eastern Region</b> | <b>30,256</b>          | <b>29,299</b> | <b>28,275</b>           | <b>27,218</b> | <b>7.0</b>  | <b>7.6</b> |

**4.3.5 North-Eastern Region:** The gap between Energy Requirement and Energy Supplied was 1.4% in the region during the year 2023-24 as compared to 0.4% during the year 2022-23 and the North-Eastern Region was able to meet the gap between Peak Demand and Peak Met during the

year 2023-24 and the year 2022-23.

Further, the comparison between Power Supply Position in terms of Energy and Peak during the year 2023-24 and the year 2022-23 is outlined below:

**Energy:**

| State / UT                  | April,2023 -March,2024 |                 | April,2022 - March,2023 |                 | % Change           |                 |
|-----------------------------|------------------------|-----------------|-------------------------|-----------------|--------------------|-----------------|
|                             | Energy Requirement     | Energy Supplied | Energy Requirement      | Energy Supplied | Energy Requirement | Energy Supplied |
|                             | ( MU )                 | ( MU )          | ( MU )                  | ( MU )          | ( % )              | ( % )           |
| Arunachal Pradesh           | 1,014                  | 1,014           | 915                     | 892             | 10.8               | 13.7            |
| Assam                       | 12,445                 | 12,341          | 11,465                  | 11,465          | 8.6                | 7.6             |
| Manipur                     | 1,023                  | 1,008           | 1,014                   | 1,014           | 0.9                | -0.6            |
| Meghalaya                   | 2,236                  | 2,066           | 2,237                   | 2,237           | -0.1               | -7.6            |
| Mizoram                     | 684                    | 684             | 645                     | 645             | 6.0                | 6.0             |
| Nagaland                    | 921                    | 921             | 926                     | 873             | -0.6               | 5.5             |
| Tripura                     | 1,691                  | 1,691           | 1,547                   | 1,547           | 9.3                | 9.3             |
| <b>North-Eastern Region</b> | <b>20,022</b>          | <b>19,733</b>   | <b>18,758</b>           | <b>18,680</b>   | <b>6.7</b>         | <b>5.6</b>      |

**Peak:**

| State / UT                  | April,2023 -March,2024 |              | April,2022 - March,2023 |              | % Change    |            |
|-----------------------------|------------------------|--------------|-------------------------|--------------|-------------|------------|
|                             | Peak Demand            | Peak Met     | Peak Demand             | Peak Met     | Peak Demand | Peak Met   |
|                             | ( MW )                 | ( MW )       | ( MW )                  | ( MW )       | ( % )       | ( % )      |
| Arunachal Pradesh           | 186                    | 186          | 166                     | 166          | 12.3        | 12.0       |
| Assam                       | 2,413                  | 2,413        | 2,379                   | 2,376        | 1.4         | 1.6        |
| Manipur                     | 258                    | 248          | 248                     | 248          | 4.0         | 0.0        |
| Meghalaya                   | 405                    | 405          | 404                     | 404          | 0.2         | 0.2        |
| Mizoram                     | 162                    | 162          | 159                     | 159          | 1.9         | 1.9        |
| Nagaland                    | 174                    | 174          | 168                     | 167          | 3.8         | 4.2        |
| Tripura                     | 362                    | 362          | 333                     | 333          | 8.7         | 8.7        |
| <b>North-Eastern Region</b> | <b>3,678</b>           | <b>3,678</b> | <b>3,603</b>            | <b>3,603</b> | <b>2.1</b>  | <b>2.1</b> |

#### 4.4 Frequency Profile of National Grid

The five regional grids of the country are operating as an integrated National Grid. The Indian Electricity Grid Code (IEGC) specified by the Central Electricity Regulatory Commission (CERC) mandates the operating band for frequency of grid as 49.90 Hz to 50.05 Hz. The percentage of time

during which the power system of the country operated below 49.90 Hz, between 49.90 to 50.05 Hz (IEGC Band) and above 50.05 Hz and the maximum and minimum frequencies of the National Grid along with the average frequency level during the year 2022-23 and 2023-24, are tabulated below:

| Frequency Profile of National Grid |                              |                                     |                |                        |                        |                        |
|------------------------------------|------------------------------|-------------------------------------|----------------|------------------------|------------------------|------------------------|
| Year                               | % of Time when Frequency was |                                     |                | Average Frequency (Hz) | Maximum Frequency (Hz) | Minimum Frequency (Hz) |
|                                    | Below 49.90 Hz               | Between 49.90-50.05 Hz (IEGC Band ) | Above 50.05 Hz |                        |                        |                        |
| 2022-23                            | 11.15                        | 69.70                               | 19.14          | 49.99                  | 50.55                  | 49.41                  |
| 2023-24                            | 7.20                         | 73.83                               | 18.97          | 50.00                  | 50.43                  | 49.48                  |

It may be seen from the above that the average grid frequency during the year 2023-24 was precisely at the nominal frequency level of 49.48 Hz.

#### 4.5 National Power Committee (NPC)

**4.5.1** National Power Committee (NPC) was established by Ministry of Power vide Order dated 25<sup>th</sup> March, 2013, to evolve a common approach on issues related to reliability and security of the grid, at national level. Chairperson, CEA is the Chairperson of NPC. Member (GO&D), CEA, Member Secretaries and Chairpersons of RPCs, the Chairpersons of Technical Co-ordination Sub Committees (TCC) of five regions, are members of NPC with Chief Engineer (NPC), CEA, as its Member Secretary.

**4.5.2** Since its formation, NPC has taken several initiatives for improving the safe and reliable operation of Indian Grid.

**4.5.3** During the year 2023-24, two (02) nos. of 13<sup>th</sup> and 14<sup>th</sup> meetings of the National Power Committee were held on 05.07.2023 and 03.02.2024 respectively. In these meetings, the important points

related to grid operation such as National Energy Account (NEA), Unified Accounting Software (UAS) for RPCs, Protection Setting Protocol, Five (5) min Interface Energy Meters along with AMR system for PAN India (for all Five regions), Guidelines for locating PMU for URTDSM Phase II project, Introduction of MPLS Technology in ISTS Communication, Automatic Under Frequency Load Shedding (AUFLS) Scheme and Mapping of Feeders were discussed

**4.5.4** As per the decisions taken in the 9<sup>th</sup> meeting and 10<sup>th</sup> meeting of NPC, following three sub-committee/ sub-group were constituted:

- (a) **Sub-committee to study the AUFLS Scheme and work out a common approach for df/dt relay settings:** Sub-committee constituted with representatives from POSOCO, all the RPCs and NPC to study the AUFLS Scheme and work out a common approach for df/dt relay settings. The Sub-Committee after due deliberations and considering the comments from members, has

submitted its report to the 13<sup>th</sup> meeting of National Power Committee. Subsequent Task Force constituted for implementation of the AUFLS and df/dt scheme and also submitted its report in the 14<sup>th</sup> NPC. The report of task force on AUFLS and df/dt scheme was approved in 14<sup>th</sup> NPC held on 03.02.2024.

**(b) Sub-group constituted to finalize a common procedure for Power System Stabilizers (PSS) Tuning:**

Sub-group constituted with representatives of Protection Sub-Committee of respective RPCs, NPC, NLDC, CTU, NTPC and NHPC to finalize a common procedure for Power System Stabilizers (PSS) Tuning and report was approved in 14<sup>th</sup> NPC meeting held on 03.02.2024.

**(c) Sub-committee on the uniform philosophy of PMU locations, new analytics and requirement of up gradation of Control Centre under URTDSM project:**

Sub-committee constituted under the Chairmanship of Member Secretary, WRPC with representatives from POSOCO/ Grid-India, CTU, POWERGRID, all RPCs and NPC on the uniform philosophy of PMU locations, new analytics and requirement of up gradation of Control Centre under URTDSM project and report was approved in 13<sup>th</sup> NPC meeting held on 05.07.2023.

## **4.6 Power System Development Fund (PSDF):**

**4.6.1** Ministry of Power vide letter No. 29/9/2010-R&R (Vol-II) dated 10<sup>th</sup> January 2014 circulated a scheme regarding the operationalization of the Power System Development Fund (PSDF) and utilization of funds deposited therein. NLDC is the Nodal Agency for PSDF.

**4.6.2** During the FY 2023-24, the following meetings related to the operation/implementation of PSDF were held:

(a) Five (05) meetings of the Appraisal Committee,

headed by Chairperson, CEA, were held. In these meetings, 10 no of projects were recommended to Monitoring Committee for sanction grant under PSDF.

(b) Three (03) meeting of the Monitoring Committee, headed by Secretary (Power) was held. In these meetings, 08 no's of projects related to Operation Gas based power plants during peak crunch period, Islanding scheme, Renovation and Upgradation protection system of Substations and Reconductoring with HTLS conductor of existing transmission line to relieve congestion, etc. were sanctioned for funding from PSDF. The 22<sup>nd</sup> Monitoring Committee approved the Revised MOP Guidelines for disbursement of Funds from PSDF. Ministry of Power vide letter No. 22/22/2023-OM dated 12.03.2024 issued Revised Guidelines/Procedure for Disbursement of Funds from PSDF.

(c) Five (05) meetings of Project Monitoring Group, headed by Member (GO&D) were held. In these five meetings, 59 no. of Time Extension requests and 6 no. of Quantity Variation requests were discussed on case-to-case basis.

**4.6.3** Till 31.03.2024, Two Hundred and five (205) nos. of projects for total grant amount of ₹16419.98 Crores have been sanctioned since the operationalization of PSDF. Out of the 205 sanctioned projects, 17 projects with an amount of ₹388.05 Crores have been De-sanctioned due to various reason.

**4.6.4** The details of sanctioned grant under PSDF are as given below:

| S. No | Project Entity                                         | During FY 2023-24 |                             | Previous Years (Up to 31.03.2023) |                             | Cumulative (Up to 31.03.2024) |                             |
|-------|--------------------------------------------------------|-------------------|-----------------------------|-----------------------------------|-----------------------------|-------------------------------|-----------------------------|
|       |                                                        | No. of Projects   | Sanctioned Grant (₹ crores) | No. of Projects                   | Sanctioned Grant (₹ crores) | No. of Projects               | Sanctioned Grant (₹ crores) |
| 1     | States/UTs                                             | 6                 | 51.22                       | 178                               | 8352.50                     | 184                           | 8403.72                     |
| 2     | RPCs                                                   | 0                 | 0.00                        | 9                                 | 115.61                      | 9                             | 115.61                      |
| 3     | BBMB                                                   | 0                 | 0.00                        | 1                                 | 23.27                       | 1                             | 23.27                       |
| 4     | DVC                                                    | 0                 | 0.00                        | 2                                 | 166.46                      | 2                             | 166.46                      |
| 5     | PGCIL                                                  | 0                 | 0.00                        | 4                                 | 4159.56                     | 4                             | 4159.56                     |
| 6     | PGCIL/RE CTPCL                                         | 0                 | 0.00                        | 1                                 | 233.03                      | 1                             | 233.03                      |
| 7     | Central                                                | 2                 | 489.35                      | 1                                 | 1320.00                     | 3                             | 1809.35                     |
|       | <b>Total</b>                                           | <b>8</b>          | <b>540.57</b>               | <b>196</b>                        | <b>14370.43</b>             | <b>204</b>                    | <b>14911.00</b>             |
|       | Proposal approved by MOP:(1) - Stranded Gas projects   |                   |                             |                                   |                             | 1                             | 1509.00                     |
|       | Total including Stranded Gas projects                  |                   |                             |                                   |                             | 205                           | 16419.98                    |
|       | <b>No. of De-sanctioned projects and Amount</b>        |                   |                             |                                   |                             | <b>17</b>                     | <b>388.05</b>               |
|       | <b>Effective No. of Sanctioned projects and Amount</b> |                   |                             |                                   |                             | <b>188</b>                    | <b>16031.93</b>             |

**4.6.5** Total 76 nos. of projects have been completed till 31.03.2024. Out of this, 11 projects were completed during the FY 2023-24.

## 4.7 PUSHP Portal (Portal for Utilizations of Surplus Power)

**4.7.1** The Portal for Utilization of Surplus Power (PUSHP) Portal, launched on 9th March, 2023, is India's first innovative platform designed to optimize the nation's power generation capacity by temporary reallocation/transfer of surplus power of Central Generating Stations/Inter State Generating Stations /Independent Power Producers /State Generating Companies (GENCOs).

**4.7.2** The temporary reallocation/transfer of surplus power has been done for one day and more than one day. There is no price discovery mechanism on the portal and power is temporary reallocated/transferred at the regulated tariff as determined by the Appropriate Commission.

**4.7.3** It facilitates flexible Power Purchase Agreements (PPAs), allowing surplus power in one state to be utilized by others in need and eventually lead to reduction in the fixed charge cost burden of the states/consumers with

surplus power capacity and also enable the available generation capacity of the country to be utilized.

**4.7.4** The beneficiaries and GENCOs can also declare the surplus power of units under Reserve Shut Down for the revival of such units.

**4.7.5** As on 31.03.2024, Twenty (20) numbers of states & UTs across all the regions have started using the portal and total 94 numbers of transactions with quantum range of 6.44 MW to 250.95 MW have been successfully completed for one day basis on the portal.

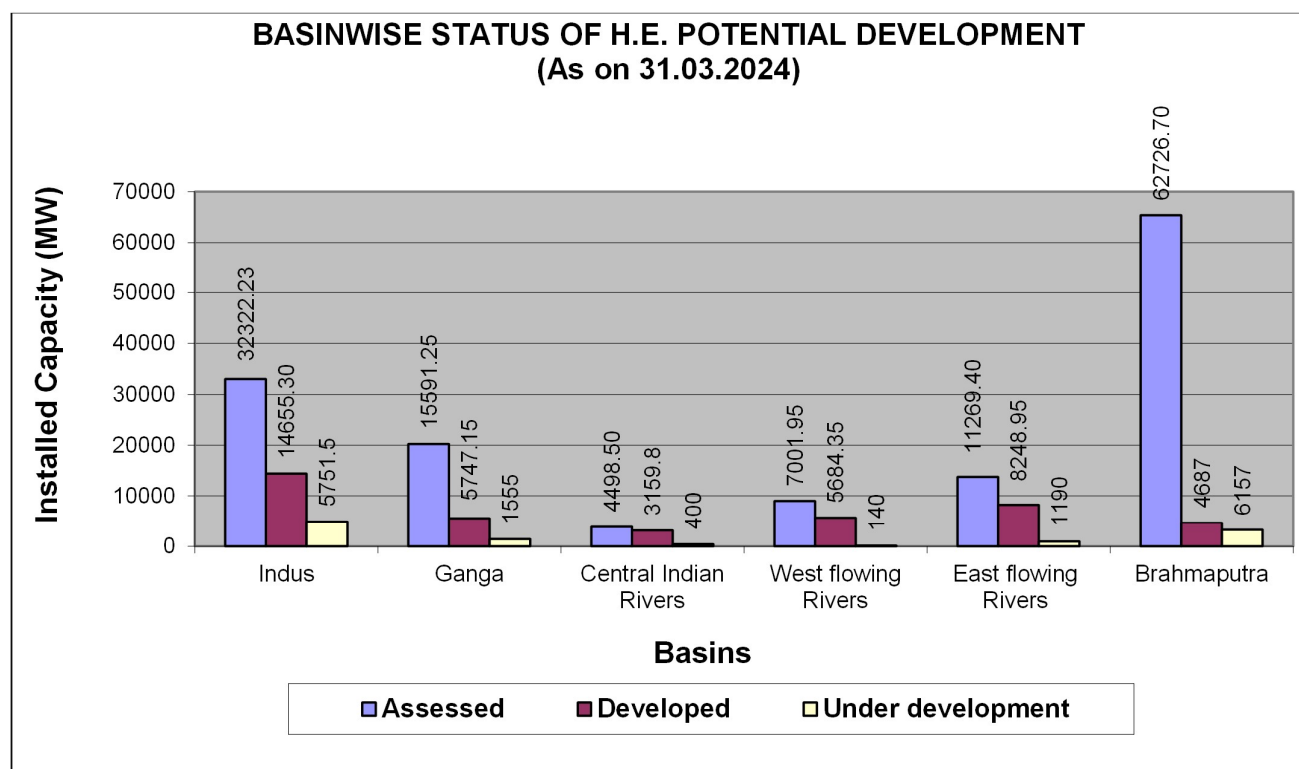
## CHAPTER – 5

### HYDRO POWER DEVELOPMENT

#### 5.1 Hydro Potential and its Development

The re-assessment studies of hydro-electric potential of the country, completed by Central Electricity Authority during 2017-23, have assessed the economically exploitable hydro power potential in terms of installed capacity as 133410 MW from schemes having capacity above 25 MW.

The basin-wise details of hydroelectric potential development in terms of Installed Capacity are indicated in the table below. As on 31.03.2024, the hydroelectric schemes in operation account for only 31.62% (42182.55 MW) and those under execution for 11.38% (15193.5 MW) of the total potential in terms of installed capacity. Thus, the bulk of the potential (57%) remains to be developed.



In addition, at present, 118 sites for development of Pumped Storage Schemes (PSS) with probable total installation of 133690.6 MW have been identified in the country. Out of this, 63 sites (59900.6 MW) are ON River PSPs and 55

sites (73790 MW) are OFF-River PSPs. 8 Nos. Pumped Storage Projects (above 25 MW) having total installed capacity of 4745.60 MW are in operation and 4 Pumped Storage project (2780 MW) are under construction.

## 5.2 50,000 MW Hydro-Electric initiative

Under the 50,000 MW Initiative, preparation of Preliminary Feasibility Reports (PFRs) for 162 hydro-electric projects spread over 16 states were taken up by CEA in the year 2003-04 as nodal agency with CPSUs/State agencies as Consultants. The role of CEA includes overall coordination, facilitating collection of data, and quality control by vetting conceptual planning, assessment of power benefits and selection of project parameters, evacuation of power and monitoring of works.

NHPC Ltd., WAPCOS, NEEPCO, SJVN Ltd. and Number of State Power Utilities were associated in preparation of these Preliminary Feasibility Reports. All the 162 Nos. of PFRs were completed in Sept., 2004 for all these projects with an installation of 47,930 MW. Details of these projects are given at [Annexure-5A](#).

Out of 162 schemes, DPRs in respect of 35 schemes (18980 MW) have already been prepared. Out of these 35 schemes, 4 schemes (513 MW) has been commissioned while 7 schemes (2035 MW) are under construction in the country. A total of 14 schemes (6831 MW) have been concurred by CEA. DPRs of 10 HEPs with aggregate capacity of 9601 MW have been prepared but returned for various reasons. A total of 8 schemes (4898 MW) are under Survey & Investigation (S&I) for preparation of DPRs while DPR in respect of remaining 119 schemes (23942 MW) is yet to be prepared due to various issues.

## 5.3 Construction Monitoring of Hydro Projects:

Hydro Project Monitoring Division is monitoring the progress of construction of on-going sanctioned hydro power projects (above 25 MW) in pursuance to following Sections of Electricity Act, 2003 which is reproduced as under:

**Section 73(f).** Promote and assist in the timely completion of schemes and projects for improving

and augmenting the electricity system;

**Section 73(i).** collect and record the data concerning the generation, transmission, trading, distribution and utilization of electricity and carry out studies relating to cost, efficiency, competitiveness and such like matters;

**Section 73(j).** Make public from time to time the information secured under this Act, and provide for the publication of reports and investigations;

The progress of each project is monitored continuously through site visits, interaction with the developers & other stake holders. Chairperson, CEA/ Member (Hydro), Chief Engineer (HPM) holds regular review meetings with the developers/contractors and other Stakeholders.

## 5.4 Hydro additions during 2022-23:

Hydro capacity addition of 120 MW was achieved against the targets of 1080 MW for the year 2022-23. Project-wise details are given at [Annexure-5B](#).

**5.4.1 Hydro capacity additions during 2023-24:** 60 MW Hydro capacity has been added against the targets of 2880 MW for the year 2023-24. Project-wise details are given at [Annexure-5C](#).

**5.4.2 Hydro capacity programme during 2024-25:** Hydro Capacity Addition Monitor able Targets planned for the Year 2024-25 is 3950 MW (2550 MW in Central Sector, 200 MW in State Sector, and 1200 MW in Private Sector.). Project-wise details are given at [Annexure-5D](#).

## 5.5 Survey & Investigation of Hydro Projects

CEA has been monitoring the progress of Survey and Investigation of all the hydro schemes (above 25 MW capacity) by conducting periodical review meetings with developers. In order to accelerate the pace of hydro development in the country, Guidelines for formulation of DPRs for Hydro Electric Schemes, their acceptance and examination for concurrence have been issued by Ministry of Power in 2014 and accordingly, CEA provides assistance to various Central/ State

agencies in the matter of survey, investigation and preparation of DPRs of hydro projects costing more than ₹1000 crores.

In line with the above Guidelines, Consultation Meetings are held by CEA, CWC, GSI and CSMRS with the project developer and guidance is provided to him for making a good quality DPR. During the year, consultation meetings were held for Singanamala PSP (800 MW) in Andhra Pradesh, Narihalla PSP (300 MW) in Karnataka, Teesta Intermediate (90 MW) in West Bengal, Sela Urthing (114 MW) in Uttarakhand, Tarali PSP (1500 MW) in Maharashtra, Veeraballi Off-stream PSP (1800 MW) in Andhra Pradesh, UP 01 (3360 MW) in Uttar Pradesh, Kandhaura PSP (1680 MW) in Uttar Pradesh, Patgaon PSP (2100 MW) in Maharashtra, Sirohi PSP (1200 MW) in Rajasthan, Reoli Dugli (456 MW) in Himachal Pradesh, Pane PSP (950 MW) in Maharashtra, Kamalapadu PSP (950 MW) in Andhra Pradesh, Vempalli (1500 MW) in Andhra Pradesh, Malshej ghat Bhorande (1500 MW) in Maharashtra, Tekwa -2 PSP (800 MW) in Madhya Pradesh, Indirasagar -Omkareshwar PSP (640 MW) in Madhya Pradesh, Upper Kolab PSP (320 MW) in Odisha, Raiwada PSP (900 MW) in Andhra Pradesh, Gujjili PSP (1500MW) in Andhra Pradesh, Yaganti PSP (1000 MW) in Andhra Pradesh, Panari PSP (1800 MW) in Madhya Pradesh, Shirwata PSP (1800 MW) in Maharashtra, Bhivpuri PSP (1000 MW) in Maharashtra, Warasgaon Warangi PSP (1500 MW) in Maharashtra, Naba (1082 MW) in Arunachal Pradesh, Sillahalla St-I (1000 MW) in Tamil Nadu, Aravetipalli PSP (1320 MW) in Andhra Pradesh, Musakhand (600 MW) in Uttar Pradesh, Kamala (1720 MW) in Arunachal Pradesh, Rayavaram PSP (1500 MW) in Andhra Pradesh, Koyna Nivakane PSP (2700 MW) in Maharashtra.

DPRs of 13 nos. of HEPs with aggregate installed capacity of 6918 MW have so far been prepared in consultation with appraising agencies since 2014 and submitted for further examination in CEA/CWC and out of which DPRs of 10 HEPs with aggregate installed capacity of 5838 MW have been concurred by CEA. DPR of Upper Sileru Pumped Storage Project (1350) have been prepared and

concurred by CEA during the period of 2023-24. In addition, a total of 17 HEPs with aggregate capacity of 11376 MW and 44 Pumped Storage Projects with aggregate capacity of 60050 MW (having cost of more than Rs.1000 Crores) are presently under Survey & Investigation in the country and DPRs of these are to be submitted to CEA for concurrence.

## 5.6 Project Planning & Optimization Studies

- Power Potential Studies of Owk PSP(800 MW), Gandikota PSP(1000 MW),Yerravaram PSP(1200 MW), Singanamala PSP (800MW), Veeraballi PSP (1800MW) and Chitravathi PSP (500 MW) in Andhra Pradesh, Purthi HEP(287 MW) and Bardang HEP(166 MW) in Himachal Pradesh, Nafra HEP (95 MW), Tawang-I HEP (600 MW), Tawang-II HEP (640 MW) in Arunachal Pradesh, Sela Urthing HEP(114 MW) in Uttarakhand, Warasagaon PSP (1200 MW), 3. Pane PSP(1500 MW), Tarali PSP (1500 MW), and Bhavali PSP (1500 MW) in Maharashtra, Upper Siang(11200 MW), Niare HEP(909 MW) in Arunachal Pradesh, Simsang Dam Project (60 MW) in Meghalaya, Arun-4(630 MW) in Nepal, and Sukhpura PSP(2560 MW) and Sirohi PSP (1200 MW), in Rajasthan, Shahpur PSP (1800 MW) and Kandhaura PSP( 1680 MW) and UP-01 PSP (3660 MW) in Uttar Pradesh and Umngot HEP(180 MW) in Meghalaya have been carried out.

## 5.7 Studies & Other Activities Related to HydroPower Planning

- A committee has been constituted by MoP under the chairmanship of Chairperson, CEA to discuss the issues related to operationalization of 1200 MW Sardar Sarovar Pumped Storage Project. In this regard, two meeting of the committee have been held.
- Implementation of Early Warning System in Hydro projects is being monitored in CEA on monthly basis.
- Matter related to Internal Committee on Upper Siang MPP constituted by DoWR, RD& GR and prepared report.
- Report on Tentative Capacity Addition of Hydro

Projects (Above 25 MW) likely to be commissioned during 2023-24 to 2031-32.

- Waiver of ISTS charges extended to PSPs for which construction work is awarded up to 30.06.2025, subject to certain conditions. Subsequently, part waiver of ISTS charges, in steps of 25% from 01.07.2025 to 01.07.2028, have been extended for PSPs for which construction work is awarded up to 30.06.2028.
- Waiver of ISTS Charges on the transmission of power from new Hydro Power Projects, for which construction work is awarded and PPA is signed on or before 30.06.2025. Subsequently, part waiver of ISTS charges, in steps of 25% from 01.07.2025 to 01.07.2028, have been extended for HEPs for which construction work is awarded and PPA is signed up to 30.06.2028.
- Ministry of Power has notified Guidelines to promote development of Pumped Storage Projects in the country on 10th April, 2023. Keeping in view the immense utility of the PSPs in grid stabilization as well as meeting the peaking power demand, guidelines have been formulated to promote PSPs and set the direction of its development. The Ministry seeks to promote the development of PSPs across the country with proactive support of the State Governments.
- CEA has also published revised Guidelines for Formulation and Concurrence of DPR of PSPs. Under the revised guidelines, the timeline for concurrence of DPR of PSPs has been reduced from 125 days to 50 days.
- The development of Digital Portal for Fast Tracking of S&I of Concurrence of DPR in association with M/s NHPC is under finalization.

### 5.8 Co-operation with Neighboring Countries in Hydro Power

During the year, following works were handled in connection with development of water resources of the common rivers of India and neighboring countries of Bhutan, Nepal etc. for mutual benefits:

- Matter relating to revision of tariff of Chukha HEP in Bhutan. Matter related to handing over of Mangdechhu HEP to RGoB.
- Tariff negotiations for Punatsangchhu-II HEP

### 5.9 Hydro Power Plants Performance & Operation Monitoring

- The report “Review of Performance of Hydro Power Stations” for the year 2023-24, is in the process of completion. Performance of 731 units in 212 Hydro Stations with aggregate Installed Capacity of 46928.17 MW (above 25 MW) has been analyzed in respect of their outages & generation in this report.
- Month-wise/station-wise hydro generation targets for year 2024-25 in respect of all the HE Stations (above 25 MW) in the country were finalized as 147.71 BU in consultation with Midterm review of generation performance of hydroelectric stations of the country for the year 2023-24 was carried out in August/September 2023 after withdrawal of South-West monsoon by interaction with Power Utilities of Central Sector and the generation programme was reviewed for the remaining part of the year 2023-24. The total generation from hydro stations in 2023-24 was 134.05 BU against original programme of 156.70 BU.

### 5.10 Hydel Generation Performance during year 2023-24

| Region           | Generation (BU) |               | Deviation (+/-) |
|------------------|-----------------|---------------|-----------------|
|                  | Programme       | Actual        | (%)             |
| Northern         | 77.80           | 73.09         | -6.05           |
| Western          | 15.49           | 16.59         | 7.08            |
| Southern         | 31.08           | 20.31         | -34.65          |
| Eastern          | 20.18           | 17.79         | -11.86          |
| N-Eastern        | 12.16           | 6.28          | -48.31          |
| <b>All India</b> | <b>156.70</b>   | <b>134.05</b> | <b>-14.45</b>   |

Against programme of 156.70 BU, the actual energy generation during the year 2023-24 was 134.05 BU, which was 14.45 % less than the target.

## 5.11 RENOVATION & MODERNISATION(R&M) OF HYDRO ELECTRIC PROJECTS

Renovation & Modernization, Uprating and Life Extension (RMU&LE) of the existing old hydroelectric power projects is considered a cost effective option to ensure optimization of resources, efficient operations, and better availability and also to augment (uprating) capacity addition in the country.

Recognizing the benefits of R&M of hydroelectric power projects, Govt. of India set up a National Committee in 1987 and a Standing Committee in 1998 and thereafter had identified the projects/schemes to be taken up for implementation under R&M. The National Perspective Plan document for R&M of hydroelectric power projects in the country was also prepared in CEA during the year 2000. The status of various projects/schemes already identified for implementation/completion till the end of XI Plan, i.e. March, 2012 has been incorporated in the National Perspective Plan.

### 5.11.1 Achievements During VIII, IX, X XI and XII Plan and Period 2017-22:

The R&M works at 118 (26 in Central and 92 in State Sector) hydro power plants (13 up to the VIII Plan, 20 in the IX Plan, 32 in the X Plan, 18 in the XI Plan, 21 in the XII Plan & 14 during 2017-22) with an aggregate installed capacity of 22634.7 MW had been completed by the end of the year 2017-22 and total a benefit of 4139.56 MW through Life Extension (LE), Uprating (U) and Restoration had been accrued.

### 5.11.2 Programme and Achievements during the period 2022-27:

The Renovation, Modernization, Uprating and Life Extension works at 57 Hydro Electric Plants (HEPs) with an aggregate installed capacity of 10583.9 MW is programmed for completion during the year 2022-27 with its break-up as 2041.8 MW through R&M at 12 HEPs, 6399.1 MW through Life Extension at 32 HEPs and 2143 MW through Life

Extension and Uprating at 13 HEPs. The 13 HEPs where both Life Extension & Uprating are envisaged, the aggregate installed capacity of 2143 MW shall get uprated after completion of R&M works to 2395.5 MW resulting in additional benefit of installed capacity of 252.5 MW. As such, the revised aggregate installed capacity after completion of RMU&LE works of these 57 projects would be 10836.4 MW. The State-wise list of hydro R&M schemes expected for completion during the year 2022-27 is given at **Annexure-5E**.

Out of these 57 schemes, Ten (10) Schemes with an aggregate installed capacity of about 2390.8 MW have been completed till March, 2024 which has resulted in benefit of 1101 MW through Life Extension and 90 MW through Uprating.

### 5.11.3 Programme for the year 2023-24

For the year 2023-24, it was programmed to complete following 7 schemes having capacity under R&M of 1770 MW. On completion of these schemes, there was to be a benefit of 930 MW through Life Extension and 96 MW through Uprating.

| S. No. | Name of Scheme, Utility/Agency  | Capacity under R&M (in MW) | Category | Cost (in Rs. Crores) | Benefit            |
|--------|---------------------------------|----------------------------|----------|----------------------|--------------------|
| 1.     | Bhakra LB, BBMB                 | 5x108                      | RMU&LE   | 570.38               | 540 (LE)<br>+90(U) |
| 2.     | Obra, UPIJNL                    | 3x33                       | RM&LE    | 46.57                | 99(LE)             |
| 3.     | Ranjit Sagar Dam, PSPCL         | 4x150                      | R&M      | 8.52                 | -                  |
| 4.     | Dhalipur, UJVNL                 | 3x17                       | RM&LE    | 88.54                | 51 (LE)            |
| 5.     | Gerusoppa Dam Power House, KPCL | 4x60                       | R&M      | 2.21                 | -                  |
| 6.     | Panchet U-1, DVC                | 1x40                       | RMU&LE   | 2.19                 | 40(LE)<br>+ 6(U)   |
| 7.     | Kopili Power Station, NEEPCO    | 4x50                       | RM&LE    | 873.8                | 200 (LE)           |

### 5.11.4 Achievements during the year 2023-24

R&M works of the following three (3) Schemes with an aggregate installed capacity of about 831 MW have been completed during the year 2023-24

which has resulted in benefit of 591 MW through Life Extension and 90 MW through Upgrading:

| S. No. | Name of Scheme, Utility/Agency                         | Capacity under R&M (in MW) | Category | Cost (in Rs. Crores) | Benefit             |
|--------|--------------------------------------------------------|----------------------------|----------|----------------------|---------------------|
| 1      | Bhakra LB, BBMB                                        | 5x108                      | RMU&LE   | 583.86               | 540.00(LE)+90.00(U) |
| 2      | Dhalipur, UJVNL                                        | 3x17                       | RM&LE    | 110.13               | 51 (LE)             |
| 3      | Gerusoppa Dam Power House (Sharavathy Tail Race), KPCL | 4x60                       | R&M      | 2.026                | -                   |

### 5.11.5 Programme during the period 2027-32

The Renovation, Modernization, Upgrading and Life Extension works at 23 Hydro Electric Plants (HEPs) with an aggregate installed capacity of 3129.20 MW is programmed for completion during 2027-32 through Life Extension and Upgrading. The State-wise list of hydro R&M schemes expected for completion during 2027-32 is given at **Annexure-5F**.

**Plan-wise summary of R&M of H.E. Projects starting from VIII Plan is given below:**

### Summary of R&M of Hydro Electric Projects

#### I Hydro R&M schemes completed up to XII Plan

| Sl. No. | Plan Period            | No. of Projects |              |            | Installed Capacity (MW) | Actual Expenditure (Rs. in Crs) | Benefit (MW)                                             |
|---------|------------------------|-----------------|--------------|------------|-------------------------|---------------------------------|----------------------------------------------------------|
|         |                        | Central Sector  | State Sector | Total      |                         |                                 |                                                          |
| 1.      | Upto VIII Plan Schemes | 2               | 11           | 13         | 1282.00                 | 127.37                          | 429.00 [39.00(U) + 54.00(LE)+ 336.00(Res.)]              |
| 2.      | IX Plan Schemes        | 8               | 12           | 20         | 4892.10                 | 570.16                          | 1093.03 [339.00(U)+ 423.00(LE) + 331.03(Res.)]           |
| 3.      | X Plan Schemes         | 5               | 27           | 32         | 4446.60                 | 1029.24                         | 827.73 [122.05(U) + 701.25 (LE) + 4.43(Res.)]            |
| 4.      | XI Plan Schemes        | 4               | 14           | 18         | 5841.20                 | 294.84                          | 735 [12 (U) + 708 (LE) + 15 (Res.)]                      |
| 5.      | XII Plan Schemes       | 2               | 19           | 21         | 4149.60                 | 1146.02                         | 549.40 [58 (U)+ 476.40 (LE)+15(Res.)]                    |
| 6.      | 2017-2022              | 5               | 9            | 14         | 2023.2                  | 848.68                          | 505.4 [479.2(LE) + 26.2(U)]                              |
|         | <b>Total</b>           | <b>26</b>       | <b>92</b>    | <b>118</b> | <b>22634.7</b>          | <b>4016.31</b>                  | <b>4139.56 [596.25 (U)+ 2841.85 (LE)+ 701.46 (Res.)]</b> |

**Abbreviations:**

MW – Mega Watt; Res. – Restoration; U – Uprating; LE – Life Extension;

**II Hydro R&M Schemes during 2022-27:**

| Sl. No. | Category                                      | No. of Projects |              |       | Capacity covered under RMU&LE (MW) | Benefit (MW)                        |
|---------|-----------------------------------------------|-----------------|--------------|-------|------------------------------------|-------------------------------------|
|         |                                               | Central Sector  | State Sector | Total |                                    |                                     |
| 1.      | Programmed                                    | 8               | 49           | 57    | 10583.9                            | 8794.60<br>[8542.1 (LE)+ 252.5 (U)] |
|         | Completed                                     | 1               | 9            | 10    | 2300.8                             | 1191<br>[1101 (LE)+ 90 (U)]         |
| 2.      | Under Implementation                          | 4               | 19           | 23    | 3599.75                            | 2905.25<br>[2857.75 (LE)+ 47.5 (U)] |
| 3.      | Under Tendering                               | 1               | 5            | 6     | 810                                | 736 [710 (LE)+ 26 (U)]              |
| 4.      | Under DPR Preparation/ Finalization/ Approval | 0               | 5            | 5     | 1385                               | 1391<br>[1385 (LE)+ 6(U)]           |
| 5.      | Under RLA Studies                             | 2               | 11           | 13    | 2488.35                            | 2571.35<br>[2488.35 (LE)+ 83 (U)]   |

**III Programme of R&M works during 2027-32:**

| Sl. No. | Category                                     | No. of Projects |              |       | Capacity covered under RMU&LE (MW) | Benefit (MW)                       |
|---------|----------------------------------------------|-----------------|--------------|-------|------------------------------------|------------------------------------|
|         |                                              | Central Sector  | State Sector | Total |                                    |                                    |
| 1.      | Programmed                                   | 3               | 20           | 23    | 3129.2                             | 3140.03<br>[3129.2 (LE)+ 10.83(U)] |
| 2.      | Under Implementation                         | 0               | 0            | 0     | 0                                  | 0                                  |
| 3.      | Under Tendering                              | 0               | 1            | 1     | 115                                | 125.83 [115(LE) + 10.83(U)]        |
| 4.      | Under DPR Preparation/Finalization/ Approval | 0               | 2            | 2     | 250                                | 250<br>[250 (LE)+ 0 (U)]           |
| 5.      | Under RLA Studies                            | 3               | 17           | 20    | 2764.2                             | 2764.2<br>[2764.2(LE)+ 0(U)]       |

**Abbreviations:**MW – Mega Watt; Res. – Restoration;  
RLA- Residual Life Assessment

U – Uprating;

LE – Life Extension;

### 5.11.6 Appraisal of DPR for existing R&M schemes based on direction/ request of MoP/ CERC/ State Power Utilities during the year 2022-23:

Examination of Detailed Project Report (DPR) for Renovation & Modernization and Life Extension (RM&LE) works of 2x23 MW Khandong Power Station, NEEPCO.

Examination of Proposal for Renovation & Modernization and Life Extension (RM&LE) of

Maithon Hydel Power Station Unit 1&3 (2x20 MW) DVC.

### 5.12 Concurrence / Appraisal of Hydro Schemes:

During the year 2023-24 (till 31.03.2024), DPR of 1 HE Schemes with an installed capacity of 2609 MW has been concurred by CEA. Details are given as under:

| S. No. | Name of Scheme/ State/ Executing Agency       | Installed Capacity (MW) | Estimated Cost (₹ in crores)       | Date of concurrence/ appraisal by CEA |
|--------|-----------------------------------------------|-------------------------|------------------------------------|---------------------------------------|
| 1      | Upper Sileru PSP in Andhra Pradesh by APGENCO | 9X150                   | 2500.59 (cost at completion level) | 13.06.2023                            |

### 5.13 Computerization in CEA

All Divisions and Sections of CEA have been equipped with the latest IT infrastructure. All computers of CEA office at Sewa Bhawan and West Block-II are interconnected through wired network. The important statistics/data/information of CEA is uploaded in bilingual (English & Hindi) website of Central Electricity Authority for global access. The content of this website is updated on daily basis. [cea.nic.in](http://cea.nic.in) is a bilingual and GIGW (*Guidelines for Indian Government Websites*) compliant website.



The website provides organizational information, online access to notified regulations of CEA, various published CEA reports and power sector specific information dashboard.

During 2023-24, following new sections/features have been implemented on CEA Website:

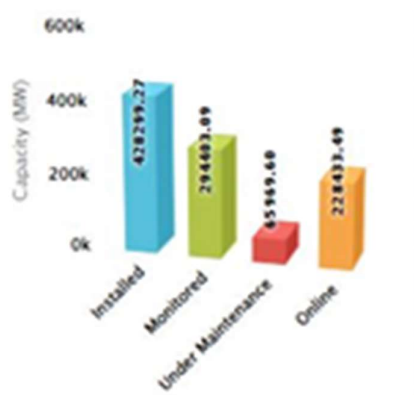
- Dedicated vigilance corner
- Dedicated webpage for Cross Border Approvals
- Implementation of Google Analytics on CEA website for tracking user behaviour.

### 5.14 National Power Data Management System (NPDMS) / National Power Portal (NPP)

- Hon'ble Minister of State (I/C) for Power and New & Renewable Energy launched the National Power Portal (NPP) on 14.11.2017 with the vision of creating a central hub for collection and dissemination of all Power Sector data.
- The NPP Dashboard has been designed and developed to disseminate the information of power sector and provide visualization chart windows on capacity, generation, transmission, and distribution at National, State, DISCOM, town, feeder level.
- The NPP facilitates online capture of input information from entire power value chain; generation, transmission, and distribution utilities in the country on daily, monthly and annual basis.

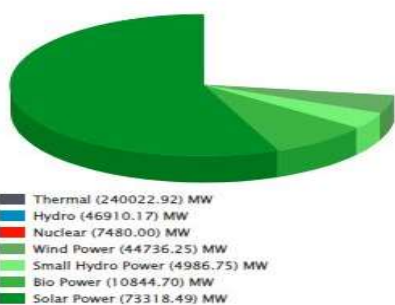
- The NPP is integrated with associated systems of Central Electricity Authority (CEA), Power Finance Corporation (PFC), Rural Electrification Corporation (REC) and other major utilities and serve as single authentic source of power sector information to apex bodies, utilities for the purpose of analysis, planning, monitoring as well as for public user. The system is available 24x7 and ensures effective and timely collection of data. It has standardized data parameters and formats for seamless exchange of data between NPP and respective systems of utilities.
- The stakeholders of NPP are Ministry of Power (MoP), CEA, PFC for Integrated Power Development Scheme (subsumed in Revamped Scheme), REC for Deen Dayal Upadhyaya Gram Jyoti Yojana (subsumed in Revamped Scheme), other power sector utilities in government as well as private sector, Apex Bodies, other government organizations and public users. Central Electricity Authority is the nodal agency for NPP.
- NPP captures data and displays following information:

**Installed Capacity:** Information pertaining to total / sector-wise / category-wise Installed Capacity of generating units.



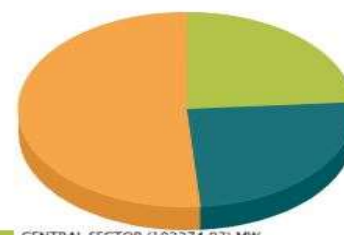
Category wise Installed Capacity

Total Capacity : 428299.28 MW

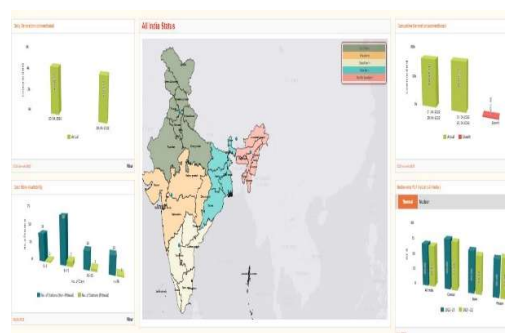


Sector-wise Installed Capacity

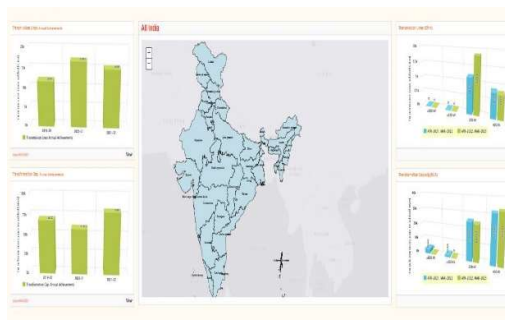
Total Capacity : 428299.27 MW



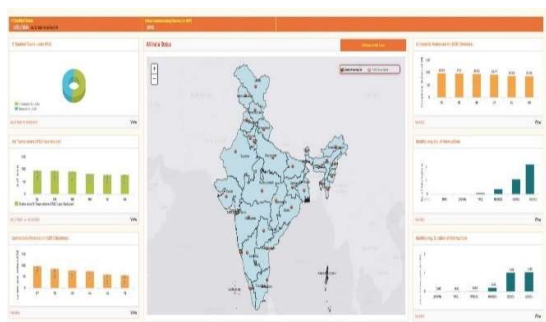
**Daily Generation:** Information pertaining to Daily Conventional generation, Cumulative Generation, Daily Coal Stock availability, Sector wise PLF.



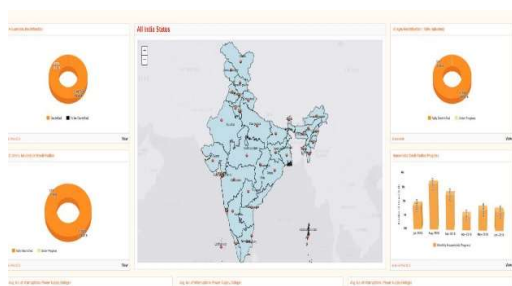
**Transmission:** Displays information on Transformation Capacity (in MVA) and total length of transmission lines (in CKM).



**Urban Distribution:** Captures and displays information of IT Enabled Towns, Towns where AT&C loss has been reduced, Connections Released, Complaints Redressed, Monthly Avg. No. of Interruptions, and Monthly Avg. Duration of Interruptions.

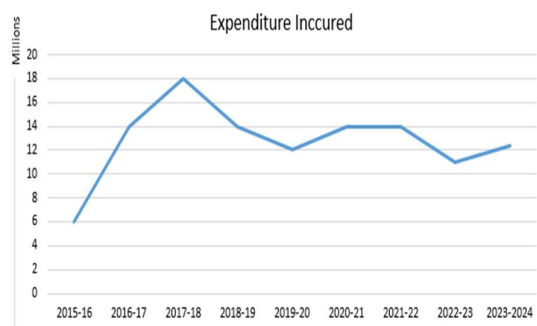


**Rural Distribution:** Captures and displays information about rural power supply. Access to backend data inputs has been given to States/UTs for further granular analysis for adoption of promotion of data-based policy making and decisions.



**Reports:** NPP facilitates various types of reports i.e. Generation Reports, Coal Reports, Transmission Line reports and Installed Capacity reports required to be published regularly.

**Expenditure incurred in NPP project:** Year-wise breakup of expenditure in NPP project is as under:



## Portals under NPP

**UPADHI Portal:** Monitoring Platform for Consumer Rules (DISCOM) provides a framework of Visible Monitoring to ensure the compliance of Consumer Rights by the Discoms. This portal is hosted under NPP and launched by Hon'ble Minister of Power & New Renewable Energy on 11.07.2023.

**DRIPS Portal:** Disaster Resources Inventory for Power Sector (DRIPS) portal will serve as a single point of contact for all the designated nodal officers of power sector departments and agencies in the affected districts, states and at national level for managing the inventory of specific Power System equipment and critical supplies much needed for an emergency response. It will be made available for public shortly.

**NPP APIs:** Datasets are also exchanged between NPP and the portals of other agencies. APIs have been developed to provide NPP data to NITI Aayog and Ministry of Coal.

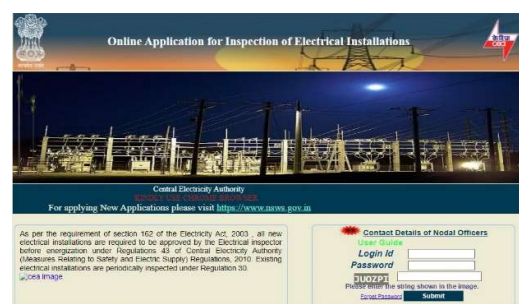
## 5.15 E-Office in CEA

For conducting file and letter handling processes in more efficient and transparent manner, e-office migration from NIC server to Railtel server has been successfully accomplished by CEA in 2023-24. During this process eOffice URL has been changed to <https://eoffice.cea.gov.in>. The eOffice application is hosted on the Cloud of Railtel which provides features like e-sign for ascertaining authentication & non-repudiation, role-based work flow, tracking and searching facility, etc. Migration of CEA eOffice from version from 6.3 to 7.x is being done in 2024-25. This upgradation is being carried out to include advanced features of eOffice and for this training has been provided to CEA officers/staff.

## 5.16 Digitization of Approvals / Clearances given by CEA

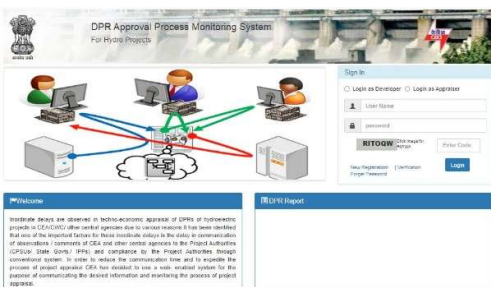
As per the direction of Project Monitoring Group of Cabinet Secretariat, online portal for the following approvals/ clearances, given by Central Electricity Authority, have been implemented:

- Online Application portal for Inspection of Electrical Installations

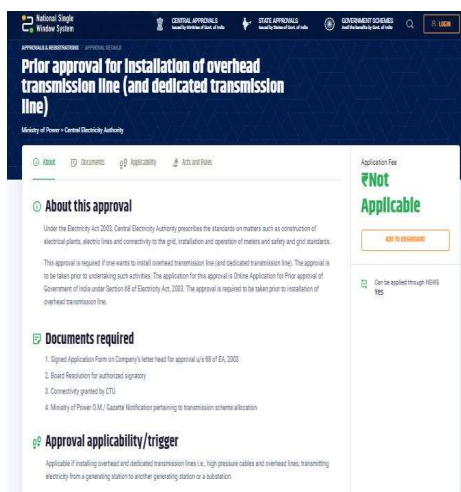


- DPR Approval Process Monitoring System for

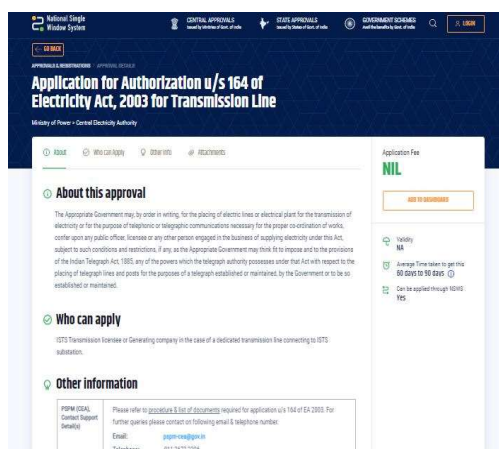
## Hydro Projects



- Prior Approval of GoI for installation of transmission lines as per Section 68 of the Electricity Act 2003



- Authorization u/s 164 of Electricity Act, 2003 for Transmission Lines



The digitization of approval/clearance processes is ensuring transparency and timely approval by CEA. This also facilitates developers to track the status of their application. Further, National Single Window System (NSWS) is a digital platform with the aim to include all Govt. approvals at one place for ease of doing business. In this regard, Central Electricity Authority has integrated its above-mentioned clearances portal with NSWS.

### 5.17 Dedicated intranet portal of CEA

Intranet portal of CEA is hosted at <https://intranet.cea.gov.in>. This portal is accessible within as well as outside the NIC network. Internal information of the organization like presentations / forms / posting orders / leave orders / internal circulars etc. are shared through this portal.

### 5.18 Procurement Related Activities

- Centralised procurement of all Regional Power Survey Offices (RPSOs), Regional Inspectorate Offices (RIOs) and CEA Headquarter is being done by IT Cell & Procurement Division of CEA.
- Mostly procurement of Goods, Works & Services is being done through GeM portal and their payments are being made via PFMS.

## CHAPTER-6

### THERMAL POWER DEVELOPMENT

#### 6.1 New Thermal Power Projects accorded Environment Clearance:

##### 6.1.1 Power Projects accorded Environment Clearance

During the year 2023-24, Environment clearance has been granted to 05 thermal power project as listed below:

|       |                                   |             |       |             |            |
|-------|-----------------------------------|-------------|-------|-------------|------------|
| 1     | Amarkantak STPP(Expansion)        | MPPGCL      | 1X660 | 660         | 01.05.2023 |
| 2     | Lara STPP (Exp. Stg-II)           | NTPC        | 2X800 | 1600        | 17.07.2023 |
| 3     | Bhandaure TPP (Mahan Energen)     | ADANI POWER | 2X800 | 1600        | 27.07.2023 |
| 4     | Satpura STPP (Expansion, Phase-V) | MPPGCL      | 1x660 | 660         | 21.11.2023 |
| 5.    | Raghunathpur TPS Stg-II           | DVC         | 2X660 | 1320        | 26.02.2024 |
| Total |                                   |             |       | <b>5840</b> |            |

##### 6.1.2 Power Projects for which order placed

During the year 2023-24, order for 8000 MW thermal capacity have been placed as listed below:

| Sl. | Project                      | Implementing Agency | Plant Configuration | Capacity (MW) | Main Plant (BTG)                |
|-----|------------------------------|---------------------|---------------------|---------------|---------------------------------|
| 1   | LARA STPP St-II (coal based) | NTPC                | 2x800               | 1600          | EPC Package awarded to M/s BHEL |
| 2   | Singrauli STPP St-III        | NTPC                | 2X800               | 1600          | EPC Package awarded to M/s BHEL |
| 3   | Talabira STPS                | NLC                 | 3X800               | 2400          | EPC Package awarded to M/s BHEL |
| 4   | Yamunanagar TPS              | HPGCL               | 1X800               | 800           | EPC Package awarded to M/s BHEL |
| 5   | Mahan Energen St-II          | ADANI POWER         | 2X800               | 1600          | EPC Package awarded to M/s BHEL |

##### 6.1.3 Thermal Power Plants expected to be commissioned by the year 2029-32

A. Candidate Brown field Thermal Power Plants identified by CEA expected to be commissioned by 2029-32 have been identified by CEA. List of the same is furnished below:

| A. Central Sector Candidate Projects |               |      |              |           |          |
|--------------------------------------|---------------|------|--------------|-----------|----------|
| 1                                    | Sipat-III     | NTPC | Chhattisgarh | 1x800=800 | Pit-Head |
| 2                                    | Darlipalli-II | NTPC | Odisha       | 1x800=800 | Pit-Head |

|    |                         |              |                |            |              |
|----|-------------------------|--------------|----------------|------------|--------------|
| 3  | Telangana St# II        | NTPC         | Telangana      | 3x800=2400 | Pit-Head     |
| 4  | New Nabi Nagar-II       | NTPC         | Bihar          | 3x800=2400 | Non Pit-Head |
| 5  | Patratu Stage II        | NTPC         | Jharkhand      | 2x800=1600 | Non Pit-Head |
| 6  | Gadarwara St#II         | NTPC         | Madhya Pradesh | 2x800=1600 | Non Pit-Head |
| 7  | Meja-II                 | U.P.-NTPC-JV | Uttar Pradesh  | 3x800=2400 | Non Pit-Head |
| 8  | Obra D TPP              | U.P.-NTPC-JV | Uttar Pradesh  | 2x800=1600 | Pit-Head     |
| 9  | Anpara E                | NTPC-UP-JV   | Uttar Pradesh  | 2x800=1600 | Pit-Head     |
| 10 | Raghunathpur TPS, PH-II | DVC          | West Bengal    | 2x660=1320 | Non Pit-Head |
| 11 | Koderma                 | DVC          | Jharkhand      | 2x800=1600 | Non Pit-Head |
| 12 | Durgapur TPS            | DVC          | West Bengal    | 1x800=800  | Non Pit-Head |
| 13 | Chandrapura Extn TPS    | DVC          | Jharkhand      | 1x800=800  | Non Pit-Head |
| 14 | TPS-II 2nd Expansion    | NLCIL        | Tamil Nadu     | 2x660=1320 | Pit-Head     |
| 15 | NLC Talabira STPS       | NLCIL        | Tamil Nadu     | 1x800=800  | Pit-Head     |
| 16 | Buxar TPP-II            | SJVN         | Bihar          | 1x660=660  | Non Pit-Head |
| 17 | Mahanadi Basin Power    | MCL          | Odisha         | 2x800=1600 | Pit-Head     |

**Subtotal Central Sector****24,100 MW****B. State Sector Candidate Projects:**

|   |                                  |        |                |            |              |
|---|----------------------------------|--------|----------------|------------|--------------|
| 1 | Super Critical TPP, Korba (W)    | CSPGCL | Chhattisgarh   | 2x660=1320 | Pit-Head     |
| 2 | IB Valley Extn St-III, (U#5 & 6) | OPGC   | Odisha         | 2x660=1320 | Pit-Head     |
| 3 | Chhabra U#7 & 8                  | RRVUNL | Rajasthan      | 2x660=1320 | Non Pit-Head |
| 4 | Kalisindh, U#3                   | RRVUNL | Rajasthan      | 1x800=800  | Non Pit-Head |
| 5 | Amarkantak                       | MPPGCL | Madhya Pradesh | 1x660=660  | Non Pit-Head |

|                               |                            |        |                |                  |              |
|-------------------------------|----------------------------|--------|----------------|------------------|--------------|
| 6                             | Satpura (Sarni) U#12       | MPPGCL | Madhya Pradesh | 1x660=660        | Non Pit-Head |
| 7                             | Koradi Replc               | MSPGCL | Maharashtra    | 2x660=1320       | Non Pit-Head |
| 8                             | Chandrapur                 | MSPGCL | Maharashtra    | 1x800=800        | Non Pit-Head |
| 9                             | Ukai, U#7, Tapi            | GSECL  | Gujarat        | 1x800=800        | Non Pit-Head |
| 10                            | Singareni U#3              | SCCL   | Telangana      | 1x800=800        | Non Pit-Head |
| 11                            | Godna TPS                  | KPCL   | Chhattisgarh   | 2x800=1600       | Pit-Head     |
| 12                            | Tenughat St II             | TVNL   | Jharkhand      | 2x660= 1320      | Pit-Head     |
| 13                            | Ukai, U#8                  | GSECL  | Gujarat        | 1x800=800        | Non Pit-Head |
| 14                            | Gandhinagar, U #6          | GSECL  | Gujarat        | 1x800=800        | Non Pit-Head |
| 15                            | Sikka, U#5                 | GSECL  | Gujarat        | 1x800=800        | Non Pit-Head |
| 16                            | Satpura, Ph-V, U#13        | MPPGCL | Madhya Pradesh | 1x660=660        | Non Pit-Head |
| 17                            | Santalidih, St-II, U#3 & 4 | WBDCL  | West Bengal    | 2x800=1600       | Non Pit-Head |
| 18                            | Barkeshwar TPS, St-II, U#3 | WBDCL  | West Bengal    | 1x660=660        | Non Pit-Head |
| 19                            | Durgapur Projects Limited. | DPL    | West Bengal    | 1x660=660        | Non Pit-Head |
| 20                            | Ropar TPP ,                | PSPGCL | Punjab         | 5x800= 4000      | Non Pit-Head |
| <b>Sub Total State Sector</b> |                            |        |                | <b>22,700 MW</b> |              |

**C. Private Sector Candidate Projects:**

|                                 |                    |             |                |                 |              |
|---------------------------------|--------------------|-------------|----------------|-----------------|--------------|
| 1                               | Raipur Extn.       | Adani Power | Chhattisgarh   | 2x800=1600      | Pit-Head     |
| 2                               | Mahan STPP, St-III | Adani Power | Madhya Pradesh | 2x800=1600      | Non Pit-Head |
| 3                               | Ind Bharat Extn.   | JSW         | Odisha         | 1x800=800       | Pit-Head     |
| 4                               | Salaya Extn        | Essar       | Gujarat        | 2x800=1600      | Non Pit-Head |
| 5                               | Lalitpur           | Bajaj       | Uttar Pradesh  | 3x660=1980      | Non Pit-Head |
| 6                               | Anuppur            | M B Power   | Madhya Pradesh | 2x660=1320      | Non Pit-Head |
| <b>Sub Total Private Sector</b> |                    |             |                | <b>8,900 MW</b> |              |

## 6.2 Captive Coal Blocks Allocated to Power Sector

As on Mar'24, 49 nos. of Coal Blocks are allocated to the Power Sector. 33 nos. of coal blocks are producing coal. 16 nos. of coal blocks are in development. The total nos. of coal blocks allotted to the Central Sector is 12 nos., State Sector is 33 nos., Private sector is 02 nos. and UMPP is 02 nos.

Captive mines allocated to Power Sector produced 101.96 MT of coal against a target of 106.71 MT for FY 2022-23. For FY 2023-24, the captive coal production was 120.88 MT against the target of 126.52 MT.

## 6.3 Linkage under SHAKTI Policy, 2017

Ministry of Coal in May 2017 formulated a new policy for the allocation of coal to the power sector named SHAKTI (Scheme for Harnessing and Allocating Koyala transparently in India), 2017. Since, the inception of the policy, coal linkage has been accorded to various Govt./Private power utilities under its various provisions/clauses. Status up to March-2024 is as under

### 6.3.1 Shakti Policy Para B (i):-

#### Policy:-

CIL/SCCL may grant Coal linkages for Central Government, State Government Gencos and JVs formed between or within CPSUs and State Govt./PSUs at the notified price of CIL/SCCL.

**Achievement:** Till Mar'24, SLC (LT) has accorded coal Linkage to 47 nos. Thermal Power Projects totaling 58,680 MW to Central /State GENCOs. In FY 23-24, 11 nos. of Thermal Projects of capacity 16,120 MW have been allocated long term linkage.

### 6.3.2 Shakti Policy Para B (ii):

#### Policy:-

CIL/SCCL may grant coal linkages on notified price on auction basis for power producers/PPs having already concluded

long term PPAs (both under section 62 and section 63 of The Electricity Act, 2003) based on domestic coal.

| Round of auction under Para B(ii) | Coal Qty booked (in G13 Grade) (MTPA) | Number of successful bidders and maximum discount offered on tariff |
|-----------------------------------|---------------------------------------|---------------------------------------------------------------------|
| 1st Round (Sep'17)                | 32.68                                 | 10 plants, Max discount 4 Paisa                                     |
| 2nd Round (May'19)                | 3.335                                 | 8 plants, Max Discount 7 Paisa                                      |
| 3rd Round (May'20)                | 3.466                                 | 5 plants, Max Discount 10 Paisa                                     |
| 4th Round (Sept'21)               | 3.819                                 | 5 plants, Max discount 12 paisa                                     |
| 5th Round (Dec'22)                | 0.058                                 | 2 plants, Max Discount 12 Paisa                                     |
| 6th Round (Dec'23)                | 2.935                                 | 7 plants, Max Discount 19 Paisa                                     |

### 6.3.3 Shakti Policy Para B (iii): -

**Policy:-** CIL/SCCL may grant future coal linkages on auction basis for power producers /IPPs without PPAs that are either commissioned or to be commissioned. All such power producers/IPPs may participate in this auction and bid for premium above the notified price of the coal. Coal drawl will be permitted only against valid long term and medium-term PPAs, which the successful bidder shall be required to procure and submit within two years of completion of auction process.

**Achievements:-** Five rounds of auctions for coal linkage under Shakti B(iii) have been held till Mar'24.

| Round of auction under Para B (iii) | Coal Qty booked (in G13 Grade) (MTPA) | Premium offered in the auction |
|-------------------------------------|---------------------------------------|--------------------------------|
| 1st Round, Feb'2020                 | 7.15                                  | Average premium is 8.5%.       |
| 2nd Round, May'2022                 | 7.01                                  | Average premium is 2%.         |
| 3rd Round, Sept'2022                | 5.57                                  | Premium is Zero.               |

|                     |       |                           |
|---------------------|-------|---------------------------|
| 4th Round, Jan'2023 | 4.87  | Average premium is 1.88%. |
| 5th Round, Dec'2023 | 7.636 | Average premium is 4.6%.  |

#### 6.3.4 Shakti Policy Para B (iv): -

**Policy:-** In this clause coal linkage may be earmarked to the states for fresh PPAs, by pre-declaring the availability of coal linkage with description. States may indicate these linkages to Discoms/State Designated Agencies (SDA). The states/Discoms may, based on such linkage, undertake tariff based competitive bidding for long-term and medium-term procurement of Power.

**Achievements:-** Till Mar'24, SLC (LT) has earmarked coal Linkage to 05 nos. of states for a capacity of 10,819 MW. In FY 2023-24, coal have been earmarked to 03 nos. of States for a capacity of 6,499 MW

#### 6.3.5 Shakti Policy Para B(v): -

**Policy:-** In this clause, Power requirement of group of States can be aggregated and procurement of such aggregated power can be made by an agency designated by Ministry of Power or authorized by such States on the basis of tariff based bidding. Coal linkages will be earmarked for such agencies by pre-declaring the availability of coal linkage with description, based on which such agency will undertake tariff based competitive bidding for long-term and medium-term procurement of power and recommend grant of these linkages to successful bidders.

**Achievements:-** Under this clause, in FY 23-24, PFCCCL (designated agency by MoP) invited bids for 4500 MW and issued LOAs for 1560.25 MW.

#### 6.3.6 Shakti Policy Para B (viii) (a):

**Policy: -**

All such All such power plants including private generators which do not have PPAs, shall be allowed coal linkage under SHAKTI Policy for a period of minimum 3 months and upto a maximum of 1 year, provided further

that the power generated through that linkage is sold through any product in power exchanges or in short term through a transparent bidding process through Discovery of Efficient Energy Price (DEEP) portal.

#### **Achievements: -**

Eighteen rounds of auctions for coal linkage under SHAKTI B (viii) (a) have been held so far upto March-2024:

| Tranches no. of auction under Para B(viii)(a) | Coal Qty booked (in G13 Grade) (MTPA) | Number of successful bidders and premium offered/ton |
|-----------------------------------------------|---------------------------------------|------------------------------------------------------|
| Tranche-1 (Apr-Jun'20)                        | 1.34                                  | 9 plants, Premium NIL                                |
| Tranche-2 (July-Sep'20)                       | 0.63                                  | 8 plants, Premium NIL                                |
| Tranche-3 (Oct-Dec'20)                        | 0.35                                  | 6 plants, Premium NIL                                |
| Tranche-4 (Jan-Mar'21)                        | 0.64                                  | 7 plants, Avg. Premium 1.4%                          |
| Tranche-5 (Apr-Jun'21)                        | 1.07                                  | 8 plants, Avg. Premium 0.4%                          |
| Tranche-6 (July-Sep'21)                       | 0.82                                  | 8 plants, Premium NIL                                |
| Tranche-7 (Oct-Dec'21)                        | 1.81                                  | 8 plants, Avg. Premium 1.2%                          |
| Tranche-8 (Jan-Mar'22)                        | 1.45                                  | 11 plants, Avg. Premium 2.95%                        |
| Tranche-9 (Apr-Jun'22)                        | 6.13                                  | 16 plants, Avg. Premium 152%                         |
| Tranche-10 (Jul-Sep'22)                       | 4.25                                  | 27 Plants, Avg. Premium 320%                         |
| Tranche-11 (Oct-Dec'22)                       | 6.01                                  | 22 Plants, Avg. Premium 133%                         |
| Tranche-12 (Jan-Mar'23)                       | 5.39                                  | 26 Plants, Avg. Premium 139%                         |
| Tranche-13 (Mar-May'23)                       | 3.67                                  | 21 Plants, Avg. Premium 38%                          |
| Tranche-14 (May-Jul'23)                       | 3.49                                  | 25 Plants, Avg. Premium 24%                          |
| Tranche-15 (Aug-Oct'23)                       | 4.82                                  | 28 Plants, Avg. Premium 60%                          |

|                                   |      |                                  |
|-----------------------------------|------|----------------------------------|
| Tranche-16<br>(Oct'23-<br>Mar'24) | 10.6 | 33 Plants, Avg.<br>Premium 42%   |
| Tranche-17<br>(Nov'23-<br>Jan'24) | 2.09 | 16 Plants, Avg.<br>Premium 15%   |
| Tranche-18<br>(Apr'24-<br>Mar'25) | 18.5 | 29 Plants, Avg.<br>Premium 2.6 % |

### 6.4 Bridge Linkage

Ministry of Coal vide Office Memorandum dated 08.02.2016, had issued policy guidelines for grant of bridge linkage to End Use Plants (EUPs) of Central and State public sector undertakings which have been allocated Coal Mines/Coal Blocks. Based on these guidelines, 35 nos. Thermal Projects totaling 39,870 MW were granted Bridge Linkage.

### 6.5 Use of Treated Sewage Water by TPS under Tariff Policy-2016:

As per Tariff Policy dated 28.01.2016, notified by Government of India, sewage treated water is to be used by Thermal Power Plants (Thermal Power Plants which are located within 50 Kms from Sewage Treatment Plants) for cooling purpose. Accordingly, MoP/ Central Electricity Authority (CEA) is exploring the feasibility for the usage of Sewage Treated Water by Thermal Power Plants for cooling purpose. Presently, 08 nos. of Thermal Power Stations with a total capacity of 9239.2 MW are utilizing 589 MLD of STP water i.e. Koradi TPS, Khaperkheda TPS & Sinnar TPP Phase-I in Maharashtra, Pragati CCGP & Pragati-III in Delhi, Bhavnagar Lignite TPS in Gujrat and Yelahanka CCP in Karnataka and Jojobera TPS in Jharkhand are utilizing STP water. Also, 02 nos. of Thermal Plants (3580 MW) have successfully placed the order for construction of the projects (Tertiary Treatment Plant and Pipeline). Quantum of STP water associated with these projects amounts to 90 MLD.

### 6.6 Mega Power Project Certificate:

Recommendation to MoP for issuance of proportionate Mega Certificate to provisional Mega Power Projects for release of Proportionate Bank Guarantee/FDRs. So far, the following 4 projects have been granted proportional Mega Certificate:

- i. MB (Madhya Pradesh) Power Ltd
- ii. DB Power Ltd
- iii. RKM Powergen Pvt. Ltd.
- iv. IL&FS Tamil Nadu Energy Ltd.

### 6.7 Ultra Mega Power Projects (UMPPs):

Government of India through Ministry of Power launched the initiative of Ultra Mega Power Projects (UMPPs) i.e. 4,000 MW super thermal power projects (both pit head and imported coal based) in November 2005 with the objective to develop large capacity power projects in India. Central Electricity Authority (CEA) has been designated as Technical Partner and Power Finance Corporation Ltd (PFC) as the Nodal Agency to facilitate the development of these projects.

#### 6.7.1 Status of UMPPs Awarded:

Initially, four UMPPs namely Sasan in Madhya Pradesh, Mundra in Gujarat, Krishnapattanam in Andhra Pradesh and Tilaiya in Jharkhand were awarded to the successful bidders. The details of these project are given as below:

| Sl. No | Name of UMPP                  | Type    | Date of Transfer | Levelling Tariff (in Rs. Per kWh) | Successful developer | Present Status |
|--------|-------------------------------|---------|------------------|-----------------------------------|----------------------|----------------|
| 1      | Mundra, Gujarat               | Coastal | 23.04.2007       | 2.26                              | Tata                 | Operational    |
| 2      | Sasan, Madhya Pradesh         | Pithead | 07.08.2007       | 1.19                              | Reliance Power Ltd.  | Operational    |
| 3      | Krishnapatnam, Andhra Pradesh | Coastal | 29.01.2008       | 2.33                              | Reliance Power Ltd.  | Terminated     |
| 4      | Tilaiya, Jharkhand            | Pithead | 07.08.2009       | 1.77                              | Reliance Power Ltd.  | Terminated     |

### 6.7.2 Present Stance of MoP, GoI on development of new UMPP:

Presently, no UMPP is under construction. As the country is undergoing energy transition from fossil fuel to non-fossil fuel, Ministry of Power vide OM dated 12.11.2021, decided to defer any action on formulation of UMPP Bidding framework as of now. Therefore, no project activities related to development of UMPP are being taken up further.

### 6.8 Construction Monitoring Of Thermal

#### Power Projects:

CEA closely monitors the progress of various construction activities of thermal power projects under construction in the country. Project monitoring related activities emerge from Section 73 (f) Electricity Act, 2003. Regular visits are made by CEA officers to the project sites for assessing the progress of various construction activities and rendering necessary advice/assistance in resolving the problems being faced by the Project Authorities to meet the schedule of commissioning. Regular Review Meetings are also held in CEA with Project Authorities, Main Plant & Equipment Manufacturers and other equipment Suppliers to review the progress status of the Projects.

#### 6.8.1 Key initiatives

Based on the past experience, there has been a significant shift in approach in the area of project monitoring. Some key initiatives taken during recent past in the role of a facilitator,

includes the following:

- Drawing up of detailed schedules for project /milestones commitments from project authorities for on-going under construction projects.
- Participation in various review meetings held in the Ministry of Power, Ministry of Heavy Industries, Project Monitoring Group and NITI Aayog etc.
- Holding review Meetings with various implementing agencies including suppliers to review the progress of work and finalizing the completion schedule of under construction thermal power projects.
- Undertaking visits to thermal projects to assess the progress of various activities at site.

During the year 2023-24, the following under construction thermal projects have been visited by CEA officers to assess the actual progress of various construction activities and the anticipated date of trial run of unit of the project:

1. Bhusawal Thermal Power Station (1x660 MW) in the State of Maharashtra being implemented by M/s MSPGCL from 30.05.2023 to 02.06.2023.
2. Ghatampur Thermal Power Project (3x660 MW) in the State of Uttar Pradesh being implemented by M/s NUPPL from 10.08.2023 to 11.08.2023
3. Panki Thermal Power Station Extn. (1x660 MW) in the State of Uttar Pradesh being implemented by M/s NUPPL from

25.08.2023 to 26.08.2023

4. Buxar Thermal Power Project (2x660 MW) in the State of Bihar being implemented by M/s STPL from 22.09.2023 to 23.09.2023

### 6.8.2 Other Important activities of construction monitoring of thermal projects:

- Monitoring the progress of construction/erection activities of the Maitree (Indo-Bangladesh (2x660 MW) Super Thermal Power Projects on monthly basis. Unit #1 & 2 of Maitree Super Thermal Power Projects have been commissioned on 23.12.2022 & 12.03.2024 respectively.
- Monitoring of progress (financial progress/ CAPEX) of National Infrastructure Pipeline (NIP) projects which are State's projects in thermal sector on monthly basis. The status of bidding of upcoming projects of State's projects in thermal sector are also prepared on Quarterly basis.
- Preparation of monthly/fortnightly reports viz. report for PMO / MIS report /Thrust Area report/ report for D.O. letter from Secretary (Power) to Cabinet Secretary/report for updates regarding Min. of Power/ Report for Urja Darpan portal in respect of under construction thermal power projects in the country.
- TPM Division in consultation with NTPC Ltd. has developed an online thermal project monitoring portal. To showcase the portal's functionality, hands on exposure training has been provided to various project developers and requested to fill their monthly project data in the portal regularly.

## 6.9 Thermal capacity addition program

### 6.9.1 Thermal capacity addition achievement during 2023-24

The thermal capacity addition target for the year 2023-24 was 14700 MW against which a capacity of 5040 MW has been achieved. During the FY 2023-24, an additional thermal capacity of 364 MW was also commissioned Sector-wise details of target and achievement during

the year 2023-24 are as follows:-

(All fig. in MW)

| Sector       | Thermal (Planned) |             | Thermal    |
|--------------|-------------------|-------------|------------|
|              | Target            | Achieved    | Achieved   |
| CENTRAL      | 6880              | 2920        | 0          |
| STATE        | 7820              | 2120        | 0          |
| PRIVATE      | 0                 | 0           | 364        |
| <b>TOTAL</b> | <b>14700</b>      | <b>5040</b> | <b>364</b> |

The details of target/ achievements for the year 2023-24 is enclosed at **Annexure 6A.**

### 6.9.2 Thermal capacity addition target during 2024-25

The target for thermal capacity addition during 2024-25 is as under:-

| SECTOR       | Target (MW)   |
|--------------|---------------|
| CENTRAL      | 6,740         |
| STATE        | 8,620         |
| PRIVATE      | 0             |
| <b>TOTAL</b> | <b>15,360</b> |

## 6.10. Thermal Engineering & Technology Development (TE&TD):

CEA has been actively associated in resolving the thermal engineering related issues and introduction of new technologies in thermal power generation. The important responsibilities entrusted to TE&TD division are:

Advice to Electricity Regulatory Commissions/Power Generating Utilities on Technical Matters, Preparation of Standard Technical Specifications for Thermal Power Stations, Issues related to the Indian Boiler Regulation, Preparations/ Amendment of CEA Regulations pertaining to Construction of Electrical Plants & Electric Lines, Safety Requirements for Construction, Operation and Maintenance of Electrical Plants & Electric Lines and Flexible Operation of Coal based Thermal Power Generating Units, etc. Investigation of Accidents/Failures at Thermal Power Stations, Examining Proposal for R&D in

Indian Power Sector, Technical Advice to various Expert/Technical Committees set up by CEA/Others organizations, Reference for expert advice on Technological matter, New and Emerging clean energy technologies in the Thermal Power Sector such as Hydrogen co-firing in Gas Turbines, Ammonia in coal based TPS, CCUS etc., Matters related to Operating Parameters of Coal/Lignite and CCGT based Thermal Power Stations, Matters related to Excellence Enhancement Center and Bureau of Energy Efficiency, Matters related to Make in India for Thermal Power Sector, Co-firing of Biomass and Exemptions to TPPs under National Mission on use of Biomass in Thermal Power Plants (SAMARTH), Safety related matters under CEA (Safety Requirements for Construction, Operation and Maintenance of Electrical Plants & Electric Lines) Regulations, 2011 as amended.

### 6.10.1 Important Activities

The following important activities have been completed/ undertaken during the financial year 2023-24:

#### A. Promoting Use of Biomass in Thermal Power Plants:

The Executive committee headed by Member (Thermal), CEA with representatives from MoP, MNRE, BHEL NTPC, BEE, CPRI, NPTI, DVC, CE(TE&TD) & SAMARTH Mission as Member of the Committee. It has been constituted by MoP to supervise the activities of SAMARTH Mission and evaluate the progress of mission directorate. With the active support of the Executive Committee, approximately 493320 metric tonnes (MT) biomass have been co-fired as on 31.03.2024 since inception of SAMARTH and in FY 23-24 approximately 374808 metric tonnes (MT) of biomass have been co-fired in thermal power plants in the country. Out of this, the biomass co-fired in the NCR region stands at 293628 MT since inception and 258772 MT in FY 23-24.

Committee under the Chairmanship of Chief Engineer (TE&TD) with representatives from Ministry of Agriculture, CPRI, NTPC, BHEL & SAMARTH Mission has been constituted by

MoP to examine the request of power plants for their exemption / relaxation from mandatory co-firing of biomass. The Committee prepared and circulated the guidelines / procedure for examining the request of the power plants for seeking exemption / relaxation from co-firing, to all the coal based thermal power plants in the country. As on 31.03.2024, total 43 no. of applications for exemption/relaxation from mandatory co-firing of biomass had been assessed. It is to mention that as on 31.03.2024, based on the information provided by above power plants, none of them was found suitable for the exemption from mandatory co-firing of biomass and as such no exemption has been given to any thermal power plant so far.

MoP through a policy addendum on 03-05-2023 has indicated the various types of agro residues such as stubble/straw/stalk/husk which are surplus and not being used as animal fodder for making the biomass pellets. This includes agro-residue obtained from crops like Paddy, Soya, Arhar, Gwar, Cotton, Gram, Jawar, Bajra, Moong, Mustard, Sesame, Til, Maize, Sunflower, Jute, Coffee, etc. as well as Groundnut Shell, Coconut Shell, Castor Seed Shell etc.

In addition, pellets made from the following agro product/crop/waste can also be used for co-firing in TPPs viz. Bamboo and its by-products, Horticulture waste such as dry leaves and trimmings obtained from maintenance & pruning of trees and plants and other biomass such as Pine Cone/Needle, Elephant Grass, Sarkanda etc. Ministry of Power issued modification on 16.06.2023 in the revised biomass policy dated 08.10.2021 and mandated 5% biomass co-firing in Thermal Power Plants (TPPs) from FY 2024-25. This obligation shall increase to 7% from FY 2025-26.

In July 2023, MoEF&CC notified Environment (Utilisation of Crop residue by Thermal Power Plants) Rules, 2023, in which the Commission for Air Quality Management (CAQM) shall impose and collect the environmental compensation from NCR Thermal Power Plants on annual basis at the rates specified in the gazette for not complying with mandatorily blend of pellets made of crop residue along with

coal.

With a view to promote and develop sustainable supply chain & at the same time to ensure faster procurement of biomass pellets for co-firing with coal in thermal power plants MoP constituted a Committee under Member (Thermal), CEA with the representatives from various states, Gencos, CE(TE&TD), SAMARTH Mission etc. to recommend a benchmark price for biomass pellets. The committee submitted its report and the benchmark price for non-torrefied biomass pellets has been issued by Ministry of Power (MoP) on 23.08.2023 for National Capital Region (NCR) and on 08.11.2023 for Northern (excluding NCR) Region and Western Region. The benchmark prices have been fixed at Rs 2.32, Rs 2.27 and Rs 2.24 per 1000 kcal (excluding GST & transportation cost from pellet manufacturing plant site to thermal power plant) for NCR, NR (excluding NCR) and WR respectively. This will also have a stability effect on the prices of raw biomass in the market.

#### **B. Safety related matters under CEA:**

Reference is invited to Central Electricity Authority (Safety requirements for Construction, Operation and Maintenance of Electrical Plants and Electric Lines) Amendment Regulations, 2022 notified in Gazette on 16th November, 2022. A new Schedule-IV was introduced in the regulation named as Elements of safety checks for electrical plants. Therefore, in compliance of above, Safety advisory was recirculated to all thermal power stations of the country.

#### **C. Recommendation to CERC for specifying Operational Norms of TPPs in Tariff Regulation, 2024**

The Tariff Policy notified by the Central Government under the Electricity Act 2003, provides that The Central Commission would, in consultation with the Central Electricity Authority, notify operating norms from time to time for generation and transmission.

In respect of thermal generating stations, in May 2023, CERC had requested CEA to furnish its recommendations on Target Availability, Target PLF, Gross Station Heat Rate, Specific Fuel Oil

Consumption, Auxiliary Energy Consumption, transit losses and GCV loss on stocking / handling etc. in due consideration of renewable capacity addition and technical minimum operation of conventional plants. The additional operational norms have also been requested for units/ stations on account of implementation of new 'Emission norms' issued by the Ministry of Environment, Forest and Climate Change.

In this regard, "Recommendation of operational norms for Thermal Power Stations for the tariff period starting from 01.04.2024" comprising of recommendation on all the aforesaid parameters was submitted to CERC on 19.12.2023. Further, the recommendations were reviewed in consideration of public comments of stakeholders on the request of CERC, and subsequently the Revised Recommendations on the same were submitted to CERC on 15.03.2024.

#### **D. National Energy Conservation Awards 2023:**

Ministry of Power has undertaken a scheme to encourage, motivate as well as give recognition through National Energy Conservation Awards to industrial units and other establishments, who take extra efforts to reduce energy intensities while maintaining the production levels. The scheme is aimed to create an environment that would spur industries and other establishment in achieving excellence in efficient use of energy and its conservation. The awards were given away for the first time in December, 14, 1991 which is now celebrated as National Energy Conservation Day throughout the country. Member (Thermal), CEA is the head of the Technical Sub-Committee to assist the Award Committee, headed by Secretary, Power, in the finalization of awards. During the year 2022-23, proposals received from three industrial sectors viz. Thermal Power Plant, Aluminium and Dairy were evaluated by CEA. The awards to the best performing industrial units in all the sectors covered during 2022-23 were given on 14th December, 2023 in New Delhi.

#### **E. Review/modification of Guidelines for Model Quality Assurance Plan:**

CEA published 'Guidelines for Model Quality Assurance Plan (MQAP) for major equipment

of Power sector' in March 2022 covering MAQP for all the segments of power sector i.e. Generation (Thermal, Hydro), Transmission and Distribution. TETD Division prepared the thermal chapter of the above MQAP. A review/modification of Model Quality Assurance Plan (MQAP) for Major Equipment in Thermal Power Plant was carried out with respect to requirements of inspection of material/equipment at Manufacturer's premise and at site.

#### **F. Review of “Guidelines for QR for Bidders of BoP for Coal/Lignite based TPS” in order to improve vendor base for BoP areas**

The CEA document “Guidelines for Qualifying Requirements for Bidders of Balance of Plant for Coal/Lignite based Thermal Power Stations, 2010”, has been a guiding document for generating utilities to define their Pre-Qualification Requirement (PQR) for bidders of Balance of Plant (BoP) for coal/lignite based Thermal Power Stations on EPC basis (Section-I) as well as on individual BoP package basis (Section-II) since, 2010.

However, recently bidders have been facing the issue of erosion of Vendor Base in critical packages due to low business opportunities in the Thermal Power Sector and stringent Pre-Qualification Requirement of customers hampering growth of new vendors. Keeping this in view, CEA “Guidelines for Qualifying Requirements for Bidders of Balance of Plant for Coal/Lignite based Thermal Power Stations, 2010” have been reviewed in September, 2023 with the aim to encourage participation and widen the vendor base for thermal power plants.

#### **G. Technical Committee on Thermal Research:**

The Technical Committee on Thermal Research convened by Central Power Research Institute (CPRI) has members from CEA, IITs, NTPC, and BHEL & Tata Power for evaluation of various research proposals/projects under the schemes of CPRI. The 14th and 15th meeting of the Committee

were held in various phases during the year 2023-24 and various thermal research projects were evaluated.

#### **H. Draft Report of the Technical Committee on CO2 Capture**

A Meeting of the Advisory Committee on "Development of a policy framework for the effective implementation of the Carbon Capture, Utilization and Storage (CCUS) initiatives in India" was held under the chairmanship of Dr. V. K. Saraswat, Member, NITI Aayog, New Delhi. The advisory committee constituted four technical committees in the areas of Carbon Capture, Utilization, Storage and Safety & Technical Standards Development.

The Technical Committee on CO2 capture has been constituted under the chairmanship of Chief Engineer (TE&TD Division). The task assigned to committee was to assess CO2 capture technologies across sectors (power, steel, cement, etc.) and examine key aspects (industrial processes, purification, monitoring) for recommendations.

The technical committee on CO2 capture projects in India submitted its report to NITI Aayog in March, 2024.

#### **I. Committee to analyse the request of M/s MSPGCL regarding evaluating the higher O&M expenses of Koradi Thermal Power Station (KTPS) (3 x 660 MW) compared to the prevailing O&M expenses norms specified by the state electricity regulator:**

M/s. MSPGCL requested CEA for Technical Consultancy for evaluating O&M expenses of Koradi Thermal Power Station (KTPS) (3x660 MW) and specifying the norm for O&M expense along with comparative analysis with norms across country. In this regard, a committee was constituted under the Chairmanship of Member (Thermal), CEA with members from UPRVUNL, NTPC, MPPGCL, MSPGCL and F&CA Division of CEA.

The committee analyzed and evaluated the higher O&M expenses of Koradi Thermal Power Station (KTPS) and compared these O&M expenses to the prevailing O&M expenses norms specified by

the State Electricity Regulator along with CERC norms. The work of the committee is under progress.

#### **J. Chairman/Members of various Committees:**

- a. Chief Engineer, TE&TD is a member of the Working Committee constituted by NITI Aayog for the development of a policy framework for the effective implementation of the Carbon Capture, Utilization and Storage (CCUS) initiatives in India.
- b. Chief Engineer, TE&TD is a member of the Standing Site Selection Committee for selection of sites for future nuclear power stations constituted by Department of Atomic Energy (Power Section).
- c. Chief Engineer, TE&TD is a member of the Technical Committee on Thermal Research for review, recommendation and monitoring of R&D proposals under IHRD, RSOP, NPP schemes of MoP, Govt. of India
- d. Chief Engineer, TE&TD has been member secretary of Sub- Committee-7 constituted under the chairmanship of CMD, NTPC to provide/update report on “Key Inputs for Power Sector” for National Electricity Plan (2022-27).
- e. Chief Engineer, TE&TD has been member secretary of Sub- Committee-1 constituted for National Electricity Plan (2022-27).
- f. Chief Engineer, TE&TD is member of Technical Committee constituted to finalize the baseline and target parameters for existing DCs under Thermal Power sector under PAT scheme.
- g. Chief Engineer, TE&TD is member of Central Boilers Board and undertakes works related to technical matters related to boilers, issues related to repeal or re-enactment of boilers act 1923, technical advice, comments on renewal or recognition of applications under IBR, 1950 etc.
- h. Director (TE&TD) is a Member on Board of Directors of Puducherry Power Corporation Limited, Puducherry.
- i. Deputy Directors of TE&TD division are the members of Subgroup-1 (R&D on properties/characteristics of Biomass) and Subgroup-2 (R&D on boiler design on biomass

co - firing and safety aspects) created under National Mission on use of biomass in coal based thermal power plants.

#### **6.11 Thermal Civil Design (TCD):**

CEA is involved in the monitoring of Ash Generation and its utilization at the Coal / Lignite based TPPs in the Country. During 2023-24 the following were published:

- Publication of Guidelines on Design, Construction, O&M and Annual Certification of Coal Ash Ponds in June 2023.
- Report on Ash Generation at Coal / Lignite based Thermal Power Station and its utilization in the Country for the year 2022-23 in November 2023.

#### **6.12 R&M/ LE Programme during (2017 - 22)**

71 thermal generating units with aggregate capacity of 14929 MW have been identified for implementation of R&M/LE works during 2017-22 period. Out of this a total of 35 nos. thermal generating units with aggregate capacity of 7570 MW for LE works and 37 nos. thermal generating units with aggregate capacity of 7359 MW for R&M works have been identified for the period 2017-22. Break-up summary of LE and R&M works of 14929 MW to be taken up during 2017-22 in terms of Central/ State sector-wise is furnished below at Table A:

| Category       | LE/R&M works identified during 2017-22 No. of units & capacity (MW) |                | Total (State Sector + Central Sector) |
|----------------|---------------------------------------------------------------------|----------------|---------------------------------------|
|                | State Sector                                                        | Central Sector |                                       |
| <b>LE</b>      | 34 (7570)                                                           | --             | 34 (7570)                             |
| <b>R&amp;M</b> | 30 (7135)                                                           | 07 (224)       | 37 (7359)                             |
| <b>Total</b>   | 64 (14705)                                                          | 07 (224)       | 71 (14929)                            |

**6.12(a) Achievements of R&M & LE Projects****upto 31-03-2024:**

Life Extension works on 5 thermal generating units with aggregate capacity of 1020 MW and R&M works on 3 thermal generating units with

aggregate capacity of 177 MW were completed upto 31-03-2024. Details of achievements is furnished below at Table:

|             | Name of the TPS | Unit No.   | Date of S/D | Capac ity (MW) | Utility    | Sector   | Date of Achievement                               |
|-------------|-----------------|------------|-------------|----------------|------------|----------|---------------------------------------------------|
| 1. 2017-18  |                 |            |             |                |            |          |                                                   |
| LE          | Ukai TPS        | 4          | 07-12-2016  | 200            | GSECL      | Stat e   | 17.05.2017                                        |
|             | Wanakbori TPS   | 3          | 25-07-2017  | 210            | GSECL      | Stat e   | 27-11-2017                                        |
| R&M         | Kathalguri CCGT | 3          | 19-06-2017  | 33.5           | NEEPCO     | Centr al | 20-07-2018                                        |
|             | Kathalguri CCGT | 6          | 19-03-2018  | 33.5           | NEEPCO     | Centr al | 31-03-2018                                        |
| Sub Total   |                 | 3 4 Units) |             | 477.00         |            |          |                                                   |
| 2. 2018-19  |                 |            |             |                |            |          |                                                   |
| LE          | Koradi TPS      | 6          | 25-08-2015  | 210            | MAHAGE NCO | Stat e   | 16-07-2018(oil firing)<br>20-08-2018(coal firing) |
|             | Obra TPS        | 12         | 01-10-2016  | 200            | UPRVUNL    | Stat e   | 24-09-2018                                        |
| R&M         | --              | --         |             | --             | ---        | --       | --                                                |
| Sub Total   |                 | 02(uni t)  |             | 410            |            |          |                                                   |
| 3. 2021-22  |                 |            |             |                |            |          |                                                   |
| LE          | ---             | --         | --          | --             | --         | --       | --                                                |
| R&M         | Barauni TPS     | 6          | 15-11-2009  | 110            | NTPC       | Centr al | 31-05-2022                                        |
| 4. 2022-23  |                 |            |             |                |            |          |                                                   |
| LE          | Obra TPS        | 13         | 16-05-2018  | 200            | UPRVU NL   | State    | 27-09-2022                                        |
| R&M         | --              |            |             |                |            |          |                                                   |
|             |                 |            |             |                |            |          |                                                   |
| Total LE    | 05 (1020)       | State      | 05(unit)    | 1020           |            |          |                                                   |
|             |                 | Centr e    | --          | --             |            |          |                                                   |
| Total R&M   | 03 (177)        | State      | --          | --             |            |          |                                                   |
|             |                 | Centr e    | 03(unit)    | 177            |            |          |                                                   |
| Grand Total |                 | 08(units)  | 1197.0      |                |            |          |                                                   |

### **6.12(b) Undergoing R&M /LE in Thermal: Power Units upto 31-03-2024:**

Total of 26 units of central and state sector, with combined capacity of 4532 MW of the identified thermal power units, are currently undergoing R&M/LE works.

### **6.13 Monitoring of R&M Projects:**

The progress of R&M and LE works being implemented at Thermal Power units are monitored by holding the review meetings and information compiled on monthly/quarterly basis. Based on data / information collected & compiled, Quarterly Review Report on status of R&M projects were prepared.

### **6.14 Implementation of FGD installation, ESP upgradation and Combustion modification in respect of new Environmental Norms**

The timelines for complying with the new emission norms (Dec 2015) were revised by MoEF&CC vide gazette notification dated 31.03.2021 which categorized thermal power plants in three categories having different timelines.

Based on the MoEF&CC notification dated 31st March 2021, a task force was constituted comprising of representatives from MoEF&CC, MOP, CEA and CPCB to categorize the thermal power plants in above mentioned three categories. CPCB vide its MoM dated 13.12.2021 categorized the 600 TPP units in category A, B & C of which 66 units (20577 MW) are under Category A, 72 units (24057 MW) are under Category B and 462 units (166885 MW) are under Category C. The criteria considered for categorisation of TPPs along with compliance timelines is given as under:

**Category A** - Within 10 km radius of NCR or cities having million plus population as per 2011 census of India. Timeline for compliance: 31.12.2022

**Category B** - Within 10 km radius of critically polluted areas or Non-Attainment cities as defined by CPCB. Timeline for compliance: 31.12.2023

**Category C** - Other than those included in category A and B. Timeline for compliance: 31.12.2024

Subsequently, based upon the recommendations of CEA, the timelines for implementation of FGD

installation for compliance with SO<sub>2</sub> norms were extended by 2 years in every category vide MoEF&CC gazette notification dated 06.09.2022. However, the timelines for compliance of parameters other than SO<sub>2</sub> remained unchanged. The revised timelines for installation of FGD in various category TPPs is as under:

**Category A** - Timeline for compliance with SO<sub>2</sub> norms: 31.12.2024

**Category B** - Timeline for compliance with SO<sub>2</sub> norms: 31.12.2025

**Category C** - Timeline for compliance with SO<sub>2</sub> norms: 31.12.2026

### **6.15 Current Status of FGD Installation, ESP Upgradation and Combustion Modification**

- Out of 600 TPP units, 42 units (21390 MW) have installed FGD systems for compliance with SO<sub>2</sub> emission norms.
- Out of 600 TPP units, 436 units (174076 MW) have upgraded ESPs for compliance with SPM emission norms.
- Out of 600 TPP units, 433 units (154661 MW) have completed combustion modification for compliance with NO<sub>x</sub> emission norms.

### **6.16 Flexible Operation of Thermal Power Stations**

India's Intended Nationally Determined Contributions (INDCs) include a reduction in the emissions intensity of its GDP by 45 percent by 2030 from 2005 level, achieve about 50 percent cumulative electric power installed capacity from non-fossil fuel-based energy resources by 2030 and to create an additional carbon sink of 2.5 to 3 billion tonnes of CO<sub>2</sub> equivalent. Generating power from renewable sources of energy is of cardinal importance if India is to meet its INDC targets. With the aim to ensure future security & reliability of power supply and stability of electricity grids while maximizing generation from renewables flexibilization of existing coal-fired power plants is

an important measure.

A comprehensive report titled “Flexibilisation of Coal Fired Power Plant - A Roadmap for Achieving 40% TML” was prepared by CEA in Feb 2023 to help utilities and understand the low load operation. The report has comprehensively touched the various aspects of flexible operation of thermal generating units including operating procedure, identification of measures of low load operation/ramp rates and cost of flexible power.

A Compensation methodology for operating a coal based power plant below 55% load was prepared by CEA and the same was forwarded to CERC by MOP.

Another report titled “OPERATING PROCEDURE AND TRAINING CURRICULUM AT 55% MINIMUM TECHNICAL LOAD OF THERMAL GENERATING UNITS “was prepared and published to provide Standard operating procedures for thermal generating utilities for 55% technical minimum load at low load operation.

As per CEA (Flexible Operation of Coal Based Thermal Power Generating Units) Regulation, 2023, the coal based thermal power generating units shall have flexible operation capability with minimum power level of 40%. Moreover, the coal based power generating units shall have flexible operation capability with minimum power level of 55% along with ramp rate of 2% between 55%-70% and ramp rate of 3% above 70% within one year of notification of the above mentioned regulations i.e. by Jan 2024. The implementation of the flexible operation (40%) shall be as per the phasing plan specified by the authority from time to time. In view of above, CEA has prepared and notified in gazette of India a roadmap for achieving the minimum load operation in a phased manner in Dec 2023. The summary of the phasing plan is as below:

- Pilot Phase (10 units, 5580 MW) : May 2023 - March 2024
- Phase 1 (91 units, 51080 MW) : July 2024 - June 2026
- Phase 2 (100 units, 46825 MW) : July 2026 - June 2028

- Phase 3 (101 units, 37215 MW) : July 2028 - December 2029
- Phase 4 (191 units, 55767 MW) : January 2030 - December 2030

In the pilot phase, 10 units of Central/State/Pvt sector shall be taken for refurbishment. The experience gained in pilot phase shall be useful for future planning.

### **6.17(a) Japan-India Co-operation for Study on Efficiency and Environmental Improvement of Coal Fired Stations**

Under Indo- Japan Co- operation for Efficiency and Environmental Improvement of Coal Fired Power Stations. Four MoUs have already been implemented between Central Electricity Authority (CEA) and Japan Coal Energy Centre (JCOAL) in the field of efficiency improvement and environmental improvement of coal fired power stations. The 4<sup>th</sup> MoU between CEA and JCOAL has been signed on 16<sup>th</sup> December, 2019 for Efficiency & Environment Improvement for Sustainable, Stable and Low Carbon Supply of Electricity. The purpose of this MoU is to address issues and barriers in expediting sustainable, stable and low carbon thermal power development by means of studies, training program and knowledge- sharing activities, outcomes of which are to be conducive to overall power development in India as well as to expedite relevant policy implementation by the Government of India. Following activities were carried out under 4<sup>th</sup> MoU:

- i) Update on the current and future policy trend in the Indian power sector and consideration of the identified issues/barriers to find out those which could be addressed through mutual collaboration.
- ii) Identification of issues to be addressed regarding both existing and upcoming facilities, and also operation and maintenance.
- iii) Implementation of studies with priorities, but not limited to environmental technologies for coal fired power generation Flexibilization measures and biomass utilization are also of high priority.
- iv) Biomass study on Co firing of biomass pellets and Waste to Energy technologies and Coal GCV loss in power plant and its remedies Implementation of an annual workshop in India and CCT Training Programme in Japan.

v) Holding joint meetings to discuss issues that have arisen or may arise in the course of implementation of the Cooperation.

Under Clean Coal Technology (CCT) Training Programme study tours to Japan have been organized in which representatives from MoP, CEA and different power utilities have participated. The participants visited the latest USC power stations and updated about various applicable technologies and equipment as well as O&M technique. During the FY22, 50 participants have undergone the Virtual CCT Training Programme from 31st Oct. 2022 to 2<sup>nd</sup> Nov., 2022.

Under Indo-Japan Cooperation, a one-day Workshop on “Project on Efficiency and Environmental Improvement for Sustainable, Stable and Low-carbon Supply of Electricity” was organized jointly by CEA and JCOAL on 14th Dec, 2023.

The 5th MoU between CEA and JCOAL for Efficiency & Environment Improvement for Sustainable, Stable and Low Carbon Supply of Electricity is under finalization.



**CEA-JCOAL Workshop FY23**

## 6.17 (b) Indo-Denmark Co-operation

A MOU on India-Denmark Energy Cooperation was signed between the two governments in June 2020. TPRM Division, CEA is coordinating the following areas/activities under this cooperation:

- i. Transfer of technology for emission control from Thermal Power plants,
- ii. Waste heat recovery from Thermal power plants,

- iii. Flexibility in operation of power plants for RE integration.

Flexible operation test (upto 40% load) has been successfully conducted in thermal generating units of Unit #7, 500MW Ramagundam TPS of NTPC and Unit #3, 210 MW Raichur TPS of KPCL under India-Denmark cooperation.

## 6.18 Fuel Management and Analysis

Central Electricity Authority (CEA) plays a pivotal role in optimal utilization of coal for the power sector. It monitors coal supply to the power plants so that plants have sufficient coal stock as per norms. CEA in association with MoP, MoC, Railways and other stakeholders closely monitors the coal supply to power plants and take necessary steps to improve supply of coal to power plants. With the concerted efforts of all stakeholders, the coal supply to power utilities is maintained to meet their coal requirement. At the beginning of the year (1<sup>st</sup> April, 2023), the coal stock available with the thermal power plants was 36.95 Million Tonnes (MT) (55% of the normative coal stock required), which was sufficient for an average of 13 days at a requirement of 85% PLF. However, as on 31<sup>st</sup> March 2024, the total coal stock available with the plants increased to 50.69 MT (73% of the normative coal stock required), which was sufficient to run these plants for an average of about 18 days at a requirement of 85% PLF.

### 6.18.1 Monitoring Mechanism

The coal stock position of all the power plants in the country are being monitored by CEA on daily and monthly basis and accordingly daily and monthly reports are published on National Power Portal (NPP) ([www.npp.gov.in](http://www.npp.gov.in)).

CEA has revised the coal stocking norms w.e.f. 6th December 2021. As per the revised norms, daily coal requirement for both Pithead and Non-Pithead plants would be estimated @85% PLF and number of days for which stock needs to be maintained would vary from 12 to 17 days for Pithead plants and 20 to 26 days for Non-Pithead plants with month-wise variation based on coal despatch/ coal consumption pattern during the year.

| Quarter | Month | Pithead | Non-pithead |
|---------|-------|---------|-------------|
| Q1      | Apr   | 17      | 26          |
|         | May   | 17      | 26          |
|         | Jun   | 17      | 26          |
| Q2      | Jul   | 14      | 22          |
|         | Aug   | 13      | 21          |
|         | Sep   | 12      | 20          |
| Q3      | Oct   | 13      | 21          |
|         | Nov   | 14      | 22          |
|         | Dec   | 15      | 23          |
| Q4      | Jan   | 16      | 24          |
|         | Feb   | 17      | 26          |
|         | Mar   | 17      | 26          |

The Gencos would be graded in three zones. Grading of a Genco/Independent Power Producers (IPP) into Red, Yellow and Green zone will be done on monthly basis based on the average coal stock maintained by the plant during previous month and its outstanding dues with coal companies.

#### Monitoring by Inter-Ministerial Committee

A Secretary level Inter Ministerial committee has been set up to ensure that the medium and long term requirements of coal are met. The IMC comprises of Chairman Railway Board, Secretary, Ministry of Coal, Secretary Ministry of Environment Forests and Climate Change as members and Secretary Ministry of Power as convener.

#### Monitoring by Subgroup:

CEA is a member of an Inter-ministerial subgroup constituted by the Infrastructure Constraints Review Committee under the Chairmanship of Joint Secretary, Ministry of Coal comprising of representatives from Ministry of Railways, Ministry of Power, Shipping, NITI Aayog, CEA, CIL and NTPC Limited. The subgroup reviews and monitors coal supply and related infrastructural constraints on day-to-day basis for adequate supply of coal to power plants.

### 6.18.2 Coal Scenario for the Power Sector during 2023-24

#### 6.18.2.1 Estimation of coal requirement for the year 2023-24

During 2023-24, based on generation target, the

coal requirement was estimated to be about 845 MT. The break-up of coal requirement during 2023-24 is given as under:

| Coal Requirement                 | 2023-24      |
|----------------------------------|--------------|
| Domestic Coal based (DCB) Plants | 821.1        |
| Imported Coal based (ICB) Plants | 23.6         |
| Total Requirement                | <b>844.7</b> |

#### 6.18.2.2 Coal Supply Position for the year 2023-24:

For the year **2023-24**, the receipt of coal was 864.3 MT against the consumption of 849.7 MT. The details of coal receipt and consumption for year **2023-24** are given as under:

|                                                       |                              |
|-------------------------------------------------------|------------------------------|
| <b>A. Estimated Requirement (Domestic + Imported)</b> | <b>844.7</b><br>(821.1+23.6) |
| <b>B. Receipt</b>                                     |                              |
| a. Domestic coal                                      | 798.6                        |
| b. Imported coal for blending                         | 23.9                         |
| c. Imported coal for ICB plants                       | 41.8                         |
| d. Total Import                                       | 65.7                         |
| <b>Total Receipt (a+d)</b>                            | <b>864.3</b>                 |
| <b>C. Consumption</b>                                 |                              |
| a. DCB Plants                                         | 807.8                        |
| b. ICB Plants                                         | 41.9                         |
| <b>c. Total</b>                                       | <b>849.7</b>                 |

During the year **2023-24**, the receipt of domestic coal by the power plants was 798.6 MT as against 731.7 MT during 2022-23 resulting in increase of about 66.9 MT (9.1%). The total coal consumption during **2023-24** was 849.7 MT as against 776.8 MT during last year, thus increasing 72.9 MT (9.4 %). Plant- wise details of coal receipt and coal consumption during **2023-24** is enclosed at **Annexure-6B**.

#### 6.18.2.3 Source-wise Receipt of coal during 2023-24

During the year 2023-24, source-wise break-up of coal receipt at the power stations is given below:

| Source | Actual Receipt (MT) |
|--------|---------------------|
| CIL    | 593.0               |
| SCCL   | 63.2                |

|                                             |              |
|---------------------------------------------|--------------|
| Captive Mines                               | 106.8        |
| E-Auction                                   | 35.6         |
| <b>Total Domestic Coal Receipt</b>          | <b>798.6</b> |
| Import Coal for Blending                    | 23.9         |
| Imported Coal by Imported Coal Based Plants | 41.8         |
| <b>Total Imported Coal Receipt</b>          | <b>65.7</b>  |
| <b>Total Receipt</b>                        | <b>864.3</b> |

#### 6.18.2.4 Import of coal during the year 2022-23.

1. Power plants designed on domestic coal, import coal in view of their cost economics and power plants designed on imported coal import coal to meet its fuel requirement.
2. Considering the anticipated shortfall in availability of domestic coal during H1 of 2023-24, Ministry of Power vide letter no. FU-21/2020-FSC(Vol-IV) dated 09.01.2023 directed all Gencos to import coal for blending @6% (by weight) up to September, 2023 so as to have adequate coal stocks at power plants end for smooth operation till September, 2023. Further, due to unprecedented increase in energy demand, supply of coal not being commensurate with the coal requirement and depletion of coal stock at domestic coal based plants by about 12 Million Tonnes (MT) during 2<sup>nd</sup> quarter (July-September, 2023), Ministry of Power vide letter no. FU- 21/2020-FSC(Vol-IV) dated 25.10.2023 directed all Gencos to continue importing coal for blending @6% (by weight) till March, 2024.
3. During 2023-24, the coal imported by domestic coal based plants was 23.9 MT and that by imported coal based plants was 41.8 MT. Thus, the total coal imported by the power plants was about 65.7 MT as against 55.6 MT during 2022-23, thus increasing by about 10.1 MT (18%).

#### 6.18.2.5 Generation Loss

During the year, 2023-24, there was no loss of generation due to shortage of coal.

#### 6.18.2.6 Specific Coal Consumption

During the year 2023-24, the Specific Coal Consumption (kg/kWh) of the Domestic Coal Based Plants was 0.686 kg/kWh as compared to 0.685 kg/kWh in 2022-23. However, for imported coal based plants, it was 0.500 kg/kWh as against 0.499 kg/kWh in 2022-23.

#### 6.18.3 Coal Quality Issues

In order to address quality concerns of the coal supplied to power plants, it was decided in the meeting dated 28.10.2015 that coal samples shall be collected and prepared by a Single Third Party Agency appointed by power utilities and coal companies. Accordingly, it was decided by the Ministry of Power and the Ministry of Coal that the power utilities would appoint a Third Party Sampler (CIMFR) for Third Party Sampling and Analysis of coal at loading-end as well as at unloading-end. Based on the Third Party Sampling analysis results furnished by CIMFR, credit/debit note are being issued by coal companies to the power plants in case of difference between declared grade of coal and analyzed grade of coal.

Subsequently, Ministry of Power vide letter dated 30.03.2021 conveyed the decision that Power Finance Corporation (PFC) shall empanel Third Party Sampling (TPS) Agencies for Power Sector, in addition to CIMFR, and consumers shall be free to take services of any of the empaneled agencies.

The Terms of Reference for empanelment for the agency was to be formulated with the following broad guidelines:

- a. Multiple Agencies should be available.
- b. Sampling only at loading end with appellate/referee provision.
- c. Choice of taking services from empaneled agencies shall be of the buyer of coal.
- d. Review mechanism to review the working of the system.

MoP, after consultation with the stakeholders including CEA, finalized the Terms of Reference for empanelment of TPS which was forwarded to PFC on 17.08.2021 for taking further necessary action.

In the 1st round of empanelment PFC had

empaneled one firm (M/s Mitra SK Private Limited) as a 'Third Party sampling Agency (TPSA) for collection, preparation and analysis of coal samples at loading end with appellate/referee provision for power sector.

Subsequently, Ministry of Power vide letter dated 20.09.2022, while amending the terms of reference dated 17.08.2021, requested PFC/CEA to finalize the Terms of Reference for second round of empanelment of Third Party Sampling (TPS) agencies for power sector and complete the empanelment process at the earliest.

In the 2nd round of empanelment, PFC vide letter dated 04.10.2023, empaneled 10 agencies (Inspectorate Griffith India Pvt Ltd, Quality Services and Solutions Pvt Ltd, Quality Austria Central Asia Pvt Ltd, Cotecna Inspection India Pvt Ltd, KCS Quality Inspection Pvt Ltd, Ravi Energie Pvt Ltd, Mitra S. K. Pvt Ltd, Superintendence Company of India Pvt Ltd, Therapeutics Chemical Research Corporation and Dr. Amin Controllers Pvt. Ltd.) as 'Third Party Sampling Agency (TPSA) for collection, preparation and analysis of coal samples at loading end with appellate / referee provision for power and non-power sector' for a period of five years from 3<sup>rd</sup> October, 2023.

#### **6.18.4 New initiatives for addressing issues related to coal supply to Power Plants**

##### **A. Flexibility in Utilization of Domestic Coal**

- a. The Government, on 04.05.2016, approved the proposal for allowing flexibility in utilization of domestic coal amongst power generating stations to reduce the cost of power generation. Under the scheme, the Annual Contracted Quantity (ACQ) of each individual coal linkage as per Fuel Supply Agreement is to be aggregated as consolidated ACQ for each State and Company owning Central Generating Stations instead of individual generating station. The State/Central Gencos have flexibility to utilize their coal in most efficient and cost effective manner in their own power plants as well as by transferring coal to other State/Central Gencos Power plants for generation of cheaper power. The methodology provides for utilizing coal amongst

State/Central Generating Stations having 4 cases- i) within state ii) one state to another state iii) one state to CGSs & vice versa and iv) within CGSs & other CGSs. The methodology in this regard has been issued by CEA on 08.06.2016.

- b. Further, the methodology for use of coal transferred by a State to Independent Power Producer (IPP) generating stations has been issued by Ministry of Power, Govt. of India on 20.02.2017. As per the methodology, the State can divert their coal and take equivalent power from IPP generating station, which is selected through an e- bidding process. The guiding principle of the methodology is that the landed cost of power from IPP generating station at the State's periphery should be lower than the variable cost of generation of the State generating station whose power is to be replaced by generation from IPP. The landed cost of power is inclusive of the transmission charges and transmission losses.
- c. Based on the experience gained, Ministry of power vide letter dated 15.06.2018 has amended clauses related to bid security, performance security coal transportation mode in the methodology for Case-4. Subsequently, Ministry of Power vide letter dated 25.10.2018 has issued 2nd amendment in the methodology allowing moisture correction while reconciliation of coal.

##### **Status of the Scheme:**

- a. All State/Central gencos have signed supplementary agreement with Coal Companies for aggregation of their ACQ. CIL, on quarterly basis, allocates coal to the plants of State /Central Gencos as per their requirement within their AACQ.
- b. Based on the methodology issued by MoP on 20.02.2017 for Case-4, Gujarat Urja Vikas Nigam Limited (GUVNL) and Maharashtra State Power Generation Company Limited (MSPGCL) invited bids for supply of power from willing IPPs.
- c. GMR Chhattisgarh Energy Limited (GCEL) emerged as successful bidder in case of bid invited by GUVNL and was awarded contract to take equivalent power of 500 MW at a tariff of Rs 2.81 per unit for a period

of 8 months starting from November 2017 to June 2018. However, power supply started from January 2018. The contract was later extended by GUVNL till November, 2018. Gujarat again invited bids and awarded contract to GCEL for supply of 1000 MW at a tariff of Rs. 3.16 per unit. The Power purchase agreement (PPA) was signed on 21.12.2018 and the contract period was upto June, 2019. However, the supply of power started from January, 2019 and the contract was extended till December, 2019.

- d. Maharashtra tied up 400 MW (185 MW) with Dhariwal Infrastructure Ltd. and 215 MW with Ideal Energy Projects Ltd. for a period of 8 months at a tariff of Rs. 2.76 per unit. The supply of power started by Dhariwal Infrastructure Ltd. from April 2018 and by Bela TPS from May 2018. Maharashtra again tied up 185 MW with Dhariwal Infrastructure Ltd. from November 2019 to October 2020.

#### B. Construction of Belt-MGR

Construction of Belt /MGR in the power plants, which are located within 100 kms of linked coal mines, is being monitored in CEA. 51 power plants have been identified to be having coal source within 100 km. The status of these 51 power plants are as under:

- a. 18 plants are having functional Belt / MGR system for receiving coal from linked mines.
- b. 4 plants are constructing Belt / Piped conveyor for receiving coal through linked mines.
- c. 13 plants are being under deliberations for construction of Belt / MGR system
- d. 16 plants are identified under “Not feasible” category.

#### 6.18.5 Gas Supply Position

1. CEA monitors 58 Nos. of gas based power stations with a total installed capacity of about 23,856 MW (As on 31<sup>st</sup> March 2024) using gas as primary fuel. The production

and supply of Gas have not been keeping pace with the growing demand of gas in the country including in power sector. Even gas allocations committed for power stations are not fulfilled due to shortage of gas in the country.

2. The domestic gas supply during 2023-24 was only about 11.9 MMSMMD against allocation of 84.2 MMSCMD. The domestic supply during 2023-24 reduced by about 10% as compared to that during 2022-23 (13.1 MMSCMD). However, the import of LNG during 2023-24 has increased to about 7.8 MMSCMD as against 2.8 MMSCMD during 2022-23.
3. The PLF achieved during 2023-24 was 14.9% against PLF of 11.5% during previous year. The increase in PLF may be attributed to higher import of LNG, mainly due to price of gas in international markets. Plant-wise details of gas allocated and supplied/consumed during 2023-24 is enclosed at **Annexure-6C**.

The gas supply position in gas based power plants during 2023-24 is as under:

|                     |                           | Allocation       | Supplied     |
|---------------------|---------------------------|------------------|--------------|
| Domestic Gas        | APM / Non - APM/ PMT      | 52.41            | 11.56        |
|                     | KGD-6                     | 31.82            | 0.32*        |
|                     | <b>Total Domestic Gas</b> | <b>84.23</b>     | <b>11.87</b> |
| Imported Gas (RLNG) | Long Term Contract        | 7.48             | 3.46         |
|                     | Spot                      | -                | 4.32         |
|                     | <b>Total Imported Gas</b> | <b>7.48</b>      | <b>7.78</b>  |
|                     |                           |                  |              |
|                     |                           | <b>Total Gas</b> | <b>91.71</b> |
|                     |                           |                  | <b>19.65</b> |

(Fig. in Million Metric Standard Cubic Meter per day (MMSCMD))

(\*: Auction gas from KGD)

## CHAPTER-7

### DISTRIBUTION SCHEMES AND INITIATIVES

#### 7.1 Initiatives in Distribution Sector

Initiatives taken by Government of India in Distribution Sector are as under:

**Village Electrification:** As per the present status of all the states, all the inhabited un-electrified census (2011) villages including villages in remote/interior areas stood electrified as on 28th April, 2018 under Deen Dayal Upadhyaya Gram Jyoti Yojana (DDUGJY).

**Household Electrification:** Government of India launched Pradhan Mantri Sahaj Bijli Har Ghar Yojana–Saubhagya on 11th October, 2017 to achieve universal household electrification (providing electricity connections to all household in rural and all poor households in urban areas across the country) by March 2019. Under the scheme, as reported by the States and UTs as on 31.03.2021, a total of 2.817 crore all the willing and un-electrified households (HHs) have been electrified since the launch of Saubhagya, thereby, 100% electrification of all the willing un-electrified households have been achieved.

Since HHs electrification is dynamic process, further, under DDUGJY (New) 4, 40,893 nos. of additional HHs have been electrified as on 15.03.2022. Accordingly, a total of 2.86 crore households have been electrified under these schemes of Govt of India.

**Distribution network:** The role of distribution Network is very significant in respect of achieving the goal of 24x7 Power for All (PFA) as it is the back bone for providing connectivity to the end user. The main scope for distribution network includes implementation of distribution network requirement (new, strengthening). This would

also include augmentation of the existing network to cater to un-electrified areas. To address some of the challenges faced like theft of power, and quality and reliable power in distribution, an effort has been made by the utilities in strengthening and augmentation of system, feeder separation, metering of unmetered connections, smart meter installation, installation of capacitor banks to improve the voltage profile and reduce line losses, smart metering, DT and Feeder metering etc. under the sanctioned components of DDUGJY, Integrated Power Development Scheme (IPDS) and Saubhagya schemes.

#### 7.2 Revamped Distribution Sector Scheme (RDSS)

Central Government launched Revamped Distribution Sector Scheme (RDSS) in July, 2021 with the objective of improving the quality and reliability of power supply to consumers through a financially sustainable and operationally efficient distribution Sector. The Scheme aims to reduce the AT&C losses to pan-India levels of 12-15% and Average Cost of Supply (ACS) - Average Revenue Realized (ARR) gap to zero by 2024-25. The Scheme has an outlay of Rs.3,03,758 crore and an estimated Gross Budgetary support of Rs 97,631 Crores from Govt. of India. The duration of the scheme is 5 Years (2021-22 to 2025-26).

The Scheme has two parts: Part 'A' - Financial support for upgradation of the Distribution Infrastructure and Prepaid Smart Metering & System Metering and Part 'B' - Training & Capacity Building and other Enabling & Supporting Activities.

Under the scheme, eligible DISCOMs (all State-owned Distribution companies and State /UT Power Departments excluding private Sector power companies) are being provided financial support for upgradation of the Distribution Infrastructure, Distribution Automation, IT intervention, implementation of SCADA/DMS & installation of 25 Crore Pre-paid smart meters for consumers and Smart Metering at Feeder and Distribution Transformers etc.

The scheme envisages installation of 25 crore prepaid smart meters for all consumers along with associated AMI & communicable meters for DTs & Feeders. Prepaid Smart Meters including System metering with communication features are important interventions in reducing Distribution losses in the Utilities and in facilitating automatic measurement of energy flows and energy accounting as well as auditing without any human intervention. This intervention will also facilitate switch over to digital pre-paid system, with recharging facility through mobile phones and enabling of Time-of-Day tariff.

Advanced ICT like Artificial Intelligence & Machine Learning Technologies is also being leveraged to analyse data generated through IT/OT devices including System Meters, prepaid Smart meters to prepare actionable MIS from system generated energy accounting reports every month so as to enable the DISCOMs to take

informed decisions on loss reduction, demand forecasting, asset management, Time of Day (ToD) tariff, Renewable Energy (RE) Integration and for other predictive analysis. This would contribute a great deal towards enhancing operational efficiency and financial sustainability of the DISCOMs.

An inter-ministerial Monitoring Committee for the Scheme has been constituted under the chairmanship of Secretary, Ministry of Power. The Monitoring Committee frames and approves all operational guidelines, sanction all Action Plans & DPRs of DISCOMs / States, review and monitor implementation of Scheme, approves scope of works and take necessary decisions for operationalization of various components of the Scheme and amendments thereof, within the framework approved by Cabinet Committee on Economic Affairs (CCEA). The Monitoring Committee is also competent to modify the scope of works under various parts of the Scheme in line with the objectives of the Scheme. Chairperson, CEA is the member of the Monitoring Committee.

The funds for a particular year in respect of Infrastructure Works are released in respect of a DISCOM for a particular year only after it is found to have fulfilled the pre qualifying criteria and its total weighted score is at least 60 marks on the result evaluation matrix after having been evaluated by the Nodal Agency and approved as such by the Monitoring Committee.

#### Sanctioned Status of RDSS as on 31<sup>st</sup> July, 2024

| State/ DISCOMs                       | Loss Reduction (LR) works outlay (Rs. Cr) | Project Management Agency (PMA) Amount for Infra works (Rs. Cr) | LR works total sanctioned Cost including PMA (Rs. Cr.) | Total Smart metering outlay in (Rs. Cr.) | PMA amount for metering (Rs. Cr.) | Total Cost of Smart metering including PMA (Rs. Cr) |
|--------------------------------------|-------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------|------------------------------------------|-----------------------------------|-----------------------------------------------------|
| <b>Andaman &amp; Nicobar Islands</b> | <b>455.18</b>                             | <b>6.83</b>                                                     | <b>462.01</b>                                          | <b>53.26</b>                             | <b>0.30</b>                       | <b>53.56</b>                                        |
| Andaman & Nicobar PD                 | 455.18                                    | 6.83                                                            | 462.01                                                 | 53.26                                    | 0.30                              | 53.56                                               |
| <b>Andhra Pradesh</b>                | <b>9,234.94</b>                           | <b>138.52</b>                                                   | <b>9,373.47</b>                                        | <b>4,112.42</b>                          | <b>15.42</b>                      | <b>4,127.85</b>                                     |
| APCPDCL                              | 1,483.80                                  | 22.26                                                           | 1,506.05                                               | 1,507.62                                 | 5.65                              | 1,513.27                                            |
| APEPDCL                              | 2,665.29                                  | 39.98                                                           | 2,705.27                                               | 947.15                                   | 3.55                              | 950.70                                              |
| APSPDCL                              | 5,085.86                                  | 76.29                                                           | 5,162.14                                               | 1,657.66                                 | 6.22                              | 1,663.87                                            |
| <b>Arunachal Pradesh</b>             | <b>929.19</b>                             | <b>13.94</b>                                                    | <b>943.13</b>                                          | <b>182.53</b>                            | <b>1.03</b>                       | <b>183.56</b>                                       |
| Arunachal PD                         | 929.19                                    | 13.94                                                           | 943.13                                                 | 182.53                                   | 1.03                              | 183.56                                              |

|                            |                  |               |                  |                  |              |                  |
|----------------------------|------------------|---------------|------------------|------------------|--------------|------------------|
| <b>Assam</b>               | <b>2,570.54</b>  | <b>38.56</b>  | <b>2,609.10</b>  | <b>4,029.08</b>  | <b>20.46</b> | <b>4,049.54</b>  |
| APDCL                      | 2,570.54         | 38.56         | 2,609.10         | 4,029.08         | 20.46        | 4,049.54         |
| <b>Bihar</b>               | <b>6,976.41</b>  | <b>104.65</b> | <b>7,081.06</b>  | <b>2,013.66</b>  | <b>7.55</b>  | <b>2,021.21</b>  |
| NBPDC                      | 3,250.89         | 48.76         | 3,299.65         | 964.46           | 3.62         | 968.07           |
| SBPDCL                     | 3,725.52         | 55.88         | 3,781.40         | 1,049.21         | 3.93         | 1,053.14         |
| <b>Chattisgarh</b>         | <b>3,577.43</b>  | <b>53.66</b>  | <b>3,631.09</b>  | <b>4,089.97</b>  | <b>15.34</b> | <b>4,105.31</b>  |
| CSPDCL                     | 3,577.43         | 53.66         | 3,631.09         | 4,089.97         | 15.34        | 4,105.31         |
| <b>Delhi</b>               | <b>318.85</b>    | <b>4.78</b>   | <b>323.63</b>    | <b>13.33</b>     | <b>0.05</b>  | <b>13.38</b>     |
| NDMC                       | 318.85           | 4.78          | 323.63           | 13.33            | 0.05         | 13.38            |
| <b>Goa</b>                 | <b>243.43</b>    | <b>3.65</b>   | <b>247.08</b>    | <b>467.42</b>    | <b>1.75</b>  | <b>469.17</b>    |
| Goa PD                     | 243.43           | 3.65          | 247.08           | 467.42           | 1.75         | 469.17           |
| <b>Gujarat</b>             | <b>5,932.49</b>  | <b>88.99</b>  | <b>6,021.48</b>  | <b>10,602.20</b> | <b>39.76</b> | <b>10,641.96</b> |
| DGVCL                      | 1,484.43         | 22.27         | 1,506.70         | 2,636.24         | 9.89         | 2,646.13         |
| MGVCL                      | 1,636.56         | 24.55         | 1,661.11         | 2,109.10         | 7.91         | 2,117.01         |
| PGVCL                      | 1,567.67         | 23.52         | 1,591.19         | 3,601.10         | 13.50        | 3,614.60         |
| UGVCL                      | 1,243.83         | 18.66         | 1,262.49         | 2,255.76         | 8.46         | 2,264.22         |
| <b>Haryana</b>             | <b>3,111.75</b>  | <b>46.68</b>  | <b>3,158.43</b>  | <b>0.00</b>      | <b>0.00</b>  | <b>0.00</b>      |
| DHBVNL                     | 1,584.21         | 23.76         | 1,607.97         | 0.00             | 0.00         | 0.00             |
| UHBVNL                     | 1,527.54         | 22.91         | 1,550.45         | 0.00             | 0.00         | 0.00             |
| <b>Himachal Pradesh</b>    | <b>2,246.74</b>  | <b>33.70</b>  | <b>2,280.44</b>  | <b>1,778.49</b>  | <b>10.00</b> | <b>1,788.49</b>  |
| HPSEBL                     | 2,246.74         | 33.70         | 2,280.44         | 1,778.49         | 10.00        | 1,788.49         |
| <b>Jammu &amp; Kashmir</b> | <b>4,567.06</b>  | <b>68.51</b>  | <b>4,635.57</b>  | <b>1,057.67</b>  | <b>5.95</b>  | <b>1,063.62</b>  |
| JPDCL                      | 2,251.52         | 33.77         | 2,285.29         | 546.47           | 3.07         | 549.54           |
| KPDCL                      | 2,315.54         | 34.73         | 2,350.27         | 511.20           | 2.88         | 514.07           |
| <b>Jharkhand</b>           | <b>3,274.03</b>  | <b>49.11</b>  | <b>3,323.14</b>  | <b>854.81</b>    | <b>3.21</b>  | <b>858.02</b>    |
| JBVNL                      | 3,274.03         | 49.11         | 3,323.14         | 854.81           | 3.21         | 858.02           |
| <b>Karnataka</b>           | <b>3.71</b>      | <b>0.06</b>   | <b>3.76</b>      | <b>0.00</b>      | <b>0.00</b>  | <b>0.00</b>      |
| MESCOM                     | 0.01             | 0.00          | 0.01             | 0.00             | 0.00         | 0.00             |
| CHESCOM                    | 3.70             | 0.06          | 3.76             | 0.00             | 0.00         | 0.00             |
| <b>Kerala</b>              | <b>2,312.98</b>  | <b>34.69</b>  | <b>2,347.67</b>  | <b>8,200.46</b>  | <b>30.75</b> | <b>8,231.21</b>  |
| KSEBL                      | 2,236.63         | 33.55         | 2,270.18         | 8,175.06         | 30.66        | 8,205.72         |
| TCED                       | 76.35            | 1.15          | 77.50            | 25.40            | 0.10         | 25.50            |
| <b>Ladakh</b>              | <b>862.84</b>    | <b>12.94</b>  | <b>875.78</b>    | <b>0.00</b>      | <b>0.00</b>  | <b>0.00</b>      |
| Laddakh                    | 862.84           | 12.94         | 875.78           | 0.00             | 0.00         | 0.00             |
| <b>Madhya Pradesh</b>      | <b>9,367.96</b>  | <b>140.52</b> | <b>9,508.48</b>  | <b>8,736.22</b>  | <b>32.76</b> | <b>8,768.98</b>  |
| MP Central                 | 3,319.98         | 49.80         | 3,369.78         | 2,776.77         | 10.41        | 2,787.18         |
| MP East                    | 3,559.84         | 53.40         | 3,613.24         | 3,387.33         | 12.70        | 3,400.03         |
| MP West                    | 2,488.13         | 37.32         | 2,525.45         | 2,572.12         | 9.65         | 2,581.76         |
| <b>Maharashtra</b>         | <b>14,148.42</b> | <b>36.11</b>  | <b>14,184.53</b> | <b>15,206.64</b> | <b>8.31</b>  | <b>15,214.95</b> |
| MSEDCL                     | 13,175.54        | 21.52         | 13,197.06        | 14,547.46        | 5.84         | 14,553.30        |
| BEST                       | 972.88           | 14.59         | 987.47           | 659.18           | 2.47         | 661.65           |
| <b>Manipur</b>             | <b>395.05</b>    | <b>5.93</b>   | <b>400.98</b>    | <b>120.48</b>    | <b>0.68</b>  | <b>121.16</b>    |
| MSPDCL                     | 395.05           | 5.93          | 400.98           | 120.48           | 0.68         | 121.16           |
| <b>Meghalaya</b>           | <b>1,213.98</b>  | <b>18.21</b>  | <b>1,232.19</b>  | <b>307.82</b>    | <b>1.73</b>  | <b>309.56</b>    |
| MePDCL                     | 1,213.98         | 18.21         | 1,232.19         | 307.82           | 1.73         | 309.56           |
| <b>Mizoram</b>             | <b>303.17</b>    | <b>4.55</b>   | <b>307.72</b>    | <b>180.59</b>    | <b>1.02</b>  | <b>181.61</b>    |
| Mizoram PD                 | 303.17           | 4.55          | 307.72           | 180.59           | 1.02         | 181.61           |
| <b>Nagaland</b>            | <b>449.54</b>    | <b>6.74</b>   | <b>456.28</b>    | <b>206.41</b>    | <b>1.16</b>  | <b>207.57</b>    |
| Nagaland PD                | 449.54           | 6.74          | 456.28           | 206.41           | 1.16         | 207.57           |
| <b>Puducherry</b>          | <b>83.14</b>     | <b>1.25</b>   | <b>84.39</b>     | <b>250.16</b>    | <b>0.94</b>  | <b>251.10</b>    |
| Puducherry PD              | 83.14            | 1.25          | 84.39            | 250.16           | 0.94         | 251.10           |

|                      |                    |                 |                    |                    |               |                    |
|----------------------|--------------------|-----------------|--------------------|--------------------|---------------|--------------------|
| <b>Punjab</b>        | <b>3,816.13</b>    | <b>57.24</b>    | <b>3,873.37</b>    | <b>5,746.95</b>    | <b>21.55</b>  | <b>5,768.50</b>    |
| PSPCL                | 3,816.13           | 57.24           | 3,873.37           | 5,746.95           | 21.55         | 5,768.50           |
| <b>Rajasthan</b>     | <b>9,279.44</b>    | <b>132.31</b>   | <b>9,411.75</b>    | <b>9,678.51</b>    | <b>36.29</b>  | <b>9,714.80</b>    |
| AVVNL                | 2,422.09           | 34.40           | 2,456.49           | 3,663.11           | 13.74         | 3,676.85           |
| JdVVNL               | 3,561.10           | 48.46           | 3,609.56           | 2,877.36           | 10.79         | 2,888.15           |
| JVVNL                | 3,296.26           | 49.44           | 3,345.70           | 3,138.04           | 11.77         | 3,149.81           |
| <b>Sikkim</b>        | <b>391.85</b>      | <b>5.88</b>     | <b>397.73</b>      | <b>96.90</b>       | <b>0.55</b>   | <b>97.45</b>       |
| Sikkim PD            | 391.85             | 5.88            | 397.73             | 96.90              | 0.55          | 97.45              |
| <b>Tamil Nadu</b>    | <b>9,426.32</b>    | <b>141.39</b>   | <b>9,567.71</b>    | <b>19,163.50</b>   | <b>71.86</b>  | <b>19,235.36</b>   |
| TANGEDCO             | 9,426.32           | 141.39          | 9,567.71           | 19,163.50          | 71.86         | 19,235.36          |
| <b>Telangana</b>     | <b>6.69</b>        | <b>0.10</b>     | <b>6.79</b>        | <b>0.00</b>        | <b>0.00</b>   | <b>0.00</b>        |
| TSSPDCL              | 0.70               | 0.01            | 0.71               | 0.00               | 0.00          | 0.00               |
| TSNPDCL              | 5.99               | 0.09            | 6.08               | 0.00               | 0.00          | 0.00               |
| <b>Tripura</b>       | <b>538.01</b>      | <b>8.07</b>     | <b>546.08</b>      | <b>316.77</b>      | <b>1.78</b>   | <b>318.55</b>      |
| TSECL                | 538.01             | 8.07            | 546.08             | 316.77             | 1.78          | 318.55             |
| <b>Uttar Pradesh</b> | <b>18,730.89</b>   | <b>280.96</b>   | <b>19,011.85</b>   | <b>18,885.47</b>   | <b>70.82</b>  | <b>18,956.29</b>   |
| DVVNL                | 3,937.24           | 59.06           | 3,996.30           | 3,663.09           | 13.74         | 3,676.82           |
| KESCO                | 614.93             | 9.22            | 624.16             | 328.40             | 1.23          | 329.63             |
| MVVNL                | 4,546.49           | 68.20           | 4,614.69           | 5,009.35           | 18.79         | 5,028.14           |
| PaVVNL               | 3,404.09           | 51.06           | 3,455.15           | 4,946.90           | 18.55         | 4,965.45           |
| PuVVNL*              | 6,228.13           | 93.42           | 6,321.56           | 4,937.73           | 18.52         | 4,956.25           |
| <b>Uttarakhand</b>   | <b>1,671.37</b>    | <b>25.07</b>    | <b>1,696.44</b>    | <b>1,045.04</b>    | <b>5.88</b>   | <b>1,050.92</b>    |
| UPCL                 | 1,671.37           | 25.07           | 1,696.44           | 1,045.04           | 5.88          | 1,050.92           |
| <b>West Bengal</b>   | <b>7,115.83</b>    | <b>106.74</b>   | <b>7,222.57</b>    | <b>12,623.11</b>   | <b>47.34</b>  | <b>12,670.45</b>   |
| WBSEDCL              | 7,115.83           | 106.74          | 7,222.57           | 12,623.11          | 47.34         | 12,670.45          |
| <b>Grand Total</b>   | <b>1,23,555.37</b> | <b>1,670.33</b> | <b>1,25,225.69</b> | <b>1,30,019.87</b> | <b>454.23</b> | <b>1,30,474.10</b> |

#### Rural Electrification works under RDSS:

Further, Central Government in line with its commitment, is supporting States under the ongoing Scheme of Revamped Distribution Sector Scheme (RDSS) for electrification of left-out households in the country. MoP requested all the States/UTs to identify the balance un-electrified villages/habitations for funding under RDSS and issued the norms for electrification of left out households. As of 31<sup>st</sup> July, 2024, electrification works amounting to Rs 2,937 Cr have been sanctioned for electrification of

6,84,112 left out households for the States of Rajasthan, Uttar Pradesh, Andhra Pradesh, Meghalaya, Mizoram, Nagaland, Jharkhand, Jammu & Kashmir, Bihar and Assam. In addition, all identified PVTG (Particularly Vulnerable Tribal Groups) Households under PM-JANMAN for on-grid electricity connection are also being sanctioned for funding under RDSS as per the Scheme guidelines. Till 31<sup>st</sup> July, 2024 works amounting to Rs 488 Cr have been sanctioned for 1,23,530 PVTG households.

### 7.3 Status of installation of Smart Consumer Meters in the Country

The state-wise status of installation of smart consumer meters in the country as on 31<sup>st</sup> July, 2024 is as given below:

| S. No. | State               | Total Sanctioned    | Achievement        |
|--------|---------------------|---------------------|--------------------|
| 1      | Andhra Pradesh      | 56,10,846           | 148099             |
| 2      | Uttarakhand         | 15,84,205           | 0                  |
| 3      | Gujarat             | 1,64,87,100         | 56161              |
| 4      | Himachal Pradesh    | 29,52,685           | 151740             |
| 5      | Jharkhand           | 13,41,306           | 0                  |
| 6      | Kerala              | 1,32,90,166         | 805                |
| 7      | Madhya Pradesh      | 1,34,44,401         | 1047267            |
| 8      | Maharashtra         | 2,35,64,747         | 267746             |
| 9      | Puducherry          | 4,03,767            | 0                  |
| 10     | Punjab              | 98,30,007           | 802826             |
| 11     | Andaman and Nicobar | 1,58,773            | 75200              |
| 12     | Arunachal Pradesh   | 2,87,446            | 0                  |
| 13     | Assam               | 69,21,329           | 2291945            |
| 14     | Bihar               | 1,72,08,939         | 3966431            |
| 15     | Chhattisgarh        | 59,62,115           | 5578               |
| 16     | Goa                 | 7,41,160            | 0                  |
| 17     | Jammu and Kashmir   | 21,34,095           | 579540             |
| 18     | Manipur             | 1,54,400            | 0                  |
| 19     | Meghalaya           | 4,60,000            | 0                  |
| 20     | Mizoram             | 2,90,039            | 656                |
| 21     | Nagaland            | 3,17,210            | 0                  |
| 22     | Rajasthan           | 1,49,00,527         | 614307             |
| 23     | Sikkim              | 1,44,680            | 0                  |
| 24     | Tamil Nadu          | 3,01,40,849         | 129641             |
| 25     | Tripura             | 5,47,489            | 0                  |
| 26     | Uttar Pradesh       | 3,09,78,280         | 1186953            |
| 27     | West Bengal         | 2,12,08,759         | 99165              |
| 28     | Chandigarh          | 29,433              | 24214              |
| 29     | Delhi               | 2,60,000            | 260000             |
| 30     | Gujarat             | 23,760              | 23760              |
| 31     | Haryana             | 10,00,000           | 847467             |
| 32     | Ladakh              | 58,930              | 42400              |
| 33     | Odisha              | 4,500               | 4500               |
| 34     | Telangana           | 8,882               | 8882               |
|        | <b>Total</b>        | <b>22,24,50,825</b> | <b>1,26,35,283</b> |

(Source: NSGM website)

## 7.4 Development of Smart Distribution in High Growth Cities

Ministry of Power is targeting the development of distribution infrastructure and smart technologies in about 60 high load growth cities during next 5 years. With the aim to improve the reliability of power supply in high growth big cities in the country, a team consisting of officers from CEA, NSGM, PFC and REC has been constituted by Ministry of

Power to assist the states for the preparation of DPRs for system augmentation and smart distribution system in the big high growth cities. Initially, cities namely Varanasi, Patna, Indore, NDMC, Guwahati, Gurugram, Noida & Kanpur have been identified for development of Smart Distribution along with availability of adequate distribution infrastructure with N-1 redundancy in the system for improving the reliability of power supply in these cities. The DPR of Varanasi city for Smart distribution projects has been

approved by the Monitoring Committee at an estimated cost of Rs 1313 Cr under RDSS. A team of CEA, PFC, REC and NSGM is handholding the concerned DISCOMs for preparing the DPRs of the other cities.

## 7.5 Examination/Technical Clearances of DPRs/PPRs:

### Technical Clearances of DPRs/PPRs:

- Commissioning of New Sub-station Control room equipment's and associated works at 66/11 kV Dabhel Sub-station, Daman, UT of DNHDD.
- Re-conductoring of 3 numbers 33 KV lines of DVC with HTLS conductor.
- Preliminary Project Report (PPR ID-12135) of MSEDCL regarding Solar Photovoltaic Water Pumping Systems and Agricultural Feeder Solarization Project/Conventional Pump.
- DPR for distribution infrastructure related works under Uttarakhand Transmission Strengthening and Distribution Improvement Program (UTSDIP) funded by ADB.

### Examination of DPRs/PPRs:

- PPR for Technical Assistance from ADB to strengthen the commercial and operational capacity of the DISCOMs for improving Energy Sector of Uttar Pradesh
- PPR for Technical Assistance for preparing comprehensive plan for Nagaland Power Sector
- PPR (PPR ID 11452) for funding by ABD reg Tamil Nadu Power Distribution Network Improvement Program
- Power Project proposal received from MEA reg Ministry of Power, Government of Myanmar
- Project proposals received from M/O DONER under NESIDS during 2023- 24 for the States of Mizoram, Nagaland and Sikkim.

## 7.6 Preparation of Distribution Perspective Plan (DPP) 2035

As per the Section 3 of the Electricity Act, 2003, the preparation of Distribution Perspective Plan-2030 was taken up to project the State-wise & All India level requirement of distribution infrastructure for the period 2022-

2030. The Draft Distribution Perspective Plan 2030 was prepared based on the data received from DISCOMs and the same was published on CEA website for stakeholders' comments in February 2024. The Draft Plan was reviewed by Hon'ble Cabinet Minister of Power & NRE in February, 2024 and as per the directions, the revision of Distribution Perspective Plan is taken upto 2035. The DPP-2035 would include projected requirement of DISCOMs regarding Power Sub-stations (66 KV, 33KV, 22KV), Feeders (66 kV, 33 kV, 22 kV and 11 kV), DTs (33/0.4 kV, 22/0.4 kV, 11/0.4 kV), LT Lines (230V & 400V), Capacitors, SCADA/RT-DAS, Consumers and Consumer metering, AT&C losses and total investment required by DISCOMs vis-à-vis total investment available with the DISCOMs etc. for the period 2024-2035. Presently, the DPP-2035 is under finalisation.

## 7.7 Release of Electricity Distribution Network Planning Criteria Document

To evolve an integrated approach for planning and strengthening of distribution system in the country, a need was felt time and again to provide a uniform Guidelines for planning of distribution system to the distribution utilities. CEA, under the guidance of MoP prepared "Electricity Distribution Network Planning Criteria" for helping the DISCOMs to plan their distribution system optimally and efficiently. This planning document was finalized in consultation with stakeholders including DISCOMs. The document was released by Hon'ble Minister of Power & NRE in January, 2024. The document was circulated to all the stakeholders and also available on CEA website for their use.

## 7.8 Cyber Security in Distribution Sector - CERT-Distribution

With the rapid implementation of IT enabled support and services in electricity distribution sector, the power sector is becoming more & more prone to various types of cyber-attacks and information security issues. In view of this, Ministry of Power constituted CERT-Distribution (CERT-D) under Chief Engineer (DP&T), CEA. CERT-D is working to improve the cyber security posture of power distribution Sector by coordinating with the DISCOMs, NCIIPC, MoP, CISO-MoP and CERT-In. The following actions

were taken by CERT-D during 2023-24: -

- CERT-D disseminated information and advisories to all the DISCOMs on cyber security issues received from NCIIPC, CERT-In & CISO-MoP and action taken report was sought from the DISCOMs.
- All Major DISCOMs have nominated their Chief Information Security Officer (CISOs) and updating their details regularly, in case of any change.
- All Major DISCOMs (except one) have on boarded Cyber Swachhta Kendra (Botnet Cleaning and Malware Analysis Centre) being operated by CERT-In. Vulnerability of DISCOMs reported in every fortnightly Power Sector Situational Report of CSK, is taken up by CERT-D with the concerned DISCOMs for closer/necessary action and closer reports are submitted to MoP/CERT-In.
- DISCOMs were advised regularly to prepare their Cyber Crisis Management Plan (CCMP). 59 DISCOMs have prepared their CCMP, out of which, CCMP of 31 DISCOMs have been approved by CERT-In. CCMPs of other DISCOMs are under various stages of preparation and approval.
- Template & guidelines issued by NCIIPC for identifying the Critical Information

Infrastructure (CII) in Distribution sector was circulated to all DISCOMs. Further, documents from NCIIPC regarding identification of CII of various DISCOMs were examined and comments were furnished. CII of major cities like Delhi, Mumbai, Kolkata, Bangalore, Trivandrum etc. have been declared and CII of other cities/DISCOMs are under examination of NCIIPC.

- DISCOMs were advised regularly to take necessary actions to ensure cyber security like quarterly review of their Cyber Security Measures, to conduct regular security audits through CERT-IN empaneled agencies, implementation of ISO 27001, conduction of mock drill etc.
- An Empowered Committee under Secretary, MoP and Standing Committee under Additional Secretary, MoP have been constituted to monitor the cyber security measures taken by Power Utilities. CERT-D is regularly participating and providing necessary inputs in the meeting of Empowered Committee & Standing Committee.

The status of DISCOMs under purview of CERT-D for various cyber security activities as on 31.03.2024 is as given below:

| S. No. | Activity                                  | Status |
|--------|-------------------------------------------|--------|
| 1      | Appointment of CSIO                       | 81/81  |
| 2      | Preparation of CCMP                       | 59/81  |
| 3      | Identification of CII                     | 64/81  |
| 4      | Implementation of ISO 27001               | 42/81  |
| 5      | Onboarding on Cyber Swachhta Kendra (CSK) | 80/81  |
| 6      | Conduction of Cyber Security Audit        | 67/81  |
| 7      | Conduction of Mock Drills                 | 35/81  |

### 7.9 Guidelines for Model Quality Assurance Plan (MQAP) for Distribution sector equipment including Type Test validity

Central Electricity Authority (CEA) issued Guidelines for Type Tests and Model Quality Assurance Plan (MQAP) for major equipment of Power sector, including hydro, thermal, transmission & Distribution sector, in March, 2022. Further, as advised by MOP, the Model Quality Assurance Plan (MQAP) including type test validity for Distribution sector was revised by delete the pre-dispatch inspection at factory site except for Power Transformers. The revised MQAP Guidelines for Distribution Sector were

issued in May, 2023 after due approval by MoP.

### 7.10 Report on Benchmarking of Operation & Maintenance (O&M) Norms for Distribution Utilities

To help the distribution utilities to adopt the best practices in operation and maintenance of distribution infrastructure to provide 24x7 reliable & quality power to the consumers, a report on the benchmarking of O&M norms for DISCOMs is under preparation covering both the financial as well as technical benchmarking of O&M. The best practices for O&M of 33 KV, 11 KV, LT infrastructure like sub-stations, DTs, lines etc including asset management, inventory management, roles and responsibilities of field

O&M staff, have been covered in the document. The Report also covered the analysis of the present practices being followed by SERCs for calculation of O&M cost of DISCOMs and to recommend a suitable model, which may be adopted by SERCs for calculation of O&M cost of DISCOMs. The Report is under finalization.

### **7.11 Guidelines for standardization /optimization of Manpower norms in DISCOMs**

To rejuvenate the Power Sector, Standardizing/optimizing the Manpower/ Workforce Norms in DISCOMs is one of the key action points and a need was felt to decide the normative Manpower in the DISCOMs. In this context, development of Guidelines for standardization/ optimization of manpower norms was taken up in consultation with various stakeholders. The recommendations include, inter-alia, total manpower required in a DISCOM, indicative organizational layering of manpower at different levels, Training and its budget, Workforce Mapping across verticals, promotion, and best practices across the DISCOMs for the benefit of all.

### **7.12 UPADHI (UPbhokta ADHIkaar) Portal**

Government of India has rolled out the Electricity (Rights of Consumers) Rules, 2020, emphasizing on the belief that power systems exist to serve the consumers and the consumers should have rights to get the proper services, and reliable, quality electricity. The Rules have an extensive coverage including areas such as Rights of consumers and Obligations of Distribution licensees, Release of connection, Metering arrangement, Billing and Payment, Disconnection and Reconnection, Reliability of supply, Consumer as Prosumer etc. To report and monitor the progress regarding the implementation of the provisions of the Electricity (Rights of Consumers) Rules, 2020, an online platform/portal UPbhokta ADHIkaar has been developed by CEA. UPADHI platform was launched by the Hon'ble Union Minister of Power and New & Renewable Energy, during the Review, Planning and Monitoring (RPM) meeting of Ministry of Power on 11th July 2023.

UPADHI encompasses the holistic framework for performance monitoring of DISCOMs & SERCs across key metrics defined in the rules and collects

state-wise information on compliance with the provision of rules on a single platform. It encapsulates the performance of DISCOMs on 75 key performance indicators (KPIs) and SERCs on 42 KPIs across key consumer service standards defined in the rules.

UPADHI can be accessed at <https://upadhi.cea.gov.in>

### **7.13 Disaster Resource Inventory for Power Sector (DRIPS) Portal**

As per the directions of Hon'ble Minister of Power, Government of India, CEA was entrusted the task of developing an electronic inventory of disaster resources for power sector in line with India Disaster Resource Network (IDRN). The same was developed in consultation with all relevant stakeholders and has been hosted on National Power Portal (NPP) for ease of convenience for all stakeholders.

The Portal is aimed to assist the disaster affected utilities in faster identification of availability of equipment and resource requisitioning across the country to mitigate the impact of disaster on an immediate basis. The portal is an online inventory of spares for all wings of Power Sector viz. Generation, Transmission and Distribution. The DRIPS Portal can be accessed by the registered users of CEA, Discoms, Gencos and Transmission utilities on <https://drips.npp.nic.in/login>.

### **7.14 Advisory for Behind-the-Meter Power Infrastructure**

E-bus Sewa scheme, aimed to increase modal share of buses in urban cities, will be deploying about 10,000 buses on PPP model alongwith providing support for development of Behind-the-Meter infrastructure. In order to sensitize the Discoms to execute the work of Behind-the-Meter power infrastructure, an advisory for Behind-the-Meter Power Infrastructure to deliver power for EV charging at Bus Depots in tier-2 and 3 cities has been issued by CEA.

### **7.15 Report of the Status of Feeder, Distribution Transformer and consumer metering**

In order to monitor the status of Feeder, Distribution

Transformer and consumer metering, a Report is being prepared regularly in CEA. The Report covers the category wise (viz. urban, rural, agricultural) status of metering of feeders & DTs and also category wise (viz. residential, industrial, agricultural etc.) status of metering of consumers. The report is being updated quarterly & annually.

### **7.16 Empanelment of Net Meter Manufacturers for solar roof top system**

To avoid the delay in availability of Net Meters for PM Surya Ghar Yojna, a Committee was constituted under Chairmanship of Member (GO&D), CEA for empanelment of Net meter manufacturers for solar roof top system. Based on the deliberations, a list of Net Meter manufacturers was finalized and uploaded on CEA website to facilitate the roof top solar installation under PM Surya Ghar Muft Bijli Yojana. Further, Common General Technical Specification for 1P, 3P Net meter/smart meters were also issued for facilitating the DISCOMs to adopt uniform specification of Net Meters for solar roof top installation.

### **7.17 Mock Test Exercise at Parliament House**

To ensure reliability of power supply to Parliament house before onset of each Parliament session, mock drill exercises at CPWD 11 KV Parliament House S/S were organized by CPWD in presence of officers of CEA, CPWD & NDMC before the Monsoon, Winter and Budget Sessions of Parliament and the reports of the Mock Test Exercise were sent to Ministry of Power, CPWD & NDMC.

### **7.18 Consultancy work**

A consultancy work regarding network planning study for its distribution system of Noida Power Company Limited (NPCL) including improvements in distribution system including requirement of smart technologies was taken up. CEA is assisting NPCL in formulation of reports & recommendations, after appointment of consultant. A draft Report has been submitted by NPCL and the same is under consideration for final recommendation.

## **7.19 National Feeder Monitoring System**

National Feeder Monitoring System (NFMS), is an automated Web-based System for Monitoring the 11 kV Distribution Feeders by integrating data from various DISCOM's and status of all outgoing 11 kV Feeders from 33/11kV substations to make the information available on near Real-Time basis through M2M communication mode.

The National Feeder Monitoring System (NFMS) aims to set up a Centralized Unified system which will be further linked to NPP. NFMS is envisaged with the following key objectives:

- Automatic monitoring for Reliability of Power parameter such as SAIFI, SAIDI and Hours of Supply etc.
- Automatic monitoring for Quality of Power such as Voltage level, Frequency, Powerfactor etc.
- Enabling Inputs for DISCOM Consumer Service Rating, Demand forecasting, Network Planning, Load Management & Energy Accounting and other operational & strategic initiatives
- Advanced reporting and generation of multi layered MIS for all stakeholders along with enablement for Advanced Data based Analytics (AI/ML based)
- Enabling seamless Machine to Machine data transfer from field equipment to the servers and data processing Unit without any manual interventions.

A Steering Committee under Chairmanship of Member (GO&D), CEA with members from PGCIL, MoP, PFC & REC is constituted by MOP for reviewing and finalizing the scope of works, corresponding cost estimates with timelines for achieving immediate milestones, release of funds, regular review and monitoring of NFMS Project implementation. The deadline for integrating all 2.5 lac feeders into the NFMS is December 31, 2024. As of now, 1.5 lac feeders have been integrated into the NFMS.

## 7.20 Monitoring of Prime Ministers development (PMDP) 2015 Distribution Projects in UT of J&K and Ladakh

CEA is the Nodal Agency for monitoring the progress of works in Distribution Sector in the UTs of J&K and Ladakh under PMDP-2015.

### Projects under Ongoing Works for PMDP 2015:

Ministry of Power on 9th Nov, 2016 has sanctioned an amount of Rs 2570.14 Crores for Strengthening of Distribution system and new

technologies in the UT of J&K and Ladakh as below:

**Rural Areas:** Projects in 21 districts amounting to Rs 1157.75 Crores including PMA charges were sanctioned for strengthening the rural distribution network including electrification of shrines, underground cable laying in tourist places and electrical infrastructure in industrial areas. JPDCL, KPDCL & PGCIL have been nominated as Project implementing Agencies (PIA) by JKPDD. Region and PIA wise Financial progress vis-à-vis surveyed and approved projects as provided by respective PIAs under the PMDP- Rural is as below:-

| Region  | PIA   | Progress |
|---------|-------|----------|
| Jammu   | JPDCL | 85%      |
|         | PGCIL | 90%      |
| Kashmir | KPDCL | 87%      |
|         | PGCIL | 100%     |
| Ladakh  | PGCIL | 67%      |

**Urban Area:** Project in 12 circles amounting to Rs 1144.59 Crores including PMA charges for strengthening the urban distribution network which includes establishment of meter testing labs were sanctioned. JPDCL, KPDCL&

RECPDCL are the PIAs. Region-wise and PIA-wise Financial progress vis-à-vis surveyed and approved projects as provided by respective PIAs under the PMDP-Urban is as below:-

| Region  | PIA     | Progress |
|---------|---------|----------|
| Jammu   | JPDCL   | 65%      |
|         | RECPDCL | 97%      |
| Kashmir | KPDCL   | 96%      |
|         | RECPDCL | 85%      |
| Ladakh  | RECPDCL | 89%      |

**Smart metering projects:** Projects for providing smart meters to 2 lakh consumers at the cost of 126.54 Crores including PMA charges were sanctioned, for which RECPDCL is the PIA. Works are under advanced stages of completion.

### Projects under Additional fund for PMDP 2015:

Ministry of Power vide sanction order dated 01-June-21 approved additional funds of Rs 1068.43 crs for completion of balance works sanctioned under PMDP2015. These works have been awarded by the respective PIAs

during November-December 2021.

**Rural Areas:** Additional Funds for projects in 21 districts amounting to Rs 565.87 Crores including PMA charges were sanctioned for strengthening the rural distribution network including electrification of shrines, underground cable laying in tourist places and electrical infrastructure in industrial areas. JPDCL, KPDCL & PGCIL have been nominated as Project implementing Agencies (PIA) by JKPDD. Region and PIA wise Financial progress vis-à-vis surveyed and approved projects as provided by respective

PIAs under the PMDP- Rural is as below:-

| Region  | PIA   | Progress |
|---------|-------|----------|
| Jammu   | JPDCL | 49%      |
|         | PGCIL | 80%      |
| Kashmir | KPDCL | 88%      |
|         | PGCIL | 98%      |
| Ladakh  | PGCIL | 71%      |

**Urban Areas:** Additional funds for Project in 12 circles amounting to Rs 486.61 Crores including PMA charges for strengthening the Urban distribution area has been sanctioned. JPDCL,

KPDCL& RECPDCL are the PIAs. Region and PIA wise Financial progress vis-à-vis surveyed and approved projects as provided by respective PIAs under the PMDP-Urban is as below:-

| Region  | PIA     | Progress |
|---------|---------|----------|
| Jammu   | JPDCL   | 41%      |
|         | RECPDCL | 100%     |
| Kashmir | KPDCL   | 87%      |
|         | RECPDCL | 88%      |
| Ladakh  | RECPDCL | 72%      |

The expected Date of completion of Projects under Additional fund is Dec, 2024.

## 7.21 Monitoring of Urban & Rural Power Supply on National Power Portal (NPP)

NPP, launched on 14th Nov, 2017, is a centralized system which facilitates online data capture/ input (daily, monthly, and annually) and to disseminate related information (operational, capacity, demand, supply, consumption etc.) through various analysed reports, graphs, statistics etc. for Indian Power Sector. The Nodal Agency for implementation of NPP and its operational control is CEA. The system has been conceptualized, designed and developed by National Informatics Centre (NIC).

In Distribution Sector, NPP captures the feeder-level Power Supply Data for rural & urban areas. CEA is Nodal for monitoring the rural and urban Power Supply in the country based on the feeder linked on NPP.

By end of March, 2024, approx. 66147 urban feeders of 62 Discoms/UTs and approx. 131675 rural feeders of 46 Discoms are integrated in NPP. As reported by the States on National Power Portal (NPP), the Average hours of Power Supply in a Day in 11 kV feeder for Rural areas and Urban areas during the FY 2023-24 is 21.9 (HH.hh) and 23.4 (HH.hh) respectively.

## CHAPTER – 8

# DESIGN & ENGINEERING SERVICES

### 8.1 Design & Engineering of Hydro Electric Projects

Central Electricity Authority (CEA) renders design & engineering services for Hydro Electric Projects under execution in the Country in Central/ State Sectors and neighboring countries. CEA provides consultancy for conventional type hydro generating units, bulb/ tubular type units, pumped storage schemes with an underground/ surface power stations. Design & Engineering includes complete design, techno-economic analysis and preparation of Technical Specifications, tender evaluation, selection and sizing of equipment, detailed layout and schematic drawings for hydro

turbine, generator, transformer, GIS, switchyard equipment and other auxiliaries.

### 8.2 Programme and Achievement during 2023-24

During 2023-24, CEA continued consultancy services for design and engineering of electrical and mechanical works of eleven (11) nos. hydroelectric projects. Out of these, nine (9) projects are in India and two (2) projects are in Bhutan. The Projects for which design & engineering services were rendered by CEA are as given below:-

| Sl. No.                                             | Name of the H.E. Project                                                                                                                                    | Executing Agency/ State | Capacity                                                   |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------|
| <b>E&amp;M Design &amp; Engineering Consultancy</b> |                                                                                                                                                             |                         |                                                            |
| <b>Main Consultancy (Neighboring Countries)</b>     |                                                                                                                                                             |                         |                                                            |
| 1.                                                  | Punatsangchhu–I HEP                                                                                                                                         | PHPA-I,Bhutan           | 6x200 MW                                                   |
| 2.                                                  | Punatsangchhu–II HEP                                                                                                                                        | PHPA-II,Bhutan          | 6x170 MW                                                   |
| <b>Main Consultancy (India)</b>                     |                                                                                                                                                             |                         |                                                            |
| 3.                                                  | Lakhwar MPP                                                                                                                                                 | UJVNL, Uttarakhand      | 3x100 MW                                                   |
| 4.                                                  | Ganol SHEP                                                                                                                                                  | MePGCL, Meghalaya       | 3x7.5 MW                                                   |
| 5.                                                  | Upper Indravati PSP                                                                                                                                         | OHPC Ltd., Odisha       | 4x150 MW                                                   |
| 6                                                   | Kharag HEP                                                                                                                                                  | OHPC Ltd., Odisha       | 3x21 MW                                                    |
| <b>Overview Consultancy (India)</b>                 |                                                                                                                                                             |                         |                                                            |
| 7.                                                  | a) Under Construction:<br>i. Tehri PSP<br>ii. Vishnugad Pipalkoti HEP<br>b) Under Operation:<br>i.. Tehri HEP<br>ii. Koteswar HEP<br>iii. Dhukwan Small HEP | THDCIL, Uttarakhand     | 4x250 MW<br>4x111 MW<br><br>4x250 MW<br>4x100 MW<br>3x8 MW |

### 8.3 Scrutiny/Examination/Preparation of DPRs of HE Projects

- a) Chapters on General Layout Aspect, Electro-Mechanical equipment, related drawings, bill of quantities, Memorandum of Changes, etc. of 18 nos. ( 17 nos. in India and 01 nos. in Nepal) of DPR of HEPs aggregating to 11176 MW including clarifications/ drawings/ documents etc. as received from time to time were examined and commented upon.
- b) Chapters on General Layout Aspect, Electro-Mechanical equipment, related drawings, bill of quantities, Memorandum of Changes, etc. of 44 nos. of DPR of PSPs aggregating to 61510 MW including clarifications/ drawings/ documents etc. as received from time to time were examined and commented upon.
- c) Chapters on E&M aspects along with related drawings & BOQ pertaining to Lift Irrigation Scheme which are referred to CEA by CWC are examined and commented upon. However, no proposal was received.
- d) Memorandum of Changes pertaining to Electro-mechanical aspect received for 05 nos. HEP of 4440 MW were examined and commented upon.
- e) Revised Cost Estimates of 02 nos. HEPs aggregating to 2600 MW, for which the concurrence was already accorded by CEA, were also examined/commented upon.
- f) Electro-mechanical chapter along with related drawings & BoQ of DPR for 3 stage Damanganga (Ekdare) - Godavari (Waghad) Link Project, Maharashtra is prepared.
- g) Electro-mechanical chapter along with related drawings & BoQ of DPR for 9 stage Damanganga-Vaitarna-Godavari Intra State Link Project, Maharashtra is prepared.
- h) Electro-mechanical chapter along with related drawings & BoQ of DPR for Pachnad Irrigation Scheme, UP for Planning Circle (CWC), Faridabad is under preparation.

The list of above projects is as given below:

#### A. List of DPRs of HEPs examined for E&M aspects during the year:

| S.No.          | Name of the Project        | State             | Installed Capacity (MW) |
|----------------|----------------------------|-------------------|-------------------------|
| 1              | Bokang Bailing HEP         | Uttarakhand       | 165                     |
| 2              | Anjaw HEP                  | Arunachal Pradesh | 270                     |
| 3              | Naba HEP                   | Arunachal Pradesh | 1105                    |
| 4              | Niare HEP                  | Arunachal Pradesh | 860                     |
| 5              | Bardang HEP                | Himachal Pradesh  | 166                     |
| 6              | Tandi Rashil HEP           | Himachal Pradesh  | 268                     |
| 7              | Emra – II HEP              | Arunachal Pradesh | 315                     |
| 8              | Dengser HEP                | Arunachal Pradesh | 640                     |
| 9              | Sach Khas HEP              | Himachal Pradesh  | 267                     |
| 10             | Dulhasti Stage II HEP      | Jammu Kashmir     | 260                     |
| 11             | Karcham Wangtoo HEP        | Himachal Pradesh  | 1200                    |
| 12             | OJU HEP                    | Arunachal Pradesh | 2230                    |
| 13             | Myntdu Leshka Stage-II HEP | Arunachal Pradesh | 210                     |
| 14             | Kirthai HEP                | Jammu Kashmir     | 390                     |
| 15             | Teesta Intermediate HEP    | Sikkim            | 90                      |
| 16             | Kamala HEP                 | Arunachal Pradesh | 1720                    |
| 17             | Demwe Upper Stage – I HEP  | Arunachal Pradesh | 270                     |
| Project abroad |                            |                   |                         |
| 18             | West Seti HEP              | Nepal             | 750                     |

**B. List of PSPs under S&I stage which were examined for E&M aspects during the year:**

| S. N o. | Name of the Project       | State          | Installed Capacity (MW) |
|---------|---------------------------|----------------|-------------------------|
| 1       | Sirohi PSP                | Rajastha n     | 1200                    |
| 2       | Narihalla PSP             | Karnatak a     | 300                     |
| 3       | Veerabali PSP             | Andhra Pradesh | 1800                    |
| 4       | Upper Sileru PSP          | Andhra Pradesh | 1350                    |
| 5       | Pane PSP                  | Maharas htra   | 1500                    |
| 6       | Gujjili PSP               | Andhra Pradesh | 1400                    |
| 7       | Vemapalli PSP             | Andhra Pradesh | 1500                    |
| 8       | Bhavali PSP               | Maharas htra   | 1500                    |
| 9       | UP-01 PSP                 | Uttar Pradesh  | 3660                    |
| 10      | Kandhaura PSP             | Uttar Pradesh  | 1680                    |
| 11      | Shahpur PSP               | Rajastha n     | 1800                    |
| 12      | Raiwada PSP               | Andhra Pradesh | 900                     |
| 13      | Tarali PSP                | Maharas htra   | 1500                    |
| 14      | Sharavathy PSP            | Karnatak a     | 2000                    |
| 15      | Yaganti PSP               | Andhra Pradesh | 1000                    |
| 16      | Chittamvalasa PSP         | Andhra Pradesh | 800                     |
| 17      | Malshej Ghat Bhorende PSP | Maharas htra   | 1440                    |
| 18      | Kamalapadu PSP            | Andhra Pradesh | 950                     |
| 19      | Shirawta PSP              | Maharas htra   | 1800                    |
| 20      | Patgaon PSP               | Maharas htra   | 2100                    |
| 21      | MP 30 Gandhi Sagar PSP    | Madhya Pradesh | 1920                    |
| 22      | Panari PSP                | Madhya Pradesh | 1800                    |
| 23      | DVC-Lugupahar PSP         | Jharkhan d     | 1500                    |
| 24      | Indirasagar PSP           | Madhya Pradesh | 525                     |
| 25      | Tekwa-2 PSP               | Madhya         | 800                     |

|    |                             |                |      |
|----|-----------------------------|----------------|------|
|    |                             | Pradesh        |      |
| 26 | Rajupalem PSP               | Gujarat        | 320  |
| 27 | Gandikota PSP               | Andhra Pradesh | 1000 |
| 28 | Saundatti PSP               | Karnatak a     | 1600 |
| 29 | Upper Indravati PSP         | Orissa         | 600  |
| 30 | Turga PSP                   | West Bengal    | 1000 |
| 31 | Warasgaon Warani PSP        | Maharas htra   | 1500 |
| 32 | Pinnapuram PSP              | Andhra Pradesh | 1680 |
| 33 | Chitravathi PSP             | Andhra Pradesh | 500  |
| 34 | Shoma PSP                   | Uttar Pradesh  | 2400 |
| 35 | Aravetipalli PSP            | Andhra Pradesh | 1320 |
| 36 | Bhivpuri PSP                | Maharas htra   | 1000 |
| 37 | Kynshi PSP                  | Meghaly a      | 1125 |
| 38 | Rayavaram PSP               | Andhra Pradesh | 1500 |
| 39 | Musakhand PSP               | Uttar Pradesh  | 600  |
| 40 | Varahi PSP                  | Karnatak a     | 1500 |
| 41 | Indirasagar Omkareshwar PSP | Madhya Pradesh | 640  |
| 42 | Savitri PSP                 | Maharas htra   | 1800 |
| 43 | Koyna Nivakane PSP          | Maharas htra   | 2700 |
| 44 | Nayagaon PSP                | Maharas htra   | 2000 |

**C. List of Lift Irrigation Schemes, referred by CWC, which were examined for E&M aspects during the year:**

No proposal received during the given period.

**D. List of HEPs Memorandum of Changes which were examined for E&M aspects during the year:**

| S. No. | Name of the Project | State            | Installed Capacity (MW) |
|--------|---------------------|------------------|-------------------------|
| 1      | Pinnapuram PSP      | Andhra Pradesh   | 1680                    |
| 2      | Luhri Stage – I HEP | Himachal Pradesh | 210                     |
| 3      | Pinnapuram IREP     | Andhra Pradesh   | 1200                    |
| 4      | Teesta VI HEP       | Sikkim           | 500                     |
| 5      | Ratle HEP           | Jammu & Kashmir  | 850                     |

**E. List of HEPs which were examined for Revised Cost Estimates for E&M aspects during the year:**

| S. No. | Name of the Project | State             | Installed Capacity (MW) |
|--------|---------------------|-------------------|-------------------------|
| 1      | Subansiri Lower HEP | Arunachal Pradesh | 2000                    |
| 2      | Kameng HEP          | Arunachal Pradesh | 600                     |

**8.4 Proposals for Foreign Assistance/Bilateral Co-operation**

Relevant material/ inputs were provided for the proposal of bilateral co-operation with various countries in the field of hydro power development (including Energy Storage) as and when received from various ministries, such as Austria, Russia, European Union, Iraq, France, Switzerland, Australia, Ghana, Tajikistan, South Africa, Tanzania, Oman, Nigeria, Czech Republic, Peru, Sri Lanka, Kenya, Japan, Germany, US, Korea etc.

**8.5 Review of Technical Standards/ Regulations:**

Participated in panel meetings of BIS for preparation of/ amendments in draft Indian Standards, namely IS: 12800 Part-I and Part-II, as and when required.

**8.6 R&D Activities:**

- Providing inputs on R&D references received from various ministries and organizations as and when received.

- Participation in various workshops, conferences and trainings conducted by CBIP, CWC, etc. pertaining to developments in Hydro Power Sector.
- Member of Technical Committee on Hydro Research for the examination of R&D proposals for Hydropower sector received from entities for funding by Government of India.

**8.7 Examination of Innovative Proposals:**

Examination/scrutiny of Innovative Proposals on generation of electricity from renewable and other sources of energy. 01 no. of innovative proposals was received in the given period and commented upon.

**8.8 Miscellaneous Works:**

- A Committee was constituted under the chairmanship of Member (Hydro), CEA in the matter of requirement of Variable Speed Machine (VSM) vis-à-vis Fixed Speed Machine (FSM) in PSPs vide OM no. 10/3/HE&TD/2023/ dated 15.02.2023 and its report published on CEA website on 27.07.2023.
- On the directives of Ministry of Power, a Committee was constituted under the chairmanship of Member (Hydro), CEA to formulate Guidelines for Field Efficiency Testing of Units in Hydro Power Plants (including PSPs), and its report Published on CEA website on 19.06.2023.  
  
A Committee was constituted under the chairmanship of Member (Hydro), CEA in the matter for finalisation of the opening and closing time ranges of guide vanes during turbine and pumping mode vide OM no. 10/3/HE&TD/2023/ dated 30.10.2023 and its report is under finalization.
- Providing inputs and participation in Technical Coordination Committee (TCC) & Authority Meetings of Punatsangchhu- I (6x200 MW) and Punatsangchhu-II (6x170 MW) HEP, Bhutan. Participation in other project related meetings like Project Level Tender Evaluation Committee (PLTEC), Pre-Bid Meetings, Tender- Evaluation Committee (TEC), etc. of Punatsangchhu-I (6x200 MW) and Punatsangchhu- II (6x170 MW) HEP, Bhutan.

- iv. Conducting inspections at manufacturer works and preparation of reports thereof for various Electro-Mechanical equipment of Punatsangchhu-I (6x200 MW), Punatsangchhu-II (6x170 MW) and Ganol (3x7.5 MW) HEP.
- v. Inputs in respect of Hydro Power Sector pertaining to Public Procurement (Preference to Make in India) [PPP-MII] Order were provided as & when required. Examination of representations from various associations/manufacturers in light of latest PPP-MII Order & other relevant MoP & DPIIT Orders.
- vi. Examination of tender documents of value more than Rs. 500cr. floated by CPSUs of Hydro Power Sector under Ministry of Power to ascertain the compliance of PPP-MII Order issued by DPIIT & MoP.
- vii. Inputs for preparation of National Electricity Plan in respect of Hydro Power Sector were provided.
- viii. Reply of various Parliament Questions, VIP references, RTI applications etc. as and when received.

## CHAPTER -9

# ECONOMIC AND COMMERCIAL ASPECTS OF POWER INDUSTRY

As per the Electricity Act, 2003, CEA has, inter-alia, been entrusted with duties and functions relating to collection/recording of data relating to generation, transmission, distribution, trading and utilization of electricity and to carry out studies relating to cost, efficiency, competitiveness etc. to evaluate the financial performance of the power sector.

### 9.1 Electricity Tariff & Duty and Average Rates of Electricity Supply in India

In-fulfillment of its obligation under section 73(i) & (j) of the Electricity Act, 2003, CEA brings out a publication titled “Electricity Tariff & Duty and Average Rates of Electricity Supply in India”. The latest edition contains information on retail electricity tariff applicable in various States / Utilities effective during the year 2022-23.

The publication provides assimilation of regulatory data on notified tariffs of various States/UTs, the estimated data on average rates of electricity supply & electricity duty for different categories of consumers, along with the summarized data on power supply schemes for special categories of consumers. It also provides the details of subsidy support given by the government to various categories of consumers. The estimated average rates of electricity published herein have been computed on the basis of tariff orders received from various State Electricity Regulatory Commissions.

The effective rates for different consumer categories have been worked out

assuming different energy consumption for various sanctioned load keeping in view the urbanization, increase in usage of electricity appliances and improvement in the standard of living. In the March 2023 edition, tariff revisions subsequent to the last edition of the publication have been incorporated and tariff applicable in 44 Distribution Utilities have been indicated.

The sanctioned load and monthly energy consumption have been assumed for each category of consumer and considering the tariff notified by the respective Regulatory Commissions, the total amount payable by a particular category of consumer is worked out for the assumed load and monthly energy consumption. The Taxes and Duties are then added to arrive at the average estimated rate of electricity supply in terms of Paise / kWh.

A statement indicating category-wise estimated average rates of electricity for various Distribution Utilities in the country is given as **Annexure-9A**.

### 9.2 Publishing of Weighted Average Rate of Sale of Power (WARSP)

CEA publishes annually “Weighted Average Rate of Sale of Power” region-wise (Western/ Eastern/ Northern/ Southern/ North-Eastern), sector-wise (Central/ State/ Private) and fuel wise (Thermal/ Nuclear/ Hydro) for conventional generating plants with capacity 25 MW and above. The Generation Tariff Data and Net Generation figures are collected from various

Conventional Generating Utilities/ Power Stations on an annual basis. From the Station-wise Rate of Sale of Power, Weighted Average Rate of Sale of Power (WARSP) figures are computed and uploaded on CEA website.

### **9.3 Assistance to DPIIT for the calculation of Whole Sale Price Index (WPI)**

CEA assists Department for Promotion of Industry and Internal Trade (DPIIT) by providing Monthly Generation Tariff data from 72 Generating Stations (CPSUs and Private) for the compilation of Whole Sale Price Index (WPI).

### **9.4 Commercial appraisal of Power Sector Projects**

#### **9.4.1 During the year 2023-24, CEA carried out commercial appraisal of following projects/proposals in India**

- RCE of Shahpurkandi of Punjab, Parbati-II HPS,
- Tariff estimation of Sitammasagar Multipurpose project in Telangana,
- Upper Sileru PSP totaling to more than Rs 5000 Cr were completed.
- Viability of Dugar HPS of NHPC due to notifications issued by state government were examined.
- Benchmark tariff of hydro stations was proposed to MoP. It is felt that a benchmark tariff would fast track the decision making process for identification of viable hydro projects
- The financial aspects of Small Hydro Kinetic technology were examined and it was suggested for separate norms for such nascent technologies

- In order to better utilize the hydro power in the country, a proposal for approval of a RERTC product by bundling non hydro RE and hydro power by the CERC was prepared and shared with the MoP

#### **9.4.2 CEA also carried out financial and commercial appraisal of following hydro projects in Bhutan**

- Assessment of financial impact to the tune of 58 Crores due to change of COD date of Mangdechhu HPS of Bhutan from mean COD to weighted mean COD. The issue was settled based on this analysis.
- IDC and Tariff for Punatsangchhu HEP-II were estimated in multiple scenarios. The total investment of GoI in this project is upwards of 120000 Cr

#### **9.4.3 Recommendation of tariff for Nuclear Power Stations**

- As per Atomic Energy Act 1962, fixation of tariff can be done by atomic power stations or through any authority or corporation established by atomic power stations or Government Company in consultation with Central Electricity Authority.
- During the year 2023-24, tariff for Kudankulam Nuclear Power Plant Unit # 1 & 2 (2 x 1000 MW) recommended to Department of Atomic Energy for the period 2014-17 & 2017-22..

#### **9.4.4 Computation of Energy Charge Rate (ECR) for Imported Coal Based Plant during Section 11 period**

In the light of rising demand and maximization of generation from the

Imported Coal Based plants (ICB), MoP had issued directions vide order dated 20.02.2023 (effective from 16<sup>th</sup> March 2023) to imported coal based power plants under section 11 of the Act with further clarifications and extensions from time to time. MoP had also constituted a committee under the Chairperson, CEA via the aforesaid directions to suggest prudent tariff for imported coal based generating stations by assessing the implication of rise in imported coal price and to suggest modalities for generation and supply from such generating stations. The validity period of the above said committee was till 30.06.2024. The committee had recommended the benchmark ECR for 7 ICB plants for the entire period from April, 2023 to March, 2024 on fortnightly basis.

## 9.5 Preparation of Bidding Documents and Guidelines

**9.5.1** Draft bidding guidelines for Pumped Storage Plants were prepared. The draft bidding guidelines are under consideration of MoP.

## 9.6 Monthly report on Indices

A summary analysis from electricity point of view of major indices like WPI, Index of 8 core industries, CPI and IIP released by DPIIT and MOSPI is prepared and sent to Ministry of Power on monthly basis.

## 9.7 The Electricity Act, 2003

### 9.7.1 Framing and Amendments of the CEA Regulations framed and notified under the Electricity Act, 2003

The Central Electricity Authority has been vested with the powers to make Regulations under Section 177 of the Electricity Act, 2003. The status of the notification of principal regulations and their subsequent amendments since the enactment of the Electricity Act, 2003, is as

under:

### A. Notified Principal Regulations:

During 2023-24 following principal regulations have been notified:

|   |                                                                                                            |                        |
|---|------------------------------------------------------------------------------------------------------------|------------------------|
| 1 | <b>**Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023</b> | Notified on 08.06.2023 |
|---|------------------------------------------------------------------------------------------------------------|------------------------|

**\*\*This regulation has repealed the Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2010, notified in year 2010.**

Apart from above, the following principal regulations have already been framed and notified by the Authority during previous years since the enactment of the Electricity Act, 2003:

| Sl. No. | Regulation                                                                                                                              | Notified on |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 1       | CEA (Installation & Operation of Meters), Regulations 2006                                                                              | 22.03.2006  |
| 2       | Central Electricity Authority (Procedure for Transaction of Business) Regulations, 2006                                                 | 22.8.2006   |
| 3       | Central Electricity Authority (Technical Standards for Connectivity to the Grid) Regulation, 2007                                       | 09.03.2007  |
| 4       | Central Electricity Authority (Furnishing of Statistics, Returns & Information) Regulation, 2007                                        | 19.04.2007  |
| 5       | Central Electricity Authority (Grid Standards) Regulation, 2010                                                                         | 26.06.2010  |
| 6       | Central Electricity Authority (Safety requirements for construction, operation and maintenance of electrical plants and electric lines) | 14.02.2011  |

|    |                                                                                                                                 |            |
|----|---------------------------------------------------------------------------------------------------------------------------------|------------|
|    | Regulations, 2011                                                                                                               |            |
| 7  | Central Electricity Authority (Technical Standards for Connectivity of the Distributed Generation Resources) Regulations, 2013  | 07.10.2013 |
| 8  | Central Electricity Authority (Technical Standards for Communication Systems in Power Systems) Regulations, 2020                | 27.02.2020 |
| 9  | Central Electricity Authority (Technical Standards for Construction of Electrical Plants and Electric Lines) Regulations, 2022* | 27.12.2022 |
| 10 | Central Electricity Authority (Flexible operation of thermal Generating Units) Regulations, 2023                                | 25.01.2023 |

#### B. Notified amendments in the Principal Regulations:

The regulations are regularly reviewed and amended by the Authority as per the requirements of various stakeholders in the power sector including general public at large. Total 12 nos. of amendments have been notified by the Authority during previous years since the enactment of the Electricity Act, 2003. These are as under:

| Sl. No. | Regulation                                                                                                   | Notified on |
|---------|--------------------------------------------------------------------------------------------------------------|-------------|
| 1       | Central Electricity Authority (Installation and Operation of meters) (Amendment) Regulations 2010            | 26.06.2010  |
| 2       | Central Electricity Authority (Technical Standards for Connectivity to the Grid) Amendment Regulations, 2013 | 15.10.2013  |
| 3       | Central Electricity Authority (Installation                                                                  | 03.12.2014  |

|   |                                                                                                                                                                       |            |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
|   | and Operation of meters) (Amendment) Regulations 2014                                                                                                                 |            |
| 4 | Central Electricity Authority (Technical Standards for Connectivity of the Distributed Generation Resources) Regulations, 2019                                        | 08.02.2019 |
| 5 | Central Electricity Authority (Technical Standards for Connectivity to the Grid) (Amendment) Regulations, 2019                                                        | 08.02.2019 |
| 6 | 3 <sup>rd</sup> Amendment to the Central Electricity Authority (Installation and Operation of Meters) (Amendment) Regulations, 2019                                   | 23.12.2019 |
| 7 | Central Electricity Authority (Installation and Operation of Meters) (Amendment) Regulations, 2022                                                                    | 28.02.2022 |
| 8 | Revision of Formats in Central Electricity Authority (Furnishing of Statistics, Returns & Information) Regulations, 2007) (Amendment)                                 | 17.03.2022 |
| 9 | Central Electricity Authority (Safety Requirements for Construction, Operation and Maintenance of Electrical Plants and Electric Lines) (Amendment) Regulations, 2022 | 16.11.2022 |

#### C. Repealed Regulations:

During 2023-24, the following regulation has been repealed:

| S.No. | Regulation                                                                                        | Repealed with effect from |
|-------|---------------------------------------------------------------------------------------------------|---------------------------|
| 1     | Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2010 | 08.06.2023                |

### 9.7.2 Court Cases

CEA is dealing with the Court Cases filed in the Hon'ble Supreme Court, High Courts, National Green Tribunal and District Courts/ Lower Courts across the Country on behalf of Government of India, Ministry of Power and Central Electricity Authority.

More than 250 court cases which are ongoing/ pending at various courts in which Ministry of Power or Central Electricity Authority have been impleaded as respondent (s) during the year 2023-2024.

### 9.8 Assistance to Ministry of Power

**Comments were furnished to the Ministry of Power on various references received. The important issues in which the comments were furnished are as under: -**

- Advice on the Electricity Haryana Amendment Bill, 2019.
- Comments on the Standing Committee's list of points on Electricity Act Amendment Bill, 2022
- Clarification on use of energy storage systems directly by consumers.
- Comments on representation regarding Direction to the "District Collectors" & "Licensees" to provide "Solatium"

amount equivalent to one hundred percent of the Compensation Amount.

### 9.9 Legal Assistance/Advice to Utilities

**The comments were furnished to various departments/organizations stakeholders/utilities on references received from them. The important issues on which the comments were furnished are as under: -**

- Reference received from Indian Railways regarding facilitating Open Access as deemed licensee.
- Reference received on Prior approval under Section 68 of the Electricity Act, 2003 for overhead lines to be constructed by Railway Electrification unit
- Reference from Serentica requesting Policy Framework clarity to ensure Energy Storage utilisation by Open Access Commercial & Industrial segment.
- Representation received Shri P. Chengal Reddy, Advocate & Chief Adviser of Consortium of Indian Farmers Associations (CIFA) on violation of provisions of Electricity Act, 2003.
- Comments on request from Online Engineers for One Nation, One License.
- Comments on representation from Sikkim Power Investment Corporation Ltd. (SPICL) about information regarding relief schemes financial measures for natural calamity hit projects in North-Eastern State of Sikkim.

### 9.10 Assistance to Ministry of Power

**The important issues in which assistance provided to MoP are as under: -**

- Deemed Grant of GNA in line with Central Electricity Regulatory Commission (Connectivity and General Network Access to the Inter State Transmission System) Regulations, 2022.
- Reference on Open Access Policy for Telecom Networks.
- Report on Over the Counter (OTC) platform by NAME.
- Report on working of Electricity Market.
- Report of the fact finding committee on frequent interruptions of power supply in urban areas of Gurugram.
- Reference on Approval for the provision of Net Metering in favour of NTPC for Green Hydrogen Mobility Project at Choglamsar, Palam, Leh UT Ladakh.
- Reference from Association of Power Producers (APP) on introducing floor price in power exchanges & other issues.
- Reference of ASSOCHAM on the subject of Deviation in Implementation of MoP direction under section 107 to CERC.
- Scheme for Pooling of Tariff of those plants whose PPAs have expired.
- Issues arising on account of the DERC (Business Plan) Regulations, 2023.
- Reference from FICCI regarding Level playing field for Captive Power Plants (CPPs).
- A report on the mitigation of impact of change in wind zone and river courses on the existing transmission

lines was prepared with stake holder comments and shared with MoP.

- In order to enable faster set up of grid scale storage, a report on business cases for storage was prepared and shared with MoP.
- The report of IIT Delhi on Development of National Scheme for Advanced grid scale energy storage technologies was examined and observations with regard to high sale price of stored energy considered in the report was forwarded.

### **9.11 References on Policy and Regulatory aspects in the Power Sector**

Following references on policy and regulatory aspects in the Power Sector have been dealt:

1. Proposal for review of existing framework of cost accounting and cost audit of distribution licensees was examined.
2. CCEA note of MoC for incentivizing underground mining was examined and suggestions were given to improve the policy such as waiver of GST compensation cess on coal mined through underground mining techniques.
3. Pass through provisions in PPAs of coastal DCB were studied for enhanced ACQ in respect of decision of CIL allowing 100% ACQ to coastal DCB. Presentation was made before the SLC-LT in this regard. The SLC LT approved the proposal for enhanced ACQ.
4. Reference for considering deemed trading licensee status for renewable power developers to procure shortfall

- electricity (5%) were studied and appropriate recommendation was made.
5. In order to ensure tariff certainty, the issue of tariff shifting of TBCB TSPs was examined in case of force majeure cases where SCOD had been shifted.
  6. Modifications to the RE procurement guidelines were made to ensure that power in case of early commissioning is not stranded.
  7. The SBD 2021 for TBCB in transmission did not have provisions of HVDC costs and Statcom costs. Suitable costs for the same were estimated in consultation with Powergrid and CTU and the same was intimated to MoP. The amendment to SBD have been issued.
  8. The proposal of escalation of Reserve Price payable to state governments by coal miners was examined and a suitable formulation was suggested to Ministry of Coal. MoC has issued the escalation formula and a long standing issue not addressed for more than 8 years has been settled.
  9. The proposal of threshold limit for DVC for introducing TBCB in DVC system was examined and appropriate threshold recommended in line with Tariff Policy. With this, DVC will be able to reap the benefits of lower cost due to competition.
  10. A proposal for rationalizing GST rates for pumped storage plants was studied and recommendations /procedure made to ensure the proposals are not misused.
  11. Draft concession agreement for regulated tariff based projects prepared by CTU was studied and recommendation for improvement of the same were shared.
  12. In order to better utilize the hydro power in the country, a proposal for approval of a RERTC product by bundling non hydro RE and hydro power by the CERC was prepared and shared with the MoP.
  13. In order to fund niche schemes of MoP and also for funding of gas based plants for generation during crunch period a study was made for levying some charges on power exchange participants.
  14. Compensation for increase in cost of Air Cooled Condensor, FGD and steel prices for BHEL projects were studied.
  15. International funding proposals such as re-powering of Hwange Thermal Power Station in Zimbabwe for \$310 million and establishment of 2000 MW Boualapha thermal power station in Lao PDR were scrutinized. While re-powering Hwange was found to be viable, in case of Boualapha it was suggested to start with a single 660 MW unit.
  16. Concerns of NPCIL on foreign exchange rate variation impact were studied and suitable addressed.
  17. Tender for 4500MW thermal power procurement with coal allocation through Shakti B.v was conducted by PFCCCL. Participated in multiple meetings of steering committee and with bidders to make the scheme successful.
  18. Electricity Amendment bill 2022 was studied and replies framed for Parliamentary Standing Committee.
  19. References on change in law acceptance by SERC's,

discontinuation of e-Reverse auction in TBCB were examined and suitably dealt with.

20. 1200 MW SECI PIB for setting up solar plant under CPSU Scheme Phase-II, Tranche-III was examined and important observations such as non tie up for complete land, non consideration of escalation, etc were shared.
21. PLI scheme for Battery Energy Storage Systems was studied and inter alia comments in minimum investment criteria, scheme design, minimum sale within the country, etc were shared.
22. Examination of PPA/contracts signed by Andaman administration with diesel generators so that the RE generation is not backed down during high RE output. Suggestions for changes in current PPAs and suggestions for design for new PPAs were shared.
23. Reference on clarification of PPA clause regarding recovery of fixed charges with respect to cumulative annual Availability were examined and recommended to MoP.
24. Ceiling fee for fast and slow chargers for charging EV at Public Charging Stations were recommended to MoP. The same were issued.
25. A lot of effort was put to get GUVNL and CGPL to come to a common understanding regarding a pass through mechanism for imported coal price in ECR. However, due to differences between the parties, the same could not be accomplished.
26. Draft CERC's (Terms and Conditions of Tariff) Regulations 2024, for the Tariff period

01.04.2024 to 31.03.2029 were studied and comments were submitted.

27. Furnished comments on working group report on safety in Indian Petroleum Sector, emphasized the importance of Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023 in Oil and gas industry.
28. Furnished comments on various amendment proposed in Electricity (Amendment) Bill, 2022 by Standing Committee on Energy.
29. Furnished comments on the CERC approach paper on Terms and Conditions of Tariff for the period commencing from 1<sup>st</sup> April, 2024.
30. Grievance redressal on NSWS portal.
31. Advisory issued to CMDs/MDs of DISCOMs for sell of surplus/unscheduled generation in electricity market under DAM/RTM contracts.
32. Reference from MoP on PAN as single log in ID on NSWS portal.
33. Reference on seeking inputs on Renewable supporting growth.

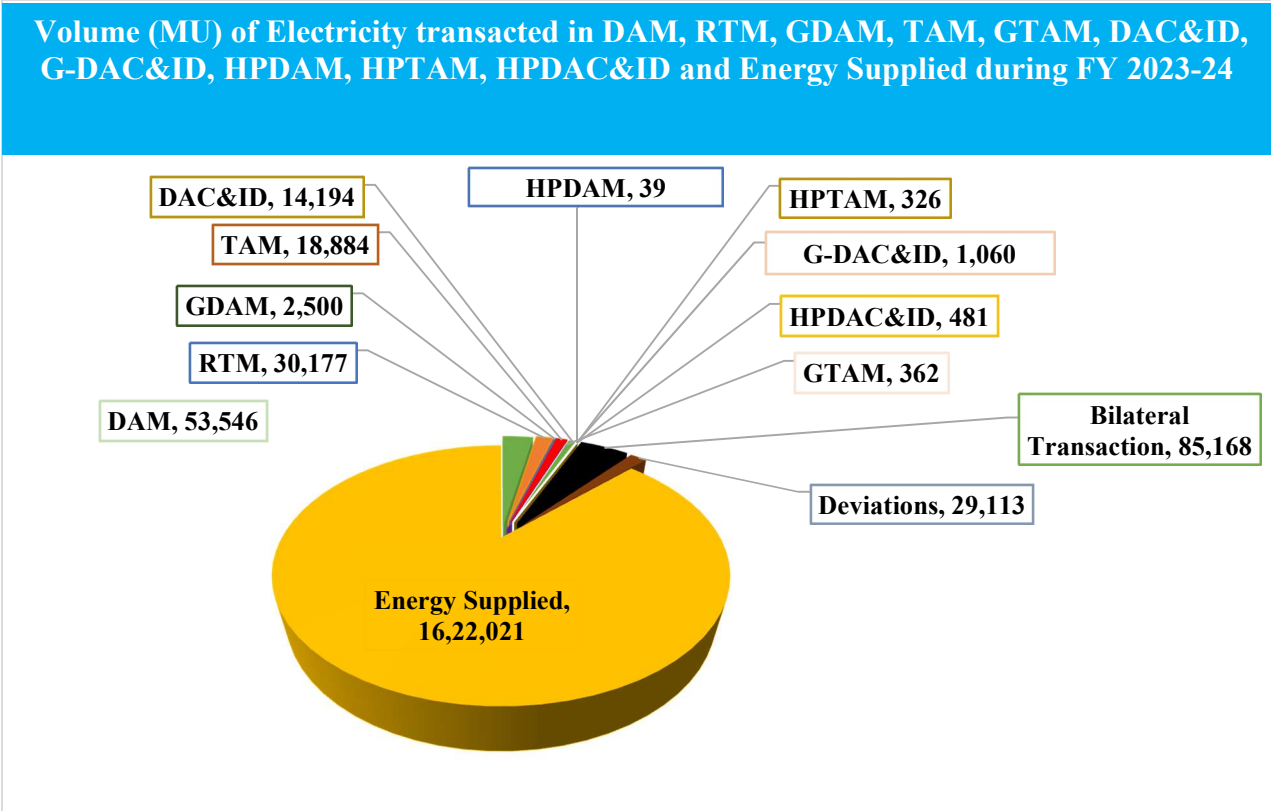
## 9.12 Market Monitoring

1. Ministry of Power in March, 2019 had entrusted CEA with the task of monitoring the transaction in the electricity market in India. As such, CEA prepares Monthly Market Monitoring Reports (MMMR) and Annual Market Monitoring Reports (AMMR) with the objective of analysis of movement of prices discovered for the electricity transacted on the power

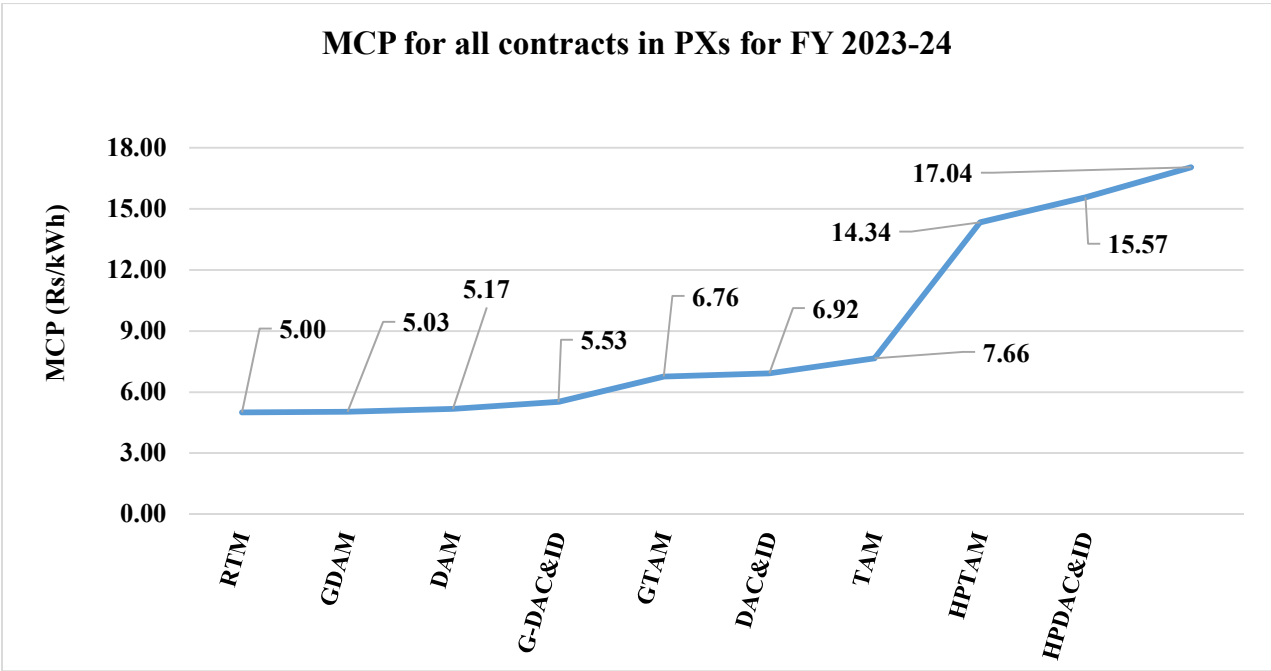
exchanges in India in the Day Ahead Market (DAM), Green Day Ahead Market (GDAM), Real Time Market (RTM), Term Ahead Market (TAM), Green Term Ahead Market (GTAM), Day Ahead Contingency & Intraday (DAC&ID), Green Day Ahead Contingency & Intraday (G-DAC&ID), High Price Day Ahead Market (HPDAM), High Price Term Ahead Market (HPTAM) and High Price Day Ahead Contingency & Intraday (HP-DAC&ID). In the year 2023-24, CEA has prepared Monthly Market Monitoring Reports (for March, 2023 to February, 2024) and Annual Market Monitoring Report (for year 2022-23). These reports are available on CEA's website (<https://cea.nic.in/market-monitoring-cell-annual-report/?lang=en>).

2. The salient points of the electricity market for FY 2023-24 are:

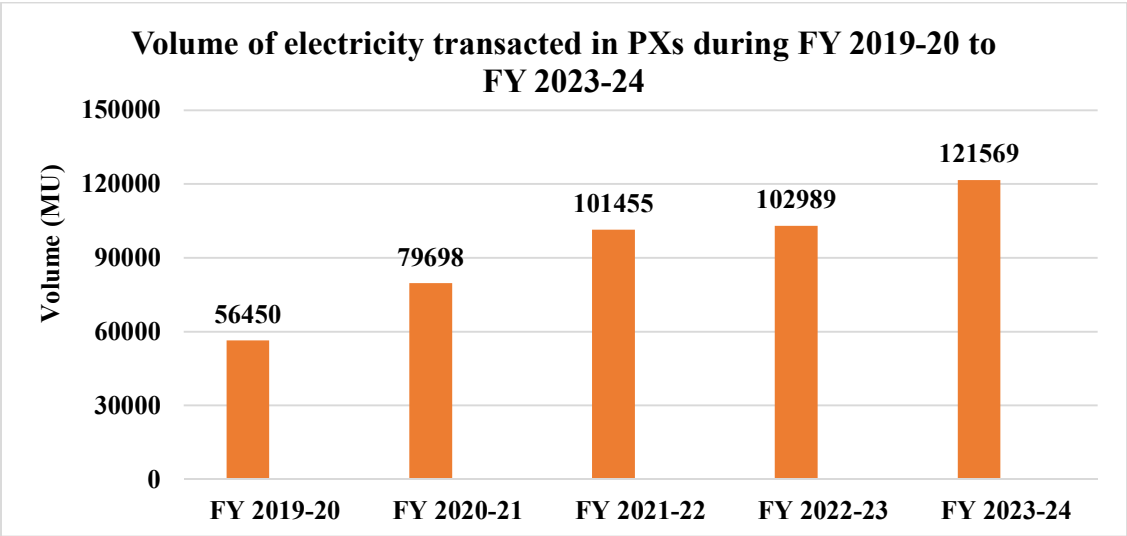
- (i) The total volume of electricity transacted during the period April, 2023 to March, 2024 in the power exchanges i.e. IEX, PXIL and HPX on delivery date basis was **121,569 MU**, which is 7.49 % of the total energy supplied i.e. **1,622,021 MU** in the country during FY 2023-24.
- (ii) The total volume of electricity transacted in the power exchanges in DAM, TAM, RTM, GTAM, GDAM, DAC&ID, G-DAC&ID, HPDAM, HPTAM and HPDAC&ID during the period April, 2023 to March, 2024 were **53546 MU, 18884 MU, 30177 MU, 3620 MU, 2500 MU, 14194 MU, 1060 MU, 39 MU, 326 MU and 481 MU**, respectively. The volume of electricity transacted in Day Ahead Market constituted about 44 % of total transacted volume of electricity in the power exchanges during the months April, 2023 to March, 2024.



(iii) The Market Clearing Price variation among all the contracts in power exchanges for the period April, 2023 to March, 2024 is as follows:



(iv) The total volume of electricity transacted in power exchanges from the period FY 2019-20 to FY 2023-24 is shown below:



\*\*\*\*\*

## CHAPTER – 10

### POWER GENERATION

#### 10.1 Power Generation

Generation of power from conventional sources (Thermal, Nuclear & Hydro) & import from Bhutan by the Central Sector, State Sector, Pvt. utilities & IPPs was about 1513256.36 million units (MUs)

during the Year 2023-24. This represents a growth of about 6.50% over the same period during previous year 2022-23 as per details given below:

**Power Generation during 2023-24**

| Category             | Program (MU)      | Actual (MU)       | Shortfall (-)/<br>Excess (+) w.r.t<br>Program (in<br>MU) | % of<br>Program | Growth (%)<br>with respect<br>to previous<br>year Actual<br>Gen. |
|----------------------|-------------------|-------------------|----------------------------------------------------------|-----------------|------------------------------------------------------------------|
| <b>Thermal</b>       | 1324110.00        | 1326548.93        | -2438.93                                                 | 100.18          | 9.98                                                             |
| <b>Nuclear</b>       | 46190.00          | 47937.41          | -1747.41                                                 | 103.78          | 4.53                                                             |
| <b>Hydro</b>         | 156700.00         | 134053.92         | 22646.08                                                 | 85.55           | -17.30                                                           |
| <b>Bhutan Import</b> | 8000.00           | 4716.1            | 3283.9                                                   | 58.95           | -30.05                                                           |
| <b>Grand Total</b>   | <b>1535000.00</b> | <b>1513256.36</b> | <b>21743.64</b>                                          | <b>98.58</b>    | <b>6.50</b>                                                      |

**Note:** Generation from stations having installed capacity less than 25MW is not being monitored in CEA since 01.04.2010.

The highlights / achievements of operation performance of generating stations in the country during the year 2023-24 are as under:

- Gross annual generation of the country was 1513.256 BU.
- The annual growth in the energy generation during the year was 6.50 %.
- Thermal, Nuclear, Hydro and Import from Bhutan achieved a growth rate of 9.98%, 4.53%, -17.30% & -30.05% respectively. The electricity generation during the year 2023-24 from coal based thermal power stations was

1260.90 BU showing a growth rate of 10.03 % over same period last year.

- In Southern Region, the growth in thermal generation was 15.66% with respect to last year, which was highest amongst all the regions.
- The national average PLF for thermal stations was 69.09 % and 103 Stations with an aggregate installed capacity of 120033.245 MW, achieved PLF above national average. 04 number of thermal power stations with an aggregate installed capacity of 5032 MW achieved above 90% PLF.

The Sector-Wise Generation and PLF during 2023-24 is given below:

| Category /              | APRIL 2023 - MAR-2024 |                   |         |
|-------------------------|-----------------------|-------------------|---------|
|                         | Program (MU)          | Actual (MU)       | PLF (%) |
| <b>CENTRAL SECTOR</b>   |                       |                   |         |
| THERMAL                 | 512190.00             | 472527.51         | 75.09   |
| NUCLEAR                 | 46190.00              | 47937.41          | 74.7    |
| HYDRO                   | 65585.00              | 58145.83          |         |
| <b>TOTAL</b>            | <b>623965.00</b>      | <b>578610.75</b>  |         |
| <b>STATE SECTOR</b>     |                       |                   |         |
| THERMAL                 | 419884.00             | 403929.70         | 64.7    |
| HYDRO                   | 75469.00              | 61667.64          |         |
| <b>TOTAL</b>            | <b>495353.00</b>      | <b>465597.34</b>  |         |
| <b>PVT. SECTOR UTL.</b> |                       |                   |         |
| THERMAL                 | 17051                 | 18414.01          | 64.17   |
| HYDRO                   | 1470.00               | 1557.60           |         |
| <b>TOTAL</b>            | <b>18521.00</b>       | <b>19971.61</b>   |         |
| <b>PVT. SECTOR IPP</b>  |                       |                   |         |
| THERMAL                 | 374985.00             | 431677.71         | 67.63   |
| HYDRO                   | 14176.00              | 12682.85          |         |
| <b>TOTAL</b>            | <b>389161.00</b>      | <b>444360.56</b>  |         |
| <b>TOTAL PVT</b>        | <b>407682.00</b>      | <b>464332.17</b>  |         |
| <b>BHUTAN IMP</b>       |                       |                   |         |
| -                       | 8000.00               | 4716.10           |         |
| <b>All India Region</b> |                       |                   |         |
| THERMAL                 | 1324110.00            | 1326548.93        | 69.09   |
| NUCLEAR                 | 46190.00              | 47937.41          | 74.7    |
| HYDRO                   | 156700.00             | 134053.92         |         |
| BHUTAN IMP              | 8000.00               | 4716.10           |         |
| <b>TOTAL</b>            | <b>1535000.00</b>     | <b>1513256.36</b> |         |

**Note:** Based on 2023-24 Actual Generation (Up to March, 24) from conventional sources Thermal/Hydro/Nuclear) with station capacity > 25 MW

## 10.2 Plant Load Factor (PLF) of Thermal Power Stations

During the year 2023-24 the average PLF of Thermal Power Stations was 69.09 % and for Nuclear Power Stations was 74.70%. 103 Thermal power plants (Coal and Lignite based) achieved PLF higher than the All India average PLF of 69.09 % as per details given in the table below:

### List of Thermal Power Stations (Coal and Lignite based) which have achieved PLF above National Average of 69.09 % during the year 2023-24

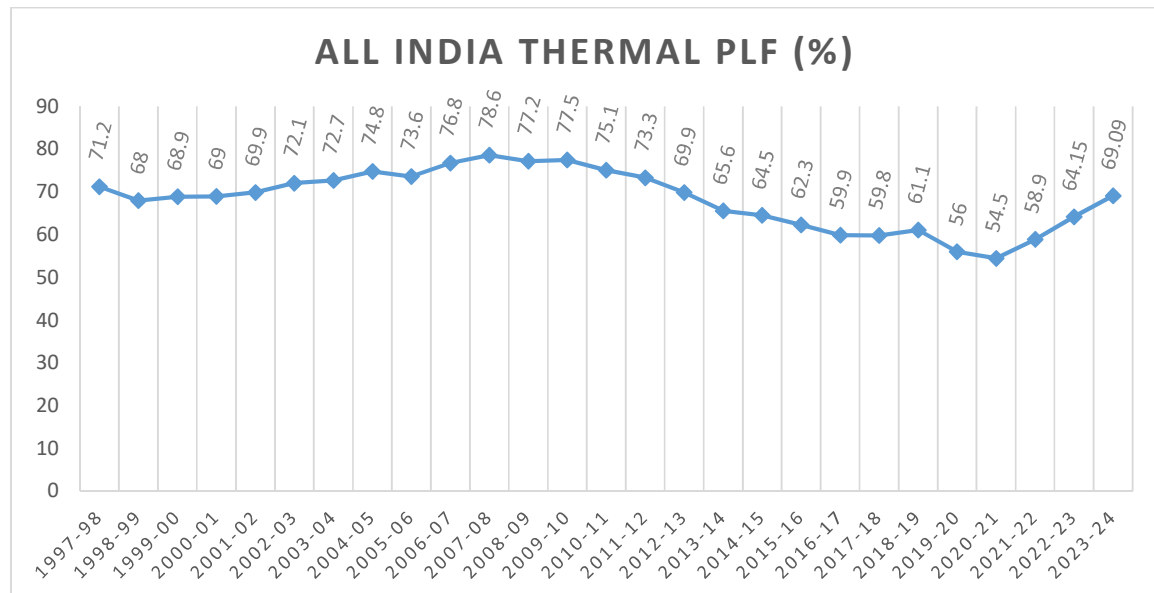
| SL. No. | Station Name                    | Capacity (MW) | Sector         | State                     | PLF (%) |
|---------|---------------------------------|---------------|----------------|---------------------------|---------|
| 1       | Adani Power Limited Kawai TPP   | 1320          | IPP Sector     | Rajasthan                 | 76.6    |
| 2       | Adani Power Limited Raigarh TPP | 600           | IPP Sector     | Chhatisgarh               | 75.36   |
| 3       | Adani Power Limited Raipur TPP  | 1370          | IPP Sector     | Chhatisgarh               | 72.19   |
| 4       | Adani Power Limited Tiroda TPP  | 3300          | IPP Sector     | Maharashtra               | 73.35   |
| 5       | Amarkantak Ext TPS              | 210           | State Sector   | Madhya Pradesh            | 97.81   |
| 6       | Amravati TPS                    | 1350          | IPP Sector     | Maharashtra               | 82.29   |
| 7       | And. Nicobar Pvt. DG            | 35.19         | PVT Sector     | Andaman & Nicobar Islands | 79.4    |
| 8       | Anpara C TPS                    | 1200          | IPP Sector     | Uttar Pradesh             | 77.43   |
| 9       | Anpara TPS                      | 2630          | State Sector   | Uttar Pradesh             | 74.7    |
| 10      | Anuppur TPP                     | 1250          | IPP Sector     | Madhya Pradesh            | 76.77   |
| 11      | Bakreswar TPS                   | 1050          | State Sector   | West Bengal               | 88.43   |
| 12      | Bandakhar TPP                   | 300           | IPP Sector     | Chhatisgarh               | 82.58   |
| 13      | Bandel TPS                      | 270           | State Sector   | West Bengal               | 74.83   |
| 14      | Baradarha TPS                   | 1200          | IPP Sector     | Chhatisgarh               | 85.43   |
| 15      | Barsingsar Lignite              | 250           | Central Sector | Rajasthan                 | 77.06   |
| 16      | Bela TPS                        | 270           | IPP Sector     | Maharashtra               | 75.49   |
| 17      | Bhadradi TPP                    | 1080          | State Sector   | Telangana                 | 73.3    |
| 18      | Bhilai TPS                      | 500           | Central Sector | Chhatisgarh               | 88.25   |
| 19      | Bhusawal TPS                    | 1210          | State Sector   | Maharashtra               | 71.27   |
| 20      | Bina TPS                        | 500           | IPP Sector     | Madhya Pradesh            | 75.8    |
| 21      | Bokaro TPS 'A' EXP              | 500           | Central Sector | Jharkhand                 | 82.15   |
| 22      | Bongaigaon TPP                  | 750           | Central Sector | Assam                     | 76.78   |
| 23      | Budge Budge TPS                 | 750           | PVT Sector     | West Bengal               | 87.05   |
| 24      | Chakabura TPP                   | 30            | IPP Sector     | Chhatisgarh               | 83.53   |
| 25      | Chandrapura(DVC) TPS            | 500           | Central Sector | Jharkhand                 | 85.33   |
| 26      | Chhabra-I PH-1 TPP              | 500           | State Sector   | Rajasthan                 | 79.27   |
| 27      | Dahanu TPS                      | 500           | PVT Sector     | Maharashtra               | 73.96   |
| 28      | Darlipali STPS                  | 1600          | Central Sector | Odisha                    | 85.77   |
| 29      | Derang TPP                      | 1200          | IPP Sector     | Odisha                    | 78.64   |
| 30      | Dhariwal TPP                    | 600           | IPP Sector     | Maharashtra               | 83.89   |
| 31      | Dr. N.Tata Rao TPS              | 2560          | State Sector   | Andhra Pradesh            | 73.5    |
| 32      | Dspm TPS                        | 500           | State Sector   | Chhatisgarh               | 92.73   |
| 33      | Durgapur Steel TPS              | 1000          | Central Sector | West Bengal               | 80.6    |
| 34      | Farakka STPS                    | 2100          | Central Sector | West Bengal               | 74.73   |
| 35      | Gadarwara TPP                   | 1600          | Central Sector | Madhya Pradesh            | 71.12   |

|    |                                 |        |                |                |       |
|----|---------------------------------|--------|----------------|----------------|-------|
| 36 | GMR Warora TPS                  | 600    | IPP Sector     | Maharashtra    | 83    |
| 37 | Haldia TPP                      | 600    | IPP Sector     | West Bengal    | 86.6  |
| 38 | Hiranmaye TPP                   | 300    | IPP Sector     | West Bengal    | 70.64 |
| 39 | IB Valley TPS                   | 1740   | State Sector   | Odisha         | 77.2  |
| 40 | ITPCL TPP                       | 1200   | IPP Sector     | Tamil Nadu     | 70.59 |
| 41 | Jalipa Kapurdi TPP              | 1080   | IPP Sector     | Rajasthan      | 74.68 |
| 42 | Jojobera TPS                    | 240    | IPP Sector     | Jharkhand      | 81.27 |
| 43 | JSW Ratnagiri TPP               | 300    | IPP Sector     | Maharashtra    | 74.96 |
| 44 | Kahalgaon TPS                   | 2340   | Central Sector | Bihar          | 80.54 |
| 45 | Kakatiya TPS                    | 1100   | State Sector   | Telangana      | 85.15 |
| 46 | Kamalanga TPS                   | 1050   | IPP Sector     | Odisha         | 82.2  |
| 47 | Karaikal CCPP                   | 32.5   | State Sector   | Puducherry     | 78.5  |
| 48 | Khaparkheda TPS                 | 1340   | State Sector   | Maharashtra    | 70.24 |
| 49 | Kodarma TPP                     | 1000   | Central Sector | Jharkhand      | 80.45 |
| 50 | Kolaghat TPS                    | 840    | State Sector   | West Bengal    | 73.1  |
| 51 | Korba STPS                      | 2600   | Central Sector | Chhatisgarh    | 89.84 |
| 52 | Korba-WEST TPS                  | 1340   | State Sector   | Chhatisgarh    | 81.62 |
| 53 | Kota TPS                        | 1240   | State Sector   | Rajasthan      | 77.88 |
| 54 | Kothagudem TPS (NEW)            | 1000   | State Sector   | Telangana      | 77.74 |
| 55 | Kothagudem TPS (STAGE-7)        | 800    | State Sector   | Telangana      | 85.54 |
| 56 | Kuttalam CCPP                   | 100    | State Sector   | Tamil Nadu     | 79.08 |
| 57 | Lakwa Replacement Power Project | 69.755 | State Sector   | Assam          | 78.16 |
| 58 | Lalitpur TPS                    | 1980   | IPP Sector     | Uttar Pradesh  | 71.03 |
| 59 | Iara TPP                        | 1600   | Central Sector | Chhatisgarh    | 83.61 |
| 60 | Mahadev Prasad STPP             | 540    | IPP Sector     | Jharkhand      | 78.85 |
| 61 | Mahatma Gandhi TPS              | 1320   | IPP Sector     | Haryana        | 71.16 |
| 62 | Maithon RB TPP                  | 1050   | IPP Sector     | Jharkhand      | 88.68 |
| 63 | Marwa TPS                       | 1000   | State Sector   | Chhatisgarh    | 84.11 |
| 64 | Mauda TPS                       | 2320   | Central Sector | Maharashtra    | 73.15 |
| 65 | Meja STPP                       | 1320   | Central Sector | Uttar Pradesh  | 71.37 |
| 66 | Mejia TPS                       | 2340   | Central Sector | West Bengal    | 76.24 |
| 67 | Mettur TPS                      | 840    | State Sector   | Tamil Nadu     | 78.68 |
| 68 | Monarchak CCPP                  | 101    | Central Sector | Tripura        | 74.45 |
| 69 | Muzaffarpur TPS                 | 390    | Central Sector | Bihar          | 84.83 |
| 70 | Nabinagar STPP                  | 1980   | Central Sector | Bihar          | 82.86 |
| 71 | Nabinagar TPP                   | 1000   | Central Sector | Bihar          | 79.06 |
| 72 | Neyveli ( EXT) TPS              | 420    | Central Sector | Tamil Nadu     | 77.82 |
| 73 | Neyveli NEW TPP                 | 1000   | Central Sector | Tamil Nadu     | 80.52 |
| 74 | Nigri TPP                       | 1320   | IPP Sector     | Madhya Pradesh | 84.87 |
| 75 | North Karanpura TPP             | 1320   | Central Sector | Jharkhand      | 88.65 |
| 76 | OP Jindal TPS                   | 1000   | IPP Sector     | Chhatisgarh    | 85.58 |
| 77 | Painampuram TPP                 | 1320   | IPP Sector     | Andhra Pradesh | 82.13 |
| 78 | Paras TPS                       | 500    | State Sector   | Maharashtra    | 81.88 |
| 79 | Pathadi TPP                     | 600    | IPP Sector     | Chhatisgarh    | 73.47 |
| 80 | Prayagraj TPP                   | 1980   | IPP Sector     | Uttar Pradesh  | 73.02 |
| 81 | Rajpura TPP                     | 1400   | IPP Sector     | Punjab         | 84.12 |
| 82 | Ramagundem STPS                 | 2600   | Central Sector | Telangana      | 74.21 |
| 83 | Ratija TPS                      | 100    | IPP Sector     | Chhatisgarh    | 73.28 |

|     |                          |       |                |                |       |
|-----|--------------------------|-------|----------------|----------------|-------|
| 84  | Rayalaseema TPS          | 1650  | State Sector   | Andhra Pradesh | 71.91 |
| 85  | Rihand STPS              | 3000  | Central Sector | Uttar Pradesh  | 88.8  |
| 86  | Rosa TPP Ph-I            | 1200  | IPP Sector     | Uttar Pradesh  | 72.19 |
| 87  | Sabarmati (D-F stations) | 362   | PVT Sector     | Gujarat        | 90.95 |
| 88  | Sagardighi TPS           | 1600  | State Sector   | West Bengal    | 82.02 |
| 89  | Santaldih TPS            | 500   | State Sector   | West Bengal    | 89.92 |
| 90  | Sasan UMTTP              | 3960  | IPP Sector     | Madhya Pradesh | 93.52 |
| 91  | SGPL TPP                 | 1320  | IPP SECTOR     | Andhra Pradesh | 80.1  |
| 92  | Singareni TPP            | 1200  | State Sector   | Telangana      | 83.99 |
| 93  | Singrauli STPS           | 2000  | Central Sector | Uttar Pradesh  | 89.69 |
| 94  | Sipat STPS               | 2980  | Central Sector | Chhatisgarh    | 85.42 |
| 95  | Surat LIG. TPS           | 500   | IPP Sector     | Gujarat        | 70.24 |
| 96  | Talcher STPS             | 3000  | Central Sector | Odisha         | 85.86 |
| 97  | Tamnar TPP               | 2400  | IPP Sector     | Chhatisgarh    | 81.31 |
| 98  | Torangallu TPS(SBU-I)    | 260   | IPP Sector     | Karnataka      | 74.94 |
| 99  | Tripura CCPP             | 726.6 | Central Sector | Tripura        | 71.66 |
| 100 | Tuticorin TPS            | 1050  | State Sector   | Tamil Nadu     | 70.31 |
| 101 | Valuthur CCPP            | 186.2 | State Sector   | Tamil Nadu     | 71.94 |
| 102 | Vindhyachal STPS         | 4760  | Central Sector | Madhya Pradesh | 89.42 |
| 103 | Wardha Warora TPP        | 540   | IPP Sector     | Maharashtra    | 81.17 |

**Note:** Based on 2023-24 Actual Generation (Up to March, 24) from conventional sources Thermal/Hydro/Nuclear) with station capacity > 25 MW

The trend in All India PLF of coal and Lignite based thermal power stations from 1997-98 onwards is shown below:



All India Sector-wise/ Organization-wise target, actual generation and PLF (%) for the year 2023-24 is at the **Annexure-10A**.

Annexure-10A**All India Sector wise/ Utility-wise Target, Actual Generation & PLF (%) for the year 2023-24**

| Category /            | APRIL 2023 - MAR-2024 |                   |              |
|-----------------------|-----------------------|-------------------|--------------|
|                       | TARGET (MU)           | ACTUAL (MU)       | PLF %        |
| <b>THERMAL</b>        |                       |                   |              |
| <b>CENTRAL SECTOR</b> |                       |                   |              |
| APCPL                 | 7600.00               | 8284.84           | 62.88        |
| BRBCL                 | 6700.00               | 6944.47           | 79.06        |
| DVC                   | 43700.00              | 44127.91          | 76.81        |
| JHAPL                 | 3800.00               | 3596.53           | 68.24        |
| K.B.U.N.L             | 3000.00               | 2906.20           | 84.83        |
| MUNPL                 | 7700.00               | 8275.82           | 71.37        |
| NEEPCO                | 3400.00               | 2997.19           |              |
| NLC                   | 22300.00              | 19547.49          | 61.14        |
| NPGCL                 | 14200.00              | 14411.96          | 82.86        |
| NSPCL                 | 3700.00               | 3875.74           | 88.25        |
| NTECL                 | 9500.00               | 6623.27           | 50.27        |
| NTPC Ltd.             | 368240.00             | 339679.40         | 76.84        |
| NTPL                  | 7000.00               | 5462.36           | 62.19        |
| NUPPL                 | 3300.00               | 0.00              | 0.00         |
| ONGC                  | 4800.00               | 4573.95           |              |
| PVUNL                 | 0.00                  | 0.00              | 0.00         |
| RGPPL                 | 1050.00               | 1220.38           |              |
| SJVNL                 | 2000.00               | 0.00              | 0.00         |
| THDC                  | 200.00                | 0.00              | 0.00         |
| <b>CENTRAL SECTOR</b> | <b>512190.00</b>      | <b>472527.51</b>  | <b>75.09</b> |
| <b>STATE SECTOR</b>   | <b>419884.00</b>      | <b>403929.70</b>  | <b>64.70</b> |
| <b>PVT SECTOR</b>     | <b>17051.00</b>       | <b>18414.01</b>   | <b>64.17</b> |
| <b>IPP SECTOR</b>     | <b>374985.00</b>      | <b>431677.71</b>  | <b>67.63</b> |
| <b>TOTAL THERMAL</b>  | <b>1324110.00</b>     | <b>1326548.93</b> | <b>69.09</b> |
| <b>NUCLEAR</b>        |                       |                   |              |
| <b>CENTRAL SECTOR</b> |                       |                   |              |
| BHAVINI               | 0.00                  | 0.00              | 0.00         |
| DAE                   | 0.00                  | 0.00              | 0.00         |
| NPCIL                 | 46190.00              | 47937.41          | 75.73        |
| <b>CENTRAL SECTOR</b> | <b>46190.00</b>       | <b>47937.41</b>   | <b>74.70</b> |
| <b>TOTAL NUCLEAR</b>  | <b>46190.00</b>       | <b>47937.41</b>   | <b>74.7</b>  |
| <b>HYDRO</b>          |                       |                   |              |
| <b>CENTRAL SECTOR</b> |                       |                   |              |
| BBMB                  | 9700.00               | 11583.17          |              |
| DVC                   | 286.00                | 176.28            |              |

|                        |                   |                   |  |
|------------------------|-------------------|-------------------|--|
| NEEPCO.                | 4937.00           | 4848.09           |  |
| NHDC                   | 3265.00           | 4470.50           |  |
| NHPC                   | 5429.00           | 0.00              |  |
| NHPC                   | 25628.00          | 21585.84          |  |
| NTPC Ltd.              | 3100.00           | 2952.05           |  |
| SJVNL                  | 9080.00           | 8088.21           |  |
| THDC                   | 4160.00           | 4441.69           |  |
| <b>CENTRAL SECTOR</b>  | <b>65585.00</b>   | <b>58145.83</b>   |  |
| <b>STATE SECTOR</b>    | <b>75469.00</b>   | <b>61667.64</b>   |  |
| <b>PVT SECTOR</b>      | <b>1470.00</b>    | <b>1557.60</b>    |  |
| <b>IPP SECTOR</b>      | <b>14176.00</b>   | <b>12682.85</b>   |  |
| <b>TOTAL HYDRO</b>     | <b>156700.00</b>  | <b>134053.92</b>  |  |
| <b>BHUTAN IMP</b>      |                   |                   |  |
|                        | 8000.00           | 4716.10           |  |
| <b>ALL INDIA TOTAL</b> | <b>1535000.00</b> | <b>1513256.36</b> |  |

**Note:** PLF is calculated for Coal & Lignite based power station only. Based on 2023-24 Actual Generation (Up to March, 24) from conventional sources Thermal/Hydro/Nuclear) with station capacity > 25 MW.

### 10.3 Generation Capacity Addition

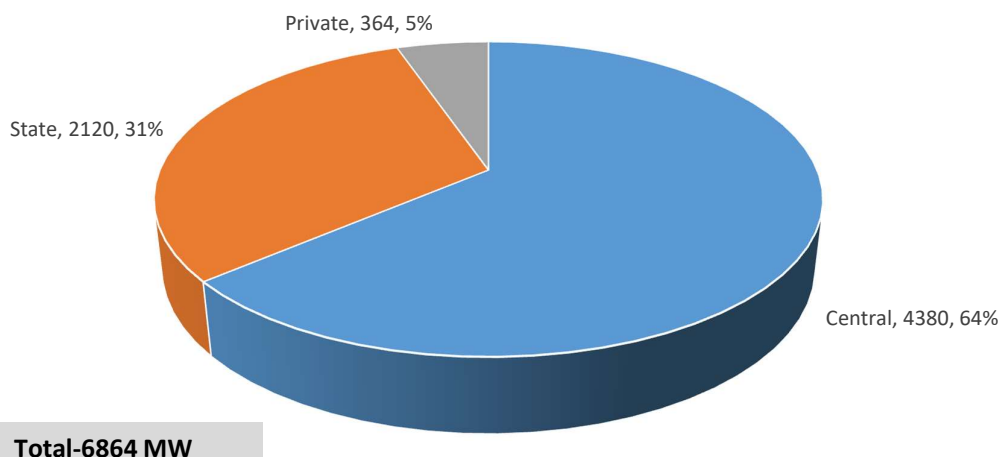
During the year 2023-24, a total of 6,864 MW generation capacity was added from conventional sources. The capacity addition during the last 10 years Sector-wise and mode-wise are given below:

#### Generation Capacity addition during the last 10 years – Sector-wise

(Figures in MW)

| Year    | Central Sector | State Sector | Private Sector | Total           |
|---------|----------------|--------------|----------------|-----------------|
| 2014-15 | 4395.21        | 4886.10      | 13285.00       | <b>22566.31</b> |
| 2015-16 | 3775.60        | 7070.00      | 13131.00       | <b>23976.60</b> |
| 2016-17 | 4310.50        | 5177.30      | 4722.00        | <b>14209.80</b> |
| 2017-18 | 3560.00        | 1960.00      | 3985.00        | <b>9505.00</b>  |
| 2018-19 | 2070.00        | 2879.755     | 972.00         | <b>5921.755</b> |
| 2019-20 | 4240.00        | 2780.00      | 45.00          | <b>7065.00</b>  |
| 2020-21 | 4380           | 957.15       | 99             | <b>5436.15</b>  |
| 2021-22 | 2370           | 1590         | 918            | <b>4878</b>     |
| 2022-23 | 660            | 920          | 0              | <b>1580</b>     |
| 2023-24 | 4380           | 2120         | 364            | <b>6864</b>     |

### Sector-wise generation capacity addition (in MW) achieved in 2023-24

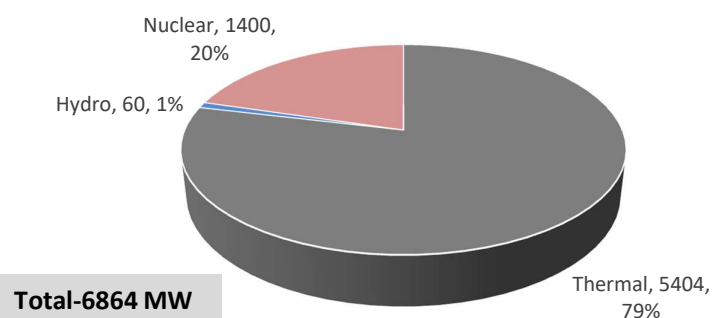


### Generation Capacity addition during the last 10 years – Mode-wise

(Figures in MW)

| Year    | Thermal  | Hydro   | Nuclear | Total    |
|---------|----------|---------|---------|----------|
| 2014-15 | 20830.30 | 736.00  | 1000.00 | 22566.31 |
| 2015-16 | 22460.60 | 1516.00 | 0.00    | 23976.60 |
| 2016-17 | 11550.80 | 1659.00 | 1000.00 | 14209.80 |
| 2017-18 | 8710.00  | 795.00  | 0.00    | 9505.00  |
| 2018-19 | 5781.755 | 140.00  | 0.00    | 5921.755 |
| 2019-20 | 6765.00  | 300.00  | 0.00    | 7065.00  |
| 2020-21 | 4926.15  | 510.00  | 0.00    | 5436.15  |
| 2021-22 | 4485     | 393     | 0.00    | 4878     |
| 2022-23 | 1460     | 120     | 0.00    | 1580     |
| 2023-24 | 5404     | 60      | 1400    | 6864     |

### Mode-wise generation capacity addition (in MW) achieved in 2023-24



## 10.4 Installed Electricity Generating Capacity

Total All India Installed Electricity Generating Capacity, as on 31.03.2024 is 441969.55 MW comprising of Thermal 243216.87 MW, Hydro 46928.17 MW, Nuclear 8180.00 MW and 143644.51 MW from Renewable Energy Sources (RES). The details are shown in the Tables given below:

### All India Installed Electricity Generating Capacity- Sector wise

| Type         | Central Sector (MW) | State Sector (MW) | Private Sector (MW) | Total (MW)      |
|--------------|---------------------|-------------------|---------------------|-----------------|
| THERMAL      | 78897.87            | 77879.87          | 86439.14            | 243216.9        |
| HYDRO        | 15742.72            | 27254.45          | 3931                | 46928.17        |
| NUCLEAR      | 8180                | 0                 | 0                   | 8180            |
| RES*         | 1632.3              | 2535.112          | 139477.1            | 143644.5        |
| <b>Total</b> | <b>104452.9</b>     | <b>107669.4</b>   | <b>229847.2</b>     | <b>441969.6</b> |

\*Sector wise breakup of RES capacity is provisional.

The growth of installed generating capacity in the country is shown in the table below:

### Growth of Installed generating capacity in the country- Mode wise

| Year        | Thermal | Nuclear | Hydro | RES* | Total  |
|-------------|---------|---------|-------|------|--------|
| Dec.1947    | 854     | -       | 508   | -    | 1362   |
| Dec.,1955   | 1755    | -       | 940   | -    | 2695   |
| March, 1961 | 2736    | -       | 1917  | -    | 4653   |
| March, 1966 | 4903    | -       | 4124  | -    | 9027   |
| March, 1974 | 9058    | 640     | 6966  | -    | 16664  |
| March, 1980 | 16424   | 640     | 11384 | -    | 28448  |
| March, 1985 | 27030   | 1095    | 14460 | -    | 42585  |
| March, 1990 | 43764   | 1565    | 18307 | -    | 63636  |
| March, 1991 | 45768   | 1565    | 18753 | -    | 66086  |
| March, 1992 | 48086   | 1785    | 19194 | -    | 69065  |
| March, 1996 | 60083   | 2225    | 20986 | -    | 83294  |
| March, 1997 | 61012   | 2225    | 21658 | 900  | 85795  |
| March, 1998 | 64005   | 2225    | 21904 | 968  | 89102  |
| March, 1999 | 67566   | 2225    | 22479 | 1024 | 93294  |
| March, 2000 | 70193   | 2680    | 23857 | 1155 | 97885  |
| March, 2001 | 72343   | 2860    | 25153 | 1270 | 101626 |
| March, 2002 | 74429   | 2720    | 26269 | 1628 | 105046 |
| March, 2003 | 76762   | 2720    | 26767 | 1628 | 107877 |
| March, 2004 | 77969   | 2720    | 29507 | 2488 | 112684 |

|             |        |      |       |        |        |
|-------------|--------|------|-------|--------|--------|
| March, 2005 | 80902  | 2770 | 30942 | 3812   | 118426 |
| March, 2006 | 82410  | 3360 | 32326 | 6191   | 124287 |
| March, 2007 | 86015  | 3900 | 34654 | 7760   | 132329 |
| March, 2008 | 91907  | 4120 | 35909 | 11125  | 143061 |
| March, 2009 | 93725  | 4120 | 36878 | 13242  | 147965 |
| March, 2010 | 102454 | 4560 | 36863 | 15521  | 159398 |
| March, 2011 | 112824 | 4780 | 37567 | 18455  | 173626 |
| March, 2012 | 131603 | 4780 | 38990 | 24504  | 199877 |
| March, 2013 | 151531 | 4780 | 39491 | 27542  | 223344 |
| March, 2014 | 168255 | 4780 | 40531 | 35850  | 249416 |
| March, 2015 | 188898 | 5780 | 41267 | 39950  | 275895 |
| March, 2016 | 210675 | 5780 | 42783 | 47091  | 306330 |
| March, 2017 | 218330 | 6780 | 44478 | 58558  | 328146 |
| March, 2018 | 222907 | 6780 | 45293 | 70651  | 345631 |
| March, 2019 | 226279 | 6780 | 45399 | 79412  | 357871 |
| March, 2020 | 230600 | 6780 | 45699 | 88255  | 371334 |
| March, 2021 | 234728 | 6780 | 46209 | 95803  | 383521 |
| March, 2022 | 236109 | 6780 | 46723 | 109885 | 399497 |
| March, 2023 | 237269 | 6780 | 46850 | 125160 | 416059 |
| March, 2024 | 243217 | 8180 | 46928 | 190573 | 441970 |

\*Renewable Energy Sources (RES) includes Wind, Small Hydro Project, Biomass Gasifier, Biomass Power, and Urban & Industrial Waste Power & Solar Power.

All India Installed Capacity (in MW) of Power Stations located in the Regions of Main Land and Islands (as on 31.03.2024) are given at **Annexure-10B**.

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## CHAPTER – 11

## POWER DEVELOPMENT IN NORTH-EASTERN REGION

## 11.1 Hydro-electric Potential in N.E. Region

As per Re-assessment studies of 2017-23 carried out by CEA, hydro potential of the North Eastern Region in terms of installed capacity has been estimated as 55929.7 MW (above 25 MW

capacity). Out of the above, 2027 MW (above 25 MW capacity) have been harnessed so far while projects amounting to 5000 MW (above 25 MW capacity) are under construction. State-wise identified hydro-electric potential (above 25 MW) of North-Eastern Region and its status of development is given below:

| Region / State     | Identified potential as per Re-assessment Study of 2017-23 (Above 25 MW) | H. E. Schemes Developed (Above 25 MW) | H.E. Schemes Under Construction (Above 25 MW) |
|--------------------|--------------------------------------------------------------------------|---------------------------------------|-----------------------------------------------|
| Meghalaya          | 2026                                                                     | 322                                   | 0                                             |
| Tripura            | 0                                                                        | 0                                     | 0                                             |
| Manipur            | 615                                                                      | 105                                   | 0                                             |
| Assam              | 643                                                                      | 350                                   | 120                                           |
| Nagaland           | 325                                                                      | 75                                    | 0                                             |
| Ar. Pradesh        | 50394                                                                    | 1115                                  | 4880                                          |
| Mizoram            | 1926.7                                                                   | 60                                    | 0                                             |
| <b>Total(NER):</b> | <b>55929.7</b>                                                           | <b>2027</b>                           | <b>5000</b>                                   |

| Region / State    | H. E. Schemes Concurred by CEA (MW) | H. E. Schemes Under Examination in CEA (MW) | H. E. Schemes Returned to Project authorities (MW) | H. E. Schemes under S&I (MW)* | H. E. Schemes for which S&I is held up (MW) | Balance Capacity (MW) |
|-------------------|-------------------------------------|---------------------------------------------|----------------------------------------------------|-------------------------------|---------------------------------------------|-----------------------|
| Meghalaya         | 85                                  | 0                                           | 0                                                  | 270                           | 170                                         | 1434                  |
| Tripura           | 0                                   | 0                                           | 0                                                  | 0                             | 0                                           | 0                     |
| Manipur           | 0                                   | 0                                           | 0                                                  | 0                             | 0                                           | 510                   |
| Assam             | 0                                   | 0                                           | 60                                                 | 0                             | 0                                           | 143                   |
| Nagaland          | 186                                 | 0                                           | 0                                                  | 0                             | 0                                           | 64                    |
| Ar. Pradesh       | 13798                               | 0                                           | 3523                                               | 19991                         | 145                                         | 9566                  |
| Mizoram           | 0                                   | 0                                           | 0                                                  | 0                             | 0                                           | 1866.70               |
| <b>Total(NER)</b> | <b>14069</b>                        | <b>0</b>                                    | <b>3583</b>                                        | <b>20261</b>                  | <b>315</b>                                  | <b>13583.70</b>       |

\* Including two multipurpose projects namely Upper Siang (11200 MW) in Arunachal Pradesh and Simsang Dam (60 MW) in Meghalaya

## 11.2 Survey & Investigation of Hydro Projects

As on 31.03.2024, 9 No. of schemes aggregating to 9001 MW are under Survey and Investigation in the North Eastern Region.

## 11.3 Status of development

Hydro Electric Projects being planned in the North Eastern Region are as under:

| S. No. | Name of Project       | Agency                                                  | State             | Present Status                                                                                                                       |
|--------|-----------------------|---------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Demwe Lower (1750 MW) | Athena Energy Venture (P) Ltd.                          | Arunachal Pradesh | Concurrence accorded by CEA on 20.11.2009. MoP vide letter dated 22.12.2021 indicated that Demwe Lower HEP to be pursued by THDC.    |
| 2      | Dibbin (120 MW)       | KSK Dibbin Hydro Power Limited                          | Arunachal Pradesh | Concurrence accorded by CEA on 04.12.2009. MoP vide letter dated 22.12.2021 indicated that Dibbin HEP to be pursued by NEEPCO.       |
| 3      | Lower Siang (2700 MW) | Jaiprakash Associates Ltd.                              | Arunachal Pradesh | Concurrence accorded by CEA on 16.02.2010. MoP vide letter dated 22.12.2021 indicated that Lower Siang HEP to be pursued by NHPC.    |
| 4      | Nafra (120 MW)        | NEEPCO                                                  | Arunachal Pradesh | Concurrence accorded by CEA on 11.02.11. MoU between NEEPCO and Govt of Arunachal Pradesh signed on 14.08.2021.                      |
| 5      | Tawang-I (600 MW)     | NHPC Ltd.                                               | Arunachal Pradesh | Concurrence accorded by CEA on 10.10.2011. MoP vide letter dated 22.12.2021 indicated that Tawang-I HEP to be pursued by NEEPCO.     |
| 6      | Tawang-II (800 MW)    | NHPC Ltd.                                               | Arunachal Pradesh | Concurrence accorded by CEA on 22.09.2011. MoP vide letter dated 22.12.2021 indicated that Tawang-II HEP to be pursued by NEEPCO.    |
| 7      | Tato-II * (186 MW)    | NEEPCO. (previous developer- Tato Hydro Power Pvt. Ltd) | Arunachal Pradesh | Concurrence accorded by CEA on 22.05.12. MoA signed by NEEPCO with GoArP on 12.08.2023.                                              |
| 8      | Hirong* (500 MW)      | NEEPCO (Previous developer- Jaiprakash Associates Ltd.) | Arunachal Pradesh | Concurrence accorded by CEA on 10.04.2013. MoA signed by NEEPCO with GoArP on 12.08.2023.                                            |
| 9      | Etalín* (3097 MW)     | SJVNL (Previous developer- Etalín H.E. Power Co. Ltd.)  | Arunachal Pradesh | Concurrence accorded by CEA on 12.07.2013. MoA signed by SJVNL with GoArP on 12.08.2023.                                             |
| 10     | Talong Londa (225 MW) | GMR                                                     | Arunachal Pradesh | Concurrence accorded by CEA on 16.08.2013. MoP vide letter dated 22.12.2021 indicated that Talong Londa HEP to be pursued by NEEPCO. |
| 11     | Naying* (1000 MW)     | NEEPCO (Previous developer- D.S. Construction Ltd)      | Arunachal Pradesh | Concurrence accorded by CEA on 11.09.2013. MoA signed by NEEPCO with GoArP on 12.08.2023.                                            |

|    |                                         |                                                                    |                      |                                                                                                                                                                                                             |
|----|-----------------------------------------|--------------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12 | Siyom<br>(1000 MW)                      | Siyota Hydro power<br>Pvt. Ltd                                     | Arunachal<br>Pradesh | Concurrence accorded by CEA on 17.12.13.                                                                                                                                                                    |
| 13 | Dikhu<br>(186 MW)                       | Naga Manu Power<br>Private Ltd.                                    | Nagaland             | Concurrence accorded by CEA on 31.03.14.                                                                                                                                                                    |
| 14 | Kalai-II*<br>(1200 MW)                  | THDCIL<br>(Previous developer-<br>Kalai Power Pvt.<br>Ltd.)        | Arunachal<br>Pradesh | Concurrence accorded by CEA on 27.03.2015.<br>MoA signed by THDCL with GoArP on<br>30.12.2023.                                                                                                              |
| 15 | Kynshi – I<br>(270 MW)                  | Athena Kynshi<br>power Pvt.Ltd.                                    | Meghalaya            | Concurrence accorded by CEA on 31.3.2015.                                                                                                                                                                   |
| 16 | Heo *<br>(240 MW)                       | NEEPCO<br>(previous developer-<br>Heo Hydro Power<br>Pvt. Ltd.)    | Arunachal<br>Pradesh | Concurrence accorded by CEA on 28.07.15.<br>MoA signed by NEEPCO with GoArP on<br>12.08.2023.                                                                                                               |
| 17 | Tato-I *<br>(186 MW)                    | NEEPCO<br>(Previous developer-<br>Siyota Hydro Power<br>Pvt. Ltd.) | Arunachal<br>Pradesh | Concurrence accorded by CEA on 28.10.15.<br>MoA signed by NEEPCO with GoArP on<br>12.08.2023.                                                                                                               |
| 18 | Dibang<br>(2880MW)                      | NHPC Ltd.                                                          | Arunachal<br>Pradesh | Concurrence accorded by CEA on 18.09.2017.                                                                                                                                                                  |
| 19 | Attunli*<br>(680 MW)                    | SJVNL<br>(Previous developer-<br>Attunli H.E. Power<br>Co. Ltd.)   | Arunachal<br>Pradesh | Concurrence accorded by CEA on 15.03.2018.<br>MoA signed by SJVNL with GoArP on 12.08.2023.                                                                                                                 |
| 20 | Wah-Umiam<br>Stage-III<br>(85 MW)       | NEEPCO                                                             | Meghalaya            | Concurrence accorded by CEA on 26.07.2021                                                                                                                                                                   |
| 21 | Ranganadi<br>St-II (130MW)              | NEEPCO                                                             | Arunachal<br>Pradesh | DPR was returned to developer for re-submission<br>after tying-up of requisite inputs.                                                                                                                      |
| 22 | Karbi Langpi<br>(U. Borpani)<br>(60 MW) | Assam State<br>Electricity Board                                   | Assam                | DPR was returned to developer for re-submission<br>after tying-up of requisite inputs.                                                                                                                      |
| 23 | Yamne St-II<br>(84 MW)                  | SS Yamne Energy<br>Ventures Private<br>Ltd.                        | Arunachal<br>Pradesh | DPR was returned to developer for re-submission<br>after tying-up of requisite inputs.                                                                                                                      |
| 24 | Pemashelphu<br>(90 MW)                  | Mechuka Hydro<br>Power pvt. Ltd.                                   | Arunachal<br>Pradesh | DPR was returned to developer for re-submission<br>after tying-up of requisite inputs.                                                                                                                      |
| 25 | Sissiri<br>(100 MW)                     | Soma Sissiri Hydro<br>Pvt. Ltd.                                    | Arunachal<br>Pradesh | DPR was returned to developer for re-submission<br>after tying-up of requisite inputs.                                                                                                                      |
| 26 | Gimliang<br>(80 MW)                     | SKI Pvt. Ltd.                                                      | Arunachal<br>Pradesh | DPR was returned to developer for re-submission<br>after tying-up of requisite inputs.                                                                                                                      |
| 27 | Raigam<br>(141 MW)                      | SKI Pvt. Ltd.                                                      | Arunachal<br>Pradesh | DPR was returned to developer for re-submission<br>after tying-up of requisite inputs.                                                                                                                      |
| 28 | Kangtang Shiri<br>(80 MW)               | Kangtang Shiri<br>Hydro Project Pvt.<br>Ltd                        | Arunachal<br>Pradesh | DPR was returned to developer for re-submission<br>after tying-up of requisite inputs.                                                                                                                      |
| 29 | Nyukcharang<br>Chu (96 MW)              | Sew Energy Ltd.                                                    | Arunachal<br>Pradesh | DPR was returned to developer for re-submission<br>after tying-up of requisite inputs.                                                                                                                      |
| 30 | Magochu<br>(96 MW)                      | Sew MagoChu<br>Power Corporation<br>Limited                        | Arunachal<br>Pradesh | DPR was returned and all the partial clearances<br>issued till date were rescinded as no progress has<br>been made by the Developer towards resolving the<br>issues pending with various appraising groups. |

|    |                                     |                                            |                   |                                                                                                                                                                                                                                                                            |
|----|-------------------------------------|--------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31 | Subansiri Middle (Kamala) (1800 MW) | Kamala HECL (Jindal Power Ltd.)            | Arunachal Pradesh | DPR was returned and all the partial clearances issued till date were rescinded as no progress has been made by the Developer towards resolving the issues pending with various appraising groups.                                                                         |
| 32 | Hutong- II (1200 MW)                | Mountain Fall India Pvt. Ltd.              | Arunachal Pradesh | DPR was returned to developer for re-submission after tying-up of requisite inputs.                                                                                                                                                                                        |
| 33 | Kalai-I (1352 MW)                   | Mountain Fall India Pvt. Ltd.              | Arunachal Pradesh | DPR was returned to developer for re-submission after tying-up of requisite inputs.                                                                                                                                                                                        |
| 34 | Tagurshit (74 MW)                   | Larsen & Toubro Arunachal Hydro power Ltd. | Arunachal Pradesh | Developer vide letter dated 28.08.2018 informed that the company has decided not to go ahead with implementation of the project. In view of this, CEA returned the DPR vide its letter dated 20.06.2019 as the scheme is no more under consideration for CEA's concurrence |

\*MoP vide letter dated 22.12.2021 indicated that projects to be pursued by central PSUs.

## 11.4 Status of Various Hydro Power Projects in North-Eastern Region Appraised by CEA

### 11.4.1 Revised Cost Estimates

#### (i) Tuirial HEP (2x30=60 MW), Mizoram, NEEPCO

The project was cleared by CEA in July, 1998 at an estimated cost of ₹368.72 crores with likely completion by 2006-07. Project was to be financed substantially under Loan assistance of 11,695 Million Japanese Yen from Japan Bank of International Co-operation (JICA). This project was under execution and subsequently put on hold since June, 2004 due to poor law & order conditions and agitation by claimants of crop compensation.

Continuation or otherwise of the project was reviewed due to increase in the project cost and resumption of work was dependent upon viability of the project. CEA on 3.11.05 informed MOP that the present day cost of the project at October 2004 price level was likely to be ₹687.80 crores (including IDC of ₹ 40.05 crores and financing charges ₹ 0.16 crores). The first year tariff at this cost being ₹ 3.69/Kwh., project at this cost/tariff appeared unviable. In the meantime, JICA discontinued loan and requested for prepayment of entire outstanding amount.

Efforts were made to revive the project and the revised cost estimates were vetted by CEA a

number of times and lastly vetted on 26.4.10 for the Hard cost of ₹ 877.06 crores at March, 10 P.L. PIB meeting was held on 4th June 2010 which recommended the project for CCEA approval.

CCEA approval was accorded to the project on 14.01.2011 for ₹913.63 crores including IDC of ₹36.57 crores at March, 2010 Price Level. The financial pattern of ₹913.63 crores comprises of (i) Equity of ₹ 137.04 Crs. (ii) Loan from financial institutions amounting to ₹ 184.63 crores (iii) Subordinate loan from Govt. of India amounting to ₹ 291.96 crores and (iv) Grant from DoNER amounting to ₹300 crores.

Cost estimates at completion level, submitted by NEEPCO, was vetted by CEA amounting to ₹1244.15 crores (Total project Cost) vide CEA letter dated 08.04.2019.

#### (ii) Pare HEP (2x55=110 MW), Arunachal Pradesh, NEEPCO

Pare HEP was accorded concurrence by CEA on 24th Sept. 2007 for an estimated cost of ₹553.25 crores including IDC & FC of ₹49.26 crores at June 2007 Price Level.

CCEA approval was accorded to the project on 04.12.2008 for ₹573.99 crores including IDC of ₹67.66 crores and FC of ₹0.40 crores at June, 2007 Price Level. The completion cost considering 44 months as construction period is estimated as ₹674.45 crores including IDC as ₹76.52 crores and FC as ₹0.47 crores.

Cost estimates at completion level, submitted by NEEPCO, was vetted by CEA amounting to ₹1640.31 crores (Total project Cost) vide CEA letter dated 25.02.2019.

**(iii) Subansiri Lower (8x250=2000 MW), Arunachal Pradesh, NHPC**

Subansiri Lower HE Project located in Lower Subansiri District of Arunachal Pradesh was accorded concurrence of CEA on 13.01.2003 for an estimated cost of ₹ 6608.68 Crores including IDC and FC of Rs705.58 Crores at December, 2002 price level.

CCEA approval was accorded to the project on 9.09.2003 for ₹6285.33 Crores including IDC and FC of ₹ 670.92 Crores at December, 2002 price level.

Memorandum of Changes (MoC) has been approved by CEA vide letter dated 15.03.2018. Revised cost estimates at April, 2017 price level, submitted by NHPC, was vetted by CEA amounting to ₹10601.16 crores (Total Hard Cost) vide CEA letter dated 18.07.2019.

**11.4.2 CEA concurred Projects, yet to be taken under construction.**

**(A) Central Sector Projects:**

**(i) Tawang H.E Project St-I (3x200= 600 MW) in Ar. Pradesh by NHPC Ltd.**

Project was accorded concurrence by CEA on 10.10.2011 at an estimated cost of ₹4824.01 Crores (including IDC & FC) at May, 2010 price level. MOP vide letter dated 22.12.2021 indicated that Tawang-I HEP to be pursued by NEEPCO.

Environment clearance was accorded on 10.06.2011. Forest clearance Stage-I & II yet to be obtained.

**(ii) Tawang H.E Project St.-II (4x200=800 MW) in Ar. Pradesh by NHPC Ltd.**

The project was concurred by CEA on 22.9.2011 at an estimated cost of ₹ 6112.3 crores (including IDC & FC) at May, 2010 price level. MOP vide letter dated 22.12.2021 indicated that Tawang-II HEP to be pursued by NEEPCO.

Project was accorded environment clearance on 10.06.2011. MoEF&CC vide letter dated 08.01.2014 has accorded Forest Clearance (Stage- I) for diversion of 116.62 ha forest land for the project. Forest clearance stage-II yet to be obtained.

**(iii) Hironag HE Project (4x125 =500MW) in Arunachal Pradesh by NEEPCO.**

Hironag H.E. Project was accorded concurrence by CEA on 10<sup>th</sup> April, 2013 at an estimated completed cost of ₹ 5532.63 Crores.

Environment clearance and Forest clearance are yet to be obtained. EIA/EMP report being revised as per Siang BSR. Project allotted to NEEPCO.

**(iv) Wah-Umum Stage-III (Erstwhile Mawphu Stage-II) H.E. Project (2x42.5= 85 MW)-Meghalaya**

The project was concurred by CEA on 26.07.2021 at an estimated cost of ₹ 965.40 crores at January, 2019 price level.

Environment clearance recommended by EAC on 26.02.2018. However, Environment clearance will be issued immediately on submission of Stage-I Forest Clearance. Forest Clearance yet to be obtained.

**(v) Naying HE Project (4x250 =1000MW) in Arunachal Pradesh by NEEPCO**

Naying H.E. Project was accorded concurrence by CEA on 11<sup>th</sup> Sept, 2013 at an estimated completed cost of ₹ 9301.11 Crores. Environment clearance and Forest clearance are yet to be obtained. Environment clearance is linked with Siang Basin Study Report. MoEF&CC stated that developer needs to apply afresh for EC online as old proposal is not valid anymore. Project allotted to NEEPCO.

Govt. of Arunachal Pradesh has approved the allotment of Naying HEP to NEEPCO Ltd. on 13.01.2023.

**(vi) Nafra HE Project (2x60= 120 MW) in Arunachal Pradesh by NEEPCO**

The project was concurred by CEA on 11.02.2011 at an estimated cost of ₹ 848.22 crores at completion cost. Project was accorded Environmental Clearance on 19.08.2013. Forest

Clearance-I accorded on 12.07.2011 & Forest Clearance-II accorded on 26.06.2012.

MoU between NEEPCO and Govt of Arunachal Pradesh signed on 14.08.2021.

## **(B) State Sector Projects**

NIL

## **(C) Private Sector Projects**

### **i) Demwe Lower HE Project (5x342 + 1x40=1750 MW), Arunachal Pradesh by ADPL**

Demwe Lower HE Project was accorded concurrence by CEA on 20.11.2009 for an estimated cost of ₹ 13144.91 Crores (Completion Cost). MOP vide letter dated 22.12.2021 indicated that Demwe lower HEP to be pursued by THDC.

MoEF&CC has accorded Environmental clearance to the project on 12.2.10. Forest clearances stage-II has been accorded on 03.05.2013. As per NGT order dated 24.10.2017, NBWL issue to be reconsidered by MoEF&CC.

Project was under litigation at NCLT. THDC had filed the Intervention Application in NCLAT. The arguments of CoC, RP, and THDCIL have been concluded. Navayuga has been requested to submit their counter reply on THDCIL's submission and the matter heard for adjudicating the reply of Navayuga on 24.01.2024. Hon'ble NCLAT has reserved the judgment.

### **ii) Lower Siang HE Project (9x300=2700 MW), Ar. Pradesh by JAPL**

Lower Siang HE Project was accorded concurrence by CEA on 15.02.2010 for an estimated cost of ₹ 19990.74 Crores (Completion Cost).

Environment clearance & Forest clearance are yet to be obtained. MoP vide letter dated 22.12.2021 indicated that Lower Siang HEP to be pursued by NHPC. Techno-commercial aspects of Siang Lower are un-certain at this stage and shall only be firmed up after fixing the parameters of Siang Upper.

### **iii) Etalin HE Project (10x307+1x9.6+1x7.4 = 3097MW) in Arunachal Pradesh by EHEPCL**

Etalin H.E. Project was accorded concurrence by CEA on 12<sup>th</sup> July, 2013 at an Estimated completed cost of ₹ 25296.95 Crores. MOP vide

letter dated 22.12.2021 indicated that Etalin HEP to be pursued by SJVNL.

Environment clearance recommended by EAC on 31.01.17. Letter will be issued after Forest clearance stage-I. Forest clearance stage-I & II are yet to be obtained. Project allotted to SJVNL.

### **iv) Talong Londa HE Project (3x75 = 225MW) in Arunachal Pradesh by GMR**

Talong Londa H.E. Project was accorded concurrence by CEA on 16<sup>th</sup> Aug, 2013 at an estimated completed cost of ₹2172.88 Crores. MOP vide letter dated 22.12.2021 indicated that Talong Londa HEP to be pursued by NEEPCO.

Environment clearance accorded on 07.08.15. Forest clearance stage-I& II are yet to be obtained.

### **v) Kalai – II HE Project (6x200 = 1200MW) in Arunachal Pradesh by KPPL**

Kalai – II H.E. Project was accorded concurrence by CEA on 27<sup>th</sup> March, 2015 at an estimated completed cost of ₹ 14199.64 Crores. MOP vide letter dated 22.12.2021 indicated Kalai-II HEP to be pursued by THDC.

Environment clearance has been accorded on 20.05.2015. Forest clearance Stage -I&II are yet to be obtained. . Project allotted to THDC.

### **vi) Heo HE Project (3x80 = 240MW) in Ar. Pradesh by HHPPL**

Heo H.E. Project was accorded concurrence by CEA on 28.07.2015 at an estimated completed cost of ₹ 1614.35 Crores. MOP vide letter dated 22.12.2021 indicated that Heo HEP to be pursued by NEEPCO. Project allotted to NEEPCO.

Environmental Clearance accorded on 10.11.15. Forest clearance stage-I accorded on 27.10.15.

EC & FC-I transferred to NEEPCO.

Forest clearance stage-II yet to be obtained.

### **vii) Tato – I HE Project (3x62 = 186MW) in Ar. Pradesh by SHPPL**

Tato – I H.E. Project was accorded concurrence by CEA on 28.10.2015 at an estimated completed cost of ₹1493.55 Crores. MOP vide letter dated 22.12.2021 indicated that Tato-I HEP to be pursued by NEEPCO.

Environmental Clearance accorded on 10.11.15. Forest clearance stage-I accorded on 27.10.15. Forest clearance stage-II yet to be obtained. Project allotted to NEEPCO.

**viii) Attunli HE Project (4x170 = 680MW) in Ar. Pradesh by AHPCL**

Attunli H.E. Project was accorded concurrence by CEA on 15.03.2018 at an estimated completed cost of ₹ 6111.28 Crores. MOP vide letter dated 22.12.2021 indicated that Attunli HEP to be pursued by SJVNL. Environmental Clearance and Forest clearance are yet to be obtained.

**ix) Dikhu HE Project (3x62= 186 MW) in Nagaland by NMPPL**

Dikhu H.E. Project was accorded concurrence by CEA on 31.03.2014 at an estimated completed cost of ₹1994.74 Crores.

Environmental Clearance yet to be obtained. FC not applicable as forest land is not involved.

**x) Tato – II HE Project (4x175 = 700 MW) in Ar. Pradesh by THPPL**

Tato – II H.E. Project was accorded concurrence by CEA on 22.05.2012 at an estimated completed cost of ₹5616.20 Crores. MOP vide letter dated 22.12.2021 indicated that Tato-II HEP to be pursued by NEEPCO. Project allotted to NEEPCO.

Environmental Clearance accorded on 27.06.11. FC is linked to Cumulative Impact Assessment Study of Siang Basin which has been carried out and accepted by MoEF&CC.

**11.5 Status of Under Construction Hydro Power Projects in North Eastern Region including Sikkim:**

**11.5.1 Central Sector Projects**

• **NHPC Projects (Hydro)**

**(i) Subansiri Lower HEP (8x250 = 2000 MW), Arunachal Pradesh**

The project is located in the districts Lower Subansiri/Dhemaji in Arunachal Pradesh/Assam on river Subansiri. The project was Techno-Economically cleared by CEA on 13.01.2003. The CCEA clearance was accorded on 09.09.2003 for an estimated cost of Rs. 6285.33 crores with the schedule commissioning of the project in September, 2010. The design energy is 7421.59 Gwh. The anticipated cost of the project is Rs. 1992.43 crores at January-2020 price level.

The Project envisages construction of concrete gravity dam, horse shoe type head race tunnels, circular steel lined pressure shaft and surface power house having Francis turbine driven generating sets.

Major civil works have been awarded to M/s. BGS-SGS-Soma Joint Venture and Larsen & Toubro. Ltd. Chennai on 19.12.2003. E&M works has been awarded to Consortium of M/s Alstom Power Hydraulique, France and Alstom Projects India Ltd. New Delhi on 11.02.2005. Hydro-Mechanical Package awarded to Texmaco on 19.06.2006.

All work except safety works were stalled from December, 2011 to October, 2019 due to agitation launched by various activists against construction of Subansiri Lower HE Project and as per directions of NGT. Works restarted w.e.f. 15.10.2019 after clearance from NGT. However, work initially remained suspended w.e.f. 24.03.2020 to 20.04.2020 due to COVID-19 lockdown and further got affected due to Monsoon Period from May 2020.

Power House Civil works package has been re-awarded to M/s Patel Engineering Ltd on 01.09.2020.

A massive landslide occurred near the inlet of the DT-1 & DT-2 on 27.10.2023. This land slide washed out both the inlets. The muck of the landslides blocked the river flow through DT-1.

The project is in advance stage of construction and three units (3x250 MW) are likely to be to be commissioned in FY 2024-25 and remaining five units in FY 2026-27.

**(ii) Teesta-VI HEP (4x125=500 MW), Sikkim**

The project is located in South Sikkim district of Sikkim state on river Teesta. The project was Techno-Economically cleared by CEA on 27.12.2006 to M/s Lanco Teesta Hydro Power Ltd (LTHPL), at an estimated cost of Rs 3283.08 Crs. The project envisages construction of 23.5m high Barrage, 2 nos. of HRT of 9.5m diameter and 11.8 Km long, 4 nos. Pressure shaft each of 5.40m dia and Power House to generate 2441 MU.

Major Civil works were awarded to M/s Lanco Infrastructure Ltd in March, 2007 and E&M works to M/s Alstom Projects, India in April, 2009. About 50% projects works were completed till March, 2014. Since April 2014, project was stalled due to financial crunch with the developer.

Accordingly, the Corporate Insolvency Resolution Process (CIRP) was initiated vide order dated 16.03.2018 of Hon'ble NCLT, Hyderabad Bench. In the Bidding process, NHPC emerged as successful bidder for acquisition of LTHPL. Subsequently, the investment proposal for an estimated cost of Rs 5748.04 crore (Jul'18 PL), which includes Bid amount of Rs 907 crore for acquisition of LTHPL; was approved by the CCEA on 08.03.2019 for investment, acquisition of M/s LTHPL and execution of balance works of Teesta-VI HE Project by NHPC.

Regarding, tendering, LOT-I (Civil works of Barrage, Desilting Basins, SFT, Intake, Part of HRT-I & HRT-II and other associated structure) awarded to M/s. Jaypee Associate Ltd on 31.03.2020 and LOT-II (Civil works of Part of

HRT-I & HRT-II, Surge shaft, Pressure shaft, Powerhouse, TRT & other associated structure) awarded to M/s. Patel Engineering Limited on 22.09.2021. HM works were awarded on 27.10.2020. EM works were awarded on 28.08.2020. The Project is expected to be commissioned by Dec, 2027.

**(iii) Rangit-IV HEP (3x40=120 MW), Sikkim**

The project is located in West Sikkim district of Sikkim state on river Rangit. The project was Techno-Economically cleared by CEA on 06.07.2007 to M/s Jal Power Corp. Ltd (JPCL), at

an estimated cost of Rs 726.16 Crs with the schedule commissioning of the project in January, 2012. The design energy is 513 Gwh. The project envisages construction of 44m high and 112.95m long Dam, 1 no. of HRT of 6.40m diameter and 6.453 Km long, Surge Shaft 16m dia and 57m height, 1 no. Pressure shaft of 5.50m dia and 241m long.

Major Civil works were awarded to M/s Coastal Project Pvt. Ltd in Nov, 2007 and E&M works to M/s Andritz, India in Aug, 2009. About 50% projects works were completed till Oct, 2013. Since Nov. 2013, project was stalled due to financial crunch with the developer. The project Lenders file application in court of Hon'ble National Company Law Tribunal (NCLT), on 24<sup>th</sup> April, 2018. Last hearing of NCLT held on 29.03.2019 and order pronounced on 9.04.2019. As per the order, IRP has been appointed. NHPC Ltd. submitted EOI on dated 08.07.2019 and was shortlisted under final list of Prospective Resolution Applicants on dated 23.08.2019. The Resolution Plan submitted by NHPC on 04.12.2019.

Lot-I balance Civil works (Diversion Channel, Coffor Dams, Dam, Spillway & Stilling Basin, Intake Structure, Desilting Channel, SFT, HRT, Surge Shaft, Pressure Shaft, Power House, TRC and other associated structures.) was awarded to M/s PES Engg. Pvt. Ltd on 28.06.2021 and to M/s Andritz Hydro Pvt. Ltd. on 08.07.2021 respectively. The Project is expected to be commissioned by May, 2025.

**(iv) Dibang Multipurpose Project (12x240=2880MW)-Arunachal Pradesh:**

The project is in initial stage of construction. The Project is expected to be commissioned by 2031-32.

**11.5.2 Sector Projects****APGCL Project (Hydro)****Lower Kopili (2x55+2x2.5+1x5 = 120 MW)**

The project is located in Dima Hasao & Karbi Anglong districts of Assam on Kopili river. Concurrence of project was given by CEA on 24.5.2016 at an estimated cost of Rs. 1115.91 crore

(at January, 2015 PL). The design energy of project is 469.58 MU and scheduled to be commissioned in 48 months from Zero date. The project has been delayed as there was delay in getting the Stage-II Forest Clearance for handing over the forest land. The Forest department of Karbi Anglong handed over the Forest land to APGCL on 3rd July, 2021. The revised cost as estimated by developer is Rs. 1795 crore. The project achieved the financial closure and loan agreement has been signed with ADB on 30th December, 2020. The Civil and Hydro-mechanical packages was awarded to L&T Ltd. and agreement was signed on 21.8.2020. The Electro-mechanical packages was awarded to Andritz Hydro Pvt. Ltd. on 9.9.2021.

The project envisages construction of concrete gravity dam with 1 no. of HRT of 7.0m diameter and 3.64 Km long, Surge Shaft of Restricted Orifice type 25m dia, Pressure shaft of 6.10m dia and 451.20m length. One surface power house for two units of 55 MW with rated head of 114 meter and one Auxilary power house for remaining capacity of 10 MW (2x2.5=5 MW + 1x5 MW). Switchyard

to main power house is at voltage level of 220 kV. The expected COD of the project is Oct, 2025.

### 11.6 Hydro Power Generation Performance

Hydro Power generation during the year 2022-24 (as on 31.03.2024) in the North Eastern Region was 6.28 BU against a target of 12.16 BU, which is about 48.31 % less.

### 11.7 R&M Schemes (Hydro) of North Eastern Region

Twelve (12) existing hydro schemes of North Eastern Region with an aggregate installed capacity of 770 MW have been identified for R&M works to accrue a benefit of 538 MW. The R&M activities of eight (8) schemes have already been completed at an actual expenditure of about Rs. 259 Crores to accrue a benefit of 121 MW. The remaining four (4) schemes having an aggregate installed capacity of 411 MW are under various stages of implementation and are likely to accrue a benefit of 417 MW at an estimated cost of about Rs. 2035.41 Crores. The scheme-wise status of the R&M works of the hydro schemes of North Eastern Region as on 31.03.2024 is given hereunder:

#### A. Schemes Completed

| S. No. | Name of Scheme, Agency, State  | Installed Cap. (MW) | Actual cost (Rs. Crs.) | Benefits (MW)  | Status                                     |
|--------|--------------------------------|---------------------|------------------------|----------------|--------------------------------------------|
| 1.     | Khandong, U-1, NEEPCO, Assam   | 1x25                | 0.62                   | 25 (Res.)      | U-1 Restoration works completed in 1991-92 |
| 2.     | Gumti, TPGL, Tripura           | 3x5                 | 17.50                  | -              | R&M works completed in 1994-95             |
| 3.     | Khandong, NEEPCO, Assam        | 2x25                | 3.35                   | -              | R&M works completed in 2003-04             |
| 4.     | Umium St.I, MePGCL, Meghalaya  | 4x9                 | 84.21                  | 36 (LE)        | RM&LE works completed in 2002-03           |
| 5.     | Loktak, NHPC, Manipur          | 3x30 (Derated)      | 17.88                  | 15(Res.)       | R&M works completed in 2011-12             |
| 6.     | Umium St.II, MePGCL, Meghalaya | 2x9                 | 55.67                  | 18(LE) + 2 (U) | R&M works completed in 2011-12             |
| 7.     | Kopili, NEEPCO, Assam          | 2x50                | 50.92                  | -              | R&M works completed in 2014-15             |
| 8.     | Khandong, NEEPCO, Assam        | 1x25                | 29.18                  | 25(LE)         | R&M works completed in 2014-15             |
|        | <b>Sub Total(A)</b>            | <b>359</b>          | <b>259.33</b>          | <b>121</b>     |                                            |

**B. Ongoing–Under Implementation**

| S. No.              | Name of Scheme, Agency, State                 | Installed Cap. (MW) | Est. cost (Rs. Crs.) | Benefits (MW) | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------|-----------------------------------------------|---------------------|----------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9.                  | Khandong Power Station, NEEPCO, Assam         | 2x23                | 278.63               | 46(LE)        | Work for E/M Package has been awarded. Procurement and manufacturing activities have been started. Civil package has been awarded. Work in progress. Work for H&M Equipment has been awarded                                                                                                                                                                                                                                                                                                    |
| 10.                 | Kopili Power Station, NEEPCO, Assam           | 4x50                | 1075.19              | 200(LE)       | Commercial operation of Unit 4 started w.e.f. 20.08.2023. Commercial operation of Unit 3 started w.e.f. 03.09.2023. Commercial operation of Unit 2 started w.e.f. 12.11.2023. Erection of Unit 1 is in progress.                                                                                                                                                                                                                                                                                |
| 11.                 | Kyrdemkulai (Umiam St.III), MePGCL, Meghalaya | 2x30                | 408                  | 60(LE) + 6(U) | Electro & Mechanical Equipments (Package-1) - 90% of design of E&M works completed and procurement has been started<br>Hydro Mechanical & Civil facility (Package-2) –under tendering.                                                                                                                                                                                                                                                                                                          |
| 12.                 | Loktak, NHPC, Manipur                         | 3x35                | 273.59               | 105 (LE)      | <b>E&amp;M:</b> Out of four packages, EM-2(Bus Duct) is awarded, the work under package EM-3 (EOT Crane) has been completed & work of EM-4 (DG Set) and EM-1 (Main) package is in progress.<br><b>Civil:</b> Three out of five Civil packages have been awarded. The work under package C1 and C2 has been completed and work under package C4 is in progress. Tendering of remaining two civil packages is under process. .<br><b>HM:</b> HM Package has been awarded and work is in progress. |
| <b>Sub Total(B)</b> |                                               | <b>411</b>          | <b>2035.41</b>       | <b>417</b>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Total(A+B)</b>   |                                               | <b>770</b>          | <b>2294.74</b>       | <b>538</b>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**Abbreviations: MW – Mega Watt; Res. – Restoration; U – Uprating; LE – Life Extension**

## CHAPTER-12

### HUMAN RESOURCE DEVELOPMENT

#### 12.0 Training of Manpower in CEA

Human Resource is essential for carrying out any business or service by an organization and the same is required to be developed through technical, functional and behavioral training. Keeping this in view, HRD Division of CEA has been organizing various training programmes in technical, functional, managerial and behavioral areas to keep officers abreast of the latest technological developments as well as to bring about attitudinal changes.

Training Policy for Group A & B officers of CPES Cadre of CEA broadly covers the various training needs for officers of all levels in CEA. The broad objectives of the Training Policy are as under: -

- To enable CPES officers of CEA to discharge their functions effectively.
- To provide practical exposure to the CPES officers in the area of construction and Operation & Maintenance (O&M) of various types of Power Plants as well as Transmission & Distribution facilities, Grid Operation, Tariff related issues, Power Market etc. which would enhance their technical competencies.
- To enable the officers to draw plans, advise and monitor Power Sector projects with the strong background knowledge/experience of the sector.
- To familiarize the officers with the best practices in the application of advanced technologies in Power Sector.
- To develop and enhance the capabilities in the CPES officers to deal with rapid developments and challenges encountered by the Power Sector from time to time.
- To enhance the managerial competencies of the officers to enable them to play a leading role in the Power Sector so that the management can channelize the expertise of CEA officers in an effective manner.

26 nos. of Assistant Director in 2023-24 of CPES Cadre was conducted for giving exposure in various aspects of power sector such as Generation, Transmission, Distribution, Grid Control, Energy Market, etc. It was conducted in various Organization like PGCIL, NTPC, GRID INDIA, CERC, BSES, TPDDL, IEX, PTC, NISE, and BHEL.

#### 12.2 Mid-Career/ Refresher Training Programmes

1. In 2023-24, a one-week mid-career training program was conducted for 18 Chief Engineers at THDC, Rishikesh. This training was aimed to enrich their existing knowledge with the latest insights and trends in the power sector, preparing the officers of CEA for leadership roles.
2. A one-week mid-career training program for 45 Directors of CEA, conducted in two batches at NPTI, Faridabad. The training focused on conflict resolution, project planning, managing effective meetings, leadership innovation, and emerging technology in the power sector.
3. A one-week mid-career training program for 103 Deputy Directors of CEA, conducted in four batches at NPTI, Faridabad. The training focused on the regulatory framework in the power sector, flexible operation of thermal power plants, innovations in distribution networks, green hydrogen, battery storage, carbon trading, communication, team building, and other related topics.
4. Various domain-related training programs for CEA officers were conducted at professional institutes such as CBIP, NPTI, NAHRD, and LBSNAA. During the year 2023-24, 120 officers were deputed to attend these programs, which included technical courses, workshops, seminars, and conferences.

#### 12.1 Induction Training programme

6 weeks Training programme for recently recruited

### **12.3 Training under Apprentice Act, 1961(Amendment rules 2015)**

As per Apprentice Act 1961, (amendment rules 2015), Apprenticeship Training is being imparted at CEA to Graduate/Diploma Engineers. As per the requirement of the Board of Apprentice Training (BOAT), six modules namely Planning of Power sector, Thermal Power Projects, Hydroelectric Power Project, Power System Planning, Power Grid Operation and Power Distribution System were developed and the Apprenticeship Training is being imparted as per these modules. During the year 2023-24, 4 apprentice trainees have joined & completed their apprenticeship under the Apprentice Act 1961.

### **12.4 Summer / Winter Internship Training**

During the financial year 2023-24, Internship training were given to a total 42 number of students in CEA from reputed institutes of India.

### **12.5 In-house Presentations**

In-house presentations are organized in collaboration with various industries and organizations to keep CEA officers updated on the latest technologies. During the year 2023-24, 13 technical presentations were conducted for CEA officers—both at Headquarters & officers posted in various Sub-Offices of CEA joined via video conferencing.

A Five one-day Thematic Capacity Building Programme, in compliance with the direction of CVC, was conducted for officers of CEA to provide general overview on Role of IO/PO in conducting enquiries, Public procurement, Ethics and Governance, Cyber hygiene and Security, Systems and procedures of the organizations etc. in which 110 officers/Officials of CEA participated respectively.

One Awareness lecture was organized on Civil services Conduct Rule 1965 (CCS Rules 1965) and role of IO/PO in Conducting enquiries procedures for officers/Officials of CEA. The lecture was attended by 160 officers.

### **12.6 Annual Capacity Building Plan:**

Under the direction of the Ministry of Power a Capacity Building Unit (CBU) was formed within the Central Electricity Authority in July 2023 to prepare the Annual Capacity Building Plan (ACBP) for CEA. The ACBP aims to identify gaps in manpower capacity, infrastructure, training needs, and other areas to effectively carry out the work of the CEA's divisions/wings.

The ACBP is being prepared in collaboration with the consultant M/s CRISIL and the Capacity Building Commission, under the guidance of the CBU.

### **12.7 Recognition of Training Institutes**

To fulfill its statutory duty specified under Section 73(g) of Electricity Act 2003 and Central Electricity Authority (measures relating to safety and electric Supply) Regulations 2023, CEA has been assessing the Power Sector training institutes in terms of infrastructure, utilization and quality of training programmes and facilitating CEA's accreditation for them in line with the CEA Guidelines for Recognition for Training Institutes for Power Sector. CEA has been advising /recommending various measures to the training institutes for improvement in the training infrastructure and methodologies for enhancing the skills and productivity of the personnel.

During the period 2023-24, the following 22 nos. training institutes were granted recognition in various fields of Power Sector: -

| S.No. | Name of the Institute                                                         | Field                                                                 |
|-------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1     | BSES Rajdhani Power Limited's Training Centre (Knowledge Centre)              | Distribution                                                          |
| 2     | Power Engineer's Training & Research Centre, Moolamattom, Idukki, Kerala      | Hydro, Transmission<br>Distribution, Power<br>Management              |
| 3     | Hot Line Training Institute – NPTI, Bangalore                                 | Transmission,<br>Distribution                                         |
| 4     | Regional Training Centre (RTC) MSETCL, Akola (Maharashtra)                    | Transmission                                                          |
| 5     | Power Management Institute (PMI), Noida (U.P.)                                | Thermal, Hydro,<br>Power Management                                   |
| 6     | DVC Training Institute at Chandrapura TPS, Distt.,Bokaro (Jharkhand)          | Thermal, Hydro,<br>Transmission,<br>Distribution, Power<br>Management |
| 7     | Power Station Training Centre (PSTC), NLC India Limited, Neyveli (Tamilnadu)  | Thermal                                                               |
| 8     | National Power Training Institute (SR), Neyveli (T.N.)                        | Thermal, Hydro,<br>Transmission,<br>Distribution, Power<br>Management |
| 9     | National Power Training Institute (ER), Durgapur, West Bengal                 | Thermal, Hydro,<br>Transmission,<br>Distribution, Power<br>management |
| 10    | TPSDI, Shahad, Thane, Maharastra                                              | Transmission,<br>Distribution                                         |
| 11    | OHPC Training Centre, Bhubaneswar, Orissa                                     | Hydro                                                                 |
| 12    | OPTCL Power Training Centre, Bhubaneshwar ,Orissa                             | Transmission,<br>Distribution                                         |
| 13    | Hitachi Energy TEC, Maneja, Vadodara (Gujarat)                                | Transmission,<br>Distribution                                         |
| 14    | Gujarat Energy Knowledge Centre, Bhavnagar, Gujarat                           | Distribution                                                          |
| 15    | Power Generation Training Institute(PGTI) of CSPGCL, Korba East (Chhatisgarh) | Thermal                                                               |
| 16    | Regional Training Center, MSETCL, Babhaleshwar, Ahmednagar, Maharashtra       | Transmission                                                          |
| 17    | TPSDI, Jojobera, Dist. Jamshedpur, Jharkhand                                  | Thermal                                                               |

|    |                                                                         |                                            |
|----|-------------------------------------------------------------------------|--------------------------------------------|
| 18 | NPTI, Faridabad (Haryana)                                               | Thermal, Hydro, Transmission, Distribution |
| 19 | Regional Training Center, Waluj, Aurangabad, EHV CC,O&M Zone Aurangabad | Transmission                               |
| 20 | NPTI, Guwahati, Khailipara, Guwahati, Assam                             | Thermal, Hydro, Transmission, Distribution |
| 21 | Tata Power Skill Development Institute, Mundra, Gujarat                 | Thermal                                    |
| 22 | Jindal Institute for Power Technology, PO-Tamnara, Dist-Raigarh         | Thermal                                    |

## 12.8 Revision of Guidelines for Recognition of Training Institute in Power Sector.

In compliance with the provisions of Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations 2023, CEA revised the existing guidelines for the "Recognition of Training Institutes 2019" and the "Statutory Guidelines for Training and Certification of Load Dispatchers & Recognition of Training Institutes 2011." In place of these two guidelines, the following guidelines have been issued in January 2024:

1. Statutory Guidelines for Training and Certification of Load Dispatchers & Recognition of Training Institutes
2. Guidelines for Recognition of Training Institutes in the Field of Generation of Electricity, Including Details of Training Curriculum for Engineers, Supervisors, and Technicians.
3. Guidelines for Recognition of Training Institutes in the Field of Transmission of Electricity, Including Details of Training Curriculum for Engineers, Supervisors, and Technicians.
4. Guidelines for Recognition of Training Institutes in the Field of Distribution of Electricity, Including Details of Training Curriculum for Engineers, Supervisors, and Technicians.

These guidelines are for the training of personnel engaged in O&M in the fields of generation, transmission, distribution, and load despatch. Further the CEA is developing a portal within the National Single Window Clearance system to all applications from training institutes online.

## 12.9 Foreign Visits/Training programmes for CEA Officers

The CEA officers were deputed to the Foreign visits/ training programmes to give them exposure to technological trends in the developed countries.

During the period of 2023-24, a total of 18 nos. officers of CEA at various levels visited foreign nations under 19 programme. The details of the foreign visits undertaken by the CEA officers is as follows:-

| S.No. | Purpose of the Visit                                                                                                                                                                                                                | Name of Officer             | Designation of Officer | Country                    |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------|----------------------------|
| 1     | Visit of CEA officers for DSM Compliance as a part of CEA/NLDC Team                                                                                                                                                                 | Shri Anjum Parvej           | Director               | Bhutan                     |
|       |                                                                                                                                                                                                                                     | Shri Manish Maurya          | Deputy Director        | Bhutan                     |
| 2     | 21st Joint steering committee(JSC)/ Joint working Group (JWG) between India and Bangladesh                                                                                                                                          | Shri Ghanshyam Prasad       | Chairperson, CEA       | Bangladesh                 |
|       |                                                                                                                                                                                                                                     | Shri Ishan Sharan           | Chief-Engineer         | Bangladesh                 |
| 3     | UNFCCC's fifty-eight session of the Subsidiary Body for Scientific and Technological Advice (SBSTA-58) and the Subsidiary Body for Implementation (SBI-58)                                                                          | Shri Vijay Menghani         | Chief Engineer         | Germany                    |
| 4     | 2023 Asia Clean Energy Forum by ADB                                                                                                                                                                                                 | Shri Annepu Sursh           | Director               | Philippines                |
| 5     | Environment Week organised by Committee for Trade and Environment (CTE) of WTO, Geneva, will hold an                                                                                                                                | Shri Narender Singh         | Chief Engineer         | Switzerland                |
| 6     | Promoting energy Cooperation in SASEC Region: "Final Dissemination Workshop and Presentation of TA Results Regional Technical Assistance No. 9584: South Asia Subregional Economic Cooperation (SASEC) Regional Energy Cooperation" | Shri Bhagwan Sahay Bairwa   | Director               | Sri Lanka                  |
|       |                                                                                                                                                                                                                                     | Shri Om Kant Shukla         | Director               | Sri Lanka                  |
| 7     | 2nd BRICS Energy Senior Officials meeting of the working group on energy sector cooperation and the 8th Annual meeting of BRICS Ministers of Energy.                                                                                | Shri Vijay Menghani         | Chief Engineer         | Johannesburg, South Africa |
| 8     | Residential Training Programme to understand European Electricity Markets                                                                                                                                                           | Shri Sharad Chandra Bhupesh | Deputy Director        | Berlin, Germany            |
| 9     | 31st TCC Meeting of Punatsangchhu-I HEP                                                                                                                                                                                             | Shri MAKP Singh             | Member (Hydro)         | Bhutan                     |
|       |                                                                                                                                                                                                                                     | Shri Manoj Tiwari           | Chief Engineer         | Bhutan                     |
|       |                                                                                                                                                                                                                                     | Shri RP Pradhan             | Director               | Bhutan                     |
| 10    | Inter-Ministerial delegation in COP 28, CMA 18 and CMA 5                                                                                                                                                                            | Shri Vijay Menghani         | Chief Engineer         | Dubai, UAE                 |
| 11    | 3 rd Quad Senior Cyber Group (QSCG) Principals Meeting                                                                                                                                                                              | Shri MAKP Singh             | Member (Hydro)         | Tokyo, Japan               |
| 12    | JTT meeting                                                                                                                                                                                                                         | Shri Ajay Talegaonkar       | Member (E&C)           | Bhutan                     |

|    |                                                                                                                                                  |                           |                    |                     |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|--------------------|---------------------|
|    |                                                                                                                                                  | Shri MAKP Singh           | Member (Hydro)     | Bhutan              |
|    |                                                                                                                                                  | Shri Bhagwan Sahay Bairwa | Director           | Bhutan              |
| 13 | 11th Meeting of Indo-Nepal Joint Steering Committee (JSC)/ Joint Working Group (JWG) on Cooperation in Power Sector                              | Shri AK Rajput            | Member (PS)        | Nepal               |
|    |                                                                                                                                                  | Shri Bhagwan Sahay Bairwa | Director           | Nepal               |
|    |                                                                                                                                                  | Shri Om Kant Shukla       | Director           | Nepal               |
| 14 | Power Secretary level meeting                                                                                                                    | Shri Ajay Talegaonkar     | Member             | Bhutan              |
|    |                                                                                                                                                  | Shri MAKP Singh           | Member (Hydro)     | Bhutan              |
|    |                                                                                                                                                  | Shri Bhagwan Sahay Bairwa | Director           | Bhutan              |
| 15 | SARIC program - Regional Power Markets Training Program                                                                                          | Shri Ajay Devedwal        | Assistant Director | Tasmania, Australia |
| 16 | Capstone and Awarding Ceremony for WePOWER SAR-100 Program                                                                                       | Ms Komal Dupare           | Assistant Director | Thailand            |
| 17 | Workshop on Cross-border Transmission Lines Optimization of the BIMSTEC Grid Interconnection Masterplan                                          | Shri Manish Maurya        | Deputy Director    | Dhaka, Bangladesh   |
|    |                                                                                                                                                  | Shri Bhagwan Sahay Bairwa | Director           | Dhaka, Bangladesh   |
|    |                                                                                                                                                  | Shri Om Kant Shukla       | Chief Engineer     | Dhaka, Bangladesh   |
|    |                                                                                                                                                  | Shri Rakesh Goyal         | Chief Engineer     | Dhaka, Bangladesh   |
| 18 | JWG meeting led by Secy (P)                                                                                                                      | Shri Rakesh Goyal         | Chief Engineer     | Sri Lanka           |
| 19 | 1st JWG meeting on MNRE with Srilanka on renewable development and opportunity under OSOWOG including discussion on cross border interconnection | Shri Rakesh Goyal         | Chief Engineer     | Sri Lanka           |

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ANNEXURES

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**Annexure-3A****Details of Inter-regional Transmission lines as on 31.03.2024**

| <b>Details of Inter-regional transmission lines</b>                                                                                           | <b>Transmission Capacity in MW<br/>(As on 31.03.2024)</b> |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>EAST-NORTH</b>                                                                                                                             |                                                           |
| Dehri-Sahupuri 220 kV S/c                                                                                                                     | 130                                                       |
| Muzaffarpur-Gorakhpur 400 kV D/c (with Series Cap+TCSC)                                                                                       | 2,000                                                     |
| Patna – Balia 400kV D/c (Quad)                                                                                                                | 1,600                                                     |
| Biharshariff – Balia 400kV D/c(Quad)                                                                                                          | 1,600                                                     |
| Barh – Balia 400kV D/c (Quad)                                                                                                                 | 1,600                                                     |
| Gaya - Balia 765kV S/c                                                                                                                        | 2,100                                                     |
| Sasaram-Allahabad/Varanasi 400kV D/C line (Sasaram HVDC back to back has been bypassed)                                                       | 1,000                                                     |
| Sasaram - Fatehpur 765kV2x S/c                                                                                                                | 4,200                                                     |
| Barh-II-Gorakhpur 400kV D/c (Quad) line                                                                                                       | 1,600                                                     |
| Gaya-Varanasi 765 kV S/c line                                                                                                                 | 2,100                                                     |
| LILO of Biswanath Chariali - Agra +/- 800 kV, 3000 MW HVDC Bi-pole at new pooling station in Alipurduar and addition of second 3000 MW module | 3,000                                                     |
| Biharsharif-Varanasi 400kV D/c line (Quad)                                                                                                    | 1,600                                                     |
| <b>Sub-total</b>                                                                                                                              | <b>22,530</b>                                             |
| <b>EAST-WEST</b>                                                                                                                              |                                                           |

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|                                                                 |               |
|-----------------------------------------------------------------|---------------|
| Budhipadar-Korba 220 kV 3 ckts.                                 | 390           |
| Rourkela-Raipur 400 kV D/c with series comp.+TCSC               | 1,400         |
| Ranchi –Sipat 400 kV D/c with series comp.                      | 1,200         |
| Rourkela-Raipur 400 kV D/c (2 <sup>nd</sup> ) with series comp. | 1,400         |
| Ranchi - Dharamjayagarh - WR Pooling Station 765kV S/c line     | 2,100         |
| Ranchi - Dharamjayagarh 765kV 2nd S/c                           | 2,100         |
| Jharsuguda-Dharamjayagarh 765kV D/c line                        | 4,200         |
| Jharsuguda-Dharamjayagarh 765kV 2nd D/c line                    | 4,200         |
| Jharsuguda- Raipur 765kV D/c line                               | 4,200         |
| <b>Sub-total</b>                                                | <b>21,190</b> |
| <b>WEST- NORTH</b>                                              |               |
| Auriya-Malanpur 220 KV D/c                                      | 260           |
| Kota - Ujjain 220 KV D/c                                        | 260           |
| Vindhyachal HVDC back-to-back                                   | 500           |
| Gwalier-Agra 765 kV 2 x S/c                                     | 4,200         |
| Zerda-Kankroli 400kV D/c                                        | 1,000         |
| Champa Pool- Kurukshetra HVDC Bipole                            | 3,000         |
| Gwalior-Jaipur 765kV 2xS/c lines                                | 4,200         |
| RAPP-Sujalpur 400kV D/c                                         | 1,000         |
| Adani(Mundra) - Mahendranagar HVDC bipole                       | 2,500         |
| Upgradation of Champa Pool- Kurukshetra HVDC Bipole             | 3,000         |
| Jabalpur - Orai 765kV D/c line                                  | 4,200         |
| LILO of Satna - Gwalior 765kV 2xS/c line at Orai                | 4,200         |
| Banaskantha-Chittorgarh 765kV D/c line                          | 4,200         |
| Vindhyachal-Varanasi 765kV D/c line                             | 4,200         |

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|--------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Neemuch PS – Chhittorgarh (PG) s/s 400 kV D/C line                                                                                         | 1,600         |
| <b>Sub-total</b>                                                                                                                           | <b>38,320</b> |
| <b>EAST- SOUTH</b>                                                                                                                         |               |
| Balimela-Upper Sileru 220kV S/c                                                                                                            | 130           |
| Gazuwaka HVDC back-to-back                                                                                                                 | 1,000         |
| Talcher-Kolar HVDC bipole                                                                                                                  | 2,000         |
| Upgradation of Talcher-Kolar HVDC Bipole                                                                                                   | 500           |
| Angul - Srikakulum                                                                                                                         | 4,200         |
| <b>Sub-total</b>                                                                                                                           | <b>7,830</b>  |
| <b>WEST- SOUTH</b>                                                                                                                         |               |
| Chandrapur HVDC back-to-back                                                                                                               | 1,000         |
| Kolhapur-Belgaum 220kV D/c                                                                                                                 | 260           |
| Ponda – Nagajhari 220kV D/c                                                                                                                | 260           |
| Raichur - Sholapur 765kV S/c line (PG)                                                                                                     | 2,100         |
| Raichur - Sholapur 765kV S/c line (Pvt. Sector)                                                                                            | 2,100         |
| Narendra - Kolhapur 765kV D/c (ch at 400kV)                                                                                                | 2,200         |
| Wardha - Hyderabad 765kV D/c line(Part of Wardha – Nizamabad line)                                                                         | 4,200         |
| Raigarh –Pugalur HVDC line with with Raigarh and Pugalur Station HVDC Terminal (Pole-I, Pole-II, Pole-III & Pole-IV each 1500 MW charged ) | 6,000         |
| Warora Pool - Warangal (New) 765kV D/c line                                                                                                | 4,200         |
| <b>Sub-total</b>                                                                                                                           | <b>22,320</b> |
| <b>EAST- NORTH EAST</b>                                                                                                                    |               |
| Birpara-Salakati 220kV D/c                                                                                                                 | 260           |
| Malda - Bongaigaon 400 kV D/c                                                                                                              | 1,000         |
| Siliguri - Bongaigaon 400 kV D/c (Quad) line                                                                                               | 1,600         |
| Reconductoring of Alipurduar (ER) - Salakati (NER) 220 KV D/C Line                                                                         | 90            |

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|                                                                                                                                                  |                |
|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Re-conductoring of Siliguri – Bongaigaon 400kV D/c line with Twin HTLS conductor under North Eastern Region Strengthening Scheme-XII (NERSS-XII) | 600            |
| <i>Sub-total</i>                                                                                                                                 | <b>3,550</b>   |
| <b>NORTH EAST-NORTH</b>                                                                                                                          |                |
| Biswanath Chariali - Agra +/- 800 kV, 3000 MW HVDC Bi-pole\$                                                                                     | 3,000          |
| <i>Sub-total</i>                                                                                                                                 | <b>3,000</b>   |
| <b>TOTAL (CUMULATIVE)</b>                                                                                                                        | <b>118,740</b> |

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**Annexure – 3B****Transmission Schemes notified to be implemented through Tariff Based Competitive Bidding (TBCB) route (as on 31.03.2024)****a) Schemes already commissioned/ready for commissioning by Transmission Service Providers:**

| Sl. No. | Transmission Scheme                                                                                                                   |
|---------|---------------------------------------------------------------------------------------------------------------------------------------|
| 1.      | Transmission system associated with IPPs of Nagapattinam/ Cuddalore Area- Package A                                                   |
| 2.      | Transmission system for Strengthening in SR for Import of Power from ER.                                                              |
| 3.      | ATS of Unchahar TPS                                                                                                                   |
| 4.      | NR System strengthening Scheme- NRSS-XXXI(Part-A)                                                                                     |
| 5.      | Transmission System associated with Gadarwara STPS (2x800 MW) of NTPC (Part-A)                                                        |
| 6.      | Transmission System associated with Gadarwara STPS (2x800 MW) of NTPC (Part-B)                                                        |
| 7.      | Transmission System Strengthening associated with Vindhyachal – V                                                                     |
| 8.      | Strengthening of Transmission system beyond Vemagiri                                                                                  |
| 9.      | Transmission system associated with LTA applications from Rajasthan SEZ Part-A                                                        |
| 10.     | New WR-NR 765 kV Inter- Regional Corridor                                                                                             |
| 11.     | Transmission system associated with LTA applications from Rajasthan SEZ Part-B                                                        |
| 12.     | Transmission system associated with LTA applications from Rajasthan SEZ Part-C                                                        |
| 13.     | System Strengthening Scheme in Eastern Region ERSS XXI                                                                                |
| 14.     | 765 kV System Strengthening Scheme in Eastern Region ERSSXVIII                                                                        |
| 15.     | Transmission System for providing connectivity to RE Projects at Bhuj-II (2000 MW ) in Gujarat                                        |
| 16.     | Transmission system strengthening scheme for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under “Phase-II Part-F |

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| Sl. No. | Transmission Scheme                                                                                                                                       |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17.     | Transmission system strengthening scheme for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under Phase II –Part A                     |
| 18.     | Transmission system for evacuation of power from Neemuch SEZ (1000 MW)                                                                                    |
| 19.     | System strengthening for WR                                                                                                                               |
| 20.     | System strengthening common for WR and NR                                                                                                                 |
| 21.     | Scheme for enabling import of NER/ER surplus by NR                                                                                                        |
| 22.     | Part ATS for RAPP U-7&8 in Rajasthan                                                                                                                      |
| 23.     | Eastern Region System Strengthening Scheme-VII                                                                                                            |
| 24.     | Northern Regional System Strengthening Scheme, NRSS-XXIX                                                                                                  |
| 25.     | Connectivity lines for Maheshwaram 765/400 kV S/S                                                                                                         |
| 26.     | Common Transmission system for phase-II generation projects in Orissa and immediate evacuation system for OPGC project (Orissa)                           |
| 27.     | Creation of new 400 kV GIS substations in Gurgaon area and Palwal as a part of ISTS                                                                       |
| 28.     | NER System Strengthening Scheme II                                                                                                                        |
| 29.     | Connectivity system for Khargone TPP (2x660MW)                                                                                                            |
| 30.     | WRSS – 21 Part – B – Transmission System Strengthening for Relieving Over Loadings Observed in Gujarat Intra-State System Due to RE injections in Bhuj PS |
| 31.     | Eastern Region System Strengthening Scheme-VI                                                                                                             |
| 32.     | Northern Region System Strengthening Scheme, NRSS-XXXI (Part-B)                                                                                           |
| 33.     | Western Region System Strengthening – II under Project – B (Maharashtra)                                                                                  |
| 34.     | Western Region System Strengthening – II under Project – C (Gujarat)                                                                                      |
| 35.     | Additional system strengthening for Sipat STPS                                                                                                            |
| 36.     | Additional system strengthening for Chhattisgarh (B)                                                                                                      |
| 37.     | System strengthening for IPPs in Chhattisgarh and other generation projects in Western Region                                                             |

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| Sl. No. | Transmission Scheme                                                                                                                                                                                                                                                   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 38.     | Transmission System for Ultra Mega Solar Park in Fatehgarh, Distt. Jaisalmer Rajasthan                                                                                                                                                                                |
| 39.     | Transmission System Associated with LTA applications from Rajasthan SEZ Part-D                                                                                                                                                                                        |
| 40.     | Transmission System for Western Region Strengthening Scheme – 21 (WRSS – 21) Part – A – Transmission System Strengthening for Relieving Over Loadings Observed in Gujarat Intra-State System Due to Re-injections in Bhuj PS                                          |
| 41.     | Transmission System for Transmission System Associated with RE Generations at Bhuj-II, Dwarka & Lakadia                                                                                                                                                               |
| 42.     | Transmission System for Jam Khambaliya Pooling Station and Interconnection of Jam Khambaliya Pooling Station for Providing Connectivity to RE Projects (1500 MW) in Dwarka (Gujarat) and Installation of 400/220 kV ICT along with associated bays at CGPL switchyard |
| 43.     | Additional inter- Regional AC link for import into Southern Region i.e Warora - Warangal and Chilakaluripeta Hyderabad- Kurnool 765 kV link                                                                                                                           |
| 44.     | Transmission Scheme for Evacuation of power from RE sources in Karur/Tirrupur Wind Energy Zone (Tamil Nadu) (1000 MW) - Phase I                                                                                                                                       |
| 45.     | Transmission scheme for evacuation of 3 GW RE injection at Khavda Pooling Station 1 (KPS 1) under Phase I                                                                                                                                                             |
| 46.     | Transmission System required for evacuation of power from Kudgi TPS (3x800 MW in Phase-I) of NTPC Ltd.                                                                                                                                                                |
| 47.     | Transmission System for Patran 400 kV S/S                                                                                                                                                                                                                             |
| 48.     | Transmission System Associated with Krishnapattanam UMPP - Synchronous interconnection between SR and WR (Part-B)                                                                                                                                                     |
| 49.     | Transmission system strengthening in Indian system for transfer of power from new HEP's in Bhutan                                                                                                                                                                     |
| 50.     | North Eastern Region Strengthening Scheme (NERSS-VI)                                                                                                                                                                                                                  |
| 51.     | Evacuation of Power from RE Sources in Koppal Wind Energy Zone (Karnataka) (2500 MW)                                                                                                                                                                                  |
| 52.     | Transmission system for evacuation of power from RE projects in Rajgarh (1500 MW) SEZ in Madhya Pradesh: Phase-I                                                                                                                                                      |
| 53.     | Transmission system for evacuation of power from RE projects in Osmanabad area (1 GW) in Maharashtra                                                                                                                                                                  |

**b) Transmission Schemes under implementation by the Transmission Service Providers:**

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| Sl. No. | Transmission Scheme                                                                                                                                                                            |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.      | Immediate evacuation for North Karanpura (3x660 MW) generation project of NTPC alongwith creation of 400/220 kV sub-station at Dhanbad (ERSS-XIX)                                              |
| 2.      | Transmission scheme for evacuation of 4.5 GW RE injection at Khavda P.S. under Phase-II – Part A                                                                                               |
| 3.      | ISTS Network Expansion scheme in Western Region & Southern Region for export of surplus power during high RE scenario in Southern Region (Narendra –Pune 765 kV D/c line and associated works) |
| 4.      | Transmission System for evacuation of additional 7 GW RE Power from Khavda RE Park under Phase-III: Part A                                                                                     |
| 5.      | Transmission system for evacuation of power from REZ in Rajasthan (20 GW) under Phase- III Part A1                                                                                             |
| 6.      | Transmission system for evacuation of power from REZ in Rajasthan (20 GW) under Phase- III Part A3                                                                                             |
| 7.      | Western Region Expansion Scheme XXXIII (WRES-XXXIII): Part B (Establishment of 765/400/220 kV Karera S/s (near Datiya) alongwith associated transmission lines)                                |
| 8.      | 400 kV Khandukhal (Srinagar) - Rampura (Kashipur) D/c line                                                                                                                                     |
| 9.      | Transmission scheme for injection beyond 3 GW RE power at Khavda PS1 (KPS1)                                                                                                                    |
| 10.     | System Strengthening Scheme for Eastern and North Eastern Regions: A. Eastern Region Strengthening Scheme-XXV (ERSS-XXV) B. North Eastern Region Strengthening Scheme-XV (NERSS-XV)            |
| 11.     | Transmission system strengthening for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under Phase II –Part B                                                                 |
| 12.     | Transmission system strengthening scheme for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under Phase-II- Part C                                                          |
| 13.     | Transmission system strengthening scheme for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under Phase-II- Part D                                                          |
| 14.     | Transmission system strengthening scheme for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under Phase-II- Part G                                                          |
| 15.     | Transmission Network Expansion in Gujarat associated with integration of RE projects from Khavda potential RE zone                                                                             |
| 16.     | Establishment of Khavda Pooling Station-2 (KPS 2) in Khavda RE Park                                                                                                                            |
| 17.     | Establishment of Khavda Pooling Station-3 (KPS 3) in Khavda RE Park                                                                                                                            |
| 18.     | Transmission system for evacuation of power from REZ in Rajasthan (20 GW) under Phase- III Part B1                                                                                             |
| 19.     | Transmission scheme for evacuation of 4.5 GW RE injection at Khavda P.S. under Phase-II – Part C                                                                                               |
| 20.     | Transmission scheme for evacuation of 4.5 GW RE injection at Khavda P.S. under Phase-II – Part B                                                                                               |

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| Sl. No. | Transmission Scheme                                                                                                                                                                                                                                                                                                       |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21.     | Transmission scheme for Solar Energy Zone in Ananthapuram (Ananthapur) (2500 MW) and Kurnool (1000 MW), Andhra Pradesh                                                                                                                                                                                                    |
| 22.     | Transmission system strengthening scheme for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under Phase-II- Part E                                                                                                                                                                                     |
| 23.     | Western Region Expansion Scheme XXVII (Raipur Pool – Dhamtari 400 kV D/c line)                                                                                                                                                                                                                                            |
| 24.     | Western Region Expansion Scheme XXVIII & XXIX (Creation of 220 kV level (GIS) at 765/400 kV Raipur Pool S/s & Creation of 220 kV level at 765/400 kV Dharamjaigarh S/s)                                                                                                                                                   |
| 25.     | Inter-regional ER-WR Interconnection (Jeypore- Jagdalpur 400 kV D/c line)                                                                                                                                                                                                                                                 |
| 26.     | Transmission system for evacuation of power from REZ in Rajasthan (20 GW) under Phase- III: Part C1                                                                                                                                                                                                                       |
| 27.     | Transmission system for evacuation of power from REZ in Rajasthan (20 GW) under Phase- III: Part H                                                                                                                                                                                                                        |
| 28.     | Transmission system for evacuation of power from REZ in Rajasthan (20 GW) under Phase- III: Part D                                                                                                                                                                                                                        |
| 29.     | Transmission Scheme for Solar Energy Zone in Bidar (2500 MW), Karnataka                                                                                                                                                                                                                                                   |
| 30.     | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part 1) (Bikaner Complex)-Part-A                                                                                                                                                                                                                    |
| 31.     | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part 1) (Bikaner Complex)-Part-D                                                                                                                                                                                                                    |
| 32.     | Transmission Scheme for integration of Renewable Energy Zone (Phase-II) in Koppal-II (Phase-A & B) and Gadag-II (Phase- A) in Karnataka                                                                                                                                                                                   |
| 33.     | Transmission System for evacuation of additional 7 GW RE Power from Khavda RE Park under Phase-III: Part B                                                                                                                                                                                                                |
| 34.     | Transmission Scheme for Solar Energy Zone in Gadag (1000 MW), Karnataka- Phase-I                                                                                                                                                                                                                                          |
| 35.     | Transmission Scheme for Solar Energy Zone in Gadag (1500 MW), Karnataka: Phase-II                                                                                                                                                                                                                                         |
| 36.     | System Strengthening Scheme in Northern Region (NRSS-XXXVI)” along with LILO of Sikar-Neemrana 400 kV D/C line at Babai (RRVNL)                                                                                                                                                                                           |
| 37.     | Transmission system for evacuation power from Pakal Dul HEP in Chenab Valley HEPs - Connectivity System                                                                                                                                                                                                                   |
| 38.     | Establishment of new 220/132 kV substation at Nangalbibra                                                                                                                                                                                                                                                                 |
| 39.     | Transmission System for 400 kV Udupi (UPCL) – Kasargode D/C Line                                                                                                                                                                                                                                                          |
| 40.     | Transmission system for evacuation of power from REZ in Rajasthan (20 GW) under Phase- III: Part F                                                                                                                                                                                                                        |
| 41.     | Western Region Strengthening Scheme-XIX (WRSS-XIX) and North Eastern Region Strengthening Scheme-IX (NERSS-IX) (LILO of 2 <sup>nd</sup> circuit of Zarda-Ranchodpura 400 kV D/c line at Banaskantha, Establishment of 400/220 kV Vapi-II S/s, Padghe-Khargar 400 kV D/c line, Pare HEP – North Lakhimpur 132 kV d/c line) |
| 42.     | Additional 400 kV feed to Goa and additional system for power evacuation from generation projects pooled at Raigarh (Tamnar) Pool                                                                                                                                                                                         |

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| Sl. No. | Transmission Scheme                                                                                                                                         |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 43.     | Transmission system for evacuation of power from REZ in Rajasthan (20 GW) under Phase- III: Part G                                                          |
| 44.     | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part 1) (Bikaner Complex)-Part B                                                      |
| 45.     | Transmission System for Evacuation of Power from RE Projects in Rajgarh 1000 MW SEZ in Madhya Pradesh Phase-II                                              |
| 46.     | Transmission scheme for evacuation of power from Dhule 2 GW REZ                                                                                             |
| 47.     | Western Region Expansion Scheme XXXIII (WRES-XXXIII): Part C (Establishment of 765/400/220 kV Ishanagar (New) S/s along with associated transmission lines) |
| 48.     | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part1) (Bikaner Complex)-Part-C                                                       |
| 49.     | Transmission system for evacuation of power from RE projects in Solapur (1500 MW) SEZ in Maharashtra                                                        |

## c) Schemes under bidding process by Bid Process Coordinators:

| Sl. No. | Transmission Schemes                                                                                                                                |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.      | Creation of 400/220 kV, 2x315 MVA S/S at Siot, Jammu & Kashmir                                                                                      |
| 2.      | Transmission system for evacuation of power from Chhatarpur SEZ (1500 MW)                                                                           |
| 3.      | Transmission system for evacuation of power from Luhri Stage-I HEP                                                                                  |
| 4.      | North Eastern Region Expansion Scheme-XVI (NERES-XVI) [Establishment of Gogamukh 400/220/132 kV substation and other associated works]              |
| 5.      | Transmission system for evacuation of power from REZ in Rajasthan (20 GW) under Phase-III: Part I                                                   |
| 6.      | Transmission system for evacuation of power from Shongtong Karcham HEP (450 MW) and Tidong HEP (150 MW)                                             |
| 7.      | Provision of Dynamic Reactive Compensation at KPS1 and KPS3                                                                                         |
| 8.      | Eastern Region Expansion Scheme-XXXIV (ERES-XXXIV) [Establishment of 765/400 kV, 2x1500 MVA GIS substation at Paradeep along with associated lines] |
| 9.      | Western Region Network Expansion scheme in Kallam area of Maharashtra                                                                               |
| 10.     | Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat                                          |

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| Sl. No. | Transmission Schemes                                                                                                                      |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------|
|         | under Phase-IV (7 GW): Part A                                                                                                             |
| 11.     | Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part B  |
| 12.     | Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part C  |
| 13.     | Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part D  |
| 14.     | Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part E2 |
| 15.     | Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-V (8 GW): Part A   |
| 16.     | Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-V (8 GW): Part C   |
| 17.     | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part A                  |
| 18.     | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part B                  |
| 19.     | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part C                  |
| 20.     | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part D                  |
| 21.     | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part E                  |
| 22.     | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part F                  |
| 23.     | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part H1                 |
| 24.     | Transmission Scheme for integration of Renewable Energy Zone in Tumkur area of Karnataka                                                  |
| 25.     | Transmission system strengthening for interconnection of Bhadla-III and Bikaner III complex                                               |

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| Sl. No. | Transmission Schemes                                                                                                                                           |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 26.     | Network Expansion scheme in Gujarat for drawl of about 3.6 GW load under Phase-I in Jamnagar area                                                              |
| 27.     | North Eastern Region Generation Scheme-I (NERGS-I) [Establishment of 400 kV switching station at Bokajan in Assam]                                             |
| 28.     | Augmentation of transformation capacity at Bhuj-II PS                                                                                                          |
| 29.     | Eastern Region Expansion Scheme-XXXIX (ERES-XXXIX) [Establishment of 765/400 kV, 2x1500 MVA GIS substation at Gopalpur along with associated lines]            |
| 30.     | Eastern Region Generation Scheme-I (ERGS-I) [LILO of both circuits of Angul –Sundargarh (Jharsuguda) 765 kV 2xS/c lines at NLC Talabira generation switchyard] |
| 31.     | Network Expansion Scheme in Navinal (Mundra) area of Gujarat for drawal of power in the area                                                                   |
| 32.     | Additional Transmission system for evacuation of power from Bhadla-III PS as part of Rajasthan REZ Phase-III scheme (20 GW)                                    |
| 33.     | Transmission Scheme for integration of Davanagere / Chitradurga and Bellary REZ in Karnataka                                                                   |
| 34.     | Transmission Scheme for integration of Bijapur REZ in Karnataka                                                                                                |
| 35.     | Transmission System under ISTS for evacuation of power from Kudankulam Unit - 3 & 4 (2x1000 MW)                                                                |
| 36.     | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part 3: 6GW) (Bikaner Complex) :Part A                                                   |
| 37.     | Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part 3: 6 GW) (Bikaner Complex) :Part B                                                  |
| 38.     | Augmentation of transformation capacity at Jam Khambhaliya PS                                                                                                  |

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**Annexure – 3C**  
(Item 3.4.1)

**Issues Pertaining to Transmission System Planning taken up with the National Committee on Transmission during 2023-24**

**1. 13<sup>th</sup> meeting of National Committee on Transmission held on 12.05.2023**

- (i). Modification in the schemes
  - a) Eastern Region Expansion Scheme-XXXIV (ERES-XXXIV)
- (ii). Establishment of State-of Art Unified Network Management System (U-NMS) for ISTS and State Utility Communication System for Southern Region
- (iii). Eastern Region Expansion Scheme-XXXVII (ERES-XXXVII)

**2. 14<sup>th</sup> meeting of National Committee on Transmission held on 09.06.2023**

- (i). Modification in the schemes
  - a) Revised timeframe of the transmission scheme “Transmission system for evacuation of power from Luhri Stage-I HEP”
  - b) Delinking of Fatehgarh-II- Bhadla-III 400 kV D/c line from transmission scheme “Transmission system for evacuation of power from REZ in Rajasthan (20 GW) under Phase-III Part B1”:
- (ii). Requirement of additional FOTE of STM-16 capacity at Bhuj PS to cater to connectivity of RE Gencos
- (iii). Requirement of additional FOTE of STM-16 capacity at Bhuj-II substation to cater to connectivity of RE Gencos
- (iv). Upgradation of STM-4 communication link of Dehgam, Ranchhodpura, Santhapur Rep, Bhachau and CGPL Mundra to STM-16 capacity
- (v). Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-V (8 GW): Part A1
- (vi). Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part H2
- (vii). Augmentation of transformation capacity by 1x1500 MVA (3rd), 765/400 kV ICT at Maheshwaram (PG) substation in Telangana
- (viii). Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part E1
- (ix). Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part E3
- (x). Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part E4

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- (xi). Western Region Network Expansion scheme in Kallam area of Maharashtra
- (xii). Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part A
- (xiii). Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part B
- (xiv). Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part C
- (xv). Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part D
- (xvi). Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part E2
- (xvii). Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-V (8 GW): Part A
- (xviii). Transmission System for Evacuation of Power from potential renewable energy zone in Khavda area of Gujarat under Phase-V (8 GW): Part C
- (xix). Transmission system for evacuation of power from Rajasthan REZ PhIV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part A
- (xx). Transmission system for evacuation of power from Rajasthan REZ PhIV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part B
- (xxi). Transmission system for evacuation of power from Rajasthan REZ PhIV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part C
- (xxii). Transmission system for evacuation of power from Rajasthan REZ PhIV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part D
- (xxiii). Transmission system for evacuation of power from Rajasthan REZ PhIV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part E
- (xxiv). Transmission system for evacuation of power from Rajasthan REZ PhIV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part F (By clubbing Part F1 & F2)
- (xxv). .Transmission system for evacuation of power from Rajasthan REZ PhIV (Part-2 :5.5 GW) (Jaisalmer/Barmer Complex): Part H1

### 3. 15<sup>th</sup> meeting of National Committee on Transmission held on 25.08.2023

- (i). Modification in the schemes
  - a) Termination of Bidar – Maheshwaram 765 kV D/c line alongwith 240 MVar switchable line reactor at Maheshwaram end under the “Transmission Scheme for Solar Energy Zone in Bidar (2500 MW), Karnataka”
  - b) Addition in scope of works in OM dated 22.02.2023 to facilitate termination of Bidar Maheshwaram 765 kV D/c line alongwith 240 MVar switchable line reactor at Maheshwaram end under the “Transmission Scheme for Solar Energy Zone in Bidar (2500 MW), Karnataka”
  - c) Delinking of Jhatikara - Dwarka 400 kV D/c line (Quad) from the Transmission scheme “Transmission system for evacuation of power from REZ in Rajasthan (20 GW) under Phase-III Part D”

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- d) De linking of augmentation of 765/400 kV, 1500 MVA transformer at Bhiwani S/s from Transmission System for evacuation of RE power from renewable energy parks in Leh (5 GW Leh- Kaithal HVDC Transmission corridor)
- e) Review of implementation timeline of “Transmission system for evacuation of power from REZ in Rajasthan (20 GW) under Phase-III Part I” based on TSP request
- f) Modification in implementation schedule of “Transmission Scheme for integration of Renewable Energy Zone (Phase-II) in Koppal-II (Phase-B) in Karnataka” from 36 months to 24 months
- g) Revision in scope of Spare 80 MVAR Reactor under Rajasthan Ph IV Part C transmission scheme
- h) Modifications in the schemes viz. OPGW on 400 kV Agra - Ballabgarh & 400 kV Kishenpur - Wagoora line approved in 11th NCT
- (ii). ISTS Network Expansion scheme “Transmission Scheme for integration of Renewable Energy Zone in Tumkur area of Karnataka”
- (iii). Transmission system strengthening for interconnections of Bhadla-III & Bikaner- III complex
- (iv). Network Expansion scheme in Gujarat for drawl of about 3.6 GW load under Phase-I in Jamnagar area
- (v). North Eastern Region Generation Scheme-I (NERGS-I)
- (vi). Implementation of Unified Network Management System (UNMS) in the Western Region

#### 4. 16<sup>th</sup> meeting of National Committee on Transmission held on 30.11.2023

- (i). Modification in the schemes
  - a) Transmission System for Evacuation of power from potential renewable energy zone in Khavda area of Gujarat under Phase-IV (7 GW): Part B scheme
  - b) Additional 1x500 MVA 400/220 kV (9th) ICT, for injection from any additional RE project (other than 4000 MW injection under SECI bids upto Tranche IV) at Bhuj PS
  - c) Spare Reactor (1-ph, 1x80 MVA<sub>r</sub>) unit at 765/400 kV Beawar S/s under Rajasthan REZ Ph-IV (Part-2: 5.5 GW) (Jaisalmer/ Barmer Complex): Part D scheme
  - d) Change in Scope of Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-1: Bikaner Complex)- Part-A package
- (ii). Augmentation of transformation capacity at 765/400 kV Indore S/s in Madhya Pradesh
- (iii). Augmentation of transformation capacity by 1x1500 MVA (3rd), 765/400 kV ICT at Kurnool New substation in Andhra Pradesh
- (iv). ISTS Network Expansion scheme “Reconductoring of Raichur – Veltloor (Mahabubnagar) 400 kV S/c line with HTLS conductor”
- (v). Augmentation of transformation capacity at Bhuj-II PS (GIS)

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- (vi). Eastern Region Generation Scheme-I (ERGS-I)
- (vii). Network Expansion Scheme in Navinal (Mundra) area of Gujarat for drawal of power in the area
- (viii). Eastern Region Expansion Scheme XXXIX (ERES-XXXIX)
- (ix). OPGW installation on existing line 765/400 kV Pune (PG) (GIS) – 400 kV Parli (PG) line which is to be LILOed at Kallam Substations under TBCB project.
- (x). OPGW installation on Itarsi-Dhule Transmission Line
- (xi). Additional FOTE at AGC locations for Redundancy in NER region

#### 5. 17<sup>th</sup> meeting of National Committee on Transmission held on 31.01.2024

- (i). Modification in the schemes
  - a) Network Expansion Scheme in Navinal (Mundra) area of Gujarat for drawal of power in the area
  - b) Transmission system for evacuation of power from Shongtong Karcham HEP (450 MW) and Tidong HEP (150 MW)
  - c) Timeline for 1500 MVA, 765/400 kV ICT Augmentation at Jhatikara S/s
  - d) Change in Scope of the transmission scheme “North Eastern Region Expansion Scheme-XVI (NERES-XVI)”
  - e) Change in Scope of transmission scheme "Eastern Region Expansion Scheme-XXXIV (ERES-XXXIV)”
  - f) Augmentation of transformation capacity at Bhuj-II PS- request for change in implementation timeframe
  - g) Change in Completion Schedule of transmission scheme “North Eastern Region Generation Scheme-I (NERGS-I)”
- (ii). Augmentation of transformation capacity by 2x500 MVA (7th & 8th), 400/220 kV ICTs at Tumkur (Pavagada) 400/220 kV Pooling Station in Karnataka
- (iii). Augmentation of transformation capacity at Jam Khambhaliya PS (JKTL)
- (iv). Augmentation of transformation capacity at 765/400 kV Lakadia S/s (WRSS XXI (A) Transco Ltd) in Gujarat

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- (v). Additional Transmission system for evacuation of power from Bhadla-III PS as part of Rajasthan REZ Phase-III scheme (20 GW)

**6. 18<sup>th</sup> meeting of National Committee on Transmission held on 05.03.2024**

- (i). Modification in the schemes
  - a) Modification in scope of work of Eastern Region Expansion Scheme-XXXIX (ERES XXXIX)
  - b) Transmission System for Evacuation of Power from REZ in Rajasthan (20 GW) under Phase-III Part-G
  - c) Transmission scheme for evacuation of 4.5 GW RE injection at Khavda PS under Phase II Part A
- (ii). Augmentation of 2x500 MVA, 400/230 kV transformation capacity (3<sup>rd</sup> & 4<sup>th</sup> ICTs) at Karur PS
- (iii). Augmentation of transformation capacity at Jam Khambhaliya PS (JKTL) – Part B
- (iv). Transmission Scheme for integration of Tumkur-II REZ in Karnataka
- (v). Transmission system for evacuation of power from Rajasthan REZ PhIV (Part 3: 6GW) (Bikaner Complex): Part A
- (vi). . Transmission system for evacuation of power from Rajasthan REZ PhIV (Part 3: 6GW) (Bikaner Complex): Part B
- (vii). Transmission Scheme for integration of Davanagere / Chitradurga and Bellary REZ in Karnataka
- (viii). Transmission Scheme for integration of Bijapur REZ in Karnataka
- (ix). Transmission System under ISTS for evacuation of power from Kudankulam Unit - 3 & 4 (2x1000 MW)
- (x). Additional FOTE for redundancy at AGC locations in ER
- (xi). OPGW laying work on 400 kV Bokaro(A)-Kodarma line

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**Annexure-3D****Transmission Lines Completed During FY- 2023-24**

| As on 31-March -2024           |                                                                                                                                                                              |              |                  |                   |                     |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------|-------------------|---------------------|
| Voltage Level (kV)             | Name of Transmission Lines                                                                                                                                                   | Circuit Type | Executing Agency | Line Length (cKM) | Month of Completion |
| 1.                             | 2.                                                                                                                                                                           | 3.           | 4.               | 5.                | 6.                  |
| <b>765 kV</b>                  |                                                                                                                                                                              |              |                  |                   |                     |
| <b><u>CENTRAL SECTOR</u></b>   |                                                                                                                                                                              |              |                  |                   |                     |
| 1                              | 765KV Rampur-Ghatampur line under CKT-2 LILO OF 765 KV Ghatampur- Hapur (WUPPTCL) Line on 765 KV GIS S/S RAMPUR (MOHANLALGANJ TRANSMISSION LIMITED) :INTRASTATE Project (UP) | S/C          | PGCIL            | 116               | Dec-23              |
| 2                              | 765KV Rampur-Hapur line under LILO OF 765 KV Ghatampur-Hapur (WUPPTCL) Line on 765 KV GIS S/S RAMPUR (MOHANLALGANJ TRANSMISSION LIMITED) :INTRASTATE Project (UP)            | S/C          | PGCIL            | 112               | Feb-24              |
| <b>Total of CENTRAL SECTOR</b> |                                                                                                                                                                              |              |                  | <b>228</b>        |                     |
| <b><u>STATE SECTOR</u></b>     |                                                                                                                                                                              |              |                  |                   |                     |
| 3                              | NCTPS Stage-III - North Chennai (PS) line                                                                                                                                    | D/C          | TANTRANSCO       | 13                | Mar-24              |
| <b>Total of STATE SECTOR</b>   |                                                                                                                                                                              |              |                  | <b>13</b>         |                     |
| <b><u>PRIVATE SECTOR</u></b>   |                                                                                                                                                                              |              |                  |                   |                     |
| 4                              | Hyderabad - Kurnool                                                                                                                                                          | D/C          | ADANI            | 335               | JUL-23              |
| 5                              | Warangal (New) - Chilakaluripeta                                                                                                                                             | D/C          | ADANI            | 392               | SEP-23              |
| 6                              | Warangal (New) - Hyderabad                                                                                                                                                   | D/C          | ADANI            | 268               | SEP-23              |
| 7                              | Warora (Pool) - Warangal (New)                                                                                                                                               | D/C          | ADANI            | 665               | OCT-23              |

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|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|-------------|--------|
| 8                              | Khavda PS GIS – Bhuj PS 765 kV D/C Line                                                                                                                           | D/C | ADANI | 218         | Jan-24 |
| <b>Total of PRIVATE SECTOR</b> |                                                                                                                                                                   |     |       | <b>1878</b> |        |
| <b>Total of 765 kV</b>         |                                                                                                                                                                   |     |       | <b>2119</b> |        |
| <b>400 kV</b>                  |                                                                                                                                                                   |     |       |             |        |
| <b><u>CENTRAL SECTOR</u></b>   |                                                                                                                                                                   |     |       |             |        |
| 9                              | Bikaner-II PS-Khetri 400kV line (phase II part F)                                                                                                                 | M/C | PGCIL | 1102        | JUN-23 |
| 10                             | Khetri-Bhiwadi 400kV line (Twin HTLS) (PHASE II Part F)                                                                                                           | D/C | PGCIL | 251         | JUN-23 |
| 11                             | Removal of LILO of one Ckt. of Bhadla-Bikaner (RVPN) at Bikaner (PG) and Extension of this LILO section from Bikaner (PG) to Bikaner-PS -Bikaner (PG)             | D/C | PGCIL | 68          | JUL-23 |
| 12                             | Reconductoring of Kolhapur(PG) - Kolhapur (MSETCL) 400kv D/C line (Ckt-2) under TS strengthening beyond Kolhapur for export of power from Solar and Wind Energy   | D/C | PGCIL | 40          | AUG-23 |
| 13                             | Fatehgarh-III (Ramgarh-II) - Jaisalmer-II (RVPNL 400kV D/C line (Twin HTLS)                                                                                       | D/C | PGCIL | 100         | SEP-23 |
| 14                             | Sitamarhi (POWERGRID) (Quad Moose) - Dhalkebar (Nepal) (Indian Portion)                                                                                           | D/C | PGCIL | 78          | SEP-23 |
| 15                             | LILO of one Ckt. on Monopole of Aaur (WUPPTCL) - Indirapuram (WUPPTCL) D/C Quad Moose on Noida under TB -202111 (Mohanlalganj Trans. Ltd. (Intrastate Project UP) | D/C | PGCIL | 21          | OCT-23 |
| 16                             | 400kV D/C Muradnagar-II – Simbhawali Line under Meerut Simbhawali :INTRASTATE Project                                                                             | D/C | PGCIL | 142         | Nov-23 |

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|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|-----|--------|
| 17 | Fatehgarh-III (Ramgarh-II) -Fatehgarh-II 400kV D/c line (Twin HTLS) under Transmission system strengthening scheme for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under phase-II- Part A<br>SPV Name: POWERGRID Ramgarh Transmission Limited | D/C | PGCIL | 88  | Dec-23 |
| 18 | 400KV D/C Lower Subhansiri - Biswanath Chariyali line -II under Part C of North East / Northern Western Interconnector -I Project                                                                                                                                   | D/C | PGCIL | 371 | Dec-23 |
| 19 | Reconductoring of Parli (PG) – Parli (M)CKT-I 400KV D/C Line with Twin HTLS Conductor & Associated Line Upgradation Activities Under Western Region Expansion Scheme-XXX (WRSS-XXX)                                                                                 | S/C | PGCIL | 5   | Dec-23 |
| 20 | Ckt-2 of Re-conductoring of Siliguri – Bongaigaon 400kV D/c line with Twin HTLS conductor under North Eastern Region Strengthening Scheme-XII (NERSS-XII)                                                                                                           | S/C | PGCIL | 218 | Jan-24 |
| 21 | Reconductoring of Parli (PG) – Parli (M)CKT-II 400KV D/C Line with Twin HTLS Conductor & Associated Line Upgradation Activities Under Western Region Expansion Scheme-XXX (WRSS-XXX)                                                                                | S/C | PGCIL | 5   | Feb-24 |
| 22 | Reconductoring of 400 kV Jodhpur (Surpura) (RVPN) – Kankroli S/c (twin moose) line with twin HTLS conductor under Scheme to relieve high loading of WR-NR Inter Regional Corridor (400 kV Bhinmal Zerda line)                                                       | S/C | PGCIL | 188 | Feb-24 |

| ANN                            |                                                                                                                                                                  |     |       |             |        |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|-------------|--------|
| 23                             | LILO of 400kV Lucknow (PGCIL) - Sultanpur S/C Line at 400kV Substation Mohanlalganj, Lucknow (MOHANLALGANJ TRANSMISSION LIMITED) :INTRASTATE Project (UP)        | S/C | PGCIL | 14          | Mar-24 |
| 24                             | LILO of 400kV Sarojnai Nagar - Unnao (765kV) S/C Line at 400kV GIS Substation Mohanlalganj, Lucknow (MOHANLALGANJ TRANSMISSION LIMITED) :INTRASTATE Project (UP) | S/C | PGCIL | 77          | Mar-24 |
| 25                             | Ckt-1 of Re-conductoring of Siliguri – Bongaigaon 400kV D/c line with Twin HTLS conductor under North Eastern Region Strengthening Scheme-XII (NERSS-XII)        | D/C | PGCIL | 218         | Mar-24 |
| 26                             | Neemuch PS – Chhittorgarh (PG) s/s 400 kV D/C line (POWERGRID Neemuch Transmission System Limited (NTL)                                                          | D/C | PGCIL | 233         | Mar-24 |
| 27                             | Neemuch PS- Mandsaur s/s 400 kV D/C line (POWERGRID Neemuch Transmission System Limited (NTL)                                                                    | D/C | PGCIL | 236         | Mar-24 |
| 28                             | CKT-1 FOR RECONDUCTORING WORK OF JHARSUGUDA/SUNDARGARH#(PG) – ROURKELA (PG) 400KV 2XD/C (ERSS-XXIX)                                                              | D/C | PGCIL | 142         | Mar-24 |
| 29                             | CKT-3 FOR RECONDUCTORING WORK OF JHARSUGUDA/SUNDARGARH#(PG) – ROURKELA (PG) 400KV 2XD/C (ERSS-XXIX) ( from Tower Loc. 881 to 1150 i.e 96 CKM)                    | D/C | PGCIL | 96          | Mar-24 |
| <b>Total of CENTRAL SECTOR</b> |                                                                                                                                                                  |     |       | <b>3693</b> |        |
| <b><u>STATE SECTOR</u></b>     |                                                                                                                                                                  |     |       |             |        |

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|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------|-----|--------|
| 30 | LILO of Krishnapatnam - Manubolu (Ckt - I) Quad Moose line at Sembcorp (M/s SEIL- Plant 2) 400KV GIS Substation                                                                                       | D/C | APTRANSCO  | 7   | Jan-24 |
| 31 | BTPP-Naubatpur Line (Twin Moose Conductor)                                                                                                                                                            | D/C | BSPTCL     | 248 | Mar-24 |
| 32 | Soja -Zerda line (Pkg-2) with Twin AL-59 Conductor                                                                                                                                                    | D/C | GETCO      | 135 | MAY-23 |
| 33 | 400 kV D/C BHOGAT - KALAWAD LINE with Twin AL-59 cond. & 1 OPGW and 1 Earth Wire.                                                                                                                     | D/C | GETCO      | 250 | Mar-24 |
| 34 | LILO of one circuit of 400 kV VADAVI - HALVAD LINE at 400 kV SANAND-II GIDC S/S with Twin AL-59 cond. & OPGW.                                                                                         | D/C | GETCO      | 85  | Mar-24 |
| 35 | Balance work of 400 kV D/C VADINAR - AMRELI LINE (PKG-III) with ACSR Twin Moose condu. & Earth Wire, M/s Jyoti                                                                                        | D/C | GETCO      | 121 | Mar-24 |
| 36 | LILO of one ckt of 400 KV Jalandhar Kurukshetra DC line at 400 KV Dhanansu.                                                                                                                           | D/C | PSTCL      | 8   | Jan-24 |
| 37 | Laying of 400 KV 1X2500 Sq.mm XLPE UG cable from perumbakkam junction to Pallikarunai OH/UG Junction in the Sholinghanallur to Guindy 400 KV Feeder for aroute length of 8.546 Km under turnkey basis | S/C | TANTRANSCO | 8   | MAY-23 |
| 38 | LILO of Karaikudi - Pugalur feeder-I at Vellalaviduthi                                                                                                                                                | D/C | TANTRANSCO | 167 | OCT-23 |
| 39 | Udangudi - Ottapidaram (QM ACSR Conductor) line                                                                                                                                                       | D/C | TANTRANSCO | 141 | Dec-23 |
| 40 | LILO of Pugalur - Kalivanthapattu Line at Ariyalur                                                                                                                                                    | D/C | TANTRANSCO | 14  | Feb-24 |
| 41 | Dindi S/s -Yedula s/s                                                                                                                                                                                 | D/C | TSTRANSCO  | 111 | JUN-23 |
| 42 | Narlapur SS - Yedula SS                                                                                                                                                                               | D/C | TSTRANSCO  | 55  | SEP-23 |
| 43 | Yedula-Vattem QMDC line                                                                                                                                                                               | D/C | TSTRANSCO  | 30  | OCT-23 |
| 44 | 400 kV Panki - PTPS Line (Ckt-II)                                                                                                                                                                     | D/C | UPPTCL     | 1   | Mar-24 |

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|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------|-------------|--------|
| <b>Total of STATE SECTOR</b>   |                                                                                                                                                |     |                                                   | <b>1381</b> |        |
| <b><u>PRIVATE SECTOR</u></b>   |                                                                                                                                                |     |                                                   |             |        |
| 45                             | Warangal(New)- Warangal(Existing)                                                                                                              | D/C | ADANI                                             | 95          | JUN-23 |
| 46                             | Kharghar -Vikhroli D/C and M/C line with bays at Kharghar and Vikhroli (with capacity of 2000 MW /Ckt.) along with 400kV Bus Extn. at Kharghar | D/C | ADANI                                             | 43          | OCT-23 |
| 47                             | LILO of both Ckts. of Pugalur-Pugalur (HVDC) (with Quad Moose ACSR conductor) at Karur PS                                                      | D/C | ADANI                                             | 9           | OCT-23 |
| 48                             | AGEL-PSS3 Khavda-I PS 400kV S/c line                                                                                                           | S/C | Adani Green Energy Limited (AGEL)                 | 5           | Mar-24 |
| 49                             | AREHFL PSS1 – Khavda (GIS) PS 400kV S/c line (with minimum power carrying capacity of 1250MW per ckt. at nominal voltage)                      | S/C | Adani Renewable Energy Holding Four Ltd. (AREHFL) | 3           | Mar-24 |
| 50                             | AREHFL PSS2 – Khavda (GIS) PS 400kV S/c line (with minimum power carrying capacity of 1250MW per ckt. at nominal voltage)                      | S/C | Adani Renewable Energy Holding Four Ltd. (AREHFL) | 5           | Mar-24 |
| 51                             | Pachora SEZ PP -Bhopal (Sterlite) 400 kV D/c line                                                                                              | D/C | G R INFRA                                         | 288         | Feb-24 |
| 52                             | LILO of both circuits of Parli(PG) – Pune(GIS) 400kV D/c line at Kallam PS                                                                     | M/C | INDIGRID                                          | 68          | Mar-24 |
| 53                             | Koppal PS - Narendra (New) Trans. Ltd.                                                                                                         | D/C | RENEW                                             | 276         | OCT-23 |
| 54                             | Babai-Bhiwani D/C Transmission Line                                                                                                            | D/C | TATA POWER                                        | 222         | Dec-23 |
| <b>Total of PRIVATE SECTOR</b> |                                                                                                                                                |     |                                                   | <b>1014</b> |        |
| <b>Total of 400 kV</b>         |                                                                                                                                                |     |                                                   | <b>6088</b> |        |
| <b>230 kV</b>                  |                                                                                                                                                |     |                                                   |             |        |

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**STATE SECTOR**

|    |                                                                                                                                                                                            |            |            |     |        |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------|-----|--------|
| 55 | Sholinganallur (Ottiyambakkam) - Omega                                                                                                                                                     | D/C        | TANTRANSCO | 66  | APR-23 |
| 56 | Erection of 230KV SC on DC tower with ZEBRA conductor from Pudukottai 230 LV SS to Vellalaviduthy 400/230 KV SS.                                                                           | S/C        | TANTRANSCO | 51  | MAY-23 |
| 57 | Erection of 230kV Dc line on DC towers with Zebra conductor by making LILO of 230 kV MTPS- Salem Feeder at the New Karuppur 230/110kV s/s                                                  | D/C        | TANTRANSCO | 25  | OCT-23 |
| 58 | Erection of 230kV S/C line on D/C tower from Vellalaviduthi 400kV S/S (Common Point-A) - Thuvakudy 230kV S/S                                                                               | S/C        | TANTRANSCO | 46  | OCT-23 |
| 59 | Erection of 230 kV S/C line on DC tower from Vellalaviduthi 400 kV S/S - Nemmeli Thippayajudy 230 kV S/S                                                                                   | S/C        | TANTRANSCO | 23  | OCT-23 |
| 60 | Vellalaviduthi - existing Pudukottaai 230 kV SS                                                                                                                                            | S/C on D/C | TANTRANSCO | 21  | OCT-23 |
| 61 | Vellaviduthi S/S (Loc. A-61 Nos. ) and 230 kV S/C line on D/C tower for Loc A - Mondipatty 230kV s/s                                                                                       | S/C        | TANTRANSCO | 92  | OCT-23 |
| 62 | LILO of 230 kV Sriperumpudur – Korattur feederfor extension of new EHT supply for a initial demand of 40000 KVA to M/s.Ctrls Data Centers Ltd.,- SIDCO Ambathur Industrial Estate, Chennai | D/C        | TANTRANSCO | 3   | Nov-23 |
| 63 | Erection of 230 kV SC line on DC towers with Zebra conductor from Nemmeli Thippiyakudy 230 kV SS to Karaikudi 230 kV SS                                                                    | S/C        | TANTRANSCO | 104 | Nov-23 |
| 64 | 230 kV SC line on DC towers with Zebra conductor from Loc.No:153 of existing 230 kV Chekkanurani - Sembatty feeder to proposed 230 kV Samayanallur SS.                                     | S/C        | TANTRANSCO | 5   | Dec-23 |

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|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------------------|------------|--------|
| 65                             | 230KV,1X1200 sq mm XLPE Aluminium UG cable from new joint bay in Royapettah high road masilamani road to 230kv Mylapore GIS SS for a route length of 1.149 kms for the deviation of existing 230KV TNEB HQ GIS SS to Mylapore feeder | S/C | TANTRANSCO             | 1          | Jan-24 |
| 66                             | Erection of 230 KV DC line on DC/MC tower with Zebra conductor from LILO of Sangneri – Abhishekapatti 230KV DC feeder between Tower No.70 – 71 to M/s.TP Solar Ltd                                                                   | D/C | TANTRANSCO             | 6          | Feb-24 |
| <b>Total of STATE SECTOR</b>   |                                                                                                                                                                                                                                      |     |                        | <b>443</b> |        |
| <b><u>PRIVATE SECTOR</u></b>   |                                                                                                                                                                                                                                      |     |                        |            |        |
| 67                             | JSW Renew Energy Limited – Karur 230kV S/c line (strung on D/c tower of M/s JSW Renew Energy Ltd )                                                                                                                                   | S/C | JSW Neo Energy Limited | 4          | Mar-24 |
| 68                             | JSW Renew Energy Limited – Tuticorin-II 230kV S/c line                                                                                                                                                                               | S/C | JSW Renew Energy Ltd   | 20         | Mar-24 |
| 69                             | Generation PS of JSW Renew Energy Ltd.- Tuticorin-II 230kV S/c line (part line will be strung as 2nd ckt. On D/c towers of JSW Neo Energy Ltd.-Tuticorin II 230kV line).                                                             | S/C | JSW Renew Energy Ltd   | 22         | Mar-24 |
| <b>Total of PRIVATE SECTOR</b> |                                                                                                                                                                                                                                      |     |                        | <b>46</b>  |        |
| <b>Total of 230 kV</b>         |                                                                                                                                                                                                                                      |     |                        | <b>489</b> |        |
| <b>220 kV</b>                  |                                                                                                                                                                                                                                      |     |                        |            |        |
| <b><u>CENTRAL SECTOR</u></b>   |                                                                                                                                                                                                                                      |     |                        |            |        |
| 70                             | LILO of MTPS-A to Durgapur line at Barjora S/S                                                                                                                                                                                       | S/C | DVC                    | 9          | Jan-24 |
| 71                             | Re-conductoring of BTPS- Salakati 220kv line with Single HTLS Conductor                                                                                                                                                              | D/C | PGCIL                  | 8          | JUN-23 |
| <b>Total of CENTRAL SECTOR</b> |                                                                                                                                                                                                                                      |     |                        | <b>17</b>  |        |
| <b><u>STATE SECTOR</u></b>     |                                                                                                                                                                                                                                      |     |                        |            |        |

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|----|----------------------------------------------------------------------------------------------------------------------|---------|-----------|-------|--------|
| 72 | 220KV DC/SC (2-Phase) line from the existing 220/11KV LIS SS Settypalli to proposed 220/25KV RTSS Veldurthy          | D/C+S/C | APTRANSCO | 6     | MAY-23 |
| 73 | 220KV Pallantla-Guddigudem DC Line at Loc.No.54 to 400/220/11KV Guddigudem SS                                        | D/C     | APTRANSCO | 25    | MAY-23 |
| 74 | Hindupur (400) - Pampanur Tanda (220)(GEC-I)                                                                         | D/C     | APTRANSCO | 140   | MAY-23 |
| 75 | LILO Pallantla -Pattiseema line at Guddigudem line                                                                   | D/C     | APTRANSCO | 23    | MAY-23 |
| 76 | 220kV TMDC from 400kV Hindupur S/S - 220kV S/S Gollapuram                                                            | D/C     | APTRANSCO | 59    | JUL-23 |
| 77 | Jammalamadugu - Porumamilla YSR Kadapa District                                                                      | D/C     | APTRANSCO | 200   | SEP-23 |
| 78 | LILO of 220kV Pallantla-Pattisema DC line at Guddigudem                                                              | D/C     | APTRANSCO | 24.00 | Dec-23 |
| 79 | Hindupur - Penukonda                                                                                                 | D/C     | APTRANSCO | 50    | Dec-23 |
| 80 | 220KV DC/SC line from 220KV SWS Inaparajupalli to M/s Shree Cement Ltd.                                              | S/C     | APTRANSCO | 2     | Jan-24 |
| 81 | Rachaganneri-Racherla                                                                                                | D/C     | APTRANSCO | 107   | Jan-24 |
| 82 | Sullurpeta-Racherla                                                                                                  | D/C     | APTRANSCO | 39    | Jan-24 |
| 83 | 220 KV Krishnagiri-Settypalli feeder (2nd circuit) (DC WORK)                                                         | S/C     | APTRANSCO | 11    | Feb-24 |
| 84 | 220KV Guddigudem-Purushothapatnam D/C line                                                                           | D/C     | APTRANSCO | 47    | Mar-24 |
| 85 | Tajpur- Goraul line                                                                                                  | D/C     | BSPTCL    | 102   | APR-23 |
| 86 | Amnour - Digha                                                                                                       | D/C     | BSPTCL    | 48    | Dec-23 |
| 87 | BTTP-Dehri-on-sone (single zebra conductor)                                                                          | D/C     | BSPTCL    | 150   | Mar-24 |
| 88 | LILO of one circuit of 220kV D/C BECL - Botad line at proposed 220kV Vallabhipur with ACSR Zebra conductor with OPGW | D/C     | GETCO     | 11    | AUG-23 |

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|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------|-----|--------|
| 89  | LILO of both Clts. of 220 kV D/C Thavar -Tharad line at proposed 220kV Ghodasr (Rah) with Al-59 conductor with OPGW                                       | M/C | GETCO  | 23  | OCT-23 |
| 90  | 220 kV D/C Kalawad - Kangasiyali Line                                                                                                                     | D/C | GETCO  | 114 | Nov-23 |
| 91  | 220 kV D/C Vadavi - Sanand DFCC Line                                                                                                                      | D/C | GETCO  | 62  | Jan-24 |
| 92  | LILO of both circuits of 220kV D/C Palanpur - Kheralu line at proposed 220kV Sisrana with AL-59 conductor with OPGW                                       | M/C | GETCO  | 51  | Feb-24 |
| 93  | LILO of 220 kV Gondal - Sadla line at Kamlapur with AL-59 conductor                                                                                       | D/C | GETCO  | 46  | Mar-24 |
| 94  | 220 kV Double Circuit Transmission Line from 220/132 kV GIS Sub-Station Mazra to 220/33 GIS Sub-Station Karian.                                           | D/C | HPPTCL | 36  | Jan-24 |
| 95  | Creation of 220kV D/C Mund-IOCL line                                                                                                                      | D/C | HVPNL  | 68  | SEP-23 |
| 96  | LILO of Both ckt of 220kV D/C Sohna Road-Rangala Rajpur at 220kV s/stn. RojKaMeo                                                                          | M/C | HVPNL  | 9   | Jan-24 |
| 97  | Panchkula (PGCIL)- Panchkula Sector-32                                                                                                                    | D/C | HVPNL  | 43  | Mar-24 |
| 98  | Take in arrangement of Madanpur-Kunihar line at Sector-32 and take in arrangement from remaining portion of Madanpur-Kunihar line at PGCIL,Naggal(400KV). | D/C | HVPNL  | 43  | Mar-24 |
| 99  | Alusteng-Harwan Transmission Line                                                                                                                         | S/C | JKPDD  | 14  | Feb-24 |
| 100 | (PMDP - Kashmir) LILO of one ckt. Alusteng - New Wanpoh at Tailbal                                                                                        | D/C | JKPDD  | 7   | Mar-24 |
| 101 | (PMDP - Kashmir) Extension of Mirbazar - Alusteng line upto New Wanpoh                                                                                    | D/C | JKPDD  | 12  | Mar-24 |
| 102 | Strengthening of existing 220kV Antharasanahalli-Nelamangala SC line by replacing Drake by HTLS conductor                                                 | S/C | KPTCL  | 59  | AUG-23 |

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| 103 | Construction from the existing Chintamani S/s - the proposed Srinivasapura S/s at Srinivasapura       | D/C  | KPTCL  | 48  | OCT-23 |
| 104 | Ramanagara (Kothipura)_MCMV Line: Construction of 220-66 kV MC line from 220 kV Bidadi S/s 18.188 kms | M/C  | KPTCL  | 75  | Nov-23 |
| 105 | Ghanagapura (Nimbarga) to Kalaburagi (Ferozbad) line                                                  | D/C  | KPTCL  | 72  | Mar-24 |
| 106 | 220kV Chintamani - Mitemari line:(Work B)                                                             | D/C  | KPTCL  | 100 | Mar-24 |
| 107 | 220/ 110 kV MCMV line from kanhirode to Mundayad                                                      | D/C  | KSEB   | 18  | Nov-23 |
| 108 | (NERPSIP- Meghalya) Byrnihat (Killing) – Mawngap – New Shillong 220 kV D/c line                       | D/C  | MeECL  | 258 | Mar-24 |
| 109 | LILO of Bina - Ganjbasoda 220 KV Line at Bina (MPPTCL) 400KV S/s                                      | DCDS | MPPTCL | 25  | MAY-23 |
| 110 | Rewa UMSP - Sidhi 220kV DCDS line                                                                     | DCDS | MPPTCL | 126 | AUG-23 |
| 111 | LILO of Gwalior - Datiya 220kV line at Gwalior(PGCIL) 765kV S/s                                       | DCDS | MPPTCL | 6   | Nov-23 |
| 112 | 220kV line from 220kV S/s Nipaniya for Power supply of 54.5 MVA to Pump house of Shamgarh - Suwasra   | S/C  | MPPTCL | 15  | Mar-24 |
| 113 | LILO of Birsinghpur-Satna 220kV line at New S/s Manpur (Bijouri)(TBCB)                                | D/C  | MPPTCL | 37  | Mar-24 |
| 114 | LILO on 220 kV Nalasopara- Boisar line at Palghar                                                     | D/C  | MSETCL | 5   | MAY-23 |
| 115 | 220kV DC line from 220kV Wani s/s to 132kV Pandharkawda                                               | D/C  | MSETCL | 65  | JUN-23 |
| 116 | 220kV Deoli (PG) -Ghatodi line upto Yawatmal LILO point                                               | D/C  | MSETCL | 99  | OCT-23 |

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|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|----|--------|
| 117 | 220 kV DC line from 400 kV Jejuri Substation to location no. 28 of existing 220 kV Jejuri-Lonand circuit                                                                | LILO | MSETCL | 14 | Dec-23 |
| 118 | 220kV SCDC line from 220kV Phaltan MIDC to 220kV Mol-Manjarwadi Solar Project of M/s. T.S.Wind Farm Developer.                                                          | S/C  | MSETCL | 20 | Jan-24 |
| 119 | LILO on 220kV Kathapur- Lonikand -I line at 220kV Khed City                                                                                                             | D/C  | MSETCL | 11 | Feb-24 |
| 120 | 220KV DC Kalmeshwar-Warud (Phase I & II) (from Kalmeshwar substation to loc no 90)                                                                                      | D/C  | MSETCL | 53 | Mar-24 |
| 121 | 220 kV DC line from 400 kV Kumbhargaoon (Nanded) - Kurunda - Pkg A (Loc 1 to 112 )                                                                                      | D/C  | MSETCL | 67 | Mar-24 |
| 122 | 220KV DC Kalmeshwar-Warud (Phase IV & V)(from loc no 168 to 313 i.e. upto 220kV Warud SS)                                                                               | D/C  | MSETCL | 84 | Mar-24 |
| 123 | 220kV Wani Pandharkawda DC line (Phase-II)                                                                                                                              | D/C  | MSETCL | 51 | Mar-24 |
| 124 | 220kV Umred - Nagbhid DCDC line Part-I (Umred SS to loc no 95).                                                                                                         | D/C  | MSETCL | 53 | Mar-24 |
| 125 | LILO of Duburi - Balasore at Dhamra                                                                                                                                     | D/C  | OPTCL  | 70 | APR-23 |
| 126 | 220kV LILO of Narendrapur-Therubali at Aska New S/s                                                                                                                     | D/C  | OPTCL  | 85 | Mar-24 |
| 127 | Bhari (proposed) - Daheru Railway TSS line                                                                                                                              | D/C  | PSTCL  | 26 | JUN-23 |
| 128 | Provision of 220 KV 2nd Ckt. Using special arrangement for connecting 400 kv S/s Dhanansu with 220 Kv S/s Gounsgarh regarding power evacuation from 400 kv S/s Dhanansu | S/C  | PSTCL  | 2  | Dec-23 |
| 129 | LILO of 220 KV Rajpura Mohali at 220 KV Switching Station for release of connection to M/s Ultratech.                                                                   | D/C  | PSTCL  | 2  | Jan-24 |

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| 130 | 2-Phase Single Circuit Transmission line on D/C towers from 220 kV S/s Jafferpur to TSS Rudraupur for M/s IRCON                                                                                                                                                                                                                                                                                                                                                                                                                      | S/C | PTCUL                                         | 10  | MAY-23 |
| 131 | <ul style="list-style-type: none"> <li>• S/C transmission line from PPS-1 to PGCIL Bhadla-II (28.626 CKM)</li> <li>• S/C Transmission Line from PPS-2 to PGCIL Bhadla-II through 220 kV D/C dead end tower outside premises of PPS-1 without entering in PPS-1 (31.612 CKM)</li> <li>• S/C Transmission Line from PPS-3 to PGCIL Bhadla-II through 220 kV D/C dead end tower outside premises of PPS-4 without entering in PPS-4 (30.990 CKM)</li> <li>• S/C Transmission Line from PPS-4 to PGCIL Bhadla-II (29.427 CKM)</li> </ul> | S/C | Rajasthan Solar Park Development Company Ltd. | 121 | Mar-24 |
| 132 | LILO of 220kV Sirohi - Bhinmal line at 220kV GSS Reodar line                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | D/C | RVPNL                                         | 50  | AUG-23 |
| 133 | Jangaon -Devannapet line                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | D/C | TSTRANSCO                                     | 109 | APR-23 |
| 134 | Existing 220/132kV Fabcity - Amazon (Fabcity) S/S                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | D/C | TSTRANSCO                                     | 4   | JUN-23 |
| 135 | LILO of Pudukurthy- Bheemghanpur line to KTPP SS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | D/C | TSTRANSCO                                     | 46  | JUN-23 |
| 136 | 220kV 1000Sq.mm XPLE Cable from 220kV SS Maheswaram-220/33kV Amazon (Pharmacity) SS (Ckt.I 1.464 ckm and Ckt.-II - 1.506ckm)                                                                                                                                                                                                                                                                                                                                                                                                         | D/C | TSTRANSCO                                     | 1   | AUG-23 |
| 137 | LILO of Salivagu- Bheemghanpur line to KTPP SS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | D/C | TSTRANSCO                                     | 42  | SEP-23 |
| 138 | 220KV line from 220KV Chandanvally SS to 220KV Amazon (Chandanvally)                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | D/C | TSTRANSCO                                     | 3   | Dec-23 |

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| 139 | LILO Mahabubabad - Warangal line to Ammavaripet                            | D/C | TSTRANSCO | 64  | Mar-24 |
| 140 | Saharanpur PG (400) - Deoband line                                         | D/C | UPPTCL    | 50  | APR-23 |
| 141 | LILO of 220kV Phulpur- Jhansi 400kV Machhalishahar                         | S/C | UPPTCL    | 99  | MAY-23 |
| 142 | 220 kV Pilipokhar Agra (400) - I.O.C.L. Mathura D/C line                   | D/C | UPPTCL    | 84  | JUN-23 |
| 143 | 220kV Budaun (400) - Dataganj DC line                                      | D/C | UPPTCL    | 55  | JUL-23 |
| 144 | LILO of 220 kV Agra(765) PG- Sikandra(220) line at Kirawali(220)           | D/C | UPPTCL    | 4   | AUG-23 |
| 145 | LILO of 220 kV Hapur(220)- Simbhaoli(220) line at Simbhaoli(400)           | D/C | UPPTCL    | 39  | AUG-23 |
| 146 | LILO of 220kV Gorakhpur (PG)- Maharajganj DC line at Annand Nagar          | D/C | UPPTCL    | 70  | SEP-23 |
| 147 | LILO of 220kV Sitapur (220) - Nighasan (220) SC line at Kanduni            | D/C | UPPTCL    | 59  | SEP-23 |
| 148 | Maharajganj - Gorakhpur PG (400) line                                      | D/C | UPPTCL    | 175 | SEP-23 |
| 149 | 220kV Shahjahanpur (PG) - Gola (220) DC line                               | D/C | UPPTCL    | 115 | OCT-23 |
| 150 | Bhadohi - Raja Ka Talab DC line                                            | D/C | UPPTCL    | 3   | OCT-23 |
| 151 | Kursi Road (PG) - Kanduni line                                             | D/C | UPPTCL    | 91  | OCT-23 |
| 152 | LILO of 220kV Parichha (TPS) - Orai Ckt.II at Moth line                    | D/C | UPPTCL    | 11  | OCT-23 |
| 153 | LILO of Modipuram- Faridnagar at Modipuram-II line                         | D/C | UPPTCL    | 12  | OCT-23 |
| 154 | 220 kV Harduaganj - Khair Line                                             | D/C | UPPTCL    | 77  | Nov-23 |
| 155 | LILO of 220 kV Auraiya (400) - Sikandara (Agra) Ckt-II @ Saifai (220) Line | D/C | UPPTCL    | 30  | Nov-23 |
| 156 | 220 kV Gr. Noida (765) - YEIDA Sec-24 Line                                 | M/C | UPPTCL    | 54  | Nov-23 |
| 157 | 220 kV Gr. Noida (765) - YEIDA Sec-18 Line                                 | M/C | UPPTCL    | 45  | Nov-23 |
| 158 | 220 kV Gonda-Basti (400)-II Line                                           | S/C | UPPTCL    | 78  | Nov-23 |
| 159 | 220 kV Azamgarh-II-Bhadohi Line                                            | D/C | UPPTCL    | 72  | Nov-23 |

|                              |                                                                                                            |     |                                            |             |        |
|------------------------------|------------------------------------------------------------------------------------------------------------|-----|--------------------------------------------|-------------|--------|
| ANN                          |                                                                                                            |     |                                            |             |        |
| 160                          | 220 kV Jehata(400)-Mallawan DC line                                                                        | D/C | UPPTCL                                     | 158         | Jan-24 |
| 161                          | LILO of 220kV R.C. Green (Gr. Noida)- Sector-148 (Noida) line at Knowledge Park-5 Noida                    | D/C | UPPTCL                                     | 14          | Jan-24 |
| 162                          | LILO of 220kV R.C. Green (Gr. Noida)- Gr. Noida(400kV Pali) line at Jalpura                                | D/C | UPPTCL                                     | 28          | Jan-24 |
| 163                          | LILO of Motiram Adda (400) - Barahuan Line @ 220kV Khorabar (2km Monopole+ 2km Cable)                      | D/C | UPPTCL                                     | 6           | Feb-24 |
| 164                          | Stringing of 220 kV Sarnath (400)- Gajokhar Line (Ckt-II)                                                  | D/C | UPPTCL                                     | 30          | Mar-24 |
| 165                          | LILO of Howrah-KTPP at Jangalpur 220 kV GIS                                                                | D/C | WBSETCL                                    | 1           | MAY-23 |
| 166                          | Diversion of existing STPS-Asansol Transmission line                                                       | D/C | WBSETCL                                    | 5           | JUL-23 |
| 167                          | LILO of Durgapur - Asansole Tr. Line at proposed Mongalpur 220kV GIS                                       | M/C | WBSETCL                                    | 9           | JUL-23 |
| 168                          | Jeerat - Krishnanagar 220 kV D/C connectivity                                                              | D/C | WBSETCL                                    | 114         | Dec-23 |
| <b>Total of STATE SECTOR</b> |                                                                                                            |     |                                            | <b>5156</b> |        |
| <b><u>PRIVATE SECTOR</u></b> |                                                                                                            |     |                                            |             |        |
| 169                          | SBE Renewables Sixteen Private Limited Power Plant – Fatehgarh-II PS 220kV S/c line                        | S/C | Adani Renewable Energy Holding Sixteen Ltd | 14          | Mar-24 |
| 170                          | AGEL - Bhuj-II line                                                                                        | S/C | Adani Green Energy Limited (AGEL)          | 79          | APR-23 |
| 171                          | AltraXergi Power Pvt. Ltd. Solar power plant – Fatehgarh-III PS 220 kV S/c high capacity line on D/c tower | S/C | Altra Xergi Power Private Limited          | 5           | Mar-24 |
| 172                          | Common Pooling Station of Amp Energy Green Four & Five Solar Power Plant - Bhadla-II PS 220 kV S/c line    | S/C | AMP Energy Green Private Limited           | 12          | Mar-24 |

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| 173 | Ayana Renewable Power Six Pvt Ltd –Koppal PS 220kV S/c line                                                                                                                  | S/C | Ayana Renewable Power Six Pvt Ltd  | 24 | Mar-24 |
| 174 | Common PS of Power Project of 100 MW each for One Volt Energy Private Limited, Amplus Ages Private Limited and Grian Energy Private Limited) – Bikaner-II PS 220 kV S/c line | S/C | Grian Energy Private Limited       | 3  | Mar-24 |
| 175 | Masaya Solar Energy Private Limited- Khandwa 220kV S/C Line                                                                                                                  | S/C | MASAYA                             | 14 | SEP-23 |
| 176 | Prerak Greentech Private Limited Project – Bikaner-II PS 220 kV S/c line on D/c tower                                                                                        | S/C | Prerak Greentech Private Limited   | 11 | Mar-24 |
| 177 | Renew Solar Power Pvt Ltd – Koppal PS 220kV S/c line                                                                                                                         | S/C | Renew Solar Power Pvt Ltd          | 54 | Mar-24 |
| 178 | ReNew Surya Aayan Private Limited solar power plant – Fatehgarh-III PS 220 kV S/c line on D/c tower                                                                          | S/C | ReNew Surya Aayan Private Limited  | 9  | Mar-24 |
| 179 | ReNew Surya Ojas Private Limited generation switchyard – Koppal 220kV S/c                                                                                                    | S/C | ReNew Surya Ojas Pvt Ltd           | 24 | Mar-24 |
| 180 | ReNew Surya Roshni Private Limited hybrid power plant – Fatehgarh-III PS 220 kV S/c high capacity line on D/c tower                                                          | S/C | Renew Surya Roshni Private Limited | 18 | Mar-24 |
| 181 | ReNew Surya Vihaan Private Ltd. – Fatehgarh-III PS 220 kV S/c line                                                                                                           | S/C | Renew Surya Vihaan Private Limited | 6  | Mar-24 |
| 182 | Agar Unit - 4 (200MW) - Pachora SEZ PP 220kV S/c line                                                                                                                        | S/C | Rewa Ultra Mega Solar Ltd.         | 1  | Mar-24 |

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|-------------------------|-----------------------------------------------------|-----|----------------------------|-------|--------|
| 183                     | Agar Unit-5 (350MW) - Pachora SEZ PP 220kV S/c line | S/C | Rewa Ultra Mega Solar Ltd. | 32    | Mar-24 |
| 184                     | TPL- Jam Khambhaliya PS line                        | S/C | TORRENT                    | 28    | SEP-23 |
| Total of PRIVATE SECTOR |                                                     |     |                            | 334   |        |
| Total of 220 kV         |                                                     |     |                            | 5507  |        |
| Grand Total             |                                                     |     |                            | 14203 |        |

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Annexure-3E**Sub-Stations Completed During FY - 2023-24**

| As on 31-March-2024          |                                                                                                                                                                                                              |                       |                  |                |                     |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------|----------------|---------------------|
| Sl No                        | Name of Sub Stations                                                                                                                                                                                         | Voltage Ratio (kV/kV) | Executing Agency | Capacity (MVA) | Month of Completion |
| 1.                           | 2.                                                                                                                                                                                                           | 3.                    | 4.               | 5.             | 6.                  |
| <b>765 kV</b>                |                                                                                                                                                                                                              |                       |                  |                |                     |
| <b><u>CENTRAL SECTOR</u></b> |                                                                                                                                                                                                              |                       |                  |                |                     |
| 1                            | 765/400 1x1500MVA T/F (3rd) at Bikaner(PG)                                                                                                                                                                   | 765/400               | PGCIL            | 1500           | Apr-23              |
| 2                            | 1x1500MVA 765/400kV ICT (3rd) at Nizamabad (PG) SS (2x1500MVA already existing)                                                                                                                              | 765/400               | PGCIL            | 1500           | Jan-24              |
| 3                            | 765/400/220kV Extension at Fatehgarh-II 765/400kV, 1500MVA 6th ICT (Transmission system strengthening Scheme for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under Phase II – Part B1) | 765/400               | PGCIL            | 1500           | Feb-24              |
| <b>TOTAL CENTRAL SECTOR</b>  |                                                                                                                                                                                                              |                       |                  | <b>4500</b>    |                     |
| <b><u>STATE SECTOR</u></b>   |                                                                                                                                                                                                              |                       |                  |                |                     |
| 4                            | Ariyalur                                                                                                                                                                                                     | 765/400               | TANTRANSCO       | 1500           | Dec-23              |
| 5                            | North Chennai Pooling Station (NCPS) 765/400 kV GIS SS (3x1500MVA)                                                                                                                                           | 765/400               | TANTRANSCO       | 4500           | Feb-24              |
| <b>TOTAL STATE SECTOR</b>    |                                                                                                                                                                                                              |                       |                  | <b>6000</b>    |                     |
| <b><u>PRIVATE SECTOR</u></b> |                                                                                                                                                                                                              |                       |                  |                |                     |
| 6                            | Warangal (New) S/S with 2X1500 MVA transformers and 2X240 MVAR bus reactors.                                                                                                                                 | 765/400               | ADANI            | 3000           | Sep-23              |

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|------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------|-------|--------------|--------|
| 7                            | Establishment of 3X1500 MVA 765/400 kV Khavda (GIS) with 1X330 MVAR 765 kV bus reactor and 1X125 MVAR 420 kV bus reactor | 765/400 | ADANI | 4500         | Feb-24 |
| <b>TOTAL PRIVATE SECTOR</b>  |                                                                                                                          |         |       | <b>7500</b>  |        |
| <b>TOTAL 765 kV</b>          |                                                                                                                          |         |       | <b>18000</b> |        |
| <b>400 kV</b>                |                                                                                                                          |         |       |              |        |
| <b><u>CENTRAL SECTOR</u></b> |                                                                                                                          |         |       |              |        |
| 8                            | Extn. at Bhadla-II PS (500MVA ICT-2nd 3rd 4th and 5th ) at Bikaner (PG) Fategarh-II and Bhadla-II)                       | 400/220 | PGCIL | 500          | May-23 |
| 9                            | 1x500MVA 400/230kV ICT-5 (4th) at Tuticorin under Evacuation of RE in Tirunelveli and Tuticorin WEZ                      | 400/230 | PGCIL | 500          | AUG-23 |
| 10                           | 765/400/220kV Extension at Fatehgarh-II (400/220kV 500MVA (9th ))                                                        | 400/220 | PGCIL | 500          | JUL-23 |
| 11                           | Aug. of 1x315MVA 400/220kV ICT to 1x500MVA 400/220kV ICT at Patiala (PG) S/S                                             | 400/220 | PGCIL | 500          | JUL-23 |
| 12                           | 1x500MVA 400/220kV ICT at Kochi SS                                                                                       | 400/220 | PGCIL | 500          | JUL-23 |
| 13                           | 1x500 MVA 400/220 kV ICT (3rd) at Sohawal (PG)                                                                           | 400/220 | PGCIL | 500          | JUN-23 |
| 14                           | Replacement of 1x250 MVA 400/220 kV ICT at 765/400/220 kV Moga (PG) S/s with 1x500 MVA 400/220kV ICT                     | 400/220 | PGCIL | 250          | MAY-23 |
| 15                           | Replacement of 1x315 MVA 400/220kV ICT (ICT-1) at 400/220 kV Ludhiana (PG) S/s with 1x500 MVA                            | 400/220 | PGCIL | 185          | MAY-23 |

| ANN |                                                                                                                                                                                                                                                             |         |       |     |        |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|-----|--------|
| 16  | 1x500MVA 400/220kV ICT(7th) at Bhadla-II SS underunder Transmission system strengthening Scheme for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under Phase II – Part B1                                                              | 400/220 | PGCIL | 500 | Apr-23 |
| 17  | 1x500MVA 400/220kV ICT (3rd) at Ranchi (PG) S/S                                                                                                                                                                                                             | 400/220 | PGCIL | 500 | MAY-23 |
| 18  | Extension 765/400/220kV Bhadla-II PS (500MVA ICT-5th)(Out of which 3 commissioned)                                                                                                                                                                          | 400/220 | PGCIL | 500 | MAY-23 |
| 19  | 1x 500MVA 400/220kV ICT (2nd) at Fatehgarh-III (Ramgarh-II) under phase-II- Part A                                                                                                                                                                          | 400/220 | PGCIL | 500 | OCT-23 |
| 20  | 1x 500MVA 400/220kV ICT (3rd) at Fatehgarh-III (Ramgarh-II) under phase-II- Part A                                                                                                                                                                          | 400/220 | PGCIL | 500 | OCT-23 |
| 21  | 1x 500MVA 400/220kV ICT (1st) at Fatehgarh-III (Ramgarh-II)                                                                                                                                                                                                 | 400/220 | PGCIL | 500 | SEP-23 |
| 22  | 1x500MVA 400/220kV ICT (3rd) at Bhatapara (PG)                                                                                                                                                                                                              | 400/220 | PGCIL | 500 | SEP-23 |
| 23  | 1x 500MVA 400/220kV ICT (4th) at Fatehgarh-III (Ramgarh-II) under Transmission system strengthening scheme for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under phase-II- Part A<br>SPV Name: POWERGRID Ramgarh Transmission Limited | 400/220 | PGCIL | 500 | Nov-23 |

| ANN |                                                                                                                                                                                                     |         |       |      |        |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|------|--------|
| 24  | 1x500MVA 400/220kV ICT along with associated bays at Kurukshetra (PG) GIS SS under Augmentation of transformation capacity at 400/220 kV Kurkshetra (PG) Substation & 400/220 kV Patiala Substation | 400/220 | PGCIL | 500  | Nov-23 |
| 25  | 1X500MVA 400/220KV ICT (3rd) at Palakkad (PG) SS under Augmentation of Transformation Capacity in Southern Region                                                                                   | 400/220 | PGCIL | 500  | Dec-23 |
| 26  | 1X500MVA 400/230KV 4TH ICT at Tuticorin-II GIS S/S Under Evacuation of RE inTirunelveli & Tuticorin WEZ (TN) 500MW                                                                                  | 400/230 | PGCIL | 500  | Dec-23 |
| 27  | 1X500MVA 400/220KV ICT-1 at BIKANER-II under RTM-Addtional Scope for PBTSL                                                                                                                          | 400/220 | PGCIL | 500  | Jan-24 |
| 28  | REP.1X315MVA 400/220KV ICT BY 1X500MVA AT TRICHY Under Add Cap Works-NLCT2_Block 2019-24                                                                                                            | 400/220 | PGCIL | 185  | Jan-24 |
| 29  | 1X500MVA 400/220KV ICT-2 at BIKANER-II (Implementation of 220 kV bays for RE generators and 400/220kV ICTs at Bikaner II PS)                                                                        | 400/220 | PGCIL | 500  | Mar-24 |
| 30  | 2X500MVA 400/220KV ICT-1 & 2 at Mohanlalganj (MOHANLALGANJ TRANSMISSION LIMITED) :INTRASTATE Project (UP)                                                                                           | 400/220 | PGCIL | 1000 | Mar-24 |
| 31  | 1x500MVA 400/220kV ICT (3rd) at Kolar (PG) SS under Augmentation of Transformation Capacity in Southern Region                                                                                      | 400/220 | PGCIL | 500  | Mar-24 |

| ANN                         |                                                                                                                                                                                                      |            |           |              |        |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|--------------|--------|
| 32                          | 1x500MVA, 400/230kV ICT (4th) at Hosur SS under Augmentation of Transformation capacity by 1x500MVA, 400/230kV ICT (4th) and common facility works at Hosur SS                                       | 400/230    | PGCIL     | 500          | Mar-24 |
| 33                          | 2x500 MVA, 400/220 kV ICT 1& 2 at Neemuch under Transmission system for evacuation of power from Neemuch Solar Park (1000 MW):<br>SPV Name:POWERGRID Neemuch Transmission System Limited             | 400/220    | PGCIL     | 1000         | Mar-24 |
| 34                          | 1X500MVA, 400/220kV ICT (6th) at Pavagada under Augmentation of Transformation capacity by 1X500MVA, 400/220kV ICT (6th) and common facility works at Pavagada (Tumkur) PS                           | 400/220    | PGCIL     | 500          | Mar-24 |
| 35                          | 1x500MVA 400/220kV ICT (3rd) ICT at Raigarh under Augmentation of transformation capacity by 1x500MVA 400/220kV ICT (3rd) ICT at Raigarh PG                                                          | 400/220    | PGCIL     | 500          | Mar-24 |
| 36                          | 1X500MVA 400/220KV ICT (8TH) at 765/400KV Bhadla-II PS under Transmission system strengthening Scheme for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under Phase II – Part B1 | 400/220    | PGCIL     | 500          | Mar-24 |
| <b>TOTAL CENTRAL SECTOR</b> |                                                                                                                                                                                                      |            |           | <b>14620</b> |        |
| <b>STATE SECTOR</b>         |                                                                                                                                                                                                      |            |           |              |        |
| 37                          | Guddigudem SS (LIS) (2x315 MVA)                                                                                                                                                                      | 400/220/11 | APTRANSCO | 630          | APR-23 |
| 38                          | Aug. at Jammalamadugu                                                                                                                                                                                | 400/220    | APTRANSCO | 500          | APR-23 |
| 39                          | Hindupur (ICT 3rd )                                                                                                                                                                                  | 400/220    | APTRANSCO | 315          | MAY-23 |

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| 40                    | 400/220kV SS Hindupur                                            | 400/220      | APTRANSCO  | 315          | Dec-23 |
| 41                    | 400KV Kasor (3X315+1X500 - 3X315)                                | 400/220      | GETCO      | 500          | Mar-24 |
| 42                    | Kalaburagi (Ferozabad) S/s (T/F - II)                            | 400/220      | KPTCL      | 500          | Mar-24 |
| 43                    | Kottayam (T/F - II)                                              | 400/220      | KSEB       | 315          | AUG-23 |
| 44                    | Kottayam (T/F - I)                                               | 400/220      | KSEB       | 315          | MAY-23 |
| 45                    | Kirnapur Addl. T/F                                               | 400/132      | MPPTCL     | 100          | OCT-23 |
| 46                    | New Koyna (Aug) (3x105 MVA)                                      | 400/220      | MSETCL     | 315          | MAY-23 |
| 47                    | 400kV Babhaleshwar Substation (3x167)                            | 400/220/33   | MSETCL     | 500          | Feb-24 |
| 48                    | Nakodar (Aug. of 1x315 MVA T/F 1 with 1x500 MVA T/F)             | 400/220      | PSTCL      | 185          | JUN-23 |
| 49                    | 400 kV S/S Dhanansu (1x500 MVA)                                  | 400/220      | PSTCL      | 500          | Jan-24 |
| 50                    | 400kV GSS Chittorgarh                                            | 400/220      | RVPNL      | 315          | Jan-24 |
| 51                    | Vellalaviduthi (2X315 )                                          | 400/230      | TANTRANSCO | 315          | OCT-23 |
| 52                    | Vellalaviduthi T/F-II                                            | 400/230      | TANTRANSCO | 315          | Dec-23 |
| 53                    | 400KV Vellore Substation                                         | 400/220      | TSTRANSCO  | 185          | AUG-23 |
| 54                    | Choutuppal (Ist Transformer)                                     | 400/220/ 132 | TSTRANSCO  | 500          | Nov-23 |
| 55                    | Suryapet                                                         | 400/220      | TSTRANSCO  | 500          | Dec-23 |
| 56                    | Dichpally (3x315 MVA to 3x315 +1x500)                            | 400/220      | TSTRANSCO  | 500          | Feb-24 |
| 57                    | Sultanpur (Capacity Aug.) T/F-III                                | 400/220      | UPPTCL     | 315          | MAY-23 |
| 58                    | Rasra Ballia (New (T/F-II )                                      | 400/220      | UPPTCL     | 500          | OCT-23 |
| 59                    | 400 kV S/s Panki, Kanpur (Capacity Augmentation) (500-315) ICT-I | 400/220      | UPPTCL     | 185          | Feb-24 |
| 60                    | Sahupuri, Chandauli (New) T/F-II                                 | 400/220      | UPPTCL     | 500          | Feb-24 |
| 61                    | Durgapur 400KV SS Tr. Aug.                                       | 400/220      | WBSETCL    | 315          | AUG-23 |
| 62                    | Gokarna 400KV                                                    | 400/220      | WBSETCL    | 315          | Dec-23 |
| 63                    | New Chanditola 400KV                                             | 400/220      | WBSETCL    | 315          | Mar-24 |
|                       | <b>TOTAL STATE SECTOR</b>                                        |              |            | <b>10065</b> |        |
| <b>PRIVATE SECTOR</b> |                                                                  |              |            |              |        |
| 64                    | 2x500MVA (400/230 kV) Karur Pooling Station                      | 400/230      | ADANI      | 1000         | OCT-23 |

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| 65                          | 3 x 500 MVA ICT GIS Buildings at Vikhroli                                           | 400/220 | ADANI      | 1500         | OCT-23 |
| 66                          | Pachora Substation ((3x500 MVA T/F)                                                 | 400/220 | G R INFRA  | 1500         | Feb-24 |
| 67                          | Establishment of 2x500MVA, 400/220kV near Kallam PS                                 | 400/220 | INDIGRID   | 1000         | Mar-24 |
| 68                          | Koppal (PS)                                                                         | 400/220 | RENEW      | 1500         | OCT-23 |
| 69                          | Augmentation of 400/220 KV Koppal Substation with 2x500 MVA ICTs                    | 400/220 | RENEW      | 1000         | Jan-24 |
| <b>TOTAL PRIVATE SECTOR</b> |                                                                                     |         |            | <b>7500</b>  |        |
| <b>TOTAL 400 kV</b>         |                                                                                     |         |            | <b>32185</b> |        |
| <b>230 kV</b>               |                                                                                     |         |            |              |        |
| <b>STATE SECTOR</b>         |                                                                                     |         |            |              |        |
| 70                          | Kadapperi S/S Aug.(160-100)                                                         | 230/110 | TANTRANSCO | 60           | APR-23 |
| 71                          | Veerapuram 230 kV                                                                   | 230/110 | TANTRANSCO | 100          | AUG-23 |
| 72                          | Enhancement of Transformer capacity from 50MVA to 100MVA in Eachangadu 230/110KV SS | 230/110 | TANTRANSCO | 50           | JUL-23 |
| 73                          | Erode (T/F - II)                                                                    | 230/110 | TANTRANSCO | 100          | JUL-23 |
| 74                          | Thirupathur 230/110 KV (Second 100 MVA ICT)                                         | 230/110 | TANTRANSCO | 100          | OCT-23 |
| 75                          | Singapuram S/S (ICT-II)                                                             | 230/110 | TANTRANSCO | 100          | OCT-23 |
| 76                          | Nokia 230kV (1x100 to 2x100)                                                        | 230/110 | TANTRANSCO | 100          | OCT-23 |
| 77                          | Kits Park 230/110 KV SS (1x80 to 1x100)                                             | 230/110 | TANTRANSCO | 20           | OCT-23 |
| 78                          | Karuppur                                                                            | 230/110 | TANTRANSCO | 100          | OCT-23 |
| 79                          | Athipattu 230kV (1x100 to 2x100)                                                    | 230/110 | TANTRANSCO | 100          | OCT-23 |
| 80                          | Thiruchuli 230/110 kV SS<br>(Enhancement from 1x50 to 1x100 MVA)                    | 230/110 | TANTRANSCO | 50           | Nov-23 |
| 81                          | Sankarapuram 230/110 kV SS<br>(Additional Transformer from 1x100 to 2x100 MVA)      | 230/110 | TANTRANSCO | 100          | Nov-23 |
| 82                          | Coimbatore 230/110 kV SS (Additional Transformer from 2x100 to 3x100 MVA)           | 230/110 | TANTRANSCO | 100          | Nov-23 |

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| 83                    | Vyasarpadi 230kV SS (2x100 to 3x100)                                                                            | 230/110 | TANTRANSCO | 100         | Feb-24 |
|                       | <b>TOTAL STATE SECTOR</b>                                                                                       |         |            | <b>1180</b> |        |
|                       | <b>TOTAL 230 kV</b>                                                                                             |         |            | <b>1180</b> |        |
| <b>220 kV</b>         |                                                                                                                 |         |            |             |        |
| <b>CENTRAL SECTOR</b> |                                                                                                                 |         |            |             |        |
| 84                    | 220/132 kV Transformer (3rd) at Salakati (NERSS-XI)                                                             | 220/132 | PGCIL      | 100         | SEP-23 |
| 85                    | Repl. of 220/132KV 1X50 MVA TO 1X160 MVA ICT at Kopili S/S Under North Eastern Region Strengthening Scheme -III | 220/132 | PGCIL      | 100         | Dec-23 |
| 86                    | 2X200MVA 220/132KV ICT-1 & 2 at Mohanlalganj (MOHANLALGANJ TRANSMISSION LIMITED) :INTRASTATE Project (UP)       | 220/132 | PGCIL      | 400         | Mar-24 |
|                       | <b>TOTAL CENTRAL SECTOR</b>                                                                                     |         |            | <b>600</b>  |        |
| <b>STATE SECTOR</b>   |                                                                                                                 |         |            |             |        |
| 87                    | Bavojipeta                                                                                                      | 220/33  | APTRANSCO  | 50          | APR-23 |
| 88                    | Piduguralla (Upgradation)                                                                                       | 220/132 | APTRANSCO  | 160         | JUN-23 |
| 89                    | 33KV features at existing 220/11KV Pattiseema LIS                                                               | 220/33  | APTRANSCO  | 50          | JUN-23 |
| 90                    | 220/132KV SS at Pampanur tanda                                                                                  | 220/132 | APTRANSCO  | 160         | Dec-23 |
| 91                    | 220/33kv Purusothapatnam (T/F-II)                                                                               | 220/33  | APTRANSCO  | 50          | Jan-24 |
| 92                    | 220/33kv Pattiseema (T/F-II)                                                                                    | 220/33  | APTRANSCO  | 50          | Jan-24 |
| 93                    | Raxaul (New) s/s (T/F I)                                                                                        | 220/132 | BSPTCL     | 200         | JUL-23 |
| 94                    | Dumraon (GIS) s/s (2nd T/F)                                                                                     | 220/132 | BSPTCL     | 160         | Dec-23 |
| 95                    | Raxaul (New) s/s (T/F -II)                                                                                      | 220/132 | BSPTCL     | 200         | Dec-23 |
| 96                    | 220/132/33 KV GSS Digha (New) (2x200 MVA T/F)                                                                   | 220/132 | BSPTCL     | 400         | Jan-24 |
| 97                    | Naubatpur GIS (Balanced) ( 220/132 kV T/Fs of 160 MVA )                                                         | 220/132 | BSPTCL     | 160         | Feb-24 |
| 98                    | 220 KV S/s Saraipali                                                                                            | 220/132 | CSPTCL     | 160         | Mar-24 |

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| 99  | 220KV Karamsad                                                                               | 220/66  | GETCO  | 110 | OCT-23 |
| 100 | 220KV Dhuvaran                                                                               | 220/132 | GETCO  | 50  | OCT-23 |
| 101 | 220 kV Sadla                                                                                 | 220/66  | GETCO  | 420 | Nov-23 |
| 102 | 220 kV Ranavav                                                                               | 220/132 | GETCO  | 400 | Nov-23 |
| 103 | 220KV Bhat (Augmentation of 100 MVA T/F to 160 MVA)                                          | 220/66  | GETCO  | 60  | Dec-23 |
| 104 | 400 kV Vav (Augmentation of 100 MVA T/F to 160 MVA)                                          | 220/66  | GETCO  | 60  | Jan-24 |
| 105 | 220kV Giyavad                                                                                | 220/66  | GETCO  | 320 | Mar-24 |
| 106 | 220kV Kamlapur Panchvada (Jasdan)                                                            | 220/66  | GETCO  | 320 | Mar-24 |
| 107 | 220kV Mera (Mithi Paldi) (Upgradation-66kV)                                                  | 220/66  | GETCO  | 160 | Mar-24 |
| 108 | 220kV Metoda (Upgradation-66kV metoda-iii)                                                   | 220/66  | GETCO  | 320 | Mar-24 |
| 109 | 220kV Sevaliya (Upgradation-66kV)                                                            | 220/66  | GETCO  | 320 | Mar-24 |
| 110 | 220KV Chhatral (3x100+1X160 - 4X100)                                                         | 220/66  | GETCO  | 60  | Mar-24 |
| 111 | Gumma ( GIS S/S )                                                                            | 220/66  | HPPTCL | 100 | MAY-23 |
| 112 | Charor (Additional T/F)                                                                      | 220/33  | HPPTCL | 100 | MAY-23 |
| 113 | Charor GIS                                                                                   | 220/132 | HPPTCL | 100 | MAY-23 |
| 114 | 220/132 kV GIS Sub-Station at Mazra, Distt. Chamba                                           | 220/132 | HPPTCL | 200 | Jan-24 |
| 115 | 33/220 kV, 25/31.5 MVA Power Transformer at 220 kV Switchyard of AD Hydro Power Ltd at Prini | 33/220  | HPPTCL | 32  | Jan-24 |
| 116 | Shahabad                                                                                     | 220/66  | HVPNL  | 100 | AUG-23 |
| 117 | 220 Kv Substation Salempur                                                                   | 220/66  | HVPNL  | 160 | JUN-23 |
| 118 | Gurgaon Sector-69 GIS (T/F II)                                                               | 220/33  | HVPNL  | 100 | JUN-23 |
| 119 | Pinjore s/s                                                                                  | 220/66  | HVPNL  | 100 | MAY-23 |
| 120 | Safidon s/s                                                                                  | 220/132 | HVPNL  | 60  | MAY-23 |
| 121 | Chajipur (Panipat) s/s                                                                       | 220/132 | HVPNL  | 60  | MAY-23 |
| 122 | 220 kV Raiwali                                                                               | 220/66  | HVPNL  | 100 | OCT-23 |

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| 123 | 220/33 Bastara s/s                                                                          | 220/33    | HVPNL  | 100 | OCT-23 |
| 124 | 220kV Bastara                                                                               | 220/132   | HVPNL  | 160 | SEP-23 |
| 125 | 220 kV Kaithal                                                                              | 220/132   | HVPNL  | 60  | SEP-23 |
| 126 | 220kV Bastara                                                                               | 220/132   | HVPNL  | 60  | Nov-23 |
| 127 | 220 kV Sagwan                                                                               | 220/33    | HVPNL  | 100 | Nov-23 |
| 128 | 220kV METL Dadri Toe                                                                        | 220/33    | HVPNL  | 100 | Jan-24 |
| 129 | 220kV Palwal                                                                                | 220/66    | HVPNL  | 60  | Mar-24 |
| 130 | 220KV GIS HSIIDC Rai, Sonapat                                                               | 220/132   | HVPNL  | 160 | Mar-24 |
| 131 | 220 KV Dadhibana                                                                            | 220/33    | HVPNL  | 100 | Mar-24 |
| 132 | Amargarh (Delina)                                                                           | 220/132   | JKPDD  | 160 | AUG-23 |
| 133 | GIS Harwan                                                                                  | 220/33    | JKPDD  | 160 | Feb-24 |
| 134 | 220/33 kV Haripora (Tailbal) (GIS) Grid S/S (earlier Batpora) (PMDP- Kashmir)               | 220/33    | JKPDD  | 160 | Mar-24 |
| 135 | 220kV Narendra S/S (Additional 1X100MVA T/F)                                                | 220/110   | KPTCL  | 100 | APR-23 |
| 136 | Sira (Kallukote)s/s                                                                         | 220/66/11 | KPTCL  | 100 | AUG-23 |
| 137 | Yerandanahalli Replacing 2x100MVA 220/66kV PT by 2x150MVA 220/66kV PT (1st T/F replacement) | 220/66    | KPTCL  | 50  | JUL-23 |
| 138 | Kushtagi (3rd T/F)                                                                          | 220/110   | KPTCL  | 100 | JUL-23 |
| 139 | Srinivasapura s/s                                                                           | 220/66    | KPTCL  | 200 | OCT-23 |
| 140 | Puttenahalli(DG Plant): (Additional 3rd T/F)                                                | 220/66    | KPTCL  | 150 | Dec-23 |
| 141 | Yerandanahalli: Replacing 2x100MVA, 220/66kV PT by 2x150MVA, 220/66kV Power Transformers    | 220/66    | KPTCL  | 50  | Feb-24 |
| 142 | Ghanagapura: Establishing 2X100MVA, 220/110kV sub-station.                                  | 220/110   | KPTCL  | 200 | Mar-24 |
| 143 | Ganjbasoda 220KV S/s additional 1x160 MVA X-mer                                             | 220/132   | MPPTCL | 160 | AUG-23 |

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| 144 | Itarsi 220 kV S/s replacement of 3x40MVA by 1x160MVA X-mer     | 220/132 | MPPTCL | 40  | AUG-23 |
| 145 | Indore -SZ 220 kV S/s Replacement of 3x40MVA by 1x160MVA X-mer | 220/132 | MPPTCL | 40  | JUN-23 |
| 146 | Julwaniya S/s (Addl. 160MVA X-mer )                            | 220/132 | MPPTCL | 160 | JUN-23 |
| 147 | Rajgarh (D) Addl. (1x160MVA) X-mer                             | 220/132 | MPPTCL | 160 | MAY-23 |
| 148 | Nimrani Addl. T/F                                              | 220/132 | MPPTCL | 160 | MAY-23 |
| 149 | Jabalpur 220 kV S/s replacement of 3x40MVA by 1x160MVA X-mer   | 220/132 | MPPTCL | 40  | MAY-23 |
| 150 | Chichli Addl. T/F                                              | 220/132 | MPPTCL | 160 | MAY-23 |
| 151 | Aug. of Mehgaon Xmer (200-160)                                 | 220/132 | MPPTCL | 40  | MAY-23 |
| 152 | Nagda 220 kV S/s Aug. (200-125)                                | 220/132 | MPPTCL | 75  | SEP-23 |
| 153 | Tikamgarh 220 KV S/s, Addl 1x160 MVA X-mer                     | 220/132 | MPPTCL | 160 | Nov-23 |
| 154 | Ashta 400KV S/s, Addl. 1x160 MVA X-mer                         | 220/132 | MPPTCL | 160 | Mar-24 |
| 155 | Mugaliyachhap 220 KV S/s, Addl. 1x160MVA X-mer                 | 220/132 | MPPTCL | 160 | Mar-24 |
| 156 | Manpur (Bijouri) 220/33kV substation (TBCB)                    | 220/33  | MPPTCL | 100 | Mar-24 |
| 157 | 220/33KV GIS at Ulwe node                                      | 220/33  | MSETCL | 100 | JUL-23 |
| 158 | Kolshet s/s                                                    | 220/110 | MSETCL | 50  | MAY-23 |
| 159 | Shendra s/s (2 x 100MVA 220/132kV ICTs)                        | 220/132 | MSETCL | 200 | MAY-23 |
| 160 | 400kV Waluj substation (1x50MVA, 220/33kV T/F)                 | 220/33  | MSETCL | 50  | Nov-23 |
| 161 | 220kV Boisar-II Substation(Aug.) (1x 200-150)                  | 220/132 | MSETCL | 50  | Feb-24 |
| 162 | 220/33kV Khed City substation (2x50 MVA, 220/33kV T/Fs)        | 220/33  | MSETCL | 100 | Feb-24 |
| 163 | 400kV Kharghar substation (1 X 50MVA, 220/33kV T/F)            | 220/33  | MSETCL | 50  | Mar-24 |

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| 164 | 220kV Pandharpur substation<br>1 X (200-100) MVA 220/132kV ICT          | 220/132 | MSETCL | 100 | Mar-24 |
| 165 | Up-gradation of existing 132 kV Pandharkawda substation to 220 kV level | 220/132 | MSETCL | 100 | Mar-24 |
| 166 | 220kV Nanded City substation<br>(1x50MVA, 220/22kV T/F)                 | 220/22  | MSETCL | 50  | Mar-24 |
| 167 | Creation of 110 kV Level at 220 kV Peth S/s (1x 100MVA, 220/110kV ICT)  | 220/110 | MSETCL | 100 | Mar-24 |
| 168 | 2X63 MVA 220/33kV Kantabada S/s                                         | 220/33  | OPTCL  | 126 | APR-23 |
| 169 | Dhamra S/s (T/F -I)                                                     | 220/132 | OPTCL  | 160 | APR-23 |
| 170 | 220/132/33kV Balimunda                                                  | 220/132 | OPTCL  | 160 | Nov-23 |
| 171 | Bamra SS T/F-II                                                         | 220/132 | OPTCL  | 160 | Dec-23 |
| 172 | 220 kV S/S G-1. (Aug. of 2x100 MVA T/F with 2x160 MVA) T/F-II Aug.      | 220/66  | PSTCL  | 60  | JUN-23 |
| 173 | 220 kV S/S G-1. (Aug. of 1x100 MVA T/F with 1x160 MVA)                  | 220/66  | PSTCL  | 60  | MAY-23 |
| 174 | Udhoke (Addl. T/F)                                                      | 220/66  | PSTCL  | 100 | SEP-23 |
| 175 | 220 kV S/S Banga (Addl. 100 MVA T/F)                                    | 220/132 | PSTCL  | 100 | SEP-23 |
| 176 | 220 kV S/S Kharar (Aug. of 1x100 MVA T/F with 1x160 MVA T/F)            | 220/66  | PSTCL  | 60  | Dec-23 |
| 177 | 220 kV S/S Dhandari Kalan-1 (Aug. of 1x100 MVA T/F with 1x160 MVA T/F)  | 220/66  | PSTCL  | 60  | Dec-23 |
| 178 | 220 kV S/S Sahnewal (Augmentation of 100 MVA 220/66 to 160 MVA)         | 220/66  | PSTCL  | 60  | Dec-23 |
| 179 | 220 kV S/S Talwandi Sabo (Addl. 100 MVA T/F)                            | 220/66  | PSTCL  | 100 | Jan-24 |
| 180 | 220kV GSS Reodar (Distt. Sirohi)                                        | 220/132 | RRVPNL | 160 | Mar-24 |
| 181 | 220kV GSS Chaksu (Additional 100 MVA T/F)                               | 220/132 | RRVPNL | 100 | Mar-24 |
| 182 | 220kV GSS Halasar                                                       | 220/132 | RVPNL  | 100 | AUG-23 |

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| 183 | 220 kV GSS Dholpur (160-100)                         | 220/132    | RVPNL     | 60  | JUL-23 |
| 184 | 220 kV GSS Dhorimanna (160-100)                      | 220/132    | RVPNL     | 60  | JUL-23 |
| 185 | 220 kV GSS Sridungargarh (160-100)                   | 220/132    | RVPNL     | 60  | JUL-23 |
| 186 | 220kV GSS Pindwara                                   | 220/132    | RVPNL     | 100 | OCT-23 |
| 187 | Devannapet LI SS (3x40 2x25 MVA)                     | 220/11     | TSTRANSCO | 170 | JUN-23 |
| 188 | Choutuppal SS                                        | 220/ 132   | TSTRANSCO | 160 | Jan-24 |
| 189 | Choutuppal SS (160 MVA T/F-II)                       | 220/ 132   | TSTRANSCO | 160 | Feb-24 |
| 190 | Ammavaripet T/F-I                                    | 220/132    | TSTRANSCO | 100 | Mar-24 |
| 191 | Barahua Aug.(160-100)                                | 220/132    | UPPTCL    | 60  | APR-23 |
| 192 | Balrampur S/s (T/F II)                               | 220/132    | UPPTCL    | 160 | APR-23 |
| 193 | Babina S/s Jhansi (T/F II)                           | 220/132    | UPPTCL    | 160 | APR-23 |
| 194 | Deoband Saharanpur (New) T/F                         | 220/33     | UPPTCL    | 60  | APR-23 |
| 195 | Moth-II Jhansi T/F-I                                 | 220/132    | UPPTCL    | 160 | APR-23 |
| 196 | Farrukhabad S/s (T/F II)                             | 220/132    | UPPTCL    | 160 | APR-23 |
| 197 | Deoband (Saharanpur) S/s                             | 220/132    | UPPTCL    | 320 | APR-23 |
| 198 | 220/132/33 KV S/s DataganjBadaun (New) T/F- I and II | 220/132    | UPPTCL    | 320 | AUG-23 |
| 199 | 220 kV S/s Morta Ghaziabad (New) T/F- II and III     | 220/33     | UPPTCL    | 120 | AUG-23 |
| 200 | Modipuram-II (T/F II)                                | 220/132    | UPPTCL    | 160 | JUN-23 |
| 201 | Kidwainagar S/s Kanpur (T/F III)                     | 220/33     | UPPTCL    | 60  | MAY-23 |
| 202 | Rukhi Syana (New) T/F-II                             | 220/132    | UPPTCL    | 160 | MAY-23 |
| 203 | Bhadrekhi (Orai) (New) T/F-II                        | 220/132    | UPPTCL    | 160 | MAY-23 |
| 204 | Kirawali Agra (New) T/F- II                          | 220/132    | UPPTCL    | 100 | OCT-23 |
| 205 | Bhadohi S/S (T/F - I)                                | 220/132    | UPPTCL    | 200 | OCT-23 |
| 206 | 220/33 kV S/s IITGNL (GIS) GB Nagar                  | 220/33     | UPPTCL    | 60  | SEP-23 |
| 207 | Anand Nagar Mahrajganj (New) T/F- I and II           | 220/132/33 | UPPTCL    | 320 | SEP-23 |
| 208 | Kirawali Agra (New) T/F- I                           | 220/132    | UPPTCL    | 100 | SEP-23 |
| 209 | Khatauli S/s (T/F-II)                                | 220/132    | UPPTCL    | 160 | SEP-23 |
| 210 | Kanduni Sitapur (New) T/F- I                         | 220/132    | UPPTCL    | 200 | SEP-23 |

| ANN |                                                           |         |         |              |        |
|-----|-----------------------------------------------------------|---------|---------|--------------|--------|
| 211 | 220/132/33 kV S/s Mahrajganj (New) T/F- I                 | 220/132 | UPPTCL  | 160          | SEP-23 |
| 212 | 220/132 kV S/s Moth (New) T/F                             | 220/132 | UPPTCL  | 160          | Nov-23 |
| 213 | 220/33 kV S/s YEIDA SEC-18, GB Nagar (New) T/F-I & II     | 220/33  | UPPTCL  | 120          | Nov-23 |
| 214 | 220/33 kV S/s Morta, Ghaziabad (New) T/F-I                | 220/33  | UPPTCL  | 60           | Nov-23 |
| 215 | 220/33 kV S/s YEIDA SEC-18, GB Nagar (New) T/F-III        | 220/33  | UPPTCL  | 60           | Dec-23 |
| 216 | 220/132 kV S/s Bhadohi (New) T/F-II                       | 220/132 | UPPTCL  | 200          | Dec-23 |
| 217 | 220 kV S/s YEIDA SEC-24, Gr. Noida (New) T/F-I            | 220/132 | UPPTCL  | 160          | Jan-24 |
| 218 | 220 kV S/s Sikandra, Agra (Capacity Augmentation) T/F-III | 220/33  | UPPTCL  | 60           | Jan-24 |
| 219 | 220 kV S/s Knowledge Park-5, Gr. Noida (New) T/F-I        | 220/132 | UPPTCL  | 160          | Jan-24 |
| 220 | 220 kV S/s Jalpura, Gr. Noida (New) T/F-I & II            | 220/132 | UPPTCL  | 320          | Jan-24 |
| 221 | 220 kV S/s Khorabar, Gorakhpur (New) T/F-I                | 220/33  | UPPTCL  | 60           | Feb-24 |
| 222 | 220 kV S/s Mahoba (Capacity Augmentation) (160-100) T/F-I | 220/33  | UPPTCL  | 60           | Feb-24 |
| 223 | Jangalpur GIS                                             | 220/132 | WBSETCL | 320          | JUL-23 |
| 224 | Mangalpur 220KV S/S (2X100 MVA T/F)                       | 220/33  | WBSETCL | 200          | Jan-24 |
| 225 | Newtown AA II C (2X160 MVA T/F)                           | 220/132 | WBSETCL | 320          | Jan-24 |
| 226 | Newtown AA II C (2X100 MVA T/F)                           | 220/33  | WBSETCL | 200          | Jan-24 |
|     | <b>Total State Sector</b>                                 |         |         | <b>18763</b> |        |
|     | <b>Total 220 kv</b>                                       |         |         | <b>19363</b> |        |
|     | <b>Grand Total</b>                                        |         |         | <b>70728</b> |        |

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**Annexure – 4A****Power Supply Position for 2023-24 (Revised)**

|                                         | Energy                 |                 |                     |            | Peak                   |               |                |            |
|-----------------------------------------|------------------------|-----------------|---------------------|------------|------------------------|---------------|----------------|------------|
| State /                                 | April,2023 -March,2024 |                 |                     |            | April,2023 -March,2024 |               |                |            |
| System /                                | Energy Requirement     | Energy Supplied | Energy not Supplied |            | Peak Demand            | Peak Met      | Demand not Met |            |
| Region                                  | ( MU )                 | ( MU )          | ( MU )              | ( % )      | ( MW )                 | ( MW )        | ( MW )         | ( % )      |
| Chandigarh                              | 1,789                  | 1,789           | 0                   | 0.0        | 411                    | 411           | 0              | 0.0        |
| Delhi                                   | 35,501                 | 35,496          | 5                   | 0.0        | 7,437                  | 7,437         | 0              | 0.0        |
| Haryana                                 | 63,983                 | 63,636          | 348                 | 0.5        | 13,088                 | 12,844        | 244            | 1.9        |
| Himachal Pradesh                        | 12,805                 | 12,767          | 38                  | 0.3        | 2,181                  | 2,181         | 0              | 0.0        |
| UT of J&K and Ladakh                    | 20,040                 | 19,763          | 277                 | 1.4        | 3,181                  | 3,133         | 48             | 1.5        |
| Punjab                                  | 69,533                 | 69,528          | 5                   | 0.0        | 15,293                 | 15,293        | 0              | 0.0        |
| Rajasthan                               | 1,07,422               | 1,06,806        | 616                 | 0.6        | 18,128                 | 18,128        | 0              | 0.0        |
| Uttar Pradesh                           | 1,48,791               | 1,48,287        | 504                 | 0.3        | 28,704                 | 28,284        | 420            | 1.5        |
| Uttarakhand                             | 15,644                 | 15,532          | 112                 | 0.7        | 2,635                  | 2,405         | 230            | 8.7        |
| <b>Northern Region (##)</b>             | <b>4,76,852</b>        | <b>4,74,946</b> | <b>1,906</b>        | <b>0.4</b> | <b>80,980</b>          | <b>80,548</b> | <b>432</b>     | <b>0.5</b> |
| Chhattisgarh                            | 39,930                 | 39,872          | 58                  | 0.1        | 6,148                  | 6,148         | 0              | 0.0        |
| Gujarat                                 | 1,45,768               | 1,45,740        | 28                  | 0.0        | 24,829                 | 24,544        | 285            | 1.1        |
| Madhya Pradesh                          | 99,301                 | 99,150          | 151                 | 0.2        | 18,252                 | 17,817        | 435            | 2.4        |
| Maharashtra                             | 2,07,108               | 2,06,931        | 176                 | 0.1        | 31,178                 | 27,996        | 3,182          | 10.2       |
| Dadra & Nagar Haveli and<br>Daman & Diu | 10,164                 | 10,164          | 0                   | 0.0        | 1,327                  | 1,327         | 0              | 0.0        |
| Goa                                     | 5,111                  | 5,111           | 0                   | 0.0        | 776                    | 776           | 0              | 0.0        |
| <b>Western Region (##)</b>              | <b>5,17,714</b>        | <b>5,17,301</b> | <b>413</b>          | <b>0.1</b> | <b>76,050</b>          | <b>72,556</b> | <b>3,494</b>   | <b>4.6</b> |

| ANN                         |                        |                 |                     |            |                        |               |                |            |
|-----------------------------|------------------------|-----------------|---------------------|------------|------------------------|---------------|----------------|------------|
|                             | Energy                 |                 |                     |            | Peak                   |               |                |            |
| State /                     | April,2023 -March,2024 |                 |                     |            | April,2023 -March,2024 |               |                |            |
| System /                    | Energy Requirement     | Energy Supplied | Energy not Supplied |            | Peak Demand            | Peak Met      | Demand not Met |            |
| Region                      | ( MU )                 | ( MU )          | ( MU )              | ( % )      | ( MW )                 | ( MW )        | ( MW )         | ( % )      |
| Andhra Pradesh              | 80,209                 | 80,151          | 57                  | 0.1        | 13,237                 | 13,237        | 0              | 0.0        |
| Telangana                   | 84,623                 | 84,613          | 9                   | 0.0        | 15,622                 | 15,622        | 0              | 0.0        |
| Karnataka                   | 94,088                 | 93,934          | 154                 | 0.2        | 17,212                 | 17,212        | 0              | 0.0        |
| Kerala                      | 30,943                 | 30,938          | 5                   | 0.0        | 5,284                  | 5,284         | 0              | 0.0        |
| Tamil Nadu                  | 1,26,163               | 1,26,151        | 12                  | 0.0        | 19,045                 | 19,045        | 0              | 0.0        |
| Puducherry                  | 3,456                  | 3,455           | 1                   | 0.0        | 524                    | 524           | 0              | 0.0        |
| Lakshadweep (#)             | 64                     | 64              | 0                   | 0.0        | 12                     | 12            | 0              | 0.6        |
| <b>Southern Region (##)</b> | <b>4,19,531</b>        | <b>4,19,293</b> | <b>238</b>          | <b>0.1</b> | <b>68,094</b>          | <b>68,094</b> | <b>0</b>       | <b>0.0</b> |
| Bihar                       | 41,514                 | 40,918          | 596                 | 1.4        | 8,049                  | 7,420         | 629            | 7.8        |
| DVC                         | 26,560                 | 26,552          | 8                   | 0.0        | 3,451                  | 3,451         | 0              | 0.0        |
| Jharkhand                   | 14,408                 | 13,858          | 550                 | 3.8        | 2,193                  | 2,020         | 173            | 7.9        |
| Odisha                      | 41,358                 | 41,333          | 25                  | 0.1        | 6,443                  | 6,443         | 0              | 0.0        |
| West Bengal                 | 67,576                 | 67,490          | 86                  | 0.1        | 11,626                 | 11,626        | 0              | 0.0        |
| Sikkim                      | 544                    | 543             | 0                   | 0.0        | 133                    | 133           | 0              | 0.3        |
| Andaman- Nicobar (#)        | 386                    | 374             | 12                  | 3.2        | 65                     | 60            | 5              | 7.7        |
| <b>Eastern Region (##)</b>  | <b>1,92,013</b>        | <b>1,90,747</b> | <b>1,266</b>        | <b>0.7</b> | <b>30,256</b>          | <b>29,299</b> | <b>957</b>     | <b>3.2</b> |
| Arunachal Pradesh           | 1,014                  | 1,014           | 0                   | 0.0        | 186                    | 186           | 0              | 0.2        |
| Assam                       | 12,445                 | 12,341          | 104                 | 0.8        | 2,413                  | 2,413         | 0              | 0.0        |

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|                                  | Energy                 |                  |                     |            | Peak                   |                 |                |            |
|----------------------------------|------------------------|------------------|---------------------|------------|------------------------|-----------------|----------------|------------|
| State /                          | April,2023 -March,2024 |                  |                     |            | April,2023 -March,2024 |                 |                |            |
| System /                         | Energy Requirement     | Energy Supplied  | Energy not Supplied |            | Peak Demand            | Peak Met        | Demand not Met |            |
| Region                           | ( MU )                 | ( MU )           | ( MU )              | ( % )      | ( MW )                 | ( MW )          | ( MW )         | ( % )      |
| Manipur                          | 1,023                  | 1,008            | 15                  | 1.5        | 258                    | 248             | 10             | 3.8        |
| Meghalaya                        | 2,236                  | 2,066            | 170                 | 7.6        | 405                    | 405             | 0              | 0.0        |
| Mizoram                          | 684                    | 684              | 0                   | 0.0        | 162                    | 162             | 0              | 0.0        |
| Nagaland                         | 921                    | 921              | 0                   | 0.0        | 174                    | 174             | 0              | 0.1        |
| Tripura (*)                      | 1,691                  | 1,691            | 0                   | 0.0        | 362                    | 362             | 0              | 0.0        |
| <b>North-Eastern Region (##)</b> | <b>20,022</b>          | <b>19,733</b>    | <b>289</b>          | <b>1.4</b> | <b>3,678</b>           | <b>3,678</b>    | <b>0</b>       | <b>0.0</b> |
| <b>All India</b>                 | <b>16,26,132</b>       | <b>16,22,020</b> | <b>4,112</b>        | <b>0.3</b> | <b>2,43,271</b>        | <b>2,39,931</b> | <b>3,340</b>   | <b>1.4</b> |

(#) Lakshadweep and Andaman & Nicobar Islands are stand- alone systems, power supply position of these, does not form part of regional requirement and supply.

(\*) Excludes the supply to Bangladesh.

**Note:**

1. Power Supply Position Report has been compiled based on the data furnished by State Utilities/ Electricity Departments. The MU & MW figures has been rounded off to nearest unit place.

2. (##) The Regional figures include data of other miscellaneous entities drawing power directly from ISTS.

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**Annexure - 4B****Allocation from Conventional Central Generating Stations and Bhutan Stations as on 31.03.2024**

| S. N o. | Region / State                   | Firm Share                               |                                     |          |                  |                   |                       | Dedic ated Power | Un-Allocated Power                              |       |                   |                                        |                                            |         | Total Share from C.G.S. and Bhutan Stations |                        |           |                     |                      |
|---------|----------------------------------|------------------------------------------|-------------------------------------|----------|------------------|-------------------|-----------------------|------------------|-------------------------------------------------|-------|-------------------|----------------------------------------|--------------------------------------------|---------|---------------------------------------------|------------------------|-----------|---------------------|----------------------|
|         |                                  | Firm Power from Regional Pool and Bhutan | Firm from Other Region s and Bhutan | Non Firm | Total Firm Power |                   |                       |                  | Unallocated Power from Regional Pool and Bhutan |       |                   | Allocatio n from other Region / Bhutan | Total Allocation of Un-Allocated Power (*) |         |                                             |                        |           |                     |                      |
|         |                                  | MW                                       | MW                                  | MW       | MW               | % Region al Total | % of All India Tot al |                  | MW                                              | MW    | % Region al Total | % of All Indi a Tot al                 | MW                                         | MW      | % Region al Total                           | % of All Indi a Tot al | MW        | % Reg ion al Tot al | % of All India Total |
|         |                                  | 1                                        | 2                                   | 3        | 4=1+2+3          | 5                 | 6                     |                  | 7                                               | 8     | 9                 | 10                                     | 11                                         | 12=8+11 | 13                                          | 14                     | 15=4+7+12 | 16                  | 17                   |
| 1       | Chandigarh                       | 166.6                                    | 3.0                                 | 0.0      | 169.6            | 0.7               | 0.2                   | 0.0              | 120.2                                           | 4.7   | 1.2               | 0.0                                    | 120.2                                      | 4.1     | 1.1                                         | 289.8                  | 1.0       | 0.3                 |                      |
| 2       | Delhi                            | 3064.0                                   | 741.7                               | 0.0      | 3805.7           | 16.3              | 4.7                   | 0.0              | 0.0                                             | 0.0   | 0.0               | 30.0                                   | 30.0                                       | 1.0     | 0.3                                         | 3835.6                 | 13.8      | 3.9                 |                      |
| 3       | Haryana                          | 1871.6                                   | 418.3                               | 48.0     | 2337.9           | 10.0              | 2.9                   | 431.0            | 8.7                                             | 0.3   | 0.1               | 15.0                                   | 23.7                                       | 0.8     | 0.2                                         | 2792.6                 | 10.0      | 2.8                 |                      |
| 4       | Himachal Pradesh                 | 1374.5                                   | 23.0                                | 0.0      | 1397.5           | 6.0               | 1.7                   | 0.0              | 119.3                                           | 4.6   | 1.2               | 0.0                                    | 119.3                                      | 4.1     | 1.1                                         | 1516.8                 | 5.5       | 1.5                 |                      |
| 5       | UT of Jammu & Kashmir and Ladakh | 1614.1                                   | 420.9                               | 35.0     | 2070.0           | 8.9               | 2.5                   | 89.0             | 1146.0                                          | 44.6  | 11.3              | 118.1                                  | 1264.1                                     | 43.3    | 11.5                                        | 3423.0                 | 12.3      | 3.4                 |                      |
| 6       | Punjab                           | 1639.9                                   | 820.3                               | 100.0    | 2560.2           | 10.9              | 3.1                   | 0.0              | 45.2                                            | 1.8   | 0.4               | 30.0                                   | 75.2                                       | 2.6     | 0.7                                         | 2635.4                 | 9.5       | 2.6                 |                      |
| 7       | Rajasthan                        | 1771.9                                   | 132.2                               | 125.0    | 2029.0           | 8.7               | 2.5                   | 550.0            | 667.5                                           | 26.0  | 6.6               | 15.0                                   | 682.5                                      | 23.4    | 6.2                                         | 3261.5                 | 11.7      | 3.3                 |                      |
| 8       | Uttar Pradesh                    | 7107.4                                   | 624.8                               | 66.0     | 7798.3           | 33.3              | 9.6                   | 440.0            | 148.7                                           | 5.8   | 1.5               | 101.9                                  | 250.6                                      | 8.6     | 2.3                                         | 8488.9                 | 30.5      | 8.5                 |                      |
| 9       | Uttarakhand                      | 941.0                                    | 28.1                                | 0.0      | 969.0            | 4.1               | 1.2                   | 0.0              | 312.8                                           | 12.2  | 3.1               | 35.4                                   | 348.2                                      | 11.9    | 3.2                                         | 1317.2                 | 4.7       | 1.3                 |                      |
| 10      | PowerGrid                        | 6.3                                      | 0.0                                 | 0.0      | 6.3              | 0.0               | 0.0                   | 0.0              | 3.3                                             | 0.1   | 0.0               | 0.0                                    | 3.3                                        | 0.1     | 0.0                                         | 9.6                    | 0.0       | 0.0                 |                      |
| 11      | Railways NR                      | 0.0                                      | 241.8                               | 0.0      | 241.8            | 1.0               | 0.3                   | 0.0              | 0.0                                             |       |                   | 0.0                                    | 0.0                                        |         |                                             | 241.8                  |           | 0.2                 |                      |
| 12      | NDMC                             | 0.0                                      | 0.0                                 | 0.0      | 0.0              | 0.0               | 0.0                   | 0.0              | 0.0                                             |       |                   | 0.0                                    | 0.0                                        |         |                                             | 0.0                    |           | 0.0                 |                      |
|         | Northern Region                  | 19557.3                                  | 3454.0                              | 374.0    | 23385.3          | 100.0             | 28.6                  | 1510.0           | 2571.7                                          | 100.0 | 25.3              | 345.3                                  | 2917.0                                     | 100.0   | 26.6                                        | 27812.3                | 100.0     | 28.0                |                      |

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| S. N o. | Region / State  | Firm Share                               |                                     |          |                  |                   |                       | Dedic ated Power | Un-Allocated Power                              |       |                   |                                        |                                            |       |                   | Total Share from C.G.S. and Bhutan Stations |       |                     |
|---------|-----------------|------------------------------------------|-------------------------------------|----------|------------------|-------------------|-----------------------|------------------|-------------------------------------------------|-------|-------------------|----------------------------------------|--------------------------------------------|-------|-------------------|---------------------------------------------|-------|---------------------|
|         |                 | Firm Power from Regional Pool and Bhutan | Firm from Other Region s and Bhutan | Non Firm | Total Firm Power |                   |                       |                  | Unallocated Power from Regional Pool and Bhutan |       |                   | Allocatio n from other Region / Bhutan | Total Allocation of Un-Allocated Power (*) |       |                   |                                             |       |                     |
|         |                 | MW                                       | MW                                  | MW       | MW               | % Region al Total | % of All India Tot al |                  | MW                                              | MW    | % Region al Total | % of All Indi a Tot al                 | MW                                         | MW    | % Region al Total | % of All Indi a Tot al                      | MW    | % Reg ion al Tot al |
|         |                 | 1                                        | 2                                   | 3        | 4=1+2+3          | 5                 | 6                     | 7                | 8                                               | 9     | 10                | 11                                     | 12=8+11                                    | 13    | 14                | 15=4+7+12                                   | 16    | 17                  |
| 13      | Chhattisgarh    | 2489.9                                   | 143.0                               | 0.0      | 2632.8           | 12.1              | 3.2                   | 50.0             | 25.0                                            | 0.8   | 0.2               | 0.0                                    | 25.0                                       | 0.8   | 0.2               | 2707.8                                      | 9.4   | 2.7                 |
| 14      | Gujarat         | 5274.4                                   | 1505.7                              | 0.0      | 6780.1           | 31.1              | 8.3                   | 160.0            | 341.1                                           | 11.3  | 3.4               | 32.7                                   | 373.8                                      | 12.1  | 3.4               | 7313.9                                      | 25.3  | 7.4                 |
| 15      | Madhya Pradesh  | 4526.5                                   | 574.0                               | 0.0      | 5100.4           | 23.4              | 6.2                   | 1520.0           | 584.1                                           | 19.4  | 5.8               | 40.0                                   | 624.1                                      | 20.2  | 5.7               | 7244.5                                      | 25.0  | 7.3                 |
| 16      | Maharashtra     | 6022.6                                   | 148.1                               | 0.0      | 6170.6           | 28.3              | 7.6                   | 2028.7           | 755.9                                           | 25.1  | 7.4               | 0.0                                    | 755.9                                      | 24.5  | 6.9               | 8955.2                                      | 30.9  | 9.0                 |
| 17      | DNH &DD         | 272.7                                    | 5.0                                 | 0.0      | 277.7            | 1.3               | 0.3                   | 248.7            | 1162.8                                          | 38.5  | 11.5              | 0.0                                    | 1162.8                                     | 37.6  | 10.6              | 1689.1                                      | 5.8   | 1.7                 |
| 18      | Goa             | 434.0                                    | 102.0                               | 0.0      | 536.0            | 2.5               | 0.7                   | 19.7             | 114.1                                           | 3.8   | 1.1               | 0.0                                    | 114.1                                      | 3.7   | 1.0               | 669.8                                       | 2.3   | 0.7                 |
| 19      | PowerGrid       | 0.0                                      | 0.0                                 | 0.0      | 0.0              | 0.0               | 0.0                   | 0.0              | 7.3                                             | 0.2   | 0.1               | 0.0                                    | 7.3                                        | 0.2   | 0.1               | 7.3                                         | 0.0   | 0.0                 |
| 20      | HWP of DAE      | 0.0                                      | 0.0                                 | 0.0      | 0.0              | 0.0               | 0.0                   | 0.0              | 14.0                                            | 0.5   | 0.1               | 0.0                                    | 14.0                                       | 0.5   | 0.1               | 14.0                                        | 0.0   | 0.0                 |
| 21      | BARC Facilities | 0.0                                      | 0.0                                 | 0.0      | 0.0              | 0.0               | 0.0                   | 0.0              | 10.0                                            | 0.3   | 0.1               | 0.0                                    | 10.0                                       | 0.3   | 0.1               | 10.0                                        | 0.0   | 0.0                 |
| 22      | Railways WR     | 0.0                                      | 334.1                               | 0.0      | 334.1            | 1.5               | 0.4                   | 0.0              | 2.3                                             | 0.1   | 0.0               | 0.0                                    | 2.3                                        | 0.1   | 0.0               | 336.4                                       | 1.2   | 0.3                 |
|         | Western Region  | 19020.0                                  | 2811.7                              | 0.0      | 21831.7          | 100.0             | 26.7                  | 4027.1           | 3016.6                                          | 100.0 | 29.7              | 72.7                                   | 3089.3                                     | 100.0 | 28.2              | 28948.0                                     | 100.0 | 29.1                |
| 23      | Andhra Pradesh  | 1854.3                                   | 0.0                                 | 0.0      | 1854.3           | 10.3              | 2.3                   | 0.0              | 135.0                                           | 5.3   | 1.3               | 0.0                                    | 135.0                                      | 4.9   | 1.2               | 1989.3                                      | 9.4   | 2.0                 |
| 24      | Telangana       | 3377.0                                   | 0.0                                 | 0.0      | 3377.0           | 18.8              | 4.1                   | 0.0              | 300.5                                           | 11.9  | 3.0               | 200.1                                  | 500.6                                      | 18.3  | 4.6               | 3877.6                                      | 18.4  | 3.9                 |
| 25      | Karnataka       | 3597.7                                   | 752.4                               | 0.0      | 4350.1           | 24.2              | 5.3                   | 0.0              | 932.7                                           | 36.8  | 9.2               | 0.0                                    | 932.7                                      | 34.1  | 8.5               | 5282.8                                      | 25.1  | 5.3                 |
| 26      | Kerala          | 1537.5                                   | 150.0                               | 0.0      | 1687.5           | 9.4               | 2.1                   | 360.0            | 243.7                                           | 9.6   | 2.4               | 0.0                                    | 243.7                                      | 8.9   | 2.2               | 2291.2                                      | 10.9  | 2.3                 |

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| S. N o. | Region / State  | Firm Share                               |                                     |          |                  |                   |                        | Dedic ated Power | Un-Allocated Power                              |       |                   |                                        |                                             |       |                   | Total Share from C.G.S. and Bhutan Stations |       |                     |
|---------|-----------------|------------------------------------------|-------------------------------------|----------|------------------|-------------------|------------------------|------------------|-------------------------------------------------|-------|-------------------|----------------------------------------|---------------------------------------------|-------|-------------------|---------------------------------------------|-------|---------------------|
|         |                 | Firm Power from Regional Pool and Bhutan | Firm from Other Region s and Bhutan | Non Firm | Total Firm Power |                   |                        |                  | Unallocated Power from Regional Pool and Bhutan |       |                   | Allocatio n from other Region / Bhutan | Total Allocation of Un- Allocated Power (*) |       |                   |                                             |       |                     |
|         |                 | MW                                       | MW                                  | MW       | MW               | % Region al Total | % of All Indi a Tot al |                  | MW                                              | MW    | % Region al Total | % of All Indi a Tot al                 | MW                                          | MW    | % Region al Total | % of All Indi a Tot al                      | MW    | % Reg ion al Tot al |
|         |                 | 1                                        | 2                                   | 3        | 4=1+2+3          | 5                 | 6                      | 7                | 8                                               | 9     | 10                | 11                                     | 12=8+11                                     | 13    | 14                | 15=4+7+12                                   | 16    | 17                  |
| 27      | Tamil Nadu      | 6104.8                                   | 35.0                                | 0.0      | 6139.9           | 34.1              | 7.5                    | 0.0              | 635.2                                           | 25.1  | 6.3               | 0.0                                    | 635.2                                       | 23.2  | 5.8               | 6775.1                                      | 32.1  | 6.8                 |
| 28      | Puducherry      | 338.6                                    | 0.0                                 | 0.0      | 338.6            | 1.9               | 0.4                    | 0.0              | 278.6                                           | 11.0  | 2.7               | 0.0                                    | 278.6                                       | 10.2  | 2.5               | 617.2                                       | 2.9   | 0.6                 |
| 29      | NLC             | 166.0                                    | 0.0                                 | 0.0      | 166.0            | 0.9               | 0.2                    | 0.0              | 0.0                                             | 0.0   | 0.0               | 0.0                                    | 0.0                                         | 0.0   | 0.0               | 166.0                                       | 0.8   | 0.2                 |
| 30      | PowerGrid       | 0.0                                      | 0.0                                 | 0.0      | 0.0              | 0.0               | 0.0                    | 0.0              | 8.3                                             | 0.3   | 0.1               | 0.0                                    | 8.3                                         | 0.3   | 0.1               | 8.3                                         | 0.0   | 0.0                 |
| 32      | Railways        | 0.0                                      | 65.9                                | 0.0      | 65.9             | 0.4               | 0.1                    | 0.0              | 0.0                                             | 0.0   | 0.0               | 0.0                                    | 0.0                                         | 0.0   | 0.0               | 65.9                                        | 0.3   | 0.1                 |
|         | Southern Region | 16976.0                                  | 1003.3                              | 0.0      | 17979.3          | 100.0             | 22.0                   | 360.0            | 2534.0                                          | 100.0 | 25.0              | 200.1                                  | 2734.1                                      | 100.0 | 25.0              | 21073.4                                     | 100.0 | 21.2                |
| 33      | Bihar           | 5446.5                                   | 0.0                                 | 0.0      | 5446.5           | 35.7              | 6.7                    | 710.0            | 1125.8                                          | 64.6  | 11.1              | 0.0                                    | 1125.8                                      | 64.6  | 10.3              | 7282.3                                      | 40.4  | 7.3                 |
| 34      | DVC             | 3190.8                                   | 0.0                                 | 0.0      | 3190.8           | 20.9              | 3.9                    | 0.0              | 8.0                                             | 0.5   | 0.1               | 0.0                                    | 8.0                                         | 0.5   | 0.1               | 3198.8                                      | 17.8  | 3.2                 |
| 35      | Jharkhand       | 1764.6                                   | 50.0                                | 0.0      | 1814.6           | 11.9              | 2.2                    | 0.0              | 180.4                                           | 10.4  | 1.8               | 0.0                                    | 180.4                                       | 10.4  | 1.6               | 1995.1                                      | 11.1  | 2.0                 |
| 36      | Odisha          | 1915.7                                   | 200.0                               | 0.0      | 2115.7           | 13.9              | 2.6                    | 0.0              | 231.9                                           | 13.3  | 2.3               | 0.0                                    | 231.9                                       | 13.3  | 2.1               | 2347.7                                      | 13.0  | 2.4                 |
| 37      | West Bengal     | 2257.5                                   | 0.0                                 | 0.0      | 2257.5           | 14.8              | 2.8                    | 292.0            | 189.7                                           | 10.9  | 1.9               | 0.0                                    | 189.7                                       | 10.9  | 1.7               | 2739.2                                      | 15.2  | 2.8                 |
| 38      | Sikkim          | 187.1                                    | 0.0                                 | 0.0      | 187.1            | 1.2               | 0.2                    | 0.0              | 4.1                                             | 0.2   | 0.0               | 0.0                                    | 4.1                                         | 0.2   | 0.0               | 191.3                                       | 1.1   | 0.2                 |
| 39      | PowerGrid       | 0.0                                      | 0.0                                 | 0.0      | 0.0              | 0.0               | 0.0                    | 0.0              | 2.5                                             | 0.1   | 0.0               | 0.0                                    | 2.5                                         | 0.1   | 0.0               | 2.5                                         | 0.0   | 0.0                 |
| 40      | Railways        | 252.8                                    | 0.0                                 | 0.0      | 252.8            |                   |                        | 0.0              | 0.0                                             | 0.0   | 0.0               | 0.0                                    | 0.0                                         | 0.0   | 0.0               | 252.8                                       | 1.4   | 0.3                 |
|         | Eastern Region  | 15015.0                                  | 250.0                               | 0.0      | 15265.0          | 100.0             | 18.7                   | 1002.0           | 1742.5                                          | 100.0 | 17.2              | 0.0                                    | 1742.5                                      | 100.0 | 15.9              | 18009.4                                     | 100.0 | 18.1                |
| 41      | Arunachal       | 284.5                                    | 0.0                                 | 0.0      | 284.5            | 9.0               | 0.3                    | 0.0              | 4.1                                             | 1.4   | 0.0               | 6.7                                    | 10.8                                        | 2.3   | 0.1               | 295.3                                       | 8.1   | 0.3                 |

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| S. N o. | Region / State | Firm Share                               |                                     |          |                  |                   |                       | Dedic ated Power | Un-Allocated Power                              |       |                   |                                        |                                            |       |                   | Total Share from C.G.S. and Bhutan Stations |       |                     |
|---------|----------------|------------------------------------------|-------------------------------------|----------|------------------|-------------------|-----------------------|------------------|-------------------------------------------------|-------|-------------------|----------------------------------------|--------------------------------------------|-------|-------------------|---------------------------------------------|-------|---------------------|
|         |                | Firm Power from Regional Pool and Bhutan | Firm from Other Region s and Bhutan | Non Firm | Total Firm Power |                   |                       |                  | Unallocated Power from Regional Pool and Bhutan |       |                   | Allocatio n from other Region / Bhutan | Total Allocation of Un-Allocated Power (*) |       |                   |                                             |       |                     |
|         |                | MW                                       | MW                                  | MW       | MW               | % Regio nal Total | % of All India Tot al |                  | MW                                              | MW    | % Region al Total | % of All Indi a Tot al                 | MW                                         | MW    | % Region al Total | % of All Indi a Tot al                      | MW    | % Reg ion al Tot al |
|         |                | 1                                        | 2                                   | 3        | 4=1+2+3          | 5                 | 6                     | 7                | 8                                               | 9     | 10                | 11                                     | 12=8+11                                    | 13    | 14                | 15=4+7+12                                   | 16    | 17                  |
|         | Pradesh        |                                          |                                     |          |                  |                   |                       |                  |                                                 |       |                   |                                        |                                            |       |                   |                                             |       |                     |
| 42      | Assam          | 1248.0                                   | 321.5                               | 0.0      | 1569.5           | 49.6              | 1.9                   | 0.0              | 140.9                                           | 49.6  | 1.4               | 160.9                                  | 301.9                                      | 64.1  | 2.8               | 1871.4                                      | 51.5  | 1.9                 |
| 43      | Manipur        | 216.9                                    | 0.0                                 | 0.0      | 216.9            | 6.9               | 0.3                   | 0.0              | 26.1                                            | 9.2   | 0.3               | 0.0                                    | 26.1                                       | 5.5   | 0.2               | 243.0                                       | 6.7   | 0.2                 |
| 44      | Meghalaya      | 258.4                                    | 0.0                                 | 0.0      | 258.4            | 8.2               | 0.3                   | 0.0              | 68.9                                            | 24.3  | 0.7               | 0.0                                    | 68.9                                       | 14.6  | 0.6               | 327.3                                       | 9.0   | 0.3                 |
| 45      | Mizoram        | 190.4                                    | 0.0                                 | 0.0      | 190.4            | 6.0               | 0.2                   | 0.0              | 27.1                                            | 9.6   | 0.3               | 4.9                                    | 32.0                                       | 6.8   | 0.3               | 222.4                                       | 6.1   | 0.2                 |
| 46      | Nagaland       | 173.3                                    | 0.0                                 | 0.0      | 173.3            | 5.5               | 0.2                   | 0.0              | 13.8                                            | 4.9   | 0.1               | 14.7                                   | 28.5                                       | 6.1   | 0.3               | 201.9                                       | 5.6   | 0.2                 |
| 47      | Tripura        | 464.7                                    | 0.0                                 | 0.0      | 464.7            | 14.7              | 0.6                   | 0.0              | 2.9                                             | 1.0   | 0.0               | 0.0                                    | 2.9                                        | 0.6   | 0.0               | 467.6                                       | 12.9  | 0.5                 |
| 48      | PowerGrid      | 1.5                                      | 0.0                                 | 0.0      | 1.5              | 0.0               | 0.0                   | 0.0              | 0.0                                             | 0.0   | 0.0               | 0.0                                    | 0.0                                        | 0.0   | 0.0               | 1.5                                         | 0.0   | 0.0                 |
| 49      | Railways       | 0.0                                      | 5.5                                 | 0.0      | 5.5              | 0.2               | 0.0                   | 0.0              | 0.0                                             | 0.0   | 0.0               | 0.0                                    | 0.0                                        | 0.0   | 0.0               | 5.5                                         | 0.2   | 0.0                 |
|         | North-Eastern  | 2837.7                                   | 327.0                               | 0.0      | 3164.7           | 100.0             | 3.9                   | 0.0              | 283.9                                           | 100.0 | 2.8               | 187.2                                  | 471.1                                      | 100.0 | 4.3               | 3635.8                                      | 100.0 | 3.7                 |
|         | All India      | 73,406                                   | 7,846                               | 374      | 81,626           | 100               | 100                   | 6,899            | 10,149                                          | 100   | 100               | 805                                    | 10,954                                     | 100   | 100               | 99,479                                      | 100   | 100                 |

Note:

1. Firm share includes merchant power and capacity allocated / diverted from other stations located within / outside the region.
  2. Grand Total power does not include power allocated to Bangladesh. Total Power allocated to Bangladesh = 250 MW (100 MW each from NR and WR and 50 MW from ER NTPC stations' unallocated power).
  3. Grand Total power does not include surrendered power by various beneficiaries.
- (\*) It includes specific allocation to various beneficiaries.

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## Annexure- 5A

**PFRS under 50 000 MW Hydroelectric Initiative**  
**Statewise List of Schemes**

| Sl. No.                         | Scheme                                         | Consultant | Installed Capacity |           |            | Head (m) | Annual Energy (GWh) | Tariff (Rs/kWh) |
|---------------------------------|------------------------------------------------|------------|--------------------|-----------|------------|----------|---------------------|-----------------|
|                                 |                                                |            | Nos of Units       | Size (MW) | Total (MW) |          |                     |                 |
| <b><u>Andhra Pradesh</u></b>    |                                                |            |                    |           |            |          |                     |                 |
| 1                               | Pondugala                                      | WAPCOS     | 3                  | 27        | 81         | 18.67    | 399.36              | 3.48            |
|                                 | <b>Total (Andhra Pradesh ) - 1 Nos. scheme</b> |            | <b>3</b>           |           | <b>81</b>  |          |                     |                 |
| <b><u>Arunachal Pradesh</u></b> |                                                |            |                    |           |            |          |                     |                 |
| 2                               | Agoline                                        | NHPC       | 3                  | 125       | 375        | 163.00   | 1267.38             | 3.51            |
| 3                               | Amulin                                         | NHPC       | 3                  | 140       | 420        | 132.00   | 1716.40             | 3.37            |
| 4                               | Ashupani                                       | NHPC       | 2                  | 15        | 30         | 395.00   | 126.45              | 8.75            |
| 5                               | Attunli                                        | NHPC       | 4                  | 125       | 500        | 264.00   | 2247.32             | 2.35            |
| 6                               | Badao                                          | NEEPCO     | 4                  | 30        | 120        | 154.50   | 441.00              | 2.32            |
| 7                               | Bhareli-I                                      | NEEPCO     | 8                  | 140       | 1120       | 97.00    | 4112.40             | 1.85            |
| 8                               | Bhareli-II                                     | NEEPCO     | 5                  | 120       | 600        | 51.00    | 2345.00             | 1.67            |
| 9                               | Chanda                                         | NEEPCO     | 4                  | 27.5      | 110        | 175.67   | 401.91              | 2.67            |
| 10                              | Demwe                                          | NHPC       | 12                 | 250       | 3000       | 138.00   | 10823.82            | 1.97            |
| 11                              | Dengser                                        | NHPC       | 4                  | 138       | 552        | 120.00   | 2666.71             | 3.26            |
| 12                              | Dibbin                                         | NEEPCO     | 2                  | 50        | 100        | 151.24   | 335.72              | 2.23            |
| 13                              | Duimukh                                        | NHPC       | 3                  | 50        | 150        | 65.00    | 551.48              | 8.50            |
| 14                              | Elango                                         | NHPC       | 3                  | 50        | 150        | 363.00   | 583.14              | 5.00            |
| 15                              | Emini                                          | NHPC       | 4                  | 125       | 500        | 125.00   | 1695.45             | 3.51            |
| 16                              | Emra-II                                        | NHPC       | 3                  | 130       | 390        | 278.00   | 1648.09             | 3.02            |
| 17                              | Etabue                                         | NHPC       | 3                  | 55        | 165        | 378.00   | 683.66              | 3.43            |
| 18                              | Etalin                                         | NHPC       | 16                 | 250       | 4000       | 385.00   | 16071.60            | 1.70            |
| 19                              | Hirong                                         | NHPC       | 4                  | 125       | 500        | 285.00   | 2535.80             | 1.62            |
| 20                              | Hutong                                         | WAPCOS     | 12                 | 250       | 3000       | 166.77   | 9901.00             | 1.28            |
| 21                              | Kalai                                          | WAPCOS     | 10                 | 260       | 2600       | 193.21   | 10608.64            | 1.01            |
| 22                              | Kameng Dam                                     | NEEPCO     | 5                  | 120       | 600        | 65.00    | 2345.55             | 2.29            |
| 23                              | Kapakleyak                                     | NEEPCO     | 4                  | 40        | 160        | 245.00   | 627.95              | 1.74            |
| 24                              | KurungI&II                                     | NHPC       | 3                  | 110       | 330        | 151.00   | 1435.40             | 4.04            |
| 25                              | Mihumdon                                       | NHPC       | 4                  | 100       | 400        | 286.00   | 1451.75             | 3.60            |
| 26                              | Mirak                                          | NHPC       | 3                  | 47        | 141        | 136.40   | 748.44              | 3.42            |
| 27                              | Naba                                           | NHPC       | 4                  | 250       | 1000       | 221.00   | 3995.25             | 2.14            |
| 28                              | Nalo                                           | NHPC       | 4                  | 90        | 360        | 221.00   | 1733.00             | 3.27            |
| 29                              | Naying                                         | NHPC       | 4                  | 250       | 1000       | 245.00   | 5077.15             | 1.18            |
| 30                              | Niare                                          | NHPC       | 4                  | 200       | 800        | 205.00   | 3356.62             | 2.02            |

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|                                                 |               |        |            |      |              |        |         |      |
|-------------------------------------------------|---------------|--------|------------|------|--------------|--------|---------|------|
| 31                                              | Oju-I         | NHPC   | 4          | 175  | 700          | 257.00 | 3291.58 | 2.08 |
| 32                                              | Oju-II        | NHPC   | 4          | 250  | 1000         | 322.00 | 4629.93 | 1.46 |
| 33                                              | Pakke         | NEEPCO | 2          | 55   | 110          | 452.50 | 335.26  | 3.33 |
| 34                                              | Papu          | NEEPCO | 2          | 100  | 200          | 238.00 | 505.00  | 2.94 |
| 35                                              | Phanchung     | NEEPCO | 2          | 30   | 60           | 157.13 | 174.83  | 3.24 |
| 36                                              | Ringong       | NHPC   | 3          | 50   | 150          | 166.50 | 659.07  | 3.61 |
| 37                                              | Sebu          | NEEPCO | 2          | 40   | 80           | 123.00 | 227.53  | 3.71 |
| 38                                              | Simang        | NHPC   | 3          | 30   | 90           | 125.00 | 417.82  | 5.43 |
| 39                                              | Talong        | NEEPCO | 3          | 100  | 300          | 171.67 | 915.50  | 2.24 |
| 40                                              | Tarangwarang  | NEEPCO | 2          | 15   | 30           | 185.55 | 93.81   | 2.88 |
| 41                                              | Tato-II       | NHPC   | 4          | 175  | 700          | 168.00 | 3465.90 | 1.48 |
| 42                                              | Tenga         | NEEPCO | 4          | 150  | 600          | 875.00 | 1046.50 | 3.52 |
| 43                                              | Utung         | NEEPCO | 3          | 33.3 | 100          | 291.00 | 359.13  | 3.10 |
| <b>Total (Arunachal Pr. ) - 42 Nos. schemes</b> |               |        | <b>182</b> |      | <b>27293</b> |        |         |      |
| <b>Chhattisgarh</b>                             |               |        |            |      |              |        |         |      |
| 44                                              | Kotri         | WAPCOS | 3          | 50   | 150          | 36.99  | 330.95  | 5.48 |
| 45                                              | Nugur-I       | WAPCOS | 5          | 34   | 170          | 24.54  | 316.13  | 4.89 |
| 46                                              | Nugur-II      | WAPCOS | 5          | 42   | 210          | 16.66  | 787.78  | 4.16 |
| 47                                              | Rehar-I       | WAPCOS | 3          | 57   | 171          | 46.84  | 264.38  | 8.70 |
| 48                                              | Rehar-II      | WAPCOS | 3          | 49   | 147          | 38.17  | 290.32  | 5.16 |
| <b>Total (Chhattisgarh ) - 5 Nos. schemes</b>   |               |        | <b>19</b>  |      | <b>848</b>   |        |         |      |
| <b>Himachal Pradesh</b>                         |               |        |            |      |              |        |         |      |
| 49                                              | Bajoli Holi   | HPSEB  | 3          | 60   | 180          | 278.00 | 762.98  | 2.03 |
| 50                                              | Bardang       | HPSEB  | 3          | 38   | 114          | 55.00  | 438.41  | 2.91 |
| 51                                              | Chamba        | HPSEB  | 3          | 42   | 126          | 110.00 | 646.82  | 1.48 |
| 52                                              | Chhatru       | HPSEB  | 3          | 36   | 108          | 160.00 | 455.72  | 2.89 |
| 53                                              | Gharopa       | HPSEB  | 3          | 38   | 114          | 169.00 | 534.25  | 2.09 |
| 54                                              | Gondhala      | HPSEB  | 3          | 48   | 144          | 134.00 | 586.08  | 1.92 |
| 55                                              | Jangi Thopan  | HPSEB  | 3          | 160  | 480          | 174.14 | 1779.45 | 2.00 |
| 56                                              | Khab-I        | SJVNL  | 3          | 150  | 450          | 170.00 | 1551.00 | 2.24 |
| 57                                              | Khab-II       | SJVNL  | 3          | 62   | 186          | 70.00  | 640.00  | 3.04 |
| 58                                              | Khoksar       | HPSEB  | 3          | 30   | 90           | 99.00  | 351.91  | 2.46 |
| 59                                              | Luhri         | HPSEB  | 3          | 155  | 465          | 88.00  | 1825.13 | 2.41 |
| 60                                              | Thopan Powari | HPSEB  | 3          | 160  | 480          | 161.14 | 1786.26 | 1.81 |
| 61                                              | Tidong-I      | HPSEB  | 2          | 30   | 60           | 511.50 | 211.65  | 2.71 |
| 62                                              | Tidong-II     | HPSEB  | 2          | 35   | 70           | 575.00 | 256.18  | 2.02 |
| 63                                              | Yangthang     | HPSEB  | 3          | 87   | 261          | 186.45 | 938.02  | 2.08 |
| <b>Total (Himachal Pr. ) - 15 Nos. schemes</b>  |               |        | <b>43</b>  |      | <b>3328</b>  |        |         |      |
| <b>Jammu &amp; Kashmir</b>                      |               |        |            |      |              |        |         |      |
| 64                                              | Barinium      | WAPCOS | 2          | 120  | 240          | 117.77 | 1170.34 | 2.54 |
| 65                                              | Bichlari      | WAPCOS | 2          | 17.5 | 35           | 462.60 | 148.29  | 1.11 |
| 66                                              | Dumkhar       | NHPC   | 3          | 15   | 45           | 27.80  | 219.18  | 4.66 |
| 67                                              | Kanyunche     | NHPC   | 3          | 15   | 45           | 28.76  | 223.02  | 4.71 |
| 68                                              | Karkit        | NHPC   | 3          | 10   | 30           | 26.90  | 153.11  | 5.40 |
| 69                                              | Kawar         | WAPCOS | 4          | 80   | 320          | 74.00  | 1426.56 | 1.09 |

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|                                                 |                           |        |           |       |             |        |         |       |
|-------------------------------------------------|---------------------------|--------|-----------|-------|-------------|--------|---------|-------|
| 70                                              | Khalsi                    | NHPC   | 3         | 20    | 60          | 33.00  | 272.60  | 4.10  |
| 71                                              | Kiru                      | WAPCOS | 4         | 107.5 | 430         | 105.33 | 1935.77 | 0.77  |
| 72                                              | Rattle                    | WAPCOS | 4         | 140   | 560         | 92.33  | 2483.37 | 1.40  |
| 73                                              | Shamnot                   | WAPCOS | 4         | 92.5  | 370         | 56.33  | 1650.19 | 1.69  |
| 74                                              | Shuas                     | WAPCOS | 2         | 115   | 230         | 115.70 | 1117.87 | 2.94  |
| 75                                              | Takmaching                | NHPC   | 3         | 10    | 30          | 18.53  | 145.52  | 5.54  |
| 76                                              | Ujh                       | WAPCOS | 4         | 70    | 280         | 143.33 | 465.06  | 5.06  |
| <b>Total (J &amp; K ) - 13 Nos. schemes</b>     |                           |        | <b>41</b> |       | <b>2675</b> |        |         |       |
| <b>Karnataka</b>                                |                           |        |           |       |             |        |         |       |
| 77                                              | Agnashini                 | KPCL   | 4         | 150   | 600         | 427.00 | 1431.00 | 1.07  |
| 78                                              | Gangavali                 | KPCL   | 2         | 200   | 400         | 378.30 | 759.00  | 1.46  |
| 79                                              | Gundia                    | KPCL   | 2         | 150   | 300         | 600.00 | 616.00  | 1.41  |
| 80                                              | Kalinadi Stage-III        | KPCL   | 2         | 150   | 300         | 407.67 | 610.00  | 1.67  |
| 81                                              | Tamankal                  | KPCL   | 2         | 150   | 300         | 87.29  | 401.00  | 3.32  |
| <b>Total (Karnataka ) - 5 Nos. schemes</b>      |                           |        | <b>12</b> |       | <b>1900</b> |        |         |       |
| <b>Kerala</b>                                   |                           |        |           |       |             |        |         |       |
| 82                                              | Karappara Kuriarkutty PH1 | WAPCOS | 2         | 18    | 66          | 390.00 | 126.10  | 7.88  |
|                                                 | PH2                       |        | 2         | 15    |             | 307.00 |         |       |
| 83                                              | Perianjakully             | WAPCOS | 2         | 30    | 60          | 282.90 | 86.30   | 6.25  |
| <b>Total (Kerala ) - 2 Nos. schemes</b>         |                           |        | <b>6</b>  |       | <b>126</b>  |        |         |       |
| <b>Madhya Pradesh</b>                           |                           |        |           |       |             |        |         |       |
| 84                                              | Basania                   | NHPC   | 3         | 30    | 90          | 38.00  | 240.00  | 17.23 |
| 85                                              | Bauras                    | NHPC   | 3         | 18.33 | 55          | 17.50  | 248.43  | 3.96  |
| 86                                              | Hoshangabad               | NHPC   | 3         | 20    | 60          | 16.50  | 288.21  | 4.10  |
| <b>Total (Madhya Pradesh ) - 3 Nos. schemes</b> |                           |        | <b>9</b>  |       | <b>205</b>  |        |         |       |
| <b>Maharashtra</b>                              |                           |        |           |       |             |        |         |       |
| 87                                              | Ghargaon                  | WAPCOS | 4         | 13    | 52          | 9.84   | 74.47   | 15.50 |
| 88                                              | Hiranyakeshi              | WAPCOS | 2         | 9     | 18          | 36.10  | 23.76   | 20.26 |
| 89                                              | Kadvi                     | WAPCOS | 2         | 11    | 22          | 36.30  | 29.59   | 34.03 |
| 90                                              | Kasari                    | WAPCOS | 2         | 12.5  | 25          | 40.67  | 33.32   | 18.16 |
| 91                                              | Kumbhi                    | WAPCOS | 2         | 8.5   | 17          | 37.48  | 22.93   | 35.19 |
| 92                                              | Kunghara                  | WAPCOS | 4         | 18    | 72          | 12.77  | 133.40  | 11.34 |
| 93                                              | Pranhita                  | WAPCOS | 2         | 24    | 48          | 25.30  | 135.96  | 10.32 |
| 94                                              | Samda                     | WAPCOS | 4         | 13    | 52          | 10.64  | 83.40   | 14.11 |
| 95                                              | Wainganga                 | WAPCOS | 5         | 21    | 105         | 19.74  | 246.15  | 3.86  |
| <b>Total (Maharashtra ) - 9 Nos. schemes</b>    |                           |        | <b>27</b> |       | <b>411</b>  |        |         |       |
| <b>Manipur</b>                                  |                           |        |           |       |             |        |         |       |
| 96                                              | Khongnum Chakka st.-II    | WAPCOS | 2         | 33.5  | 67          | 281.25 | 192.84  | 4.59  |
| 97                                              | Nunglieban                | WAPCOS | 2         | 52.5  | 105         | 82.42  | 268.93  | 5.16  |
| 98                                              | Pabaram                   | WAPCOS | 2         | 95    | 190         | 116.67 | 474.77  | 4.33  |
| <b>Total (Manipur ) - 3 Nos. schemes</b>        |                           |        | <b>6</b>  |       | <b>362</b>  |        |         |       |
| <b>Meghalaya</b>                                |                           |        |           |       |             |        |         |       |
| 99                                              | Mawblei                   | WAPCOS | 2         | 70    | 140         | 400.33 | 303.66  | 4.44  |
| 100                                             | Mawhu                     | WAPCOS | 3         | 40    | 120         | 438.15 | 482.96  | 1.40  |

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|                                             |                           |        |           |      |             |         |         |      |
|---------------------------------------------|---------------------------|--------|-----------|------|-------------|---------|---------|------|
| 101                                         | Mawput                    | WAPCOS | 3         | 7    | 21          | 93.42   | 83.95   | 4.07 |
| 102                                         | Nongkolait                | WAPCOS | 2         | 60   | 120         | 463     | 332.87  | 1.97 |
| 103                                         | Nongnam                   | WAPCOS | 2         | 25   | 50          | 215.17  | 212.59  | 2.44 |
| 104                                         | Rangmaw                   | WAPCOS | 2         | 32.5 | 65          | 321.00  | 229.60  | 2.32 |
| 105                                         | Selim                     | WAPCOS | 2         | 85   | 170         | 433.67  | 534.68  | 2.02 |
| 106                                         | Sushen                    | WAPCOS | 2         | 32.5 | 65          | 114.58  | 220.6   | 3.85 |
| 107                                         | Umduna                    | WAPCOS | 3         | 19   | 57          | 253.17  | 231.24  | 1.68 |
| 108                                         | Umjaut                    | WAPCOS | 3         | 23   | 69          | 375.20  | 276.70  | 1.51 |
| 109                                         | Umngi                     | WAPCOS | 2         | 27   | 54          | 304.75  | 89.65   | 2.86 |
| <b>Total (Meghalaya ) - 11 Nos. schemes</b> |                           |        | <b>26</b> |      | <b>931</b>  |         |         |      |
| <b><u>Mizoram</u></b>                       |                           |        |           |      |             |         |         |      |
| 110                                         | Boinu                     | WAPCOS | 4         | 160  | 640         | 158.67  | 1118.93 | 4.83 |
| 111                                         | Lungleng                  | WAPCOS | 5         | 163  | 815         | 219.67  | 1169.06 | 4.17 |
| 112                                         | Tlawng                    | WAPCOS | 2         | 22.5 | 45          | 123.67  | 151.67  | 5.84 |
| <b>Total (Mizoram ) - 3 Nos. schemes</b>    |                           |        | <b>11</b> |      | <b>1500</b> |         |         |      |
| <b><u>Nagaland</u></b>                      |                           |        |           |      |             |         |         |      |
| 113                                         | Dikhu                     | NEEPCO | 4         | 35   | 140         | 79.44   | 513.41  | 2.80 |
| 114                                         | Tizu                      | NEEPCO | 3         | 50   | 150         | 64.19   | 568.41  | 2.56 |
| 115                                         | Yangnyu                   | NEEPCO | 2         | 20   | 40          | 115     | 176.45  | 4.48 |
| <b>Total (Nagaland) - 3 Nos. schemes</b>    |                           |        | <b>9</b>  |      | <b>330</b>  |         |         |      |
| <b><u>Orissa</u></b>                        |                           |        |           |      |             |         |         |      |
| 116                                         | Baljori                   | WAPCOS | 2         | 89   | 178         | 165.75  | 479.8   | 5.90 |
| 117                                         | Lower Kolab               | WAPCOS | 3         | 155  | 465         | 196.9   | 845.86  | 7.10 |
| 118                                         | Naraj                     | WAPCOS | 7         | 41   | 287         | 16.14   | 759.31  | 4.92 |
| 119                                         | Tikarpara                 | WAPCOS | 7         | 37   | 259         | 16.97   | 828.37  | 3.69 |
| <b>Total (Orissa ) - 4 Nos. schemes</b>     |                           |        | <b>19</b> |      | <b>1189</b> |         |         |      |
| <b><u>Sikkim</u></b>                        |                           |        |           |      |             |         |         |      |
| 120                                         | Dikchu                    | NHPC   | 3         | 35   | 105         | 352     | 469     | 2.15 |
| 121                                         | Lachen                    | NHPC   | 3         | 70   | 210         | 350     | 865.94  | 2.35 |
| 122                                         | Lingza                    | NHPC   | 3         | 40   | 120         | 736     | 477.51  | 2.85 |
| 123                                         | Panan                     | NHPC   | 4         | 50   | 200         | 312     | 762     | 2.15 |
| 124                                         | Rangyong                  | NHPC   | 3         | 47   | 141         | 723.18  | 639.52  | 2.70 |
| 125                                         | Ringpi                    | NHPC   | 2         | 35   | 70          | 1106.37 | 317.41  | 3.17 |
| 126                                         | Rongni Storage            | NHPC   | 3         | 65   | 195         | 442     | 510.35  | 8.60 |
| 127                                         | Rukel                     | NHPC   | 3         | 11   | 33          | 537.1   | 149.41  | 5.48 |
| 128                                         | Talem                     | NHPC   | 3         | 25   | 75          | 393.19  | 305.48  | 4.34 |
| 129                                         | Teesta-I                  | NHPC   | 4         | 80   | 320         | 576.85  | 1298.12 | 1.80 |
| <b>Total (Sikkim) - 10 Nos. schemes</b>     |                           |        | <b>31</b> |      | <b>1469</b> |         |         |      |
| <b><u>Uttaranchal</u></b>                   |                           |        |           |      |             |         |         |      |
| 130                                         | Arakot Tiuni              | UJVNL  | 3         | 24   | 72          | 250.2   | 382.9   | 1.00 |
| 131                                         | Badrinath                 | WAPCOS | 2         | 70   | 140         | 459.67  | 702.7   | 0.81 |
| 132                                         | Bagoli Dam                | UJVNL  | 3         | 24   | 72          | 139.5   | 340.7   | 4.10 |
| 133                                         | Bhaironghati              | WAPCOS | 2         | 32.5 | 65          | 108.9   | 293.18  | 1.80 |
| 134                                         | Bogudiyar - Sirkari Bhyal | WAPCOS | 2         | 85   | 170         | 344.47  | 744     | 1.99 |
| 135                                         | Bokang Baling             | WAPCOS | 3         | 110  | 330         | 455.2   | 1124.62 | 1.68 |
| 136                                         | Chhunger - Chal           | WAPCOS | 2         | 120  | 240         | 292.83  | 853.28  | 1.13 |

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|                                               |                              |        |            |      |              |        |         |      |
|-----------------------------------------------|------------------------------|--------|------------|------|--------------|--------|---------|------|
| 137                                           | Deodi                        | WAPCOS | 2          | 30   | 60           | 560.3  | 296.76  | 1.37 |
| 138                                           | Devsari                      | WAPCOS | 3          | 100  | 300          | 227.5  | 878.5   | 2.77 |
| 139                                           | Gangotri                     | WAPCOS | 1          | 55   | 55           | 336.33 | 264.76  | 1.62 |
| 140                                           | Garba Tawaghat               | WAPCOS | 3          | 210  | 630          | 470.97 | 2483.11 | 0.90 |
| 141                                           | Gohana Tal                   | WAPCOS | 2          | 30   | 60           | 584.52 | 269.35  | 1.64 |
| 142                                           | Harsil                       | WAPCOS | 3          | 70   | 210          | 281.33 | 920.57  | 1.10 |
| 143                                           | Jadh Ganga                   | WAPCOS | 2          | 25   | 50           | 142.6  | 220.88  | 2.19 |
| 144                                           | Jakhol Sankri                | UJVNL  | 3          | 11   | 33           | 364    | 144.24  | 1.71 |
| 145                                           | Jelam Tamak                  | WAPCOS | 2          | 30   | 60           | 195.58 | 268.12  | 1.71 |
| 146                                           | Kalika Dantu                 | WAPCOS | 2          | 115  | 230          | 99.75  | 1067.3  | 2.95 |
| 147                                           | Karmoli                      | WAPCOS | 2          | 70   | 140          | 419.67 | 621.31  | 1.30 |
| 148                                           | Khartoi Lumti Talli          | WAPCOS | 2          | 27.5 | 55           | 56.6   | 241.51  | 3.00 |
| 149                                           | Lata Tapovan                 | UJVNL  | 4          | 77.5 | 310          | 265    | 1123    | 2.21 |
| 150                                           | Maleri Jelam                 | WAPCOS | 2          | 27.5 | 55           | 200.33 | 243.07  | 1.80 |
| 151                                           | Mapang - Bogidiyar           | WAPCOS | 2          | 100  | 200          | 465.07 | 882.04  | 1.30 |
| 152                                           | Naitwar-Mori                 | UJVNL  | 3          | 11   | 33           | 76     | 151     | 1.85 |
| 153                                           | Nand Prayag                  | UJVNL  | 3          | 47   | 141          | 72     | 794     | 2.05 |
| 154                                           | Ramganga                     | UJVNL  | 3          | 22   | 66           | 100.1  | 327     | 3.25 |
| 155                                           | Rishi Ganga - I              | WAPCOS | 2          | 35   | 70           | 536.17 | 327.3   | 1.18 |
| 156                                           | Rishi Ganga - II             | WAPCOS | 1          | 35   | 35           | 236.96 | 164.64  | 2.22 |
| 157                                           | Rupsiabagar<br>Khasiyabara   | WAPCOS | 2          | 130  | 260          | 449.47 | 1195.63 | 1.59 |
| 158                                           | Sela Urthing                 | WAPCOS | 2          | 115  | 230          | 255.5  | 816.73  | 1.40 |
| 159                                           | Sirkari Bhyol<br>Rupsiabagar | WAPCOS | 3          | 70   | 210          | 388.97 | 967.97  | 1.55 |
| 160                                           | Taluka Sankri                | UJVNL  | 2          | 70   | 140          | 564.9  | 559.47  | 1.33 |
| 161                                           | Tamak Lata                   | UJVNL  | 4          | 70   | 280          | 291.4  | 1040.7  | 2.30 |
| 162                                           | Urthing Sobla                | UJVNL  | 4          | 70   | 280          | 414.96 | 1360.2  | 1.49 |
| <b>Total (Uttaranchal ) - 33 Nos. schemes</b> |                              |        | <b>81</b>  |      | <b>5282</b>  |        |         |      |
| <b>Grand Total - 162 Nos. schemes</b>         |                              |        | <b>525</b> |      | <b>47930</b> |        |         |      |

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**Annexure-5B****Hydro Capacity Addition vis-à-vis Target during the Year 2023-24**

| Sl. No. | Particular                                  | Unit Nos.           | Capacity (MW) |          | Capacity Addition |                           | Remarks/Critical Issues                                                               |
|---------|---------------------------------------------|---------------------|---------------|----------|-------------------|---------------------------|---------------------------------------------------------------------------------------|
|         |                                             |                     | Target        | Actual   | As Programmed     | Actual(A)/ Anticipated    |                                                                                       |
| A       | Central Sector                              |                     |               |          |                   |                           |                                                                                       |
| 1       | Naitwar mori<br>2X30=60MW                   | Unit #1<br>Unit # 2 | 30<br>30      |          | Dec'22<br>Jan'23  | Slipped to<br>2023-24     | -COVID-19 related restrictions.<br>- Delay in Environment Clearance for Quarry site.- |
| 2       | Subansiri Lower<br>8x250= 2000 MW           | Unit #1<br>Unit #2  | 250<br>250    |          | Jan'23<br>Feb'23  | Slipped to<br>2023-24     | -Extended Monsoon & damage to protection wall of power house / collapse of DT-2       |
| 3       | Tehri PSS<br>4x250= 1000 MW                 | Unit #1             | 250           |          | Mar'23            | Slipped to<br>2023-24     | -Geological constraints                                                               |
|         | Sub total (A)                               |                     | 810 MW        |          |                   |                           |                                                                                       |
| B       | State Sector                                |                     |               |          |                   |                           |                                                                                       |
| 1       | Pallivasal<br>KSEB, Kerala<br>2x30=60 MW    | Unit #1<br>Unit # 2 | 30<br>30      |          | Mar'23<br>Mar'23  | Slipped to<br>2023-24     | -Extended Monsoon<br>-Slow progress of works                                          |
| 2       | Thottiyar<br>KSEB, Kerala<br>1x30+1x10=40MW | Unit #1<br>Unit # 2 | 10<br>30      |          | Feb'23<br>Mar'23  | Slipped to<br>2023-24     | -Extended Monsoon<br>-Slow progress of works                                          |
| 3       | Vyasi<br>2x60=120 MW                        | Unit #1<br>Unit # 2 | 60<br>60      | 60<br>60 | Apr'22<br>May'22  | May'22 (A)<br>April'22(A) | -U#1 Commissioned on 24.05.2022<br>-U# 2 Commissioned on 22.04.2022                   |
|         | Sub- total (B):                             |                     | 220 MW        | 120 MW   |                   |                           |                                                                                       |
| C.      | Private Sector                              |                     |               |          |                   |                           |                                                                                       |
| 1       | Tidong-I<br>2x50= 100 MW                    | Unit #1             | 50            |          | Mar'23            | Slipped to<br>2023-24     | -Delay in clearance of transmission line<br>-Pressure shaft misalignment issues       |
|         | Sub- total (C):                             |                     | 50MW          |          |                   |                           |                                                                                       |
|         | Total (A+B +C)                              |                     | 1080MW        | 120MW    |                   |                           |                                                                                       |

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**Annexure-5C****Hydro Capacity addition Programme vis-a-vis achievement for 2023-24**

| Sl. No. | Name of Project                  | State/ Implem. Agency        | Unit No.                 | Capacity (MW)            |          | Capacity Addition                     |                                                                                          | Reason for slippage                                                                                                                                                                                                                                                                   |
|---------|----------------------------------|------------------------------|--------------------------|--------------------------|----------|---------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |                                  |                              |                          | Target                   | Actual   | As Program med                        | Actual(A)/ Anticipated                                                                   |                                                                                                                                                                                                                                                                                       |
|         | <b>Central Sector</b>            |                              |                          |                          |          |                                       |                                                                                          |                                                                                                                                                                                                                                                                                       |
| 1       | Naitwar Mori<br>2x30=60 MW       | Uttarakhand /SJVNL           | U-1<br>U-2               | 30<br>30                 | 30<br>30 | June'23<br>July'23                    | 24.11.2023<br>04.12.2023                                                                 | <b>Commissioned</b>                                                                                                                                                                                                                                                                   |
| 2       | Subansiri Lower<br>8x250=2000 MW | Arunachal Pradesh/N HPC      | U-1<br>U-2<br>U-3<br>U-4 | 250<br>250<br>250<br>250 |          | Aug'23<br>Sept'23<br>Jan'24<br>Mar'24 | Slipping to 2025-26<br>Slipping to 2025-26<br>Slipping to 2025-26<br>Slipping to 2025-26 | <ul style="list-style-type: none"> <li>• Diversion tunnel-1 collapsed.</li> <li>• Slope stability of left bank.</li> <li>• Erection of balance 6 nos. of radial gates, DT plugging</li> </ul>                                                                                         |
| 3       | Tehri PSS<br>4x250= 1000 MW      | Uttarakhand/ THDC            | U-1<br>U-2<br>U-3<br>U-4 | 250<br>250<br>250<br>250 |          | Oct'23<br>Nov'23<br>Jan'24<br>Feb'24  | Slipping to 2024-25<br>Slipping to 2024-25<br>Slipping to 2024-25<br>Slipping to 2024-25 | <ul style="list-style-type: none"> <li>• River joining works are critical</li> <li>• Permission from Up irrigation Dept from 1-30th Jun,2024</li> <li>• Plug breaking of HRT</li> </ul>                                                                                               |
|         |                                  | <b>Sub- total (Central):</b> |                          | <b>2060</b>              |          | <b>2060</b>                           |                                                                                          |                                                                                                                                                                                                                                                                                       |
|         | <b>State Sector</b>              |                              |                          |                          |          |                                       |                                                                                          |                                                                                                                                                                                                                                                                                       |
| 4       | Pallivasal<br>2x30= 60 MW        | Kerala/ KSEB Ltd.            | U-1<br>U-2               | 30<br>30                 |          | Dec'23<br>Dec'23                      | Slipping to 2024-25<br>Slipping to 2024-25                                               | <ul style="list-style-type: none"> <li>• Delay in land acquisition.</li> <li>• Poor geology strata in HRT.</li> <li>• Works stopped by contractor since 28.1.15 to 11.04.2017 due to contractual issues. Contract terminated on 13.09.2018 and re-awarded in August, 2019.</li> </ul> |
| 5       | Thottiyar<br>1x30 + 1x10= 40 MW  | Kerala/ KSEB Ltd.            | U-1<br>U-2               | 30<br>10                 |          | Dec'23<br>Mar'24                      | Slipping to 2024-25<br>Slipping to 2024-25                                               | <ul style="list-style-type: none"> <li>• Land acquisition issue.</li> <li>• The works of weir and approach channel stopped from 2010 to 2012 by local people.</li> <li>• The work stopped by Court from</li> </ul>                                                                    |

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|                         |                                   |                                      |                   |                   |           |                            |                                                                   |                                                                                                                                                                                                                  |
|-------------------------|-----------------------------------|--------------------------------------|-------------------|-------------------|-----------|----------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |                                   |                                      |                   |                   |           |                            |                                                                   | 12.12.2012 to April-2013.<br><ul style="list-style-type: none"> <li>Financial crunch with contractor leading to foreclosure of contract in April, 2017 and balance works re-awarded in January, 2018.</li> </ul> |
|                         |                                   | <b>Sub- total (State):</b>           |                   | <b>100</b>        |           | <b>100</b>                 |                                                                   |                                                                                                                                                                                                                  |
|                         | <b>Private Sector</b>             |                                      |                   |                   |           |                            |                                                                   |                                                                                                                                                                                                                  |
| 6                       | Pinnapuram<br>4X240+2X120=1200 MW | M/s Greenko<br>Energies Pvt.<br>Ltd. | U-1<br>U-2<br>U-3 | 240<br>240<br>240 |           | Jan'24<br>Feb'24<br>Mar'24 | Slipping to 2024-25<br>Slipping to 2024-25<br>Slipping to 2024-25 | -Slow progress of works                                                                                                                                                                                          |
|                         |                                   | <b>Sub- total (Private):</b>         |                   | <b>720</b>        |           | <b>720</b>                 | -                                                                 |                                                                                                                                                                                                                  |
| <b>Total (CS+SS+PS)</b> |                                   |                                      |                   | <b>2880</b>       | <b>60</b> | <b>2880</b>                |                                                                   |                                                                                                                                                                                                                  |

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**Annexure-5D****Hydro Capacity addition Programme for 2024-25**

| Sl. No. | Name of Project                           | State/ Implem. Agency         | Unit No. | Capacity (MW) |          | Capacity Addition          |
|---------|-------------------------------------------|-------------------------------|----------|---------------|----------|----------------------------|
|         |                                           |                               |          | Target        | Actual   | As Programmed/ Anticipated |
|         | <b>Central Sector</b>                     |                               |          |               |          |                            |
| 1       | Parbati-II /NHPC<br>(4*200=800 MW)        | Himachal Pradesh/NHPC         | U-1      | 200           | 0        | Dec'24                     |
|         |                                           |                               | U-2      | 200           |          | Dec'24                     |
|         |                                           |                               | U-3      | 200           |          | Dec'24                     |
|         |                                           |                               | U-4      | 200           |          | Dec'24                     |
| 2       | Tehri PSS/THDC<br>(4*250=1000 MW)         | Uttarakhand/THDC              | U-1      | 250           | 0        | Sept'24                    |
|         |                                           |                               | U-2      | 250           |          | Sept'24                    |
|         |                                           |                               | U-3      | 250           |          | Oct'24                     |
|         |                                           |                               | U-4      | 250           |          | Oct'24                     |
| 3       | Subansiri Lower / NHPC<br>(8x250=2000 MW) | Arunachal Pradesh/Assam/ NHPC | U-1      | 250           | 0        | Mar'25                     |
|         |                                           |                               | U-2      | 250           |          | Mar'25                     |
|         |                                           |                               | U-3      | 250           |          | Mar'25                     |
|         |                                           | <b>Sub- total (Central):</b>  |          | <b>2550</b>   | <b>0</b> |                            |
|         | <b>State Sector</b>                       |                               |          |               |          |                            |
| 4       | Pallivasal<br>(2x30= 60 MW)               | Kerala/ KSEB Ltd.             | U-1      | 30            | 0        | Sept'24                    |
|         |                                           |                               | U-2      | 30            |          | Sept'24                    |
| 5       | Thottiyar<br>(1x30 + 1x10 = 40 MW)        | Kerala/ KSEB Ltd.             | U-1 U-2  | 10<br>30      | 0        | Aug'24<br>Aug'24           |
| 6       | Uhl-III<br>(3*33.3=100 MW)                | Himachal Pradesh/BVPCL        | U-1      | 33.3          | 0        | Oct'24                     |
|         |                                           |                               | U-2      | 33.3          |          | Nov'24                     |
|         |                                           |                               | U-3      | 33.3          |          | Dec'24                     |

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|                  |                                       |                                                    |                                        |                                        |   |                                                          |
|------------------|---------------------------------------|----------------------------------------------------|----------------------------------------|----------------------------------------|---|----------------------------------------------------------|
|                  |                                       | Sub- total<br>(State):                             |                                        | 200                                    | 0 |                                                          |
|                  | Private Sector                        |                                                    |                                        |                                        |   |                                                          |
| 7                | Pinnapuram<br>4X240+2X120<br>=1200 MW | Andhrapradesh/M/s<br>Greenko Energies<br>Pvt. Ltd. | U-1<br>U-2<br>U-3<br>U-4<br>U-5<br>U-6 | 240<br>240<br>240<br>240<br>120<br>120 | 0 | Dec'24<br>Dec'24<br>Jan'25<br>Feb'25<br>Feb'25<br>Mar'25 |
|                  |                                       | Sub-total<br>(Private):                            |                                        | 1200                                   | 0 |                                                          |
| Total (CS+SS+PS) |                                       |                                                    |                                        | 3950                                   | 0 |                                                          |

Total capacity likely to be achieved by 31.03.2024 – 2880 MW

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## Annexure- 5E

## State-wise List of Hydro RMU&amp;LE schemes programmed for completion during 2022-27

| Sl. No                                    | Name of Project, Agency Cap. (No.x MW)                        | Inst. | CS/ SS | Capacity Covered Under RMU&LE (No.x MW) | Est. Cost     | Actual Exp. | Benefits (MW)          | Capacity after RMU&LE | Category | Year of Completion   |
|-------------------------------------------|---------------------------------------------------------------|-------|--------|-----------------------------------------|---------------|-------------|------------------------|-----------------------|----------|----------------------|
|                                           |                                                               |       |        |                                         | (Rs. in Crs.) |             |                        |                       |          |                      |
| A. Completed Schemes                      |                                                               |       |        |                                         |               |             |                        |                       |          |                      |
| Himachal Pradesh                          |                                                               |       |        |                                         |               |             |                        |                       |          |                      |
| 1                                         | Bhabha Power House, HPSEB (3x40)                              |       | SS     | 3x40                                    | 90.14         | 43.01       | 120 (LE)               | 120                   | RM&LE    | Completed in 2022-23 |
| 2                                         | Bhakra LB, BBMB (5x108)                                       |       | CS     | 5x108                                   | 489.77        | 583.86      | 540.00(LE)+ 90.00 (U)  | 630                   | RMU&LE   | Completed in 2023-24 |
| Uttarakhand                               |                                                               |       |        |                                         |               |             |                        |                       |          |                      |
| 3                                         | Tiloth (Maneri Bhali - I), UJVNL (3x30)                       |       | SS     | 3x30                                    | 384.66        | 189.45      | 90 (LE)                | 90                    | RM&LE    | Completed in 2022-23 |
| 4                                         | Dhalipur, UJVNL (3x17)                                        |       | SS     | 3x17                                    | 152.65        | 110.13      | 51 (LE)                | 51                    | RM&LE    | Completed in 2023-24 |
| Uttar Pradesh                             |                                                               |       |        |                                         |               |             |                        |                       |          |                      |
| 5                                         | Rihand, UPJVNL (6x50)                                         |       | SS     | 6x50                                    | 132.20        | 129.67      | 300 (LE)               | 300                   | RM&LE    | Completed in 2022-23 |
| Telangana                                 |                                                               |       |        |                                         |               |             |                        |                       |          |                      |
| 6                                         | Nagarjuna Sagar Phase-II works, TSGENCO (1x110+7x100.8)       |       | SS     | 1x110+7x100.8                           | 21.67         | 14.34       | -                      | 815.6                 | R&M      | Completed in 2022-23 |
| 7                                         | Nagarjuna Sagar Left Canal Power House, TSGENCO (2x30.6)      |       | SS     | 2x30.6                                  | 30.99         | 1.50        | -                      | 61.2                  | R&M      | Completed in 2022-23 |
| Karnataka                                 |                                                               |       |        |                                         |               |             |                        |                       |          |                      |
| 8                                         | Munirabad Dam Power House, KPCL (2x9 + 1x10)                  |       | SS     | 2x9 + 1x10                              | 4.60          | 2.20        | -                      | 28                    | R&M      | Completed in 2022-23 |
| 9                                         | Linganamakki Dam Power House, KPCL (2x27.5)                   |       | SS     | 2x27.5                                  | 2.75          | 2.75        | -                      | 55                    | R&M      | Completed in 2022-23 |
| 10                                        | Gerusoppa Dam Power House (Sharavathy Tail Race), KPCL (4x60) |       | SS     | 4x60                                    | 59.66         | 2.026       | -                      | 240                   | R&M      | Completed in 2023-24 |
| Sub Total(A)                              |                                                               |       |        | 2300.80                                 | 1369.09       | 1078.94     | 1191 [1101(LE)+ 90(U)] | 2390.80               |          |                      |
| B. Ongoing Schemes – Under Implementation |                                                               |       |        |                                         |               |             |                        |                       |          |                      |
| Uttarakhand                               |                                                               |       |        |                                         |               |             |                        |                       |          |                      |

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|                |                                                        |    |                  |         |       |                      |       |        |         |
|----------------|--------------------------------------------------------|----|------------------|---------|-------|----------------------|-------|--------|---------|
| 11             | Chilla (Ph B), UJVNL (4x36)                            | SS | 4x36             | 490.56  | 7.39  | 144(LE)+ 12(U)       | 156   | RMU&LE | 2025-26 |
| 12             | Dhakrani, UJVNL (3x11.25)                              | SS | 3x11.25          | 137.31  | 10.69 | 33.75 (LE)           | 33.75 | RM&LE  | 2025-26 |
| Uttar Pradesh  |                                                        |    |                  |         |       |                      |       |        |         |
| 13             | Obra, UPJVNL (3x33)                                    | SS | 3x33             | 58.80   | 48.15 | 99 (LE)              | 99    | RM&LE  | 2024-25 |
| Telangana      |                                                        |    |                  |         |       |                      |       |        |         |
| 14             | Pochampad HPS Stage -1, TSGENCO (3x9)                  | SS | 3x9              | 8.44    | -     | -                    | 27    | R&M    | 2026-27 |
| Andhra Pradesh |                                                        |    |                  |         |       |                      |       |        |         |
| 15             | Upper Sileru Power House, APGENCO (4x60)               | SS | 4x60             | 10.53   | 4.53  | -                    | 240   | R&M    | 2026-27 |
| 16             | Nagarjunasagar Right Canal Power House, APGENCO (3x30) | SS | 3x30             | 10.1    | 4.54  | -                    | 90    | R&M    | 2025-26 |
| 17             | Tungabhadra Dam, APGENCO (4x9)                         | SS | 4x9              | 6.65    | 2.517 | 36 (LE)              | 36    | RM&LE  | 2025-26 |
| 18             | Hampi Canal PH, APGENCO (4x9)                          | SS | 4x9              | -       | -     | 36 (LE)              | 36    | RM&LE  | 2025-26 |
| Karnataka      |                                                        |    |                  |         |       |                      |       |        |         |
| 19             | Nagjhari (Unit-1 to 3) KPCL (6x150)                    | SS | 3x150 (U-1 to 3) | 266.00  | 80.5  | 450 (LE)             | 450   | RM&LE  | 2025-26 |
| 20             | Shivasamudram, KPCL (6x3+4x6)                          | SS | 6x3+4x6          | 169.18  | 3.27  | 42 (LE)              | 42    | RM&LE  | 2024-25 |
| 21             | Kadra Dam Power House, KPCL(3x50)                      | SS | 3x50             | 44.47   | 2.627 | -                    | 150   | R&M    | 2024-25 |
| 22             | Kodasalli Dam Power House, KPCL (3x40)                 | SS | 3x40             | 50.60   | 2.654 | -                    | 120   | R&M    | 2024-25 |
| 23             | Sharavathy Generating Station, KPCL (10x103.5)         | SS | 10x103.5         | 196.56  | -     | 1035 (LE)            | 1035  | RM&LE  | 2025-26 |
| Jharkhand      |                                                        |    |                  |         |       |                      |       |        |         |
| 24             | Panchet U-1, DVC (2x40)                                | CS | 1x40 (U-1)       | 121.85  | 2.19  | 40(LE) + 6(U)        | 46    | RMU&LE | 2025-26 |
| Tamil Nadu     |                                                        |    |                  |         |       |                      |       |        |         |
| 25             | Moyar PH, TNPGL (3x12)                                 | SS | 3x12             | 121.127 | 74.02 | 36 (LE)+ 6(U)        | 42    | RMU&LE | 2024-25 |
| 26             | Kodayar PH-I, TNPGL (1x60)                             | SS | 1x60             | 80.96   | 2.3   | 60 (LE)+ 10 (U)      | 70    | RMU&LE | 2025-26 |
| Kerala         |                                                        |    |                  |         |       |                      |       |        |         |
| 27             | Kuttiyadi, KSEB (3x25)                                 | SS | 3x25             | 377.41  | 31.43 | 75.00 (LE) + 7.5 (U) | 82.5  | RMU&LE | 2025-26 |

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|                                      |                                                 |    |              |         |         |                                    |         |        |         |
|--------------------------------------|-------------------------------------------------|----|--------------|---------|---------|------------------------------------|---------|--------|---------|
| 28                                   | Sabarigiri (Unit- 6 & Unit 2), KSEB (4x55+2x60) | SS | 1x60 +1 x 55 | -       | -       | -                                  | 115     | R&M    | 2024-25 |
| Odisha                               |                                                 |    |              |         |         |                                    |         |        |         |
| 29                                   | Balimela, OHPCL (6x60)                          | SS | 6x60         | 382.91  | 167.60  | 360(LE)                            | 360     | RM&LE  | 2024-25 |
| Assam                                |                                                 |    |              |         |         |                                    |         |        |         |
| 30                                   | Kopili Power Station, NEEPCO (4x50)             | CS | 4x50         | 1075.19 | 1207.25 | 200(LE)                            | 200     | RM&LE  | 2024-25 |
| 31                                   | Khandong Power Station, NEEPCO (2x23)           | CS | 2x23         | 277.74  | 84.5    | 46 (LE)                            | 46      | RM&LE  | 2024-25 |
| Manipur                              |                                                 |    |              |         |         |                                    |         |        |         |
| 32                                   | Loktak, NHPC (3x35)                             | CS | 3x35         | 273.59  | 69.72   | 105 (LE)                           | 105     | RM&LE  | 2025-26 |
| Meghalaya                            |                                                 |    |              |         |         |                                    |         |        |         |
| 33                                   | Umiam St.III, (Kyrdemkulai) MePGCL (2x30)       | SS | 2x30         | 408.00  | 53.15   | 60(LE) + 6(U)                      | 66      | RMU&LE | 2026-27 |
| Sub Total (B)                        |                                                 |    | 3599.75      | 4567.98 | 1859.03 | 2905.25<br>[2857.75(LE)+ 47.50(U)] | 3647.25 |        |         |
| C. Ongoing Schemes – Under Tendering |                                                 |    |              |         |         |                                    |         |        |         |
| Himachal Pradesh                     |                                                 |    |              |         |         |                                    |         |        |         |
| 34                                   | Giri, HPSEBL (2x30)                             | SS | 2x30         | 440.12  | -       | 60.00 (LE)                         | 60      | RM&LE  | 2025-26 |
| Uttarakhand                          |                                                 |    |              |         |         |                                    |         |        |         |
| 35                                   | Ramganaga, UJVNL (3x66)                         | SS | 3x66         | 455.20  | -       | 198 (LE)                           | 198     | RM&LE  | 2026-27 |
| Gujarat                              |                                                 |    |              |         |         |                                    |         |        |         |
| 36                                   | Kadana PSS, GSECL (4x60)                        | SS | 4x60         | 750.25  | -       | 240 (LE) + 20 (U)                  | 260     | RMU&LE | 2025-26 |
| Rajasthan                            |                                                 |    |              |         |         |                                    |         |        |         |
| 37                                   | Rana Pratap Sagar RRVUNL (4x43)                 | SS | 4x43         | 548.11  | -       | 172 (LE) +6 (U)                    | 178     | RMU&LE | 2026-27 |
| Karnataka                            |                                                 |    |              |         |         |                                    |         |        |         |
| 38                                   | Supa Dam Power House, KPCL (2x50)               | SS | 2x50         | 47.91   | -       | -                                  | 100     | R&M    | 2024-25 |
| West Bengal                          |                                                 |    |              |         |         |                                    |         |        |         |
| 39                                   | Maithon, DVC (2x20+1x23.2-U#2)                  | CS | 2x20 (U-1&3) | 109.29  | 7.76    | 40.00 (LE)                         | 40      | RM&LE  | 2025-26 |
| Sub Total (C)                        |                                                 |    | 810          | 2350.88 | 7.76    | 736<br>[710(LE)+26(U)]             | 836.00  |        |         |

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| D. Ongoing Schemes – Under DPR Preparation/ Finalisation/ Approval |                                                                          |    |                            |        |      |                         |         |        |         |
|--------------------------------------------------------------------|--------------------------------------------------------------------------|----|----------------------------|--------|------|-------------------------|---------|--------|---------|
| Uttarakhand                                                        |                                                                          |    |                            |        |      |                         |         |        |         |
| 40                                                                 | Kulhal, UJVNL (3x10)                                                     | SS | 3x10                       | 115.24 | -    | 30(LE)                  | 30      | RM&LE  | 2026-27 |
| Tamil Nadu                                                         |                                                                          |    |                            |        |      |                         |         |        |         |
| 41                                                                 | Kodayar PH-II, TNPGL (1x40 )                                             | SS | 1x40                       | -      | -    | 40.0(LE)+ 6(U)          | 46      | RMU&LE | 2026-27 |
| Kerala                                                             |                                                                          |    |                            |        |      |                         |         |        |         |
| 42                                                                 | Idukki 1 <sup>st</sup> and 2 <sup>nd</sup> stage, KSEB (6x130)           | SS | 6x130                      | -      | -    | 780 (LE)                | 780     | RM&LE  | 2026-27 |
| 43                                                                 | Idamalayar, KSEB (2x37.5)                                                | SS | 2x37.5                     | -      | -    | 75 (LE)                 | 75      | RM&LE  | 2026-27 |
| Andhra Pradesh                                                     |                                                                          |    |                            |        |      |                         |         |        |         |
| 44                                                                 | Lower Sileru, APGENCO (4x115)                                            | SS | 4x115                      | 698.94 | 1.8  | 460(LE)                 | 460     | RM&LE  | 2026-27 |
| Sub Total (D)                                                      |                                                                          |    | 1385.00                    | 814.18 | 1.80 | 1391<br>1385(LE)+ 6(U)] | 1391.00 |        |         |
| E. Ongoing Schemes – Under RLA Studies                             |                                                                          |    |                            |        |      |                         |         |        |         |
| Jammu & Kashmir (UT)                                               |                                                                          |    |                            |        |      |                         |         |        |         |
| 45                                                                 | Salal Stage-I, (Unit 1,2 &3) NHPC (3x115)                                | CS | 3x115                      | -      | -    | 345 (LE)                | 345     | RM&LE  | 2026-27 |
| Himachal Pradesh                                                   |                                                                          |    |                            |        |      |                         |         |        |         |
| 46                                                                 | Pong Power House, BBMB (6x66)                                            | CS | 6x66                       | 402.00 | -    | 396 (LE) + 54 (U)       | 450     | RMU&LE | 2026-27 |
| Punjab                                                             |                                                                          |    |                            |        |      |                         |         |        |         |
| 47                                                                 | Anandpur Sahib Hydel Project, PSPCL (4x33.5)                             | SS | 4x33.5                     | -      | -    | 134 (LE)                | 134     | RM&LE  | 2026-27 |
| 48                                                                 | Mukerian St.I, St.II, St.III & St.IV, PSPCL (3x15, 3x15, 3x19.5& 3x19.5) | SS | 3x15, 3x15, 3x19.5& 3x19.5 | 2.5    | -    | 207 (LE)                | 207     | RM&LE  | 2026-27 |
| 49                                                                 | Shanan HEP, PSPCL (1x50+ 4x15)                                           | SS | 1x50+ 4x15                 | 8.02   | -    | 110 (LE)                | 110     | RM&LE  | 2026-27 |
| 50                                                                 | UBDC St.I & St.II, PSPCL (3x15+ 3x15.45 )                                | SS | 3x15+ 3x15.45              | 1.71   | -    | 91.35 (LE)              | 91.35   | RM&LE  | 2026-27 |
| Madhya Pradesh                                                     |                                                                          |    |                            |        |      |                         |         |        |         |
| 51                                                                 | Bansagar Ton-I, MPPGCL (3x105)                                           | SS | 3x105                      | -      | -    | 315 (LE)                | 315     | RM&LE  | 2026-27 |
| Maharashtra                                                        |                                                                          |    |                            |        |      |                         |         |        |         |
| 52                                                                 | Vaitarna, MSPGCL (1x60)                                                  | SS | 1x60                       | -      | -    | 60 (LE)                 | 60      | RM&LE  | 2026-27 |

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|                          |                                                 |    |                           |                |                |                                             |                 |        |         |
|--------------------------|-------------------------------------------------|----|---------------------------|----------------|----------------|---------------------------------------------|-----------------|--------|---------|
| 53                       | Koyna Dam foot (Right Bank), MSPGCL (2x20)      | SS | 2x20                      | -              | -              | 40 (LE)                                     | 40              | RM&LE  | 2026-27 |
| 54                       | Koyna St-3, MSPGCL (4x80)                       | SS | 4x80                      | -              | -              | 320 (LE)                                    | 320             | RM&LE  | 2026-27 |
| <b>Andhra Pradesh</b>    |                                                 |    |                           |                |                |                                             |                 |        |         |
| 55                       | Machkund St.I & St.II, APGENCO (3x17+ 3x23)     | SS | 3x17+ 3x23                | 1.98           | -              | 120 (LE) +9 (U)                             | 129             | RMU&LE | 2026-27 |
| <b>Kerala</b>            |                                                 |    |                           |                |                |                                             |                 |        |         |
| 56                       | Sabarigiri, (Unit-1,2,3, & 5 ) KSEB (4x55+2x60) | SS | 4x55 (Unit-1,2, ,3, & 5 ) | -              | -              | 220(LE) + 20 (U)                            | 240             | RMU&LE | 2026-27 |
| <b>Jharkhand</b>         |                                                 |    |                           |                |                |                                             |                 |        |         |
| 57                       | Subernrekha, JUUNL (2x65)                       | SS | 2x65                      | -              | -              | 130(LE)                                     | 130             | RM&LE  | 2026-27 |
| <b>Sub Total (E)</b>     |                                                 |    | <b>2488.35</b>            | <b>416.21</b>  | <b>0.00</b>    | <b>2571.35<br/>[2488.35(LE)+ 83(U)]</b>     | <b>2571.35</b>  |        |         |
| <b>Total (A+B+C+D+E)</b> |                                                 |    | <b>10583.90</b>           | <b>9518.34</b> | <b>2947.53</b> | <b>8794.60<br/>[8542.10(LE)+ 252.50(U)]</b> | <b>10836.40</b> |        |         |

Abbreviations: R&M – Renovation & Modernisation;. U – Uprating; LE – Life Extension; Res – Restoration;

MW – Mega Watt; CS-Central Sector: SS- State Sector

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## Annexure- 5F

## State-wise List of Hydro RMU&amp;LE schemes programmed for completion during 2027-32

| Sl. No                                                             | Name of Project, Agency Inst. Cap. (No.X MW) | CS/ SS | Capacity Covered Under RMU&LE (No.x MW) | Est. Cost     | Actual Exp. | Benefits (MW)                  | Capacity after RMU& LE | Category | Completion Target |
|--------------------------------------------------------------------|----------------------------------------------|--------|-----------------------------------------|---------------|-------------|--------------------------------|------------------------|----------|-------------------|
|                                                                    |                                              |        |                                         | (Rs. in Crs.) |             |                                |                        |          |                   |
| A.Ongoing Schemes – Under Tendering                                |                                              |        |                                         |               |             |                                |                        |          |                   |
| Madhya Pradesh                                                     |                                              |        |                                         |               |             |                                |                        |          |                   |
| 1                                                                  | Gandhi Sagar, MPPGCL (5x23)                  | SS     | 5x23                                    | 433.68        | 4.17        | 115 (LE) + 10.83 (U)           | 125.83                 | RMU&LE   | 2029-30           |
| Sub Total(A)                                                       |                                              |        | 115                                     | 433.68        | 4.17        | 125.83<br>115 (LE) + 10.83 (U) | 125.83                 |          |                   |
| B. Ongoing Schemes – Under DPR Preparation/ Finalisation/ Approval |                                              |        |                                         |               |             |                                |                        |          |                   |
| Madhya Pradesh                                                     |                                              |        |                                         |               |             |                                |                        |          |                   |
| 2                                                                  | Pench, MPPGCL (2x80)                         | SS     | 2x80                                    | -             | -           | 160 (LE)                       | 160                    | RM&LE    | 2026-28           |
| 3                                                                  | Bargi, MPPGCL (2x45)                         | SS     | 2x45                                    | -             | -           | 90 (LE)                        | 90                     | RM&LE    | 2026-28           |
| Sub Total(B)                                                       |                                              |        | 250                                     | 0             | 0           | 250                            | 250                    |          |                   |
| C. Ongoing Schemes – Under RLA Studies                             |                                              |        |                                         |               |             |                                |                        |          |                   |
| Jammu & Kashmir (UT)                                               |                                              |        |                                         |               |             |                                |                        |          |                   |
| 4                                                                  | Salal Stage-II, (Unit 4,5 &6) NHPC (6x115)   | CS     | 3x115                                   | -             | -           | 345 (LE)                       | 345                    | RM&LE    | 2027-32           |
| Himachal Pradesh                                                   |                                              |        |                                         |               |             |                                |                        |          |                   |
| 5                                                                  | Chamera-I, NHPC (3x180)                      | CS     | 3x180                                   | -             | -           | 540 (LE)                       | 540                    | RM&LE    | 2027-32           |
| Uttarakhand                                                        |                                              |        |                                         |               |             |                                |                        |          |                   |
| 6                                                                  | Tanakpur, NHPC (3x31.4)                      | CS     | 3x31.4                                  | -             | -           | 94.2 (LE)                      | 94.2                   | RM&LE    | 2027-32           |
| 7                                                                  | Chibro, UJVNL (4x60)                         | SS     | 4x60                                    | 184.88        | -           | 240 (LE)                       | 240                    | RM&LE    | 2027-32           |
| 8                                                                  | Khodri, UJVNL (4x30)                         | SS     | 4x30                                    | 169.63        | -           | 120 (LE)                       | 120                    | RM&LE    | 2027-32           |

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| Tamil Nadu    |                                |    |         |        |      |                                        |         |       |         |
|---------------|--------------------------------|----|---------|--------|------|----------------------------------------|---------|-------|---------|
| 9             | Kundah-I, TNPGL (3x20)         | SS | 3x20    | -      | -    | 60 (LE)                                | 60      | RM&LE | 2027-32 |
| 10            | Kundah-II, TNPGL (5x35)        | SS | 5x35    | -      | -    | 175 (LE)                               | 175     | RM&LE | 2027-32 |
| 11            | Kundah-III, TNPGL (3x60)       | SS | 3x60    | -      | -    | 180 (LE)                               | 180     | RM&LE | 2027-32 |
| 12            | Kundah-IV, TNPGL (2x50)        | SS | 2x50    | -      | -    | 100 (LE)                               | 100     | RM&LE | 2027-32 |
| 13            | Kundah-V, TNPGL (2x20)         | SS | 2x20    | -      | -    | 40 (LE)                                | 40      | RM&LE | 2027-32 |
| 14            | Mettur Tunnel, TNPGL (4x50)    | SS | 4x50    | -      | -    | 200 (LE)                               | 200     | RM&LE | 2027-32 |
| 14            | Sarkarpathy, TNPGL (1x30)      | SS | 1x30    | -      | -    | 30 (LE)                                | 30      | RM&LE | 2027-32 |
| 16            | Sholayar-II, TNPGL (1x25)      | SS | 1x25    | -      | -    | 25 (LE)                                | 25      | RM&LE | 2027-32 |
| 17            | Suruliyar, TNPGL (1x35)        | SS | 1x35    | -      | -    | 35 (LE)                                | 35      | RM&LE | 2027-32 |
| 18            | Kadamparai PH, TNPGL (4x100)   | SS | 4x100   | -      | -    | 400 (LE)                               | 400     | RM&LE | 2027-32 |
| 18            | Aliyar, TNPGL (1x60)           | SS | 1x60    | -      | -    | 60 (LE)                                | 60      | RM&LE | 2027-32 |
| 20            | Lower Mettur-I, TNPGL (2x15)   | SS | 2x15    | -      | -    | 30 (LE)                                | 30      | RM&LE | 2027-32 |
| 21            | Lower Mettur-II, TNPGL (2x15)  | SS | 2x15    | -      | -    | 30 (LE)                                | 30      | RM&LE | 2027-32 |
| 22            | Lower Mettur-III, TNPGL (2x15) | SS | 2x15    | -      | -    | 30 (LE)                                | 30      | RM&LE | 2027-32 |
| 23            | Lower Mettur-IV, TNPGL (2x15)  | SS | 2x15    | -      | -    | 30 (LE)                                | 30      | RM&LE | 2027-32 |
| Sub Total (C) |                                |    | 2764.20 | 354.51 | 0.00 | 2764.20<br>[2764.20 (LE)+<br>0(U)]     | 2764.20 |       |         |
| Total (A+B+C) |                                |    | 3129.20 | 788.19 | 4.17 | 3140.03<br>[3129.20 (LE)+<br>10.83(U)] | 3140.03 |       |         |

Abbreviations: R&amp;M – Renovation &amp; Modernisation; U – Uprating; LE – Life Extension; Res – Restoration;

MW – Mega Watt; CS-Central Sector; SS- State Sector

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## Annexure-6A

## Thermal Capacity Addition Programme for the year 2023-24

| S.No | Project Name                | Unit No. | Targeted Cap. (MW) | Commissioning Schedule at the beginning of FY 2023-24 | Cap. Achieved / likely to be achieved during 2023-24 (MW) | Anticipated/ Actual(A) date of capacity Addition | Remarks/ Reasons                                                                                                                                                                                                                                                |
|------|-----------------------------|----------|--------------------|-------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Barh STPP St-I (NTPC)       | U-2      | 660                | Apr-2023                                              | 660                                                       | Jun-2023                                         | Unit commissioned on 30.06.2023                                                                                                                                                                                                                                 |
| 2    | Telangana STPP St- I (NTPC) | U-1      | 800                | Jun-2023                                              | 800                                                       | Sep-2023                                         | Unit commissioned on 05.09.2023                                                                                                                                                                                                                                 |
|      |                             | U-2      | 800                | Jul-2023                                              | 800                                                       | Feb-2024                                         | Unit commissioned on 11.02.2024                                                                                                                                                                                                                                 |
| 3    | North Karanpura STPP (NTPC) | U-2      | 660                | Nov-2023                                              | 660                                                       | Feb-2024                                         | Unit commissioned on 28.02.2024                                                                                                                                                                                                                                 |
| 4    | Buxar TPP (SJVN)            | U-1      | 660                | Dec-2023                                              | 0                                                         | Jun-2024                                         | Slipped due to slow work progress in BoP areas i.e. AHP, CHP, FOPH, CWPH, WPT Plant, Railway siding & Chimney; Local residents obstruct labor entry at the plant; to ensure uninterrupted 6 km stringing from Jan 4-6, district administration will be present. |
|      |                             | U-2      | 660                | Mar-2024                                              | 0                                                         | Sep-2024                                         |                                                                                                                                                                                                                                                                 |
| 5    | Ghatampur TPP (NUPPL)       | U-1      | 660                | Jul-2023                                              | 0                                                         | Jun-2024                                         | Slipped due to slow work progress in BoP areas i.e. CHP, AHP areas; Delay in material supply in CHP (Wagon Tippler, S/R)                                                                                                                                        |
|      |                             | U-2      | 660                | Oct-2023                                              | 0                                                         | Nov-2024                                         |                                                                                                                                                                                                                                                                 |
|      |                             | U-3      | 660                | Dec-2023                                              | 0                                                         | Mar-2025                                         |                                                                                                                                                                                                                                                                 |
| 6    | Khurja SCTPP (THDC)         | U-1      | 660                | Feb-2024                                              | 0                                                         | Jun-2024                                         | Slipped due to slow progress in BoP areas – CHP, AHP, Chimney, Air Compressor plant.                                                                                                                                                                            |

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|              |                                          |     |              |          |             |           |                                                                                                                                                                                                            |
|--------------|------------------------------------------|-----|--------------|----------|-------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7            | Dr. Narla Tata Rao TPS St-V<br>(APGENCO) | U-8 | 800          | Jun-2023 | 800         | Dec-2023  | Unit commissioned on 20.12.2023                                                                                                                                                                            |
| 8            | Jawaharpur STPP<br>(UPRVUNL)             | U-1 | 660          | Jun-2023 | 660         | Feb-2024  | Unit commissioned on 09.02.2024                                                                                                                                                                            |
|              |                                          | U-2 | 660          | Dec-2023 | 0           | Apr-2024  | Slipped due to delay in readiness of Ash Handling plant (M/s Indure); delay in readiness of Wagon Tippler; Inadequate manpower.                                                                            |
| 9            | Obra-C STPP<br>(UPRVUNL)                 | U-1 | 660          | Jun-2023 | 660         | Feb-2024  | Unit commissioned on 02.02.2024                                                                                                                                                                            |
|              |                                          | U-2 | 660          | Jan-2024 | 0           | Jun-2024  | Slipped due to financial dispute of M/s DPSI (main contractor) with its sub-vendor; Delay in readiness of AHP.                                                                                             |
| 10           | Panki TPS Extn.<br>(UPRVUNL)             | U-1 | 660          | Jan-2024 | 0           | Sep-2024  | Slipped due to delay in completion of CWPH, ACW system, Raw water intake system; Delay in AHP, CHP, DM Plant; Inadequate manpower at site (BHEL).                                                          |
| 11           | North Chennai TPP, St-III<br>(TANGEDCO)  | U-1 | 800          | Sep-2023 | 0           | June-2024 | Slipped due to delay in completion of AHP works; Delay in completion of CHP: ICHS Works.                                                                                                                   |
| 12           | Udangudi STPP St-I<br>(TANGEDCO)         | U-1 | 660          | Mar-2024 | 0           | Sep-2024  | Slipped due to slow progress in execution of works by M/s BHEL in various areas viz. DM Water System, Sea water system; GIS switchyard, NDCTs and critical piping erection; FGD work is yet to be awarded. |
| 13           | Yadadri TPS<br>(TSGENCO)                 | U-1 | 800          | Dec-2023 | 0           | Dec-2024  | Slipped due to delay in issuing fresh EC , hence all commissioning works were put on hold.                                                                                                                 |
|              |                                          | U-2 | 800          | Dec-2023 | 0           | Dec-2024  |                                                                                                                                                                                                            |
| 14           | Bhusawal TPS<br>(MAHAGENCO)              | U-6 | 660          | Jul-2023 | 0           | Jun-2024  | Slipped due to mal operation of boiler and deformation in spiral wall tubes of boiler.                                                                                                                     |
| <b>Total</b> |                                          |     | <b>14700</b> |          | <b>5040</b> |           |                                                                                                                                                                                                            |

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**Additional Capacity Achieved**

| S.No | Project Name                                             | Unit No. | Targeted Cap. (MW) | Commissioning Schedule at the beginning of FY 2023-24 | Cap. Achieved during 2023-24 (MW) | Anticipated/ Actual(A) date of capacity Addition | Remarks/ Reasons                                  |
|------|----------------------------------------------------------|----------|--------------------|-------------------------------------------------------|-----------------------------------|--------------------------------------------------|---------------------------------------------------|
| 1    | Kashipur CCPP, Ph-II<br>(M/s Sravanthi Energy Pvt. Ltd.) | Mod-II   | -                  | -                                                     | 214                               | Jun-2023                                         | Unit commissioned retrospectively from 13.06.2023 |
| 2    | Shirpur Thermal Power Plant (M/s Jindal Power Ltd.)      | U-2      | -                  | -                                                     | 150                               | Nov-2023                                         | Unit commissioned retrospectively from 22.11.2023 |
|      | <b>Total</b>                                             |          |                    |                                                       | <b>364</b>                        |                                                  |                                                   |

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**ANNEXURE-6B**  
**PLANT-WISE COAL RECEIPT AND CONSUMPTION IN 2023-24**

| S.No | Name of TPS        | Capacity<br>(MW) | Figures in Thousand Tonnes |        |       |             |
|------|--------------------|------------------|----------------------------|--------|-------|-------------|
|      |                    |                  | Receipt                    |        |       | Consumption |
|      |                    |                  | Indigeno<br>us             | Import | Total |             |
| 1    | PANIPAT TPS        | 710              | 2804                       | 0      | 2804  | 2726        |
| 2    | RAJIV GANDHI TPS   | 1200             | 3453                       | 0      | 3453  | 3207        |
| 3    | YAMUNA NAGAR TPS   | 600              | 2665                       | 0      | 2665  | 2603        |
| 4    | GH TPS (LEH.MOH.)  | 920              | 2991                       | 0      | 2991  | 2926        |
| 5    | ROPAR TPS          | 840              | 2883                       | 0      | 2883  | 2749        |
| 6    | CHHABRA-I PH-1 TPP | 500              | 2970                       | 2      | 2972  | 2577        |
| 7    | KOTA TPS           | 1240             | 6101                       | 0      | 6101  | 6078        |
| 8    | SURATGARH TPS      | 1500             | 5155                       | 20     | 5174  | 5074        |
| 9    | KALISINDH TPS      | 1200             | 4370                       | 0      | 4370  | 4458        |
| 10   | SURATGARH STPS     | 1320             | 3878                       | 229    | 4107  | 4122        |
| 11   | CHHABRA-I PH-2 TPP | 500              | 1371                       | 2      | 1373  | 1697        |
| 12   | CHHABRA-II TPP     | 1320             | 4197                       | 249    | 4445  | 4428        |
| 13   | ANPARA TPS         | 2630             | 11709                      | 0      | 11709 | 11873       |
| 14   | HARDUAGANJ TPS     | 1265             | 4935                       | 0      | 4935  | 4758        |
| 15   | OBRA TPS           | 1660             | 4165                       | 0      | 4165  | 3791        |
| 16   | PARICHHA TPS       | 920              | 4051                       | 0      | 4051  | 3935        |
| 17   | DSPM TPS           | 500              | 2977                       | 0      | 2977  | 2978        |
| 18   | KORBA-WEST TPS     | 1340             | 6750                       | 0      | 6750  | 7015        |

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|    |                                  |      |       |     |       |       |
|----|----------------------------------|------|-------|-----|-------|-------|
| 19 | MARWA TPS                        | 1000 | 5584  | 0   | 5584  | 5357  |
| 20 | GANDHI NAGAR TPS                 | 630  | 2133  | 32  | 2164  | 2099  |
| 21 | UKAI TPS                         | 1110 | 3914  | 28  | 3942  | 3732  |
| 22 | WANAKBORI TPS                    | 2270 | 8698  | 52  | 8750  | 8376  |
| 23 | AMARKANTAK EXT TPS               | 210  | 1097  | 0   | 1097  | 1100  |
| 24 | SANJAY GANDHI TPS                | 1340 | 5634  | 0   | 5634  | 5493  |
| 25 | SATPURA TPS                      | 1330 | 2503  | 0   | 2503  | 2481  |
| 26 | SHREE SINGAJI TPP                | 2520 | 10662 | 0   | 10662 | 10443 |
| 27 | BHUSAWAL TPS                     | 1210 | 6107  | 414 | 6521  | 6094  |
| 28 | CHANDRAPUR(MAHARA<br>SHTRA) STPS | 2920 | 11700 | 914 | 12614 | 12894 |
| 29 | KHAPARKHEDA TPS                  | 1340 | 6347  | 394 | 6741  | 6948  |
| 30 | KORADI TPS                       | 2190 | 9811  | 8   | 9818  | 9725  |
| 31 | NASIK TPS                        | 630  | 2462  | 142 | 2604  | 2494  |
| 32 | PARAS TPS                        | 500  | 2788  | 0   | 2788  | 2708  |
| 33 | PARLI TPS                        | 750  | 2944  | 0   | 2944  | 2961  |
| 34 | DAMODARAM<br>SANJEEVAIAH TPS     | 2400 | 5664  | 542 | 6206  | 6237  |
| 35 | Dr. N.TATA RAO TPS               | 2560 | 10569 | 0   | 10569 | 10412 |
| 36 | RAYALASEEMA TPS                  | 1650 | 8275  | 0   | 8275  | 8163  |
| 37 | BELLARY TPS                      | 1700 | 6630  | 51  | 6681  | 6511  |
| 38 | RAICHUR TPS                      | 1720 | 7020  | 36  | 7055  | 6739  |
| 39 | YERMARUS TPP                     | 1600 | 4166  | 87  | 4254  | 4168  |
| 40 | METTUR TPS                       | 840  | 4739  | 83  | 4822  | 4797  |
| 41 | METTUR TPS - II                  | 600  | 2222  | 433 | 2655  | 2555  |

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|    |                             |      |       |     |       |       |
|----|-----------------------------|------|-------|-----|-------|-------|
| 42 | NORTH CHENNAI TPS           | 1830 | 9033  | 768 | 9802  | 9494  |
| 43 | TUTICORIN TPS               | 1050 | 5695  | 0   | 5695  | 5771  |
| 44 | BHADRADRI TPP               | 1080 | 5305  | 0   | 5305  | 5231  |
| 45 | KAKATIYA TPS                | 1100 | 4802  | 0   | 4802  | 4815  |
| 46 | KOTHAGUDEM TPS<br>(NEW)     | 1000 | 4698  | 0   | 4698  | 4792  |
| 47 | KOTHAGUDEM TPS<br>(STAGE-7) | 800  | 3064  | 0   | 3064  | 3046  |
| 48 | RAMAGUNDEM-B TPS            | 62.5 | 224   | 0   | 224   | 230   |
| 49 | SINGARENI TPP               | 1200 | 5429  | 0   | 5429  | 5381  |
| 50 | TENUGHAT TPS                | 420  | 1710  | 0   | 1710  | 1731  |
| 51 | IB VALLEY TPS               | 1740 | 8728  | 0   | 8728  | 8701  |
| 52 | BAKRESWAR TPS               | 1050 | 5246  | 0   | 5246  | 5184  |
| 53 | BANDEL TPS                  | 270  | 1233  | 0   | 1233  | 1233  |
| 54 | D.P.L. TPS                  | 550  | 1737  | 0   | 1737  | 1535  |
| 55 | KOLAGHAT TPS                | 840  | 4294  | 0   | 4294  | 4295  |
| 56 | SAGARDIGHI TPS              | 1600 | 7868  | 0   | 7868  | 7698  |
| 57 | SANTALDIH TPS               | 500  | 2654  | 0   | 2654  | 2672  |
| 58 | DADRI (NCTPP)               | 1820 | 6632  | 812 | 7445  | 6948  |
| 59 | RIHAND STPS                 | 3000 | 14471 | 0   | 14471 | 14551 |
| 60 | SINGRAULI STPS              | 2000 | 9858  | 0   | 9858  | 9878  |
| 61 | TANDA TPS                   | 1760 | 6440  | 810 | 7250  | 6771  |
| 62 | UNCHAHAHAR TPS              | 1550 | 5307  | 481 | 5788  | 5473  |
| 63 | KORBA STPS                  | 2600 | 13637 | 0   | 13637 | 13859 |
| 64 | LARA TPP                    | 1600 | 8379  | 0   | 8379  | 8407  |

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|    |                        |      |       |      |       |       |
|----|------------------------|------|-------|------|-------|-------|
| 65 | SIPAT STPS             | 2980 | 14784 | 0    | 14784 | 14642 |
| 66 | GADARWARA TPP          | 1600 | 5689  | 616  | 6305  | 6320  |
| 67 | KHARGONE STPP          | 1320 | 4486  | 461  | 4947  | 4694  |
| 68 | VINDHYACHAL STPS       | 4760 | 22365 | 0    | 22365 | 23350 |
| 69 | MAUDA TPS              | 2320 | 9711  | 980  | 10691 | 10322 |
| 70 | SOLAPUR STPS           | 1320 | 4516  | 392  | 4908  | 4687  |
| 71 | SIMHADRI               | 2000 | 8050  | 917  | 8966  | 8663  |
| 72 | KUDGI STPP             | 2400 | 7120  | 1265 | 8385  | 7970  |
| 73 | RAMAGUNDEM STPS        | 2600 | 11659 | 0    | 11659 | 11511 |
| 74 | TELANGANA STPP PH-1    | 1600 | 1819  | 0    | 1819  | 1870  |
| 75 | JAWAHARPUR STPP        | 660  | 19    | 0    | 19    | 0     |
| 76 | BARAUNI TPS            | 710  | 2375  | 0    | 2375  | 2244  |
| 77 | BARH STPS              | 2640 | 9645  | 571  | 10216 | 9861  |
| 78 | KAHALGAON TPS          | 2340 | 12494 | 534  | 13028 | 12879 |
| 79 | NORTH KARANPURA<br>TPP | 1320 | 3976  | 0    | 3976  | 3707  |
| 80 | DARLIPALI STPS         | 1600 | 9071  | 0    | 9071  | 9129  |
| 81 | TALCHER STPS           | 3000 | 16480 | 243  | 16723 | 16586 |
| 82 | FARAKKA STPS           | 2100 | 8654  | 597  | 9250  | 8647  |
| 83 | BONGAIGAON TPP         | 750  | 3217  | 159  | 3377  | 3144  |
| 84 | INDIRA GANDHI STPP     | 1500 | 5821  | 360  | 6181  | 5594  |
| 85 | MEJA STPP              | 1320 | 5112  | 96   | 5208  | 4984  |
| 86 | BHILAI TPS             | 500  | 2828  | 107  | 2935  | 2813  |
| 87 | VALLUR TPP             | 1500 | 4902  | 228  | 5129  | 4823  |
| 88 | MUZAFFARPUR TPS        | 390  | 2139  | 0    | 2139  | 1991  |

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|     |                               |      |       |     |       |       |
|-----|-------------------------------|------|-------|-----|-------|-------|
| 89  | NABINAGAR STPP                | 1980 | 9000  | 547 | 9547  | 8815  |
| 90  | NABINAGAR TPP                 | 1000 | 5000  | 116 | 5116  | 5093  |
| 91  | BOKARO TPS 'A' EXP            | 500  | 2359  | 13  | 2372  | 2178  |
| 92  | CHANDRAPURA(DVC) TPS          | 500  | 2539  | 49  | 2589  | 2523  |
| 93  | KODARMA TPP                   | 1000 | 4415  | 168 | 4583  | 4408  |
| 95  | DURGAPUR STEEL TPS            | 1000 | 4438  | 156 | 4594  | 4289  |
| 96  | MEJIA TPS                     | 2340 | 10482 | 357 | 10839 | 10663 |
| 97  | RAGHUNATHPUR TPP              | 1200 | 4656  | 116 | 4772  | 4425  |
| 98  | MAHATMA GANDHI TPS            | 1320 | 5286  | 100 | 5385  | 5157  |
| 99  | GOINDWAL SAHIB TPP            | 540  | 1841  | 0   | 1841  | 1707  |
| 100 | RAJPURA TPP                   | 1400 | 5926  | 39  | 5964  | 5997  |
| 101 | TALWANDI SABO TPP             | 1980 | 7766  | 8   | 7774  | 7680  |
| 102 | ADANI POWER LIMITED KAWAI TPP | 1320 | 4050  | 812 | 4862  | 5140  |
| 103 | ANPARA C TPS                  | 1200 | 5134  | 0   | 5134  | 5140  |
| 104 | BARKHERA TPS                  | 90   | 343   | 0   | 343   | 287   |
| 105 | KHAMBARKHERA TPS              | 90   | 314   | 0   | 314   | 285   |
| 106 | KUNDARKI TPS                  | 90   | 276   | 0   | 276   | 276   |
| 107 | LALITPUR TPS                  | 1980 | 7800  | 0   | 7800  | 7729  |
| 108 | MAQSOODPUR TPS                | 90   | 320   | 0   | 320   | 287   |
| 109 | PRAYAGRAJ TPP                 | 1980 | 8149  | 0   | 8149  | 7772  |
| 110 | ROSA TPP Ph-I                 | 1200 | 4800  | 0   | 4800  | 4717  |
| 111 | UTRAULA TPS                   | 90   | 273   | 0   | 273   | 286   |
| 112 | AKALTARA TPS                  | 1800 | 7063  | 0   | 7064  | 6965  |

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|     |                                |      |       |      |       |       |
|-----|--------------------------------|------|-------|------|-------|-------|
| 113 | BALCO TPS                      | 600  | 2339  | 38   | 2377  | 2412  |
| 114 | BANDAKHAR TPP                  | 300  | 1512  | 0    | 1512  | 1594  |
| 115 | BARADARHA TPS                  | 1200 | 6230  | 0    | 6230  | 6311  |
| 116 | BINJKOTE TPP                   | 600  | 2387  | 0    | 2387  | 2213  |
| 117 | NAWAPARA TPP                   | 600  | 2606  | 0    | 2606  | 2527  |
| 118 | PATHADI TPP                    | 600  | 2721  | 0    | 2721  | 2742  |
| 119 | TAMNAR TPP                     | 2400 | 12272 | 351  | 12623 | 12819 |
| 120 | UCHPINDA TPP                   | 1440 | 5597  | 0    | 5597  | 5553  |
| 121 | SABARMATI (D-F STATIONS)       | 362  | 1216  | 367  | 1583  | 1639  |
| 122 | ANUPPUR TPP                    | 1250 | 5923  | 0    | 5923  | 5866  |
| 123 | BINA TPS                       | 500  | 2375  | 0    | 2375  | 2359  |
| 124 | SEIONI TPP                     | 600  | 2790  | 0    | 2790  | 2951  |
| 125 | AMRAVATI TPS                   | 1350 | 6587  | 0    | 6587  | 6459  |
| 126 | BUTIBORI TPP                   | 600  | 0     | 0    | 0     | 0     |
| 127 | DAHANU TPS                     | 500  | 2217  | 72   | 2289  | 2154  |
| 128 | DHARIWAL TPP                   | 600  | 3065  | 0    | 3065  | 2979  |
| 129 | DISHERGARH TPP                 | 12   | 54    | 0    | 54    | 49    |
| 130 | GMR WARORA TPS                 | 600  | 2891  | 0    | 2891  | 2814  |
| 131 | ADANI POWER LIMITED TIRODA TPP | 3300 | 13624 | 0    | 13624 | 13733 |
| 132 | WARDHA WARORA TPP              | 540  | 2672  | 0    | 2672  | 2454  |
| 133 | PAINAMPURAM TPP                | 1320 | 4211  | 1697 | 5908  | 5886  |
| 134 | SGPL TPP                       | 1320 | 2203  | 3404 | 5607  | 5565  |
| 135 | VIZAG TPP                      | 1040 | 3773  | 515  | 4289  | 4160  |

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|     |                                         |      |       |      |       |       |
|-----|-----------------------------------------|------|-------|------|-------|-------|
| 136 | JOJOBERA TPS                            | 240  | 1124  | 0    | 1124  | 1169  |
| 137 | MAHADEV PRASAD STPP                     | 540  | 2686  | 0    | 2686  | 2711  |
| 138 | MAITHON RB TPP                          | 1050 | 4890  | 0    | 4890  | 4820  |
| 139 | DERANG TPP                              | 1200 | 6673  | 0    | 6673  | 5937  |
| 140 | KAMALANGA TPS                           | 1050 | 5603  | 87   | 5690  | 5534  |
| 141 | VEDANTA TPP                             | 600  | 2261  | 0    | 2261  | 2346  |
| 142 | BUDGE BUDGE TPS                         | 750  | 3492  | 0    | 3492  | 3527  |
| 143 | HALDIA TPP                              | 600  | 3255  | 0    | 3255  | 3115  |
| 144 | HIRANMAYE TPP                           | 300  | 1513  | 0    | 1513  | 1519  |
| 145 | SOUTHERN REPL. TPS                      | 135  | 459   | 0    | 459   | 454   |
| 146 | ADANI POWER LIMITED RAIGARH TPP         | 600  | 2895  | 78   | 2973  | 3014  |
| 147 | OP JINDAL TPS                           | 1000 | 6116  | 0    | 6116  | 5619  |
| 148 | ADANI POWER LIMITED RAIPUR TPP          | 1370 | 5594  | 270  | 5863  | 5779  |
| 149 | MAHAN TPP                               | 1200 | 4188  | 42   | 4230  | 4231  |
| 150 | NIGRI TPP                               | 1320 | 5603  | 0    | 5603  | 5662  |
| 151 | SASAN UMTTP                             | 3960 | 18077 | 0    | 18077 | 17992 |
| 152 | NTPL TUTICORIN TPP                      | 1000 | 3760  | 366  | 4126  | 4294  |
| 153 | SIKKA REP. TPS                          | 500  | 0     | 429  | 429   | 403   |
| 154 | ADANI POWER LIMITED MUNDRA TPP - I & II | 2640 | 0     | 5910 | 5910  | 5767  |
| 155 | ADANI POWER LIMITED MUNDRA TPP - III    | 1980 | 0     | 6774 | 6774  | 6707  |
| 156 | MUNDRA UMTTP                            | 4000 | 0     | 8305 | 8305  | 8496  |

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|     |                                         |        |      |      |      |      |
|-----|-----------------------------------------|--------|------|------|------|------|
| 157 | SALAYA TPP                              | 1200   | 0    | 1781 | 1781 | 1949 |
| 158 | JSW RATNAGIRI TPP                       | 300    | 0    | 773  | 773  | 816  |
| 159 | TROMBAY TPS                             | 750    | 0    | 2665 | 2665 | 2563 |
| 160 | SIMHAPURI TPS                           | 600    | 0    | 1578 | 1578 | 1646 |
| 161 | THAMMINAPATNAM<br>TPS                   | 300    | 0    | 0    | 0    | 0    |
| 162 | TORANGALLU TPS(SBU-<br>I)               | 260    | 0    | 562  | 562  | 562  |
| 163 | TORANGALLU TPS(SBU-<br>II)              | 600    | 0    | 1065 | 1065 | 1031 |
| 164 | ADANI POWER LIMITED<br>UDUPI TPP        | 1200   | 39   | 2813 | 2852 | 2612 |
| 165 | ITPCL TPP                               | 1200   | 0    | 3989 | 3989 | 4170 |
| 166 | MUTHIARA TPP                            | 1200   | 0    | 3462 | 3462 | 3452 |
| 167 | TUTICORIN TPP ST-IV                     | 525    | 0    | 1556 | 1556 | 1555 |
| 168 | SHREE CEMENT LTD TPS                    | 300    | 0    | 110  | 110  | 119  |
| 169 | OPG Power Generation<br>Private Limited | 414    | 0    | 34   | 34   | 34   |
| 170 | CHAKABURA TPP                           | 30     | 760  | 0    | 760  | 719  |
| 171 | KASAIPALLI TPP                          | 270    | 2348 | 0    | 2348 | 1736 |
| 172 | NIWARI TPP                              | 90     | 487  | 0    | 487  | 443  |
| 173 | RATIJA TPS                              | 100    | 886  | 0    | 886  | 858  |
| 174 | BELA TPS                                | 270    | 1313 | 0    | 1313 | 1226 |
| 175 | MIHAN TPS                               | 246.01 | 0    | 0    | 0    | 0    |
| 176 | NASIK (P) TPS                           | 1350   | 0    | 0    | 0    | 0    |
| 177 | SALORA TPP                              | 135    | 0    | 0    | 0    | 0    |

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|                        |                           |               |               |              |               |               |
|------------------------|---------------------------|---------------|---------------|--------------|---------------|---------------|
| 178                    | SHIRPUR TPP               | 300           | 200           | 43           | 243           | 148           |
| 179                    | TUTICORIN (P) TPP         | 300           | 0             | 0            | 0             | 0             |
| 180                    | UTKAL TPP (IND<br>BARATH) | 350           | 0             | 0            | 0             | 0             |
| 181                    | TITAGARH TPS              | 240           | 0             | 0            | 0             | 0             |
| 182                    | GEPL TPP Ph-I             | 120           | 0             | 0            | 0             | 0             |
| 183                    | KATGHORA TPP              | 35            | 0             | 0            | 0             | 0             |
| 184                    | SVPL TPP                  | 63            | 401           | 0            | 401           | 315           |
| 185                    | SWASTIK KORBA TPP         | 25            | 0             | 0            | 0             | 0             |
| <b>TOTAL ALL INDIA</b> |                           | <b>210970</b> | <b>798590</b> | <b>65729</b> | <b>864319</b> | <b>849717</b> |

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Annexure-9A

**CENTRAL ELECTRICITY AUTHORITY**  
**FINANCIAL STUDIES & ASSISTANCE DIVISION**  
**STATEMENT SHOWING ESTIMATED AVERAGE RATES OF ELECTRICITY (FY 2022-23)**

(Rates in Paise/KWh)

| S. No. | Name of State/Utility          | Tariff effective from | Domestic             |                      |                        | Commercial           |                        |                        |                        | Agriculture          |                       |                        | Small Industry 10KW (1500 KWh/ Month) | Medium Industry 50KW (7500 KWh/ Month) | Large Industry (11KV) 1000KW 60%L.F. (438000 KWh/ Month) | Heavy Industry (11KV) 1000KW 60%L.F. (438000 KWh/ Month) | Heavy Industry (33KV) 2000KW 60%L.F. (876000 KWh/ Month) | Railway Traction 12500KW (2500000 KWh/ Month) |                |
|--------|--------------------------------|-----------------------|----------------------|----------------------|------------------------|----------------------|------------------------|------------------------|------------------------|----------------------|-----------------------|------------------------|---------------------------------------|----------------------------------------|----------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------|----------------|
|        |                                |                       | 1KW (100 KWh/ Month) | 4KW (400 KWh/ Month) | 10KW (1000 KWh/ Month) | 2KW (300 KWh/ Month) | 10KW (1500 KWh/ Month) | 30KW (4500 KWh/ Month) | 50KW (7500 KWh/ Month) | 2HP (400 KWh/ Month) | 5HP (1000 KWh/ Month) | 10HP (2000 KWh/ Month) |                                       |                                        |                                                          |                                                          |                                                          |                                               |                |
| 1      | Andaman & Nicobar Islands      | 01.08.2022            | 265.00               | 604.00               | 789.00                 | 905.00               | 1282.00                | 1440.00                | 1471.00                | 196.00               | 196.00                | 196.00                 | 952.00                                | 1100.00                                | 1062.00                                                  | 1063.00                                                  | 1063.00                                                  | -                                             | -              |
| 2      | Andhra Pradesh                 | 01.04.2022            | 321.00               | 653.00               | 856.00                 | 971.00               | 1119.00                | 1150.00                | 1156.00                | 450.00 #             | 450.00 #              | 450.00 #               | 820.00                                | 820.00                                 | 877.00                                                   | 877.00                                                   | 830.00                                                   | 763.00                                        | -              |
| 3      | Arunachal Pradesh              | 01.06.2018            | 400.00               | 400.00               | 500.00                 | 500.00               | 500.00                 | 500.00                 | 500.00                 | 310.00               | 310.00                | 310.00                 | 430.00                                | 430.00                                 | 385.00                                                   | 385.00                                                   | 350.00                                                   | -                                             | -              |
| 4      | Assam                          | 01.04.2022            | 620.00               | 757.00               | 814.00                 | 896.00               | 896.00                 | 963.00                 | 963.00                 | 510.00               | 510.00                | 510.00                 | 616.00 U                              | 813.00                                 | 755.00                                                   | 755.00                                                   | 755.00                                                   | 858.00                                        | -              |
| 5      | Bihar                          | 01.04.2022            | 689.00               | 815.00 U             | 863.00 U               | 855.00 U             | 900.00 U               | 908.00 U               | 909.00 U               | 574.00               | 574.00                | 574.00                 | 867.00 R                              | 895.00                                 | 831.00                                                   | -                                                        | 819.00 at 132kV                                          | 900.00 at 132 kV                              | -              |
| 6      | Chandigarh                     | 01.07.2022            | 299.00               | 393.00               | 451.00                 | 488.00               | 568.00                 | 574.00                 | 576.00                 | 260.00               | 260.00                | 260.00                 | 461.00                                | 564.00                                 | 507.00                                                   | 507.00                                                   | 507.00                                                   | -                                             | -              |
| 7      | Chhattisgarh                   | 01.04.2022            | 431.00               | 525.00               | 718.00                 | 796.00               | 1002.00                | 1017.00                | 1017.00                | 555.00               | 555.00                | 555.00                 | 616.00                                | 720.00                                 | 1126.00                                                  | 1126.00                                                  | 1069.00                                                  | 714.00                                        | -              |
| 8      | Dadra & Nagar Haveli           | 01.04.2022            | 205.00               | 256.00               | 313.00                 | 415.00               | 442.00                 | 446.00                 | 447.00                 | 90.00                | 90.00                 | 90.00                  | 432.00                                | 499.00                                 | 579.00                                                   | 579.00                                                   | -                                                        | -                                             | -              |
| 9      | Daman & Diu                    | 01.04.2022            | 170.00               | 248.00               | 309.00                 | 415.00               | 442.00                 | 443.00                 | 447.00                 | 90.00                | 90.00                 | 90.00                  | 472.00                                | 472.00                                 | 579.00                                                   | 579.00                                                   | -                                                        | -                                             | -              |
| 10     | Delhi (BYPL/BRPL/NDPL)         | 01.10.2021            | 335.00               | 444.00               | 678.00                 | 885.00               | 1177.00                | 1177.00                | 1177.00                | 204.00               | 204.00                | 204.00                 | 1089.00                               | 1089.00                                | 942.00                                                   | 942.00                                                   | 933.00                                                   | 870.00                                        | -              |
| 11     | Delhi (NDMC)                   | 01.10.2021            | 335.00               | 444.00               | 678.00                 | 885.00               | 1177.00                | 1177.00                | 1177.00                | 204.00               | 204.00                | 204.00                 | 1089.00                               | 1089.00                                | 942.00                                                   | 942.00                                                   | 933.00                                                   | 870.00                                        | -              |
| 12     | Goa                            | 01.04.2022            | 200.00               | 310.00               | 418.00                 | 522.00               | 599.00                 | 627.00                 | 630.00                 | 179.00               | 179.00                | 179.00                 | 479.00                                | 497.00                                 | 667.00                                                   | 667.00                                                   | -                                                        | -                                             | -              |
| 13     | Gujarat                        | 01.04.2022            | 394.00 U             | 505.00 U             | 586.00 U               | 562.00               | 562.00                 | 617.00                 | 672.00                 | 90.00                | 90.00                 | 90.00                  | 582.00                                | 588.00                                 | 551.00                                                   | 626.00                                                   | 630.00                                                   | 600.00 at 132 kV                              | -              |
| 14     | Torrent Power Ltd. (Ahmedabad) | 01.04.2022            | 440.00               | 511.00               | 554.00                 | 608.00               | 624.00                 | 716.00                 | 716.00                 | 340.00               | 340.00                | 340.00                 | 572.00                                | 656.00                                 | 600.00                                                   | 600.00                                                   | -                                                        | -                                             | -              |
| 15     | Torrent Power Ltd. (Surat)     | 01.04.2022            | 423.00               | 507.00               | 556.00                 | 578.00               | 578.00                 | 720.00                 | 720.00                 | 340.00               | 340.00                | 340.00                 | 572.00                                | 656.00                                 | 600.00                                                   | 600.00                                                   | -                                                        | -                                             | -              |
| 16     | Haryana                        | 01.04.2022            | 235.00               | 481.00               | 616.00                 | 716.00               | 716.00                 | 828.00                 | 828.00                 | 667.00               | 667.00                | 667.00                 | 716.00                                | 828.00                                 | 791.00                                                   | 791.00                                                   | 780.00                                                   | 841.00 at 11kV                                | -              |
| 17     | Himachal Pradesh               | 01.04.2022            | 512.00               | 526.00               | 560.00                 | 621.00               | 582.00                 | 687.00                 | 687.00                 | 455.00               | 440.00                | 434.00                 | 515.00                                | 660.00                                 | 663.00                                                   | 663.00                                                   | 630.00                                                   | 821.00 at 66 kV                               | -              |
| 18     | Jammu & Kashmir                | 13.10.2022            | 238.00               | 324.00               | 397.00                 | 458.00               | 792.00                 | 792.00                 | 792.00                 | 102.00               | 102.00                | 102.00                 | 541.00                                | 541.00                                 | 534.00                                                   | 534.00                                                   | 521.00                                                   | 729.00 at 11/33 kV                            | -              |
| 19     | Jharkhand                      | 01.10.2020            | 738.00 U             | 681.00 U             | 670.00 U               | 688.00 U             | 703.00 U               | 703.00 U               | 703.00 U               | 510.00               | 510.00                | 510.00                 | 795.00                                | 795.00                                 | 696.00                                                   | 741.00                                                   | 719.00                                                   | 824.00                                        | -              |
| 20     | Karnataka                      | 01.04.2022            | 631.00 D             | 882.00 D             | 955.00 D               | 1090.00 D            | 1104.00 D              | 1107.00 D              | 1107.00 D              | 0.00                 | 0.00                  | 0.00                   | 843.00 D                              | 961.00 D                               | 912.00 D                                                 | 919.00 D                                                 | 918.00 D                                                 | 872.00                                        | -              |
| 21     | Kerala                         | 26.06.2022            | 446.00               | 984.00               | 1160.00                | 777.00               | 1087.00                | 1141.00                | 1141.00                | 259.00               | 259.00                | 259.00                 | 646.00                                | 767.00                                 | 719.00                                                   | 719.00                                                   | 719.00                                                   | 729.00 at 110 kV                              | -              |
| 22     | Ladakh                         | 01.10.2022            | 238.00               | 324.00               | 397.00                 | 451.00               | 784.00                 | 784.00                 | 784.00                 | 135.00               | 135.00                | 135.00                 | 537.00                                | 537.00                                 | 534.00                                                   | 534.00                                                   | 521.00                                                   | 729.00 at 11/33 kV                            | -              |
| 23     | Lakshadweep                    | 01.04.2022            | 180.00               | 455.00               | 614.00                 | 860.00               | 988.00                 | 1009.00                | 1014.00                | -                    | -                     | -                      | 687.00                                | 687.00                                 | 958.00                                                   | 958.00                                                   | -                                                        | -                                             | -              |
| 24     | Madhya Pradesh                 | 08.04.2022            | 643.00 U             | 963.00 U             | 1020.00 U              | 860.00 U             | 866.00 U               | 958.00 U               | 958.00 U               | 542.00               | 599.00                | 625.00                 | 952.00 U                              | 952.00 U                               | 822.00                                                   | 822.00                                                   | 877.00                                                   | 762.00 at 132/220kV                           | -              |
| 25     | Maharashtra                    | 01.04.2022            | 668.00               | 1011.00              | 1307.00                | 1210.00              | 1072.00                | 1871.00                | 1871.00                | 352.00               | 352.00                | 352.00                 | 748.00                                | 1070.00                                | 1032.00 B                                                | 1032.00 B                                                | 1032.00                                                  | 1076.00                                       | -              |
| 26     | Mumbai (B.E.S.T.)              | 01.04.2022            | 488.00               | 741.00               | 1007.00                | 977.00               | 840.00                 | 1129.00                | 1129.00                | 387.00               | 387.00                | 387.00                 | 717.00                                | 998.00                                 | 839.00                                                   | 839.00                                                   | -                                                        | 859.00 at 33/11kV                             | -              |
| 27     | Mumbai (Adani Electricity)     | 01.04.2022            | 643.00               | 804.00               | 973.00                 | 1043.00              | 906.00                 | 1256.00                | 1256.00                | 547.00               | 547.00                | 547.00                 | 819.00                                | 1114.00                                | 908.00                                                   | 908.00                                                   | -                                                        | 887.00 at 33/11kV                             | -              |
| 28     | Mumbai (TATA'S)                | 01.04.2022            | 524.00               | 788.00               | 1035.00                | 1015.00              | 878.00                 | 1168.00                | 1168.00                | 389.00               | 389.00                | 389.00                 | 746.00                                | 1020.00                                | 864.00                                                   | 864.00                                                   | -                                                        | 859.00 at 33/11kV                             | -              |
| 29     | Manipur                        | 01.04.2022            | 575.00               | 679.00               | 716.00                 | 772.00               | 812.00                 | 818.00                 | 820.00                 | 479.00               | 479.00                | 479.00                 | 607.00                                | 967.00                                 | 1038.00                                                  | 1038.00                                                  | -                                                        | -                                             | -              |
| 30     | Mizoram                        | 01.04.2022            | 540.00               | 760.00               | 826.00                 | 886.00               | 896.00                 | 898.00                 | 898.00                 | 399.00               | 399.00                | 399.00                 | 833.00                                | 853.00                                 | 1086.00                                                  | 1086.00                                                  | -                                                        | -                                             | -              |
| 31     | Meghalaya                      | 01.04.2022            | 475.00               | 585.00               | 639.00                 | 803.00               | 834.00                 | 835.00                 | 847.00                 | 347.00               | 347.00                | 347.00                 | 816.00                                | 816.00                                 | 850.00                                                   | 850.00                                                   | 817.00                                                   | -                                             | -              |
| 32     | Nagaland                       | 01.07.2022            | 545.00               | 631.00               | 673.00                 | 835.00               | 879.00                 | 886.00                 | 888.00                 | 320.00               | 320.00                | 320.00                 | 653.00                                | 687.00                                 | 729.00                                                   | 730.00                                                   | -                                                        | -                                             | -              |
| 33     | Odisha                         | 01.04.2022            | 426.00               | 548.00               | 618.00                 | 710.00               | 790.00                 | 804.00                 | 806.00                 | 161.00               | 158.00                | 158.00                 | 677.00                                | 723.00                                 | 720.00                                                   | 720.00                                                   | 695.00                                                   | 724.00 at 33/22 kV                            | -              |
| 34     | Puducherry                     | 01.04.2022            | 220.00               | 436.00               | 580.00                 | 703.00               | 781.00                 | 794.00                 | 796.00                 | 0.00 S               | 0.00 S                | 0.00 S                 | 638.00                                | 638.00                                 | 695.00                                                   | -                                                        | 694.00                                                   | -                                             | -              |
| 35     | Punjab                         | 01.04.2022            | 434.00               | 710.00               | 900.00                 | 834.00               | 870.00                 | 881.00                 | 881.00                 | 566.00 WS            | 566.00 WS             | 566.00 WS              | 741.00                                | 829.00                                 | 842.00                                                   | 869.00                                                   | 869.00                                                   | 963.00 at 132 kV                              | -              |
| 36     | Rajasthan                      | 01.09.2022            | 833.00               | 830.00               | 845.00                 | 1022.00              | 1046.00                | 1155.00                | 1157.00                | 574.00               | 574.00                | 574.00                 | 793.00                                | 868.00                                 | 763.00                                                   | -                                                        | 745.00                                                   | 710.00                                        | -              |
| 37     | Sikkim                         | 01.04.2022            | 150.00               | 288.00               | 355.00                 | 483.00               | 599.00                 | 698.00                 | 702.00                 | -                    | -                     | -                      | 622.00 U                              | 697.00                                 | 840.00                                                   | 840.00                                                   | -                                                        | -                                             | -              |
| 38     | Tamil Nadu                     | 10.09.2022            | 113.00               | 569.00               | 878.00                 | 1138.00              | 1138.00                | 1138.00                | 1138.00                | 0.00                 | 0.00                  | 0.00                   | 730.00                                | 866.00                                 | 855.00                                                   | 855.00                                                   | 855.00                                                   | 1056.00                                       | -              |
| 39     | Telangana                      | 01.04.2022            | 339.00               | 719.00               | 881.00                 | 1031.00              | 1124.00                | 1143.00                | 1147.00                | 258.00 \$            | 253.00 \$             | 252.00 \$              | 833.00                                | 842.00                                 | 926.00                                                   | 925.00                                                   | 873.00                                                   | 782.00 at 33 kV                               | -              |
| 40     | Tripura                        | 01.09.2022            | 548.00               | 643.00               | 793.00                 | 743.00               | 826.00                 | 826.00                 | 826.00                 | 394.00               | 394.00                | 394.00                 | 796.00                                | 821.00                                 | -                                                        | -                                                        | -                                                        | -                                             | -              |
| 41     | Uttarakhand                    | 01.04.2022            | 365.00               | 553.00               | 675.00                 | 678.00               | 678.00                 | 794.00                 | 794.00                 | 215.00               | 215.00                | 215.00                 | 620.00                                | 693.00                                 | 787.00                                                   | 787.00                                                   | 787.00                                                   | 744.00                                        | -              |
| 42     | Uttar Pradesh                  | 29.07.2022            | 693.00 U             | 739.00 U             | 774.00 U               | 1043.00 U            | 1174.00 U              | 1233.00 U              | 1245.00 U              | 665.00 U             | 665.00 U              | 665.00 U               | 993.00 U                              | 993.00 U                               | 985.00 U                                                 | 985.00 U                                                 | 944.00                                                   | 1235.00                                       | Below 132 kV   |
| 43     | West Bengal (WBSEDCL)          | 01.04.2022            | 446.00 R             | 574.00 R             | 633.00 R               | 670.00 R             | 670.00 R               | 670.00 R               | 670.00 R               | 235.00 R             | 235.00 R              | 235.00 R               | 918.00 R                              | 918.00 R                               | 911.00 R                                                 | 911.00 R                                                 | 958.00                                                   | 936.00 at 25 kV                               | 132 kV & above |
| 44     | D.V.C. Jharkhand Area          | 01.02.2023            | 526.00               | 469.00               | 458.00                 | 545.00               | 545.00                 | 545.00                 | 545.00                 | 315.00               | 315.00                | 315.00                 | 641.00                                | 641.00                                 | 609.00                                                   | 642.00                                                   | 625.00                                                   | 659.00                                        | -              |

B : General Industry C : Seasonal Industry D : Bangalore, Devangere & Other City Municipal Corp. F : Areas under Village Panchayats U : Urban R : Rural O : Other Areas  
 WS: Without Subsidy # Without Demand side management measures \$ For Corporate Farmers S : With Subsidy  
 Tariffs notified have varying parameters for various categories of consumers. The above comparison is based on certain assumed loads and electricity consumption levels in a month.

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## Annexure-10A

## All India Sector wise/ Utility-wise Target, Actual Generation &amp; PLF (%) for the year 2023-24

| Category /     | APRIL 2023 - MAR-2024 |             |       |
|----------------|-----------------------|-------------|-------|
|                | TARGET (MU)           | ACTUAL (MU) | PLF % |
| THERMAL        |                       |             |       |
| CENTRAL SECTOR |                       |             |       |
| APCPL          | 7600.00               | 8284.84     | 62.88 |
| BRBCL          | 6700.00               | 6944.47     | 79.06 |
| DVC            | 43700.00              | 44127.91    | 76.81 |
| JHAPL          | 3800.00               | 3596.53     | 68.24 |
| K.B.U.N.L      | 3000.00               | 2906.20     | 84.83 |
| MUNPL          | 7700.00               | 8275.82     | 71.37 |
| NEEPCO         | 3400.00               | 2997.19     |       |
| NLC            | 22300.00              | 19547.49    | 61.14 |
| NPGCL          | 14200.00              | 14411.96    | 82.86 |
| NSPCL          | 3700.00               | 3875.74     | 88.25 |
| NTECL          | 9500.00               | 6623.27     | 50.27 |
| NTPC Ltd.      | 368240.00             | 339679.40   | 76.84 |
| NTPL           | 7000.00               | 5462.36     | 62.19 |
| NUPPL          | 3300.00               | 0.00        | 0.00  |
| ONGC           | 4800.00               | 4573.95     |       |
| PVUNL          | 0.00                  | 0.00        | 0.00  |

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|                       |                   |                   |              |
|-----------------------|-------------------|-------------------|--------------|
| RGPPL                 | 1050.00           | 1220.38           |              |
| SJVNL                 | 2000.00           | 0.00              | 0.00         |
| THDC                  | 200.00            | 0.00              | 0.00         |
| <b>CENTRAL SECTOR</b> | <b>512190.00</b>  | <b>472527.51</b>  | <b>75.09</b> |
| <b>STATE SECTOR</b>   | <b>419884.00</b>  | <b>403929.70</b>  | <b>64.70</b> |
| <b>PVT SECTOR</b>     | <b>17051.00</b>   | <b>18414.01</b>   | <b>64.17</b> |
| <b>IPP SECTOR</b>     | <b>374985.00</b>  | <b>431677.71</b>  | <b>67.63</b> |
| <b>TOTAL THERMAL</b>  | <b>1324110.00</b> | <b>1326548.93</b> | <b>69.09</b> |
| <b>NUCLEAR</b>        |                   |                   |              |
| <b>CENTRAL SECTOR</b> |                   |                   |              |
| BHAVINI               | 0.00              | 0.00              | 0.00         |
| DAE                   | 0.00              | 0.00              | 0.00         |
| NPCIL                 | 46190.00          | 47937.41          | 75.73        |
| <b>CENTRAL SECTOR</b> | <b>46190.00</b>   | <b>47937.41</b>   | <b>74.70</b> |
| <b>TOTAL NUCLEAR</b>  | <b>46190.00</b>   | <b>47937.41</b>   | <b>74.7</b>  |
| <b>HYDRO</b>          |                   |                   |              |
| <b>CENTRAL SECTOR</b> |                   |                   |              |
| BBMB                  | 9700.00           | 11583.17          |              |
| DVC                   | 286.00            | 176.28            |              |
| NEEPCO.               | 4937.00           | 4848.09           |              |
| NHDC                  | 3265.00           | 4470.50           |              |
| NHPC                  | 5429.00           | 0.00              |              |
| NHPC                  | 25628.00          | 21585.84          |              |
| NTPC Ltd.             | 3100.00           | 2952.05           |              |
| SJVNL                 | 9080.00           | 8088.21           |              |
| THDC                  | 4160.00           | 4441.69           |              |
| <b>CENTRAL SECTOR</b> | <b>65585.00</b>   | <b>58145.83</b>   |              |

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|                 |            |            |  |
|-----------------|------------|------------|--|
| STATE SECTOR    | 75469.00   | 61667.64   |  |
| PVT SECTOR      | 1470.00    | 1557.60    |  |
| IPP SECTOR      | 14176.00   | 12682.85   |  |
| TOTAL HYDRO     | 156700.00  | 134053.92  |  |
| BHUTAN IMP      |            |            |  |
|                 | 8000.00    | 4716.10    |  |
| ALL INDIA TOTAL | 1535000.00 | 1513256.36 |  |

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## Annexure-10B

| ALL INDIA INSTALLED CAPACITY (IN MW) OF POWER STATIONS<br>LOCATED IN THE REGIONS OF MAIN LAND AND ISLANDS |                      |                   |         |          |        |          |         |           |            |           |                |
|-----------------------------------------------------------------------------------------------------------|----------------------|-------------------|---------|----------|--------|----------|---------|-----------|------------|-----------|----------------|
| (As on 31.03.2024)                                                                                        |                      |                   |         |          |        |          |         |           |            |           |                |
| (UTILITIES)                                                                                               |                      |                   |         |          |        |          |         |           |            |           |                |
| Region                                                                                                    | Ownership/<br>Sector | Mode wise breakup |         |          |        |          |         |           |            |           | Grand<br>Total |
|                                                                                                           |                      | Thermal           |         |          |        |          | Nuclear | Renewable |            |           |                |
|                                                                                                           |                      | Coal              | Lignite | Gas      | Diesel | Total    |         | Hydro     | RES*(MNRE) | Total     |                |
| Northern<br>Region                                                                                        | State                | 18985.00          | 250.00  | 2878.90  | 0.00   | 22113.90 | 0.00    | 6008.25   | 747.70     | 6755.95   | 28869.85       |
|                                                                                                           | Private              | 22624.33          | 1080.00 | 772.00   | 0.00   | 24476.33 | 0.00    | 3241.00   | 37366.22   | 40607.22  | 65083.55       |
|                                                                                                           | Central              | 15341.62          | 250.00  | 2344.06  | 0.00   | 17935.68 | 1620.00 | 11580.51  | 379.00     | 11959.51  | 31515.19       |
|                                                                                                           | Sub Total            | 56950.95          | 1580.00 | 5994.96  | 0.00   | 64525.91 | 1620.00 | 20829.76  | 38492.92   | 59322.68  | 125468.59      |
| Western<br>Region                                                                                         | State                | 21290.00          | 900.00  | 2849.82  | 0.00   | 25039.82 | 0.00    | 5446.50   | 602.23     | 6048.73   | 31088.55       |
|                                                                                                           | Private              | 31762.17          | 500.00  | 4676.00  | 0.00   | 36938.17 | 0.00    | 481.00    | 47440.01   | 47921.01  | 84859.18       |
|                                                                                                           | Central              | 21610.42          | 0.00    | 3280.67  | 0.00   | 24891.09 | 3240.00 | 1635.00   | 666.30     | 2301.30   | 30432.39       |
|                                                                                                           | Sub Total            | 74662.59          | 1400.00 | 10806.49 | 0.00   | 86869.08 | 3240.00 | 7562.50   | 48708.54   | 56271.04  | 146380.12      |
| Southern<br>Region                                                                                        | State                | 22192.50          | 0.00    | 791.98   | 159.96 | 23144.44 | 0.00    | 11827.48  | 633.08     | 12460.56  | 35605.00       |
|                                                                                                           | Private              | 13572.50          | 250.00  | 5340.24  | 273.70 | 19436.45 | 0.00    | 0.00      | 52643.84   | 52643.84  | 72080.29       |
|                                                                                                           | Central              | 14129.40          | 3390.00 | 359.58   | 0.00   | 17878.98 | 3320.00 | 0.00      | 541.90     | 541.90    | 21740.88       |
|                                                                                                           | Sub Total            | 49894.40          | 3640.00 | 6491.80  | 433.66 | 60459.86 | 3320.00 | 11827.48  | 53818.82   | 65646.30  | 129426.16      |
| Eastern<br>Region                                                                                         | State                | 6970.00           | 0.00    | 80.00    | 0.00   | 7050.00  | 0.00    | 3550.22   | 278.11     | 3828.33   | 10878.33       |
|                                                                                                           | Private              | 5553.00           | 0.00    | 0.00     | 0.00   | 5553.00  | 0.00    | 209.00    | 1721.15    | 1930.15   | 7483.15        |
|                                                                                                           | Central              | 15989.50          | 0.00    | 0.00     | 0.00   | 15989.50 | 0.00    | 1005.20   | 10.00      | 1015.20   | 17004.70       |
|                                                                                                           | Sub Total            | 28512.50          | 0.00    | 80.00    | 0.00   | 28592.50 | 0.00    | 4764.42   | 2009.26    | 6773.68   | 35366.18       |
| North Eastern<br>Region                                                                                   | State                | 0.00              | 0.00    | 411.36   | 36.00  | 447.36   | 0.00    | 422.00    | 268.75     | 690.75    | 1138.10        |
|                                                                                                           | Private              | 0.00              | 0.00    | 0.00     | 0.00   | 0.00     | 0.00    | 0.00      | 276.09     | 276.09    | 276.09         |
|                                                                                                           | Central              | 949.02            | 0.00    | 1253.60  | 0.00   | 2202.62  | 0.00    | 1522.01   | 30.00      | 1552.01   | 3754.63        |
|                                                                                                           | Sub Total            | 949.02            | 0.00    | 1664.96  | 36.00  | 2649.98  | 0.00    | 1944.01   | 574.84     | 2518.85   | 5168.82        |
| Islands                                                                                                   | State                | 0.00              | 0.00    | 0.00     | 84.35  | 84.35    | 0.00    | 0.00      | 5.25       | 5.25      | 89.60          |
|                                                                                                           | Private              | 0.00              | 0.00    | 0.00     | 35.19  | 35.19    | 0.00    | 0.00      | 29.78      | 29.78     | 64.97          |
|                                                                                                           | Central              | 0.00              | 0.00    | 0.00     | 0.00   | 0.00     | 0.00    | 0.00      | 5.10       | 5.10      | 5.10           |
|                                                                                                           | Sub Total            | 0.00              | 0.00    | 0.00     | 119.54 | 119.54   | 0.00    | 0.00      | 40.13      | 40.13     | 159.67         |
| ALL INDIA                                                                                                 | State                | 69437.50          | 1150.00 | 7012.06  | 280.31 | 77879.87 | 0.00    | 27254.45  | 2535.11    | 29789.56  | 107669.43      |
|                                                                                                           | Private              | 73512.00          | 1830.00 | 10788.24 | 308.89 | 86439.14 | 0.00    | 3931.00   | 139477.09  | 143408.09 | 229847.23      |

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|                                                      |                                                                                                                       |            |                 |          |             |           |                |           |           |           |           |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------|-----------------|----------|-------------|-----------|----------------|-----------|-----------|-----------|-----------|
|                                                      | Central                                                                                                               | 68019.96   | 3640.00         | 7237.91  | 0.00        | 78897.87  | 8180.00        | 15742.72  | 1632.30   | 17375.02  | 104452.89 |
|                                                      | Total                                                                                                                 | 210969.46  | 6620.00         | 25038.21 | 589.20      | 243216.87 | 8180.00        | 46928.17  | 143644.51 | 190572.68 | 441969.55 |
| Figures at decimal may not tally due to rounding off |                                                                                                                       |            |                 |          |             |           |                |           |           |           |           |
| Abbreviation:-                                       | SHP=Small Hydro Project (≤ 25 MW), BP=Biomass Power, U&I=Urban & Industrial Waste Power, RES=Renewable Energy Sources |            |                 |          |             |           |                |           |           |           |           |
| Note : -                                             | 1. RES include SHP, BP, U&I, Solar and Wind Energy. Installed capacity in respect of RES (MNRE) as on 31.03.2023      |            |                 |          |             |           |                |           |           |           |           |
|                                                      | (As per latest information available with MNRE)                                                                       |            |                 |          |             |           |                |           |           |           |           |
|                                                      | *Break up of RES all India as on 31.03.2024 is given below (in MW) :                                                  |            |                 |          |             |           |                |           |           |           |           |
|                                                      | Small Hydro Power                                                                                                     | Wind Power | Bio-Power       |          | Solar Power |           | Total Capacity |           |           |           |           |
|                                                      |                                                                                                                       |            | BM Power/Cogen. |          |             |           |                |           |           |           |           |
|                                                      | 5003.25                                                                                                               | 45886.51   | 10355.35        |          | 585.80      |           | 81813.60       | 143644.51 |           |           |           |
|                                                      |                                                                                                                       |            |                 |          |             |           |                |           |           |           |           |
| A.                                                   | Capacity during                                                                                                       | Added      | March., 2024    |          | 700 MW      |           |                |           |           |           |           |
|                                                      | 1. NPCIL's Kakrapar Atomic Power Project Unit-4 of 700 MW has been commissioned on 31.03.2024..                       |            |                 |          |             |           |                |           |           |           |           |
|                                                      |                                                                                                                       |            |                 |          |             |           |                |           |           |           |           |
| B.                                                   | Capacity during                                                                                                       | Retired    | March., 2024    |          | 0 MW        |           |                |           |           |           |           |
|                                                      |                                                                                                                       |            |                 |          |             |           |                |           |           |           |           |
|                                                      |                                                                                                                       |            |                 |          |             |           |                |           |           |           |           |
| C.                                                   | Net Conv. Capacity Added during                                                                                       |            | March., 2024    | A-B      | 700 MW      |           |                |           |           |           |           |
|                                                      |                                                                                                                       |            |                 |          |             |           |                |           |           |           |           |
| D.                                                   | Net RES Capacity Added during                                                                                         |            | March., 2024    |          | 7074.42 MW  |           |                |           |           |           |           |
|                                                      |                                                                                                                       |            |                 |          |             |           |                |           |           |           |           |
| E.                                                   | Net Capacity Added during                                                                                             |            | March., 2024    | C+D      | 7774.42 MW  |           |                |           |           |           |           |
|                                                      |                                                                                                                       |            |                 |          |             |           |                |           |           |           |           |
| *                                                    | Off-grid RES Capacity has been included from July-2021 onwards                                                        |            |                 |          |             |           |                |           |           |           |           |
|                                                      | Sector wise breakup of RES capacity as shown is provisional.                                                          |            |                 |          |             |           |                |           |           |           |           |
|                                                      | Allocation from central sector stations has been updated till 29.02.2024.                                             |            |                 |          |             |           |                |           |           |           |           |
|                                                      | Share from private sector generating stations has been updated as per latest information available.                   |            |                 |          |             |           |                |           |           |           |           |
|                                                      | NPCIL's Kakrapar Atomic Power Project Unit-4 of 700 MW has been commissioned on 31.03.2024.                           |            |                 |          |             |           |                |           |           |           |           |

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| INSTALLED CAPACITY (IN MW) OF POWER UTILITIES IN THE STATES/UTS LOCATED IN NORTHERN REGION |                      |                   |         |         |        |         |         |           |                    |          |                |
|--------------------------------------------------------------------------------------------|----------------------|-------------------|---------|---------|--------|---------|---------|-----------|--------------------|----------|----------------|
| INCLUDING ALLOCATED SHARES IN JOINT & CENTRAL SECTOR UTILITIES                             |                      |                   |         |         |        |         |         |           |                    |          |                |
|                                                                                            |                      |                   |         |         |        |         |         |           | (As on 31.03.2024) |          |                |
| State                                                                                      | Ownership/<br>Sector | Mode wise breakup |         |         |        |         |         |           |                    |          | Grand<br>Total |
|                                                                                            |                      | Thermal           |         |         |        |         | Nuclear | Renewable |                    |          |                |
|                                                                                            |                      | Coal              | Lignite | Gas     | Diesel | Total   |         | Hydro     | RES*(MNRE)         | Total    |                |
| Delhi                                                                                      | State                | 0.00              | 0.00    | 1800.40 | 0.00   | 1800.40 | 0.00    | 0.00      | 0.00               | 0.00     | 1800.40        |
|                                                                                            | Private              | 878.22            | 0.00    | 108.00  | 0.00   | 986.22  | 0.00    | 0.00      | 340.51             | 340.51   | 1326.73        |
|                                                                                            | Central              | 2771.28           | 0.00    | 207.01  | 0.00   | 2978.29 | 102.83  | 723.09    | 0.00               | 723.09   | 3804.21        |
|                                                                                            | Sub-Total            | 3649.50           | 0.00    | 2115.41 | 0.00   | 5764.91 | 102.83  | 723.09    | 340.51             | 1063.60  | 6931.34        |
| Haryana                                                                                    | State                | 2510.00           | 0.00    | 150.00  | 0.00   | 2660.00 | 0.00    | 200.00    | 69.30              | 269.30   | 2929.30        |
|                                                                                            | Private              | 4561.78           | 0.00    | 0.00    | 0.00   | 4561.78 | 0.00    | 539.00    | 1758.62            | 2297.62  | 6859.40        |
|                                                                                            | Central              | 1566.61           | 0.00    | 431.59  | 0.00   | 1998.20 | 100.94  | 1591.73   | 5.00               | 1596.73  | 3695.86        |
|                                                                                            | Sub-Total            | 8638.39           | 0.00    | 581.59  | 0.00   | 9219.98 | 100.94  | 2330.73   | 1832.92            | 4163.65  | 13484.56       |
| Himachal Pradesh                                                                           | State                | 0.00              | 0.00    | 0.00    | 0.00   | 0.00    | 0.00    | 805.60    | 256.61             | 1062.21  | 1062.21        |
|                                                                                            | Private              | 0.00              | 0.00    | 0.00    | 0.00   | 0.00    | 0.00    | 1219.40   | 818.53             | 2037.93  | 2037.93        |
|                                                                                            | Central              | 144.67            | 0.00    | 0.00    | 0.00   | 144.67  | 28.95   | 1223.88   | 0.00               | 1223.88  | 1397.50        |
|                                                                                            | Sub-Total            | 144.67            | 0.00    | 0.00    | 0.00   | 144.67  | 28.95   | 3248.88   | 1075.14            | 4324.02  | 4497.64        |
| Jammu & Kashmir and Ladakh                                                                 | State                | 0.00              | 0.00    | 175.00  | 0.00   | 175.00  | 0.00    | 1230.00   | 148.67             | 1378.67  | 1553.67        |
|                                                                                            | Private              | 0.00              | 0.00    | 0.00    | 0.00   | 0.00    | 0.00    | 0.00      | 137.49             | 137.49   | 137.49         |
|                                                                                            | Central              | 870.13            | 0.00    | 129.07  | 0.00   | 999.21  | 67.98   | 1091.88   | 0.00               | 1091.88  | 2159.07        |
|                                                                                            | Sub-Total            | 870.13            | 0.00    | 304.07  | 0.00   | 1174.21 | 67.98   | 2321.88   | 286.16             | 2608.04  | 3850.23        |
| Punjab                                                                                     | State                | 1760.00           | 0.00    | 150.00  | 0.00   | 1910.00 | 0.00    | 1243.40   | 127.80             | 1371.20  | 3281.20        |
|                                                                                            | Private              | 5014.00           | 0.00    | 0.00    | 0.00   | 5014.00 | 0.00    | 288.00    | 1939.82            | 2227.82  | 7241.82        |
|                                                                                            | Central              | 1440.00           | 0.00    | 0.00    | 0.00   | 1440.00 | 196.81  | 2296.04   | 0.00               | 2296.04  | 3932.85        |
|                                                                                            | Sub-Total            | 8214.00           | 0.00    | 150.00  | 0.00   | 8364.00 | 196.81  | 3827.44   | 2067.62            | 5895.06  | 14455.87       |
| Rajasthan                                                                                  | State                | 7580.00           | 250.00  | 603.50  | 0.00   | 8433.50 | 0.00    | 433.00    | 23.85              | 456.85   | 8890.35        |
|                                                                                            | Private              | 3257.00           | 1080.00 | 0.00    | 0.00   | 4337.00 | 0.00    | 104.00    | 26325.04           | 26429.04 | 30766.04       |
|                                                                                            | Central              | 1031.56           | 250.00  | 171.13  | 0.00   | 1452.69 | 556.74  | 1407.67   | 344.00             | 1751.67  | 3761.10        |

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|                                |                    |                 |                |                |             |                 |                |                 |                 |                 |                  |
|--------------------------------|--------------------|-----------------|----------------|----------------|-------------|-----------------|----------------|-----------------|-----------------|-----------------|------------------|
|                                | <b>Sub-Total</b>   | <b>11868.56</b> | <b>1580.00</b> | <b>774.63</b>  | <b>0.00</b> | <b>14223.19</b> | <b>556.74</b>  | <b>1944.67</b>  | <b>26692.89</b> | <b>28637.56</b> | <b>43417.49</b>  |
| <b>Uttar Pradesh</b>           | State              | 7135.00         | 0.00           | 0.00           | 0.00        | 7135.00         | 0.00           | 724.10          | 49.10           | 773.20          | 7908.20          |
|                                | Private            | 8814.33         | 0.00           | 0.00           | 0.00        | 8814.33         | 0.00           | 842.40          | 5116.47         | 5958.87         | 14773.20         |
|                                | Central            | 5538.42         | 0.00           | 1029.51        | 0.00        | 6567.93         | 289.48         | 1857.52         | 30.00           | 1887.52         | 8744.93          |
|                                | <b>Sub-Total</b>   | <b>21487.75</b> | <b>0.00</b>    | <b>1029.51</b> | <b>0.00</b> | <b>22517.26</b> | <b>289.48</b>  | <b>3424.02</b>  | <b>5195.57</b>  | <b>8619.59</b>  | <b>31426.33</b>  |
| <b>Uttarakhand</b>             | State              | 0.00            | 0.00           | 0.00           | 0.00        | 0.00            | 0.00           | 1372.15         | 72.37           | 1444.52         | 1444.52          |
|                                | Private            | 99.00           | 0.00           | 664.00         | 0.00        | 763.00          | 0.00           | 248.20          | 864.22          | 1112.42         | 1875.42          |
|                                | Central            | 503.10          | 0.00           | 69.66          | 0.00        | 572.76          | 31.24          | 535.54          | 0.00            | 535.54          | 1139.54          |
|                                | <b>Sub-Total</b>   | <b>602.10</b>   | <b>0.00</b>    | <b>733.66</b>  | <b>0.00</b> | <b>1335.76</b>  | <b>31.24</b>   | <b>2155.89</b>  | <b>936.59</b>   | <b>3092.48</b>  | <b>4459.48</b>   |
| <b>Chandigarh</b>              | State              | 0.00            | 0.00           | 0.00           | 0.00        | 0.00            | 0.00           | 0.00            | 0.00            | 0.00            | 0.00             |
|                                | Private            | 0.00            | 0.00           | 0.00           | 0.00        | 0.00            | 0.00           | 0.00            | 65.52           | 65.52           | 65.52            |
|                                | Central            | 44.83           | 0.00           | 15.03          | 0.00        | 59.86           | 8.01           | 101.71          | 0.00            | 101.71          | 169.57           |
|                                | <b>Sub-Total</b>   | <b>44.83</b>    | <b>0.00</b>    | <b>15.03</b>   | <b>0.00</b> | <b>59.86</b>    | <b>8.01</b>    | <b>101.71</b>   | <b>65.52</b>    | <b>167.23</b>   | <b>235.09</b>    |
| <b>Central - Unallocated</b>   |                    | 1431.03         | 1431.03        | 0.00           | 291.05      | 0.00            | 1722.08        | 237.03          | 751.45          | 0.00            | 751.45           |
| <b>Total (Northern Region)</b> | State              | 18985.00        | 250.00         | 2878.90        | 0.00        | 22113.90        | 0.00           | 6008.25         | 747.70          | 6755.95         | 28869.85         |
|                                | Private            | 22624.33        | 1080.00        | 772.00         | 0.00        | 24476.33        | 0.00           | 3241.00         | 37366.22        | 40607.22        | 65083.55         |
|                                | Central            | 15341.62        | 250.00         | 2344.06        | 0.00        | 17935.68        | 1620.00        | 11580.51        | 379.00          | 11959.51        | 31515.19         |
|                                | <b>Grand Total</b> | <b>56950.95</b> | <b>1580.00</b> | <b>5994.96</b> | <b>0.00</b> | <b>64525.91</b> | <b>1620.00</b> | <b>20829.76</b> | <b>38492.92</b> | <b>59322.68</b> | <b>125468.59</b> |
|                                |                    |                 |                |                |             |                 |                |                 |                 |                 |                  |
|                                |                    |                 |                |                |             |                 |                |                 |                 |                 |                  |

### INSTALLED CAPACITY (IN MW) OF POWER UTILITIES IN THE STATES/UTS LOCATED IN WESTERN REGION

### INCLUDING ALLOCATED SHARES IN JOINT & CENTRAL SECTOR UTILITIES

| (As on 31.03.2024) |                      |                   |         |         |        |         |         |           |            |        |                |
|--------------------|----------------------|-------------------|---------|---------|--------|---------|---------|-----------|------------|--------|----------------|
| State              | Ownership/<br>Sector | Mode wise breakup |         |         |        |         |         |           |            |        | Grand<br>Total |
|                    |                      | Thermal           |         |         |        |         | Nuclear | Renewable |            |        |                |
|                    |                      | Coal              | Lignite | Gas     | Diesel | Total   |         | Hydro     | RES*(MNRE) | Total  |                |
| Goa                | State                | 0.00              | 0.00    | 0.00    | 0.00   | 0.00    | 0.00    | 0.00      | 0.05       | 0.05   | 0.05           |
|                    | Private              | 0.00              | 0.00    | 48.00   | 0.00   | 48.00   | 0.00    | 0.00      | 45.42      | 45.42  | 93.42          |
|                    | Central              | 492.29            | 0.00    | 19.67   | 0.00   | 511.96  | 41.68   | 2.00      | 0.00       | 2.00   | 555.64         |
|                    | Sub-Total            | 492.29            | 0.00    | 67.67   | 0.00   | 559.96  | 41.68   | 2.00      | 45.47      | 47.47  | 649.11         |
| Gujarat            | State                | 4510.00           | 900.00  | 2177.82 | 0.00   | 7587.82 | 0.00    | 772.00    | 95.04      | 867.04 | 8454.86        |

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**INCLUDING ALLOCATED SHARES IN JOINT & CENTRAL SECTOR UTILITIES**

|  |                    |
|--|--------------------|
|  | (As on 31.03.2024) |
|--|--------------------|

| State | Ownership/<br>Sector | Mode wise breakup |         |           | Grand<br>Total |
|-------|----------------------|-------------------|---------|-----------|----------------|
|       |                      | Thermal           | Nuclear | Renewable |                |

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|                                |                    | Coal     | Lignite | Gas     | Diesel | Total    |         | Hydro    | RES*(MNRE) | Total    |           |
|--------------------------------|--------------------|----------|---------|---------|--------|----------|---------|----------|------------|----------|-----------|
| <b>Andhra Pradesh</b>          | State              | 6610.00  | 0.00    | 235.40  | 0.00   | 6845.40  | 0.00    | 1673.60  | 57.38      | 1730.98  | 8576.38   |
|                                | Private            | 3873.88  | 0.00    | 3831.32 | 36.80  | 7742.00  | 0.00    | 0.00     | 9111.95    | 9111.95  | 16853.96  |
|                                | Central            | 1546.95  | 189.34  | 0.00    | 0.00   | 1736.29  | 127.27  | 0.00     | 250.00     | 250.00   | 2113.56   |
|                                | <b>Sub-Total</b>   | 12030.83 | 189.34  | 4066.72 | 36.80  | 16323.69 | 127.27  | 1673.60  | 9419.33    | 11092.93 | 27543.89  |
| <b>Telangana</b>               | State              | 6242.50  | 0.00    | 0.00    | 0.00   | 6242.50  | 0.00    | 2479.93  | 41.22      | 2521.15  | 8763.65   |
|                                | Private            | 1389.45  | 0.00    | 831.82  | 0.00   | 2221.27  | 0.00    | 0.00     | 5147.58    | 5147.58  | 7368.85   |
|                                | Central            | 3166.85  | 61.30   | 0.00    | 0.00   | 3228.15  | 148.73  | 0.00     | 10.00      | 10.00    | 3386.88   |
|                                | <b>Sub-Total</b>   | 10798.80 | 61.30   | 831.82  | 0.00   | 11691.92 | 148.73  | 2479.93  | 5198.80    | 7678.73  | 19519.38  |
| <b>Karnataka</b>               | State              | 5020.00  | 0.00    | 0.00    | 0.00   | 5020.00  | 0.00    | 3631.60  | 193.89     | 3825.49  | 8845.49   |
|                                | Private            | 2050.00  | 0.00    | 0.00    | 25.20  | 2075.20  | 0.00    | 0.00     | 17558.86   | 17558.86 | 19634.06  |
|                                | Central            | 3180.16  | 486.42  | 0.00    | 0.00   | 3666.58  | 698.00  | 0.00     | 0.00       | 0.00     | 4364.58   |
|                                | <b>Sub-Total</b>   | 10250.16 | 486.42  | 0.00    | 25.20  | 10761.78 | 698.00  | 3631.60  | 17752.74   | 21384.34 | 32844.12  |
| <b>Kerala</b>                  | State              | 0.00     | 0.00    | 0.00    | 159.96 | 159.96   | 0.00    | 1864.15  | 217.90     | 2082.05  | 2242.01   |
|                                | Private            | 832.50   | 0.00    | 174.00  | 0.00   | 1006.50  | 0.00    | 0.00     | 1097.41    | 1097.41  | 2103.91   |
|                                | Central            | 1226.30  | 325.33  | 359.58  | 0.00   | 1911.21  | 362.00  | 0.00     | 50.00      | 50.00    | 2323.21   |
|                                | <b>Sub-Total</b>   | 2058.80  | 325.33  | 533.58  | 159.96 | 3077.67  | 362.00  | 1864.15  | 1365.31    | 3229.46  | 6669.13   |
| <b>Tamil Nadu</b>              | State              | 4320.00  | 0.00    | 524.08  | 0.00   | 4844.08  | 0.00    | 2178.20  | 122.70     | 2300.90  | 7144.98   |
|                                | Private            | 5426.67  | 250.00  | 503.10  | 211.70 | 6391.47  | 0.00    | 0.00     | 19678.13   | 19678.13 | 26069.60  |
|                                | Central            | 3202.33  | 1709.16 | 0.00    | 0.00   | 4911.49  | 1448.00 | 0.00     | 231.90     | 231.90   | 6591.39   |
|                                | <b>Sub-Total</b>   | 12949.00 | 1959.16 | 1027.18 | 211.70 | 16147.04 | 1448.00 | 2178.20  | 20032.73   | 22210.93 | 39805.97  |
| <b>NLC</b>                     | State              | 0.00     | 0.00    | 0.00    | 0.00   | 0.00     | 0.00    | 0.00     | 0.00       | 0.00     | 0.00      |
|                                | Private            | 0.00     | 0.00    | 0.00    | 0.00   | 0.00     | 0.00    | 0.00     | 0.00       | 0.00     | 0.00      |
|                                | Central            | 0.00     | 66.00   | 0.00    | 0.00   | 66.00    | 0.00    | 0.00     | 0.00       | 0.00     | 66.00     |
|                                | <b>Sub-Total</b>   | 0.00     | 66.00   | 0.00    | 0.00   | 66.00    | 0.00    | 0.00     | 0.00       | 0.00     | 66.00     |
| <b>Puducherry</b>              | State              | 0.00     | 0.00    | 32.50   | 0.00   | 32.50    | 0.00    | 0.00     | 0.00       | 0.00     | 32.50     |
|                                | Private            | 0.00     | 0.00    | 0.00    | 0.00   | 0.00     | 0.00    | 0.00     | 49.91      | 49.91    | 49.91     |
|                                | Central            | 140.80   | 118.35  | 0.00    | 0.00   | 259.15   | 86.00   | 0.00     | 0.00       | 0.00     | 345.15    |
|                                | <b>Sub-Total</b>   | 140.80   | 118.35  | 32.50   | 0.00   | 291.65   | 86.00   | 0.00     | 49.91      | 49.91    | 427.56    |
| <b>Central - Unallocated</b>   |                    | 1426.00  | 1666.00 | 434.10  | 0.00   | 0.00     | 2100.10 | 450.00   | 0.00       | 0.00     | 0.00      |
| <b>Total (Southern Region)</b> | State              | 22192.50 | 0.00    | 791.98  | 159.96 | 23144.44 | 0.00    | 11827.48 | 633.08     | 12460.56 | 35605.00  |
|                                | Private            | 13572.50 | 250.00  | 5340.24 | 273.70 | 19436.45 | 0.00    | 0.00     | 52643.84   | 52643.84 | 72080.29  |
|                                | Central            | 14129.40 | 3390.00 | 359.58  | 0.00   | 17878.98 | 3320.00 | 0.00     | 541.90     | 541.90   | 21740.88  |
|                                | <b>Grand Total</b> | 49894.40 | 3640.00 | 6491.80 | 433.66 | 60459.86 | 3320.00 | 11827.48 | 53818.82   | 65646.30 | 129426.16 |

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| INSTALLED CAPACITY (IN MW) OF POWER UTILITIES IN THE STATES/UTS LOCATED IN EASTERN REGION |                      |                   |         |       |        |         |         |           |                    |         |                |
|-------------------------------------------------------------------------------------------|----------------------|-------------------|---------|-------|--------|---------|---------|-----------|--------------------|---------|----------------|
| INCLUDING ALLOCATED SHARES IN JOINT & CENTRAL SECTOR UTILITIES                            |                      |                   |         |       |        |         |         |           |                    |         |                |
|                                                                                           |                      |                   |         |       |        |         |         |           | (As on 31.03.2024) |         |                |
| State                                                                                     | Ownership/<br>Sector | Mode wise breakup |         |       |        |         |         |           |                    |         | Grand<br>Total |
|                                                                                           |                      | Thermal           |         |       |        |         | Nuclear | Renewable |                    |         |                |
|                                                                                           |                      | Coal              | Lignite | Gas   | Diesel | Total   |         | Hydro     | RES*(MNRE)         | Total   |                |
| Bihar                                                                                     | State                | 0.00              | 0.00    | 0.00  | 0.00   | 0.00    | 0.00    | 0.00      | 70.70              | 70.70   | 70.70          |
|                                                                                           | Private              | 700.00            | 0.00    | 0.00  | 0.00   | 700.00  | 0.00    | 0.00      | 379.45             | 379.45  | 1079.45        |
|                                                                                           | Central              | 6253.75           | 0.00    | 0.00  | 0.00   | 6253.75 | 0.00    | 110.00    | 0.00               | 110.00  | 6363.75        |
|                                                                                           | Sub-Total            | 6953.75           | 0.00    | 0.00  | 0.00   | 6953.75 | 0.00    | 110.00    | 450.15             | 560.15  | 7513.90        |
| Jharkhand                                                                                 | State                | 420.00            | 0.00    | 0.00  | 0.00   | 420.00  | 0.00    | 130.00    | 4.05               | 134.05  | 554.05         |
|                                                                                           | Private              | 580.00            | 0.00    | 0.00  | 0.00   | 580.00  | 0.00    | 0.00      | 181.50             | 181.50  | 761.50         |
|                                                                                           | Central              | 1607.31           | 0.00    | 0.00  | 0.00   | 1607.31 | 0.00    | 61.00     | 0.00               | 61.00   | 1668.31        |
|                                                                                           | Sub-Total            | 2607.31           | 0.00    | 0.00  | 0.00   | 2607.31 | 0.00    | 191.00    | 185.55             | 376.55  | 2983.86        |
| West Bengal                                                                               | State                | 4810.00           | 0.00    | 80.00 | 0.00   | 4890.00 | 0.00    | 986.00    | 121.95             | 1107.95 | 5997.95        |
|                                                                                           | Private              | 2437.00           | 0.00    | 0.00  | 0.00   | 2437.00 | 0.00    | 0.00      | 518.98             | 518.98  | 2955.98        |
|                                                                                           | Central              | 1436.34           | 0.00    | 0.00  | 0.00   | 1436.34 | 0.00    | 410.00    | 0.00               | 410.00  | 1846.34        |
|                                                                                           | Sub-Total            | 8683.34           | 0.00    | 80.00 | 0.00   | 8763.34 | 0.00    | 1396.00   | 640.93             | 2036.93 | 10800.27       |
| DVC                                                                                       | State                | 0.00              | 0.00    | 0.00  | 0.00   | 0.00    | 0.00    | 0.00      | 0.00               | 0.00    | 0.00           |
|                                                                                           | Private              | 150.00            | 0.00    | 0.00  | 0.00   | 150.00  | 0.00    | 0.00      | 0.00               | 0.00    | 150.00         |
|                                                                                           | Central              | 2887.02           | 0.00    | 0.00  | 0.00   | 2887.02 | 0.00    | 186.20    | 0.00               | 186.20  | 3073.21        |
|                                                                                           | Sub-Total            | 3037.02           | 0.00    | 0.00  | 0.00   | 3037.02 | 0.00    | 186.20    | 0.00               | 186.20  | 3223.21        |
| Odisha                                                                                    | State                | 1740.00           | 0.00    | 0.00  | 0.00   | 1740.00 | 0.00    | 2074.22   | 26.30              | 2100.52 | 3840.52        |
|                                                                                           | Private              | 1686.00           | 0.00    | 0.00  | 0.00   | 1686.00 | 0.00    | 0.00      | 634.18             | 634.18  | 2320.18        |
|                                                                                           | Central              | 1865.21           | 0.00    | 0.00  | 0.00   | 1865.21 | 0.00    | 89.00     | 10.00              | 99.00   | 1964.21        |
|                                                                                           | Sub-Total            | 5291.21           | 0.00    | 0.00  | 0.00   | 5291.21 | 0.00    | 2163.22   | 670.48             | 2833.70 | 8124.91        |
| Sikkim                                                                                    | State                | 0.00              | 0.00    | 0.00  | 0.00   | 0.00    | 0.00    | 360.00    | 55.11              | 415.11  | 415.11         |
|                                                                                           | Private              | 0.00              | 0.00    | 0.00  | 0.00   | 0.00    | 0.00    | 209.00    | 7.04               | 216.04  | 216.04         |
|                                                                                           | Central              | 103.54            | 0.00    | 0.00  | 0.00   | 103.54  | 0.00    | 64.00     | 0.00               | 64.00   | 167.54         |
|                                                                                           | Sub-Total            | 103.54            | 0.00    | 0.00  | 0.00   | 103.54  | 0.00    | 633.00    | 62.15              | 695.15  | 798.69         |

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|                               |                    |                 |             |              |             |                 |             |                |                |                |                 |
|-------------------------------|--------------------|-----------------|-------------|--------------|-------------|-----------------|-------------|----------------|----------------|----------------|-----------------|
| <b>Central - Unallocated</b>  |                    | 1638.33         | 1836.33     | 0.00         | 0.00        | 0.00            | 1836.33     | 0.00           | 85.01          | 0.00           | 85.01           |
| <b>Total (Eastern Region)</b> | State              | 6970.00         | 0.00        | 80.00        | 0.00        | 7050.00         | 0.00        | 3550.22        | 278.11         | 3828.33        | 10878.33        |
|                               | Private            | 5553.00         | 0.00        | 0.00         | 0.00        | 5553.00         | 0.00        | 209.00         | 1721.15        | 1930.15        | 7483.15         |
|                               | Central            | 15989.50        | 0.00        | 0.00         | 0.00        | 15989.50        | 0.00        | 1005.20        | 10.00          | 1015.20        | 17004.70        |
|                               | <b>Grand Total</b> | <b>28512.50</b> | <b>0.00</b> | <b>80.00</b> | <b>0.00</b> | <b>28592.50</b> | <b>0.00</b> | <b>4764.42</b> | <b>2009.26</b> | <b>6773.68</b> | <b>35366.18</b> |
|                               |                    |                 |             |              |             |                 |             |                |                |                |                 |
|                               |                    |                 |             |              |             |                 |             |                |                |                |                 |

**INSTALLED CAPACITY (IN MW) OF POWER UTILITIES IN THE STATES/UTS LOCATED IN NORTH-EASTERN REGION**

**INCLUDING ALLOCATED SHARES IN JOINT & CENTRAL SECTOR UTILITIES**

| (As on 31.03.2024) |                      |                   |         |        |        |         |         |           |            |        |                |
|--------------------|----------------------|-------------------|---------|--------|--------|---------|---------|-----------|------------|--------|----------------|
| State              | Ownership/<br>Sector | Mode wise breakup |         |        |        |         |         |           |            |        | Grand<br>Total |
|                    |                      | Thermal           |         |        |        |         | Nuclear | Renewable |            |        |                |
|                    |                      | Coal              | Lignite | Gas    | Diesel | Total   |         | Hydro     | RES*(MNRE) | Total  |                |
| Assam              | State                | 0.00              | 0.00    | 306.36 | 0.00   | 306.36  | 0.00    | 100.00    | 5.01       | 105.01 | 411.37         |
|                    | Private              | 0.00              | 0.00    | 0.00   | 0.00   | 0.00    | 0.00    | 0.00      | 162.28     | 162.28 | 162.28         |
|                    | Central              | 581.52            | 0.00    | 435.56 | 0.00   | 1017.08 | 0.00    | 422.08    | 25.00      | 447.08 | 1464.16        |
|                    | Sub-Total            | 581.52            | 0.00    | 741.92 | 0.00   | 1323.44 | 0.00    | 522.08    | 192.29     | 714.37 | 2037.81        |
| Arunachal Pradesh  | State                | 0.00              | 0.00    | 0.00   | 0.00   | 0.00    | 0.00    | 0.00      | 109.11     | 109.11 | 109.11         |
|                    | Private              | 0.00              | 0.00    | 0.00   | 0.00   | 0.00    | 0.00    | 0.00      | 35.79      | 35.79  | 35.79          |
|                    | Central              | 37.05             | 0.00    | 46.82  | 0.00   | 83.87   | 0.00    | 544.55    | 0.00       | 544.55 | 628.42         |
|                    | Sub-Total            | 37.05             | 0.00    | 46.82  | 0.00   | 83.87   | 0.00    | 544.55    | 144.90     | 689.45 | 773.32         |
| Meghalaya          | State                | 0.00              | 0.00    | 0.00   | 0.00   | 0.00    | 0.00    | 322.00    | 55.03      | 377.03 | 377.03         |
|                    | Private              | 0.00              | 0.00    | 0.00   | 0.00   | 0.00    | 0.00    | 0.00      | 18.04      | 18.04  | 18.04          |
|                    | Central              | 51.60             | 0.00    | 109.69 | 0.00   | 161.29  | 0.00    | 95.38     | 0.00       | 95.38  | 256.67         |
|                    | Sub-Total            | 51.60             | 0.00    | 109.69 | 0.00   | 161.29  | 0.00    | 417.38    | 73.07      | 490.45 | 651.74         |
| Tripura            | State                | 0.00              | 0.00    | 105.00 | 0.00   | 105.00  | 0.00    | 0.00      | 16.01      | 16.01  | 121.01         |
|                    | Private              | 0.00              | 0.00    | 0.00   | 0.00   | 0.00    | 0.00    | 0.00      | 13.46      | 13.46  | 13.46          |
|                    | Central              | 56.00             | 0.00    | 381.94 | 0.00   | 437.94  | 0.00    | 68.49     | 5.00       | 73.49  | 511.43         |
|                    | Sub-Total            | 56.00             | 0.00    | 486.94 | 0.00   | 542.94  | 0.00    | 68.49     | 34.47      | 102.96 | 645.90         |
| Manipur            | State                | 0.00              | 0.00    | 0.00   | 36.00  | 36.00   | 0.00    | 0.00      | 5.45       | 5.45   | 41.45          |
|                    | Private              | 0.00              | 0.00    | 0.00   | 0.00   | 0.00    | 0.00    | 0.00      | 13.04      | 13.04  | 13.04          |

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|                                     |                    |               |             |                |              |                |             |                |               |                |                |
|-------------------------------------|--------------------|---------------|-------------|----------------|--------------|----------------|-------------|----------------|---------------|----------------|----------------|
|                                     | Central            | 47.10         | 0.00        | 81.58          | 0.00         | 128.68         | 0.00        | 87.24          | 0.00          | 87.24          | 215.92         |
|                                     | <b>Sub-Total</b>   | <b>47.10</b>  | <b>0.00</b> | <b>81.58</b>   | <b>36.00</b> | <b>164.68</b>  | <b>0.00</b> | <b>87.24</b>   | <b>18.49</b>  | <b>105.73</b>  | <b>270.41</b>  |
| <b>Nagaland</b>                     | State              | 0.00          | 0.00        | 0.00           | 0.00         | 0.00           | 0.00        | 0.00           | 32.67         | 32.67          | 32.67          |
|                                     | Private            | 0.00          | 0.00        | 0.00           | 0.00         | 0.00           | 0.00        | 0.00           | 3.17          | 3.17           | 3.17           |
|                                     | Central            | 32.10         | 0.00        | 73.93          | 0.00         | 106.03         | 0.00        | 66.33          | 0.00          | 66.33          | 172.36         |
|                                     | <b>Sub-Total</b>   | <b>32.10</b>  | <b>0.00</b> | <b>73.93</b>   | <b>0.00</b>  | <b>106.03</b>  | <b>0.00</b> | <b>66.33</b>   | <b>35.84</b>  | <b>102.17</b>  | <b>208.20</b>  |
| <b>Mizoram</b>                      | State              | 0.00          | 0.00        | 0.00           | 0.00         | 0.00           | 0.00        | 0.00           | 45.47         | 45.47          | 45.47          |
|                                     | Private            | 0.00          | 0.00        | 0.00           | 0.00         | 0.00           | 0.00        | 0.00           | 30.31         | 30.31          | 30.31          |
|                                     | Central            | 31.05         | 0.00        | 60.46          | 0.00         | 91.51          | 0.00        | 97.94          | 0.00          | 97.94          | 189.45         |
|                                     | <b>Sub-Total</b>   | <b>31.05</b>  | <b>0.00</b> | <b>60.46</b>   | <b>0.00</b>  | <b>91.51</b>   | <b>0.00</b> | <b>97.94</b>   | <b>75.78</b>  | <b>173.72</b>  | <b>265.23</b>  |
| <b>Central - Unallocated</b>        |                    | 112.50        | 112.60      | 0.00           | 63.62        | 0.00           | 176.22      | 0.00           | 140.00        | 0.00           | 140.00         |
| <b>Total (North-Eastern Region)</b> | State              | 0.00          | 0.00        | 411.36         | 36.00        | 447.36         | 0.00        | 422.00         | 268.75        | 690.75         | 1138.10        |
|                                     | Private            | 0.00          | 0.00        | 0.00           | 0.00         | 0.00           | 0.00        | 0.00           | 276.09        | 276.09         | 276.09         |
|                                     | Central            | 949.02        | 0.00        | 1253.60        | 0.00         | 2202.62        | 0.00        | 1522.01        | 30.00         | 1552.01        | 3754.63        |
|                                     | <b>Grand Total</b> | <b>949.02</b> | <b>0.00</b> | <b>1664.96</b> | <b>36.00</b> | <b>2649.98</b> | <b>0.00</b> | <b>1944.01</b> | <b>574.84</b> | <b>2518.85</b> | <b>5168.82</b> |
|                                     |                    |               |             |                |              |                |             |                |               |                |                |
|                                     |                    |               |             |                |              |                |             |                |               |                |                |

### INSTALLED CAPACITY (IN MW) OF POWER UTILITIES IN THE STATES/UTS LOCATED IN ISLANDS

### INCLUDING ALLOCATED SHARES IN JOINT & CENTRAL SECTOR UTILITIES

| INCLUDING ALLOCATED SHARES IN JOINT & CENTRAL SECTOR & REFINES |                      |                   |         |      |        |       |         |           |            |       |                |
|----------------------------------------------------------------|----------------------|-------------------|---------|------|--------|-------|---------|-----------|------------|-------|----------------|
| (As on 31.03.2024)                                             |                      |                   |         |      |        |       |         |           |            |       |                |
| State                                                          | Ownership/<br>Sector | Mode wise breakup |         |      |        |       |         |           |            |       | Grand<br>Total |
|                                                                |                      | Thermal           |         |      |        |       | Nuclear | Renewable |            |       |                |
|                                                                |                      | Coal              | Lignite | Gas  | Diesel | Total |         | Hydro     | RES*(MNRE) | Total |                |
| Andaman & Nicobar                                              | State                | 0.00              | 0.00    | 0.00 | 57.52  | 57.52 | 0.00    | 0.00      | 5.25       | 5.25  | 62.77          |
|                                                                | Private              | 0.00              | 0.00    | 0.00 | 35.19  | 35.19 | 0.00    | 0.00      | 24.81      | 24.81 | 60.00          |
|                                                                | Central              | 0.00              | 0.00    | 0.00 | 0.00   | 0.00  | 0.00    | 0.00      | 5.10       | 5.10  | 5.10           |
|                                                                | Sub-Total            | 0.00              | 0.00    | 0.00 | 92.71  | 92.71 | 0.00    | 0.00      | 35.16      | 35.16 | 127.87         |
| Lakshadweep                                                    | State                | 0.00              | 0.00    | 0.00 | 26.83  | 26.83 | 0.00    | 0.00      | 0.00       | 0.00  | 26.83          |
|                                                                | Private              | 0.00              | 0.00    | 0.00 | 0.00   | 0.00  | 0.00    | 0.00      | 4.97       | 4.97  | 4.97           |
|                                                                | Central              | 0.00              | 0.00    | 0.00 | 0.00   | 0.00  | 0.00    | 0.00      | 0.00       | 0.00  | 0.00           |
|                                                                | Sub-Total            | 0.00              | 0.00    | 0.00 | 26.83  | 26.83 | 0.00    | 0.00      | 4.97       | 4.97  | 31.80          |

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|                 |             |      |      |      |        |        |      |      |       |       |        |
|-----------------|-------------|------|------|------|--------|--------|------|------|-------|-------|--------|
| Total (Islands) | State       | 0.00 | 0.00 | 0.00 | 84.35  | 84.35  | 0.00 | 0.00 | 5.25  | 5.25  | 89.60  |
|                 | Private     | 0.00 | 0.00 | 0.00 | 35.19  | 35.19  | 0.00 | 0.00 | 29.78 | 29.78 | 64.97  |
|                 | Central     | 0.00 | 0.00 | 0.00 | 0.00   | 0.00   | 0.00 | 0.00 | 5.10  | 5.10  | 5.10   |
|                 | Grand Total | 0.00 | 0.00 | 0.00 | 119.54 | 119.54 | 0.00 | 0.00 | 40.13 | 40.13 | 159.67 |
|                 |             |      |      |      |        |        |      |      |       |       |        |
|                 |             |      |      |      |        |        |      |      |       |       |        |

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